

# Specifications

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## **C-10 and C-12 Truck Engines**

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MBJ1-Up (Engine)  
MBL1-Up (Engine)



## Important Safety Information

Most accidents that involve product operation, maintenance and repair are caused by failure to observe basic safety rules or precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards. This person should also have the necessary training, skills and tools to perform these functions properly.

**Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.**

**Do not operate or perform any lubrication, maintenance or repair on this product, until you have read and understood the operation, lubrication, maintenance and repair information.**

Safety precautions and warnings are provided in this manual and on the product. If these hazard warnings are not heeded, bodily injury or death could occur to you or to other persons.

The hazards are identified by the "Safety Alert Symbol" and followed by a "Signal Word" such as "DANGER", "WARNING" or "CAUTION". The Safety Alert "WARNING" label is shown below.



The meaning of this safety alert symbol is as follows:

**Attention! Become Alert! Your Safety is Involved.**

The message that appears under the warning explains the hazard and can be either written or pictorially presented.

Operations that may cause product damage are identified by "NOTICE" labels on the product and in this publication.

**Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. If a tool, procedure, work method or operating technique that is not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that the product will not be damaged or be made unsafe by the operation, lubrication, maintenance or repair procedures that you choose.**

The information, specifications, and illustrations in this publication are on the basis of information that was available at the time that the publication was written. The specifications, torques, pressures, measurements, adjustments, illustrations, and other items can change at any time. These changes can affect the service that is given to the product. Obtain the complete and most current information before you start any job. Caterpillar dealers have the most current information available.



**When replacement parts are required for this product Caterpillar recommends using Caterpillar replacement parts or parts with equivalent specifications including, but not limited to, physical dimensions, type, strength and material.**

**Failure to heed this warning can lead to premature failures, product damage, personal injury or death.**

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## Specifications Section

# Engine Design

**SMCS Code:** 1201

**S/N:** MBJ1-Up

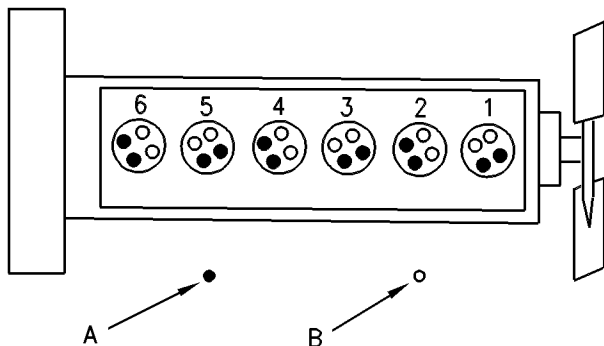


Illustration 1 g00386924

## Cylinder And Valve Location

(A) Exhaust valves and (B) Inlet valves

Bore ..... 125.0 mm (4.92 inch)

Stroke ..... 140.0 mm (5.51 inch)

Displacement ..... 10.3 L (629 in<sup>3</sup>)

Cylinder arrangement ..... In-line

Valves per cylinder ..... 4

In order to check the engine valve lash setting, the engine must be cold and the engine must be stopped. Engine valve lash settings

|                                |                        |
|--------------------------------|------------------------|
| Inlet .....                    | 0.38 mm (.015 inch)    |
| Exhaust .....                  | 0.64 mm (.025 inch)    |
| Engine compression brake ..... | 0.64 mm<br>(.025 inch) |

Type of combustion ..... Direct Injection

Firing Order ..... 1 - 5 - 3 - 6 - 2 - 4

The crankshaft rotation is viewed from the flywheel end of the engine. Crankshaft rotation ..... Counterclockwise

**Note:** The front end of the engine is opposite of the flywheel end of the engine. The left side of the engine and the right side of the engine are viewed from the flywheel end of the engine. The No. 1 cylinder is the front cylinder.

## Engine Design

**SMCS Code:** 1201

**S/N:** MBL1-Up

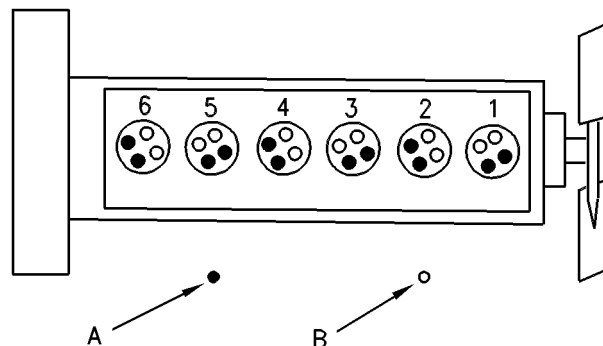


Illustration 2 g00386924

### Cylinder And Valve Location

(A) Exhaust valves and (B) Inlet valves

Bore ..... 130.0 mm (5.12 inch)

Stroke ..... 150.0 mm (5.91 inch)

Displacement ..... 12 L (732 in<sup>3</sup>)

Cylinder arrangement ..... In-line

Valves per cylinder ..... 4

In order to check the engine valve lash setting, the engine must be cold and the engine must be stopped. Engine valve lash settings

|                                |                        |
|--------------------------------|------------------------|
| Inlet .....                    | 0.38 mm (.015 inch)    |
| Exhaust .....                  | 0.64 mm (.025 inch)    |
| Engine compression brake ..... | 1.02 mm<br>(.040 inch) |

Type of combustion ..... Direct Injection

Firing Order ..... 1 - 5 - 3 - 6 - 2 - 4

The crankshaft rotation is viewed from the flywheel end of the engine. Crankshaft rotation ..... Counterclockwise

**Note:** The front end of the engine is opposite of the flywheel end of the engine. The left side of the engine and the right side of the engine are viewed from the flywheel end of the engine. The No. 1 cylinder is the front cylinder.

i01796722

## Fuel Transfer Pump

SMCS Code: 1256

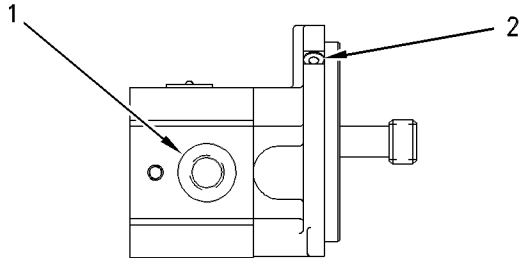


Illustration 3

g00280095

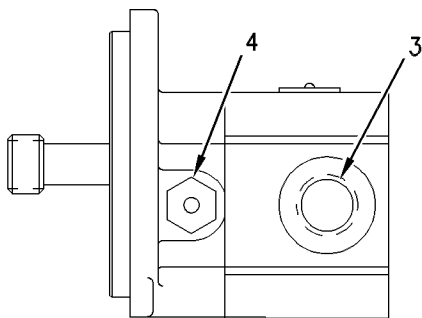


Illustration 4

g00280096

- (1) Outlet port
- (2) Drain hole
- (3) Inlet port
- (4) Pressure regulating valve

When the fuel transfer pump is using diesel fuel, the fuel transfer pump has the following specifications.

Run the pump at 2538 rpm. The full bypass pressure should be the following value. .... 682 to 751 kPa  
(99 to 109 psi)

Run the pump at 846 rpm. The flow at 482 kPa (70 psi) must be 3.07 L/min (0.811 US gpm).

Run the pump at 3384 rpm. The flow at 585.8 to 654.6 kPa (85 to 90 psi) must be 7.09 L/min (1.87 US gpm).

Run the pump at 147 rpm with a blocked inlet. Run the pump for 30 seconds. The suction of the pump is required to be 60.84 kPa (18 inches hg).

When the pump is viewed from the drive end the rotation of the pump is counterclockwise.

i01774922

## Fuel Lines

SMCS Code: 1274

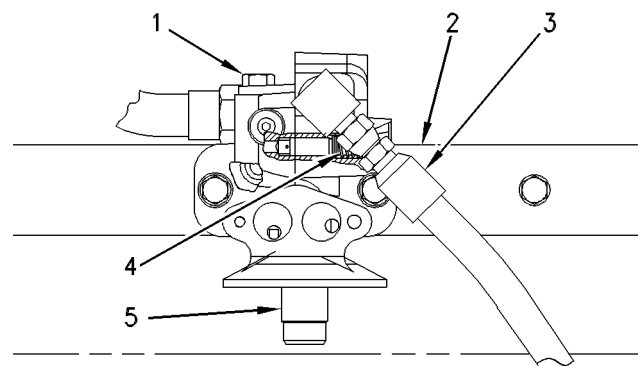


Illustration 5

g00281513

Typical example

- (1) Plug
- (2) Fuel manifold
- (3) Supply hose assembly
- (4) 104-1729 Spring

Assembled length ..... 18.4 mm (0.72 inch)  
Operating length (minimum) ..... 15.4 mm  
(0.61 inch)  
Free length after test ..... 30.2 ± 0.7 mm  
(1.19 ± 0.03 inch)  
Outside diameter ..... 11.3 mm (0.44 inch)

- (5) Stud

Apply 9S-3263 Thread Lock Compound on the taperlock end of the stud to the following length. .... 7.6 mm (0.30 inch)  
Tighten the stud to the following torque. .... 70 ± 15 N·m (50 ± 11 lb ft)

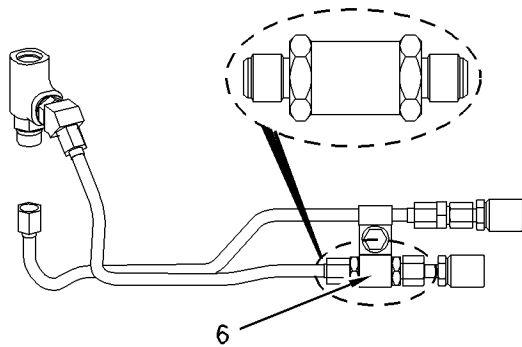


Illustration 6

g00907296

- (6) The return fuel line has a fuel regulator valve that is operated by the temperature of the fuel.

The valve starts to move at the following temperature. .... 70 °C (158 °F)

The valve is at the maximum movement at the following temperature. .... 80 °C (176 °F)

The maximum operating temperature of the valve is at the following temperature. .... 120 °C (248 °F)

i00582898

## Fuel Filter and Water Separator

**SMCS Code:** 1261; 1263

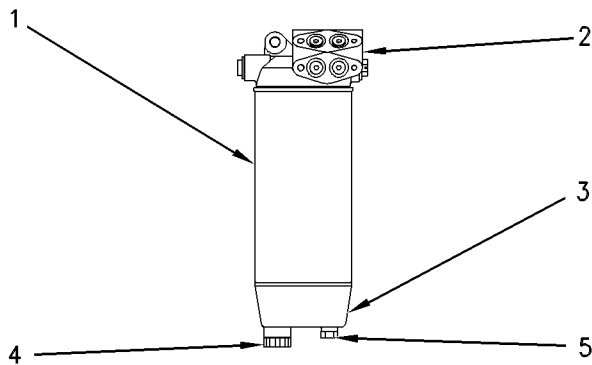


Illustration 7

g00281520

- (1) Ten micron filter element  
(2) Fuel filter base assembly

Coat the threads that are on the stud 7.7 mm (0.30 inch) from the taperlock. Use **4C-4032** Bearing Mount Compound on the threads. Torque for stud ..... 70 ± 15 N·m (52 ± 11 lb ft)

- (3) Hand tighten the bowl.

- (4) To open the self-ventilated drain, turn the plug counterclockwise.

- (5) Plug

i01197996

## Fuel Priming Pump and Primary Fuel Filter

**SMCS Code:** 1258; 1261

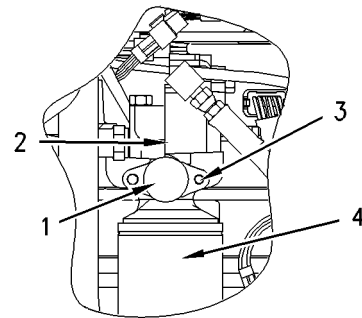


Illustration 8

g00640576

- (1) Fuel priming pump  
(2) Primary fuel filter base  
(3) Tighten the bolt to the following torque: ..... 28 ± 7 N·m (21 ± 5 lb ft)  
(4) Primary filter element

i01819464

## Electronic Unit Injector

**SMCS Code:** 1290

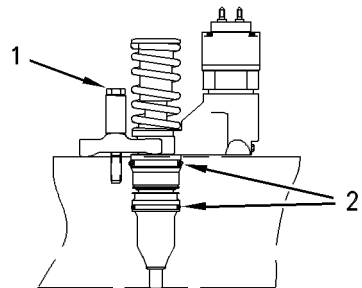


Illustration 9

g00403738

- (1) Bolt

1. Tighten the bolt to the following torque  
30 ± 7 N·m (22 ± 5 lb ft).
  2. Loosen the bolt until the clamp is loose.
  3. Tighten the bolt to the following torque  
30 ± 7 N·m (22 ± 5 lb ft).
- (2) Lubricate the seals and the bore with a 50 percent mix of **8T-2998** Lubricant and engine oil.

i01175866

## Electronic Unit Injector Mechanism

**SMCS Code:** 1290

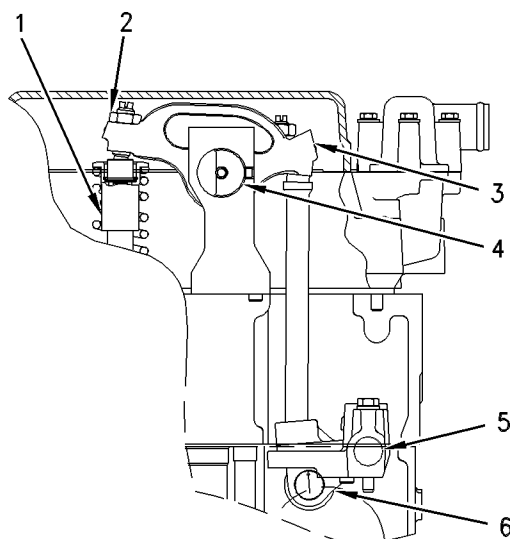


Illustration 10

g00280175

- (1) Electronic unit injector

The adjusting instructions for the electronic unit injector are listed below.

1. Loosen locknut (2).
2. Turn the electronic unit injector rocker arm adjusting screw clockwise. Stop turning the adjusting screw when contact is made with the electronic unit injector.
3. Turn the adjusting screw in the clockwise direction.  
  
Turn ..... 180 degrees (1/2 of a turn)
4. Hold the adjusting screw in this position. Tighten locknut (2).

Tighten the locknut to the following torque. .... 55 ± 10 N·m (40 ± 7 lb ft)

- (3) Rocker arm
- (4) Rocker arm shaft

Diameter of new rocker arm shaft ..... 34.000 ± 0.010 mm (1.3386 ± .0004 inch)

- (5) Lifter group shaft

Diameter of new lifter group shaft ..... 20.000 ± 0.010 mm (.7874 ± .0004 inch)

The bore diameter in the body of the electronic unit injector lifter assembly for the lifter group shaft is the following value. .... 20.050 ± 0.015 mm (0.7894 ± 0.0006 inch)

- (6) Electronic unit injector lifter

i01175875

## Electronic Unit Injector Rocker Arm

**SMCS Code:** 1123

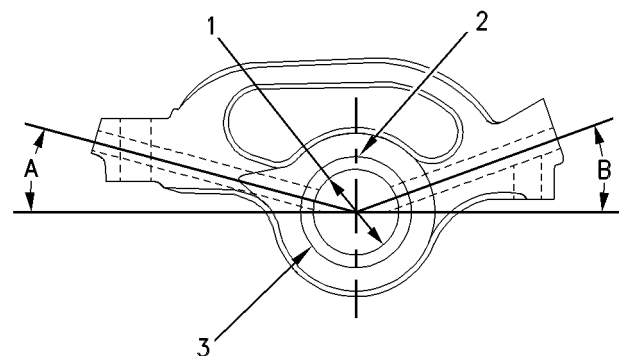


Illustration 11

g00280315

- (1) Bore in rocker arm bearing for shaft ..... 34.050 ± 0.015 mm (1.3405 ± 0.0006 inch)

The rocker arm bearing must not extend beyond either face of the rocker arm.

- (2) Bearing joint

Oil passage angle (A) ..... 15 degrees

Oil passage angle (B) ..... 20 degrees

Assemble the bearing. The oil hole openings that are in the bearing must be centered over the oil passages that are in the rocker arm. Centers must be within the following value. .... 2.3 mm (.09 inch)

- (3) Bore in the rocker arm for the bearing .....  $37.000 \pm 0.020$  mm  
( $1.4567 \pm 0.0008$  inch)

i01237756

## Electronic Unit Injector Wiring

SMCS Code: 1290

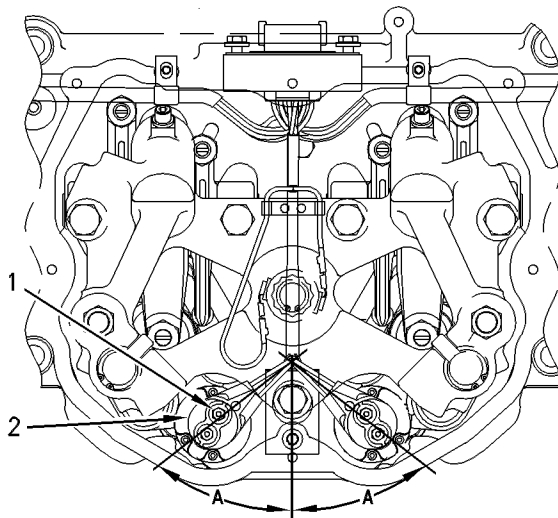


Illustration 12

g00280453

- (1) Tighten the nuts that are on the cap assembly to the following torque. ....  $2.5 \pm 0.25$  N·m  
( $22 \pm 2$  lb in)

(2) Cap assembly

Angle A ..... 52 degrees

Position connectors so that the wires are not contacting any other surfaces.

## Lifter Group

SMCS Code: 1209

i01175911

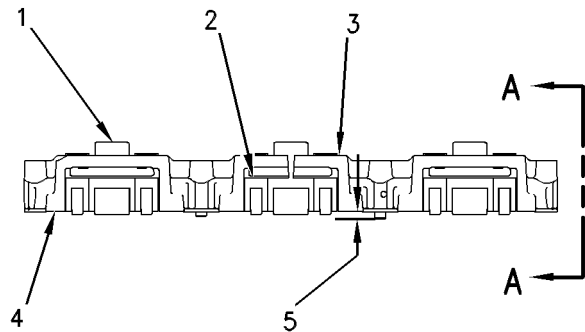


Illustration 13

g00280305

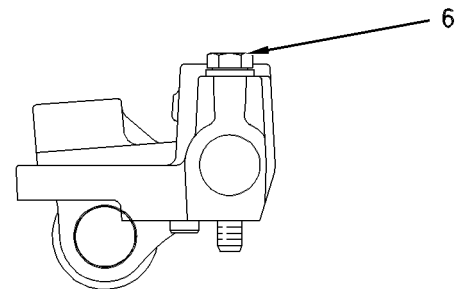


Illustration 14

g00288322

View A-A

- (1) Electronic unit injector lifter
- (2) Valve lifter
- (3) Oil supply lifter bracket assembly
- (4) Lifter bracket assembly
- (5) The dowels extend below the surface of the lifter bracket assembly. Distance .....  $4.0 \pm 1.0$  mm  
( $0.16 \pm 0.04$  inch)
- (6) Tighten the bolts to the following torque. ....  $30 \pm 3$  N·m ( $22 \pm 2$  lb ft)

i00887793

## Rocker Shaft

SMCS Code: 1102

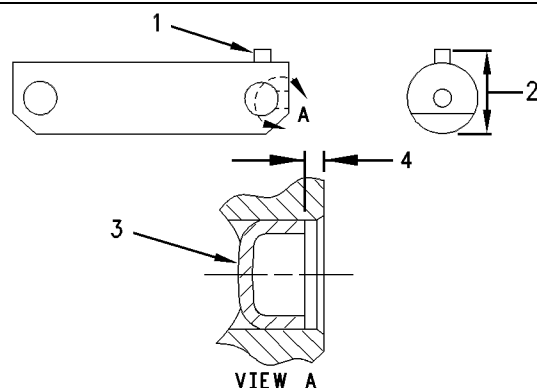


Illustration 15

g00280329

- (1) Pin
- (2) Distance from the top surface of the pin to the opposite side of the shaft .....  $40.0 \pm 1.0$  mm ( $1.57 \pm 0.04$  inch)
- (3) The plug is installed into the end of the shaft.
- (4) Install the plug to the following depth. ....  $1.25 \pm 0.25$  mm ( $0.049 \pm 0.010$  inch)

i01896192

## Valve Mechanism

SMCS Code: 1102

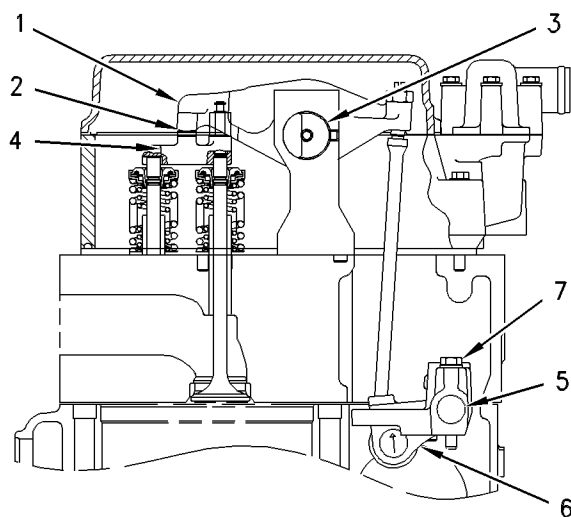


Illustration 16

g00986490

- (1) Valve rocker arm

- (2) Valve lash (engine stopped)

Inlet valves .....  $0.38 \pm 0.08$  mm ( $0.015 \pm 0.003$  inch)

Exhaust valves .....  $0.64 \pm 0.08$  mm ( $0.025 \pm 0.003$  inch)

**Note:** After setting the valve lash, tighten the adjusting screw locknut. Recheck the valve lash.

Torque for the adjusting screw locknut ..  $30 \pm 7$  N·m ( $22 \pm 5$  lb ft)

- (3) Rocker arm shaft

Diameter of new rocker arm shaft .....  $34.000 \pm 0.010$  mm ( $1.3386 \pm 0.0004$  inch)

- (4) Valve bridge

- (5) Lifter group shaft

Diameter of new lifter group shaft .....  $20.000 \pm 0.010$  mm ( $0.7874 \pm 0.0004$  inch)

Bore diameter in the body of the valve lifter .....  $20.050 \pm 0.015$  mm ( $0.7894 \pm 0.0006$  inch)

- (6) Valve lifter

- (7) Tighten the bolts to the following torque. ....  $30 \pm 3$  N·m ( $22 \pm 2$  lb ft)

i00573207

## Valve Mechanism Cover

SMCS Code: 1107

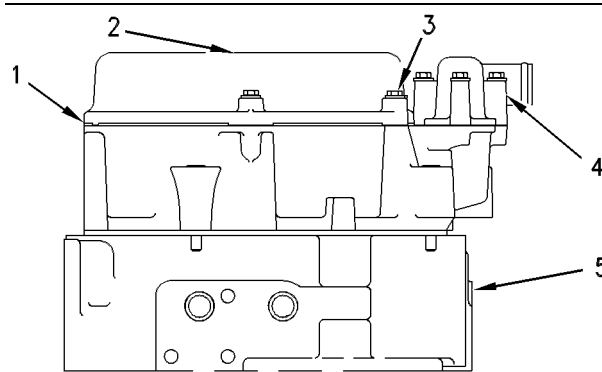


Illustration 17

g00280491

- (1) Seal

Cut the seal to length for each of the three valve mechanism covers. Coat the joint surface with **3S-6252 Sealant**.

(2) Cover

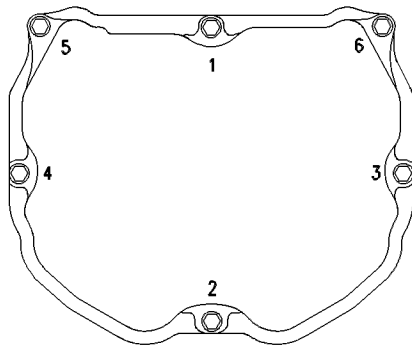


Illustration 18

g00280493

(3) Tighten bolts 1 through 6 in a numerical sequence to the following torque. ...  $12 \pm 3$  N·m  
( $9 \pm 2$  lb ft)

Again tighten bolts 1 through 6 in a numerical sequence to the following torque. ....  $12 \pm 3$  N·m  
( $9 \pm 2$  lb ft)

(4) Breather assembly

The conditions that follow will allow the crankcase pressure to be less than 0.4 kPa (0.06 psi).

Inside diameter of fumes disposal  
tube ..... 31.8 mm (1.25 inch)  
Blowby per hour ..... 19.82 cubic meters

(5) Cylinder head

# Cylinder Head Valves

SMCS Code: 1105

i00953496

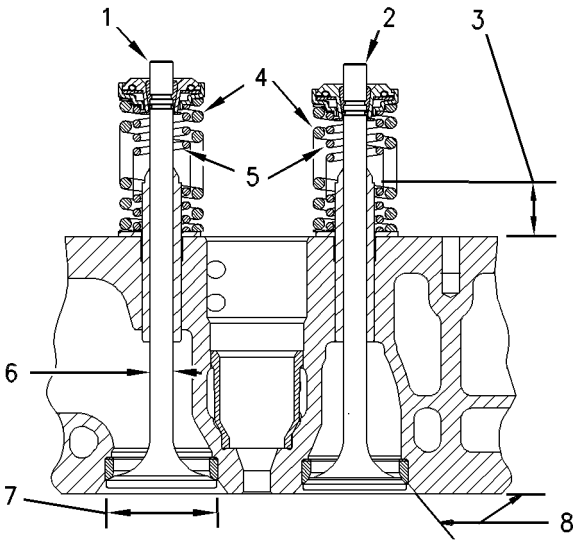


Illustration 19

g00448911

**Note:** Coat the inlet valve stems and the exhaust valve stems with **8T-2998** Lubricant prior to installation into the cylinder head.

- (1) Inlet valve
- (2) Exhaust valve
- (3) Height to the step that is in the valve guide .....  $22.00 \pm 0.50$  mm ( $.866 \pm .020$  inch)
- (4) **7W-7082** Spring
- Assembled Length ..... 51.69 mm (2.035 inch)  
Load at assembled length .....  $240 \pm 24$  N  
( $54 \pm 5$  lb)  
Operating length (minimum) ..... 37.72 mm  
(1.485 inch)  
Load at minimum operating length ..  $650 \pm 32$  N  
( $146 \pm 7$  lb)  
Free length after test ..... 59.89 mm (2.358 inch)  
Outside diameter ..... 34.00 mm (1.339 inch)
- (5) **7W-7083** Spring
- Assembled Length ..... 49.19 mm (1.937 inch)  
Load at assembled length .....  $136 \pm 14$  N  
( $31 \pm 3$  lb)  
Operating length (minimum) ..... 35.22 mm  
(1.387 inch)  
Load at minimum operating length ..  $292 \pm 15$  N  
( $66 \pm 3$  lb)  
Free length after test ..... 61.39 mm (2.417 inch)

Outside diameter ..... 23.24 mm (.915 inch)

- (6) New valve stem diameter ....  $9.441 \pm 0.008$  mm  
(.3717  $\pm$  .0003 inch)

The valves can be reused if the minimum dimensions for the valve stems are met.

Inlet ..... 9.408 mm (.3704 inch)  
Exhaust ..... 9.408 mm (.3704 inch)

#### Specifications for the valve guides

After installation into the cylinder head, the valve guide bore is the following value. ..  $9.484 \pm 0.026$  mm (.3734  $\pm$  .0010 inch)  
The valve guide can be reused. The valve guide bore must not be larger than the following value. .... 9.538 mm (.3755 inch)

- (7) Diameter of valve head

Inlet valve .....  $45.00 \pm 0.13$  mm  
(1.772  $\pm$  .005 inch)  
Exhaust valve .....  $42.00 \pm 0.13$  mm  
(1.654  $\pm$  .005 inch)

- (8) Angle of face of valve seat insert

Inlet valve .....  $29 \frac{3}{4} \pm 1/4$  degrees  
Exhaust valve .....  $44 \frac{3}{4} \pm 1/4$  degrees

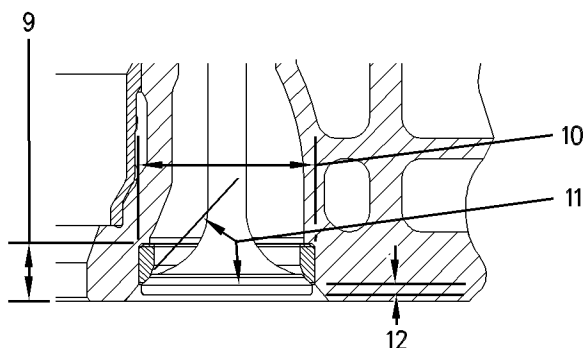


Illustration 20

g00282169

- (9) Depth of the bore in the cylinder head for the valve seat insert

Inlet .....  $15.00 \pm 0.05$  mm (.591  $\pm$  .004 inch)  
Exhaust .....  $14.10 \pm 0.05$  mm (.555  $\pm$  .004 inch)

**Note:** Shrink the valve seat inserts by reducing the temperature. Shrinking the valve seat inserts allows placement into the counterbore.

- (10) Valve seat insert

Diameter of valve seat insert (inlet valve) .....  $46.025 \pm 0.013$  mm  
(1.8120  $\pm$  .0005 inch)

Bore in cylinder head for valve seat insert (inlet valve) .....  $45.961 \pm 0.013$  mm  
(1.8095  $\pm$  .0005 inch)

Diameter of valve seat insert (exhaust valve) .....  $43.390 \pm 0.015$  mm  
(1.7083  $\pm$  .0006 inch)

Bore in cylinder head for valve seat insert (exhaust valve) .....  $43.320 \pm 0.025$  mm  
(1.7055  $\pm$  .0010 inch)

- (11) Angle of valve face

Angle of the face of the inlet valve .....  $29 \frac{1}{4} \pm 1/4$  degrees  
Angle of the face of the exhaust valve .....  $44 \frac{1}{4} \pm 1/4$  degrees

- (12) Minimum thickness of the valve lip

Inlet valve ..... 2.51 mm (.099 inch)  
Exhaust valve ..... 2.03 mm (.080 inch)

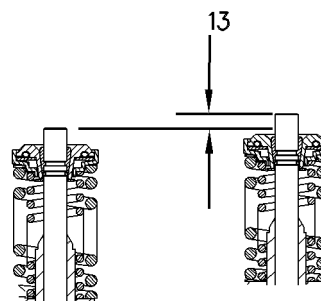


Illustration 21

g00472047

- (13) Maximum variation of the valve stem length

Inlet valve to inlet valve ..... 0.66 mm (.026 inch)  
Exhaust valve to exhaust valve ..... 0.66 mm  
(.026 inch)

i01234015

## Cylinder Head

SMCS Code: 1100

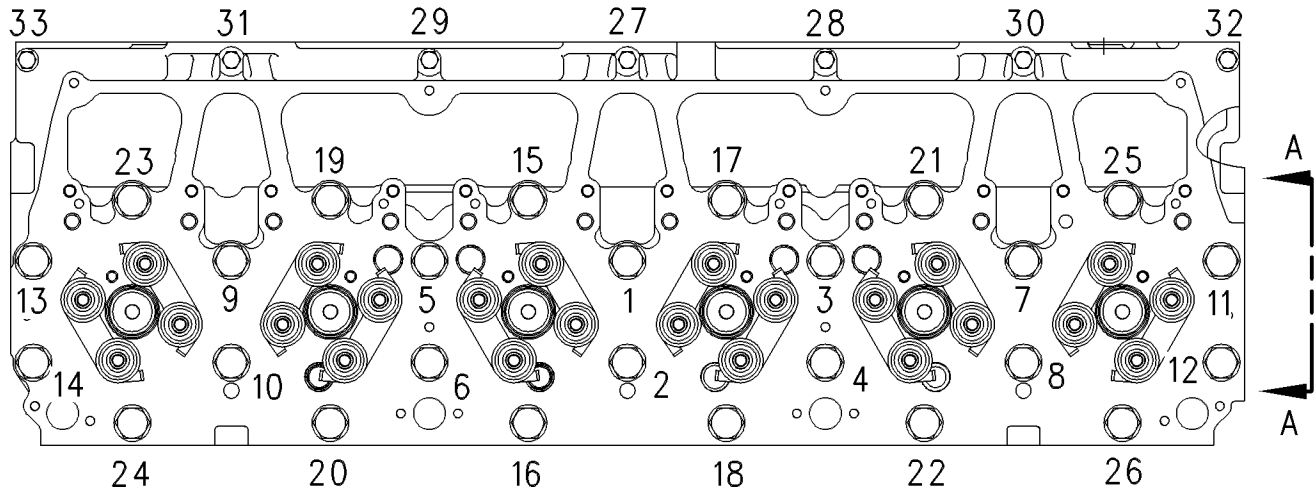


Illustration 22

g00280510

**Note:** Lubricate the bolt threads, the underside of the bolt heads, and the washers with **6V-4876** Lubricant prior to assembly.

Use the following procedure in order to tighten the cylinder head bolts:

1. Tighten bolt (1) through bolt (26) in a numerical sequence.

Tighten the bolts to the following torque. ....  $160 \pm 15 \text{ N}\cdot\text{m}$  ( $120 \pm 11 \text{ lb ft}$ )

2. Tighten bolt (1) through bolt (26) again in a numerical sequence.

Tighten the bolts again to the following torque. ....  $160 \pm 15 \text{ N}\cdot\text{m}$  ( $120 \pm 11 \text{ lb ft}$ )

3. Place a mark on bolt (1) through bolt (26). Rotate bolt (1) through bolt (26) in a numerical sequence.

Rotate the bolts in the clockwise direction. .... 90 degrees (1/4 turn)

4. **Loosen bolt (1) through bolt (26) until the washers are loose under the bolt heads.**

5. Tighten bolt (1) through bolt (26) in a numerical sequence.

Tighten the bolts to the following torque. ....  $160 \pm 15 \text{ N}\cdot\text{m}$  ( $120 \pm 11 \text{ lb ft}$ )

6. Tighten bolt (1) through bolt (26) again in a numerical sequence.

Tighten the bolts again to the following torque. ....  $160 \pm 15 \text{ N}\cdot\text{m}$  ( $120 \pm 11 \text{ lb ft}$ )

7. Place a mark on bolt (1) through bolt (26). Rotate bolt (1) through bolt (26) in a numerical sequence.

Rotate the bolts in the clockwise direction. .... 90 degrees (1/4 turn)

8. Tighten bolt (27) through bolt (33) in a numerical sequence.

Tighten the bolts to the following torque. ....  $28 \pm 7 \text{ N}\cdot\text{m}$  ( $20 \pm 5 \text{ lb ft}$ )

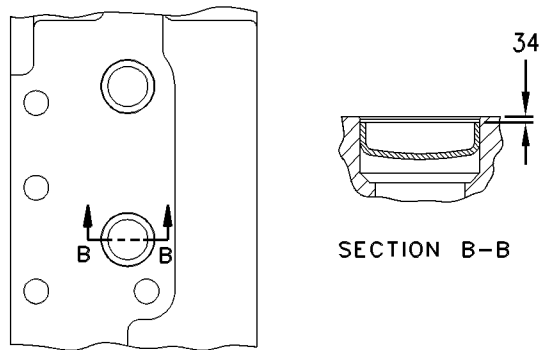


Illustration 23  
View A-A

g00284618

(34) The cup plugs are measured from the head face to the top edge of the plug. Depth of installation .....  $1.25 \pm 0.25$  mm ( $0.049 \pm 0.010$  inch)

Height of new cylinder head .....  $105.00 \pm 0.15$  mm ( $4.134 \pm 0.006$  inch)

Minimum permissible thickness of cylinder head .....  $104.35$  mm ( $4.108$  inch)

**Note:** The flatness of the cylinder head should be within  $0.15$  mm ( $0.006$  inch). The flatness should also be a maximum of  $0.05$  mm ( $0.002$  inch) for any  $150.00$  mm ( $5.906$  inch) span.

i01895140

## Compression Brake

**SMCS Code:** 1119; 1129

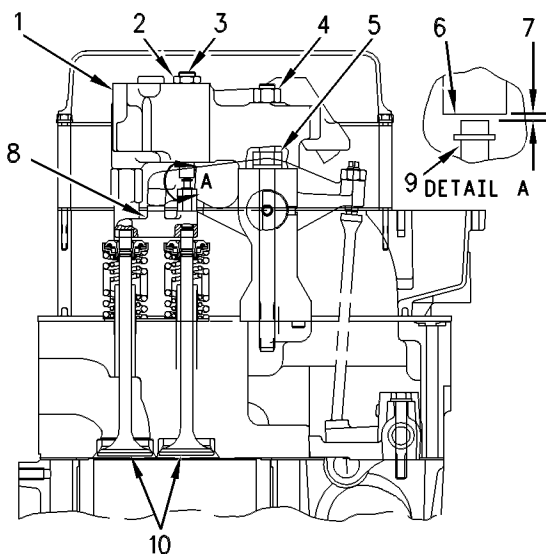


Illustration 24

g00517802

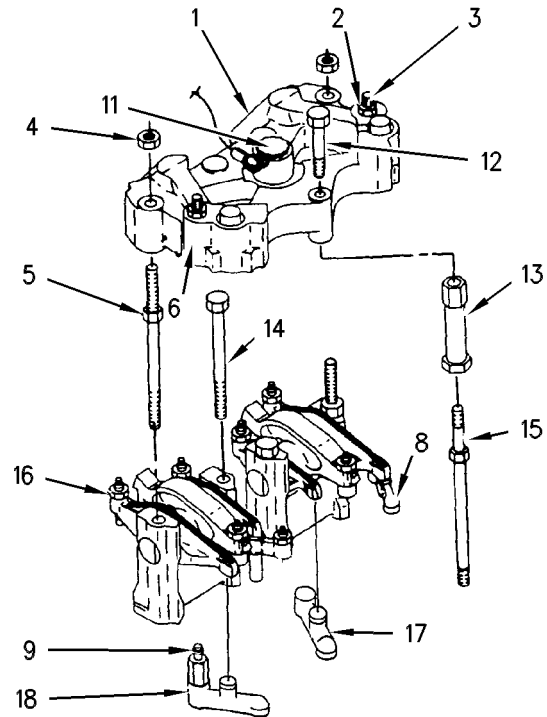


Illustration 25

g00517803

(1) Compression brake

(2) Locknut

Tighten the locknut to the following torque. ....  $35 \pm 8$  N·m ( $26 \pm 6$  lb ft)

(3) Slave piston adjustment screw

(4) Hold-down nut

Tighten the hold-down nuts to the following torque. ....  $80 \pm 15$  N·m ( $60 \pm 11$  lb ft)

(5) Stud

Tighten the studs to the following torque. ....  $100 \pm 20$  N·m ( $75 \pm 15$  lb ft)

**Note:** Prior to installation of the compression brake, install stud (5) and tighten stud (5) to the recommended torque.

(6) Slave piston

(7) Slave piston lash setting

**Note:** The valve lash must be adjusted before adjusting the slave piston lash.

Table 1

| Slave Piston Lash Setting<br>Jake Brake Compression Brakes |                                        |
|------------------------------------------------------------|----------------------------------------|
| Serial Number Prefix                                       | Slave Piston Lash Setting              |
| 3CS1-UP<br>2PN1-UP<br>8YS1-UP                              | 0.64 ± 0.08 mm<br>(0.025 ± 0.003 inch) |
| 2KS1-UP<br>9NS1-UP                                         | 1.02 ± 0.08 mm<br>(0.040 ± 0.003 inch) |
| 1YN1-UP                                                    | 1.14 ± 0.08 mm<br>(0.045 ± 0.003 inch) |
| MBJ1-UP<br>MBL1-UP                                         | 1.40 ± 0.08 mm<br>(0.055 ± 0.003 inch) |

Adjust the slave piston lash while the engine is stopped and the exhaust valves are closed.

Refer to the Testing and Adjusting, "Compression Brake" topic for the adjustment procedure.

(8) Exhaust valve bridge

(9) Screw assembly (if equipped)

Apply **4C-9507** Retaining Compound to the threads of the screw assembly.

Tighten the locknut to the following torque. .... 25 ± 7 N·m (18 ± 5 lb ft)

(10) Exhaust valves

(11) Solenoid valve

There are two different designs of solenoid valves. The difference can be determined by the number of points on the head of the solenoid valve. Use the correct torque value for your solenoid valve.

Tighten the solenoid valve with a twelve point head to the following torque. ... 12.5 N·m (9 lb ft)

Tighten the solenoid valve with a six point head to the following torque. .... 20 N·m (15 lb ft)

(12) Bolt

Tighten the bolt to the following torque. .... 55 ± 10 N·m (41 ± 7 lb ft)

(13) Spacer

Tighten the spacer to the following torque. .... 100 ± 20 N·m (75 ± 15 lb ft)

(14) Bolt

Tighten the bolt to the following torque. .... 100 ± 20 N·m (75 ± 15 lb ft)

(15) Head bolt stud

Refer to the Specifications, "Cylinder Head" topic for the tightening procedure.

(16) Adjusting screw locknut for the valve lash clearance adjustment

Refer to the Specifications, "Valve Mechanism" topic for the tightening procedure.

(17) Inlet valve bridge

(18) Exhaust valve bridge

i01764249

# Turbocharger

**SMCS Code:** 1052

**Part Numbers Are For The Following Illustrations:**  
**222-4753, 222-4754, 222-4756, 222-4757, and 224-7102**

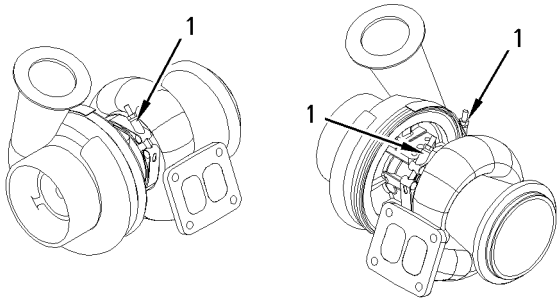


Illustration 26

g00859291

Typical example

(1) Clamp

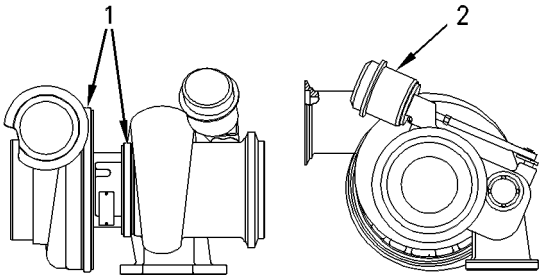


Illustration 27

g00509170

Typical example of a turbocharger with a wastegate

(1) Clamp

(2) Wastegate

### (1) Clamps

To tighten clamps (1) on the turbocharger, use the following procedure:

#### 1. Tighten the clamp.

Tighten the clamp to the following torque. ....  $14 \pm 1 \text{ N}\cdot\text{m}$  ( $10.3 \pm 0.7 \text{ lb ft}$ )

#### 2. Gently tap the clamp with a soft faced hammer.

#### 3. Again tighten the clamp.

Tighten the clamp to the following torque. ....  $14 \pm 1 \text{ N}\cdot\text{m}$  ( $10.3 \pm 0.7 \text{ lb ft}$ )

### (2) Wastegate

Opening pressure of the wastegate ..... 138 kPa  
(20 psi)

**Part Numbers Are For The Following Illustrations:**  
**123-0460, 132-3168, 144-0816, 148-2016, 151-3026,**  
**154-8677, 155-1801, 155-7953, 164-3710, 165-7311,**  
**172-6472, 185-5732, 189-9169, 189-9170, 189-9171,**  
**189-9172, 191-0095, 191-2925, 192-2076, 192-2078,**  
**195-6593, and 208-6641**

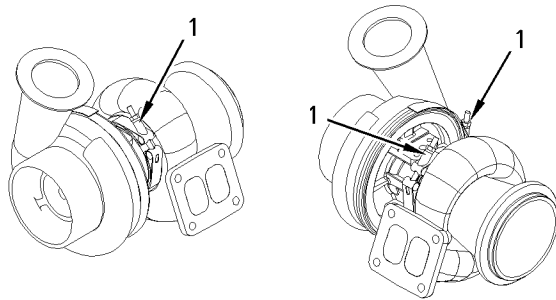


Illustration 28  
Typical example  
(1) Clamp

g00859291

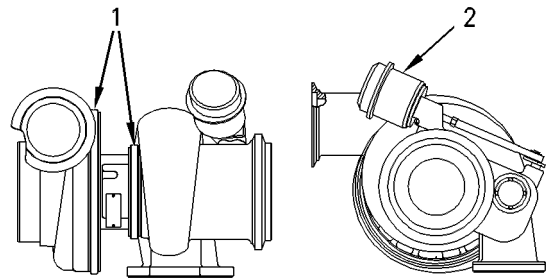


Illustration 29

g00509170

Typical example of a turbocharger with a wastegate

- (1) Clamp
- (2) Wastegate

### (1) Clamps

To tighten clamps (1) on the turbocharger, use the following procedure:

#### 1. Tighten the clamp.

Tighten the clamp to the following torque. ....  $14 \pm 1 \text{ N}\cdot\text{m}$  ( $10.3 \pm 0.7 \text{ lb ft}$ )

#### 2. Gently tap the clamp with a soft faced hammer.

#### 3. Again tighten the clamp.

Tighten the clamp to the following torque. ....  $14 \pm 1 \text{ N}\cdot\text{m}$  ( $10.3 \pm 0.7 \text{ lb ft}$ )

### (2) Wastegate

Opening pressure of the wastegate ..... 193 kPa  
(28 psi)

**Part Numbers Are For The Following Illustrations:**  
**148-2044, 189-9209, 190-6209, 191-5108,**  
**192-2079, 219-2225, 219-2226, 219-2227, 219-2228,**  
**and 222-1448**

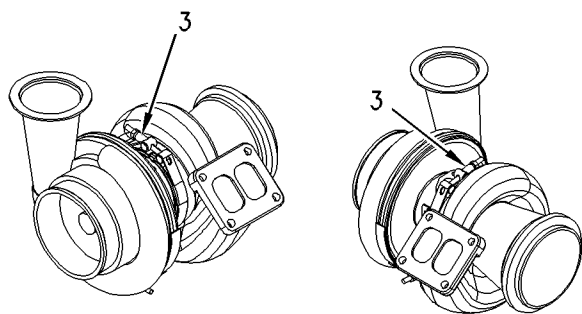


Illustration 30

g00859334

Typical example

(3) Clamp

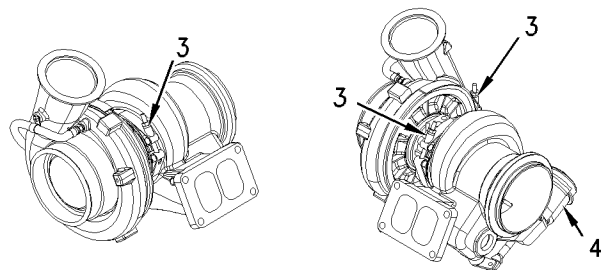


Illustration 31

g00859319

Typical example of a turbocharger with a wastegate

(3) Clamp

(4) Wastegate

(3) Clamp

To tighten the clamp (3) on the turbocharger, use the procedure that follows.

**1. Tighten the clamp.**

Tighten the clamp to the following torque. .... 18 N·m (13.3 lb ft)

**2. Loosen the clamp.**

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

**3. Tighten the clamp.**

Tighten the clamp to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(4) Wastegate

Opening pressure of the wastegate ..... 156 kPa (23 psi)

**Part Numbers Are For The Following Illustration:  
107-3580, 107-5775, 157-0075, 133-7027, 136-4835,  
and 195-6592**

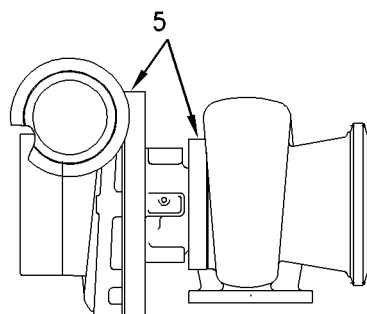


Illustration 32

g00859564

Typical example

(5) Clamp

(5) Clamps

To tighten clamps (5) on the turbocharger, use the following procedure:

**1. Tighten the clamp.**

Tighten the clamp to the following torque. .... 18 ± 3 N·m (13.3 ± 2.2 lb ft)

**2. Gently tap the clamp with a soft faced hammer.**

**3. Again tighten the clamp.**

Tighten the clamp to the following torque. .... 18 ± 3 N·m (13.3 ± 2.2 lb ft)

**Part Numbers Are For The Following Illustration:  
130-9201, and 117-2687**

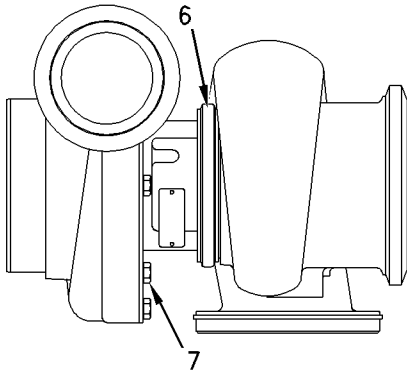


Illustration 33 g00859569

Typical example

- (6) Clamp  
(7) Bolt

#### (6) Clamp

To tighten the clamp (6) on the turbocharger, use the following procedure:

##### 1. Tighten the clamp.

Tighten the clamp nut to the following torque. .... 18 N·m (13.3 lb ft)

##### 2. Loosen the clamp.

Loosen the clamp nut to the following torque. .... 6 N·m (53.1 lb in)

##### 3. Tighten the clamp.

Tighten the clamp nut to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

#### (7) Bolt

Tighten the bolts (7) for the compressor clamp assembly in opposite pairs. Tighten the bolts for the compressor clamp assembly to the following torque 17.5 ± 1 N·m (12.90 ± 0.70 lb ft).

Apply 5P-3931 Anti-Seize Compound to the nuts that are used to mount the turbocharger. Tighten the nuts to the following torque 70 ± 15 N·m (51.6 ± 11.0 lb ft).

**Part Numbers Are For The Following Illustration:**  
129-2879, 132-3174, 135-5392, 135-5367, 135-5394, 145-5184, 151-3082, 151-5937, 155-7952, 162-6495, 162-7603, 164-9693, 164-9490, 164-9496, 169-4675, 169-4271, 173-002, 189-9210, 189-9211, 189-9212, 189-9213, 190-6205, 190-6206, 190-6207, 190-6208, 190-6210, 190-6211, 190-6212, 190-6213, 190-6215, 191-2926, 192-2077, 194-1116, 194-1117, 196-2775, and 196-2776

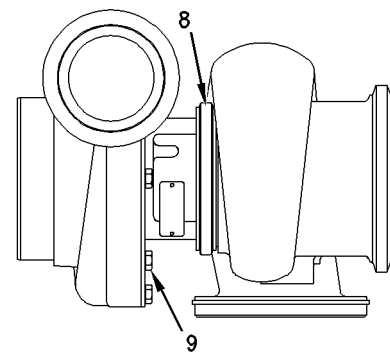


Illustration 34 g00859582

Typical example

- (8) Clamp  
(9) Bolt

#### (8) Clamp

To tighten the clamp (8) on the turbocharger, use the following procedure:

##### 1. Tighten the clamp.

Tighten the clamp to the following torque. .... 18 N·m (13.3 lb ft)

##### 2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

##### 3. Tighten the clamp.

Tighten the clamp to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

#### (9) Bolt

Tighten the bolt to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

**Part Numbers Are For The Following Illustration:**  
155-7952, 164-9490, 164-9493, 164-9496, 189-9210, 164-9490, 189-9210, 189-9211, 189-9212, 189-9213, 190-6210, 190-6211, 190-6212, 190-6213, and 208-5969

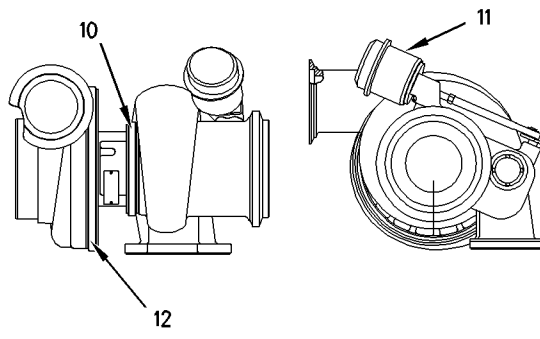


Illustration 35 g00853577

Typical example of a turbocharger with a wastegate

- (10) Clamp
- (11) Wastegate
- (12) Clamp

(10) Clamp

To tighten the clamp (10) on the turbocharger, use the following procedure:

- 1. Tighten the clamp.

Tighten the clamp to the following torque. .... 18 N·m (13.3 lb ft)

- 2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

- 3. Tighten the clamp.

Tighten the clamp to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(11) Wastegate

Opening pressure of the wastegate ..... 156 kPa (23 psi)

(12) Clamp

Tighten the clamp (12) to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

**Part Numbers Are For The Following Illustration:**  
**162-7603, 190-6205, 190-6206, 190-6207, 190-6208, 162-6495, and 162-7603**

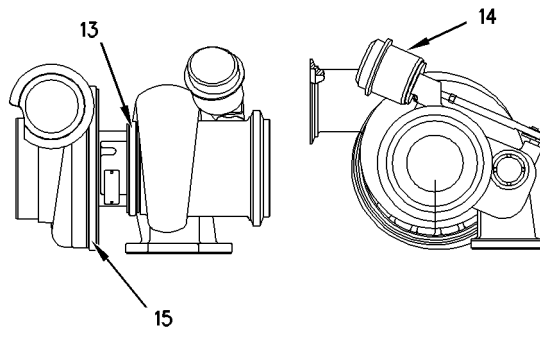


Illustration 36 g00859590

Typical example of a turbocharger with a wastegate

- (13) Clamp
- (14) Wastegate
- (15) Clamp

(13) Clamp

To tighten the clamp (13) on the turbocharger, use the following procedure:

- 1. Tighten the clamp.

Tighten the clamp to the following torque. .... 18 N·m (13.3 lb ft)

- 2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

- 3. Tighten the clamp.

Tighten the clamp to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

(14) Wastegate

Opening pressure of the wastegate ..... 165 kPa (24 psi)

(15) Clamp

Tighten the clamp to the following torque. .... 13.5 ± 1 N·m (9.9 ± 0.7 lb ft)

**Part Numbers Are For The Following Illustration:**  
**143-0821, 144-0816, and 190-6216**

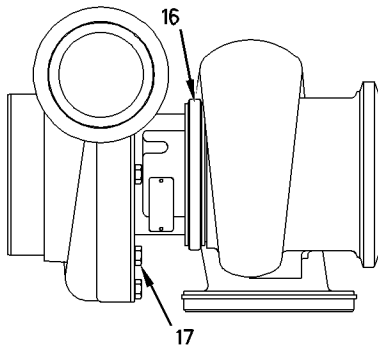


Illustration 37 g00859594

Typical example

(16) Clamp  
(17) Bolt

(16) Clamp

To tighten the clamp (16) on the turbocharger, use the following procedure:

1. Tighten the clamp.

Tighten the clamp to the following torque. .... 18 N·m (13.3 lb ft)

2. Loosen the clamp.

Loosen the clamp to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the clamp.

Tighten the clamp to the following torque. .... 13 ± 1 N·m (9.6 ± 0.7 lb ft)

(17) Bolt

To tighten the Bolt (17) on the turbocharger, use the following procedure:

1. Tighten the bolt.

Tighten the bolt to the following torque. ... 18 N·m (13.3 lb ft)

2. Loosen the bolt.

Loosen the bolt to the following torque. ... 6 N·m (53.1 lb in)

3. Tighten the bolt.

Tighten the bolt to the following torque. .... 13 ± 1 N·m (9.6 ± 0.7 lb ft)

**Part Numbers Are For The Following Illustration:  
127-8076**

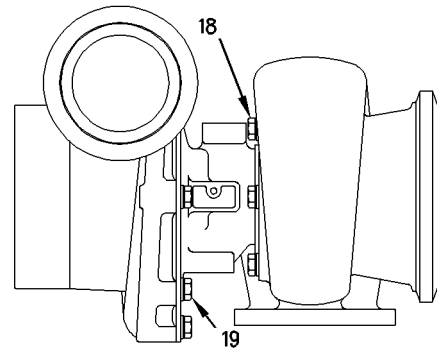


Illustration 38 g00859599

Typical example

(18) Bolt  
(19) Bolt

(18) Bolt

1. Tighten the bolt.

Tighten the turbine clamp plate bolts (18) to the following torque. .... 15.8 ± 0.6 N·m (11.6 ± 0.4 lb ft)

(19) Bolt

1. Tighten the bolt.

Tighten the compressor clamp plate bolts (19) in opposite pairs. Tighten the compressor clamp plate bolts to the following torque. ... 17.5 ± 1 N·m (12.9 ± 0.7 lb ft)

Apply **5P-3931** Anti-Seize Compound to the nuts that are used to mount the turbocharger. Tighten the nuts to the following torque 70 ± 15 N·m (51.6 ± 11.0 lb ft).

i01892348

## Wastegate Solenoid

SMCS Code: 1057

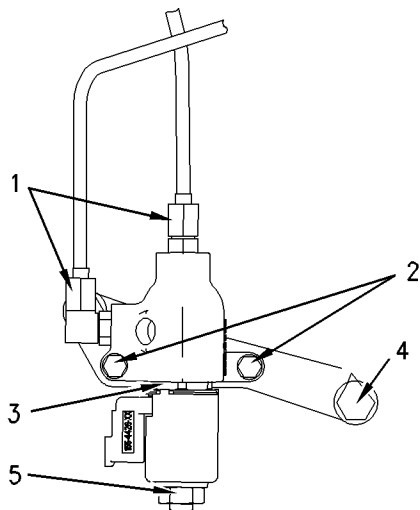


Illustration 39

g00918346

- (1) Tighten the nuts to the following torque. ....  $22 \pm 2 \text{ N}\cdot\text{m}$   
( $16.0 \pm 1.5 \text{ lb ft}$ )
- (2) Tighten the bolts to the following torque. ....  $12 \pm 3 \text{ N}\cdot\text{m}$   
( $9 \pm 2 \text{ lb ft}$ )
- (3) Tighten the solenoid body into the manifold to the following torque. ....  $50 \pm 5 \text{ N}\cdot\text{m}$   
( $37.0 \pm 4.0 \text{ lb ft}$ )
- (4) Tighten the bolt to the following torque. ....  $55 \pm 10 \text{ N}\cdot\text{m}$   
( $41 \pm 7 \text{ lb ft}$ )
- (5) Tighten the nut to the following torque. ....  $13 \pm 1.5 \text{ N}\cdot\text{m}$   
( $10 \pm 1.11 \text{ lb ft}$ )

i01193540

## Inlet Manifold

SMCS Code: 1058

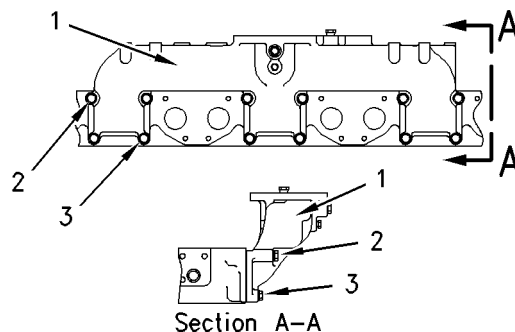


Illustration 40

g00638757

- (1) Air inlet manifold
- (2) Long bolts
- (3) Short bolts

Two different lengths of mounting bolts are used. There are six long bolts and six shorter bolts that are used on this engine. Be sure that the correct length of bolt is installed in the correct hole.

Tighten the mounting bolts to the following torque: ....  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $40 \pm 7 \text{ lb ft}$ )

i01832313

## Exhaust Manifold

SMCS Code: 1059

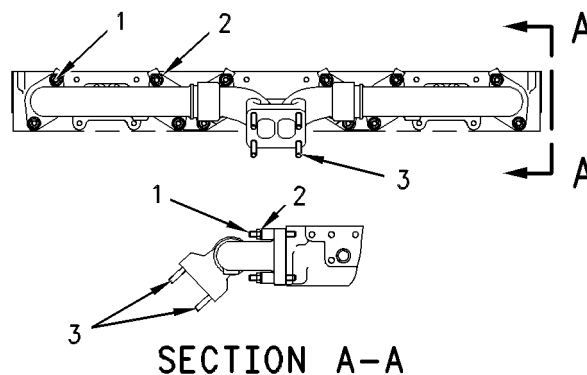


Illustration 41

g00389694

Typical example of an exhaust manifold with a flange with four studs

**Note:** The exhaust manifold gaskets have tabs. The tabs should point to the oil pan.

- (1) Apply **5P-3931** Anti-Seize Compound to the stud threads. Tighten the studs to the following torque. ....  $35 \pm 5$  N·m  
( $26 \pm 4$  lb ft)
- (2) Tighten the nuts to the following torque. ....  $55 \pm 10$  N·m  
( $41 \pm 7$  lb ft)
- (3) Apply **5P-3931** Anti-Seize Compound to the stud threads. Tighten the studs to the following torque. ....  $35 \pm 5$  N·m  
( $26 \pm 4$  lb ft)

i01746489

## Camshaft

**SMCS Code:** 1210

### C-10

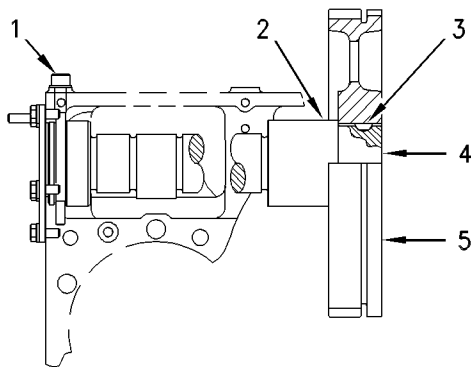


Illustration 42

g00466006

- (1) Torque for thrust pin ...  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft)
- (2) Diameter of camshaft bearing journal ...  $74.85 \pm 0.025$  mm ( $2.9468 \pm 0.0010$  inch)

The lobe on the camshaft for the injector has the most lift when the cam is at zero degrees of rotation. ....  $11.9382$  mm ( $0.47000$  inch)

The lobe on the camshaft for the injector has zero lift through the given rotation. .... 72 degrees to 254 degrees

When the injector lobe has a lift height of  $4.855 \pm 0.038$  mm ( $0.1911 \pm 0.0015$  inch), the camshaft should have an angle of rotation. ....  $35 \pm 0.25$  degrees

### C-12

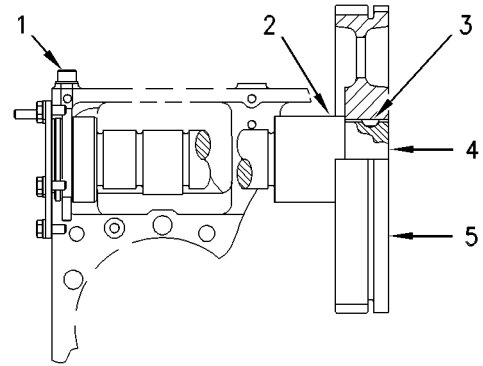


Illustration 43

g00466006

- (1) Torque for thrust pin ...  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft)
- (2) Diameter of camshaft bearing journal ...  $74.85 \pm 0.025$  mm ( $2.9468 \pm 0.0010$  inch)

The lobe on the camshaft for the injector has the most lift when the cam is at zero degrees of rotation. ....  $11.9382$  mm ( $0.47000$  inch)

The lobe on the camshaft for the injector has zero lift through the given rotation. .... 70 degrees to 254 degrees

When the injector lobe has a lift height of  $5.365 \pm 0.038$  mm ( $0.2112 \pm 0.0015$  inch), the camshaft should have an angle of rotation. ....  $30 \pm 0.25$  degrees

i01671996

## Engine Oil Filter Base (166-1329 Oil Filter Base)

**SMCS Code:** 1306

The part number for the oil filter base is located under the oil filter. The filter needs to be removed in order to view the part number. The **166-1329** Oil Filter Base is adaptable to engines that are listed in Table 2. The **166-1329** Oil Filter Base is a direct replacement for **122-0725** Oil Filter Base. The part number of the oil filter base needs to be verified before you proceed.

Table 2

| <b>166 - 1329 Oil Filter Base</b> |                                     |                                        |                                     |
|-----------------------------------|-------------------------------------|----------------------------------------|-------------------------------------|
|                                   | <b>Serial<br/>Number<br/>Prefix</b> | <b>Serial<br/>Number<br/>Beginning</b> | <b>Serial<br/>Number<br/>Ending</b> |
| <b>Industrial</b>                 | 2AW                                 | 1                                      | Up                                  |
|                                   | 1DW                                 | 1                                      | Up                                  |
|                                   | BCX                                 | 1                                      | Up                                  |
|                                   | BDL                                 | 1                                      | Up                                  |
|                                   | 5DZ                                 | 1                                      | Up                                  |
|                                   | 1WZ                                 | 1                                      | Up                                  |
|                                   | CPN                                 | 1                                      | Up                                  |
| <b>Machine</b>                    | 3PD                                 | 1                                      | Up                                  |
|                                   | 7ZR                                 | 1                                      | Up                                  |
|                                   | 5ED                                 | 1                                      | Up                                  |
|                                   | 6AR                                 | 1                                      | Up                                  |
|                                   | JAC                                 | 1                                      | Up                                  |
| <b>Marine</b>                     | 6BW                                 | 1                                      | Up                                  |
|                                   | 2XR                                 | 1                                      | Up                                  |
|                                   | 9HP                                 | 1                                      | Up                                  |
| <b>Truck</b>                      | 2PN                                 | 1                                      | Up                                  |
|                                   | 8YS                                 | 1                                      | Up                                  |
|                                   | 3CS                                 | 1                                      | Up                                  |
|                                   | 1YN                                 | 1                                      | Up                                  |
|                                   | CPD                                 | 1                                      | Up                                  |
|                                   | 9NS                                 | 1                                      | Up                                  |
|                                   | 2KS                                 | 1                                      | Up                                  |
|                                   | 9SM                                 | 1                                      | Up                                  |

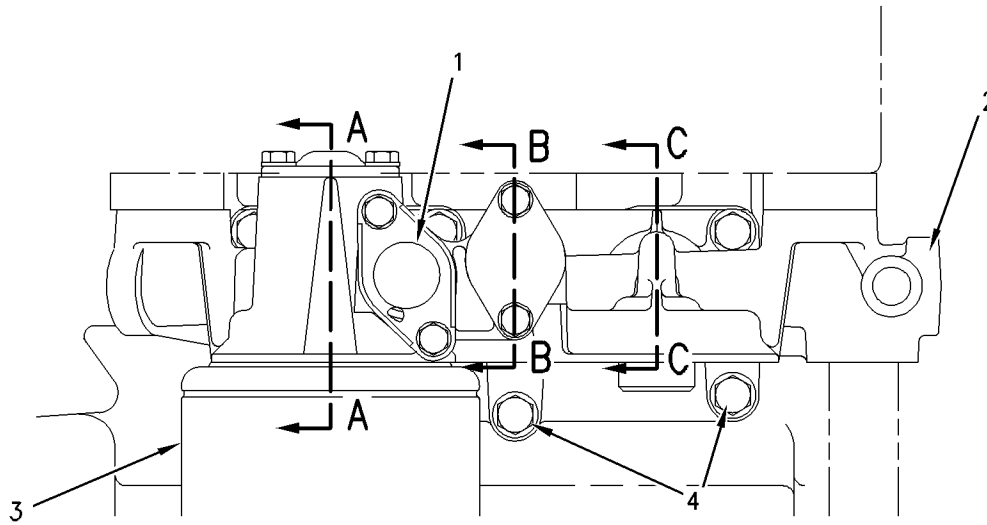


Illustration 44

g00842293

- (1) Cover
- (2) Oil filter base
- (3) Oil filter
- (4) Bolts (six total)

- (9) Seal
- (10) Stud

Tighten the stud that holds the oil filter to the oil filter base to the following torque. ...  $68 \pm 7 \text{ N}\cdot\text{m}$   
( $50 \pm 5 \text{ lb}\cdot\text{ft}$ )

Apply **9S-3263** Thread Lock to each of the six bolts.

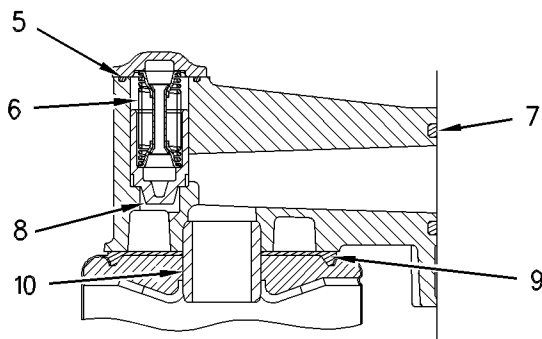


Illustration 45  
Section A-A

g00473681

- (5) Seal
- (6) 8M-3182 Spring

Assembled length ..... 63.5 mm (2.50 inch)  
Load at assembled length ..... 39.7 N (8.92 lb)  
Free length after test ..... 91.7 mm (3.61 inch)  
Outside diameter ..... 20.6 mm (.81 inch)

- (7) Seal
- (8) Plunger

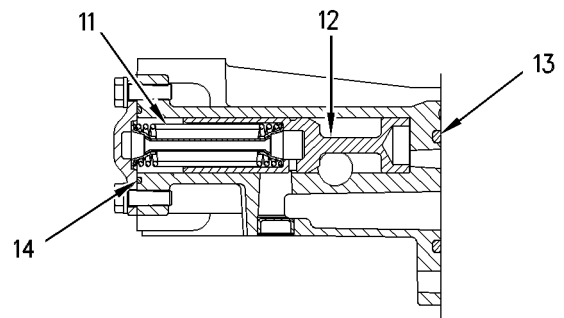


Illustration 46  
Section B-B

g00473768

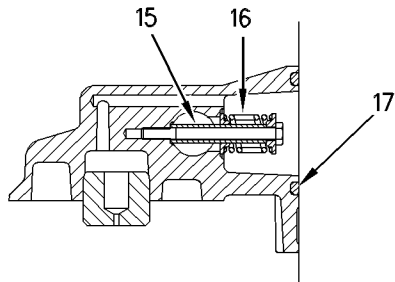
- (11) 2N-6005 Spring

Assembled length ..... 96.5 mm (3.80 inch)  
Load at assembled length ..... 92.5 N (20.80 lb)  
Minimum operating length ..... 45.7 mm  
(1.80 inch)  
Load at minimum operating length ..... 257.6 N  
(57.91 lb)  
Free length after test ..... 124.7 mm (4.91 inch)  
Outside diameter ..... 21.8 mm (.86 inch)

- (12) Valve
- (13) Seal

(14) Seal

---



---

Illustration 47

g00473770

Section C-C

Relief pressure ..... 696 kPa (101 psi)

(15) Spacer

(16) 1A-2170 Spring

Assembled length ..... 21.4 mm (.84 inch)

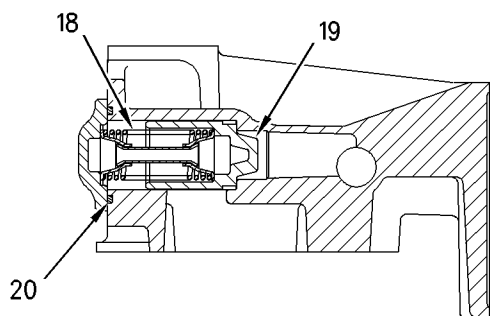
Load at assembled length .....  $198.0 \pm 13.4$  N  
( $45 \pm 3.5$  lb)

Free length after test ..... 31.0 mm (1.22 inch)

Outside diameter ..... 19.6 mm (.77 inch)

(17) Seal

---



---

Illustration 48

g00794005

Section D-D

(18) 8M-3182 Spring

Assembled length ..... 63.5 mm (2.50 inch)

Load at assembled length ..... 39.7 N (8.92 lb)

Free length after test ..... 91.7 mm (3.61 inch)

Outside diameter ..... 20.6 mm (.81 inch)

(19) Plunger

(20) Seal

i01532608

## Engine Oil Cooler

SMCS Code: 1378

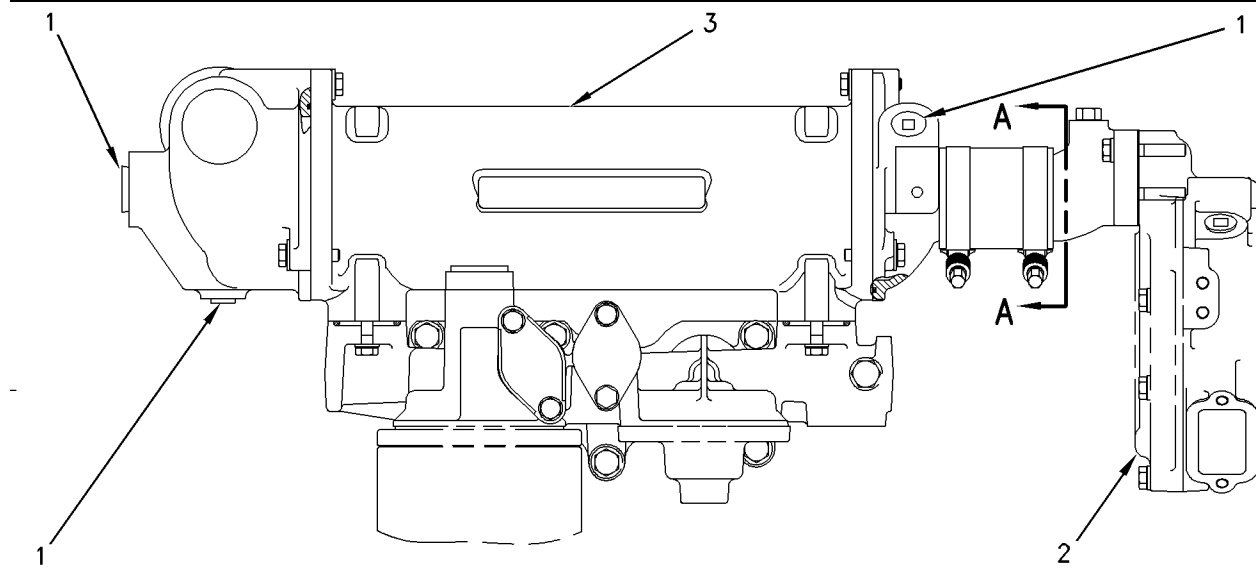


Illustration 49

g00795586

- (1) Apply 5P-3413 Pipe Sealant to the plug threads.
- (2) Water pump
- (3) Core assembly

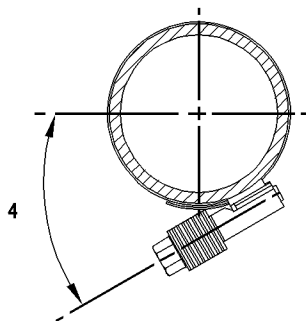


Illustration 50  
Section A-A

g00534638

- (4) Install the hose clamps at the following angle. ....  $30 \pm 5$  degrees

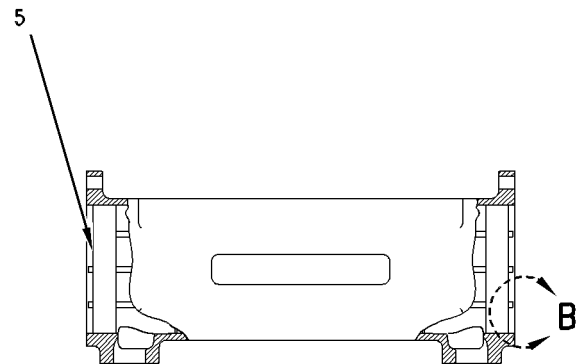


Illustration 51

g00534648

- (5) Core assembly

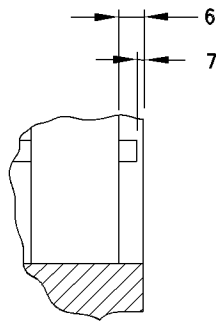


Illustration 52

g00534649

Detail B

The oil flow rate through the cooler assembly is 123 L/min (32 US gpm). The maximum water flow through the cooler assembly is 262 L/min (69 US gpm).

Lubricate the core with engine oil. Press the core into the housing to the following dimensions at both ends.

- (6) The depth of the plate is the following value. ....  $5.5 \pm 0.5$  mm ( $0.22 \pm 0.02$  inch)
- (7) The depth of the tube is the following value. ....  $1.5 \pm 0.5$  mm ( $0.06 \pm 0.02$  inch)

i01039752

# Engine Oil Pump

SMCS Code: 1304

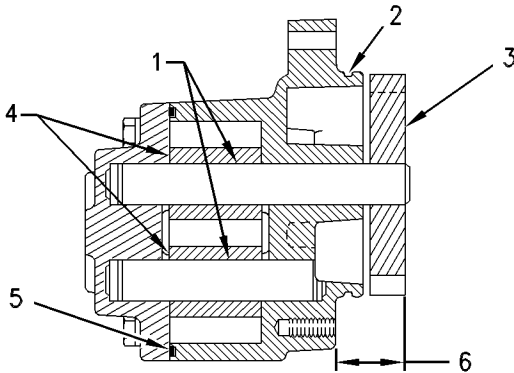


Illustration 53

g00280778

## NOTICE

Before operating the engine, the oil pump must be lubricated with clean engine oil. The oil pump must turn freely by hand. Damage to the drive gear and internal pump damage can occur if the oil pump is not lubricated with clean engine oil.

### Engine oil pump

|                        |                       |
|------------------------|-----------------------|
| Oil type .....         | SAE 30                |
| Oil temperature .....  | 50°C (122°F)          |
| Pump speed .....       | 3000 rpm              |
| Oil pressure .....     | 359 kPa (52 psi)      |
| Minimum oil flow ..... | 185 L/min (49 US gpm) |

- (1) Diameter of two shafts .....  $18.000 \pm 0.005$  mm ( $.7087 \pm .0002$  inch)

Bores in pump body for two shafts .....  $18.050 \pm 0.010$  mm ( $.7106 \pm .0004$  inch)

- (2) Seal groove

Prior to installation, lubricate the bore in the engine front cover. This will allow the seal groove and the seal to slide.

- (3) Drive gear

In order to install the drive gear, heat the drive gear to the following temperature. ....  $316^{\circ}\text{C}$  ( $600^{\circ}\text{F}$ )

- (4) Pump gears

In order to install the pump gears, heat the pump gears to the following temperature. ....  $316^{\circ}\text{C}$  ( $600^{\circ}\text{F}$ )

Length of new gears .....  $50.000 \pm 0.025$  mm ( $1.9685 \pm .0010$  inch)

Depth of bores in pump body for the gears .....  $50.13 \pm 0.02$  mm ( $1.974 \pm .001$  inch)

- (5) Seal

- (6) Distance between front face of gear and housing ...  $29.40 \pm 0.50$  mm ( $1.158 \pm .020$  inch)

i00817005

i00947454

## Engine Oil Pressure

**SMCS Code:** 1924

Operate the engine in order to achieve the normal operating temperature. When the engine has achieved the normal operating temperature, the engine oil temperature should be no greater than 110°C (230°F). The engine oil temperature should be measured after the engine oil has passed through the engine oil cooler.

Check the engine oil pressure in the oil gallery that is located on the right side of the cylinder block.

Engine oil pressure under full load conditions (1400 rpm to 1800 rpm) ..... 275 to 414 kPa (40 to 60 psi)

Minimum engine oil pressure at low idle (600 rpm to 800 rpm) ..... 68 kPa (10 psi)

i00887123

## Engine Oil Pan

**SMCS Code:** 1302

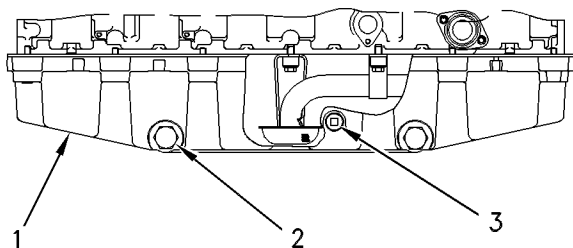


Illustration 54  
Center Sump

g00280790

- (1) Oil pan
- (2) Tighten the plugs to the following torque. ....  $70 \pm 15 \text{ N}\cdot\text{m}$  ( $50 \pm 11 \text{ lb ft}$ )
- (3) Apply **9S-3263** Thread Lock Compound to plug threads. Tighten the plug to the following torque. ....  $80 \pm 11 \text{ N}\cdot\text{m}$  ( $60 \pm 8 \text{ lb ft}$ )

## Engine Oil Pan

**SMCS Code:** 1302

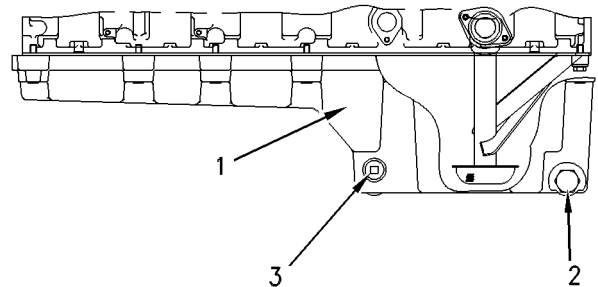


Illustration 55  
Shallow Front Sump

g00483624

- (1) Engine oil pan
- (2) Tighten the plugs to the following torque. ....  $70 \pm 15 \text{ N}\cdot\text{m}$  ( $50 \pm 11 \text{ lb ft}$ )
- (3) Apply **9S-3263** Thread Lock Compound to the plug threads. Tighten the plugs to the following torque. ....  $80 \pm 11 \text{ N}\cdot\text{m}$  ( $60 \pm 8 \text{ lb ft}$ )

i00817425

## Engine Oil Pan

**SMCS Code:** 1302

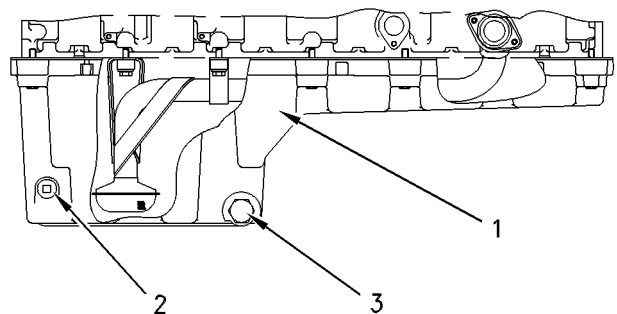


Illustration 56  
Rear Sump

g00390433

- (1) Engine oil pan
- (2) Apply **9S-3263** Thread Lock Compound to the plug threads. Tighten the plugs to the following torque. ....  $80 \pm 11 \text{ N}\cdot\text{m}$  ( $60 \pm 8 \text{ lb ft}$ )
- (3) Tighten the plugs to the following torque. ....  $70 \pm 15 \text{ N}\cdot\text{m}$  ( $50 \pm 11 \text{ lb ft}$ )

i00818610

## Coolant Conditioner Base

SMCS Code: 1352

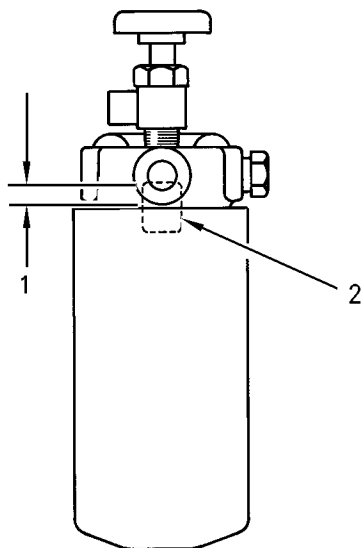


Illustration 57

g00391440

- (1) Apply **9S-3263** Thread Lock Compound on the tapered end of the stud to the following distance. .... 7.6 mm (.30 inch)
- (2) Tighten the stud to the following torque. ....  $60 \pm 7$  N·m ( $44 \pm 5$  lb ft)

i00817649

## Water Temperature Regulator

SMCS Code: 1355

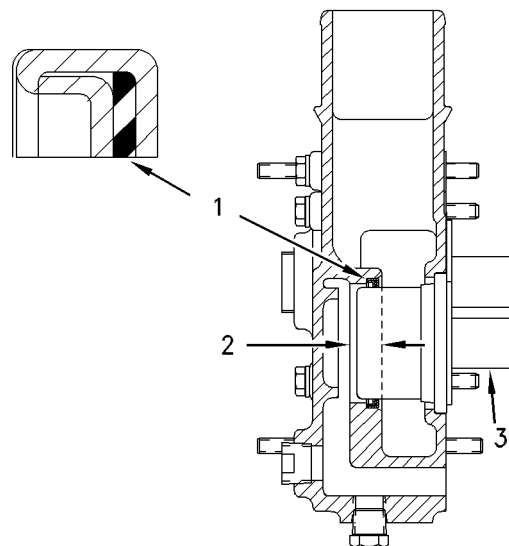


Illustration 58

g00390592

- (1) Install the lip type seal into the water temperature regulator housing. Lubricate the sealing lip on the lip type seal with a small amount of antifreeze.
- (2) Minimum stroke of the water temperature regulator at fully open temperature ..... 10.4 mm (0.41 inch)
- (3) Fully open temperature of the water temperature regulator .....  $98^{\circ}\text{C}$  ( $208^{\circ}\text{F}$ )

i01741707

## Water Pump

SMCS Code: 1361

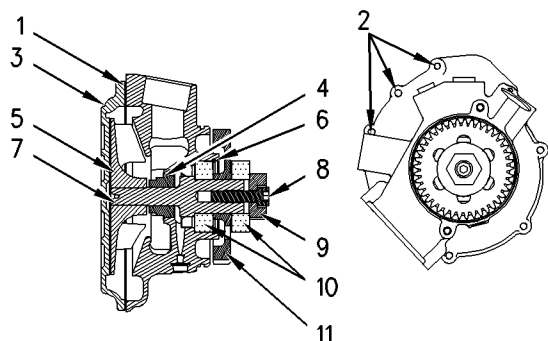


Illustration 59

g00891866

Water pump

- (1) Gasket
- (2) Mounting holes for the impeller cover
- (3) Impeller cover
 

Tighten the bolts (3) on the impeller cover to the following torque: .....  $18 \pm 3 \text{ N}\cdot\text{m}$  ( $13 \pm 2 \text{ lb ft}$ )
- (4) Seal assembly
- (5) Impeller
- (6) Retaining ring
- (7) Shaft
- (8) Tighten the bolt to the following torque: .....  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $41 \pm 7 \text{ lb ft}$ )
- (9) Bearing retaining ring
- (10) Bearing
- (11) Water pump drive gear

**Note:** The impeller is pressed onto shaft (7). The face of the impeller is flush to the end of the shaft.

The clearance between the impeller and the water pump housing is  $0.94 \pm 0.47 \text{ mm}$  ( $0.037 \pm 0.019 \text{ inch}$ ).

**Note:** Clearance between the impeller and the water pump housing is preset once the impeller is pressed onto the shaft.

The clearance between the water pump housing and the water pump drive gear is  $2.8 \text{ mm}$  ( $0.11 \text{ inch}$ ).

**Note:** Clearance between the water pump housing and the water pump drive gear is preset once the bearings are tightened against the water pump drive gear.

i01773390

## Cylinder Block

SMCS Code: 1201

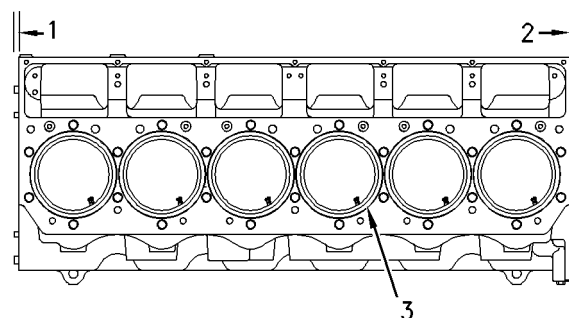


Illustration 60

g00906741

- (1) The dowels extend past the end of the cylinder block by the following distance. ...  $10 \pm 0.5 \text{ mm}$  ( $0.39 \pm 0.02 \text{ inch}$ )
- (2) The dowels extend past the end of the cylinder block by the following distance. ...  $10.0 \pm 0.5 \text{ mm}$  ( $0.39 \pm 0.02 \text{ inch}$ )
- (3) Cylinder liner

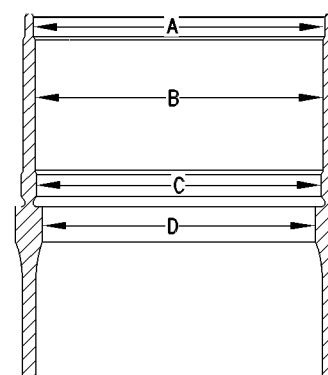


Illustration 61

g00488509

Section A-A

Bore diameters in the cylinder block for the cylinder block liners:

- |                    |                                                                    |
|--------------------|--------------------------------------------------------------------|
| Position (A) ..... | $151.50 \pm 0.03 \text{ mm}$<br>( $5.965 \pm 0.001 \text{ inch}$ ) |
| Position (B) ..... | $149.8 \pm 0.3 \text{ mm}$<br>( $5.898 \pm 0.012 \text{ inch}$ )   |
| Position (C) ..... | $148.00 \pm 0.03 \text{ mm}$<br>( $5.827 \pm 0.001 \text{ inch}$ ) |

Position (D) ..... 141.90 ± 0.15 mm  
(5.587 ± 0.006 inch)

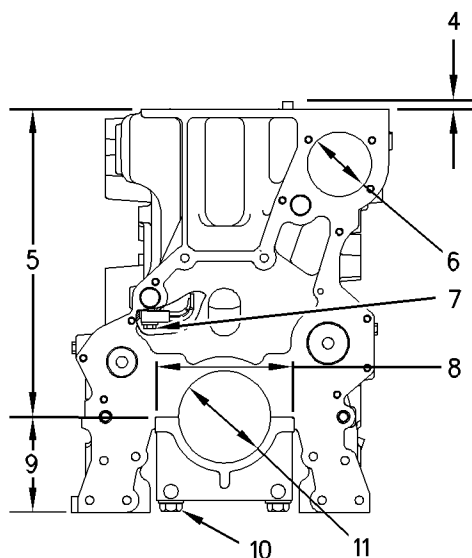


Illustration 62

g00906865

- (4) The dowels extend past the top surface of the cylinder block by the following distance. .... 10.0 ± 0.5 mm (0.39 ± 0.02 inch)
- (5) Distance from the centerline of the crankshaft bore to the top surface of the cylinder block ..... 387.00 mm (15.236 inch)
- (6) Diameter of the camshaft bores ..... 81.000 ± 0.015 mm (3.1890 ± 0.0006 inch)
- (7) Bolt
- Tighten the bolt that holds the piston cooling jet to the following torque. .... 35 ± 3 N·m (26 ± 2 lb ft)

The flatness of the top contact surface of the cylinder block must be within 0.05 mm (0.002 inch) for any 150 mm (5.9 inch) section of the surface.

- (8) Width of the main bearing cap ..... 175.000 ± 0.020 mm (6.8898 ± 0.0008 inch)

Width in cylinder block for the main bearing cap .... 175.000 ± 0.018 mm (6.8898 ± 0.0007 inch)

- (9) Distance from the centerline of the crankshaft to the bottom surface of the cylinder block ..... 120.0 mm (4.72 inch)

- (10) Main bearing cap bolts

Use the following procedure in order to install the main bearing cap bolts:

1. Orient the main bearing cap correctly. The part number on the main bearing cap must face to the right and to the front face of the block. Also, the tab slots that are in the block and the main bearing caps must be adjacent.

**Note:** The main bearing caps are marked with identification numbers 1 through 7. Install the main bearing caps into the correct positions.

2. Lubricate the main bearing cap bolts. Use “SAE 30W” oil or molybdenum grease to lubricate the threads and the washer face.

3. Tighten the main bearing cap bolts.

Tighten bolts to the following torque. .. 95 ± 5 N·m (70 ± 4 lb ft)

4. Put an alignment mark on each cap and bolt.

Rotate the bolts in the clockwise direction. .... 90 ± 5 degrees

- (11) Bore in the cylinder block for the seven main bearings ..... 116.000 ± 0.013 mm (4.5669 ± 0.0005 inch)

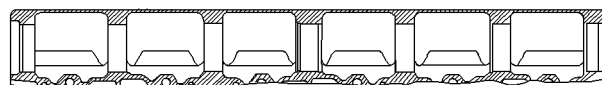


Illustration 63

g00906901

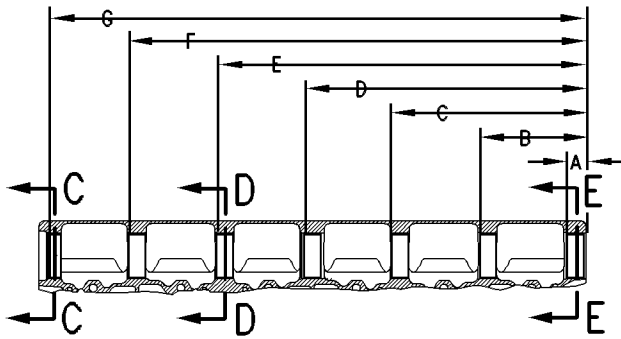


Illustration 64

g00283161

View B-B

The camshaft bearings are installed into the cylinder block at the values that follow.

|                    |                         |
|--------------------|-------------------------|
| Position (A) ..... | 35.20 mm (1.386 inch)   |
| Position (B) ..... | 191.20 mm (7.528 inch)  |
| Position (C) ..... | 347.20 mm (13.669 inch) |
| Position (D) ..... | 503.20 mm (19.811 inch) |
| Position (E) ..... | 659.20 mm (29.953 inch) |
| Position (F) ..... | 815.20 mm (32.094 inch) |
| Position (G) ..... | 954.10 mm (37.563 inch) |

**NOTICE**

Camshaft bearings must be installed into their correct position. Failure to do so will result in engine damage.

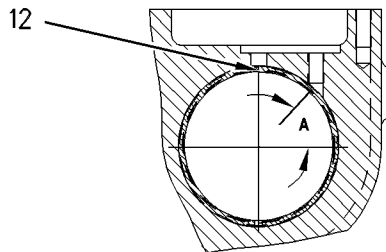


Illustration 65

g00283162

View C-C

**Note:** Illustration 65 represents camshaft bearing 7.

(12) Bearing joint

(A) Bearing oil hole at a 45 degree angle

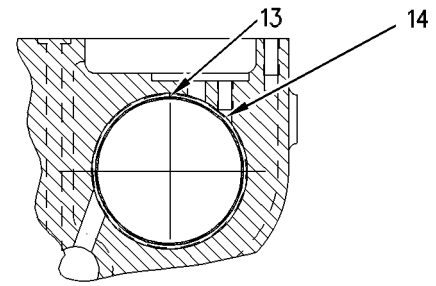


Illustration 66

g00391637

View D-D

**Note:** Illustration 66 represents camshaft bearings 2 through 6.

(13) Bearing joint

(14) Oil hole

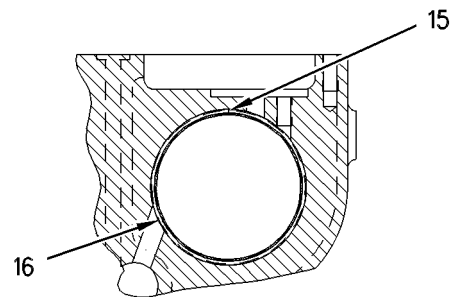


Illustration 67

g00391638

View E-E

**Note:** Illustration 67 represents camshaft bearing 1.

(15) Bearing joint

(16) Oil hole

**Note:** The width of the number 1 through number 6 camshaft bearing is  $33.4 \pm 0.5$  mm ( $1.32 \pm 0.02$  inch). The width of the number 7 camshaft bearing is  $28.0 \pm 0.5$  mm ( $1.10 \pm 0.02$  inch). The number 1 camshaft bearing has a  $5.0 \pm 0.5$  mm ( $0.20 \pm 0.02$  inch) wide single groove on the inside diameter of the camshaft bearing. The width of the single groove on the outside diameter of camshaft bearings 2 through 6 is  $8.0 \pm 0.5$  mm ( $0.32 \pm 0.02$  inch). A single groove is on the outside diameter of the number 7 camshaft bearing.

i01796787

i00819393

# Cylinder Liner

SMCS Code: 1216

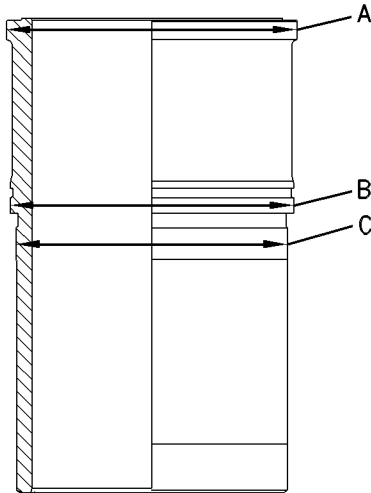


Illustration 68

g00517737

Outside diameters of the cylinder liner

Table 3

| Cylinder Liner      | C-10                                         | C-12                                         |
|---------------------|----------------------------------------------|----------------------------------------------|
| Position A          | 151.25 ± 0.05 mm (5.955 ± 0.002 inch)        |                                              |
| Position B          | 147.90 ± 0.03 mm (5.823 ± 0.001 inch)        |                                              |
| Position C          | 141.38 ± 0.08 mm (5.566 ± 0.003 inch)        |                                              |
| Inner Bore Diameter | 125.037 ± 0.037 mm<br>(4.9227 ± 0.0015 inch) | 130.037 ± 0.037 mm<br>(5.1196 ± 0.0015 inch) |

Note: Apply 5P-3975 Rubber Lubricant to the cylinder liner seals prior to assembly.

# Cylinder Liner Projection

SMCS Code: 1216

Table 4

| Specifications                                    |                                            |
|---------------------------------------------------|--------------------------------------------|
| Liner Projection                                  | 0.040 to 0.200 mm<br>(.0016 to .0079 inch) |
| Maximum Variation in Each Liner                   | 0.050 mm (.0020 inch)                      |
| Maximum Average Variation Between Adjacent Liners | 0.050 mm (.0020 inch)                      |
| Maximum Variation Between All Liners              | 0.100 mm (.0040 inch)                      |

For more information on the cylinder liner projection, refer to Testing and Adjusting, "Basic Block".

i01797715

# Crankshaft

SMCS Code: 1202

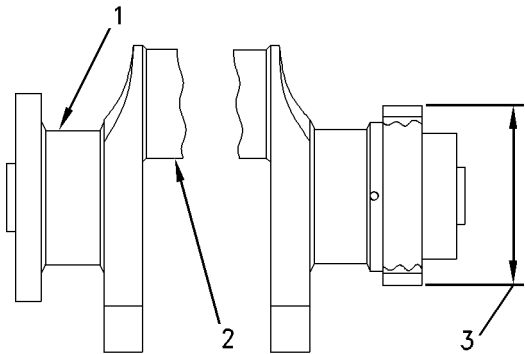


Illustration 69

g00920561

(1) Main bearing journals

The thrust plate is used on the center main bearing only.

(2) Connecting rod bearing journals

(3) Average diameter of gear after assembly ..... 136.20 ± 0.11 mm  
(5.362 ± 0.004 inch)

Maximum diameter of gear after assembly ..... 136.36 mm (5.368 inch)

**Note:** After the crankshaft is assembled into the block, the crankshaft end play should be a maximum of 0.50 mm (0.020 inch) and a minimum of 0.10 mm (0.004 inch).

i00959950

## Crankshaft Seals

**SMCS Code:** 1160; 1161

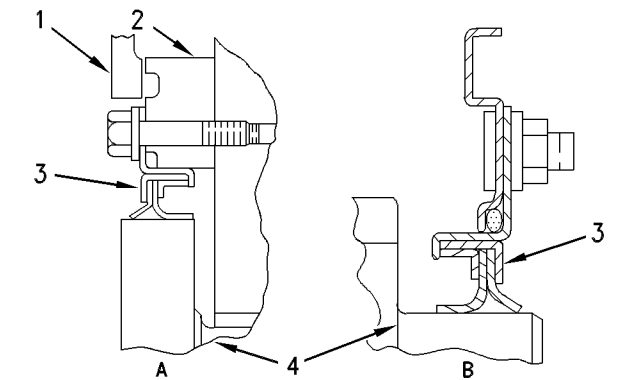


Illustration 70

g00281016

(A) Rear  
(B) Front

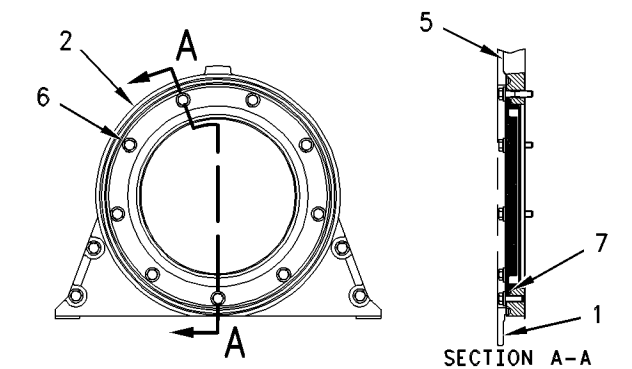


Illustration 71

g00489560

Rear seal group

**Note:** Install the front crankshaft seals and the rear crankshaft seals dry.

- (1) Flywheel housing
- (2) Seal group cover
- (3) Seal group
- (4) Crankshaft
- (5) Rear face of cylinder block
- (6) Bolt

Apply **7M-7456** Compound to the bolt threads.

Tighten the bolts to the following torque. ....  $12 \pm 3$  N·m ( $9 \pm 2$  lb ft)

(7) O-ring seal

Lightly lubricate the O-ring seal. Use the lubricant that is being sealed.

i00826094

## Vibration Damper and Pulley

**SMCS Code:** 1205

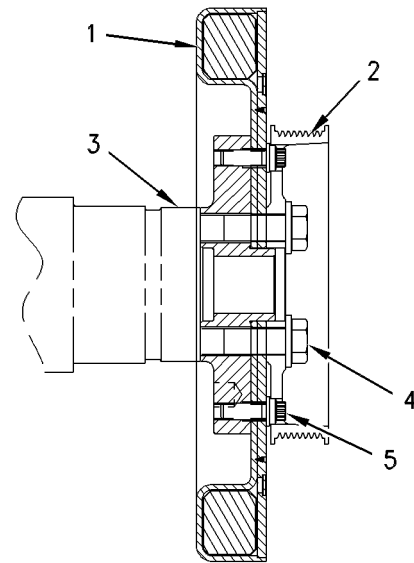


Illustration 72

g00398558

Typical Example

- (1) Damper assembly
- (2) Pulley
- (3) Crankshaft
- (4) Tighten the bolts to the following torque. ....  $240 \pm 40$  N·m ( $175 \pm 30$  lb ft)
- (5) Tighten the bolts to the following torque. ....  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft)

i01003512

i01764734

# Connecting Rod Bearing Journal

**SMCS Code:** 1202; 1219; 1225

Table 5

| Connecting Rod Bearing Journal             |                                            |
|--------------------------------------------|--------------------------------------------|
| Original size journal                      | 89.000 ± 0.020 mm<br>(3.5039 ± .0008 inch) |
| Undersize journal<br>0.508 mm (.0200 inch) | 88.492 ± 0.020 mm<br>(3.4839 ± .0008 inch) |
| Undersize journal<br>0.762 mm (.0300 inch) | 87.238 ± 0.020 mm<br>(3.4346 ± .0008 inch) |

The clearance between a new bearing and a new journal is the following value. .... 0.062 to 0.132 mm  
(.0024 to .0052 inch)

i00819737

# Main Bearing Journal

**SMCS Code:** 1202; 1203

Table 6

| Main Bearing Journal                       |                                             |
|--------------------------------------------|---------------------------------------------|
| Original size journal                      | 108.000 ± 0.020 mm<br>(4.2520 ± .0008 inch) |
| Undersize journal<br>0.508 mm (.0200 inch) | 107.492 ± 0.020 mm<br>(4.2320 ± .0008 inch) |
| Undersize journal<br>0.762 mm (.0300 inch) | 107.238 ± 0.020 mm<br>(4.2220 ± .0008 inch) |

The clearance between a new bearing and a new journal is the following value. .... 0.081 to 0.181 mm  
(.0032 to .0071 inch)

Table 7

| Main Bearing Bore                               |                                             |
|-------------------------------------------------|---------------------------------------------|
| Main bearing bore (original size)               | 116.000 ± 0.013 mm<br>(4.5669 ± .0005 inch) |
| Oversize bore in block<br>0.508 mm (.0200 inch) | 116.508 ± 0.013 mm<br>(4.5869 ± .0005 inch) |

# Connecting Rod

**SMCS Code:** 1218

**S/N:** MBJ1-Up

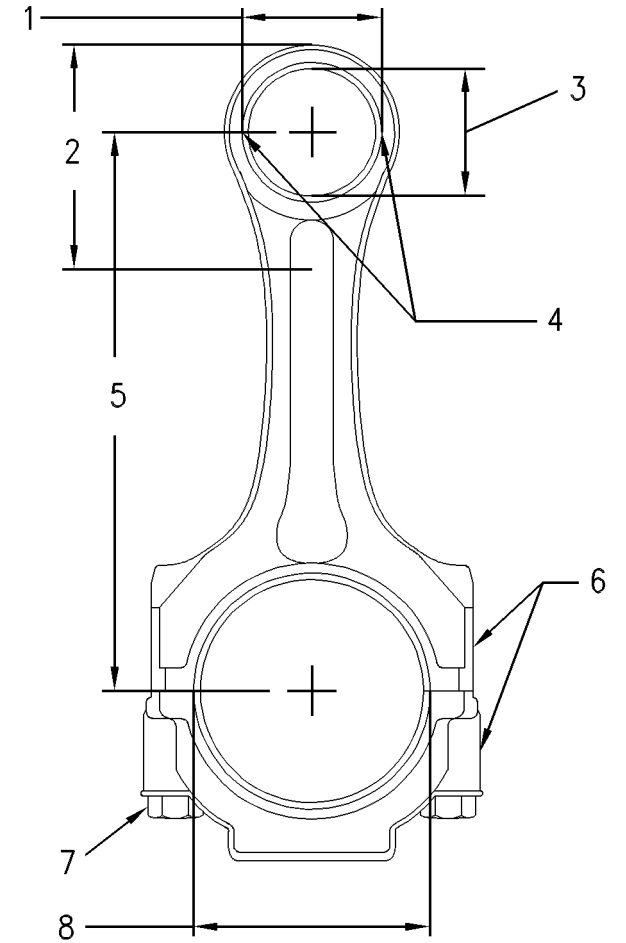


Illustration 73

g00490254

- (1) Bore in the connecting rod for the piston pin bearing ..... 55.435 ± 0.013 mm  
(2.1825 ± .0005 inch)

**Note:** The connecting rod must be heated for the installation of the piston pin bearing. Do not use a torch.

- (2) The connecting rod may be heated from 175°C to 260°C (347°F to 500°F) for the installation of the piston pin bearing. Maximum distance for heating the connecting rod ..... 85.0 mm (3.35 inch)

- (3) Bore in the bearing for the piston pin ... 50.830 ± 0.008 mm (2.0012 ± .0003 inch)

i01765082

Diameter of the piston pin .....  $50.795 \pm 0.005$  mm  
( $2.0000 \pm .0002$  inch)

Thoroughly lubricate the piston pin with clean engine oil prior to assembly of the piston and connecting rod.

- (4) The bearing joint must be within  $\pm 12.5$  degrees of either location.
- (5) Distance between the center of the bearings ..... 247.00 mm (9.724 inch)
- (6) Etch the cylinder number on the connecting rod and the cap in this location. Mark the connecting rod and the cap with a number 1 through 6. Mark the numbers on the same side of the connecting rod as the bearing retainer notch.

**Note:** Install the connecting rod in the engine with the part number to the rear of the engine.

- (7) Tighten the connecting rod bolts. Use the instructions that follow.

- 1.** Before installing the connecting rod bolts, lubricate the bolt threads and the seating faces of the caps with **2P-2506** Thread Compound.

- 2.** Tighten the connecting rod bolts.

Tighten the connecting rod bolts to the following torque. ....  $130 \pm 7$  N·m ( $95 \pm 5$  lb ft)

- 3.** Tighten each connecting rod bolt for an additional amount.

Rotate each connecting rod bolt in the clockwise direction. ....  $60 \pm 5$  degrees ( $1/6$  of a turn)

- (8) Bore in the connecting rod for the crankshaft bearing .....  $86.800 \pm 0.013$  mm  
( $3.4173 \pm .0005$  inch)

## Connecting Rod

**SMCS Code:** 1218

**S/N:** MBL1-Up

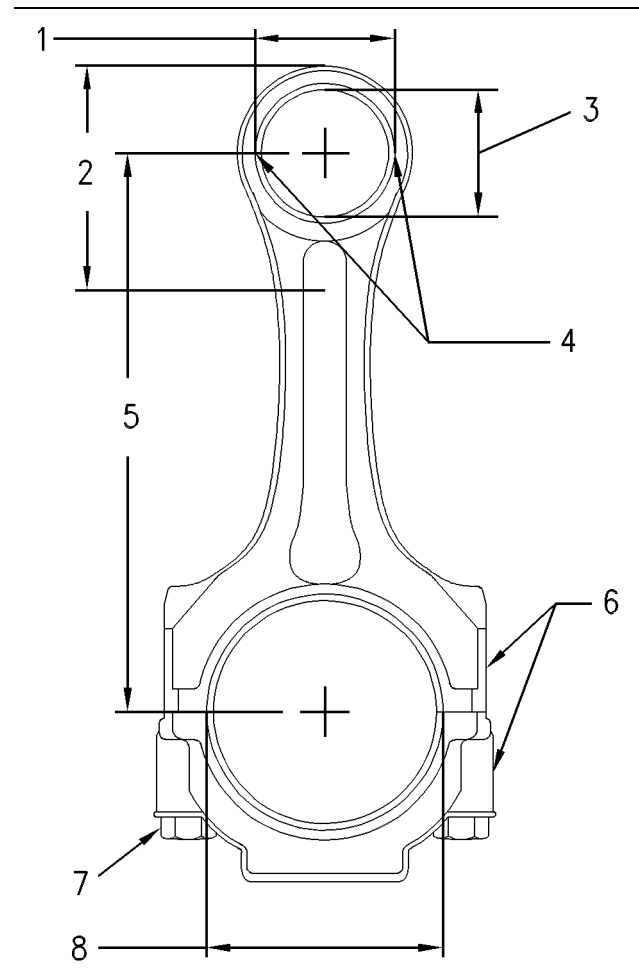


Illustration 74

g00490254

- (1) Bore in the connecting rod for the piston pin bearing .....  $57.810 \pm 0.013$  mm  
( $2.2760 \pm 0.0005$  inch)

**Note:** The connecting rod must be heated for the installation of the piston pin bearing. Do not use a torch.

- (2) The connecting rod may be heated from  $175^{\circ}\text{C}$  to  $260^{\circ}\text{C}$  ( $347^{\circ}\text{F}$  to  $500^{\circ}\text{F}$ ) for the installation of the piston pin bearing. Maximum distance for heating the connecting rod ..... 85.0 mm (3.35 inch)

- (3) Bore in the bearing for the piston pin .....  $53.205 \pm 0.008$  mm  
( $2.0947 \pm 0.0003$  inch)

Diameter of the piston pin .....  $53.170 \pm 0.005$  mm  
( $2.0933 \pm 0.0002$  inch)

Thoroughly lubricate the piston pin with clean engine oil prior to assembly of the piston and connecting rod.

- (4) The bearing joint must be within  $\pm 12.5$  degrees of either location.
- (5) Distance between the center of the bearings ..... 242.50 mm (9.547 inch)
- (6) Etch the cylinder number on the connecting rod and the cap in this location. Mark the connecting rod and the cap with a number 1 through 6. Mark the numbers on the same side of the connecting rod as the bearing retainer notch.

**Note:** Install the connecting rod in the engine with the part number to the rear of the engine.

- (7) Tighten the connecting rod bolts. Use the instructions that follow.

1. Before installing the connecting rod bolts, lubricate the bolt threads and the seating faces of the caps with **2P-2506** Thread Compound.

2. Tighten the connecting rod bolts.

Tighten the connecting rod bolts to the following torque. ....  $130 \pm 7$  N·m ( $95 \pm 5$  lb ft)

3. Tighten each connecting rod bolt for an additional amount.

Rotate each connecting rod bolt in the clockwise direction. ....  $60 \pm 5$  degrees (1/6 of a turn)

- (8) Bore in the connecting rod for the crankshaft bearing .....  $93.800 \pm 0.013$  mm ( $3.6929 \pm 0.0005$  inch)

i00959054

## Piston and Rings

**SMCS Code:** 1214; 1215

**S/N:** MBJ1-Up

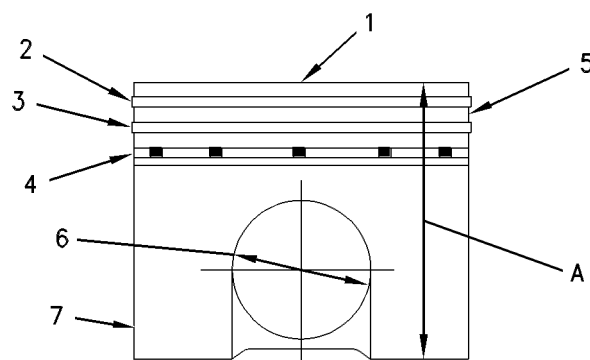


Illustration 75

g00283705

### NOTICE

The piston pin plug prevents the piston pin from scuffing the cylinder wall. The piston pin plug does not retain the piston pin into the assembly. Once the piston assembly is removed from the cylinder liner, the piston pin is free to fall out of the piston assembly at any time.

- (1) The piston is symmetrical with a center crater.

Lubricate the entire piston in zone (A) prior to assembly into the cylinder block. Use clean engine oil.

- (2) Top piston ring

Install the piston ring with the side marked "UP-1" toward the top of the piston. The red stripe faces to the right of the piston ring end gap.

The ends of the piston ring have a clearance when the piston ring is installed in a cylinder liner with a bore of 125.000 mm (4.9213 inch).  
Clearance .....  $0.625 \pm 0.185$  mm ( $.0250 \pm .0073$  inch)

The piston ring has an increase in clearance for each 0.03 mm (.001 inch) increase in the cylinder liner bore. Increase ..... 0.09 mm (.004 inch)

Minimum depth of the groove in new piston for the top ring ..... 5.43 mm (.214 inch)

- (3) Intermediate piston ring

Install the piston ring with the side marked "UP-2" toward the top of the piston. The blue stripe faces to the right of the piston ring end gap.

Width of groove in new piston for intermediate piston ring .....  $3.061 \pm 0.013$  mm  
(.1205  $\pm$  .0005 inch)

Minimum depth of groove in new piston for intermediate piston ring .... 5.43 mm (.214 inch)

Thickness of new intermediate piston ring .....  $2.988 \pm 0.013$  mm (.1176  $\pm$  .0005 inch)

Clearance between groove and intermediate piston ring ..... 0.047 to 0.099 mm  
(.0019 to .0039 inch)

The ends of the piston ring have a clearance when the piston ring is installed in a cylinder liner with a bore of 125.000 mm (4.9213 inch).  
Clearance .....  $0.625 \pm 0.125$  mm  
(.0246  $\pm$  .0049 inch)

The intermediate piston ring has an increase in clearance for each 0.03 mm (.001 inch) increase in the cylinder liner bore. Increase ..... 0.09 mm  
(.004 inch)

#### (4) Oil control piston ring

The ends of the oil control piston ring should be a distance of 180 degrees from the ring end gap when the oil control piston ring is assembled. The orange colored portion of the piston ring must be visible at the ring end gap.

Width of groove in new piston for oil control piston ring .....  $4.033 \pm 0.013$  mm  
(.1588  $\pm$  .0005 inch)

Minimum depth of groove in new piston for oil control piston ring ..... 3.727 mm (.1467 inch)

Thickness of new oil control piston ring .....  $3.987 \pm 0.013$  mm (.1570  $\pm$  .0005 inch)

Clearance between groove and oil control piston ring .... 0.020 to 0.072 mm (.0008 to .0028 inch)

The ends of the piston ring have a clearance when the piston ring is installed in a cylinder liner with a bore of 125.000 mm (4.9213 inch).  
Clearance ...  $0.44 \pm 0.15$  mm (.017  $\pm$  .006 inch)

The piston ring has an increase in clearance for each 0.03 mm (0.001 inch) increase in the cylinder liner bore. Increase ..... 0.09 mm  
(.004 inch)

After the piston rings have been installed, rotate the piston rings so that the end gaps are 120 degrees from each other.

#### (5) Crown assembly

(6) Piston pin bore diameter that is in the piston skirt ..  $50.815 \pm 0.005$  mm (2.0006  $\pm$  .0002 inch)

Piston pin bore diameter that is in the bearing of the crown assembly .....  $50.830 \pm 0.008$  mm  
(2.0012  $\pm$  .0003 inch)

Thoroughly lubricate the piston pin with clean engine oil prior to assembly.

#### (7) Piston skirt

i00959202

## Piston and Rings

**SMCS Code:** 1214; 1215

**S/N:** MBL1-Up

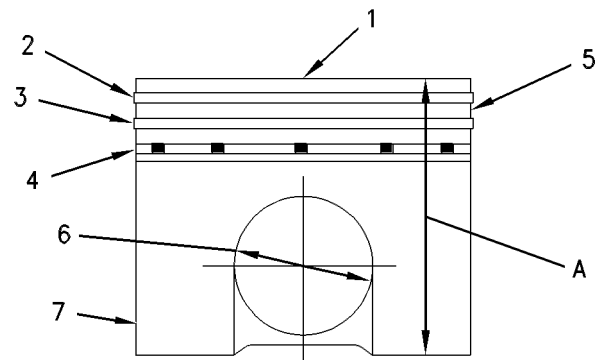


Illustration 76

g00283705

### NOTICE

The piston pin plug prevents the piston pin from scuffing the cylinder wall. The piston pin plug does not retain the piston pin into the assembly. Once the piston assembly is removed from the cylinder, the piston pin is free to fall out of the piston assembly at any time.

(1) The piston is symmetrical with a center crater.

Lubricate the entire piston in zone (A) prior to assembly into the cylinder block. Use clean engine oil.

#### (2) Top piston ring

Install the piston ring with the side marked "UP-1" toward the top of the piston. The green stripe faces to the right of the piston ring end gap.

The ends of the piston ring have a clearance when the piston ring is installed in a cylinder liner with a bore of 130.000 mm (5.1181 inch).  
Clearance .....  $0.520 \pm 0.125$  mm  
(.0205  $\pm$  .0049 inch)

The piston ring has an increase in clearance for each 0.03 mm (.001 inch) increase in the cylinder liner bore. Increase ..... 0.09 mm  
(.004 inch)

Minimum depth of the groove in new piston for the top ring ..... 5.43 mm (.214 inch)

#### (3) Intermediate piston ring

Install the piston ring with the side marked "UP-2" toward the top of the piston. The red stripe faces to the right of the piston ring end gap.

Width of groove in new piston for intermediate piston ring .....  $3.052 \pm 0.013$  mm  
(.1202  $\pm$  .0005 inch)

Minimum depth of groove in new piston for intermediate piston ring ....  $5.43$  mm (.214 inch)

Thickness of new intermediate piston ring ....  $2.950 \pm 0.013$  mm (.1161  $\pm$  .0005 inch)

Clearance between groove and intermediate piston ring .....  $0.076$  to  $0.128$  mm  
(.0030 to .0050 inch)

The ends of the piston ring have a clearance when the piston ring is installed in a cylinder liner with a bore of  $130.000$  mm ( $5.1181$  inch).  
Clearance .....  $0.625 \pm 0.125$  mm  
(.0246  $\pm$  .0049 inch)

The intermediate piston ring has an increase in clearance for each  $0.03$  mm (.001 inch) increase in the cylinder liner bore. Increase .....  $0.09$  mm  
(.004 inch)

#### (4) Oil control piston ring

The ends of the oil control piston ring should be a distance of 180 degrees from the ring end gap when the oil control piston ring is assembled. The white colored portion of the piston ring must be visible at the ring end gap.

Width of groove in new piston for oil control piston ring .....  $4.042 \pm 0.012$  mm  
(.1591  $\pm$  .0005 inch)

Minimum depth of groove in new piston for oil control piston ring .....  $3.727$  mm (.1467 inch)

Thickness of new oil control piston ring ....  $3.987 \pm 0.013$  mm (.1570  $\pm$  .0005 inch)

Clearance between groove and oil control piston ring ....  $0.030$  to  $0.080$  mm (.0012 to .0031 inch)

The ends of the piston ring have a clearance when the piston ring is installed in a cylinder liner with a bore of  $130.000$  mm ( $5.1181$  inch).  
Clearance ...  $0.44 \pm 0.15$  mm (.017  $\pm$  .006 inch)

The piston ring has an increase in clearance for each  $0.03$  mm (.001 inch) increase in the cylinder liner bore. Increase .....  $0.09$  mm  
(.004 inch)

After the piston rings have been installed, rotate the piston rings so that the end gaps are 120 degrees from each other.

#### (5) Crown assembly

#### (6) Piston pin bore diameter that is in the piston skirt .. $53.190 \pm 0.005$ mm ( $2.0941 \pm .0002$ inch)

Piston pin bore diameter that is in the bearing of the crown assembly .....  $53.205 \pm 0.010$  mm  
( $2.0947 \pm .0004$  inch)

Thoroughly lubricate the piston pin with clean engine oil prior to assembly.

#### (7) Piston skirt

i01196515

## Piston Cooling Jet

SMCS Code: 1331

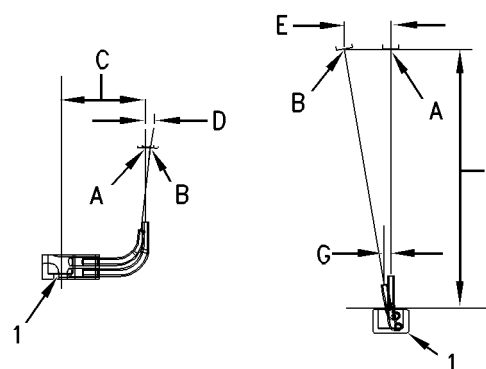


Illustration 77

g00640160

#### (1) Piston cooling jet

The piston cooling jets (1) must be checked for the location of the stream of oil. Insert a drill rod with a diameter of  $2$  mm ( $0.08$  inch) into the orifice. This drill rod simulates the stream of oil under normal operating pressure. The drill rod must pass through a circle with a diameter of  $10$  mm ( $0.40$  inch) at point (A). Place the drill rod in the second orifice. The drill rod must pass through a circle with a diameter of  $10$  mm ( $0.40$  inch) at point (B). The circles are located at dimension (F).

Use the following dimensions in order to locate point (A) and point (B).

Dimension (C) .....  $53$  mm ( $2.09$  inch)  
Dimension (D) .....  $10$  mm ( $0.40$  inch)  
Dimension (E) .....  $29$  mm ( $1.14$  inch)  
Dimension (F) .....  $162$  mm ( $6.4$  inch)  
Dimension (G) .....  $4.4$  mm ( $0.17$  inch)

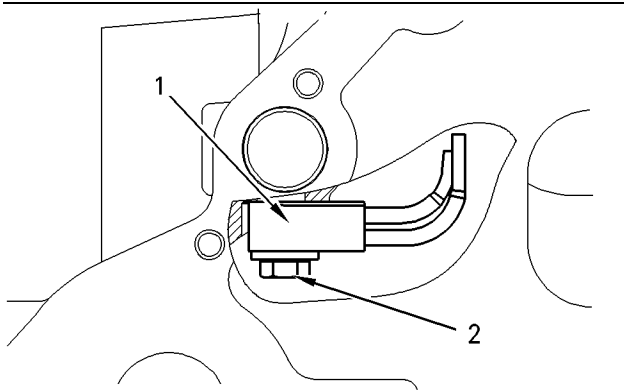


Illustration 78

g00640162

- (1) Piston cooling jet
- (2) Tighten the bolt to the following torque: .....  $35 \pm 3 \text{ N}\cdot\text{m}$  ( $26 \pm 2 \text{ lb ft}$ )

i01741394

## Accessory Drive (Air Compressor)

**SMCS Code:** 1207

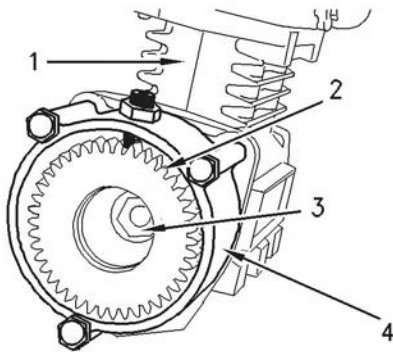


Illustration 79

g00795314

Typical example

- (1) Air compressor
- (2) Accessory drive gear (Air compressor)

**Note:** Caterpillar may use air compressors that have a tapered shaft or a straight shaft. The torque for the nut on the air compressor crankshaft is different for each type of crankshaft. See the following specifications for the proper torque values.

**Note:** Identify the manufacturer of the air compressor for your particular application.

- (3) Tighten the air compressor shaft nut to the following torque.

### Bendix Air Compressors

Tapered shaft

Torque the nut for the air compressor crankshaft (1) to the following torque when the air compressor crankshaft is tapered.  $160 \pm 15 \text{ N}\cdot\text{m}$  ( $118 \pm 11 \text{ lb ft}$ )  
Tap the gear axially with a soft faced hammer in order to seat the gear onto the tapered crankshaft.  
Torque the nut for the air compressor crankshaft again to  $160 \pm 15 \text{ N}\cdot\text{m}$  ( $118 \pm 11 \text{ lb ft}$ ).

Straight shaft

Torque the nut for the air compressor crankshaft (1) to the following torque when the air compressor crankshaft is straight.  $460 \pm 60 \text{ N}\cdot\text{m}$  ( $340 \pm 44 \text{ lb ft}$ )

### Wabco 250 Air Compressors

Torque the nut for the air compressor crankshaft (1) to the following torque.  $150 \pm 5 \text{ N}\cdot\text{m}$  ( $111 \pm 4 \text{ lb ft}$ )

Place an index mark on the nut and turn an additional  $60 \pm 5$  degrees.

- (4) Air compressor gear holder

i01794153

## Housing (Front)

**SMCS Code:** 1151

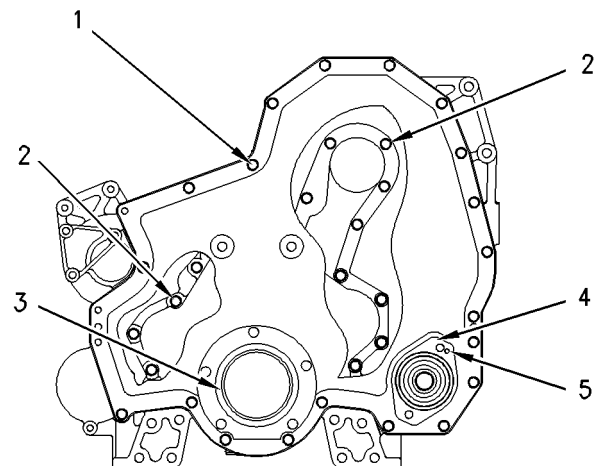


Illustration 80

g00918180

- (1) Bolt

Tighten the bolts to the following torque. ....  $35 \pm 8 \text{ N}\cdot\text{m}$  ( $26 \pm 6 \text{ lb ft}$ )

- (2) Bolt

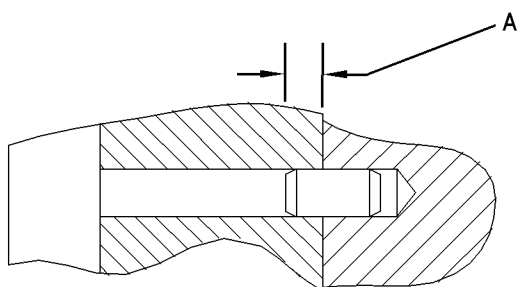
Tighten the Bolt to the following  
torque. ....  $35 \pm 8 \text{ N}\cdot\text{m}$  ( $26 \pm 6 \text{ lb ft}$ )

(3) O-ring seal and crankshaft front seal

The O-ring seal and the crankshaft front seal are part of the front housing. Lubricate the O-ring seal lightly. Use the lubricant that is being sealed.

(4) Accessory drive assembly

(5) Alignment dowel for the accessory drive



---

Illustration 81

g00837947

View A-A

(A) The dowel extends past the surface  
of the housing by the following  
distance. ....  $5.0 \pm 0.5 \text{ mm}$  ( $0.20 \pm .02 \text{ inch}$ )

**Note:** Apply **8T-9022** Silicone Gasket in order to fill the space at the connecting joint of the front housing, engine oil pan, and cylinder block.

i01796876

## Gear Group (Front)

SMCS Code: 1206

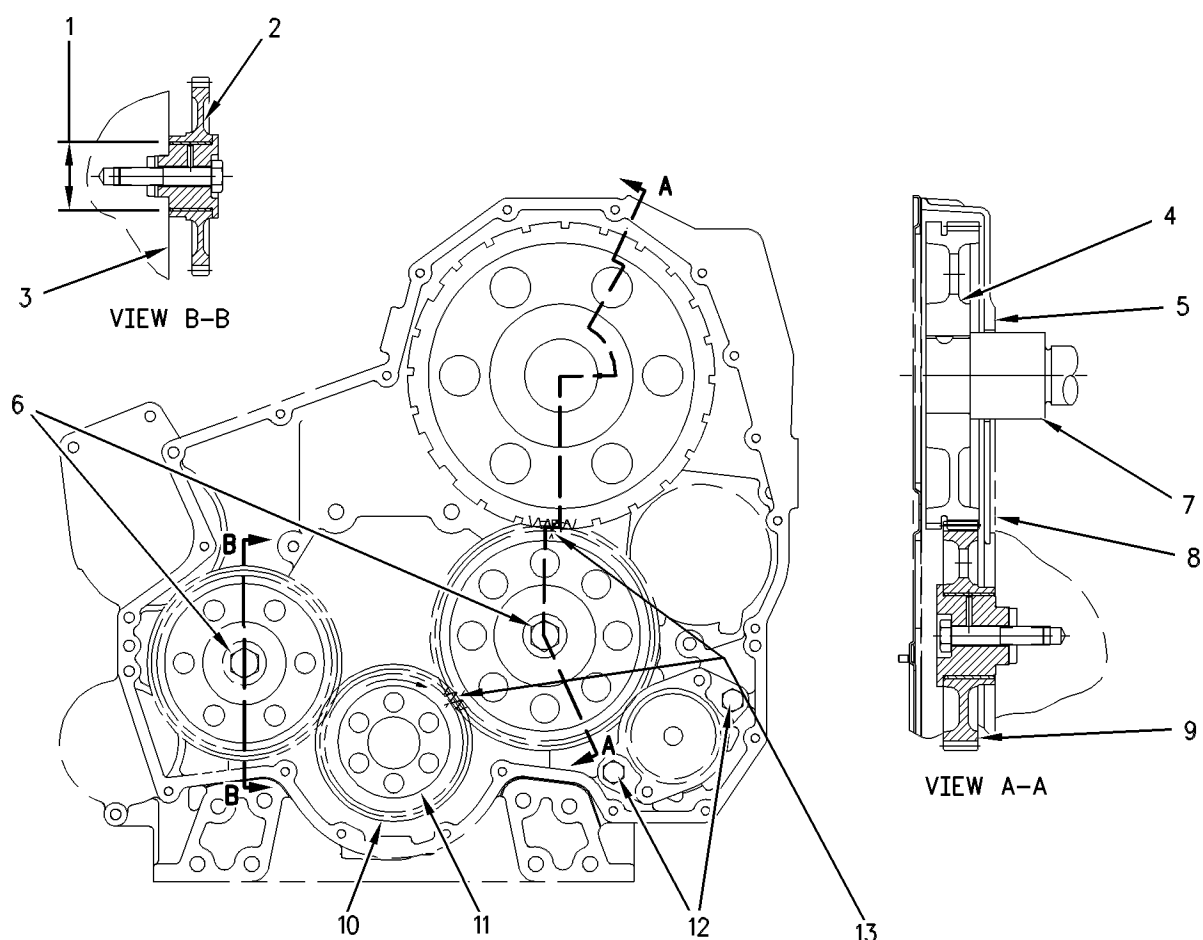


Illustration 82

g00281161

(1) Diameter of gear bore for bearing .....  $60.163 \pm 0.015$  mm  
( $2.3686 \pm 0.0006$  inch)

Diameter of shaft for idler gear ..  $55.047 \pm 0.020$  mm  
( $2.1672 \pm 0.0008$  inch)

(2) Idler gear

(3) Front face of the block

(4) Camshaft gear

(5) Front timing gear housing

(6) Tighten the bolts to the following torque. ....  $240 \pm 40$  N·m ( $175 \pm 30$  lb ft)

(7) Camshaft

(8) Front face of the block

(9) Camshaft idler gear

Diameter of gear bore for bearing .....  $74.452 \pm 0.015$  mm  
( $2.9312 \pm 0.0006$  inch)

Diameter of shaft for idler gear ..  $69.141 \pm 0.008$  mm  
( $2.7221 \pm 0.0003$  inch)

(10) Crankshaft gear

(11) Crankshaft

(12) Tighten the bolts to the following torque. ....  $100 \pm 20$  N·m ( $75 \pm 15$  lb ft)

(13) Align the timing marks that are on the idler gear with the timing marks that are on the crankshaft gear and the camshaft gear.

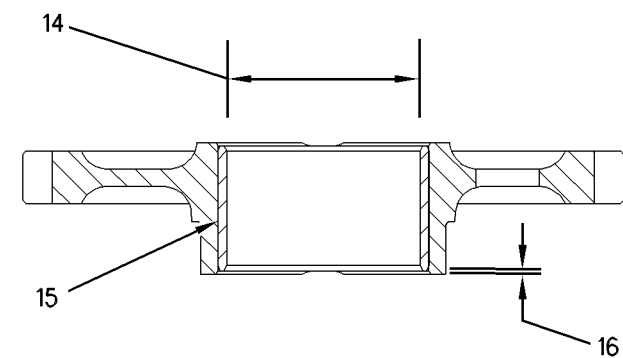


Illustration 83 g00281162

Typical example of the gear assembly for the idler gears

(14) Inside diameter of bearing that is installed in idler gear (2) .....  $55.163 \pm 0.041$  mm  
( $2.1718 \pm 0.0016$  inch)

Inside diameter of bearing that is installed in camshaft idler gear (9) .....  $69.205 \pm 0.013$  mm  
( $2.7246 \pm 0.0005$  inch)

(15) Bearing

(16) Install the bearing in the idler gear (2) to the following depth. ....  $1.00 \pm 0.25$  mm  
( $0.039 \pm 0.010$  inch)

Install the bearing in the camshaft idler gear (9) to the following depth. ....  $1.00 \pm 0.25$  mm  
( $0.039 \pm 0.010$  inch)

## Rear Power Take-Off (RPTO)

SMCS Code: 1165-RE

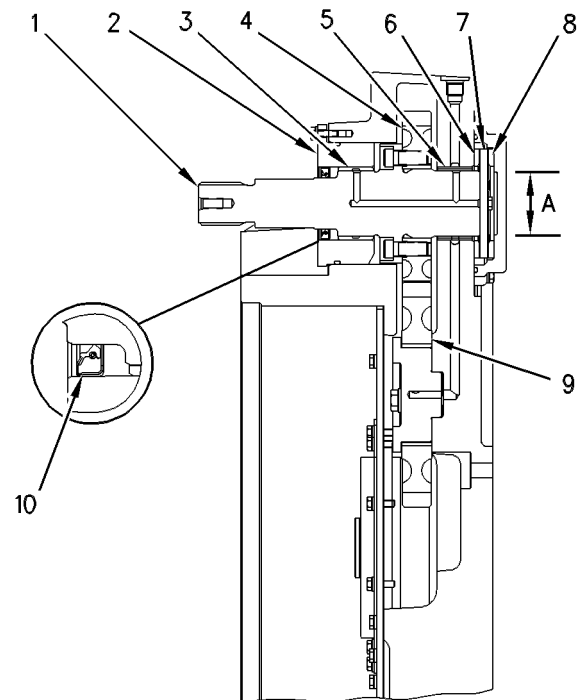


Illustration 84

g00490893

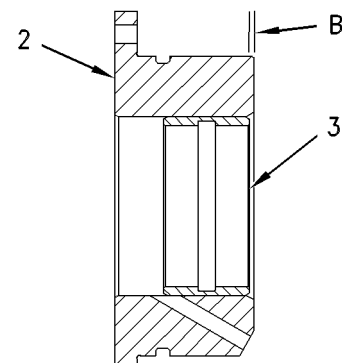


Illustration 85

g00403946

(1) Output shaft (PTO)

Diameter of the output shaft  
(A) ..  $60.245 \pm 0.025$  mm ( $2.3718 \pm 0.0010$  inch)  
End play of the output shaft ....  $0.97 \pm 0.43$  mm  
( $0.038 \pm 0.017$  inch)

(2) Carrier bearing housing

(3) Carrier bearing

The carrier bearing should be installed to the following depth (B) in the carrier bearing housing. ....  $1.6 \pm 0.5$  mm ( $0.06 \pm 0.02$  inch)  
When the engine is viewed from the rear, the split in the bearing should be located counterclockwise from the center of the top bolt at the following angle. ....  $112 \pm 10$  degrees

(4) Drive gear

Inside diameter of the drive gear .....  $60.320 \pm 0.025$  mm ( $2.3750 \pm 0.0010$  inch)

The clearance between the output shaft (1) and the drive gear (4) .....  $0.025$  to  $0.125$  mm ( $0.0010$  to  $0.0050$  inch)

(5) Bearing

(6) Thrust washer

Thickness of the thrust washer .....  $4.690 \pm 0.025$  mm ( $0.1846 \pm 0.0010$  inch)

(7) Thrust plate

Thickness of the thrust plate .....  $6.75$  mm ( $0.266$  inch)

(8) Thrust washer

Thickness of the thrust washer .....  $4.690 \pm 0.025$  mm ( $0.1846 \pm 0.0010$  inch)

(9) Idler gear

Inside diameter of the idler gear ..  $88.90 \pm 0.013$  mm ( $3.500 \pm 0.0005$  inch)

(10) Seal

Lubricate the sealing lip lightly with the lubricant that is being sealed.

Install the seal to the following depth. ....  $2.5 \pm 0.5$  mm ( $0.10 \pm 0.02$  inch)

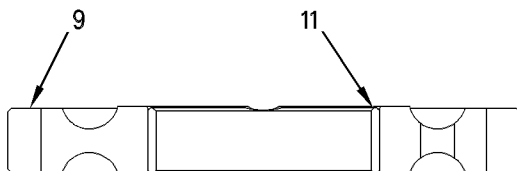


Illustration 86

g00490894

(11) Bearing

The bearing must not extend beyond either face of the idler gear.

The relief groove in the bearing must line up with the relief grooves in the idler gear within the following value. ....  $2$  degrees

i01012861

## Flywheel

SMCS Code: 1156

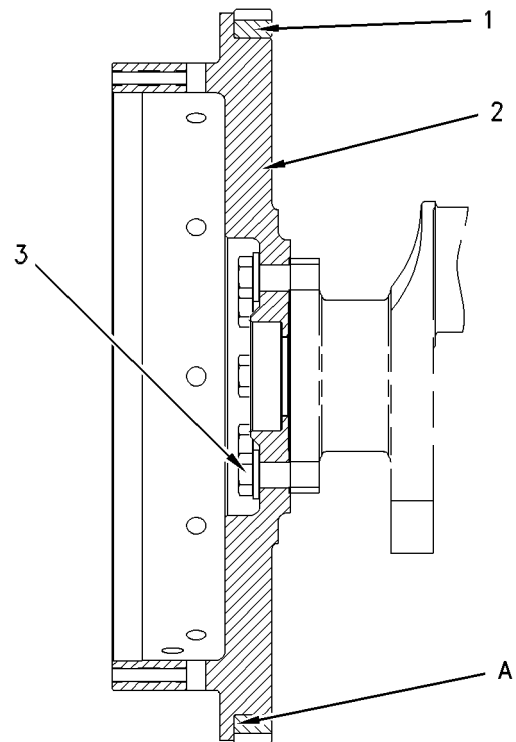


Illustration 87

g00394357

Refer to the Testing And Adjusting, "Basic Block" topic for the proper procedure to check flywheel runout.

(1) Flywheel ring gear

The flywheel ring gear must be assembled against the shoulder (A) of the flywheel. The maximum temperature of the ring gear for installation .....  $300$  °C ( $572$  °F)

(2) Flywheel

(3) Bolt

Apply **7M-7456** Compound to the bolt threads.

Tighten the bolts to the following torque. ....  $300 \pm 40$  N·m ( $220 \pm 30$  lb ft)

i01401694

## Flywheel

SMCS Code: 1156

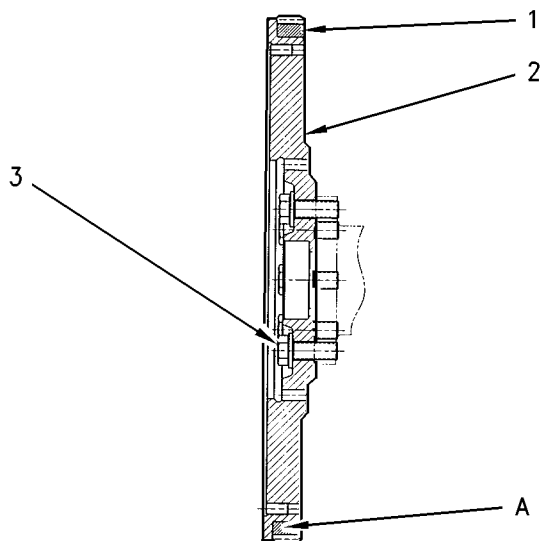


Illustration 88

g00738537

Typical Example

Refer to the Testing And Adjusting, "Basic Block" topic for the proper procedure to check flywheel runout.

### (1) Flywheel ring gear

The flywheel ring gear must be assembled against shoulder (A) of the flywheel. The maximum temperature of the ring gear for installation ..... 315 °C (599 °F)

### (2) Flywheel

### (3) Bolt

Apply **7M-7456** Compound to the bolt threads.

Tighten the bolts to the following torque. .... 300 ± 40 N·m (220 ± 30 lb ft)

i01693054

## Flywheel Housing

SMCS Code: 1157

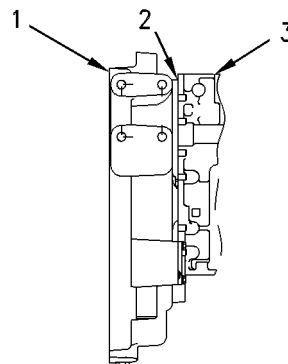


Illustration 89

g00494785

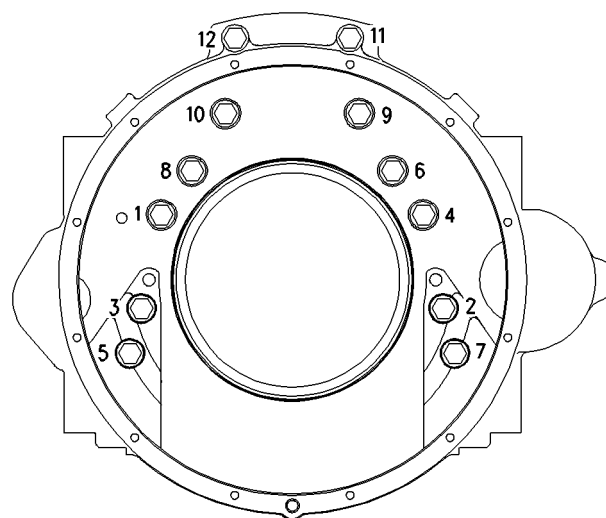


Illustration 90

g00871606

Torque sequence

### (1) Flywheel housing

(2) Apply **1U-8846** Sealant on the gasket surface of the flywheel housing. Assemble the flywheel housing to the cylinder block and tighten within ten minutes of applying the sealant. Use the torque sequence that is shown in Illustration 90 to tighten the bolts.

### (3) Cylinder block

i00971769

## Flywheel Housing (Rear Power Take-Off (RPTO))

SMCS Code: 1157

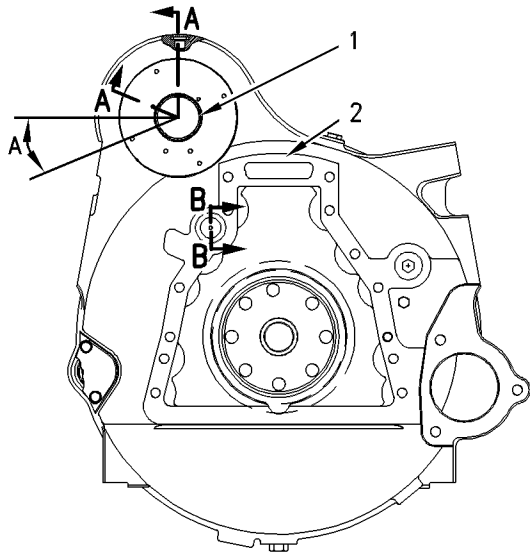


Illustration 91

g00495004

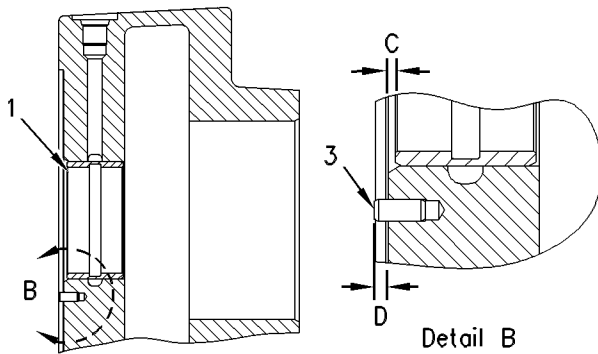


Illustration 92

g00496522

Section A-A

### (1) Bearing

The split in the bearing should be located at angle (A). .....  $23 \pm 10$  degrees  
The bearing should be installed to depth (C). .....  $1.5 \pm 0.5$  mm ( $.06 \pm .02$  inch)

### (2) Flywheel housing

### (3) Dowel

The dowels should extend beyond the surface of the flywheel housing by distance (D). .....  $3.0 \pm 0.2$  mm ( $.12 \pm .01$  inch)

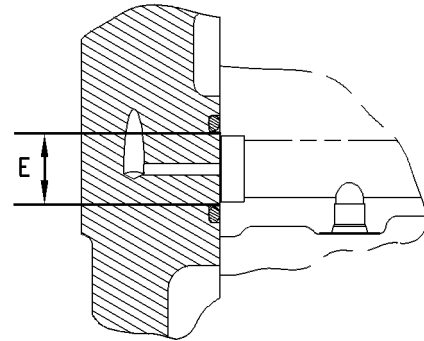


Illustration 93

g00496481

Section B-B

Apply **1U-8846** Sealant on the gasket surface of the flywheel housing except in area (E). Assemble the flywheel housing to the cylinder block and tighten within ten minutes of applying the sealant.

i00826037

## Engine to Transmission Adapter

SMCS Code: 1001

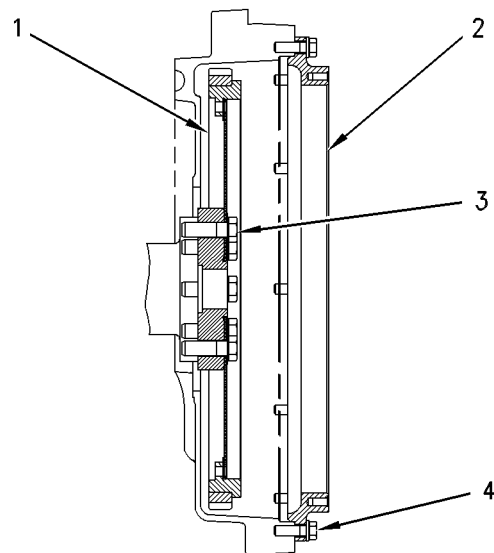


Illustration 94

g00398205

Typical Example

### (1) Flywheel

### (2) Adapter

(3) Apply **9S-3263** Thread Lock Compound to the bolts. Tighten the bolts to the following torque. ....  $300 \pm 40$  N·m ( $220 \pm 30$  lb ft)

- (4) Tighten the bolts to the following torque. ....  $70 \pm 15 \text{ N}\cdot\text{m}$  ( $51 \pm 11 \text{ lb ft}$ )

i00972031

## Belt Tensioner

**SMCS Code:** 1358

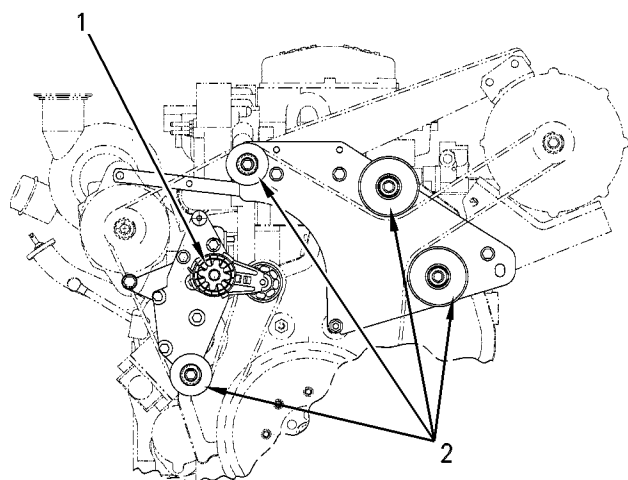


Illustration 95

g00495198

### (1) Belt tensioner

Tighten the bolt that fastens the belt tensioner to the bracket to the following torque. ...  $50 \pm 5 \text{ N}\cdot\text{m}$  ( $37 \pm 4 \text{ lb ft}$ )

### (2) Idler pulley assembly

Tighten the idler pulley assembly to the following torque. ....  $50 \pm 5 \text{ N}\cdot\text{m}$  ( $37 \pm 4 \text{ lb ft}$ )

## Belt Tensioner

**SMCS Code:** 1358

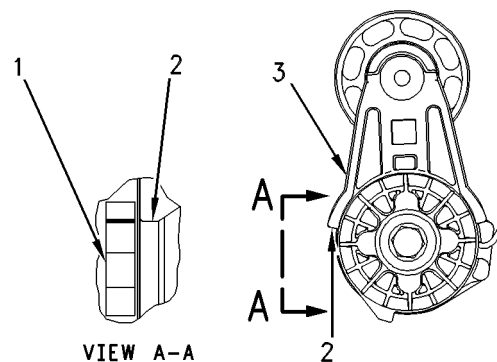


Illustration 96

g00390952

Use the following procedure in order to adjust the belt tensioner:

1. Remove the belt so that there is no tension on the belt tensioner.
2. Align the thick black line on **134-1018 Film (1)** with shoulder (2) of the tensioner's arm (3).
3. Install the belt. Shoulder (2) should be in the green zone of **134-1018 Film (1)**.

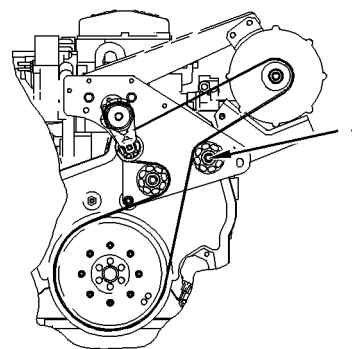


Illustration 97

g00903404

### (4) Idler pulley assembly

Tighten the bolt to the following value. ...  $50 \pm 5 \text{ N}\cdot\text{m}$  ( $36.8 \pm 3.7 \text{ lb ft}$ )

i00826186

## Auxiliary Drive Pulley

SMCS Code: 1205

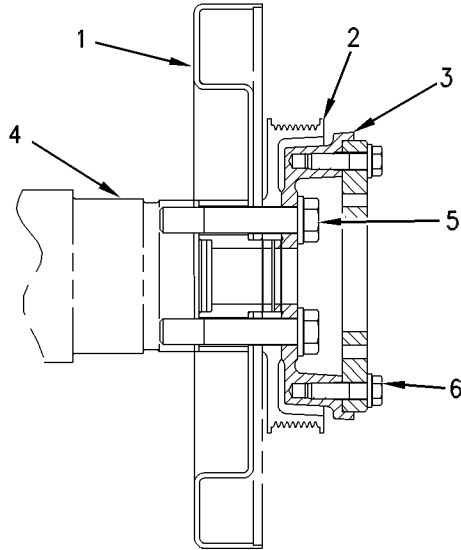


Illustration 98

g00398438

Typical Example

- (1) Damper assembly
- (2) Pulley
- (3) Adapter
- (4) Crankshaft
- (5) Tighten the bolts to the following torque. ....  $300 \pm 40 \text{ N}\cdot\text{m}$  ( $220 \pm 30 \text{ lb ft}$ )
- (6) Tighten the bolts to the following torque. ....  $70 \pm 15 \text{ N}\cdot\text{m}$  ( $50 \pm 11 \text{ lb ft}$ )

i00826442

## Alternator and Regulator

SMCS Code: 1405; 1410

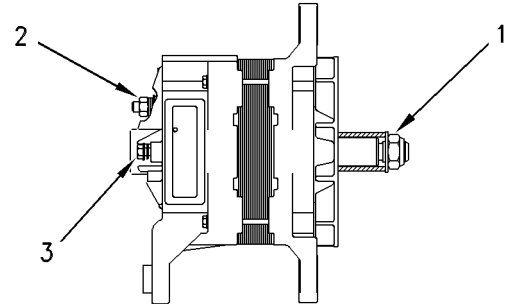


Illustration 99

g00289617

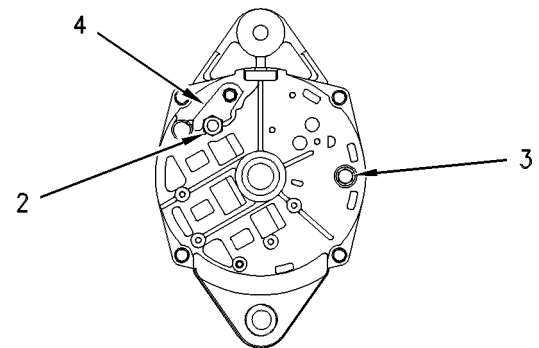


Illustration 100

g00289618

**Note:** Load the battery with a carbon pile **4C-4911** Battery Load Tester in order to get the maximum alternator output.

Polarity ..... Negative Ground

Rotation ..... Clockwise

Minimum full load current at 5000 rpm ..... 107 Amp

Minimum full load current at 1500 rpm ..... 45 Amp

Output voltage .....  $14.0 \pm 0.5 \text{ V}$

(1) Shaft nut torque .....  $102 \pm 7 \text{ N}\cdot\text{m}$  ( $75 \pm 5 \text{ lb ft}$ )

(2) Positive terminal torque .....  $11.3 \pm 2.3 \text{ N}\cdot\text{m}$   
( $100 \pm 20 \text{ lb in}$ )

(3) Negative terminal torque .....  $6.2 \pm 0.6 \text{ N}\cdot\text{m}$   
( $55 \pm 5 \text{ lb in}$ )

(4) Regulator

Voltage setting ..... No Adjustment

Permissible voltage range ..... 13.5 to 14.5 V

i01584501

## Electric Starting Motor

**SMCS Code:** 1453

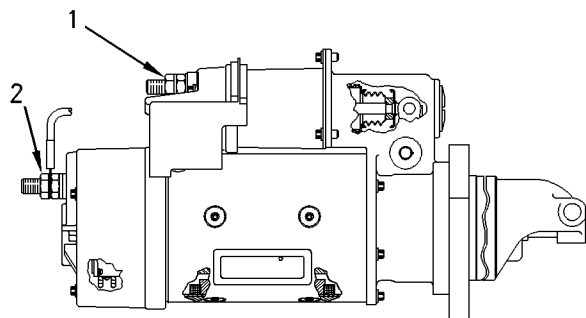


Illustration 101

g00282744

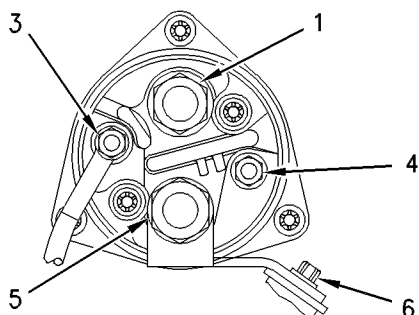


Illustration 102

g00493900

When the electric starting motor is viewed from the drive end, the motor rotates in the following direction. .... Clockwise

No load conditions at 25 °C (77 °F)

Speed ..... 6900 ± 1800 rpm  
Current draw ..... 127.5 ± 27.5 amp  
Voltage ..... 12 V

(1) Battery terminal nut

A maximum of three cable or wire terminals should be between the nuts.

Torque ..... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(2) Ground terminal nut

Torque ..... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(3) Ground terminal nut

Torque ..... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(4) Switch terminal nut

Torque ..... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(5) Motor terminal nut

Torque ..... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(6) Motor frame nut

Torque ..... 8 ± 3 N·m (72 ± 24 lb in)

Clearance between the pinion and the housing ..... 9.1 mm (0.36 inch)

## Solenoid

Current draw at 10 V and 25 °C (77 °F)

Pull-in windings ..... 92.5 ± 7.5 Amp

Hold-in windings ..... 23 Amp maximum

i00970077

## Electric Starting Motor

**SMCS Code:** 1453

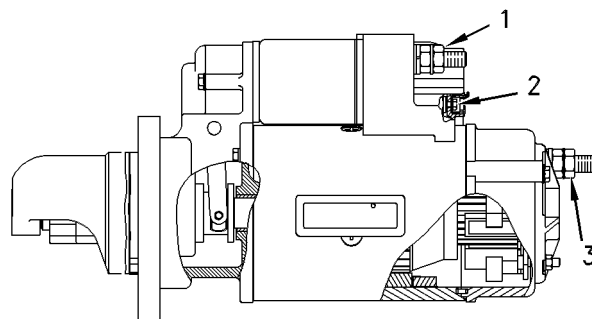


Illustration 103

g00282453

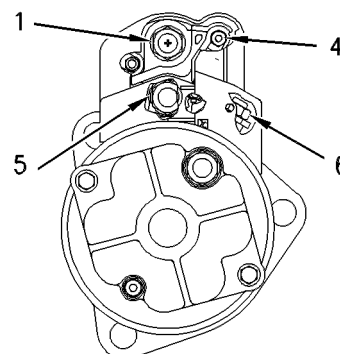


Illustration 104

g00494167

i01584471

When the electric starting motor is viewed from the drive end, the motor rotates in the following direction. .... Clockwise

No load conditions at 25 °C (77 °F)

Minimum speed with no load ..... 3300 rpm  
Maximum current with no load ..... 80 amp  
Voltage ..... 23.5 V

(1) Battery terminal

A maximum of three cable or wire terminals should be between the nuts.

Tighten the nut on the battery terminal to the following torque. ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(2) Ground terminal

Tighten the nut on the ground terminal to the following torque. .... 1.5 ± 0.5 N·m (13 ± 4 lb in)

(3) Ground terminal

Tighten the nut on the ground terminal to the following torque. ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(4) Switch terminal

The wire terminal that is on the switch terminal must be insulated with heat shrink tubing. Do not use molded terminals.

Tighten the nut on the switch terminal to the following torque. .... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(5) Motor terminal

Tighten the nut on the motor terminal to the following torque. ... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(6) Motor frame terminal

Tighten the nut on the motor frame terminal to the following torque. .... 17.5 ± 2.5 N·m (13 ± 2 lb ft)

## Solenoid

Current draw at 24 V and 25 °C (77 °F)

Pull-in windings ..... 103 Amp

Hold-in windings ..... 19 Amp maximum

## Electric Starting Motor

SMCS Code: 1453

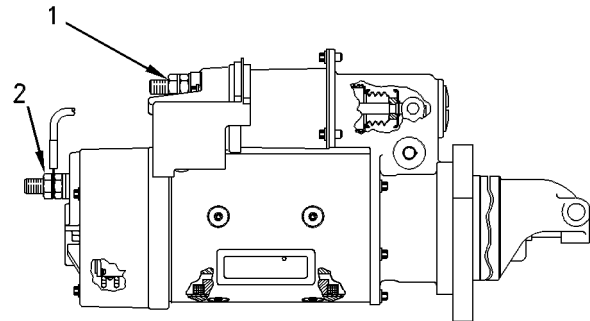


Illustration 105

g00282688

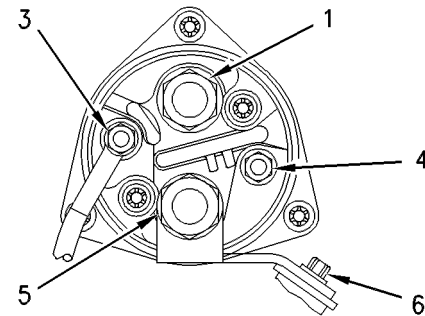


Illustration 106

g00493900

When the electric starting motor is viewed from the drive end, the motor rotates in the following direction. .... Clockwise

No load conditions at 25 °C (77 °F)

Speed ..... 7643 ± 1683 rpm  
Current draw ..... 67.5 ± 7.5 amp  
Voltage ..... 23 V

(1) Battery terminal nut

A maximum of three cable or wire terminals should be between the nuts.

Torque ..... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(2) Ground terminal nut

Torque ..... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(3) Ground terminal nut

Torque ..... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(4) Switch terminal nut

The wire terminal that is on the switch terminal must be insulated with heat shrink tubing. Do not use molded terminals.

Torque ..... 2.25 ± 0.25 N·m (20 ± 2 lb in)

(5) Motor terminal nut

Torque ..... 30.5 ± 3.5 N·m (22 ± 3 lb ft)

(6) Motor frame nut

Torque ..... 8 ± 3 N·m (72 ± 24 lb in)

Clearance between the pinion and the housing ..... 9.1 mm (0.36 inch)

## Solenoid

Current draw at 20 V and 25 °C (77 °F)

Pull-in windings ..... 56.5 ± 4.5 Amp

Hold-in windings ..... 14.6 Amp maximum

i01193848

## Electrical Ground Stud

**SMCS Code:** 1408

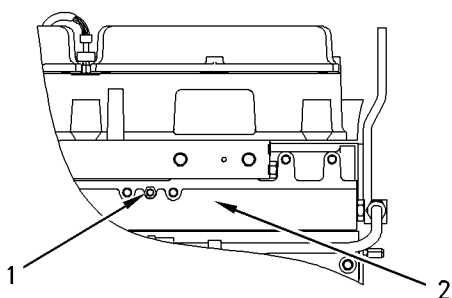


Illustration 107

g00638966

The electrical grounding stud is located on the left side of the engine.

(1) Nut

Tighten the outer nut on the ground stud to the following torque. .... 10 ± 3 N·m (89 ± 27 lb in)

(2) Cylinder block

i01884178

## Coolant Temperature Sensor

**SMCS Code:** 1906

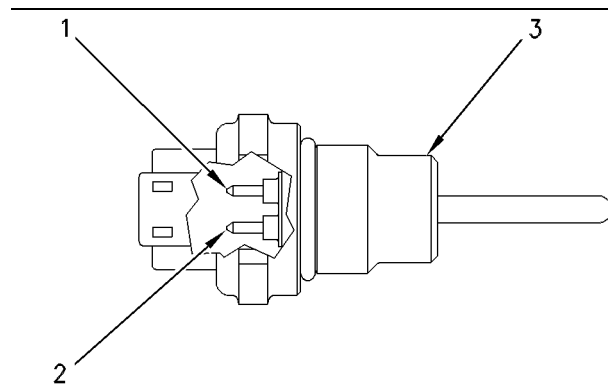


Illustration 108

g00493454

(1) Pin 1

(2) Pin 2

(3) Sensor assembly

Tighten the sensor assembly to the following torque. .... 20 ± 5 N·m (15 ± 4 lb ft)

i00968767

## Fuel Temperature Sensor

**SMCS Code:** 1922

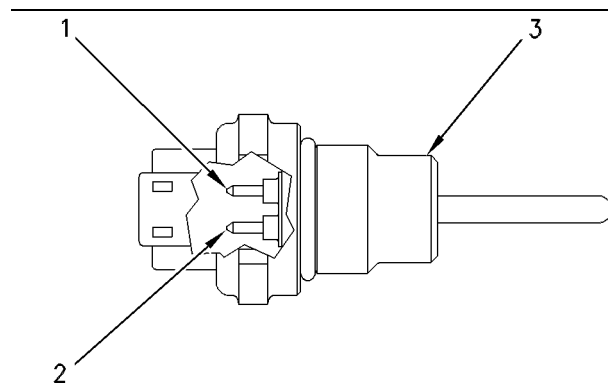


Illustration 109

g00493454

(1) Pin 1

(2) Pin 2

(3) Sensor assembly

Tighten the sensor assembly to the following torque. .... 20 ± 5 N·m (15 ± 4 lb ft)

i00627515

## Engine Oil Pressure Sensor

SMCS Code: 1924

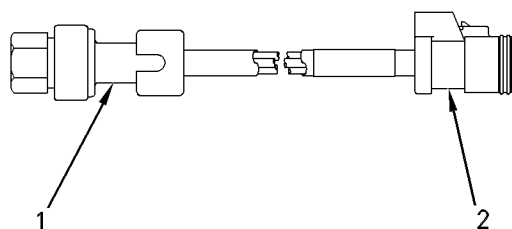


Illustration 110

g00286048

(1) Sensor assembly

Tighten sensor assembly to the following torque. ....  $10 \pm 2$  N·m ( $88 \pm 18$  lb in)

(2) Receptacle

## Boost Pressure Sensor

SMCS Code: 1917

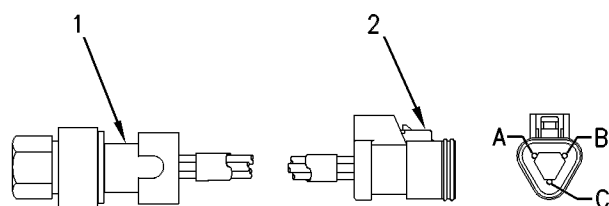


Illustration 111

g00490140

(1) Sensor assembly

Tighten the sensor assembly to the following torque. ....  $10 \pm 2$  N·m ( $88 \pm 18$  lb in)

(2) Plug with socket A, socket B, and socket C

i00974651

## Atmospheric Pressure Sensor

SMCS Code: 1923

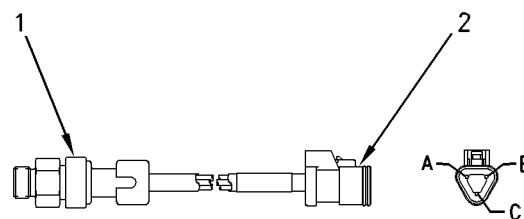


Illustration 112

g00281665

(1) Sensor assembly

Tighten the sensor assembly to the following torque. ....  $10 \pm 2$  N·m ( $88 \pm 18$  lb in)

(2) The plug has socket A, socket B, and socket C.

## Speed and Timing Sensor

SMCS Code: 1907; 1912

## Camshaft Position Sensor

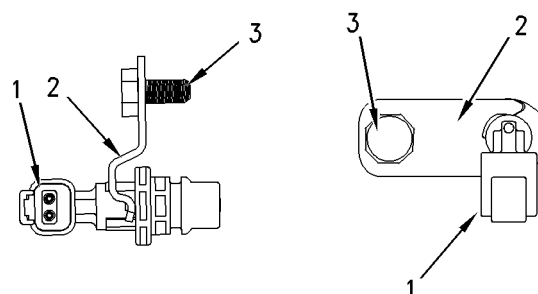


Illustration 113

g00640258

Camshaft position sensor

(1) Sensor

(2) Bracket

Ensure that the bracket is installed in the orientation that is shown.

(3) Tighten the bolt to the following torque: ....  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft)

Ensure that the sensor is seated before the bolt is tightened.

## Crankshaft Position Sensor

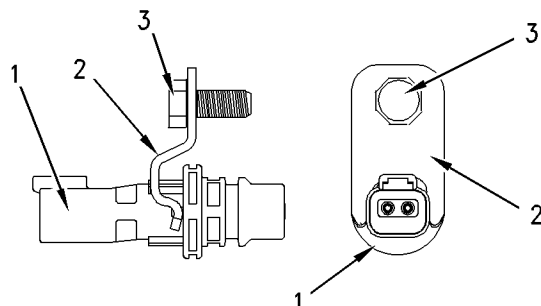


Illustration 114  
Crankshaft position sensor

g00640259

- (1) Sensor
- (2) Bracket

Ensure that the bracket is installed in the orientation that is shown.

- (3) Tighten the bolt to the following torque: .....  $28 \pm 7 \text{ N}\cdot\text{m}$  ( $21 \pm 5 \text{ lb ft}$ )

Ensure that the sensor is seated before the bolt is tightened.

i00996748

## Accelerator Pedal Position Sensor

**SMCS Code:** 1913

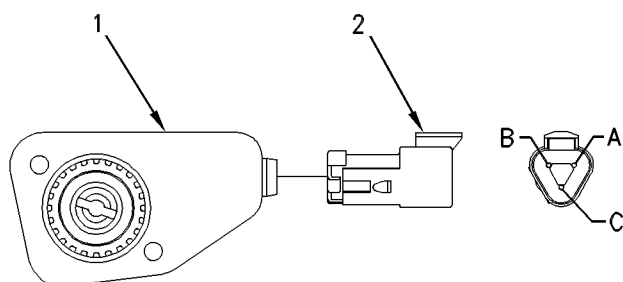


Illustration 115

g00496388

- (1) Sensor assembly

- (2) The plug has socket A, socket B, and socket C.

i01193346

## Air Compressor

**SMCS Code:** 1803

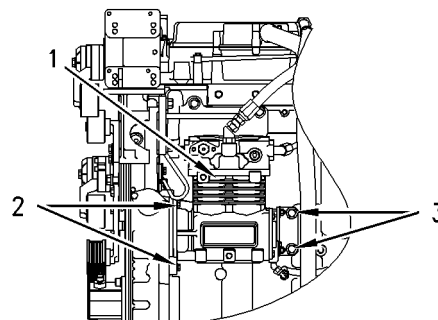


Illustration 116

g00638646

- (1) Air compressor
- (2) Bolts

Apply **9S-3263** Thread Lock Compound to the threads of the bolts. Tighten the bolts to the following torque. ....  $100 \pm 20 \text{ N}\cdot\text{m}$  ( $75 \pm 15 \text{ lb ft}$ )

- (3) Bolts

Tighten the bolts to the following torque. ....  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $41 \pm 7 \text{ lb ft}$ )

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# Systems Operation Testing and Adjusting

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## **C-10 and C-12 Truck Engines**

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MBJ1-Up (Engine)  
MBL1-Up (Engine)



## Important Safety Information

Most accidents that involve product operation, maintenance and repair are caused by failure to observe basic safety rules or precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards. This person should also have the necessary training, skills and tools to perform these functions properly.

**Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.**

**Do not operate or perform any lubrication, maintenance or repair on this product, until you have read and understood the operation, lubrication, maintenance and repair information.**

Safety precautions and warnings are provided in this manual and on the product. If these hazard warnings are not heeded, bodily injury or death could occur to you or to other persons.

The hazards are identified by the "Safety Alert Symbol" and followed by a "Signal Word" such as "DANGER", "WARNING" or "CAUTION". The Safety Alert "WARNING" label is shown below.



The meaning of this safety alert symbol is as follows:

**Attention! Become Alert! Your Safety is Involved.**

The message that appears under the warning explains the hazard and can be either written or pictorially presented.

Operations that may cause product damage are identified by "NOTICE" labels on the product and in this publication.

**Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. If a tool, procedure, work method or operating technique that is not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that the product will not be damaged or be made unsafe by the operation, lubrication, maintenance or repair procedures that you choose.**

The information, specifications, and illustrations in this publication are on the basis of information that was available at the time that the publication was written. The specifications, torques, pressures, measurements, adjustments, illustrations, and other items can change at any time. These changes can affect the service that is given to the product. Obtain the complete and most current information before you start any job. Caterpillar dealers have the most current information available.



**When replacement parts are required for this product Caterpillar recommends using Caterpillar replacement parts or parts with equivalent specifications including, but not limited to, physical dimensions, type, strength and material.**

**Failure to heed this warning can lead to premature failures, product damage, personal injury or death.**

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## Systems Operation Section

i00980518

# Engine Design

**SMCS Code:** 1000

**S/N:** MBJ1-Up

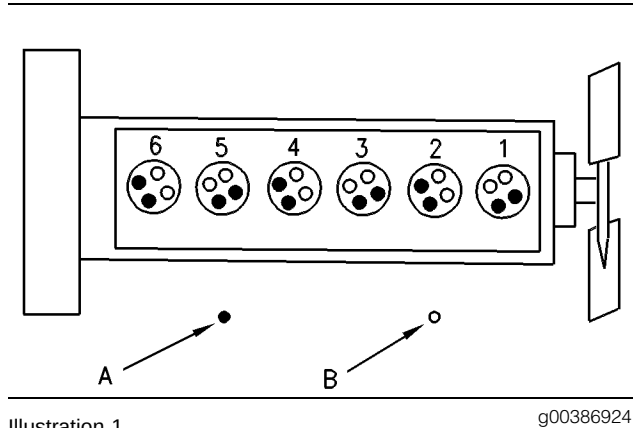


Illustration 1  
Cylinder and Valve Location  
(A) Exhaust valves  
(B) Inlet valves

|                            |                               |
|----------------------------|-------------------------------|
| Bore .....                 | 125.0 mm (4.92 inch)          |
| Stroke .....               | 140.0 mm (5.51 inch)          |
| Displacement .....         | 10.3 L (629 in <sup>3</sup> ) |
| Cylinder arrangement ..... | In-line                       |
| Valves per cylinder .....  | 4                             |

In order to check the engine valve lash setting, the engine must be cold and the engine must be stopped. Engine valve lash settings

|                                |                        |
|--------------------------------|------------------------|
| Inlet .....                    | 0.38 mm (.015 inch)    |
| Exhaust .....                  | 0.64 mm (.025 inch)    |
| Engine compression brake ..... | 0.64 mm<br>(.025 inch) |

Type of combustion ..... Direct Injection

Firing Order ..... 1 - 5 - 3 - 6 - 2 - 4

The crankshaft rotation is viewed from the flywheel end of the engine. Crankshaft rotation ..... Counterclockwise

**Note:** The front end of the engine is opposite of the flywheel end of the engine. The left side of the engine and the right side of the engine are viewed from the flywheel end of the engine. The No. 1 cylinder is the front cylinder.

i00980375

## Engine Design

**SMCS Code:** 1000

**S/N:** MBL1-Up

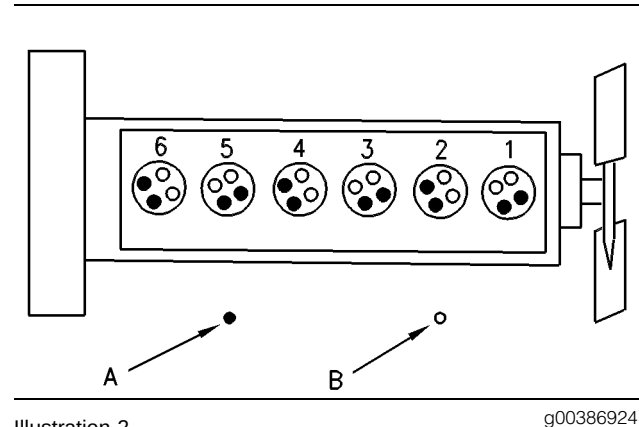


Illustration 2  
Cylinder and Valve Location  
(A) Exhaust valves  
(B) Inlet valves

|                            |                             |
|----------------------------|-----------------------------|
| Bore .....                 | 130.0 mm (5.12 inch)        |
| Stroke .....               | 150.0 mm (5.91 inch)        |
| Displacement .....         | 12 L (732 in <sup>3</sup> ) |
| Cylinder arrangement ..... | In-line                     |
| Valves per cylinder .....  | 4                           |

In order to check the engine valve lash setting, the engine must be cold and the engine must be stopped. Engine valve lash settings

|                                |                        |
|--------------------------------|------------------------|
| Inlet .....                    | 0.38 mm (.015 inch)    |
| Exhaust .....                  | 0.64 mm (.025 inch)    |
| Engine compression brake ..... | 1.02 mm<br>(.040 inch) |

Type of combustion ..... Direct Injection

Firing Order ..... 1 - 5 - 3 - 6 - 2 - 4

The crankshaft rotation is viewed from the flywheel end of the engine. Crankshaft rotation ..... Counterclockwise

**Note:** The front end of the engine is opposite of the flywheel end of the engine. The left side of the engine and the right side of the engine are viewed from the flywheel end of the engine. The No. 1 cylinder is the front cylinder.

i01797236

## General Information

**SMCS Code:** 1000

C-10 Truck Engines and C-12 Truck Engines are in-line six cylinder arrangements. C-10 Truck Engines have a bore of 125.0 mm (4.92 inch) and a stroke of 140.0 mm (5.51 inch). The displacement is 10.3 L (629 in<sup>3</sup>). C-12 Truck Engines have a bore of 130.0 mm (5.12 inch) and a stroke of 150.0 mm (5.91 inch). The displacement is 12 L (732 in<sup>3</sup>). Both engines have a firing order sequence: 1, 5, 3, 6, 2, and 4. The rotation of these engines is counterclockwise when the engine is viewed from the flywheel end of the engine. These engines utilize a turbocharger and an air-to-air aftercooler.

The Electronic Unit Injector system (EUI) eliminates many of the mechanical components that are traditionally used in the fuel injector assembly. The EUI also provides increased control of the timing and increased control of the fuel air mixture. The timing advance is achieved by precise control of the fuel injection timing. Engine rpm is controlled by adjusting the injection duration.

Engines (S/N: 3CS1-11999) and (S/N: 2KS1-39999) have a special pulse wheel, which is mounted to the crankshaft gear. The special pulse wheel provides information to the Engine Control Module (ECM) for detection of cylinder position and engine rpm.

Engines (S/N: 3CS12000-UP), (S/N: 2KS40000-UP), and (S/N: CPD1-UP) have a special crankshaft gear that provides information to the Engine Control Module (ECM) for detection of cylinder position and engine rpm.

The engine has built-in diagnostics in order to ensure that all of the components are operating properly. In the event of a system component failure, the operator will be alerted to the condition by a check engine light. The check engine light is located on the dashboard. An electronic service tool can be used to read the numerical code of the faulty component or condition. Also, the cruise control switches can be used to flash the code on the check engine light. Intermittent faults are logged and stored in memory.

## Starting The Engine

The engine's ECM will automatically provide the correct amount of fuel in order to start the engine. Do not hold the throttle down while the engine is cranking. If the engine fails to start in 30 seconds, release the starting switch. Allow the starting motor to cool for two minutes before the starting motor is used again.

---

### NOTICE

Excessive ether (starting fluid) can cause piston and ring damage. Use ether for cold weather starting purposes only.

---

## Cold Mode Operation

The ECM will set the cold start strategy when the coolant temperature is below 18 °C (64 °F).

When the cold start strategy is activated, low idle rpm will be increased to 1000 rpm and the engine's power will be limited.

Cold mode operation will be deactivated when any of the following conditions have been met:

- Coolant temperature reaches 18 °C (64 °F).
- The engine has been running for fourteen minutes.

Cold mode operation varies the fuel injection amount for white smoke cleanup. Cold mode operation also varies the timing for white smoke cleanup. The engine operating temperature is usually reached before the walk-around inspection is completed. The engine will idle at the programmed low idle rpm in order to be put in gear.

---

### NOTICE

Do not move the vehicle with the engine in the cold mode condition. Engine power could be noticeably reduced. At a vehicle speed above 8 km/h (5 mph), low idle rpm will be reduced to the customer programmed low idle and the power will still be reduced.

---

After the cold mode is completed, the engine should be operated at low rpm until normal operating temperature is reached. The engine will reach normal operating temperature faster when the engine is operated at low rpm and low power demand.

## Customer Specified Parameters

The engine is capable of being programmed for several customer specified parameters. For a brief explanation of each of the customer specified parameters, see the Operation and Maintenance Manual.

i01797283

## Glossary of Electronic Control Terms

**SMCS Code:** 1900

**ATA Data Link** – The ATA Data Link is an electrical connection that is used for communication between microprocessor based devices. The devices must be compatible with the American Trucking Association and SAE Standards (J1587 and J1708) such as trip recorders, electronic dashboards, and maintenance systems. The ATA Data Link is the communication medium that is used for the programming and the troubleshooting with Caterpillar electronic service tools.

**Aftermarket Device** – An aftermarket device is a device or an accessory that is installed by the customer after the vehicle is delivered.

**Air-To-Air Aftercooler** – An air-to-air aftercooler is a device that is used on turbocharged engines in order to cool inlet air that has undergone compression. The inlet air is cooled after the inlet air passes through the turbocharger. The inlet air is passed through an aftercooler (heat exchanger) that uses ambient air for cooling. The inlet air that has been cooled advances to the inlet manifold.

**Alternating Current (AC)** – The direction of the current flow changes. The current flow alternates constantly.

**American Wire Gauge (AWG)** – AWG is a measure of the diameter of electrical wire. AWG is also a measure of the current carrying capacity of electrical wire. When the AWG number is smaller, the diameter of the wire is larger. When the AWG number is larger, the diameter of the wire is smaller.

**Atmospheric Pressure Sensor** – The atmospheric pressure sensor measures barometric pressure. The sensor sends a signal to the Engine Control Module (ECM). The signal is used in engine control and in engine operation.

**Before Top Center (BTC)** – BTC is the 180 degrees of crankshaft rotation before the piston reaches the top center position in the normal direction of rotation.

**Boost Pressure Sensor** – The boost pressure sensor measures inlet manifold air pressure. The boost pressure sensor sends a signal to the ECM.

**Bypass Circuit** – A bypass circuit is a circuit that is used as a substitute circuit for an existing circuit. A bypass circuit is typically used as a test circuit.

**Calibration** – Calibration is an electronic adjustment of a sensor signal.

**CAT Data Link** – The data link is an electrical connection that is used to communicate with other devices that utilize microprocessors. The microprocessor based devices are compatible with the proposed American Trucking Association and the SAE standard. These devices include trip recorders, electronic dashboards, and maintenance systems. The data link is also the communication medium that is used for programming with the electronic service tool. The data link is also used for troubleshooting with the electronic service tool.

**Caterpillar Electronic Technician (Cat ET)** – Cat ET is a Caterpillar electronic service tool that is used for diagnosing and programming a variety of electronic controls.

**Caterpillar Engine Monitoring** – The part of the Caterpillar Electronic Engine Control that monitors coolant temperature, oil pressure, inlet manifold air temperature, and coolant level. Caterpillar engine monitoring then informs the operator of any problems that are detected. The Coolant Temperature, Inlet Manifold Air Temperature, and Oil pressure Sensors are supplied by Caterpillar and Monitored by the ECM. The coolant level sensor is installed by the OEM factory but still monitored by the ECM. This is opposed to an aftermarket Engine Monitoring System. An aftermarket Engine Monitoring System does not interface with the Caterpillar Electronic Engine Control.

**Clutch Switch** – The clutch switch is supplied by the OEM factory. The clutch switch is a limit switch. The usually adjustable switch is mounted near the pedal. When the pedal is released, the switch is closed. Depressing the clutch will open the circuit.

**Code** – See the Diagnostic Code.

**Coolant Fan Relay** – The relay is supplied by the OEM manufacturer. The installed relay is controlled by the ECM. The ECM controls the cooling fan relay. The ECM uses the following information to control the cooling fan relay: coolant temperature and the high pressure switch on the air conditioner.

**Coolant Level Sensor** – The sensor is installed by the OEM factory. The sensor detects the absence of coolant at the probe. The sensor also detects the presence of coolant at the probe. The sensor then sends the appropriate signal to the ECM.

**Coolant Temperature Sensor** – The coolant temperature sensor measures the engine coolant temperature. The sensor sends a signal to the ECM. The engine's coolant temperature is used in Cold Mode operation. Coolant temperature is also used in order to optimize engine performance.

**Cruise Control Range** – The cruise control range establishes the vehicle speed range for the operation of cruise control. The range is usually limited to the speed range that is anticipated on the open road.

**Custom Data** – The custom data is for monitoring purposes. The ECM stored trip data holds the custom data. The custom data allows the owner to specify operating parameters of the engine application.

**Customer Specified Parameters** – A customer specified parameter is a parameter that can be changed. A customer specified parameter's value is set by the customer. These parameters are protected by customer passwords.

**Dash Check Engine Lamp** – The dash check engine lamp is sometimes called the diagnostic lamp. The dash check engine lamp is used to warn the operator of the presence of an active diagnostic code.

**Desired RPM** – The desired rpm is input to the electronic governor within the ECM. The electronic governor uses the signal from the Accelerator Pedal Position Sensor, the Engine Speed Sensor, the Cruise Control, and the Customer Parameters in order to determine desired rpm.

**Diagnostic Code** – A diagnostic code is sometimes called a fault code. A diagnostic code is an indication of a problem or event in the electrical engine systems.

**Diagnostic Event Code** – These codes indicate an event. Diagnostic event codes are not necessarily an indication of problem within the electronic system.

**Diagnostic Fault Code** – Diagnostic fault codes indicate an electronic system malfunction that indicates a problem with the electronic system.

**Diagnostic Flash Code** – Diagnostic flash codes are used to indicate a malfunction within the electronic system. Diagnostic flash codes are also used to indicate an event that is detected by the electronic system. The codes are flashed on the dash check engine lamp.

**Diagnostic Lamp** – A diagnostic lamp is sometimes called the check engine light. The diagnostic lamp is used to warn the operator of the presence of an active diagnostic code.

**Direct Current (DC)** – Direct current is the type of current that flows consistently in only one direction.

**Dual Coil Speed Sensor** – The movement of the teeth on the output shaft of the transmission is detected by the dual coil speed sensor. The sensor contains two coils. One coil is used to supply a signal to the speedometer. The second coil is used to supply a signal to the speed buffer.

**Duty Cycle** – See Pulse Width Modulation.

**Engine Control Module (ECM)** – The ECM is the engine's control computer. The ECM provides power to the electronics. The ECM monitors data that is input from the engine's sensors. The ECM acts as a governor in order to control engine rpm.

**Electronic Unit Injector** – The unit injector is mechanically actuated and the unit injector is electronically controlled. The EUI combines the pumping, electronic fuel metering, and injection into a single unit.

**ECM Fan Control** – The ECM controls the cooling fan relay. The ECM uses the following information to control the cooling fan relay: coolant temperature and the high pressure switch on the air conditioner.

**Electronic Engine Control** – The electronic engine control is a complete electronic system. The electronic engine control monitors the engine operation under all conditions. The electronic engine control also controls the engine operation under all conditions.

**Engine Retarder Solenoids** – The engine retarder solenoids are installed by Caterpillar. The solenoids are driven directly from the ECM unlike the auxiliary brake. The auxiliary brake utilizes a relay that is installed by the OEM factory.

**Engine Speed/Timing Sensor** – The engine speed/timing sensor provides a pulse width modulated signal to the ECM. The ECM interprets the signal as the crankshaft position and the engine speed.

**Failure Mode Identifier (FMI)** – The FMI describes the type of failure that was experienced by the component. The codes for the FMI were adopted from the standard practices of SAE (J1587 diagnostics).

**Flash Code** – The Caterpillar code numbers that are displayed on the diagnostic lamp.

**Fuel Ratio Control (FRC)** – The FRC is a limit that is based on the control of the fuel to the air ratio. The FRC is used for emission control. When the ECM senses a higher turbocharger outlet pressure, the ECM increases the limit for the FRC in order to allow more fuel into the cylinders.

**Fuel Position** – The fuel position is a signal within the ECM. The signal is from the electronic governor. The signal goes to the fuel injection control. The signal is based on the desired engine speed, the FRC, the rated position, and the actual engine speed.

**Fuel Temperature Sensor** – This sensor detects the fuel temperature. The ECM monitors the fuel temperature. The ECM then calculates the needed fuel rate.

**Full Load Setting (FLS)** – This is the fuel system adjustment that is made at the factory. The adjustment is used to fine tune the fuel system. The correct value for this parameter is stamped on the **9L-6531** Information Plate. This parameter must be programmed. If the parameter is not programmed, a Diagnostic Code 253-02 Check Customer or a System Parameters Fault Code 56 will be generated.

**Full Torque Setting (FTS)** – This is similar to Full Load Setting. The parameter for full torque setting must be programmed or a Diagnostic Code 253-02 Check Customer or a System Parameters Fault Code 56 will be generated.

**Gear Down Protection** – High gear limits are used to promote driving in higher gears for increased fuel economy. The limits are programmable to different settings.

**Harness** – The harness is the bundle of wiring that connects all the components of the electrical engine system.

**Hertz (Hz)** – Hz is the measure of frequency in cycles per second.

**Histogram** – A histogram is a bar graph that indicates the relative frequency of the engine operation in specific operating ranges.

**Integrated Electronic Controls** – The engine is designed with the electronic controls as a necessary part of the system. The engine will not operate without the electronic controls.

**Inlet Manifold Temperature Sensor** – The inlet air temperature sensor is a sensor that measures the inlet air temperature. The sensor also sends a signal to the ECM.

**Oil Pressure Sensor** – The oil pressure sensor measures engine oil pressure and the oil pressure sensor sends a signal to the ECM as part of Caterpillar Engine Monitoring.

**Open Circuit** – An open circuit is a broken electrical wire connection. The signal or the supply voltage cannot reach the intended destination.

**Original Equipment Manufacturer (OEM)** – An OEM is the manufacturer of a vehicle that utilizes a Caterpillar engine.

**Parameter** – A parameter is a programmable value which affects the characteristics or the behavior of the engine and/or vehicle.

**Parameter Identifier (PID)** – A PID is a numerical code that contains two digits or three digits. A numerical code is assigned to each component. The numerical code identifies data via the data link to the ECM.

**Passive Magnetic Speed Sensor** – This speed sensor does not require a power connection and a ground connection. This speed sensor produces a signal that is based on the change in the magnetic flux of a ferrous metal gear that is near the sensing tip.

**Password** – A password is a group of numeric characters or alphanumeric characters. A password is designed to restrict the changing of information in the ECM. The electrical engine systems require correct customer passwords in order to change customer specified parameters. The electrical engine systems require correct factory passwords in order to clear certain logged events. Factory passwords are also required in order to change certain engine specifications.

**Pedal Mounted Throttle Position Sensor** – The pedal mounted throttle position sensor measures the pedal position and the pedal mounted throttle position sensor sends a signal to the ECM. The sensor is mounted on a throttle pedal assembly.

**Personality Module** – A personality module refers to the software that is downloaded into the ECM. The module contains all the instructions (software) for the ECM and performance maps for a specific horsepower family.

**Power Take-Off (PTO)** – The power take-off is operated with the cruise control switches or with the dedicated PTO switches. This mode of operation permits setting constant engine rpm when the vehicle is not moving or when the vehicle is moving at slow speeds.

**Pro-Link** – The Pro-Link is a hand-held service tool that is electronic. The Pro-Link is manufactured by Micro Processor Systems, Inc. (MPSI). This is available with a Caterpillar cartridge in order to service Caterpillar electronic engines.

**Pulse Width Modulation (PWM)** – A PWM is a digital type of electronic signal that corresponds to a measured variable. The length of the pulse (signal) is controlled by the measured variable. The variable is quantified by a certain ratio. This ratio is the percent of “on-time” that is divided by the percent of “off-time”. A PWM signal is generated by the Throttle Position Sensor.

**Rated Fuel Position (“Rated Fuel Pos” on ECAP)** – The rated fuel position indicates the maximum allowable fuel position (longest injection pulse). The rated fuel position will produce rated power for this engine configuration.

**Reference Voltage** – The reference voltage is a regulated voltage that is used by the sensor in order to generate a signal voltage.

**Remote Mounted Throttle Position Sensor** – The remote mounted throttle position sensor measures the pedal position and the remote mounted throttle position sensor sends a signal to the ECM. The sensor is mounted to the throttle pedal assembly and near the fire wall in the engine compartment. The throttle pedal is connected through an adjustable linkage to the pedal.

**Remote Station Operation** – Remote station operation is the ability to control the engine speed at a location that is outside of the truck cab.

**Sensor** – A sensor is used to detect a change in the pressure, in the temperature, or in mechanical movement. When any of these changes are detected, a sensor converts the change into an electrical signal.

**Service Brake Switch** – The service brake switch is installed by the OEM factory. The service brake switch is normally a pressure switch. The switch is normally closed when the pedal is released. Depressing the brake will open the circuit.

**Service Program Module (SPM)** – The SPM is a software program on a computer chip that was programmed at the factory. The computer chip is designed to adapt an ECAP for a particular application.

**Short Circuit** – A short circuit is an electrical circuit that is mistakenly connected to an undesirable point. For example, an electrical contact is made with the frame whenever an exposed wire rubs against a vehicle’s frame.

**Signal** – A signal is a voltage or a wave that is used to transmit information that is typically from a sensor to the ECM.

**Speed Burp** – A speed burp is a sudden brief change in engine rpm.

**Speed Circuit** – The speed circuit includes the speed sensor, the harness and the ECM.

**Speed Sensor** – The speed sensor is an electromagnetic pickup that measures speed from the rotation of gear teeth in the drive train.

**Standard SAE Diagnostic Communications Data Link** – For more information, refer to the ATA Data Link.

**Subsystem** – A subsystem is a part of the engine system that relates to a particular function.

**Supply Voltage** – Supply voltage is a constant voltage that is supplied to a component in order to provide electrical power for operation. Supply voltage may be generated by the ECM. Supply voltage may also be the battery voltage of the vehicle that is supplied by the vehicle wiring.

**“T” Harness** – This harness is a test harness that is designed to permit normal circuit operation and the measurement of the voltage simultaneously. Typically, the harness is inserted between the two ends of a connector.

**Throttle Position** – The throttle position is interpreted by the ECM. The accelerator pedal position sensor sends the ECM signals. The throttle position may be used as part of a power take-off control.

**Throttle Position Sensor** – The throttle position sensor is an electronic sensor that is connected to the throttle pedal. The throttle position sensor sends a pulse width modulated signal to the ECM.

**Total Tattletale** – The total tattletale is the total number of changes to all system parameters.

**Transducer** – A transducer is a device that converts a mechanical signal to an electrical signal.

**Trip Recorder** – The trip recorder records the operating parameters of the truck and of the engine. The trip recorder is used to analyze the operating habits of the operator. The trip recorder is used to produce the operator's logs.

**Vehicle Speed Circuit** – The vehicle speed circuit includes the vehicle speed sensor, the harness, and the ECM.

**Vehicle Speed Sensor** – A vehicle speed sensor is an electromagnetic pickup that measures vehicle speed from the rotation of gear teeth in the drive train of the vehicle.

**Warning Lamp** – The warning lamp is used to warn the operator of the presence of a problem that was detected with the Caterpillar engine monitoring.

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## Electronic Control System Components

**SMCS Code:** 1900

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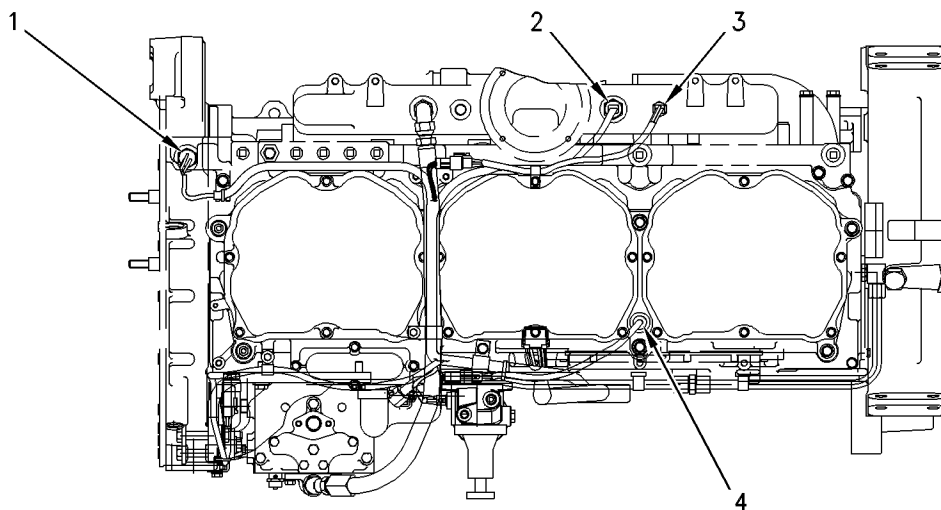


Illustration 3  
Top view

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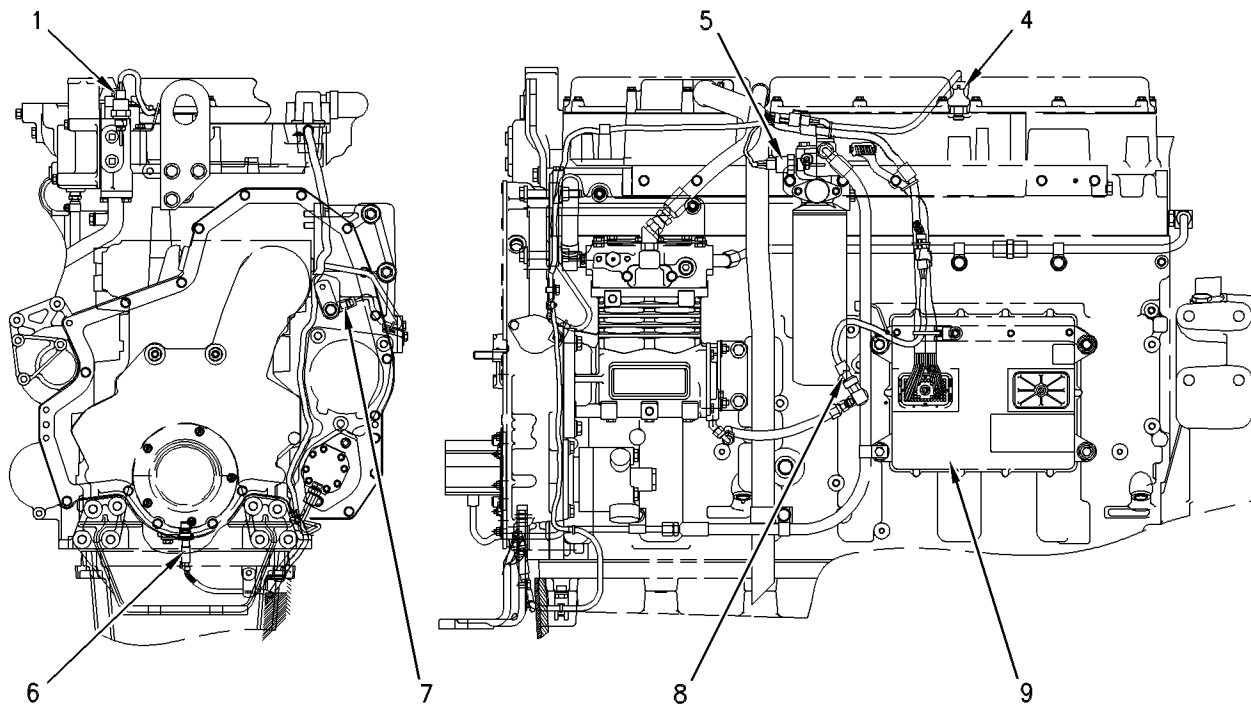


Illustration 4

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Front view and left side view

- (1) Coolant temperature sensor
- (2) Inlet air temperature sensor
- (3) Boost pressure sensor

- (4) Atmospheric pressure sensor
- (5) Fuel temperature sensor
- (6) Crankshaft position sensor

- (7) Camshaft position sensor
- (8) Oil pressure sensor
- (9) Engine control module (ECM)

The electronic control system is integrally designed into the engine's fuel system and the engine's air inlet and exhaust system in order to electronically control the fuel delivery and the injection timing. The electronic control system provides increased timing control and fuel air ratio control in comparison to conventional mechanical engines. Injection timing is achieved by precise control of injector firing time, and engine rpm is controlled by adjusting the firing duration. The ECM energizes the solenoid in the unit injector in order to start the injection of fuel. Also, the ECM de-energizes the unit injector solenoids in order to stop injection of fuel. Refer to the Systems Operation, "Fuel System" topic for a complete explanation of the fuel injection process.

The engine uses the following types of electronic components:

- Inputs
- Controls
- Outputs

An input component is one that sends an electrical signal to the ECM. The signal that is sent varies in one of the following ways:

- Voltage
- Frequency
- Pulse width

The variation of the signal is in response to a change in some specific system of the vehicle. The ECM sees the input sensor signal as information about the condition, environment, or operation of the vehicle.

A control component (ECM) receives the input signals. Electronic circuits inside the control component evaluate the signals from the input components. These electronic circuits also supply electrical energy to the output components of the system. The electrical energy that is supplied to the output components is based on predetermined combinations of input signal values.

An output component is one that is operated by a control module. The output component receives electrical energy from the control component. The output component uses that electrical energy in one of two ways. The output component can use that electrical energy in order to perform work. The output component can use that electrical energy in order to provide information.

1. As an example, a moving solenoid plunger will perform work. By performing work, the component has functioned in order to regulate the vehicle.
2. As an example, a dash panel light or an alarm will provide information to the operator of the vehicle.

These electronic components provide the ability to electronically control the engine operation. Engines with electronic controls offer the following advantages:

- Improved performance
- Improved fuel consumption
- Reduced emissions levels

Table 1

| Electrical Connectors and Functions |                                                     |
|-------------------------------------|-----------------------------------------------------|
| Connector                           | Function                                            |
| J1/P1                               | ECM connector (70 pin OEM harness)                  |
| J2/P2                               | ECM connector (70 pin engine harness)               |
| J100/P100                           | Coolant temperature sensor connector (2 pin)        |
| J103/P103                           | Temperature sensor for the inlet air (2 pin)        |
| J105/P105                           | Fuel temperature sensor connector (2 pin)           |
| J200/P200                           | Boost pressure sensor connector (3 pin)             |
| J201/P201                           | Oil pressure sensor connector (3 pin)               |
| J203/P203                           | Atmospheric pressure sensor connector (3 pin)       |
| J300/P300                           | Fuel injector connector (12 pin)                    |
| J400/P400                           | Engine timing calibration probe connector (2 pin)   |
| J401/P401                           | Crankshaft position sensor connector (2 pin)        |
| J402/P402                           | Camshaft position sensor connector (2 pin)          |
| J403/P403                           | Accelerator pedal position sensor connector (3 pin) |

## Fuel System

SMCS Code: 1250

i01777379

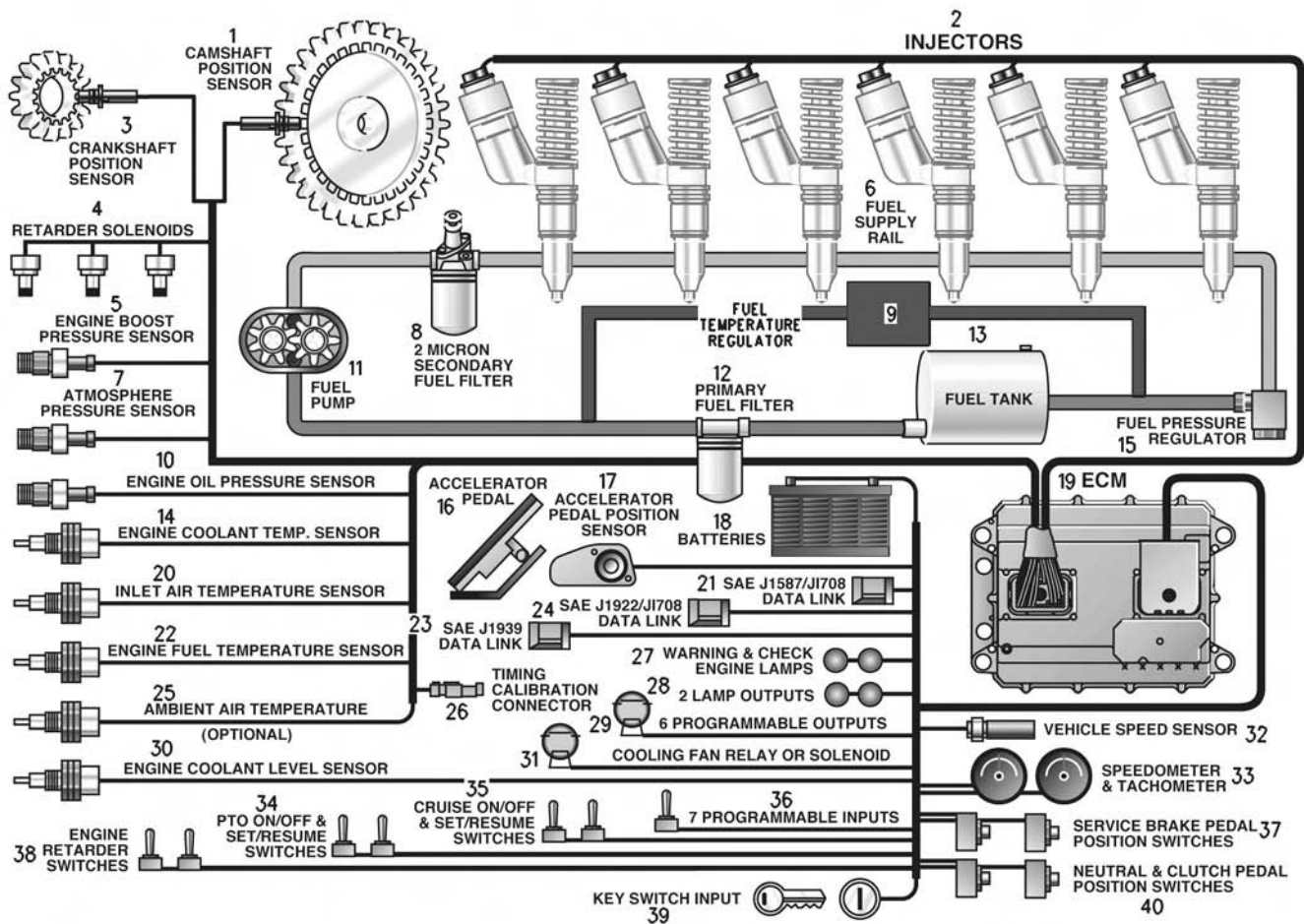


Illustration 5

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- |                                        |                                        |                                                        |
|----------------------------------------|----------------------------------------|--------------------------------------------------------|
| (1) Camshaft position sensor           | (15) Fuel pressure regulator           | (29) 6 Programmable outputs                            |
| (2) Injectors                          | (16) Accelerator pedal                 | (30) Engine coolant level sensor                       |
| (3) Crankshaft position sensor         | (17) Accelerator pedal position sensor | (31) Cooling fan relay or solenoid                     |
| (4) Retarder solenoid                  | (18) Batteries                         | (32) Vehicle speed sensor                              |
| (5) Boost pressure sensor              | (19) ECM                               | (33) Speedometer & tachometer                          |
| (6) Fuel supply rail                   | (20) Inlet air temperature sensor      | (34) PTO on/off switch and PTO set/resume switch       |
| (7) Atmospheric pressure sensor        | (21) SAE J1587/J1708 Data link         | (35) Cruise on/off switch and cruise set/resume switch |
| (8) 2 Micron secondary fuel filter     | (22) Fuel temperature sensor           | (36) 7 Programmable inputs                             |
| (9) Fuel temperature regulator         | (23) SAE J1939 Data link               | (37) Service brake pedal position switches.            |
| (10) Engine oil pressure sensor        | (24) SAE J1922/J1708 Data link         | (38) Engine retarder switch                            |
| (11) Fuel pump                         | (25) Ambient air temperature           | (39) Keyswitch                                         |
| (12) Primary fuel filter               | (26) Timing calibration connector      | (40) Neutral & clutch pedal position switches          |
| (13) Fuel tank                         | (27) Warning and check engine lamps    |                                                        |
| (14) Engine coolant temperature sensor | (28) 2 Lamp outputs                    |                                                        |

The Electronic Unit Injector system consists of the following systems: the mechanical system and the electronic system. The mechanical system is made up of the low pressure fuel supply system and the electronic unit injectors. The electronic system provides complete electronic control of all engine functions. The electronic control system consists of the following three types of components: input, control, and output.

Most changes to engine horsepower or to the performance are accomplished by installing new electronic software or upgrading mechanical components.

There are five major components of the Electronic Unit Injector fuel system:

- Electronic Unit Injectors
- Fuel transfer pump
- Electronic Control Module
- Sensors
- Actuators

The Electronic Unit Injectors produce fuel injection pressures up to 207000 kPa (30000 psi). The Electronic Unit Injectors also fire up to 19 times per second at rated speed. The fuel transfer pump supplies the injectors by drawing fuel from the tank and by pressurizing the tank between 60 and 125 PSI. The ECM is a powerful computer which controls all major engine functions. Sensors are electronic devices which monitor engine performance parameters. Engine performance parameters measure pressure, temperature and speed. This information is sent to the ECM via a signal voltage. Actuators are electronic devices which use electronic currents from the ECM to change engine performance. An example of an actuator is the Injector solenoid.

## Low Pressure Fuel System

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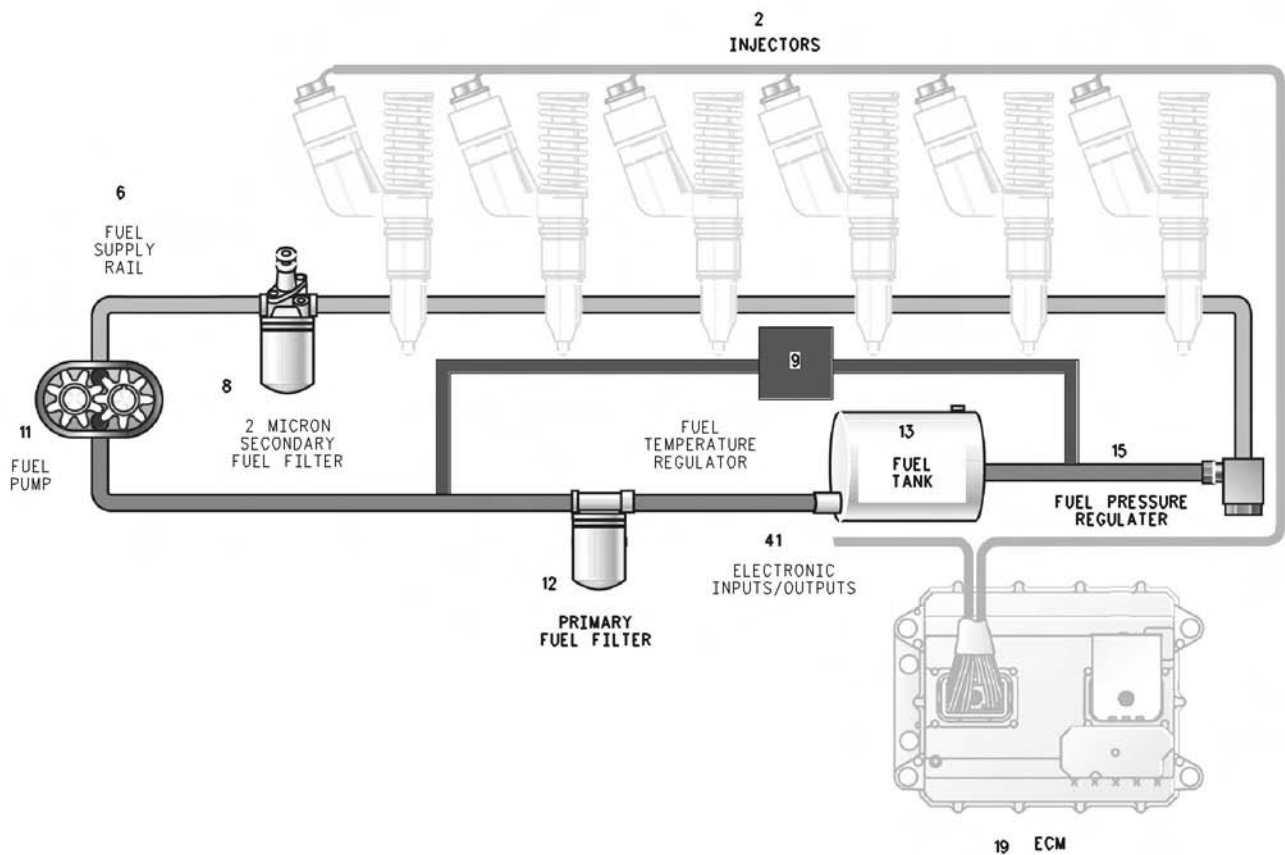


Illustration 6

g00915523

(2) Injectors  
(6) Fuel supply rail  
(8) 2 Micron secondary fuel filter

(9) Fuel temperature regulator  
(11) Fuel pump  
(12) Primary fuel filter

(13) Fuel tank  
(15) Fuel pressure regulator  
(41) Electronic sensors and switches

The low pressure fuel system supplies fuel from the fuel tank to the injectors. The low pressure fuel system has four basic functions:

- The low pressure fuel system supplies fuel to the injectors for combustion.

- The low pressure fuel system supplies fuel to the injectors for cooling.
- The low pressure fuel system supplies fuel to the fuel system in order to remove air.
- The low pressure fuel system combines cold fuel from the fuel tank and warm fuel from the fuel return line during cold starts.

The major parts in a low pressure fuel system consist of the following components:

- Fuel tank
- Fuel transfer lines
- Primary fuel filter or water separator
- Fuel transfer pump
- Secondary fuel filter
- Fuel priming pump
- Fuel pressure regulator valve
- Fuel temperature regulator valve

The electronic unit injectors, the fuel transfer pump, the ECM, sensors, and actuators are part of the low pressure fuel system.

In the low pressure fuel system, the fuel is pulled from the fuel tank to the primary fuel filter or to the water separator. The primary fuel filter removes large debris from the fuel before the fuel flows into the transfer pump. The fuel transfer pump is a gear pump that contains a pressure relief valve. Fuel flows from the outlet port of the transfer pump to the secondary fuel filter. All 1999 and newer engines use a 2 micron fuel filter. The 2 micron filter removes small abrasive contaminants from the fuel system, which can cause damage to the unit injectors.

The fuel filter base contains a hand operated fuel priming pump. The fuel priming pump removes air from the system when a fuel filter has been changed or a unit injector has been changed. The priming pump pulls fuel from the tank, around the transfer pump and into the filter. The transfer pump pushes fuel through the supply passage in the cylinder head and back to the tank.

The fuel pressure regulator consists of a spring loaded check valve. The pressure relief valve opens at approximately 60 to 125 PSI. When the engine is in the off position and the fuel pressure drops below 60 PSI, the check valve closes. The check valve closes in order to prevent the fuel in the cylinder head from draining back into the fuel tank. Retaining the fuel in the head maintains a supply of fuel for the injectors during start-up requirement.

The fuel temperature regulator valve is used to supply the injectors with warm fuel during cold operation. The injectors pick up the fuel from a fuel passage in the cylinder head. The injectors are supplied with an excess of fuel that is returned to the fuel tank. This excess fuel that is returned to the fuel tank is warm. There is a temperature difference between the return fuel and the fuel in the tank. The temperature difference decreases as the two begin to mix. The MBL1-Up and MBL1-Up engines are equipped with an extra fuel line that provides a connection for the fuel return line and the fuel transfer pump. The valve in the extra fuel line is controlled by temperature. The temperature control valve opens when the fuel is cold. This allows the excess warm fuel from the cylinder head to mix with the cold fuel from the tank. The warm fuel increases injector life.

The ECM controls major engine functions. Sensors are electronic devices that monitor engine performance parameters. The pressure sensor, the temperature sensor and the speed sensor provide information to the ECM by a signal voltage. Actuators are electronic devices which use electrical currents from the ECM to change engine performance. An example of an actuator is an injector solenoid.

## Electronic Controls

The electronic control system provides complete electronic control of all engine functions. The electronic control system consists of the following three types of components: input, control, and output. Sensors monitor engine operating conditions. This information is sent to the ECM. The ECM has three main functions. The ECM provides power for the engine electronics and monitors input signals from the engine sensors. The ECM also acts as a governor to control engine rpm. The ECM stores active faults, logged faults, and logged events. The Personality Module is the software in the ECM which contains the specific maps that define power, torque, and RPM of the engine. The ECM sends electrical current to the output components in order to control engine operation. The ECM has the following connectors: two 70 pin harness connectors, one engine harness connector, and one vehicle harness connector. The vehicle harness connects the ECM to the engine control portion of the vehicle harness. The engine control portion includes the following components.

- Accelerator pedal position sensor
- Vehicle speed sensor
- Transmission
- Brake

- Clutch switches
- Cruise control
- PTO switch
- Data links
- Check engine light
- Warning light
- Engine retarder switch
- Speedometer
- Tachometer
- Cooling fan solenoid

The following list of features are part of the electronic control system:

- Cold start strategy
- Oil pressure
- Coolant temperature warning indicator
- Automatic altitude compensation
- Variable injection timing
- Electronic engine speed governing

These features result in the following items: precise engine speed control, very little smoke, faster cold starting, and built-in engine protection.

The ECM consists of the following two main components: the ECM and the personality module.

The ECM is a computer and the personality module is the software for the computer. The personality module contains the operating maps. The operating maps define the following characteristics of the engine:

- Horsepower
- Torque curves
- Rpm
- Other characteristics

The ECM, the personality module, the sensors, and the unit injectors work together in order to control the engine. The ECM, the personality module, the sensors, and the unit injectors can not control the engine alone.

The ECM determines a desired rpm that is based on the following criteria:

- Throttle signal
- Certain diagnostic codes
- Vehicle speed signal

The ECM maintains the desired engine rpm by sensing the actual engine rpm. The ECM calculates the fuel amount that needs to be injected in order to achieve the desired rpm.

## **Fuel Injection Timing and Delivery**

The ECM controls the injected fuel amount by varying the signals to the unit injectors. The unit injectors will inject fuel ONLY if the unit injector solenoid is energized. The ECM sends a 90 volt signal to the solenoid for energizing the solenoid. By controlling the timing of the 90 volt signal, the ECM controls injection timing. By controlling the duration of the 90 volt signal, the ECM controls the injected fuel amount.

Injection timing is determined by engine rpm, and other engine data. The ECM senses the top center position of cylinder number 1 from the signal that is provided by the engine speed sensor. The ECM decides when the injection should occur relative to the top center position. The ECM provides the signal to the unit injector at the desired time.

## Unit Injector Mechanism

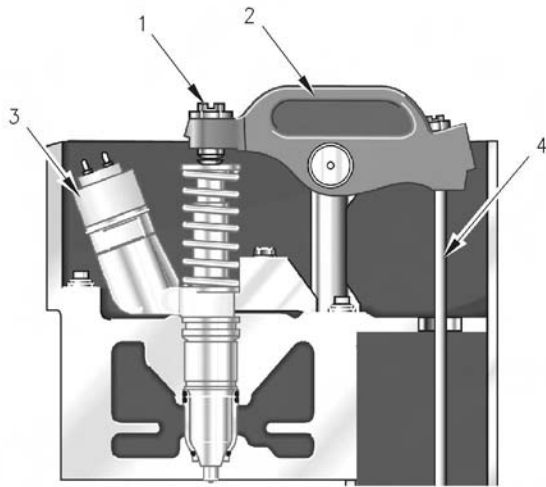


Illustration 7

g00908926

Typical examples of Electronic Unit Injector fuel systems.

- (1) Adjusting nut
- (2) Rocker arm assembly
- (3) Unit injector
- (4) Pushrod

## Injection Actuation System

The unit injector pressurizes the fuel. The correct amount of fuel is then injected into the cylinder block at precise times. The ECM determines the injection timing and the amount of fuel that is delivered. The unit injector is operated by a camshaft lobe and a rocker arm. The camshaft has three camshaft lobes for each cylinder. Two lobes operate the inlet and exhaust valves, and the other lobe operates the unit injector mechanism. Force is transferred from the unit injector lobe on the camshaft through the lifter to the pushrod (4). The force of the pushrod is transferred through rocker arm assembly (2) and to the top of the unit injector. The adjusting nut (1) allows setting of the unit injector adjustment. Refer to Systems Operation/Testing and Adjusting, "Electronic Unit Injector - Adjust" for the proper setting of the unit injector adjustment.

## Unit Injector

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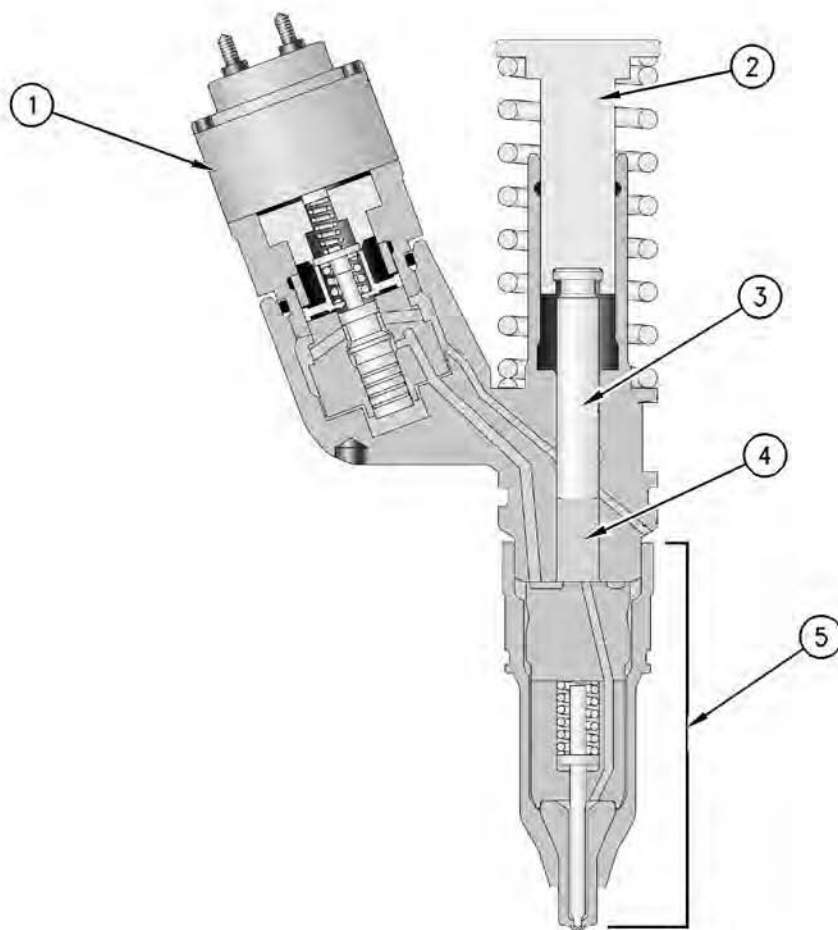


Illustration 8

g00626752

(1) Solenoid  
(2) Tappet

(3) Plunger  
(4) Barrel

(5) Nozzle Assembly

### Operation of the Fuel Injector

The operation of the Electronic Control Unit (EUI) consists of the following four stages: Pre-injection, Injection, End of injection, and Fill. Unit injectors use a plunger and barrel to pump high pressure fuel into the combustion chamber. Components of the injector include the tappet, the plunger, the barrel and nozzle assembly. Components of the nozzle assembly include the spring, the nozzle check, and a nozzle tip. The cartridge valve is made up of the following components: solenoid, armature, poppet valve, and poppet spring.

The injector is mounted in an injector bore in the cylinder head which has an integral fuel supply passage. The injector sleeve separates the injector from the engine coolant in the water jacket. Some engines use a stainless steel sleeve. The stainless steel sleeve fits into the cylinder head with a light press fit.

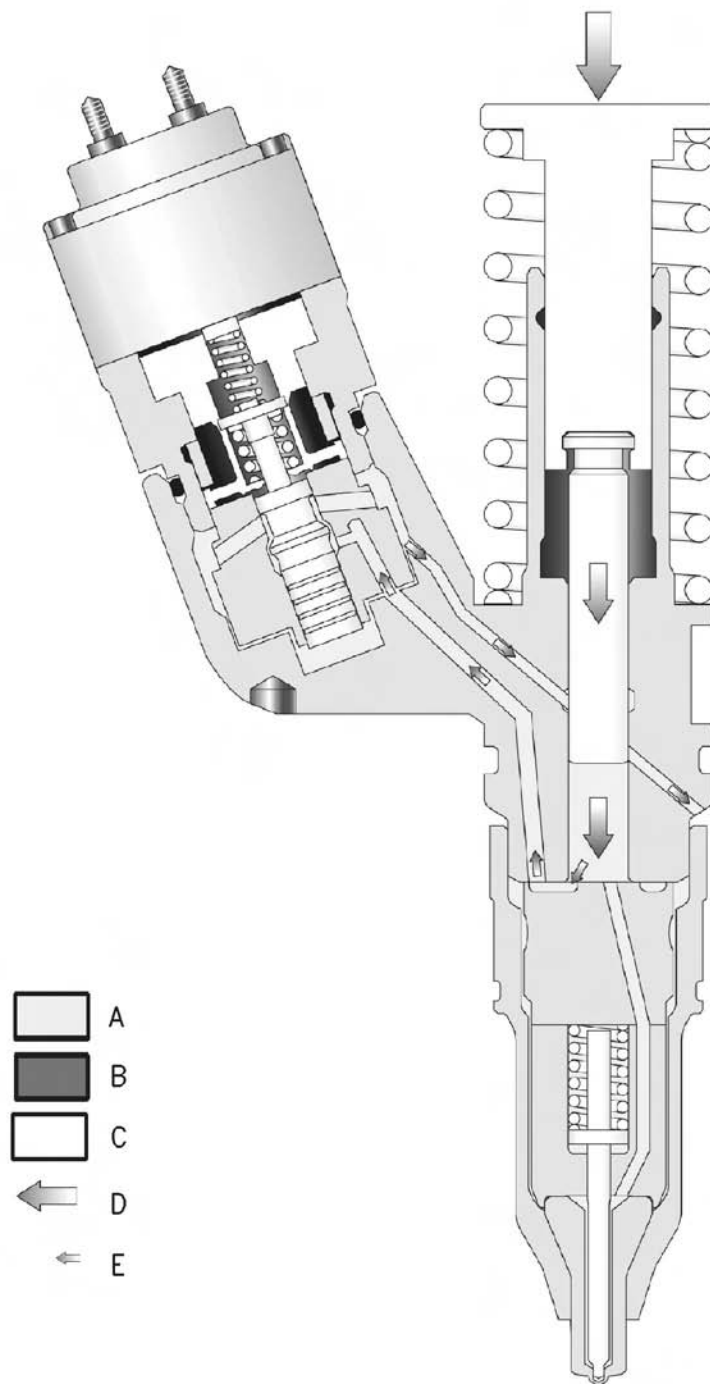


Illustration 9

g00660656

**Pre-injection**

- (A) Fuel supply pressure
- (B) Injection pressure
- (C) Moving parts
- (D) Mechanical movement
- (E) Fuel movement.

Pre-injection metering starts with the injector plunger and the injector tappet at the top of the fuel injection stroke. When the plunger cavity is full of fuel, the poppet valve is in the open position and the nozzle check is in the open position. Fuel leaves the plunger cavity when the rocker arm pushes down on the tappet and the plunger. Fuel flow that is blocked by the closed nozzle check valve flows past the open poppet valve to the fuel supply passage in the cylinder head. If the solenoid is energized, the poppet valve remains open and the fuel from the plunger cavity continues flowing into the fuel supply passage.

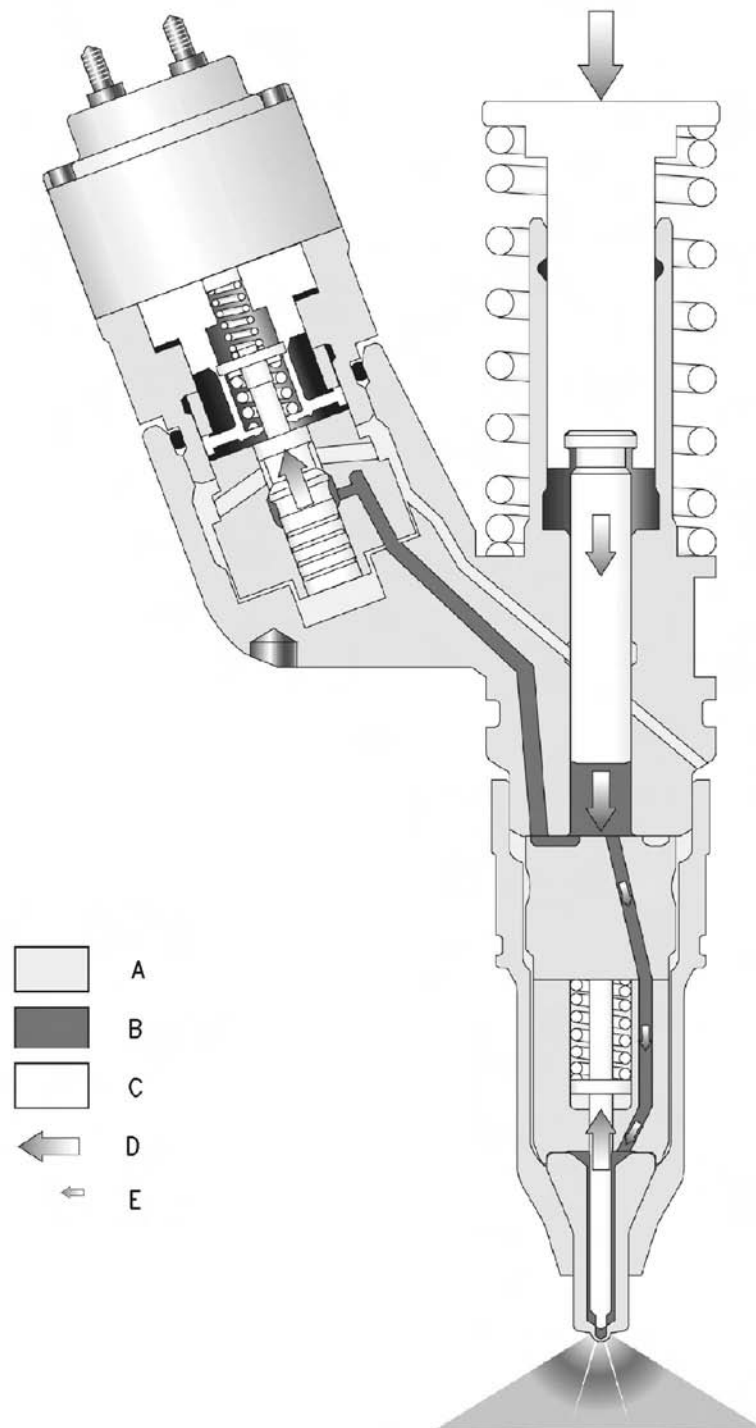


Illustration 10

g00660658

### Injection

- (A) Fuel supply pressure.
- (B) Injection pressure
- (C) Moving parts
- (D) Mechanical movement
- (E) Fuel movement.

To start injection, the ECM sends a current to the solenoid on the cartridge valve. The solenoid creates a magnetic field which attracts the armature. When the solenoid is energized, the armature assembly will lift the poppet valve so the poppet valve contacts the poppet seat. This is the closed position. Once the poppet valve closes, the flow path for the fuel that is leaving the plunger cavity is blocked. The plunger continues to push fuel from the plunger cavity and the fuel pressure builds up. When the fuel pressure reaches approximately 34500 kPa (5000 psi), the force of the high pressure fuel overcomes the spring force. This holds the nozzle check in the closed position. The nozzle check moves off the nozzle seat and the fuel flows out of the injector tip. This is the start of injection.

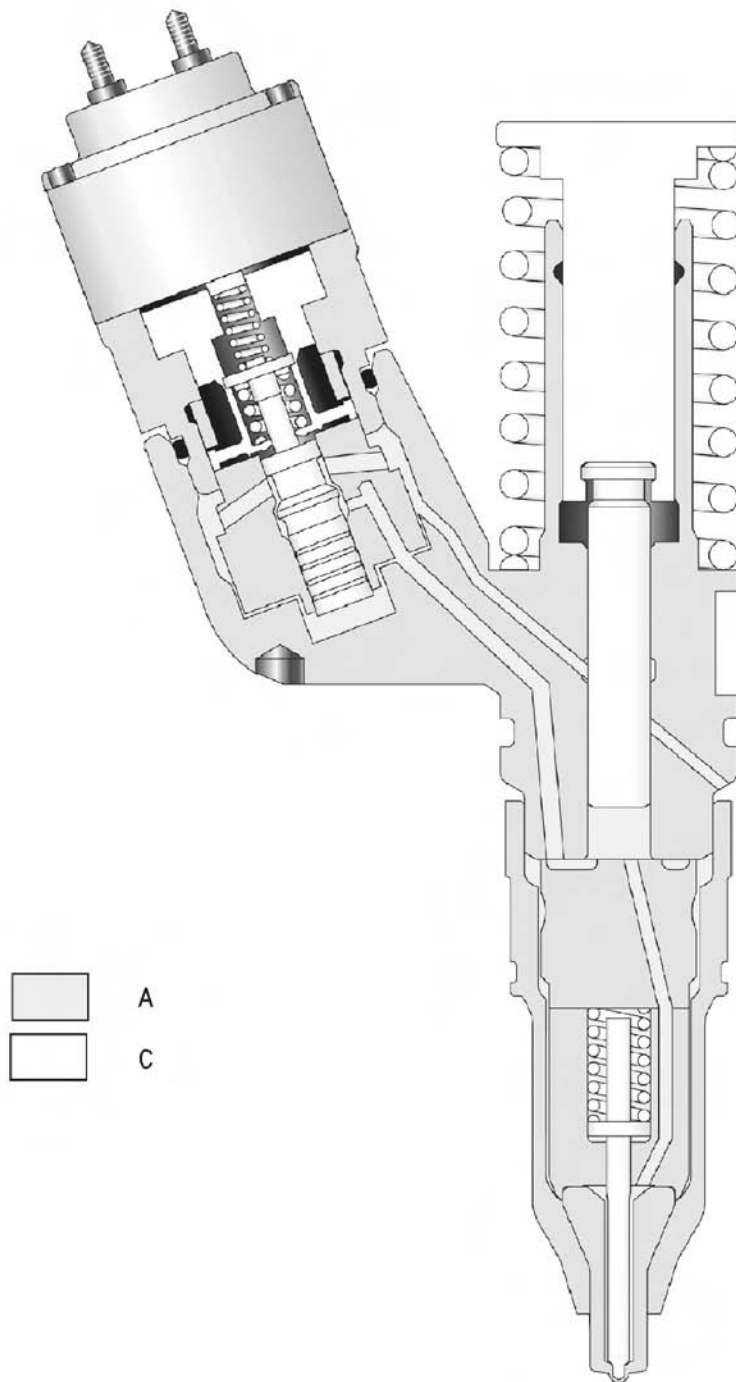


Illustration 11

g00660660

**End of injection**

(A) Fuel supply pressure

(C) Moving parts

Injection is continuous while the injector plunger moves in a downward motion and the energized solenoid holds the poppet valve closed. When injection pressure is no longer required, the ECM stops current flow to the solenoid. When the current flow to the solenoid stops, the poppet valve opens. The poppet valve is opened by the fuel injector spring and the fuel pressure. High pressure fuel can now flow around the open poppet valve and into the fuel supply passage. This results in a rapid drop in injection pressure. When the injection pressure drops to approximately 24000 kPa (3500 psi), the nozzle check closes and injection stops. This is the end of injection.

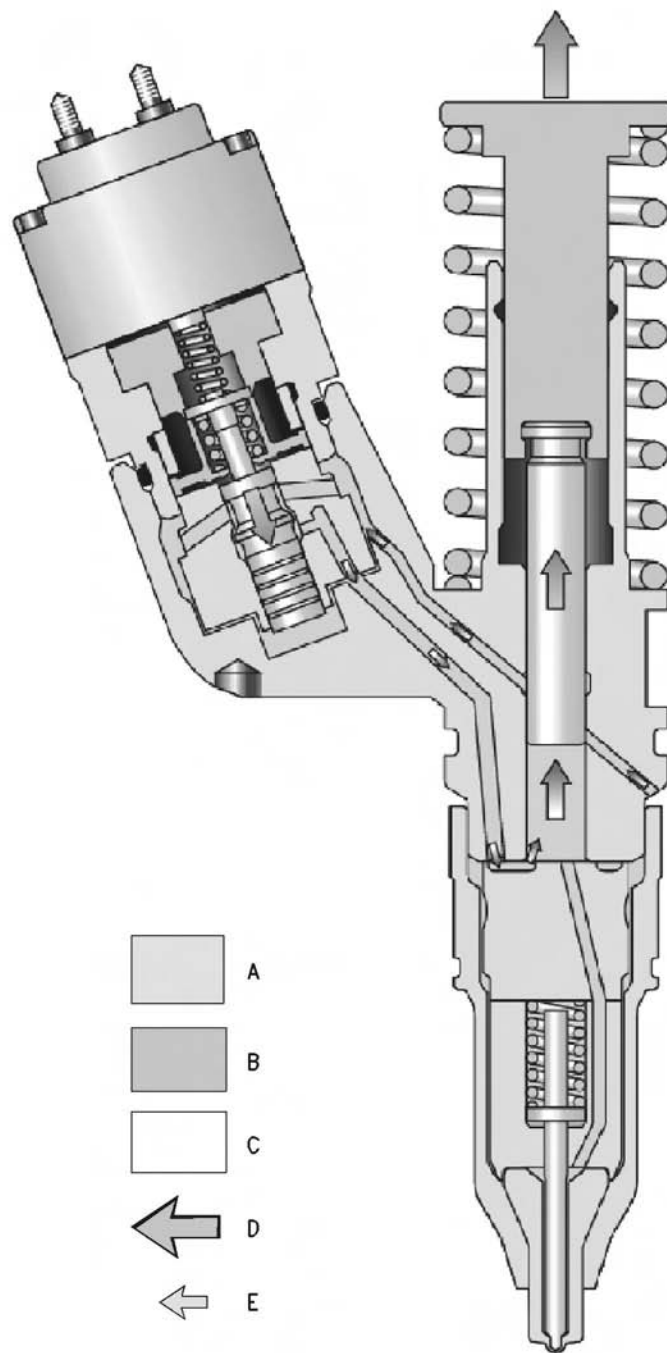


Illustration 12

g00795649

**Fill**

- (A) Fuel supply pressure
- (B) Injection pressure
- (C) Moving parts
- (D) Mechanical movement
- (E) Fuel movement.

When the plunger reaches the bottom of the barrel, fuel is no longer forced from the plunger cavity. The plunger is pulled up by the tappet and the tappet spring. The upward movement of the plunger causes the pressure in the plunger cavity to drop below fuel supply pressure. Fuel flows from the fuel supply passage around the open poppet and into the plunger cavity as the plunger travels upward. When the plunger reaches the top of the stroke, the plunger cavity is full of fuel and fuel flow into the plunger cavity stops. This is the beginning of pre-injection.

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## Air Inlet and Exhaust System

SMCS Code: 1050

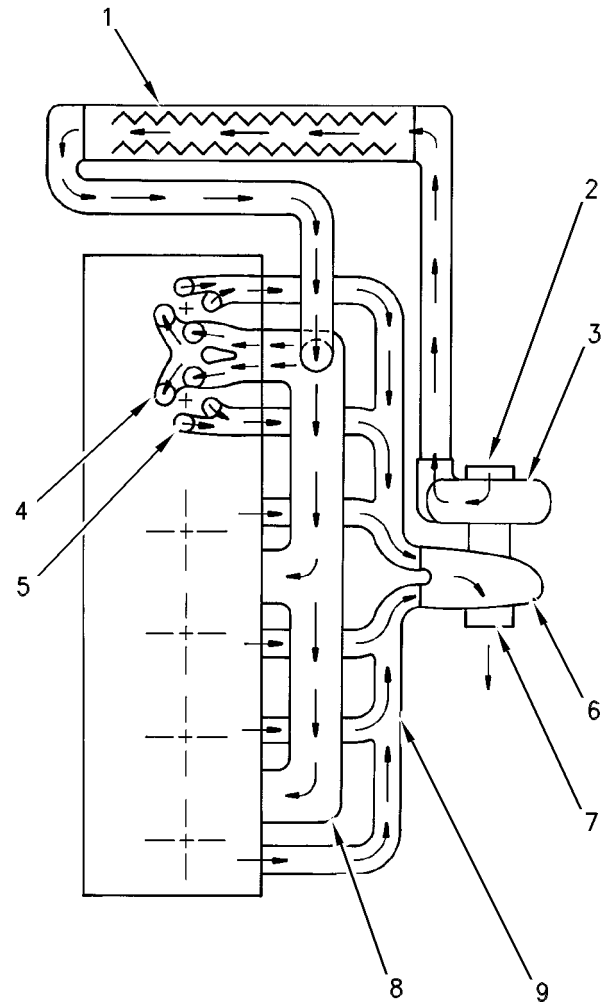


Illustration 13

g00411724

Air inlet and exhaust system components

- (1) Aftercooler
- (2) Air inlet
- (3) Turbocharger compressor wheel
- (4) Inlet valves
- (5) Exhaust valves
- (6) Turbocharger turbine wheel
- (7) Exhaust outlet
- (8) Inlet manifold
- (9) Exhaust manifold

The components of the air inlet and exhaust system control the quality of air and the amount of air that is available for combustion. The components of the air inlet and exhaust system are the following components:

- Air cleaner
- Turbocharger
- Aftercooler
- Cylinder head
- Valves and valve system components
- Piston and cylinder
- Exhaust manifold

Inlet air is pulled through the air cleaner into air inlet (2) by turbocharger compressor wheel (3). The air is compressed and heated to about 150 °C (300 °F) before the air is forced to aftercooler (1). As the air flows through the aftercooler the temperature of the compressed air lowers to about 43 °C (110 °F). Cooling of the inlet air increases combustion efficiency. Increased combustion efficiency helps achieve the following benefits:

- Lower fuel consumption
- Increased horsepower output

Aftercooler (1) is a separate cooler core that is mounted in front of the engine radiator. The engine fan moves ambient air across both cores. This cools the turbocharged inlet air and the engine coolant.

From the aftercooler, air is forced into inlet manifold (8). Air flow from the inlet chambers into the cylinders is controlled by inlet valves (4). There are two inlet valves and two exhaust valves (5) for each cylinder. The inlet valves open when the piston moves down on the intake stroke. When the inlet valves open, cooled compressed air from the inlet port is pulled into the cylinder. The inlet valves close and the piston begins to move up on the compression stroke. The air in the cylinder is compressed. When the piston is near the top of the compression stroke, fuel is injected into the cylinder. The fuel mixes with the air and combustion starts. During the power stroke, the combustion force pushes the piston downward. The exhaust valves open and the exhaust gases are pushed through the exhaust port into exhaust manifold (9) as the piston rises on the exhaust stroke. After the exhaust stroke, the exhaust valves close and the cycle starts again. The complete cycle consists of four strokes:

- Inlet
- Compression

- Power
- Exhaust

Exhaust gases from exhaust manifold (9) enter the turbine side of the turbocharger in order to turn turbocharger turbine wheel (6). The turbine wheel is connected to the shaft that drives the compressor wheel. Exhaust gases from the turbocharger pass through exhaust outlet (7), a muffler and an exhaust stack.

## Turbocharger

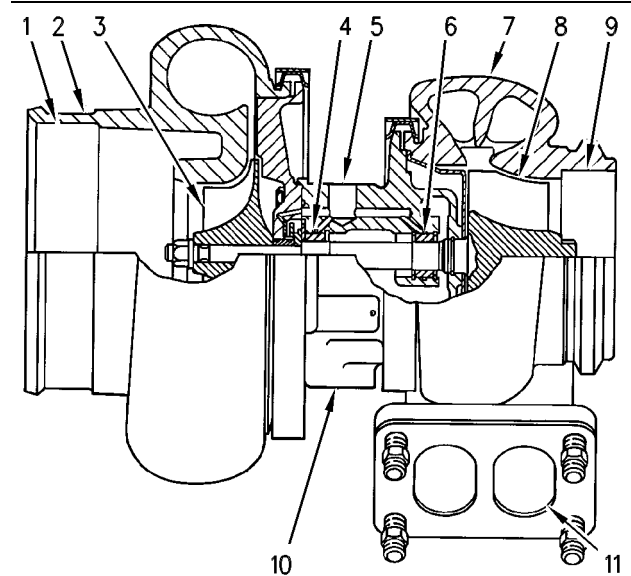


Illustration 14

g00294193

### Turbocharger

- (1) Air inlet
- (2) Compressor housing
- (3) Compressor wheel
- (4) Bearing
- (5) Oil Inlet port
- (6) Bearing
- (7) Turbine housing
- (8) Turbine wheel
- (9) Exhaust outlet
- (10) Oil outlet port
- (11) Exhaust inlet

The turbocharger is installed on the center section of the exhaust manifold. All the exhaust gases from the engine go through the turbocharger. The compressor side of the turbocharger is connected to the aftercooler by pipe.

The exhaust gases enter turbine housing (7) through exhaust inlet (11). The exhaust gases then rotate the blades of turbine wheel (8). The turbine wheel is connected by a shaft to compressor wheel (3).

Clean air from the air cleaners is pulled through compressor housing air inlet (1) by the rotation of compressor wheel (3). The action of the compressor wheel blades causes a compression of the inlet air. This compression gives the engine more power by allowing the engine to burn more air and more fuel during combustion.

When the load on the engine increases, more fuel is injected into the cylinders. The combustion of this additional fuel produces more exhaust gases. The additional exhaust gases cause the turbine and the compressor wheels of the turbocharger to rotate faster. As the compressor wheel rotates faster, more air is forced into the cylinders. The increased flow of air gives the engine more power by allowing the engine to burn the additional fuel with greater efficiency.

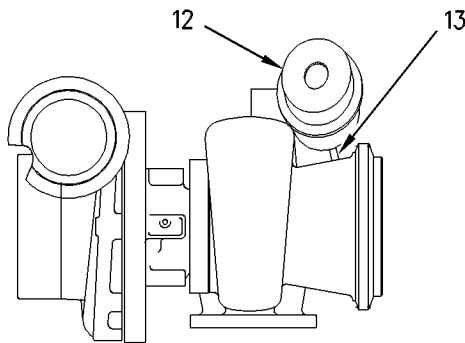


Illustration 15  
Typical example of a turbocharger with a wastegate  
(12) Canister  
(13) Actuating lever

The wastegate on the turbocharger is controlled through the use of an electric solenoid. The canister that actuates the wastegate receives the pressurized air from the inlet manifold. The air lines from the inlet manifold and the canister are joined in a manifold block. The manifold block has the following components: Inlet air line, Outlet air line, Solenoid valve, and Air bleed passage. The ECM provides a signal to the solenoid valve. The signal determines the pressure that will go to the canister. As the pressure changes the canister moves the position of the wastegate. This directly affects the boost of the engine. With this arrangement, the engine software can be designed to be more effective at lower engine speeds.

As the boost pressure increases against the diaphragm in canister (12), the valve of the wastegate is opened. When the valve of the wastegate is opened, the rpm of the turbocharger is limited by bypassing a portion of the exhaust gases around the turbine wheel of the turbocharger.

The solenoid for the wastegate will be actuated during extended idle conditions. Actuation of the solenoid valve will be every thirty minutes. The solenoid is actuated in order to reduce the chances of a frozen solenoid. Tampering with the system will result in a permanent derate. The Caterpillar dealer will be required to reset the system with the factory passwords.

**Note:** The turbocharger with a wastegate is preset at the factory and no adjustment can be made.

Bearings (4) and (6) for the turbocharger use engine oil under pressure for lubrication. The oil comes in through oil inlet port (5). The oil then goes through passages in the center section in order to lubricate the bearings. Oil from the turbocharger goes out through oil outlet port (10) in the bottom of the center section. The oil then goes back to the engine lubrication system.

## Valve System Components

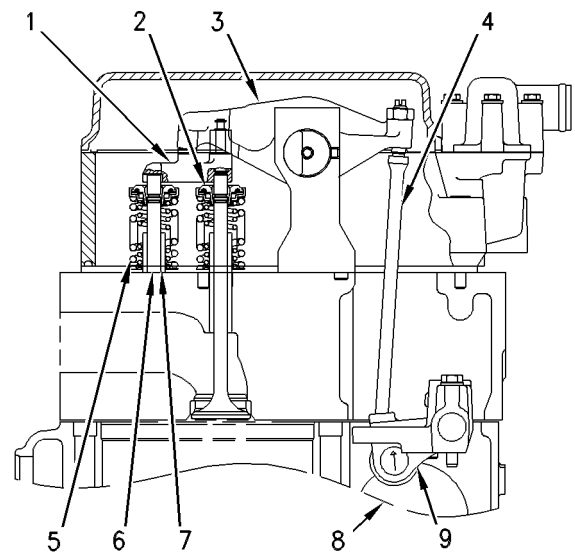


Illustration 16  
Valve system components

- (1) Valve bridge
- (2) Valve rotator
- (3) Rocker arm
- (4) Pushrod
- (5) Valve springs
- (6) Valves
- (7) Valve guide
- (8) Camshaft
- (9) Lifter

The valve system components control the flow of inlet air into the cylinders during engine operation. The valve system components also control the flow of exhaust gases out of the cylinders during engine operation.

The crankshaft gear drives the camshaft gear through an idler gear. Camshaft (8) must be timed to the crankshaft in order to get the correct relation between the piston movement and the valve movement.

The camshaft has three camshaft lobes for each cylinder. Two lobes operate the inlet and exhaust valves, and one operates the unit injector mechanism. As the camshaft turns, the camshaft lobes cause lifter (9) to move pushrod (4) up and down. Upward movement of the pushrod against rocker arm (3) results in downward movement (opening) of valves (6).

Each cylinder has two inlet valves and two exhaust valves. Valve springs (5) close the valves when the lifters move down. Valve rotators (2) cause the valves to rotate while the engine is running. The rotation of the valves keeps the carbon deposits on the valves to a minimum. Also, the rotation gives the valves longer service life.

## Lubrication System

SMCS Code: 1300

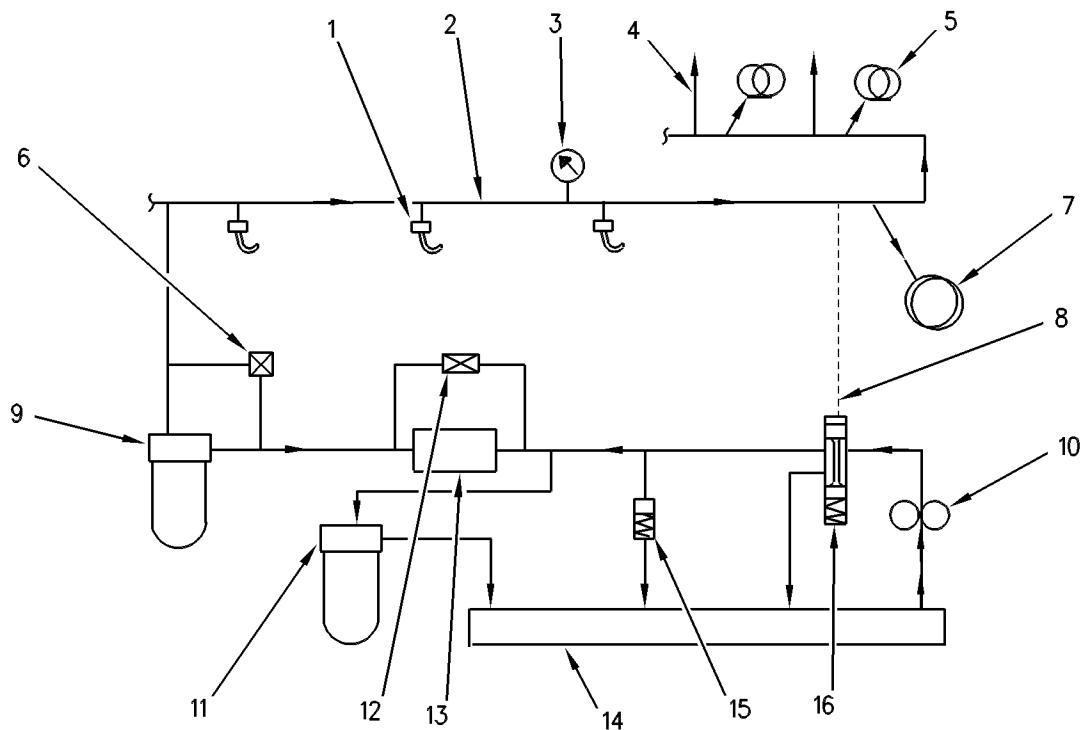


Illustration 17

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Lubrication system schematic

- |                                        |                                 |
|----------------------------------------|---------------------------------|
| (1) Piston cooling jets                | (9) Primary engine oil filter   |
| (2) Main oil gallery in cylinder block | (10) Engine oil pump            |
| (3) Engine oil pressure sensor         | (11) Secondary oil filter       |
| (4) Oil flow to valve mechanism        | (12) Oil cooler bypass valve    |
| (5) Camshaft journals                  | (13) Engine oil cooler          |
| (6) Oil filter bypass valve            | (14) Oil pan sump               |
| (7) Main bearings                      | (15) High pressure relief valve |
| (8) Signal line                        | (16) Oil pump bypass valve      |

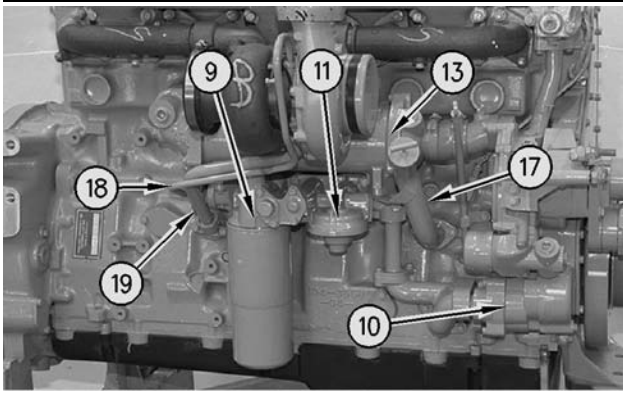


Illustration 18 g00421818

#### Right side of engine

- (9) Primary engine oil filter
- (10) Engine oil pump
- (11) Secondary oil filter (not shown)
- (13) Engine oil cooler
- (17) Engine oil filler
- (18) Oil supply line to turbocharger
- (19) Oil drain line from turbocharger

The lubrication system supplies 110 °C (230 °F) filtered oil at approximately 275 kPa (40 psi) at rated engine operating conditions. Oil pump bypass valve (16) is controlled by the engine oil manifold pressure, rather than the oil pump pressure. The engine oil manifold pressure is independent of the pressure drop that is caused by the engine oil filter and the engine oil cooler.

Oil cooler bypass valve (12) maintains the engine oil temperature to 110 °C (230 °F). High pressure relief valve (15), which is located in the filter base, protects the filters and other components during cold starts. The opening pressure of the high pressure relief valve is 695 kPa (100 psi). Secondary oil filter (11) is a five micron filter which filters five percent of the oil flow before returning the oil to the sump. The opening pressure of the oil filter bypass valve is 170 kPa (25 psi). Engine oil pressure sensor (3) is part of the engine protection system.

The turbocharger cartridge bearings are lubricated by oil supply line (18) from the main oil gallery, and oil drain line (19) returns the oil flow to the sump.

## Oil Flow Through The Lubrication System

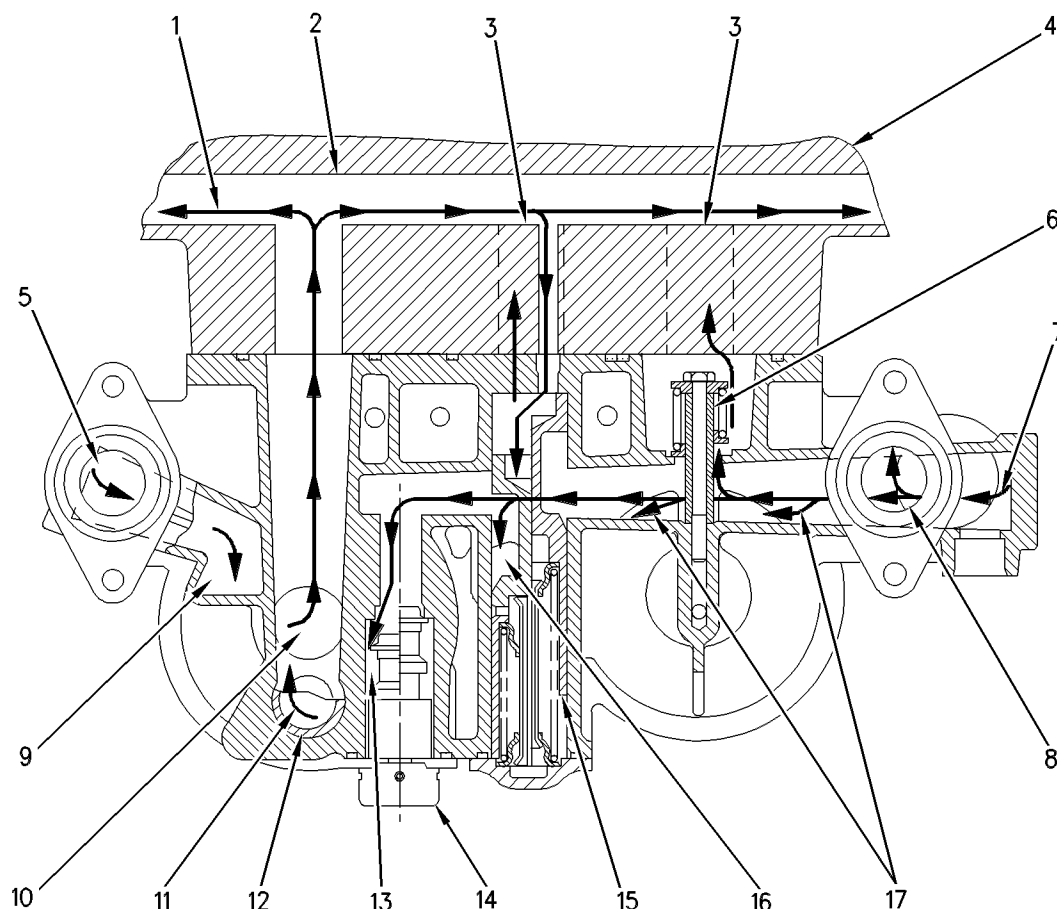


Illustration 19  
Oil flow through engine

q00421838

- |                                                                                                                                     |                                          |                                           |
|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|
| (1) Oil flow to the piston, piston cooling jets, valve mechanism, camshaft journals, crankshaft main bearings, and the turbocharger | (5) Oil from engine oil cooler           | (12) Oil filter bypass valve              |
| (2) Main oil gallery in cylinder block                                                                                              | (6) High pressure relief valve           | (13) Passage to primary engine oil filter |
| (3) Oil drains to sump                                                                                                              | (7) Oil from engine oil pump             | (14) Oil cooler bypass valve              |
| (4) Cylinder block                                                                                                                  | (8) Oil to engine oil cooler             | (15) Oil pump bypass valve                |
|                                                                                                                                     | (9) Passage to primary engine oil filter | (16) Oil pump bypass drain                |
|                                                                                                                                     | (10) Filtered oil                        | (17) Passages to secondary oil filter     |
|                                                                                                                                     | (11) Bypassed oil                        |                                           |

The engine oil pump is mounted to the back of the front gear train on the lower right hand side of the engine. The engine oil pump is driven by an idler gear from the crankshaft gear. Oil is pulled from the sump through oil pump bypass valve (15) on the way to the engine oil cooler. The bypass valve controls the oil pressure from the engine oil pump. The engine oil pump can supply excess oil for the lubricating system. When this situation is present, the oil pressure increases and the bypass valve opens. The open bypass valve allows the excess oil to return to the sump.

High pressure relief valve (6) regulates high pressure in the system. The high pressure relief valve will allow the oil to return to the sump when the oil pressure reaches 695 kPa (100 psi). The oil then flows through the engine oil cooler. The engine oil cooler uses engine coolant in order to cool the oil. The oil cooler bypass valve (14) directs the oil flow through the engine oil cooler by two different methods.

Oil cooler bypass valve (14) will close when the oil temperature exceeds the following temperatures 100 °C to 103 °C (212 °F to 217 °F). Closing of the bypass valve will direct the oil through the engine oil cooler (13). If the oil reaches a temperature of 127 °C (260 °F) the bypass valve would close directing the oil flow through the engine oil cooler (13).

The oil bypass valve (14) is normally closed if the pressure across the engine oil cooler is less than  $155 \pm 17$  kPa ( $22 \pm 3$  psi). This will direct the oil through the engine oil cooler (13).

Approximately five percent of the oil flow is directed through an orificed passage that leads to secondary oil filter (17) (if equipped). The oil flows through the bypass filter and to the engine oil sump. The main oil flow now flows toward the primary engine oil filter. When the oil pressure differential across oil filter bypass valve (12) reaches 170 kPa (25 psi), the valve allows the oil flow to bypass the primary engine oil filter in order to lubricate the engine parts. The bypass valve provides immediate lubrication to the engine components when there is a restriction in the primary engine oil filter due to the following conditions:

- Cold oil with high viscosity
- Plugged primary engine oil filter

**Note:** Refer to Specifications, "Engine Oil Filter Base" for a cross section of the valves in the engine oil filter base.

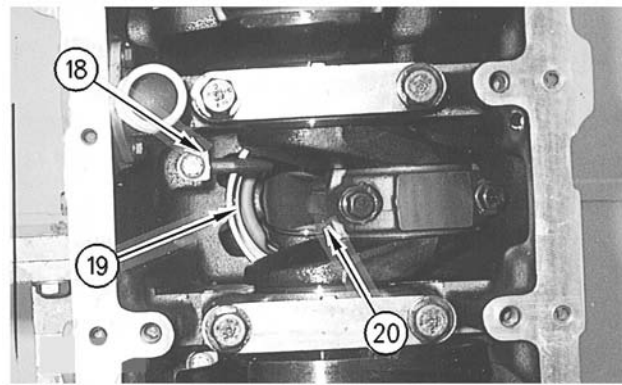


Illustration 20  
Interior of cylinder block  
(18) Piston cooling jet  
(19) Piston  
(20) Connecting rod

Filtered oil flows through main oil gallery (2) in the cylinder block to the following components:

- Piston cooling jets (18)
- Valve mechanism

- Camshaft bearings
- Crankshaft main bearings
- Turbocharger

An oil cooling chamber is formed by the following pieces: the lip forge at the top of the skirt of the piston (19) and the cavity behind the ring grooves in the crown. Oil flow from the piston cooling jet enters the cooling chamber through a drilled passage in the skirt. The oil then returns to the sump through the clearance gap between the crown and the skirt. Four holes that are drilled from the piston oil ring groove to the interior of the piston drain excess oil from the oil ring.

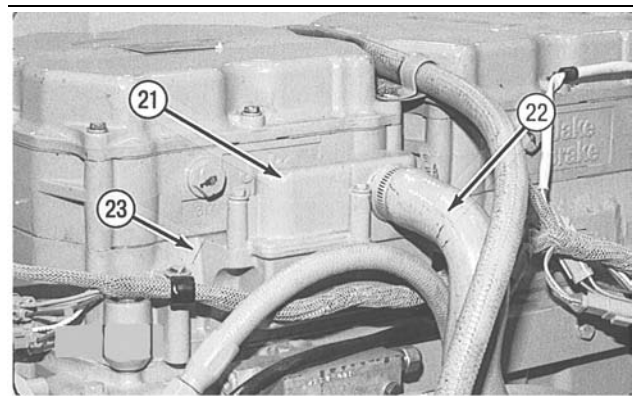


Illustration 21  
Left front side of engine  
(21) Breather  
(22) Hose  
(23) Cylinder head

g00481403

Breather (21) allows engine blowby to escape from the crankcase. The engine blowby is discharged into the atmosphere through hose (22). This prevents pressure from building up that could cause seals or gaskets to leak.

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## Cooling System

SMCS Code: 1350

### Coolant Flow

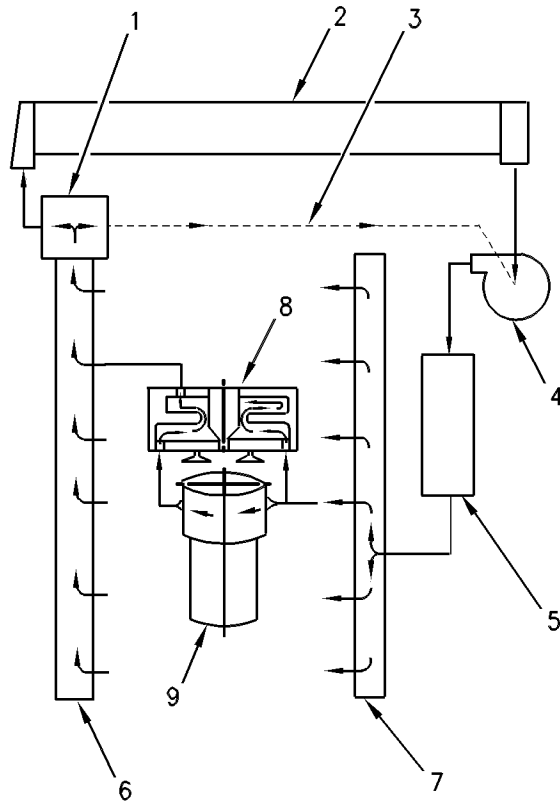


Illustration 22

g00423398

Cooling system schematic

- (1) Temperature regulator housing
- (2) Radiator
- (3) Bypass tube
- (4) Water pump
- (5) Engine oil cooler
- (6) Return manifold
- (7) Supply manifold in the block
- (8) Cylinder head
- (9) Cylinder liner

The water pump is driven by a gear. The water pump is located on the right hand side of the engine. The water pump supplies the coolant for the engine cooling system. The coolant is supplied to the following components:

- Engine oil cooler (5)

- Cylinder head (8)
- Cylinder liners (9)
- Air compressor (not shown)
- Coolant conditioner element (not shown)

**Note:** In air-to-air aftercooled systems, a coolant mixture with a minimum of 30 percent ethylene glycol base antifreeze must be used for efficient water pump performance. This mixture keeps the cavitation temperature range of the coolant high enough for efficient performance.

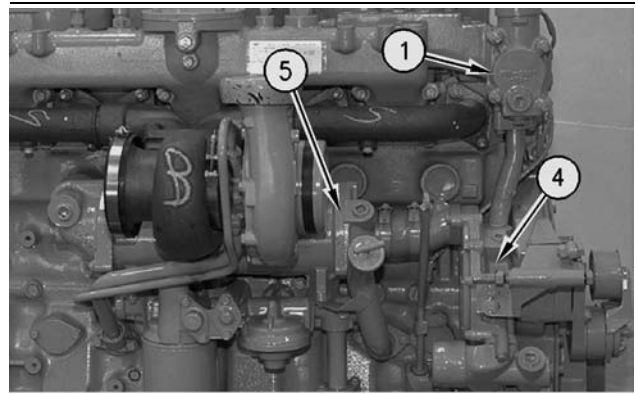


Illustration 23

g00423399

Typical right side of engine

- (1) Temperature regulator housing
- (4) Water pump
- (5) Engine oil cooler

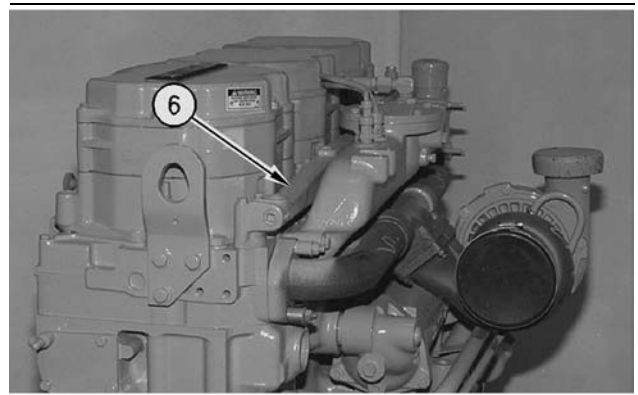


Illustration 24

g00423400

Typical rear of engine

- (6) Return manifold

Water pump (4) pulls the coolant from the bottom of radiator (2) by using the impeller's rotation. The water pump is located on the right hand side of the front timing gear housing.

The water pump impeller is rotated at 1.17 times the engine speed by an idler gear. The idler gear is turned by the crankshaft gear. The water pump shaft is supported by two ball bearings. One ball bearing is located in the water pump housing. The other ball bearing is located in the front timing gear housing. The water pump impeller face is open. The impeller vane is radial. The impeller is made out of cast iron. The rear cover is an aluminum die casting. The water pump seal is a cartridge seal that is located on the inlet side of the water pump in order to provide good water flow around the seal for cooling.

The coolant is pumped through engine oil cooler (5) to supply manifold (7). The supply manifold, which is located in the cylinder block, distributes coolant around the upper portion of the cylinder liners. At each cylinder, the coolant flows from the cylinder liner to the cylinder head. The cylinder head is divided into single cylinder cooling sections. In the cylinder head, the coolant flows across the center of the cylinder and across the injector seat boss. At the center of the cylinder, the coolant flows around the injector sleeve over the exhaust port. The coolant then exits into return manifold (6). The return manifold collects the coolant from each cylinder and the return manifold directs the flow to temperature regulator housing (1). When the coolant temperature regulator is in the closed position, the coolant flows through the coolant temperature regulator. This allows the coolant to flow directly back to the water pump for recirculation by bypassing the radiator. When the coolant temperature regulator is in the open position, the coolant is directed through the radiator and back to the water pump inlet.

## Supply Manifold

Cooling is provided for only the portion of the cylinder liner above the seal in the cylinder block. The coolant enters the cylinder block at each cylinder through slits in the supply manifold. The supply manifold is an integral casting in the cylinder block. The coolant flows around the circumference of the cylinder liner and into the cylinder head through a single drilled passage for each liner. The coolant flow is split at each cylinder liner so that 60 percent flows around the cylinder liner and the remainder flows directly to the cylinder head.

## Temperature Regulator Housing

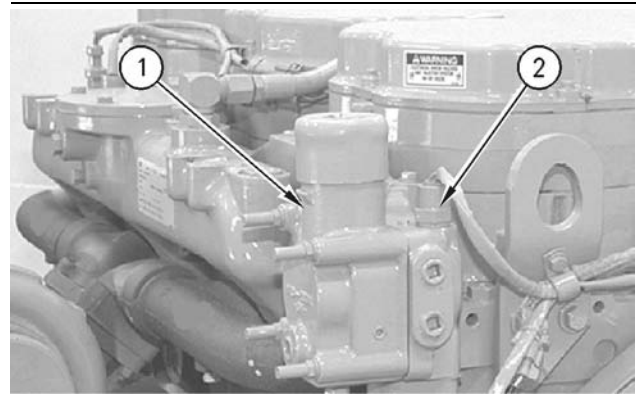


Illustration 25

g00500620

Typical right front side of engine

- (1) Temperature regulator housing
- (2) Coolant temperature sensor

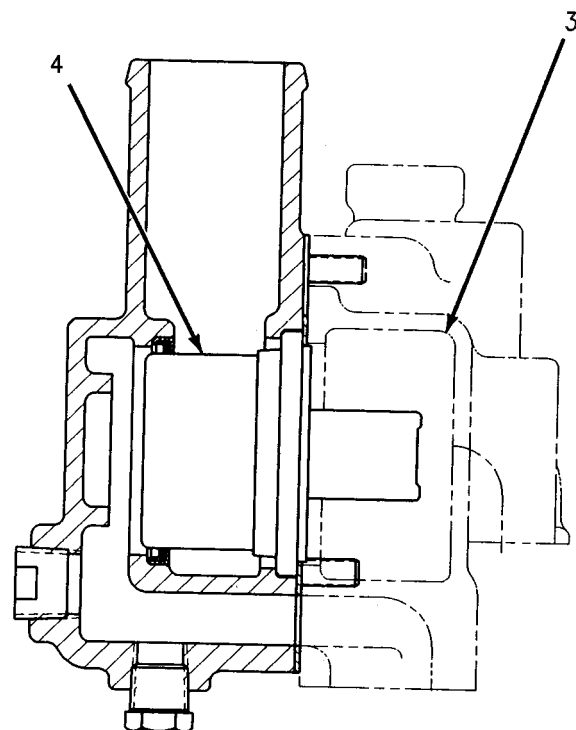


Illustration 26

g00423403

Temperature regulator housing

- (3) Return manifold
- (4) Coolant temperature regulator

The coolant temperature regulator is a full flow bypass type that is used to control the outlet temperature of the coolant. When the engine is cold, coolant temperature regulator (4) is in the closed position. This allows the coolant to flow through the coolant temperature regulator from return manifold (3). This allows the coolant to bypass the radiator. The coolant goes directly to the water pump for recirculation. As the coolant temperature increases, the coolant temperature regulator begins to open directing some of the coolant to the radiator and bypassing the remainder to the water pump inlet. At the full operating temperature of the engine, the coolant temperature regulator moves to the open position. This allows all the coolant flow to be directed to the radiator. The coolant then goes to the water pump. This route provides the maximum heat release from the coolant. A vent line is recommended from the manifold to the radiator overflow tank in order to provide venting for the cooling system. The recommended vent line is a #4 Aeroquip.

## Coolant Conditioner (If Equipped)

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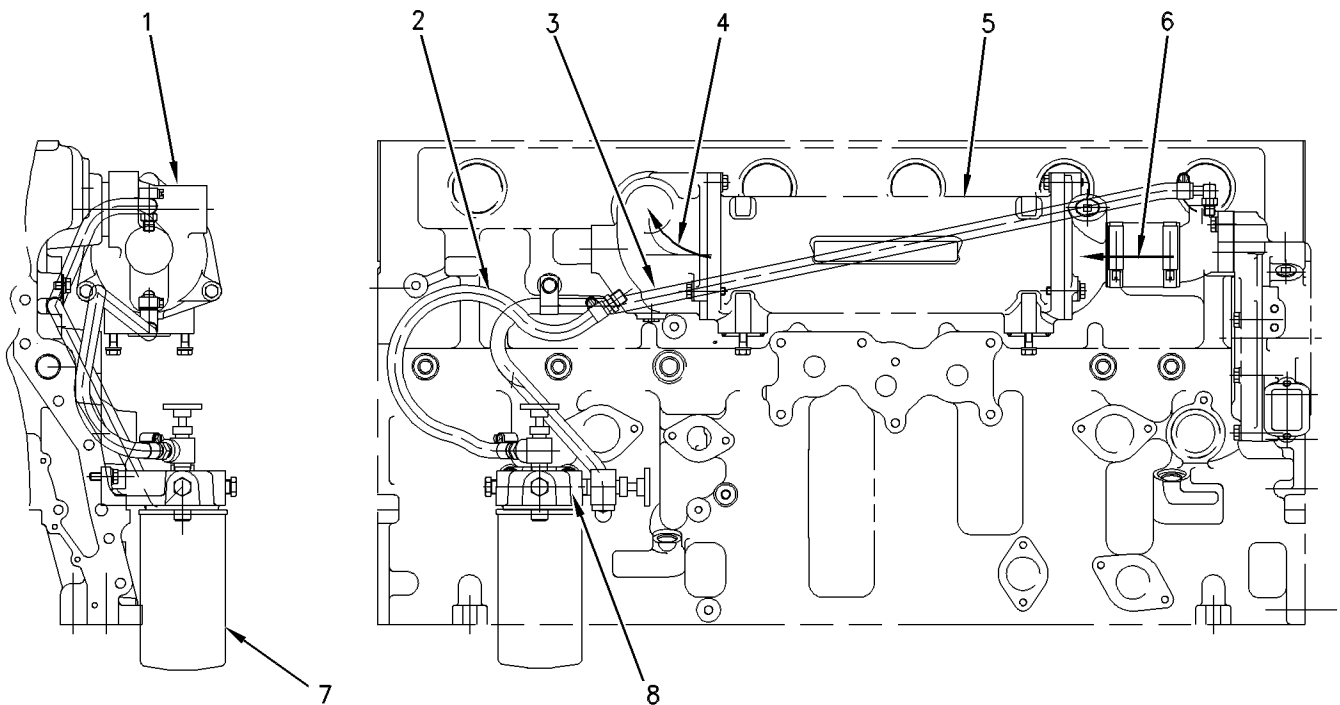


Illustration 27

g00423404

- |                                   |                                  |
|-----------------------------------|----------------------------------|
| (1) Engine oil cooler elbow       | (5) Engine oil cooler            |
| (2) Outlet hose                   | (6) Coolant flow from water pump |
| (3) Inlet hose                    | (7) Coolant conditioner element  |
| (4) Coolant flow to cylinder head | (8) Coolant conditioner base     |

Some conditions of operation can cause pitting on critical engine components. This pitting is caused by corrosion or by cavitation erosion. The addition of a corrosion inhibitor can keep this type of damage to a minimum.

Coolant conditioner element (7) is a spin-on element that is similar to the fuel filter and to the engine oil filter elements. The coolant conditioner element attaches to coolant conditioner base (8) that is mounted on the engine. Coolant flows from the water pump through inlet hose (3) and into the coolant conditioner base. The coolant that is conditioned then flows through outlet hose (2) into engine oil cooler elbow (1). There is a constant flow through the coolant conditioner element.

The element has a specific amount of inhibitor for acceptable cooling system protection. As the coolant flows through the element, the corrosion inhibitor, which is a dry material, disperses into the coolant. The coolant and the inhibitor are mixed to the correct concentration. Two basic types of elements are used for the cooling system, the precharge and the maintenance elements. Each type of element has a specific use. Each type of element must be used correctly to get the necessary concentration for cooling system protection. The elements also contain a filter. Even after the conditioner material is dispersed, the elements should be left in the system so the coolant flows through the filter.

The precharge coolant conditioner element has more than the normal amount of corrosion inhibitor. The precharge coolant conditioner element is used when a system is first filled with new coolant. This element must add enough corrosion inhibitor in order to bring the complete cooling system up to the correct concentration.

The maintenance elements have a normal amount of inhibitor and the maintenance elements are installed at each change interval. The maintenance elements provide enough inhibitor in order to keep the corrosion protection at an acceptable level. In order to provide the cooling system with protection, maintenance elements are installed at specific intervals.

## Coolant for Air Compressor

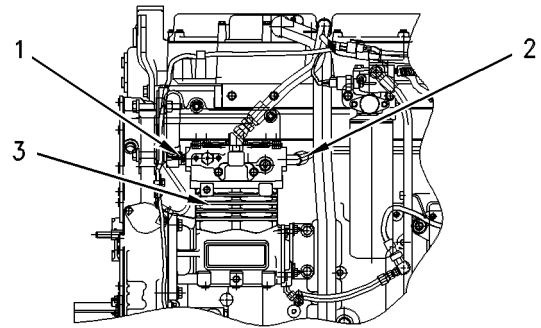


Illustration 28

g00501085

Typical air compressor

- (1) Outlet hose
- (2) Inlet hose
- (3) Air compressor

The coolant that is used for air compressor (3) comes from the cylinder head through inlet hose (2). The coolant exits the air compressor through outlet hose (1) and flows back to the cylinder head.

i01781028

## Basic Engine

**SMCS Code:** 1200

### Cylinder Block

The cylinder block is a unique design with a deep counterbore that supports the cylinder liner. The cylinder block also forms the coolant jacket. Two oil manifolds are provided in the cylinder block for engine lubrication. The manifold on the lower right side of the cylinder block provides oil to the following components:

- Piston cooling jets
- Crankshaft bearings
- Oil filter base

The manifold on the upper left side of the cylinder block provides oil to the following components:

- Camshaft bearings
- Valve mechanism

The manifold on the right supplies oil to the manifold on the left. The oil travels through the cut above the number one main bearing and the cut above the number four main bearing.

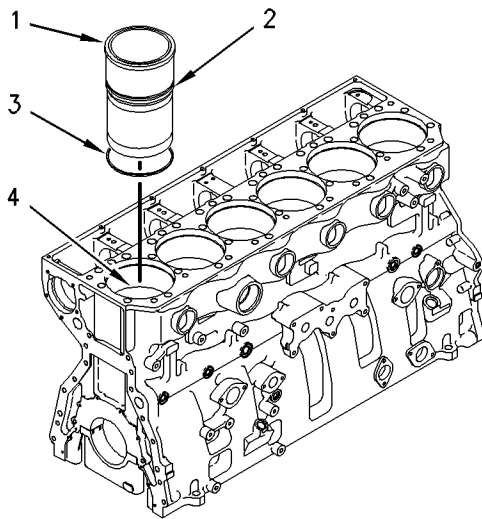


Illustration 29

g00762739

Cylinder liners (1) are seated on a ridge (4) in the middle of the cylinder wall between the crankcase and the coolant jacket. The ridge is created by a counterbore in the cylinder block. The cylinder liners have a lip (2) which rests on the ridge. The seals of the coolant jacket are located in the upper regions and middle regions of the cylinder liners. The lower barrier uses a D-ring seal (3) that is located above the seating surface of the cylinder liner. The upper barrier is the head gasket which is above the coolant jacket.

The cylinder block has seven main bearings in order to support the crankshaft. The main bearing caps are fastened to the cylinder block with two bolts per cap.

## Pistons, Rings, and Connecting Rods

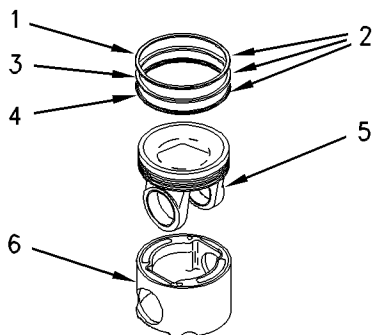


Illustration 30

g00762765

The piston is a two-piece articulated piston. The piston has a forged steel crown (5) and a cast aluminum skirt (6). Both parts are retained by the piston pin to the small end of the connecting rod.

The pistons have three rings:

- Compression ring
- Intermediate ring
- Oil ring

Rings (2) are located in grooves in steel crown (5). The rings seal the crankcase from the combustion gases and the rings also provide control of the engine oil. The design of compression ring (1) is a barrel face with a plasma face coating. The design of intermediate ring (3) is a tapered shape and a chrome finish. Oil ring (4) is double railed with a coil spring expander. The oil ring has a ground profile and a chrome finish.

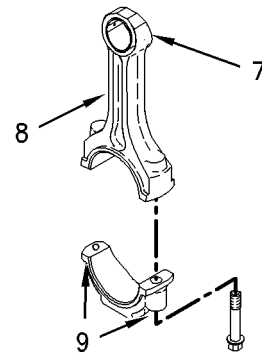


Illustration 31

g00762786

Connecting rod (8) is a conventional design. The cap is fastened to the shank by two bolts (9) that are threaded into the shank. Each side of the small end of the connecting rod (7) is machined at an angle of 12 degrees in order to fit within the piston cavity. This allows a larger surface area on the piston, and connecting rod in order to minimize bearing load.

## Crankshaft

The crankshaft converts the linear motion of the pistons into rotational motion. The crankshaft drives a group of gears (front gear train) on the front of the engine. The front gear train provides power for the following components:

- Camshaft
- Water pump
- Engine oil pump

- Air compressor
- Fuel transfer pump
- Accessory drive

The crankshaft is held in place by seven main bearings. The oil holes and the oil grooves in the shell of the upper bearing supply oil to the connecting rod bearings. The oil holes for the connecting rod bearings are located at the following main bearing journals: 2, 3, 5, and 6.

Hydrodynamic seals are used at both ends of the crankshaft to control oil leakage. The hydrodynamic grooves in the seal lip move lubrication oil back into the crankcase as the crankshaft turns. The front seal is located in the front housing. The rear seal is installed in the flywheel housing.

## Camshaft

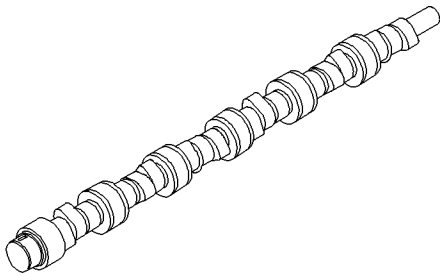


Illustration 32

g00762808

The camshaft has three lobes at each cylinder in order to operate the unit injector, the exhaust valves, and the inlet valves. Seven bearings support the camshaft. The camshaft is driven by an idler gear that is turned by the crankshaft in the front gear train. Each bearing journal is lubricated from the oil manifold in the cylinder block. A thrust pin that is located at the rear of the block positions the camshaft through a circumferential groove. The groove is machined at the rear of the camshaft. Timing of the camshaft is accomplished by aligning marks on the crankshaft gear, idler gear, and camshaft gear with each other.

The injector lobe on the camshaft has a modified profile. The modified profile produces multiple injections.

## Vibration Damper

The force from combustion in the cylinders and from driveline components will cause the crankshaft to twist. This is called torsional vibration. If the vibration is too great, the crankshaft will be damaged. Driveline components can excite torsional stress. This stress will cause damage to components. The vibration damper limits the torsional vibrations to an acceptable amount in order to prevent damage to the crankshaft.

The viscous vibration damper is installed on the front of the crankshaft. The viscous vibration damper has a weight in a case. The space between the weight and the case is filled with a viscous fluid. The weight moves in the case in order to limit the torsional vibration.

i00984742

## Rear Power Take-Off (RPTO)

**SMCS Code:** 1165-RE

The Rear Power Take-Off (RPTO) is an integral part of the flywheel housing. The rear power take-off provides continuous live power through the following direct drive gears:

- Crankshaft gear
- Idler gear
- Output shaft gear

These gears are driven off the rear of the crankshaft.

i01797547

## Compression Brake

**SMCS Code:** 1119; 1129

The compression brake is powered from the ECM. The compression brake helps the operator to slow the vehicle on grades, in curves, or for a necessary speed reduction. The service brakes should not be used continuously on long descending grades. The service brakes are assisted by the compression brake. The engine crankshaft is turned by the rear wheels during downhill operation or during any slow down condition. The engine crankshaft is turned through the differential, through the drive shaft, through the transmission, and through the clutch. An application of a braking force can be made to the engine piston in order to reduce the speed of the vehicle.

When the compression brake is activated, braking power is accomplished by opening the engine's exhaust valves. The exhaust valves are opened near the top of the compression stroke in order to release the highly compressed air into the exhaust system. The compression brake can only be activated when the engine is in the no-fuel position. Thus, combustion does not occur and no positive force is produced on the piston. The compressed air pressure that is released to the atmosphere prevents the energy from returning to the engine piston on the power stroke. The result is a loss of energy since the work that is done by the compression of the cylinder charge is not returned by the expansion process. This loss of energy is taken from the rear wheels. The rear wheels provide the braking action for the vehicle.

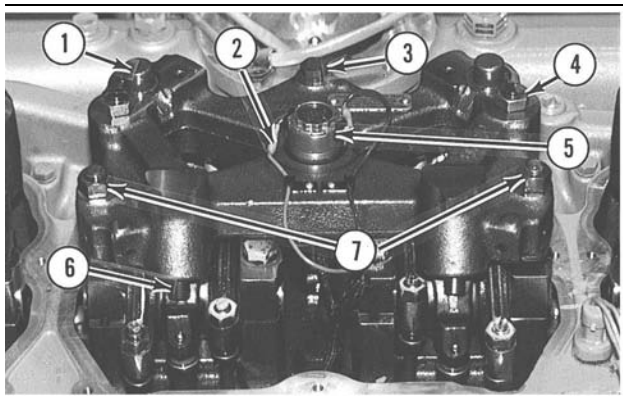


Illustration 33  
Compression brake

- (1) Control valve
- (2) Lead wire
- (3) Bolt assembly
- (4) Slave piston
- (5) Solenoid valve
- (6) Master piston
- (7) Stud and nut

The compression brake consists of three identical housing assemblies. A housing assembly is installed in each of the valve mechanism compartments above the rocker arms and above the rocker arm shaft. Each housing assembly is positioned over two cylinders. The housing assembly is mounted to the supports for the rocker arm shaft with studs and nuts (7). The housing assemblies are supported on the cylinder head with bolt assembly (3). The exhaust bridge assembly is used to transfer force from slave piston (4) to the exhaust valve. The brake logic signal for the compression brake is carried to solenoid valve (5) by lead wire (2). This is done in order to activate the compression brake on the two cylinders of the engine.

A spacer is used on the top of the valve cover base in order to provide space for the installation of the compression brake and of the valve cover. When the compression brake is installed, the height increases approximately 63.5 mm (2.50 inch).

The control circuit for the compression brake permits the operation of either one, two, or all three of the compression brake housing assemblies. This provides progressive braking capabilities with the retarding effect of two cylinders, of four cylinders, or of all six cylinders in the engine.

## Performance of the Compression Brake

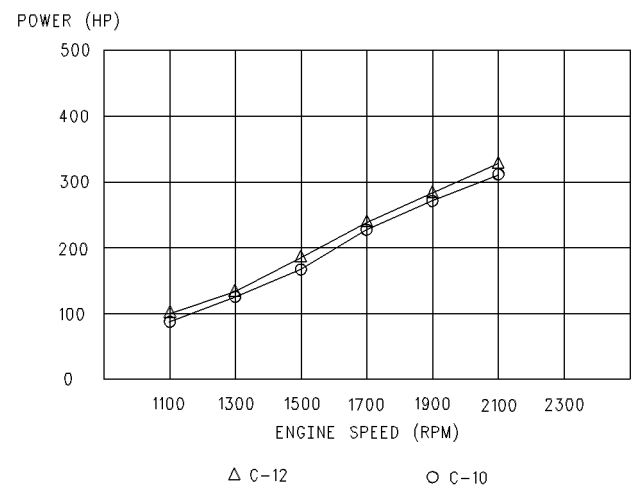


Illustration 34

g00427859

### Performance of the compression brake

The performance of the compression brake, which is shown in the graph, represents an engine with the three housing assemblies that are activated in order to stop the vehicle. The compression brake should not be activated when the engine rpm is above 2300. The maximum engine power rating is 2100 rpm. At this level, the amount of braking that is produced by the compression brake is approximately 315 to 325 horsepower.

## Operation of the Compression Brake

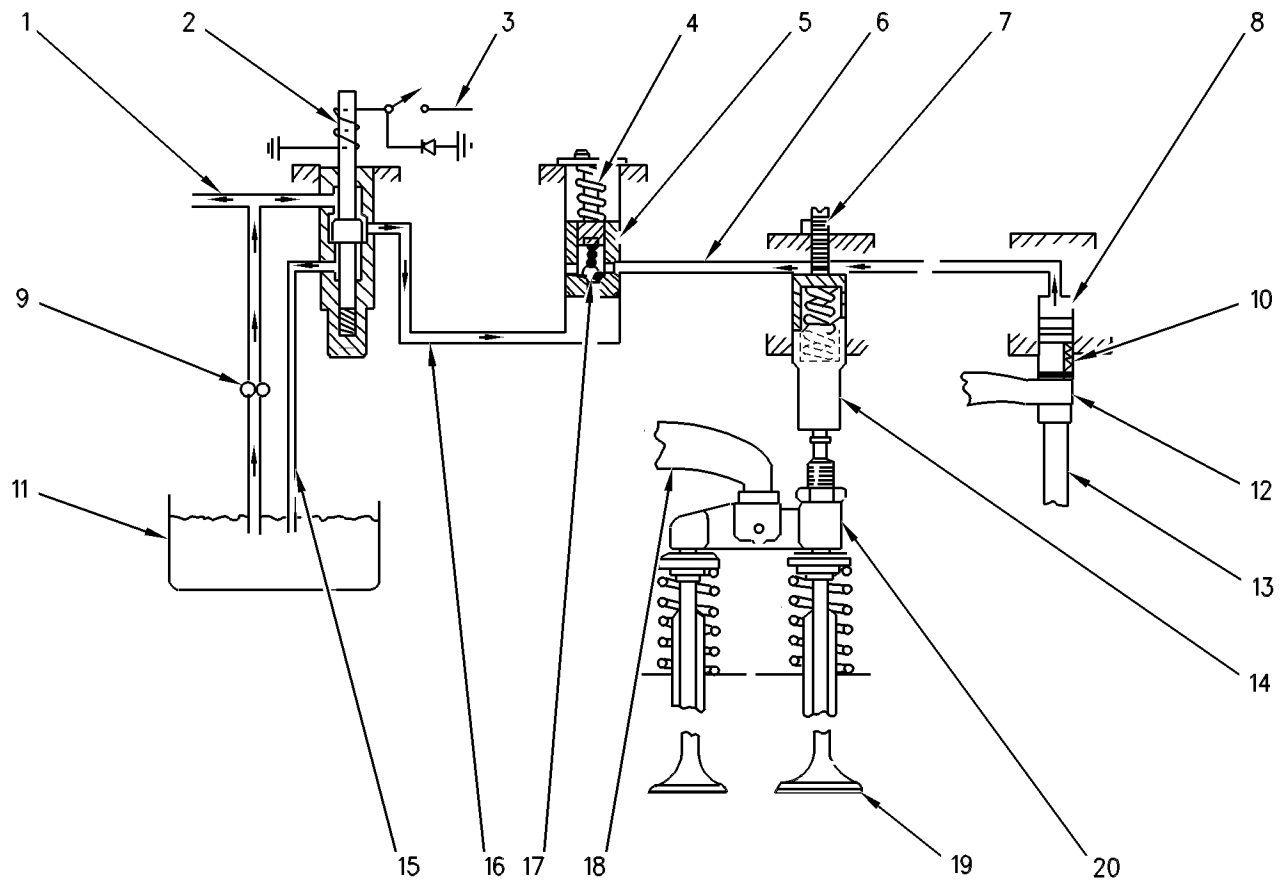


Illustration 35

g00501985

Schematic for master-slave circuit

- |                                   |                               |                               |
|-----------------------------------|-------------------------------|-------------------------------|
| (1) Rocker arm shaft oil passage  | (8) Master piston             | (15) Oil drain passage        |
| (2) Solenoid valve                | (9) Engine oil pump           | (16) Low pressure oil passage |
| (3) Lead wire                     | (10) Spring                   | (17) Ball check valve         |
| (4) Spring                        | (11) Engine oil pan           | (18) Exhaust valve rocker arm |
| (5) Control valve                 | (12) Fuel injector rocker arm | (19) Exhaust valve            |
| (6) High pressure oil passage     | (13) Fuel injector pushrod    | (20) Exhaust bridge assembly  |
| (7) Slave piston adjustment screw | (14) Slave piston             |                               |

The compression brake is operated by engine oil that is distributed around the studs. This oil is supplied through the supports for the rocker arm shaft. Solenoid valve (2) controls the oil flow in the compression brake housing.

When the solenoid is activated by a signal from the logic for the compression brake, solenoid valve (2) moves downward. This causes oil drain passage (15) to engine oil pan (11) to be closed. At the same time, low pressure oil passage (16) to control valve (5) is opened. As low pressure oil passage (16) is filled with engine oil, the control valve is pushed upward in the chamber against the force of spring (4). At this position, a groove in control valve (5) is in alignment with high pressure oil passage (6). Slave piston (14) and master piston (8) are supplied by the high pressure oil passage. Engine oil pressure will now lift ball check valve (17). High pressure oil passage (6) and the chambers behind the slave pistons and behind the master pistons will be filled with engine oil pressure. This pressure moves the master piston downward until contact is made with fuel injector rocker arm (12). When upward motion is initiated on the master piston, the pressure increases above the current level of the engine supply pressure. This causes ball check valve (17) to seat. The system is now operating in conjunction with the exhaust valve and with the injector rocker mechanism. When the solenoid is activated, the compression brake could be operable within 1/5 of a second.

Fuel injector pushrod (13) will begin to move upward on the pumping stroke of the electronically controlled unit injector. When this occurs, fuel injector rocker arm (12) makes contact with extending master piston (8). As the master piston begins to move upward, the oil pressure increases in high pressure oil passage (6). This happens because ball check valve (17) will not allow oil to exit. The upward movement of the fuel injector rocker arm creates a constant increase in pressure. This forces the slave piston downward against the screw assembly in exhaust bridge assembly (20). The slave piston moves downward with enough force to open exhaust valve (19).

**Note:** Only one of the two exhaust valves for each cylinder on this engine is used in the operation of the compression brake.

This master-slave circuit is designed to move the master piston (8). The master piston is moved when the engine is on the compression stroke. The master-slave circuit is designed so that slave piston (14) opens one exhaust valve of the same cylinder only on the compression stroke. This occurs slightly before the piston reaches the top center position. The braking force is constant. The operation of the compression brake of a cylinder is caused by the motion of the valve mechanism of that cylinder. This causes the firing sequence of the valves to be identical to the firing order of the engine.

When solenoid valve (2) is in the Off position, the engine oil supply passage is closed and oil drain passage (15) is opened. This allows oil to drain from underneath control valve (5). This also allows spring (4) to push the control valve to the bottom of the chamber. This position allows oil from high pressure oil passage (6) to drain into the chamber above the control valve's piston. This chamber vents to the outside of the compression brake housing. Spring (10) now moves master piston (8) to the retracted position away from fuel injector rocker arm (12). The time that is necessary for the system to stop operation is approximately 1/10 of a second. The compression brake will not be able to operate until solenoid (2) is activated again.

i01459456

## Electrical System

**SMCS Code:** 1400; 1550; 1900

### Grounding Practices

Proper grounding for the machine electrical system and engine electrical systems is necessary for proper machine performance and reliability. Improper grounding will result in uncontrolled electrical circuit paths and unreliable electrical circuit paths.

Uncontrolled engine electrical circuit paths can result in damage to main bearings, crankshaft bearing journal surfaces, and aluminum components.

To ensure proper functioning of the vehicle and engine electrical systems, an engine-to-frame ground strap with a direct path to the negative battery post must be used. This may be provided by way of a starting motor ground, a frame to starting motor ground, or a direct frame to engine ground.

An engine-to-frame ground strap must be used in order to connect the grounding stud of the engine to the frame of the vehicle and to the negative battery post.

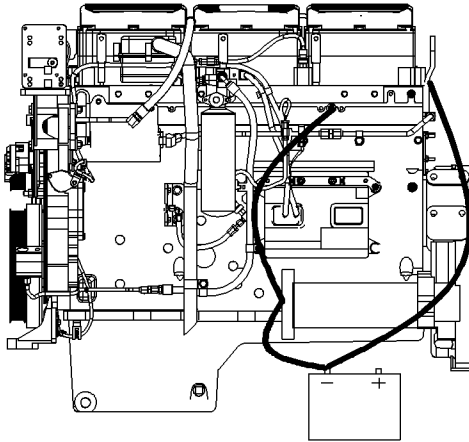


Illustration 36

g00766579

Typical example

Grounding Stud To Battery Ground ("—")

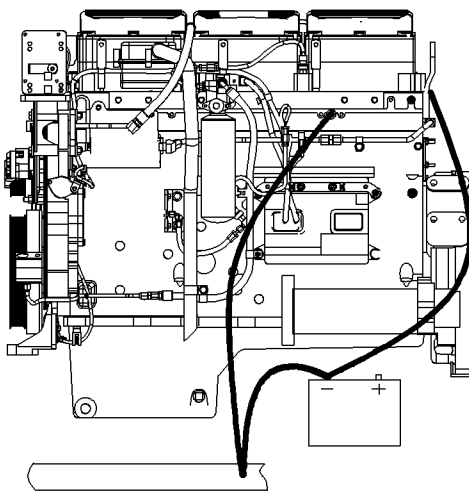


Illustration 37

g00766660

Typical example

Alternate Grounding Stud To Battery Ground ("—")

The engine must have a wire ground to the battery.

Ground wires or ground straps should be combined at ground studs that are only for ground use. All of the grounds should be tight and free of corrosion.

All of the ground paths must be capable of carrying any likely current faults. An AWG #0 or larger wire is recommended for the grounding strap to the cylinder head.

The engine alternator should be battery ground with a wire size that is capable of managing the full charging current of the alternator.

#### NOTICE

When boost starting an engine, the instructions in Systems Operation, "Engine Starting" should be followed in order to properly start the engine.

This engine may be equipped with a 12 volt starting system or a 24 volt starting system. Only equal voltage for boost starting should be used. The use of a higher voltage will damage the electrical system.

The Electronic Control Module (ECM) must be disconnected at the "J1/P1" and "J2/P2" locations before welding on the vehicle.

The engine has several input components which are electronic. These components require an operating voltage.

Unlike many electronic systems of the past, this engine is tolerant to common external sources of electrical noise. Buzzers that use electrical energy can cause disruptions in the power supply. If buzzers are used anywhere on the machine, the engine electronics should be powered directly from the battery system through a dedicated relay. The engine electronics should not be powered through a common power bus with other key switch activated devices.

## Engine Electrical System

The electrical system has the following separate circuits:

- Charging
- Starting (If equipped)
- Low amperage accessories

Some of the electrical system components are used in more than one circuit. The following components are common in more than one circuit:

- Battery or batteries

- Circuit breakers
- Battery cables
- Ammeter

The charging circuit is in operation when the engine is running. An alternator makes electricity for the charging circuit. A voltage regulator in the circuit controls the electrical output in order to keep the battery at full charge.

The starting circuit is activated only when the start switch is activated.

The low amperage accessory circuit and the charging circuit are connected through the ammeter. The starting circuit is not connected through the ammeter.

## Charging System Components

### Alternator

The alternator is driven by a belt from the crankshaft pulley. This alternator is a three-phase, self-rectifying charging unit, and the regulator is part of the alternator.

The alternator design has no need for slip rings and the only part that has movement is the rotor assembly. All conductors that carry current are stationary. The following conductors are in the circuit:

- Field winding
- Stator windings
- Six rectifying diodes
- Regulator circuit components

The rotor assembly has many magnetic poles that look like fingers with air space between each of the opposite poles. The poles have residual magnetism. The residual magnetism produces a small magnetic field between the poles. As the rotor assembly begins to turn between the field winding and the stator windings, a small amount of alternating current (AC) is produced. The AC current is produced in the stator windings from the small magnetic field. The AC current is changed to direct current (DC) when the AC current passes through the diodes of the rectifier bridge. The current is used for the following applications:

- Charging the battery
- Supplying the low amperage accessory circuit

- Strengthening the magnetic field

The first two applications use the majority of the current. As the DC current increases through the field windings, the strength of the magnetic field is increased. As the magnetic field becomes stronger, more AC current is produced in the stator windings. The increased speed of the rotor assembly also increases the current and voltage output of the alternator.

The voltage regulator is a solid-state electronic switch. The voltage regulator senses the voltage in the system. The voltage regulator switches ON and OFF many times per second in order to control the field current for the alternator. The alternator uses the field current in order to generate the required voltage output.

#### NOTICE

Never operate the alternator without the battery in the circuit. Making or breaking an alternator connection with heavy load on the circuit can cause damage to the regulator.

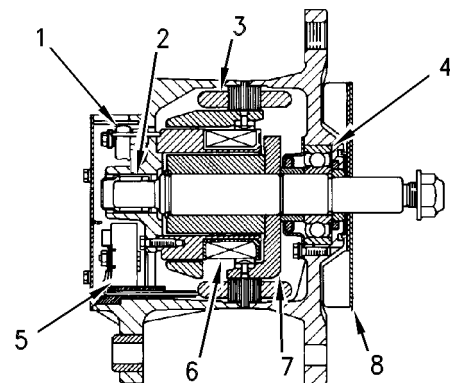


Illustration 38

g00425518

#### Typical alternator components

- (1) Regulator
- (2) Roller bearing
- (3) Stator winding
- (4) Ball bearing
- (5) Rectifier bridge
- (6) Field winding
- (7) Rotor assembly
- (8) Fan

## Starting System Components

### Starting Solenoid

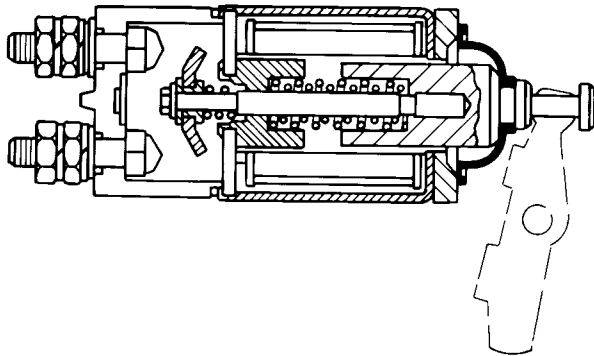


Illustration 39  
Typical starting solenoid

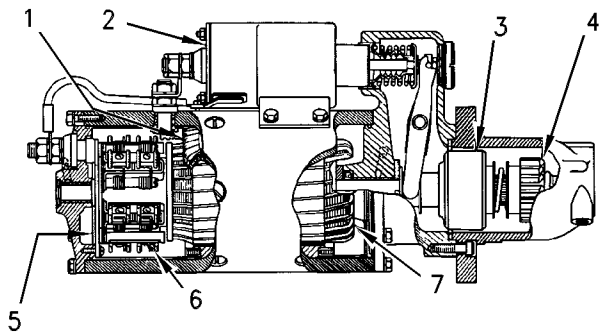


Illustration 40  
Typical starting motor components

- (1) Field
- (2) Solenoid
- (3) Clutch
- (4) Pinion
- (5) Commutator
- (6) Brush assembly
- (7) Armature

The starting solenoid (2) is an electromagnetic switch that performs the following basic operations:

- The starting solenoid (2) closes the high current starting motor circuit with a low current start switch circuit.
- The starting solenoid (2) engages the starter motor pinion (4) with the ring gear.

Solenoid (2) has windings (one or two sets) around a hollow cylinder. A spring loaded plunger (core) is inside of the cylinder. The plunger can move forward and backward. When the start switch is closed and electricity is sent through the windings, a magnetic field (1) is made. The magnetic field (1) pulls the plunger forward in the cylinder. This moves the shift lever in order to engage the pinion drive gear with the ring gear. The front end of the plunger then makes contact across the battery and motor terminals of solenoid (2). Next, the starting motor begins to turn the flywheel of the engine.

When the start switch is opened, current no longer flows through the windings. The spring now pushes the plunger back to the original position. At the same time, the spring moves the pinion gear away from the flywheel.

When two sets of solenoid windings are used, the windings are called the hold-in winding and the pull-in winding. Both sets of windings have the same number of turns around the cylinder, but the pull-in winding uses a wire with a larger diameter. The wire with a larger diameter produces a greater magnetic field (1). When the start switch is closed, part of the current flows from the battery through the hold-in windings. The rest of the current flows through the pull-in windings to the motor terminal. The current then flows through the motor to ground. Solenoid (2) is fully activated when the connection across the battery and the motor terminal is complete. When solenoid (2) is fully activated, the current is shut off through the pull-in windings. At this point, only the smaller hold-in windings are in operation. The hold-in windings operate for the duration of time that is required in order to start the engine. Solenoid (2) will now draw less current from the battery, and the heat that is generated by solenoid (2) will be kept at an acceptable level.

# Testing and Adjusting Section

## Fuel System

i01881094

### Fuel System - Inspect

**SMCS Code:** 1250-040

A problem with the components that send fuel to the engine can cause low fuel pressure. This can decrease engine performance.

1. Check the fuel level in the fuel tank. Ensure that the vent in the fuel cap is not filled with dirt.
2. Check all fuel lines for fuel leakage. The fuel lines must be free from restrictions and faulty bends. Verify that the fuel return line is not collapsed.
3. Install a new fuel filter.
4. Cut the old filter open with the **175-7546** Oil Filter Cutter. Inspect the filter for excess contamination. Determine the source of the contamination. Make the necessary repairs.
5. Service the primary fuel filter (if equipped).
6. Operate the hand priming pump (if equipped). If excessive resistance is felt, inspect the fuel pressure regulating valve. If uneven resistance is felt, test for air in the fuel. Refer to Testing and Adjusting, "Air in Fuel - Test" for more information.
7. Remove any air that may be in the fuel system. Refer to Testing and Adjusting, "Fuel System - Prime".

i01459442

### Air in Fuel - Test

**SMCS Code:** 1280-081

This procedure checks for air in the fuel. This procedure also assists in finding the source of the air.

1. Examine the fuel system for leaks. Ensure that the fuel line fittings are properly tightened. Check the fuel level in the fuel tank. Air can enter the fuel system on the suction side between the fuel transfer pump and the fuel tank.

2. Install a **2P-8278** Fuel Flow Tube (Sight Gauge) in the fuel return line. When possible, install the sight gauge in a straight section of the fuel line that is at least 304.8 mm (12 inches) long. Do not install the sight gauge near the following devices that create turbulence:

- Elbows
- Relief valves
- Check valves

Observe the fuel flow during engine cranking. Look for air bubbles in the fuel. If there is no fuel in the sight gauge, prime the fuel system. Refer to Testing and Adjusting, "Fuel System - Prime" for more information. If the engine starts, check for air in the fuel at varying engine speeds. When possible, operate the engine under the conditions which have been suspect of air in the fuel.

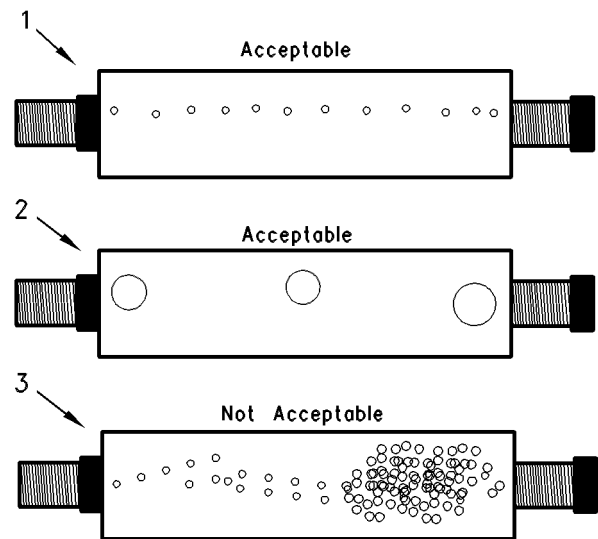


Illustration 41

g00578151

#### 2P-8278 Fuel Flow Tube (Sight Gauge)

- (1) A steady stream of small bubbles with a diameter of approximately 1.60 mm (0.063 inch) is an acceptable amount of air in the fuel.
- (2) Bubbles with a diameter of approximately 6.35 mm (0.250 inch) are also acceptable if there is two seconds to three seconds intervals between bubbles.
- (3) Excessive air bubbles in the fuel are not acceptable.

i01860816

3. If excessive air is seen in the sight gauge in the fuel return line, install a second sight gauge at the inlet to the fuel transfer pump. If a second sight gauge is not available, move the sight gauge from the fuel return line and install the sight gauge at the inlet to the fuel transfer pump. Observe the fuel flow during engine cranking. Look for air bubbles in the fuel. If the engine starts, check for air in the fuel at varying engine speeds.

If excessive air is not seen at the inlet to the fuel transfer pump, the air is entering the system after the fuel transfer pump. Proceed to Step 6.

If excessive air is seen at the inlet to the fuel transfer pump, air is entering through the suction side of the fuel system.

### WARNING

**To avoid personal injury, always wear eye and face protection when using pressurized air.**

### NOTICE

To avoid damage, do not use more than 55 kPa (8 psi) to pressurize the fuel tank.

4. Pressurize the fuel tank to 35 kPa (5 psi). Do not use more than 55 kPa (8 psi) in order to avoid damage to the fuel tank. Check for leaks in the fuel lines between the fuel tank and the fuel transfer pump. Repair any leaks that are found. Check the fuel pressure in order to ensure that the fuel transfer pump is operating properly. For information about checking the fuel pressure, see Testing and Adjusting, "Fuel System Pressure - Test".
5. If the source of the air is not found, disconnect the supply line from the fuel tank and connect an external fuel supply to the inlet of the fuel transfer pump. If this corrects the problem, repair the fuel tank or the stand pipe in the fuel tank.
6. If the injector sleeve is worn or damaged, combustion gases may be leaking into the fuel system. Also, if the O-rings on the injector sleeves are worn, missing, or damaged, combustion gases may leak into the fuel system.

## Electronic Unit Injector - Adjust

SMCS Code: 1290-025

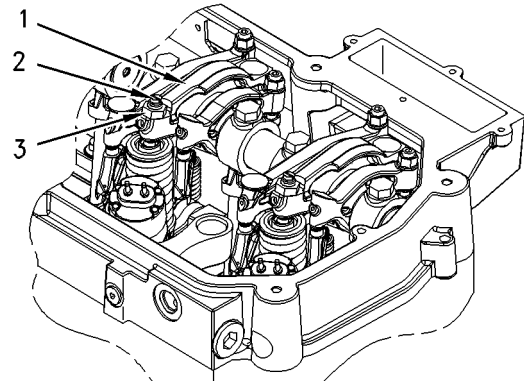


Illustration 42

g00948810

Injector Mechanism

- (1) Rocker arm
- (2) Adjusting screw
- (3) Locknut

Follow the procedure in order to adjust your Electronic unit injectors:

1. Put the No. 1 piston at the top center position on the compression stroke. Refer to Systems Operation/Testing and Adjusting, "Finding Top Center Position for No. 1 Piston".
  - a. Cylinder 3, 5, and 6 can be adjusted with cylinder 1 at TC compression stroke.
  - b. Loosen the jam nut.
  - c. Turn the adjusting screw until the screw makes contact with the electronic unit injector.
  - d. Turn the adjusting screw through 180 degrees in a clockwise direction.
  - e. Tighten the jam nut to a torque of  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $41 \pm 7 \text{ lb}\cdot\text{ft}$ ).
2. Rotate the engine in the normal operating direction by 360 degrees. Cylinder 1 will now be on TC exhaust stroke.
  - a. Cylinder 1, 2, and 4 can be adjusted with cylinder 1 at TC exhaust stroke.
  - b. Loosen the jam nut.
  - c. Turn the adjusting screw until the screw makes contact with the electronic unit injector.
  - d. Turn the adjusting screw through 180 degrees in a clockwise direction.

- e. Tighten the jam nut to a torque of  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft).

i01779471

## Electronic Unit Injector - Test

**SMCS Code:** 1290-081

This procedure assists in identifying the cause for an injector misfiring. Perform this procedure only after performing the Cylinder Cutout Test. Refer to Troubleshooting for more information.

1. Check for air in the fuel, if this procedure has not already been performed. Refer to Testing and Adjusting, "Air in Fuel - Test".

### WARNING

**Electrical shock hazard. The electronic unit injector system uses 90-120 volts.**

2. Remove the valve cover and look for broken parts. Repair any broken parts or replace any broken parts that are found. Inspect all wiring to the solenoids. Look for loose connections. Also look for frayed wires or broken wires. Ensure that the connector for the unit injector solenoid is properly connected. Perform a pull test on each of the wires. Refer to Troubleshooting, "Electrical Connectors - Inspect". Inspect the posts of the solenoid for arcing. If arcing or evidence of arcing is found, remove the cap assembly. Refer to Disassembly and Assembly Manual, "Electronic Unit Injector - Remove". Clean the connecting posts. Reinstall the cap assembly and tighten the solenoid nuts to a torque of  $2.5 \pm 0.25$  N·m ( $22 \pm 2$  lb in). Refer to Disassembly and Assembly Manual, "Electronic Unit Injector - Install".
3. Check the valve lash setting for the cylinder of the suspect unit injector. Refer to Testing and Adjusting, "Engine Valve Lash - Inspect/Adjust".
4. Ensure that the bolt that holds the unit injector is tightened to the proper torque. If necessary, loosen the bolt that holds the unit injector and tighten the bolt to a torque of  $30 \pm 7$  N·m ( $22 \pm 5$  lb ft).

5. Remove the suspect unit injector and check the unit injector for signs of exposure to coolant. Refer to Disassembly and Assembly Manual, "Electronic Unit Injector - Remove". Exposure to coolant will cause rust to form on the injector. If the unit injector shows signs of exposure to coolant, remove the injector sleeve and inspect the injector sleeve. Refer to Disassembly and Assembly Manual, "Electronic Unit Injector Sleeve - Remove". Replace the injector sleeve if the injector sleeve is damaged. Check the unit injector for an excessive brown discoloration that extends beyond the injector tip. If excessive discoloration is found, check the quality of the fuel. Refer to Testing and Adjusting, "Fuel Quality - Test". Replace the seals on the injector and reinstall the injector. Refer to Disassembly and Assembly Manual, "Electronic Unit Injector - Install". Also refer to Disassembly and Assembly Manual, "Electronic Unit Injector Sleeve - Install".

6. If the problem is not resolved, replace the suspect injector with a new injector.

i01901780

## Finding Top Center Position for No. 1 Piston

**SMCS Code:** 1105-531

### Standard Flywheel Housing

Table 2

| Required Tools |                     |          |
|----------------|---------------------|----------|
| Part Number    | Part Name           | Quantity |
| 214-7340       | Reverse Ratchet     | 1        |
| 9S-9082        | Engine Turning Tool | 1        |
| 5P-7305        | Engine Turning Tool | 1        |
| 136-4632       | Timing Pin          | 1        |
| 139-7063       | Timing Pin Adapter  | 1        |
| 139-7064       | Timing Pin Adapter  | 1        |

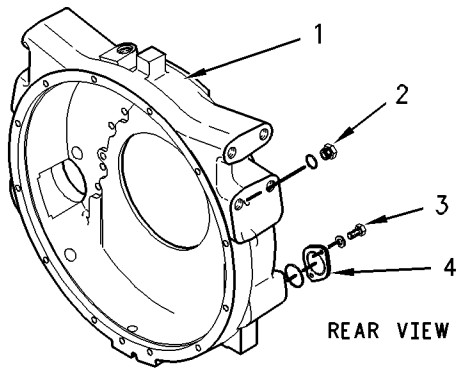


Illustration 43

g00780199

Typical Example

- (1) Flywheel Housing
- (2) Timing Hole Plug
- (3) Cover Bolt
- (4) Cover

1. Remove two bolts (3) and remove cover (4) from flywheel housing (1) in order to open the turning hole.
2. Thread the **136-4632** Timing Pin with the proper Timing Pin Adapter into the timing hole. The timing hole is located approximately 127 to 152 mm (5.0 to 6.0 inch) above the turning hole for the engine turning tool in the flywheel housing. Use the **9S-9082** Engine Turning Tool and a **214-7340** Reverse Ratchet to turn the engine flywheel. Turn the flywheel in the direction of engine rotation. The direction of engine rotation is counterclockwise, as the engine is viewed from the flywheel end. Turn the flywheel until the Timing Pin engages with the hole in the flywheel.

**Note:** **5P-7305** Engine Turning Tool may be used in certain applications if the **9S-9082** Engine Turning Tool will not fit the flywheel housing.

**Note:** If the flywheel is turned beyond the point of engagement, the flywheel must be turned in the direction that is reverse of normal engine rotation. Turn the flywheel by approximately 30 degrees. Then turn the flywheel in the direction of normal rotation until the Timing Pin engages with the hole in the flywheel. This procedure removes the play from the gears when the No. 1 piston is at the top center position.

3. Remove the front valve mechanism cover from the engine.

4. The inlet and exhaust valves for the No. 1 cylinder are fully closed if No. 1 piston is on the compression stroke and the rocker arms can be moved by hand. If the rocker arms cannot be moved and the valves are slightly open, the No. 1 piston is on the exhaust stroke.

**Note:** When the actual stroke position is identified, and the other stroke position is needed, remove the Timing Pin from the hole in the flywheel. Then turn the flywheel by 360 degrees in the direction of normal engine rotation and reinstall the Timing Pin into the hole in the flywheel.

**Note:** Never turn the engine by the crankshaft vibration damper. The crankshaft vibration damper is a precision part. Major engine failure may be caused by damage to the crankshaft vibration damper.

## Procedure for Engines that have a Rear Power Take-Off

Table 3

| Required Tools |                     |          |
|----------------|---------------------|----------|
| Part Number    | Part Name           | Quantity |
| 9U-6639        | Ratchet Wrench      | 1        |
| 208-0888       | Engine Turning Tool | 1        |
| 208-9387       | Pin                 | 1        |
| 208-9388       | Adapter             | 1        |

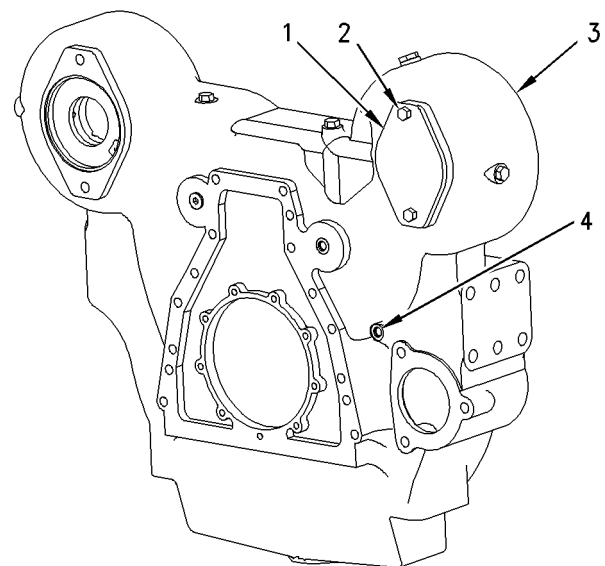


Illustration 44

g00837908

Typical Rear Power Take-Off

- (1) Cover
- (2) Bolt
- (3) Flywheel housing
- (4) Timing hole

1. Remove two bolts (2) in order to remove cover (1) from flywheel housing (3). This will allow you to use **208-0888** Engine Turning Tool to rotate the engine.
2. Remove the plug in the timing hole (4). Place the **208-9387** Pin through the hole of the **208-9388** Adapter. This will allow you to install the tool without removing the ECM. Thread the **208-9387** Adapter into the flywheel housing. The timing hole is located approximately 127 to 152 mm (5.0 to 6.0 inch) above the bore for the starter motor in the flywheel housing.
3. Install the **208-0888** Engine Turning Tool into the splines of the hydraulic pump drive gear in flywheel housing (3). Turn the engine in the direction of normal rotation. Rotation of the engine is viewed from the flywheel housing. Rotation of the engine is counterclockwise. The **208-0888** Engine Turning Tool needs to rotate the same direction as the crankshaft. Rotate the engine with the **208-0888** Engine Turning Tool and the **9U-6639** Ratchet Wrench.

**Note:** If the flywheel is turned beyond the point of engagement, the flywheel must be turned in the direction that is reverse of normal engine rotation. Turn the flywheel by approximately 30 degrees. Then turn the flywheel in the direction of normal rotation until the Timing Pin engages with the hole in the flywheel. This procedure removes the play from the gears when the No. 1 piston is at the top center position.

4. Remove the front valve mechanism cover from the engine.
5. The inlet and exhaust valves for the No. 1 cylinder are fully closed if No. 1 piston is on the compression stroke and the rocker arms can be moved by hand. If the rocker arms cannot be moved and the valves are slightly open, the No. 1 piston is on the exhaust stroke.

**Note:** When the actual stroke position is identified, and the other stroke position is needed, remove the Timing Pin from the hole in the flywheel. Then turn the flywheel by 360 degrees in the direction of normal engine rotation and reinstall the Timing Pin into the hole in the flywheel.

**Note:** Never turn the engine by the crankshaft vibration damper. The crankshaft vibration damper is a precision part. Major engine failure may be caused by damage to the crankshaft vibration damper.

## Checking and Calibrating the Electronic Injection Timing With the Electronic Service Tool

Refer to Troubleshooting, "Engine Speed/Timing Sensor - Calibrate" for the proper procedure to calibrate the electronic injection timing.

i01370330

## Fuel Quality - Test

**SMCS Code:** 1280-081

This test checks for problems regarding fuel quality. Refer to Diesel Fuels and Your Engine, SEBD0717 for additional details.

Use the following procedure to test for problems regarding fuel quality:

1. Determine if water and/or contaminants are present in the fuel. Check the water separator (if equipped). If a water separator is not present, proceed to Step 2. Drain the water separator, if necessary. A full fuel tank minimizes the potential for overnight condensation.

**Note:** A water separator can appear to be full of fuel when the water separator is actually full of water.

2. Determine if contaminants are present in the fuel. Remove a sample of fuel from the bottom of the fuel tank. Visually inspect the fuel sample for contaminants. The color of the fuel is not necessarily an indication of fuel quality. However, fuel that is black, brown, and/or similar to sludge can be an indication of the growth of bacteria or oil contamination. In cold temperatures, cloudy fuel indicates that the fuel may not be suitable for operating conditions. The following methods can be used to prevent wax from clogging the fuel filter:

- Fuel heaters
- Blending fuel with additives
- Utilizing fuel with a low cloud point such as kerosene

Refer to Operation and Maintenance Manual, "Fuel Recommendations" for more information.

3. Check fuel API with a **9U-7840** Fluid and Fuel Calibration Gp for low power complaints. The acceptable range of the fuel API is 30 to 45 when the API is measured at 15 °C (60 °F), but there is a significant difference in energy within this range. Refer to Tool Operating Manual, NEHS0607 for API correction factors when a low power problem is present and API is high.

**Note:** A correction factor that is greater than 1.000 may be the cause of low power and/or poor fuel consumption.

4. If fuel quality is still suspected as a possible cause to problems regarding engine performance, disconnect the fuel inlet line, and temporarily operate the engine from a separate source of fuel that is known to be good. This will determine if the problem is caused by fuel quality. If fuel quality is determined to be the problem, drain the fuel system and replace the fuel filters. Engine performance can be affected by the following characteristics :

- Cetane number of the fuel
- Air in the fuel
- Other fuel characteristics

i01909716

## Fuel System - Prime

**SMCS Code:** 1258-548

### **WARNING**

**Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire. To help prevent possible injury, turn the start switch off when changing fuel filters or water separator elements. Clean up fuel spills immediately.**

#### NOTICE

Use a suitable container to catch any fuel that might spill. Clean up any spilled fuel immediately.

#### NOTICE

Do not allow dirt to enter the fuel system. Thoroughly clean the area around a fuel system component that will be disconnected. Fit a suitable cover over disconnected fuel system component.

If the fuel system runs out of fuel or if air is introduced into the fuel system, the system must be primed prior to starting the engine.

Before you begin this procedure, check that the following conditions are satisfied:

- Ensure that the fuel tanks are filled with clean diesel fuel.
- Turn the key switch to the OFF position.

Review the following illustrations in order to locate the components for your application. Locate the air purge plug (1), fuel priming pump (2) (if equipped), and the ball valve assembly (3) (if equipped).

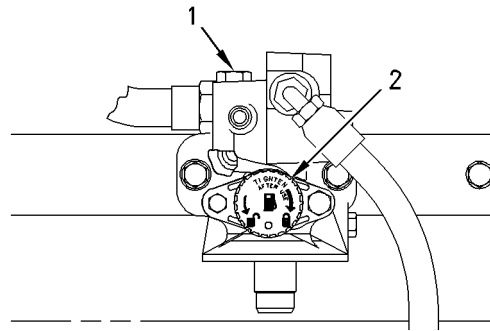


Illustration 45

g00995175

Fuel filter base that is mounted directly to the fuel supply manifold

Typical example

- (1) Air purge plug  
(2) Hand priming pump (if equipped)

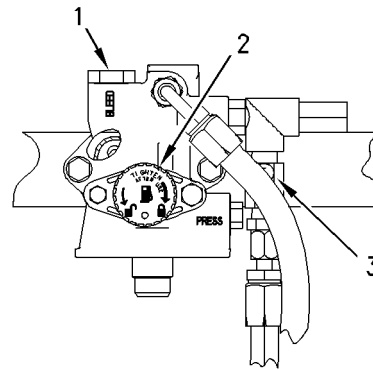


Illustration 46

g00995236

Fuel filter base that is mounted directly to the fuel supply manifold

Typical example

- (1) Air purge plug  
(2) Hand priming pump (if equipped)  
(3) Ball valve assembly (if equipped)

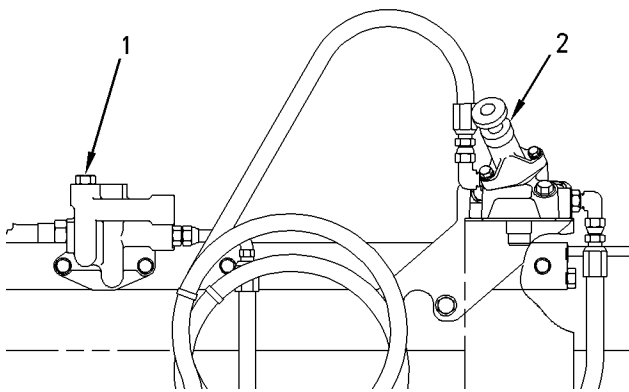


Illustration 47

g00995241

Fuel filter base that is mounted remotely

Typical example

- (1) Air purge plug
- (2) Hand priming pump (if equipped)

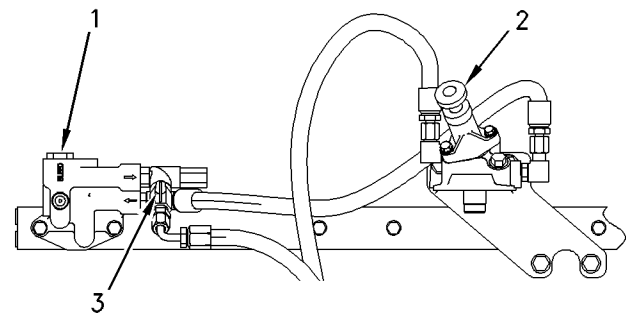


Illustration 48

g00995289

Fuel filter base that is mounted remotely

Typical example

- (1) Air purge plug
- (2) Hand priming pump (if equipped)
- (3) Ball valve assembly (if equipped)

Either of the following procedures may be followed in order to prime the fuel system.

## Hand Priming the Fuel System

1. Open air purge plug (1) for three full turns. Do not remove this plug.

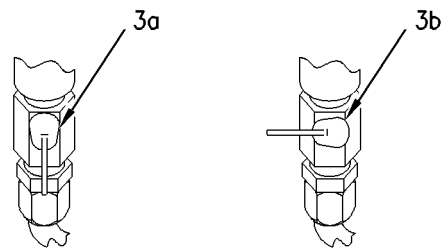


Illustration 49

g00995967

Ball valve assembly

- (3a) The valve is in the open position for normal engine operation.
- (3b) The valve is in the closed position in order to prime the fuel system.

2. If the fuel system is equipped with a ball valve assembly (3), close the valve (3b).
3. Operate the hand priming pump. While you operate the hand priming pump, observe the air purge plug for the presence of fuel.
  - a. When a small drop of fuel appears at the threads of the air purge plug, close and tighten the plug.
- Note:** Failure to tighten all fittings could result in serious fuel leaks.
- b. Clean any residual fuel from the engine components.
4. Continue to operate the fuel priming pump until a strong resistance is felt.
5. Secure the hand priming pump.
6. Attempt to start the engine.

### NOTICE

Do not crank the engine continuously for more than 30 seconds. Allow the starting motor to cool for two minutes before cranking the engine again.

- a. Crank the engine for 30 seconds. Allow the starter motor to cool for 2 minutes after each 30 second interval.
- b. Repeat Step 6.a until the engine starts and runs. If the engine runs rough, continue to operate the engine at low idle until the engine runs smoothly.

- c. If the fuel system is equipped with a ball valve assembly that has been closed, allow the engine to operate for 30 seconds and then open the valve (3a).

**Note:** If the ball valve is left in the closed position for extended periods of time, damage to the fuel injectors may occur.

If the engine will not start, further priming is necessary. If the engine starts but the engine continues to misfire or smoke, further priming is necessary.

## Priming Procedure by Engine Cranking

1. Open air purge plug (1) for three full turns. Do not remove this plug.
2. If the fuel system is equipped with a ball valve assembly (3), close the valve (3b).

**Note:** Use a suitable container to catch the fuel while you crank the engine.

---

### NOTICE

Do not crank the engine continuously for more than 30 seconds. Allow the starting motor to cool for two minutes before cranking the engine again.

---

3. Use engine cranking to prime the fuel system.
    - a. Crank the engine for 30 seconds. Allow the starter motor to cool for 2 minutes after each 30 second interval.
- Note:** After four or five intervals, most of the air should be purged from the system.
- b. While you crank the engine, observe air purge plug (1). When a small drop of fuel appears at the threads of the air purge plug, close and tighten the plug.

**Note:** Failure to tighten all fittings could result in serious fuel leaks.

- c. Clean any residual fuel from the engine components.
  - d. Repeat Step 3.a until the engine starts and runs. If the engine runs rough, continue to operate the engine at low idle until the engine runs smoothly.
4. If the fuel system is equipped with a ball valve assembly that has been closed, allow the engine to operate for 30 seconds and then open the valve (3a).

**Note:** If the ball valve is left in the closed position for extended periods of time, damage to the fuel injectors may occur.

5. Once the engine runs smoothly, stop the engine. Turn the key switch to the OFF position.

If the engine will not start, further priming is necessary. If the engine starts but the engine continues to misfire or smoke, further priming is necessary.

i01911798

## Fuel System Pressure - Test

**SMCS Code:** 1250-081; 1256-081

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### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

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### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

---

## Low Fuel Pressure

Low fuel pressure can cause low power. Low fuel pressure can also cause cavitation of the fuel which can damage the fuel injectors.

The following conditions can cause low fuel pressure:

- Plugged fuel filters
- Debris in the check valves for the fuel priming pump
- Sticking or worn fuel pressure relief valve in the fuel transfer pump
- Severe wear on return fuel pressure regulating valve in the fuel filter base or adapter assembly
- Worn gears in the fuel transfer pump

- Pinched fuel lines or undersized fuel lines
- Old fuel lines that have a reduced interior diameter that was caused by swelling
- Fuel lines with deteriorating interior surfaces
- Pinched fuel line fittings or undersized fuel line fittings
- Debris in the fuel tank, fuel lines, or fuel system components that may create restrictions

## High Fuel Pressure

Excessive fuel pressure can cause fuel filter gaskets to rupture. The following conditions can cause high fuel pressure:

- Plugged orifices in the fuel pressure regulating valve
- Stuck fuel pressure relief valve in the fuel transfer pump
- Pinched fuel return line

## Checking Fuel Pressure

Table 4

| Required Toolings |                           |                                                           |          |
|-------------------|---------------------------|-----------------------------------------------------------|----------|
| Tool              | Part Number               | Part Name                                                 | Quantity |
| A                 | 1U-5470<br>or<br>198-4240 | Engine Pressure Group<br>or<br>Digital Pressure Indicator | 1        |

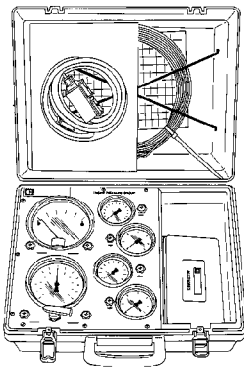


Illustration 50  
1U-5470 Engine Pressure Group

Reference: Special Instruction, SEHS8907, "Using the 1U-5470 Engine Pressure Group"

Reference: Operation Manual, NEHS0818, "Using the 198-4240 Digital Pressure Indicator"

## Fuel System Identification

Refer to the following illustrations in order to identify the fuel system for your application. Locate the pressure testing ports in order to measure the different fuel system pressures.

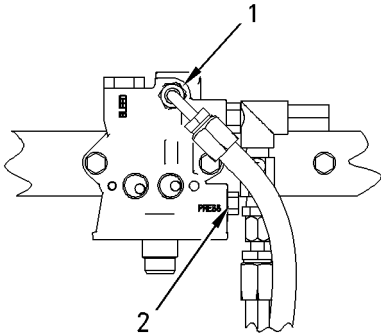


Illustration 51

g00996187

Fuel filter base that is mounted directly to the fuel manifold  
Typical example

- (1) Test location for unfiltered fuel pressure  
(2) Test location for filtered fuel pressure

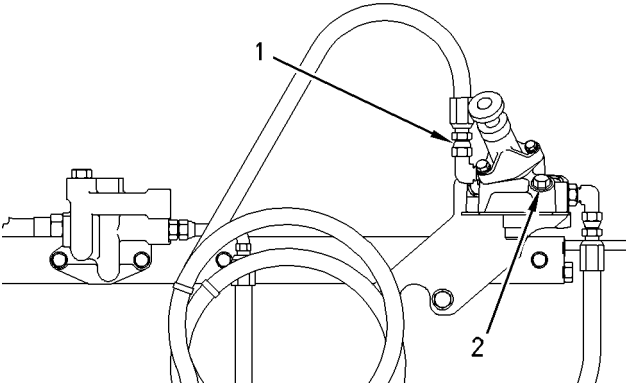


Illustration 52

g00996311

Applications with a fuel adapter  
Typical example

- (1) Test location for unfiltered fuel pressure  
(2) Test location for filtered fuel pressure

## Applications with a Quick Connect Coupler

Some applications may be equipped with a quick connect coupler. Refer to these illustrations for the location and the purpose of the fuel pressure tap.

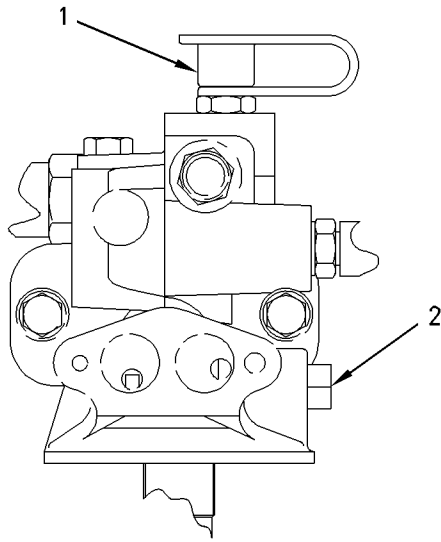


Illustration 53  
g00996062  
Fuel filter base that is mounted directly to the fuel manifold  
Typical example  
(1) Test location for unfiltered fuel pressure  
(2) Test location for filtered fuel pressure

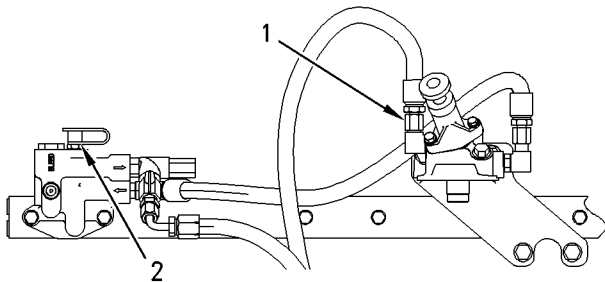


Illustration 54  
g00996149  
Applications with a fuel adapter  
Typical example  
(1) Test location for unfiltered fuel pressure  
(2) Test location for filtered fuel pressure

## Measuring Unfiltered Fuel Pressure

### **⚠ WARNING**

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire. Clean up fuel spills immediately.

### **⚠ WARNING**

A high pressure fuel line must be disconnected. To avoid personal injury or fire from fuel spray, the engine must be stopped before the fuel line is disconnected.

Use the following procedure to measure the unfiltered fuel pressure:

1. Refer to "Fuel System Identification" in order to identify the correct location for measuring the unfiltered fuel pressure. Install Tooling (A) into pressure test location (1) in order to measure the unfiltered fuel pressure.

**Note:** A fuel sensor may be installed in one of the ports that are indicated in the illustrations. If a fuel sensor is installed in the port, install a tee fitting into the port. Install the sensor and Tooling (A) onto this tee prior to operating the engine. Ensure that the tee and all fuel fittings are securely tightened. Failure to tighten all fittings could result in serious fuel leaks.

Clean any residual fuel from the engine components.

2. Start the engine.
3. Record the pressure reading that is on Tooling (A) for engine speeds of 1800 rpm and 2100 rpm. The test should be performed with no load on the engine.
4. Stop the engine and remove Tooling (A) from the fuel system.

## Measuring Filtered Fuel Pressure

### **⚠ WARNING**

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire. Clean up fuel spills immediately.

### **⚠ WARNING**

A high pressure fuel line must be disconnected. To avoid personal injury or fire from fuel spray, the engine must be stopped before the fuel line is disconnected.

1. Refer to "Fuel System Identification" in order to identify the correct location for measuring the filtered fuel pressure. Install Tooling (A) into pressure test location (2) in order to measure the filtered fuel pressure.

**Note:** A fuel sensor may be installed in one of the ports that are indicated in the illustrations. If a fuel sensor is installed in the port, install a tee fitting into the port. Install the sensor and Tooling (A) onto this tee prior to operating the engine. Ensure that the tee and all fuel fittings are securely tightened. Failure to tighten all fittings could result in serious fuel leaks.

Clean any residual fuel from the engine components.

2. Start the engine.
3. Record the pressure reading that is on Tooling (A) for engine speeds of 600 rpm and 1800 rpm. The test should be performed with no load on the engine.

**Note:** Excessive needle movement at the gauge may be present. Fuel pressure readings near the fuel supply manifold will be affected by pressure spikes. The pressure spikes are caused by excess fuel that is returning to the fuel system from the injectors. If the gauge is connected with a suitable length of hose, the air in the hose will absorb the spikes. This will give you an average reading and a steady needle. Ensure that the gauge is elevated above the pressure measuring port during the test.

4. Stop the engine and remove Tooling (A) from the fuel system.

## Fuel Filter Differential Pressure

Calculate the fuel filter's differential pressure by subtracting the filtered fuel pressure from the unfiltered fuel pressure. For best results, use the pressures that have been measured at an engine speed of 1800 rpm.

Typically, the differential pressure for a new fuel filter will not exceed the 35 kPa (5.0 psi).

As abrasive particles collect in the fuel filter, the pressure differential across the filter will increase. When a filter becomes plugged, the pressure differential may increase as much as 69 kPa (10.0 psi) before a significant power loss is detected by the operator. Low filtered fuel pressure will cause cavitation of the fuel and internal damage to the unit injectors. **The pressure differential across the fuel filter should not exceed 69 kPa (10.0 psi).**

If a high differential pressure exists, replace the fuel filter. Refer to Testing and Adjusting, "Fuel System - Prime" for information on priming the fuel system after you replace the fuel filter.

## Fuel Pressure Relief Valve

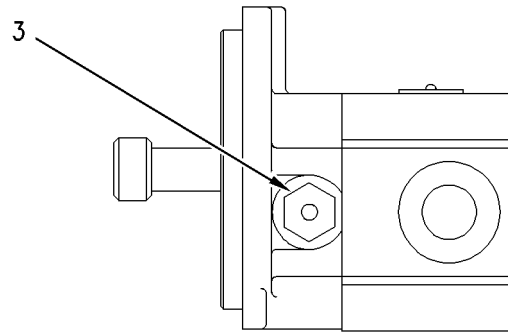


Illustration 55

g00996532

Fuel transfer pump

Typical example

(3) Fuel pressure relief valve

The fuel pressure relief valve (3) is an internal component that is located in the fuel transfer pump. The relief valve is used to regulate the maximum pressure for the fuel system. Refer to Illustration 55 in order to locate the pressure relief valve.

The maximum unfiltered fuel pressure at an engine speed of 2100 rpm ..... 715 kPa to 785 kPa  
(103.7 psi to 113.9 psi)

The fuel pressure relief valve is not a serviceable component. If the operation of the fuel pressure relief valve is suspect, replace the fuel transfer pump.

## Fuel Pressure Regulating Valve

The pressure regulator valve is used in order to maintain an optimum operating pressure within the low pressure fuel system. If the filtered fuel pressure is low, the pressure regulator valve may be worn or stuck in the open position.

There are two types of fuel pressure regulating valves that are available for this engine. The integral pressure regulator valve has been replaced with the new pressure regulator assembly that is self-contained.

## Integral Pressure Regulator Valve

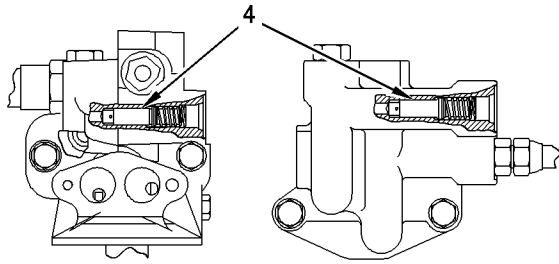


Illustration 56

g00996544

Location of the fuel pressure regulator valve

Typical example

(4) Fuel pressure regulator valve

The integral pressure regulator valve (4) is an internal component that is located in the fuel filter base or in the fuel adapter. The pressure regulator valve is used in order to maintain an optimum operating pressure within the low pressure fuel system. Refer to Illustration 56 in order to locate the pressure regulator valve.

The filtered fuel pressure at an engine speed of 600 rpm ..... 450 kPa to 600 kPa (65.3 psi to 87.0 psi)

If the filtered fuel pressure is not within specifications, remove the fuel pressure regulator valve and the valve spring. Inspect the components for wear or damage. If there is visible signs of wear or damage to the valve, replace the valve. Check the spring for distortion. Refer to Specifications, "Fuel Lines" for information that relates to the specifications for the spring. If necessary, replace the spring. Retest the filtered fuel pressure after the replacement of any components.

## Pressure Regulator Assembly that is Self-Contained

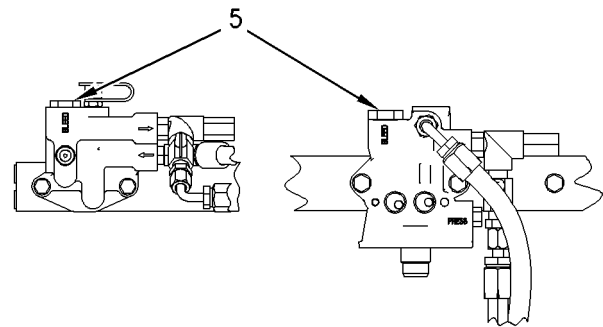


Illustration 57

g00996664

Location of the pressure regulator assembly

Typical example

(5) Pressure regulator assembly

This pressure regulator assembly (5) is a serviceable component. The pressure regulator valve is located in the fuel filter base or in the fuel adapter. This regulator valve is used in order to maintain an optimum operating pressure within the low pressure fuel system. Refer to Illustration 57 in order to locate the pressure regulator assembly that is self-contained.

The filtered fuel pressure at an engine speed of 600 rpm ..... 450 kPa to 600 kPa (65.3 psi to 87.0 psi)

This type of fuel pressure regulator valve is a serviceable component. If the operation of the fuel pressure regulator valve is suspect, replace the pressure regulator assembly.

i01779472

## Gear Group (Front) - Time

**SMCS Code:** 1206-531

**Note:** The electronic injection timing must be calibrated after reassembly of the front gear train. Refer to Troubleshooting, “ Engine Speed/Timing Sensor - Calibrate”.

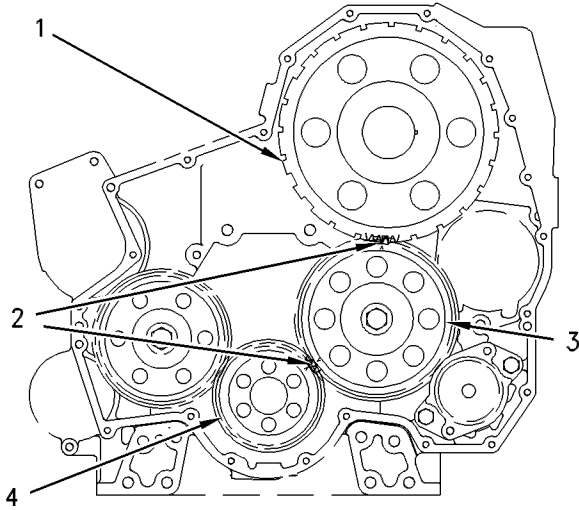


Illustration 58

g00430184

### Front Gear Group

- (1) Camshaft gear and timing reference ring
- (2) Timing marks
- (3) Idler gear
- (4) Crankshaft gear

The basis for correct fuel injection timing and valve mechanism operation is determined by the timing reference ring and the alignment of the front gear group. The timing reference ring is located on the end of the camshaft. The timing reference ring is used to measure crankshaft rotation. During installation of the front gear, timing marks (2) on idler gear (3) must be in alignment with the timing marks on crankshaft gear (4) and the timing marks on camshaft gear (1).

Refer to Disassembly and Assembly, “ Gear Group (Front) - Remove” and Disassembly and Assembly, “ Gear Group (Front) - Install”.

**Note:** If timing reference ring (1) is installed backward the engine will not start.

Check for proper alignment of the camshaft gear and timing reference ring (1) on the camshaft assembly. Inspect the key between the timing reference ring and the camshaft gear. Check the teeth on the timing ring. The teeth should not be defaced. The teeth should have sharp clean edges and the teeth should be free of contaminants.

# Air Inlet and Exhaust System

i01778359

## Air Inlet and Exhaust System - Inspect

SMCS Code: 1050-040

### Air Inlet Restriction

There will be a reduction in the performance of the engine if there is a restriction in the air inlet system or the exhaust system.

Table 5

| Required Tools            |                                                           |          |
|---------------------------|-----------------------------------------------------------|----------|
| Part Number               | Part Name                                                 | Quantity |
| 1U-5470<br>or<br>198-4240 | Engine Pressure Group<br>or<br>Digital Pressure Indicator | 1        |

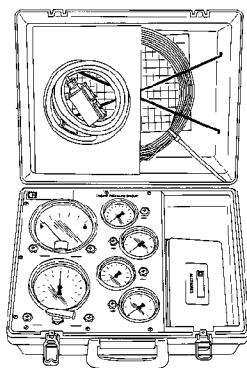


Illustration 59

g00293196

1U-5470 Engine Pressure Group

Refer to Special Instruction, SEHS8907, "Using the 1U-5470 Engine Pressure Group" for the instructions that are needed to use the 1U-5470 Engine Pressure Group. Refer to Operation Manual, NEHS0818, "Using the 198-4240 Pressure Indicator Tool Gp" for the instructions that are needed to use the 198-4240 Pressure Indicator Tool Gp.

1. Inspect the engine air cleaner inlet and ducting in order to ensure that the passageway is not blocked or collapsed.
2. Inspect the engine air cleaner element. Replace a dirty engine air cleaner element with a clean engine air cleaner element.

3. Check for dirt tracks on the clean side of the engine air cleaner element. If dirt tracks are observed, contaminants are flowing past the engine air cleaner element and/or the seal for the engine air cleaner element.

### WARNING

Hot engine components can cause injury from burns. Before performing maintenance on the engine, allow the engine and the components to cool.

### WARNING

Making contact with a running engine can cause burns from hot parts and can cause injury from rotating parts.

When working on an engine that is running, avoid contact with hot parts and rotating parts.

4. Use the differential pressure gauge of the 1U-5470 Engine Pressure Group.

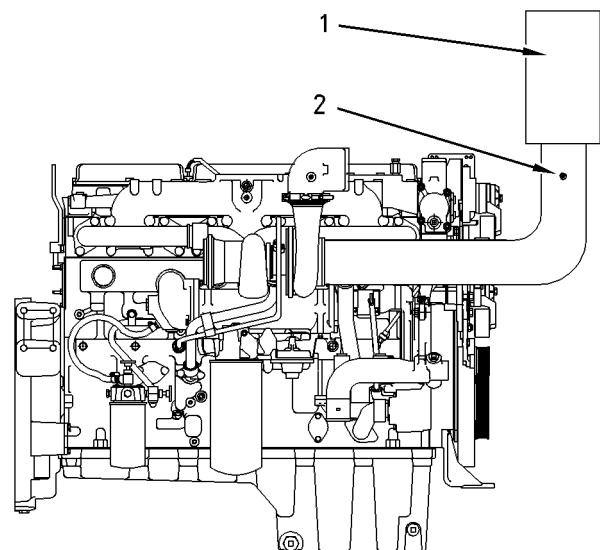


Illustration 60

g00580368

- (1) Air cleaner  
(2) Test location

- a. Connect the vacuum port of the differential pressure gauge to test location (2). Test location (2) can be located anywhere along the air inlet piping after the engine air cleaner but before the turbocharger.
- b. Leave the pressure port of the differential pressure gauge open to the atmosphere.
- c. Start the engine. Run the engine in the no-load condition at high idle.

- d. Record the value.
- e. Multiply the value from Step 4.d by 1.8.
- f. Compare the result from Step 4.e to the appropriate values that follow.

The air flow through a used engine air cleaner may have a restriction. The air flow through a plugged engine air cleaner will be restricted to some magnitude. In either case, the restriction must not be more than the following amount:

Maximum restriction ..... 6.2 kPa (25 in of H<sub>2</sub>O)

The air flow through a new engine air cleaner element must not have a restriction of more than the following amount:

Maximum restriction ..... 3.7 kPa (15 in of H<sub>2</sub>O)

### Exhaust Restriction

Back pressure is the difference in the pressure between the exhaust at the outlet elbow and the atmospheric air.

Table 6

| Required Tools            |                                                           |          |
|---------------------------|-----------------------------------------------------------|----------|
| Part Number               | Part Name                                                 | Quantity |
| 1U-5470<br>or<br>198-4240 | Engine Pressure Group<br>or<br>Digital Pressure Indicator | 1        |

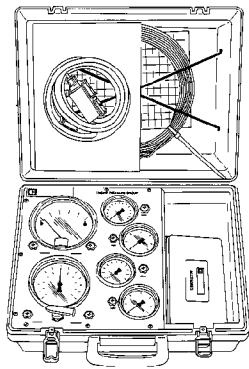


Illustration 61  
1U-5470 Engine Pressure Group

Refer to Special Instruction, SEHS8907, "Using the 1U-5470 Engine Pressure Group" for the instructions that are needed to use the 1U-5470 Engine Pressure Group. Refer to Operation Manual, NEHS0818, "Using the 198-4240 Pressure Indicator Tool Gp" for the instructions that are needed to use the 198-4240 Pressure Indicator Tool Gp.

### WARNING

Hot engine components can cause injury from burns. Before performing maintenance on the engine, allow the engine and the components to cool.

### WARNING

Making contact with a running engine can cause burns from hot parts and can cause injury from rotating parts.

When working on an engine that is running, avoid contact with hot parts and rotating parts.

Use the differential pressure gauge of the 1U-5470 Engine Pressure Group in order to measure back pressure from the exhaust. Use the following procedure in order to measure back pressure from the exhaust:

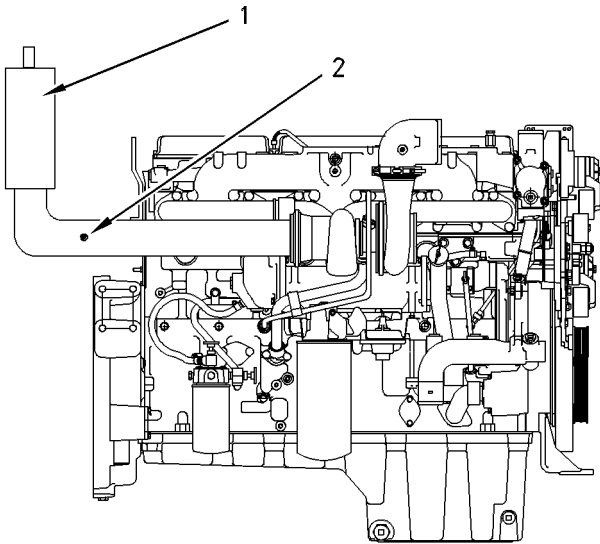


Illustration 62

g00583322

- (1) Muffler
- (2) Test location

1. Connect the pressure port of the differential pressure gauge to test location (2). Test location (2) can be located anywhere along the exhaust piping after the turbocharger but before the muffler.
2. Leave the vacuum port of the differential pressure gauge open to the atmosphere.
3. Start the engine. Run the engine in the no-load condition at high idle.
4. Record the value.

5. Multiply the value from Step 4 by 1.8.
6. Compare the result from Step 5 to the value that follows.

Back pressure from the exhaust must not be more than the following amount:

Maximum back pressure ..... 10.0 kPa  
(40 inch of H<sub>2</sub>O)

## Catalytic Converter/Muffler

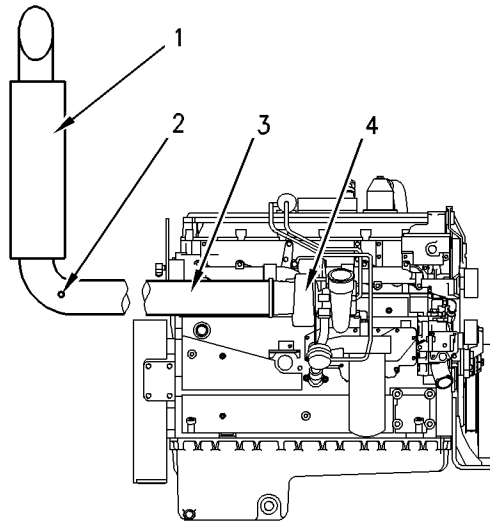


Illustration 63 g00678914

Typical example

- (1) Catalytic converter/muffler
- (2) Test location
- (3) Exhaust piping
- (4) Turbocharger

1. Connect the pressure port of the differential pressure gauge to test location. The test location should be located on a straight length of exhaust pipe. This back pressure tap should be located as close as possible to the turbocharger or the exhaust manifold.
2. Leave the vacuum port of the differential pressure gauge open to the atmosphere.
3. Start the engine. Run the engine in the no-load condition at high idle until the engine reaches normal operating temperature.
4. Record the value.
5. Multiply the value from Step 4 by 1.8.
6. Compare the result from Step 5 to the value that follows.

Back pressure from the exhaust must not be more than the following amount:

Maximum back pressure for engines with a  
Catalytic converter/muffler ..... 14.9 kPa  
(60 inch of H<sub>2</sub>O)

If the maximum back pressure is within the allowable limits, refer to Troubleshooting, "Low Power/Poor or No Response to Throttle".

If the back pressure exceeds 14.9 kPa (60 inch of H<sub>2</sub>O), replace the catalytic converter/muffler.

i01896277

## Turbocharger - Inspect

SMCS Code: 1052-040

### WARNING

Hot engine components can cause injury from burns. Before performing maintenance on the engine, allow the engine and the components to cool.

### WARNING

Personal injury can result from rotating and moving parts.

Stay clear of all rotating and moving parts.

Never attempt adjustments while the machine is moving or the engine is running unless otherwise specified.

The machine must be parked on a level surface and the engine stopped.

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

---

Before you begin inspection of the turbocharger, be sure that the inlet air restriction is within the specifications for your engine. Be sure that the exhaust system restriction is within the specifications for your engine. Refer to Systems Operation/Testing and Adjusting, "Air Inlet and Exhaust System - Inspect".

The condition of the turbocharger will have definite effects on engine performance. Use the following inspections and procedures to determine the condition of the turbocharger.

- Inspection of the Compressor and the Compressor Housing
- Inspection of the Turbine Wheel and the Turbine Housing
- Inspection of the Wastegate

## Inspection of the Compressor and the Compressor Housing

Remove air piping from the compressor inlet.

1. Inspect the compressor wheel for damage from a foreign object. If there is damage, determine the source of the foreign object. As required, clean the inlet system and repair the intake system. Replace the turbocharger. If there is no damage, go to Step 3.
2. Clean the compressor wheel and clean the compressor housing if you find buildup of foreign material. If there is no buildup of foreign material, go to Step 3.

3. Turn the rotating assembly by hand. While you turn the assembly, push the assembly sideways. The assembly should turn freely. The compressor wheel should not rub the compressor housing. Replace the turbocharger if the compressor wheel rubs the compressor wheel housing. If there is no rubbing or scraping, go to Step 4.
4. Inspect the compressor and the compressor wheel housing for oil leakage. An oil leak from the compressor may deposit oil in the aftercooler. Drain and clean the aftercooler if you find oil in the aftercooler.
  - a. Check the oil level in the crankcase. If the oil level is too high, adjust the oil level.
  - b. Inspect the air cleaner element for restriction. If restriction is found, correct the problem.
  - c. Inspect the engine crankcase breather. Clean the engine crankcase breather or replace the engine crankcase breather if the engine crankcase breather is plugged.
  - d. Remove the turbocharger oil drain line. Inspect the drain opening. Inspect the oil drain line. Inspect the area between the bearings of the rotating assembly shaft. Look for oil sludge. Inspect the oil drain hole for oil sludge. Inspect the oil drain line for oil sludge in the drain line. If necessary, clean the rotating assembly shaft. If necessary, clean the oil drain hole. If necessary, clean the oil drain line.
  - e. If Steps 4.a through 4.d did not reveal the source of the oil leakage, the turbocharger has internal damage. Replace the turbocharger.

## Inspection of the Turbine Wheel and the Turbine Housing

Remove the air piping from the turbine housing.

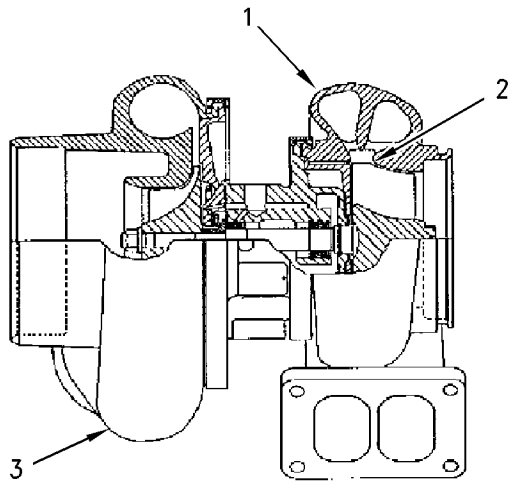


Illustration 64

g00763164

- (1) Turbine Housing
- (2) Turbine Wheel
- (3) Turbocharger

1. Inspect the turbine for damage by a foreign object. If there is damage, determine the source of the foreign object. Replace turbocharger (3). If there is no damage, go to Step 2.
2. Inspect turbine wheel (2) for buildup of carbon and other foreign material. Inspect turbine housing (1) for buildup of carbon and foreign material. Clean turbine wheel (2) and clean turbine housing (1) if you find buildup of carbon or foreign material. If there is no buildup of carbon or foreign material, go to Step 3.
3. Turn the rotating assembly by hand. While you turn the assembly, push the assembly sideways. The assembly should turn freely. Turbine wheel (2) should not rub turbine wheel housing (1). Replace turbocharger (3) if turbine wheel (2) rubs turbine housing (1). If there is no rubbing or scraping, go to Step 4.
4. Inspect the turbine and turbine housing (1) for oil leakage. Inspect the turbine and turbine housing (1) for oil coking. Some oil coking may be cleaned. Heavy oil coking may require replacement of the turbocharger. If the oil is coming from the turbocharger center housing go to Step 4.a. Otherwise go to "Inspection of the Wastegate".

- a. Remove the turbocharger oil drain line. Inspect the drain opening. Inspect the area between the bearings of the rotating assembly shaft. Look for oil sludge. Inspect the oil drain hole for oil sludge. Inspect the oil drain line for oil sludge. If necessary, clean the rotating assembly shaft. If necessary, clean the drain opening. If necessary, clean the drain line.
- b. If crankcase pressure is high, or if the oil drain is restricted, pressure in the center housing may be greater than the pressure of turbine housing (1). Oil flow may be forced in the wrong direction and the oil may not drain. Check the crankcase pressure and correct any problems.
- c. If the oil drain line is damaged, replace the oil drain line.
- d. Check the routing of the oil drain line. Eliminate any sharp restrictive bends. Make sure that the oil drain line is not too close to the engine exhaust manifold.
- e. If Steps 4.a through 4.d did not reveal the source of the oil leakage, turbocharger (3) has internal damage. Replace turbocharger (3).

## Inspection of the Wastegate

The wastegate controls the amount of exhaust gas that is allowed to bypass the turbine side of the turbocharger. This valve then controls the rpm of the turbocharger.

When the engine operates in conditions of low boost (lug), a spring presses against a diaphragm in the canister. The actuating rod will move and the wastegate will close. Then, the turbocharger can operate at maximum performance.

When the boost pressure increases against the diaphragm in the canister, the wastegate will open. The rpm of the turbocharger becomes limited. The rpm limitation occurs because a portion of the exhaust gases bypass the turbine wheel of the turbocharger.

The following levels of boost pressure indicate a possible problem with the following components: Wastegate, Wastegate solenoid, and Air lines

- too high at full load conditions
- too low at all lug conditions

**Note:** The housing assembly for the wastegate is preset at the factory and no adjustments can be made.

**NOTICE**

If the high idle rpm or the engine rating is higher than given in the Technical Marketing Information (TMI) for the height above sea level at which the engine is operated, there can be damage to engine or to turbocharger parts. Damage will result when increased heat and/or friction due to the higher engine output goes beyond the engine cooling and lubrication system's abilities.

The boost pressure controls the maximum rpm of the turbocharger, because the boost pressure controls the position of the wastegate. The following factors also affect the maximum rpm of the turbocharger:

- the engine rating
- the horsepower demand on the engine
- the high idle rpm
- the height above sea level for engine operation
- inlet air restriction
- exhaust system restriction

i01778375

# **Inlet Manifold Pressure - Test**

**SMCS Code:** 1058-081

The efficiency of an engine can be checked by making a comparison of the pressure in the inlet manifold with the information given in the TMI (Technical Marketing Information). The information is also listed on the Fuel Setting and the Service Information System. This test is used when there is a decrease of horsepower from the engine, yet there is no real sign of a problem with the engine.

The correct pressure for the inlet manifold is listed in the TMI (Technical Marketing Information). The correct pressure is also on the Fuel Setting And the related Service Information System. Development of this information is performed under the following conditions:

- 99 kPa (29.7 in Hg) dry barometric pressure
- 29 °C (85 °F) outside air temperature
- 35 API rated fuel

On a turbocharged, aftercooled engine, a change in the fuel rating will change the horsepower. A change in the fuel rating will change the inlet manifold pressure. If the fuel is rated above 35 API, the inlet manifold pressure can be less than the pressure given in the TMI (Technical Marketing Information). The pressure will also be less than the pressure that is listed on the Fuel Setting And the Related Service Information System. If the fuel is rated below 35 API, the inlet manifold pressure can be more than the pressure listed in the TMI (Technical Marketing Information). The pressure will also be more than the pressure that is listed on the Fuel Setting And the Related Service Information System.

**Note:** Ensure that the air inlet and the exhaust are not restricted when you are checking the inlet manifold pressure.

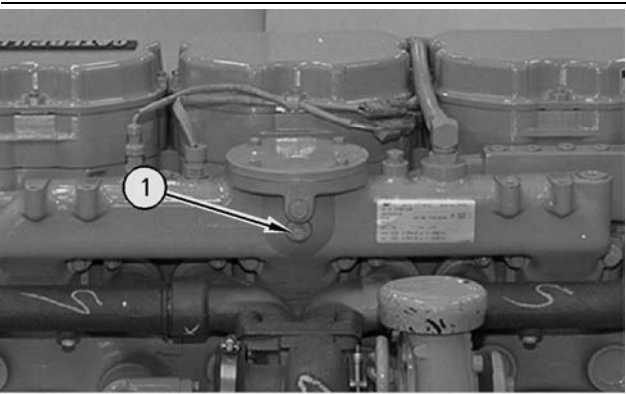


Illustration 65

g00434159

Pressure test location on inlet manifold

(1) Plug

Table 7

| Required Tools            |                                                           |          |
|---------------------------|-----------------------------------------------------------|----------|
| Part Number               | Part Name                                                 | Quantity |
| 1U-5470<br>or<br>198-4240 | Engine Pressure Group<br>or<br>Digital Pressure Indicator | 1        |

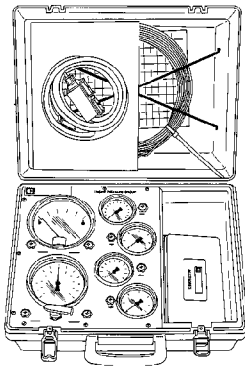


Illustration 66

g00293196

**1U-5470 Engine Pressure Group**

Refer to Special Instruction, SEHS8907, "Using the **1U-5470** Engine Pressure Group" for the instructions that are needed to use the **1U-5470** Engine Pressure Group. Refer to Operation Manual, NEHS0818, "Using the **198-4240** Pressure Indicator Tool Gp" for the instructions that are needed to use the **198-4240** Pressure Indicator Tool Gp.

Use the following procedure in order to measure the inlet manifold pressure:

1. Remove plug (1) from the inlet manifold.
2. Connect the **1U-5470** Engine Pressure Group to the inlet manifold at the pressure test location.
3. Record the value.
4. Compare the value that was recorded in Step 3 to the pressure that is given in the TMI (Technical Marketing Information). The correct pressure is also given in the Fuel Setting And Related Information Microfiche.

i01582969

## Exhaust Temperature - Test

**SMCS Code:** 1088-081

Table 8

| Required Tools |                      |     |
|----------------|----------------------|-----|
| Part Number    | Part Name            | Qty |
| 164-3310       | Infrared Thermometer | 1   |

When the engine runs at low idle, the temperature of an exhaust manifold port can indicate the condition of a fuel injection nozzle.

A low temperature indicates that no fuel is flowing to the cylinder. An inoperative fuel injection nozzle or a problem with the fuel injection pump could cause this low temperature.

A very high temperature can indicate that too much fuel is flowing to the cylinder. A malfunctioning fuel injection nozzle could cause this very high temperature.

Use the **164-3310** Infrared Thermometer to check exhaust temperature. The Operator's Manual, NEHS0630 contains the complete operating and maintenance instructions for the **164-3310** Infrared Thermometer.

i01099953

## Aftercooler - Test

**SMCS Code:** 1063-081

Table 9

| Tools Needed |                                |          |
|--------------|--------------------------------|----------|
| Part Number  | Part Name                      | Quantity |
| 1U-5470      | Engine Pressure Group          | 1        |
| FT-1984      | Aftercooler Testing Group      | 1        |
| FT-1438      | Aftercooler (Dynamometer Test) | 1        |

## Visual Inspection

Inspect the following parts at each oil change:

- Air lines
- Hoses
- Gasket joints

### WARNING

**Pressurized air can cause personal injury. When pressurized air is used for cleaning, wear a protective face shield, protective clothing, and protective shoes.**

Ensure that the constant torque hose clamps are tightened to the correct torque. Check the truck manufacturer's specifications for the correct torque. Check the welded joints for cracks. Ensure that the brackets are tightened in the correct positions. Ensure that the brackets are in good condition. Use compressed air to clean any debris or any dust from the aftercooler core assembly. Inspect the cooler core fins for the following conditions:

- Damage
- Debris
- Corrosion

Use a stainless steel brush to remove any corrosion. Ensure that you use soap and water.

**Note:** When parts of the air-to-air aftercooler system are repaired, a leak test is recommended. When parts of the air-to-air aftercooler system are replaced, a leak test is recommended.

The use of winter fronts or shutters is discouraged with air-to-air aftercooled systems. Winter fronts can only be used on certain truck models. On these trucks, tests have shown that the engine jacket water will overheat before the inlet manifold air temperature is excessive. These trucks use sensors and indicators that are installed in order to indicate engine operating conditions before excessive inlet manifold air temperatures are reached. Check with the truck manufacturer about the use of both winter fronts and shutters.

### Inlet Manifold Pressure

Normal inlet manifold pressure with high exhaust temperature can be caused by blockage of the fins of the aftercooler core. Clean the fins of the aftercooler core. Refer to "Visual Inspection" for the cleaning procedure.

Low inlet manifold pressure and high exhaust manifold temperature can be caused by any of the following conditions:

**Plugged air cleaner** – Clean the air cleaner or replace the air cleaner, as required. Refer to the Operation and Maintenance Manual, "Engine Air Cleaner Element - Clean/Replace".

**Blockage in the air lines** – Blockage in the air lines between the air cleaner and the turbocharger must be removed.

**Aftercooler core leakage** – Aftercooler core leakage should be pressure tested. Refer to "Aftercooler Core Leakage" topic for the testing procedure.

**Leakage of the induction system** – Any leakage from the pressure side of the induction system should be repaired.

**Inlet manifold leak** – An inlet manifold leak can be caused by the following conditions: loose fittings and plugs, missing fittings and plugs, damaged fittings and plugs, and leaking inlet manifold gasket.

## Aftercooler Core Leakage

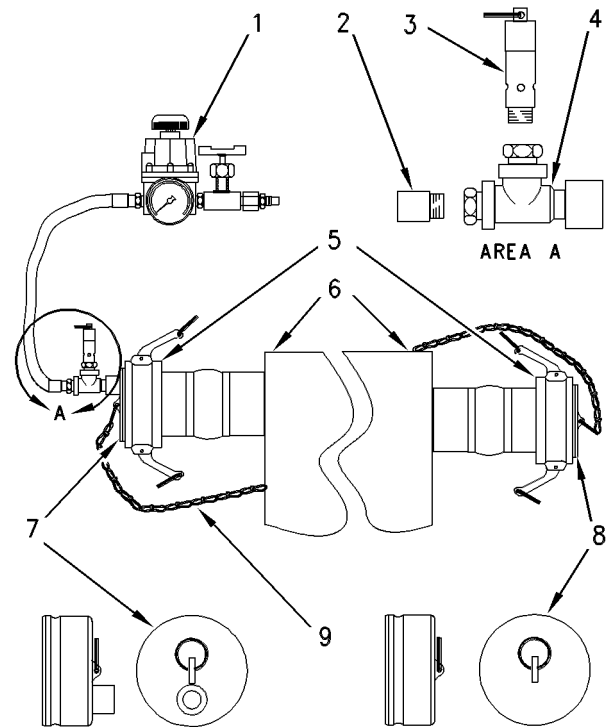


Illustration 67

g00295702

### FT - 1984 Aftercooler Testing Group

- (1) Regulator and valve assembly
- (2) Nipple
- (3) Relief valve
- (4) Tee
- (5) Coupler
- (6) Aftercooler
- (7) Dust plug
- (8) Dust plug
- (9) Chain

A low power problem in the engine can be the result of aftercooler leakage. Aftercooler system leakage can result in the following problems:

- Low power
- Low boost pressure
- Black smoke
- High exhaust temperature

### NOTICE

Remove all air leaks from the system to prevent engine damage. In some operating conditions, the engine can pull a manifold vacuum for short periods of time. A leak in the aftercooler or air lines can let dirt and other foreign material into the engine and cause rapid wear and/or damage to engine parts.

A large leak of the aftercooler core can often be found by making a visual inspection. To check for smaller leaks, use the following procedure:

1. Disconnect the air pipes from the inlet and outlet side of the aftercooler core.

### **WARNING**

**Dust plug chains must be installed to the aftercooler core or to the radiator brackets to prevent possible injury while you are testing. Do not stand in front of the dust plugs while you are testing.**

2. Install couplers (5) on each side of the aftercooler core. Also, install dust plugs (7) and (8). These items are included with the FT-1984 Aftercooler Testing Group.

**Note:** Installation of additional hose clamps on the hump hoses is recommended in order to prevent the hoses from bulging while the aftercooler core is being pressurized.

### **NOTICE**

**Do not use more than 240 kPa (35 psi) of air pressure or damage to the aftercooler core can be the result.**

3. Install the regulator and valve assembly (1) on the outlet side of the aftercooler core assembly. Also, attach the air supply.
4. Open the air valve and pressurize the aftercooler to 205 kPa (30 psi). Shut off the air supply.
5. Inspect all connection points for air leakage.
6. The aftercooler system's pressure should not drop more than 35 kPa (5 psi) in 15 seconds.
7. If the pressure drop is more than the specified amount, use a solution of soap and water to check all areas for leakage. Look for air bubbles that will identify possible leaks. Replace the aftercooler core, or repair the aftercooler core, as needed.

### **WARNING**

**To help prevent personal injury when the tooling is removed, relieve all pressure in the system slowly by using an air regulator and a valve assembly.**

8. After the testing, remove the FT-1984 Aftercooler Testing Group. Reconnect the air pipes on both sides of the aftercooler core assembly.

## Air System Restriction

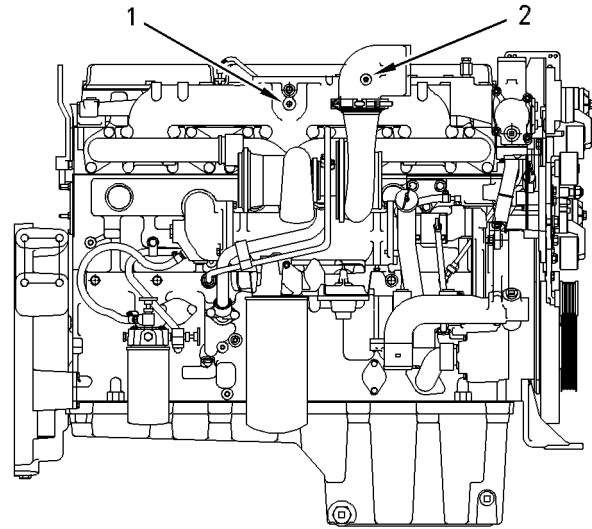


Illustration 68

g00581364

Pressure measurements should be taken at the inlet manifold (1) and at the turbocharger outlet (2).

Use the differential pressure gauge of the 1U-5470 Engine Pressure Group. Use the following procedure in order to measure the restriction of the aftercooler:

1. Connect the vacuum port of the differential pressure gauge to port (1).
2. Connect the pressure port of the differential pressure gauge to port (2).
3. Record the value.

The air lines and the cooler core must be inspected for internal restriction when both of the following conditions are met:

- Air flow is at a maximum level.
- Total air pressure drop of the charged system exceeds 13.5 kPa (4 in Hg).

If a restriction is discovered, proceed with the following tasks, as required:

- Clean
- Repair
- Replacement

# Turbocharger Failure



Personal injury can result from air pressure.

Personal injury can result without following proper procedure. When using pressure air, wear a protective face shield and protective clothing.

Maximum air pressure at the nozzle must be less than 205 kPa (30 psi) for cleaning purposes.

If a turbocharger failure occurs, remove the air-to-air aftercooler core. Internally flush the air-to-air aftercooler core with a solvent that removes oil and other foreign substances. Shake the air-to-air aftercooler core in order to eliminate any trapped debris. Wash the aftercooler with hot, soapy water. Thoroughly rinse the aftercooler with clean water and blow dry the aftercooler with compressed air. Blow dry the assembly in the reverse direction of normal air flow. To make sure that the whole system is clean, carefully inspect the system.

## NOTICE

Do not use caustic cleaners to clean the air-to-air aftercooler core.

Caustic cleaners will attack the internal metals of the core and cause leakage.

# Dynamometer Test

In hot ambient temperatures, chassis dynamometer tests for models with an air-to-air aftercooler can add a greater heat load to the jacket water cooling system. Therefore, the jacket water cooling system's temperature must be monitored. The following measurements may also need a power correction factor:

- Inlet air temperature
- Fuel API rating
- Fuel temperature
- Barometric pressure

With dynamometer tests for engines, use the FT-1438 Aftercooler (Dynamometer Test). This tool provides a water cooled aftercooler in order to control the inlet air temperature to 43 °C (110 °F).

i01096015

# Engine Crankcase Pressure (Blowby) - Test

SMCS Code: 1215; 1317

Table 10

| Tools Needed |                           |          |
|--------------|---------------------------|----------|
| Part Number  | Part Name                 | Quantity |
| 8T-2700      | Blowby/Air Flow Indicator | 1        |

Damaged pistons or rings can cause too much pressure in the crankcase. This condition will cause the engine to run rough. There will be more than the normal amount of fumes (blowby) rising from the crankcase breather. The breather can then become restricted in a very short time, causing oil leakage at gaskets and seals that would not normally have leakage. Blowby can also be caused by worn valve guides or by a failed turbocharger seal.

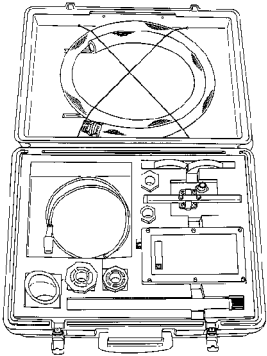


Illustration 69

g00286269

8T-2700 Blowby/Air Flow Indicator

The 8T-2700 Blowby/Air Flow Indicator is used to check the amount of blowby. Refer to Special Instruction, SEHS8712, "Using the 8T-2700 Blowby/Air Flow Indicator" for the test procedure for checking the blowby.

i01120286

## Compression - Test

**SMCS Code:** 1215

An engine that runs roughly can have a leak at the valves. An engine that runs roughly can also have valves that need an adjustment. Remove the head and inspect the valves and valve seats. This is necessary to find those small defects that would not normally cause problems. Repairs of these problems are normally performed when you are reconditioning the engine.

i01896573

## Engine Valve Lash - Inspect/Adjust

**SMCS Code:** 1102-025

### WARNING

To prevent possible injury, do not use the starter to turn the flywheel.

Hot engine components can cause burns. Allow additional time for the engine to cool before measuring valve clearance.

### WARNING

This engine uses high voltage to control the fuel injectors.

Disconnect electronic fuel injector enable circuit connector to prevent personal injury.

Do not come in contact with the fuel injector terminals while the engine is running.

**Note:** Valve lash is measured between the rocker arm and the valve bridge. All measurements and adjustments must be made with the engine stopped and the valves fully closed.

## Valve Lash Check

An adjustment is not necessary if the measurement of the valve lash is in the acceptable range. Check the valve lash while the engine is stopped. The valve lash setting is specified in Table 11.

Table 11

| Quick Reference for Engine Valve Lash Setting<br>C-10 and C-12 Engines |                                        |                                        |
|------------------------------------------------------------------------|----------------------------------------|----------------------------------------|
|                                                                        | Inlet Valves                           | Exhaust Valves                         |
| Valve Lash Setting                                                     | 0.38 ± 0.08 mm<br>(0.015 ± 0.003 inch) | 0.64 ± 0.08 mm<br>(0.025 ± 0.003 inch) |
| TC Compression Stroke                                                  | 1-2-4                                  | 1-3-5                                  |
| TC Exhaust Stroke <sup>(1)</sup>                                       | 3-5-6                                  | 2-4-6                                  |
| Firing Order                                                           | 1-5-3-6-2-4 <sup>(2)</sup>             |                                        |

<sup>(1)</sup> 360° from TC compression stroke

<sup>(2)</sup> The No. 1 cylinder is at the front of the engine.

If the measurement is not within this range, an adjustment is necessary. Refer to "Valve Lash Adjustment" for the proper procedure.

## Valve Lash Adjustment

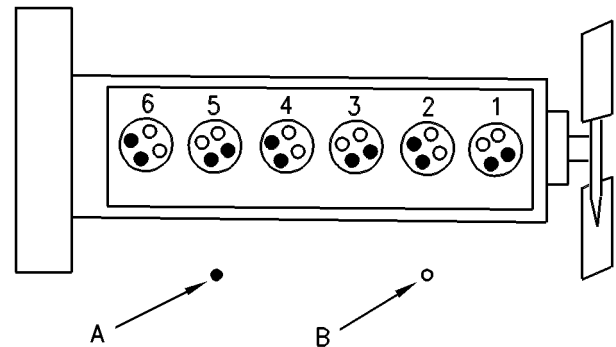


Illustration 70

Cylinder and valve location

(A) Exhaust valves

(B) Inlet valves

g00386924

Table 12

| Engine Valve Lash Setting |                                        |
|---------------------------|----------------------------------------|
| Valves                    | Valve Lash Setting                     |
| Inlet                     | 0.38 ± 0.08 mm<br>(0.015 ± 0.003 inch) |
| Exhaust                   | 0.64 ± 0.08 mm<br>(0.025 ± 0.003 inch) |

Adjust the valve lash while the engine is stopped. Use the following procedure to adjust the valve lash:

1. Put the No. 1 piston at the top center position. Refer to Testing and Adjusting, "Finding the Top Center Position For No. 1 Piston".

**Note:** If the engine is equipped with an engine compression brake, loosen the adjusting screw for the lash on the slave piston for the compression brake prior to adjusting the engine valve lash. Refer to the Testing and Adjusting, "Slave Piston Lash - Adjust" for information that relates to the adjustment of the slave piston lash for the engine compression brake.

2. Before any adjustments are made, lightly tap the rocker arm at the top of the adjustment screw with a soft mallet. This will ensure that the lifter roller seats against the camshaft's base circle.

**Note:** Refer to Table 12 for the appropriate engine valve lash setting.

3. Make an adjustment to the valve lash on the inlet valves for cylinders 1, 2, and 4.

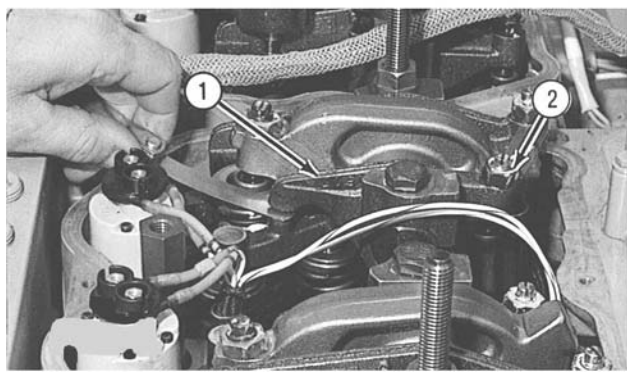


Illustration 71

g00481639

Engine valve lash adjustment

Typical example

- (1) Rocker arm
- (2) Adjustment locknut

- a. Loosen adjustment locknut (2).
- b. Place the appropriate feeler gauge between rocker arm and the valve bridge. Then, turn the adjustment screw in a clockwise direction. Slide the feeler gauge between the rocker arm and the valve bridge. Continue turning the adjustment screw until a slight drag is felt on the feeler gauge. Remove the feeler gauge.
- c. Tighten the adjustment locknut to a torque of  $30 \pm 7$  N·m ( $22 \pm 5$  lb ft). Do not allow the adjustment screw to turn while you are tightening the adjustment locknut. Recheck the valve lash after tightening the adjustment locknut.
4. Make an adjustment to the valve lash on the exhaust valves for cylinders 1, 3, and 5.
  - a. Loosen adjustment locknut (2).

- b. Place the appropriate feeler gauge between rocker arm and the valve bridge. Then, turn the adjustment screw in a clockwise direction. Slide the feeler gauge between the rocker arm and the valve bridge. Continue turning the adjustment screw until a slight drag is felt on the feeler gauge. Remove the feeler gauge.
- c. Tighten the adjustment locknut to a torque of  $30 \pm 7$  N·m ( $22 \pm 5$  lb ft). Do not allow the adjustment screw to turn while you are tightening the adjustment locknut. Recheck the valve lash after tightening the adjustment locknut.
5. Remove the timing bolt and turn the flywheel by 360 degrees in the direction of engine rotation. This will put the No. 6 piston at the top center position on the compression stroke. Install the timing bolt in the flywheel.
6. Before any adjustments are made, lightly tap the rocker arm at the top of the adjustment screw with a soft mallet. This will ensure that the lifter roller seats against the camshaft's base circle.
7. Make an adjustment to the valve lash on the inlet valves 3, 5, and 6.
  - a. Lightly tap the rocker arm at the top of the adjustment screw with a soft mallet. This will ensure that the lifter roller seats against the camshaft's base circle.
  - b. Loosen the adjustment locknut.
  - c. Place the appropriate feeler gauge between rocker arm and the valve bridge. Then, turn the adjustment screw in a clockwise direction. Slide the feeler gauge between the rocker arm and the valve bridge. Continue turning the adjustment screw until a slight drag is felt on the feeler gauge. Remove the feeler gauge.
  - d. Tighten the adjustment locknut to a torque of  $30 \pm 7$  N·m ( $22 \pm 5$  lb ft). Do not allow the adjustment screw to turn while you are tightening the adjustment locknut. Recheck the valve lash after tightening the adjustment locknut.
8. Make an adjustment to the valve lash on the exhaust valves for cylinders 2, 4, and 6.
  - a. Loosen adjustment locknut (2).

- b.** Place the appropriate feeler gauge between rocker arm and the valve bridge. Then, turn the adjustment screw in a clockwise direction. Slide the feeler gauge between the rocker arm and the valve bridge. Continue turning the adjustment screw until a slight drag is felt on the feeler gauge. Remove the feeler gauge.
- c.** Tighten the adjustment locknut to a torque of  $30 \pm 7$  N·m ( $22 \pm 5$  lb ft). Do not allow the adjustment screw to turn while you are tightening the adjustment locknut. Recheck the valve lash after tightening the adjustment locknut.
- 9.** Remove the timing bolt from the flywheel after all adjustments to the valve lash have been made. Reinstall the timing cover on the flywheel housing.

For information that relates to the adjustment of the electronic unit injector, refer to Testing and Adjusting, "Electronic Unit Injector - Adjust".

# Lubrication System

i01885008

## Engine Oil Pressure - Test

**SMCS Code:** 1304-081

The engine oil pressure may be checked electronically by using the electronic service tool. The engine oil pressure can be measured with the electronic service tool. Refer to Troubleshooting for information on the use of the electronic technician.

### Measuring Engine Oil Pressure

#### **WARNING**

**Work carefully around an engine that is running. Engine parts that are hot, or parts that are moving, can cause personal injury.**

#### **NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### **NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

Table 13

| Required Tools |                           |                                                              |          |
|----------------|---------------------------|--------------------------------------------------------------|----------|
| Tool           | Part Number               | Part Name                                                    | Quantity |
| A              | 1U-5470<br>or<br>198-4240 | Engine Pressure Group<br>or<br>Pressure Indicator Tool Group | 1        |

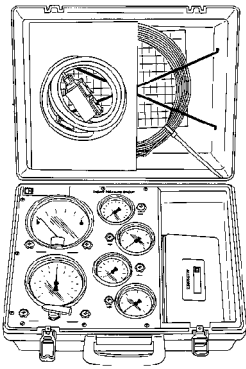


Illustration 72

g00296486

1U-5470 Engine Pressure Group

Tool (A) measures the oil pressure in the system. This engine tool group can read the oil pressure inside the oil manifold.

**Note:** Refer to Special Instruction, SEHS8907, "Using the 1U-5470 Engine Pressure Group" for more information. Refer to Operating Manual, NEHS0818, "Using the 198-4240 Pressure Indicator Tool Group" for more information.

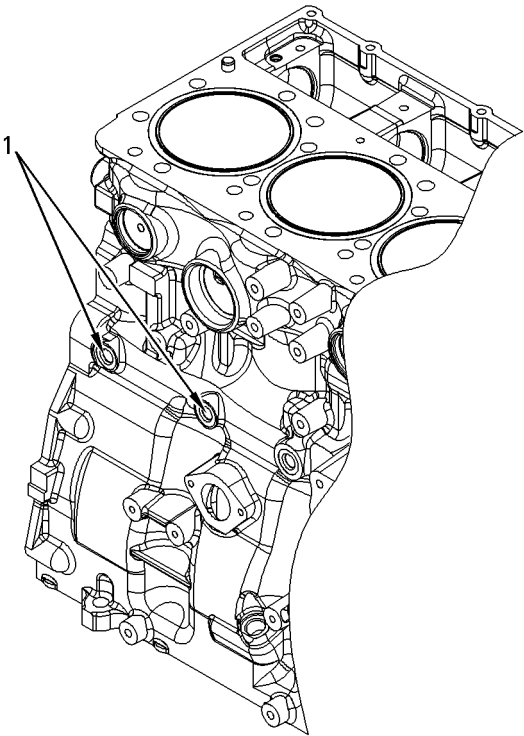


Illustration 73

g00977330

Oil gallery plug  
(1) Plug

**1.** Install Tool (A) into oil gallery plug (1).

**Note:** Engine oil pressure to the camshaft and main bearings should be checked on each side of the cylinder block at oil gallery plug (1).

2. Start the engine. Run the engine with SAE 15W40 oil. Refer to Operation and Maintenance Manual, "Engine Oil" for the recommendations of engine oil.
3. Record the value of the engine oil pressure when the engine is at operating temperature 100 °C (212 °F).

The minimum engine oil pressure at 1800 rpm should be approximately 275 to 414 kPa (40 to 59 psi). Minimum engine oil pressure at low idle rpm (600 to 800 rpm) should be approximately 68 kPa (10 psi).

4. Compare the recorded engine oil pressure with the oil pressure indicators on the instrument panel and the engine oil pressure that is displayed on the electronic service tool.
5. An engine oil pressure indicator that has a defect or an engine oil pressure sensor that has a defect can give a false indication of a low oil pressure or a high oil pressure. If there is a notable difference between the engine oil pressure readings make necessary repairs.
6. If low engine oil pressure is determined, refer to "Reasons for Low Engine Oil Pressure".
7. If high engine oil pressure is determined, refer to "Reason for High Engine Oil Pressure".

## Reasons for Low Engine Oil Pressure

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

- Engine oil level is low. Refer to Step 1.
  - Engine oil is contaminated. Refer to Step 2.
  - The engine oil bypass valves are open. Refer to Step 3.
  - The engine lubrication system is open. Refer to Step 4.
  - The oil pickup tube has a leak or a restricted inlet screen. Refer to Step 5.
  - The engine oil pump is faulty. Refer to Step 6.
  - Engine Bearings have excessive clearance. Refer to Step 7.
1. Check the engine oil level in the crankcase. The oil level can possibly be too far below the oil pump supply tube. This will cause the oil pump not to have the ability to supply enough lubrication to the engine components. If the engine oil level is low add engine oil in order to obtain the correct engine oil level. Refer to Operation and Maintenance Manual, "Engine Oil" for the recommendations of engine oil.
  2. Engine oil that is contaminated with fuel or coolant will cause low engine oil pressure. High engine oil level in the crankcase can be an indication of contamination. Determine the reason for contamination of the engine oil and make the necessary repairs. Replace the engine oil with the approved grade of engine oil. Refer to Operation and Maintenance Manual, "Engine Oil" for the recommendations of engine oil.

### NOTICE

Caterpillar oil filters are built to Caterpillar specifications. Use of an oil filter not recommended by Caterpillar could result in severe engine damage to the engine bearings, crankshaft, etc., as a result of the larger waste particles from unfiltered oil entering the engine lubricating system. Only use oil filters recommended by Caterpillar.

3. If the engine oil bypass valves are held in the open position, a reduction in the oil pressure can be the result. This may be due to debris in the engine oil. If the engine oil bypass valves are stuck in the open position, remove each engine oil bypass valve and clean each bypass valve in order to correct this problem. You must also clean each bypass valve bore. Install new engine oil filters. New engine oil filters will prevent more debris from causing this problem. For information on the repair of the engine oil bypass valves, refer to Disassembly and Assembly, "Engine Oil Filter Base - Disassemble".

4. An oil line or an oil passage that is open, broken, or disconnected will cause low engine oil pressure. An open lubrication system could be caused by a piston cooling jet that is missing or damaged. Determine the reason for an open lubrication system of the engine and make the necessary repairs.

**Note:** The piston cooling jets direct engine oil toward the bottom of the piston in order to cool the piston. This also provides lubrication for the piston pin. Breakage, a restriction or incorrect installation of the piston cooling jets will cause seizure of the piston.

5. The inlet screen of the oil pickup tube for the engine oil pump can have a restriction. This restriction will cause cavitation and a loss of engine oil pressure. Check the inlet screen on the oil pickup tube and remove any material that may be restricting engine oil flow. Low engine oil pressure may also be the result of the oil pickup tube that is drawing in air. Check the joints of the oil pickup tube for cracks or a damaged O-ring seal. Remove the engine oil pan in order to gain access to the oil pickup tube and the oil screen. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install" for more information.
6. Check the following problems that may occur to the engine oil pump.
  - a. Air leakage in the supply side of the oil pump will also cause cavitation and loss of oil pressure. Check the supply side of the oil pump and make necessary repairs. For information on the repair of the engine oil pump, refer to Disassembly and Assembly, "Engine Oil Pump - Remove".
  - b. Oil pump gears that have too much wear will cause a reduction in oil pressure. Repair the engine oil pump. For information on the repair of the engine oil pump, refer to Disassembly and Assembly, "Engine Oil Pump - Remove".
7. Excessive clearance at engine bearings will cause low engine oil pressure. Check the engine components that have excessive bearing clearance and make the necessary repairs.

## Reason for High Engine Oil Pressure

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### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

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### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

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Engine oil pressure will be high if the engine oil bypass valves become stuck in the closed position and the engine oil flow is restricted. Foreign matter in the engine oil system could be the cause for the restriction of the oil flow and the movement of the engine oil bypass valves. If the engine oil bypass valves are stuck in the closed position, remove each bypass valve and clean each bypass valve in order to correct this problem. You must also clean each bypass valve bore. Install new engine oil filters. New engine oil filters will prevent more debris from causing this problem. For information on the repair of the engine oil filter bypass valve, refer to Disassembly and Assembly, "Engine Oil Filter Base - Disassemble".

### NOTICE

Caterpillar oil filters are built to Caterpillar specifications. Use of an oil filter not recommended by Caterpillar could result in severe engine damage to the engine bearings, crankshaft, etc., as a result of the larger waste particles from unfiltered oil entering the engine lubricating system. Only use oil filters recommended by Caterpillar.

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i01398317

## Engine Oil Pump - Inspect

**SMCS Code:** 1304-040

The inlet screen of the supply tube for the engine oil pump can have a restriction. This will cause cavitation and a loss of oil pressure. Air leakage in the supply side of the engine oil pump will also cause cavitation and loss of oil pressure. If the bypass valve for the engine oil pump is held in the open position, the lubrication system cannot achieve maximum pressure. Oil pump gears that have too much wear will cause a reduction in the oil pressure.

If any part of the engine oil pump is worn enough in order to affect the performance of the engine oil pump, the engine oil pump must be replaced. Refer to Specifications, "Engine Oil Pump" for clearances.

i01126690

## Excessive Bearing Wear - Inspect

**SMCS Code:** 1203-040; 1211-040; 1219-040

When some components of the engine show bearing wear in a short time, the cause can be a restriction in an oil passage.

An engine oil pressure indicator may show that there is enough oil pressure, but a component is worn due to a lack of lubrication. In such a case, look at the passage for the oil supply to the component. A restriction in an oil supply passage will not allow enough lubrication to reach a component. This will result in early wear.

i01794028

## Excessive Engine Oil Consumption - Inspect

**SMCS Code:** 1348-040

### Engine Oil Leaks on the Outside of the Engine

Check for leakage at the seals at each end of the crankshaft. Look for leakage at the gasket for the engine oil pan and all lubrication system connections. Look for any engine oil that may be leaking from the crankcase breather. This can be caused by combustion gas leakage around the pistons. A dirty crankcase breather will cause high pressure in the crankcase. A dirty crankcase breather will cause the gaskets and the seals to leak.

### Engine Oil Leaks into the Combustion Area of the Cylinders

Engine oil that is leaking into the combustion area of the cylinders can be the cause of blue smoke. There are several possible ways for engine oil to leak into the combustion area of the cylinders:

- Leaks between worn valve guides and valve stems
- Worn components or damaged components (pistons, piston rings, or dirty return holes for the engine oil)

- Incorrect installation of the compression ring and/or the intermediate ring
- Leaks past the seal rings in the turbocharger shaft
- Overfilling of the crankcase
- Wrong dipstick or guide tube
- Sustained operation at light loads

Excessive consumption of engine oil can also result if engine oil with the wrong viscosity is used. Engine oil with a thin viscosity can be caused by fuel leakage into the crankcase or by increased engine temperature.

i01366050

## Increased Engine Oil Temperature - Inspect

**SMCS Code:** 1348-040

When the engine is at operating temperature and the engine is using SAE 15W40 oil, the maximum oil temperature should be 110 °C (230 °F). This is the temperature of the oil after passing through the oil cooler.

If the oil temperature is high, then check for a restriction in the oil passages of the oil cooler. A restriction in the oil cooler will not cause low oil pressure in the engine.

Determine if the oil cooler bypass valve is held in the open position. This condition will allow the oil to pass through the valve instead of the oil cooler. The oil temperature will increase.

## Cooling System

i01460276

### Cooling System - Check (Overheating)

**SMCS Code:** 1350-535

Above normal coolant temperatures can be caused by many conditions. Use the following procedure to determine the cause of above normal coolant temperatures:

#### **WARNING**

**Personal injury can result from escaping fluid under pressure.**

**If a pressure indication is shown on the indicator, push the release valve in order to relieve pressure before removing any hose from the radiator.**

---

1. Check the coolant level in the cooling system. Refer to Operation and Maintenance Manual, "Cooling System Coolant Level - Check". If the coolant level is too low, air will get into the cooling system. Air in the cooling system will cause a reduction in coolant flow and bubbles in the coolant. Air bubbles will keep coolant away from the engine parts, which will prevent the transfer of heat to the coolant. Low coolant level is caused by leaks or incorrectly filling the radiator.
2. Check the mixture of antifreeze and water. The mixture should be approximately 50 percent water and 50 percent antifreeze with 3 to 6 percent coolant conditioner. Refer to Operation and Maintenance Manual, "General Coolant Information". If the coolant mixture is incorrect, drain the system. Put the correct mixture of water, antifreeze and coolant conditioner in the cooling system.
3. Check for air in the cooling system. Air can enter the cooling system in different ways. The most common causes of air in the cooling system are not filling the cooling system correctly and combustion gas leakage into the cooling system. Combustion gas can get into the system through inside cracks, a damaged cylinder head, or a damaged cylinder head gasket. Air in the cooling system causes a reduction in coolant flow and bubbles in the coolant. Air bubbles keep coolant away from the engine parts, which prevents the transfer of heat to the coolant.
4. Check the fan clutch, if equipped. A fan clutch or a hydraulic driven fan that is not turning at the correct speed can cause improper air speed across the radiator core. The lack of proper air flow across the radiator core can cause the coolant not to cool to the proper temperature differential.
5. Check the water temperature gauge. A water temperature gauge which does not work correctly will not show the correct temperature. Refer to Testing and Adjusting, "Cooling System - Inspect".
6. Check the sending unit. In some conditions, the temperature sensor in the engine sends signals to a sending unit. The sending unit converts these signals to an electrical impulse which is used by a mounted gauge. If the sending unit malfunctions, the gauge can show an incorrect reading. Also if the electric wire breaks or if the electric wire shorts out, the gauge can show an incorrect reading.
7. Check the radiator.
  - a. Check the radiator for a restriction to coolant flow. Check the radiator for debris, dirt, or deposits on the inside of the radiator core. Debris, dirt, or deposits will restrict the flow of coolant through the radiator.
  - b. Check for debris or damage between the fins of the radiator core. Debris between the fins of the radiator core restricts air flow through the radiator core. Refer to Testing and Adjusting, "Cooling System - Inspect".
  - c. Ensure that the radiator size is according to the OEM's specifications. An undersized radiator does not have enough area for the effective release of heat. This may cause the engine to run at a temperature that is higher than normal. The normal temperature is dependent on the ambient temperature.
8. Check the filler cap. A pressure drop in the radiator can cause the boiling point to be lower. This can cause the cooling system to boil. Refer to Testing and Adjusting, "Cooling System - Test".
9. Check the fan and/or the fan shroud.
  - a. The fan must be large enough to send air through most of the area of the radiator core. Ensure that the size of the fan and the position of the fan are according to the OEM specifications.

- b. The fan shroud must be the proper size and the fan shroud must be positioned correctly. Ensure that the size of the fan shroud and the position of the fan shroud are according to the OEM specifications.
10. Check for loose drive belts. A loose fan drive belt will cause a reduction in the air flow across the radiator. Check the fan drive belt for proper belt tension. Adjust the tension of the fan drive belt, if necessary. Refer to Operation and Maintenance Manual, "Belt - Inspect".
11. Check the cooling system hoses and clamps. Damaged hoses with leaks can normally be seen. Hoses that have no visual leaks can soften during operation. The soft areas of the hose can become kinked or crushed during operation. These areas of the hose can cause a restriction in the coolant flow. Hoses become soft and/or get cracks after a period of time. The inside of a hose can deteriorate, and the loose particles of the hose can cause a restriction of the coolant flow. Refer to Operation and Maintenance Manual, "Hoses and Clamps - Inspect/Replace".
12. Check for a restriction in the air inlet system. A restriction of the air that is coming into the engine can cause high cylinder temperatures. High cylinder temperatures cause higher than normal temperatures in the cooling system. Refer to Testing and Adjusting, "Air Inlet and Exhaust System - Inspect".
  - a. If the measured restriction is higher than the maximum permissible restriction, remove the foreign material from the engine air cleaner element or install a new engine air cleaner element. Refer to Operation and Maintenance Manual, "Engine Air Cleaner Element - Clean/Replace".
  - b. Check for a restriction in the air inlet system again.
  - c. If the measured restriction is still higher than the maximum permissible restriction, check the air inlet piping for a restriction.
13. Check for a restriction in the exhaust system. A restriction of the air that is coming out of the engine can cause high cylinder temperatures.
  - a. Make a visual inspection of the exhaust system. Check for damage to exhaust piping or for a damaged muffler. If no damage is found, check the exhaust system for a restriction. Refer to Testing and Adjusting, "Air Inlet and Exhaust System - Inspect".
  - b. If the measured restriction is higher than the maximum permissible restriction, there is a restriction in the exhaust system. Repair the exhaust system, as required.
14. Check the shunt line, if the shunt system is used. The shunt line must be submerged in the expansion tank. A restriction of the shunt line from the radiator top tank to the engine water pump inlet will cause a reduction in water pump efficiency. A reduction in water pump efficiency will result in low coolant flow and overheating.
15. Check the water temperature regulator. A water temperature regulator that does not open, or a water temperature regulator that only opens part of the way can cause overheating. Refer to Testing and Adjusting, "Water Temperature Regulator - Test".
16. Check the water pump. A water pump with a damaged impeller does not pump enough coolant for correct engine cooling. Remove the water pump and check for damage to the impeller. Refer to Testing and Adjusting, "Water Pump - Test".
17. Check the air flow through the engine compartment. The air flow through the radiator comes out of the engine compartment. Ensure that the filters, air conditioner, and similar items are not installed in a way that prevents the free flow of air through the engine compartment.
18. Check the aftercooler. A restriction of air flow through the air to air aftercooler (if equipped) can cause overheating. Check for debris or deposits which would prevent the free flow of air through the aftercooler. Refer to Testing and Adjusting, "Aftercooler - Test".
19. Consider high outside temperatures. When outside temperatures are too high for the rating of the cooling system, there is not enough of a temperature difference between the outside air and coolant temperatures.
20. Consider high altitude operation. The cooling capacity of the cooling system goes down as the engine is operated at higher altitudes. A pressurized cooling system that is large enough to keep the coolant from boiling must be used.

**21.** The engine may be running in the lug condition. When the load that is applied to the engine is too large, the engine will run in the lug condition. When the engine is running in the lug condition, engine rpm does not increase with an increase of fuel. This lower engine rpm causes a reduction in air flow through the radiator. This lower engine rpm also causes a reduction in coolant flow through the system. This combination of less air and less coolant flow during high input of fuel will cause above normal heating.

i01390078

## Cooling System - Inspect

**SMCS Code:** 1350-040

Cooling systems that are not regularly inspected are the cause for increased engine temperatures. Make a visual inspection of the cooling system before any tests are performed.

### WARNING

**Personal injury can result from escaping fluid under pressure.**

**If a pressure indication is shown on the indicator, push the release valve in order to relieve pressure before removing any hose from the radiator.**

1. Check the coolant level in the cooling system. Refer to Operation and Maintenance Manual, "Cooling System Coolant Level - Check".
2. Check the quality of the coolant. The coolant should have the following properties:
  - Color that is similar to new coolant
  - Odor that is similar to new coolant
  - Free from dirt and debris

If the coolant does not have these properties, drain the system and flush the system. Refill the cooling system with the correct mixture of water, antifreeze, and coolant conditioner. Refer to Operation and Maintenance Manual, "General Coolant Information".

3. Look for leaks in the system.

**Note:** A small amount of coolant leakage across the surface of the water pump seals is normal. This leakage is required in order to provide lubrication for this type of seal. A hole is provided in the water pump housing in order to allow this coolant/seal lubricant to drain from the pump housing. Intermittent leakage of small amounts of coolant from this hole is not an indication of water pump seal failure.

4. Ensure that the airflow through the radiator does not have a restriction. Look for bent core fins between the folded cores of the radiator. Also, look for debris between the folded cores of the radiator.
5. Inspect the drive belts for the fan.
6. Check for damage to the fan blades.
7. Look for air or combustion gas in the cooling system.
8. Inspect the filler cap, and check the surface that seals the filler cap. This surface must be clean.

i01786951

## Cooling System - Test

**SMCS Code:** 1350-040; 1350-081

This engine has a pressure type cooling system. A pressure type cooling system has two advantages. The cooling system can be operated in a safe manner at a temperature higher than the normal boiling point (steam) of water.

This type of system prevents cavitation in the water pump. Cavitation is the forming of low pressure bubbles in liquids that are caused by mechanical forces. It is more difficult to create an air pocket or a steam pocket with this type of cooling system.

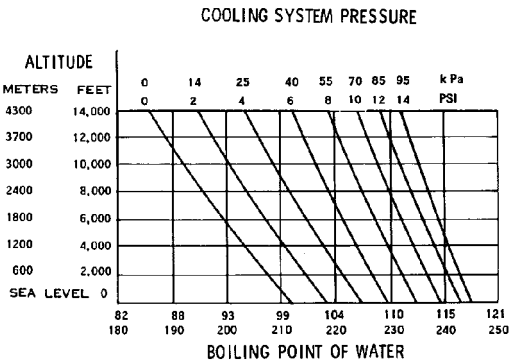


Illustration 74

g00286266

Boiling point of water

Remember that temperature and pressure work together. When a diagnosis is made of a cooling system problem, temperature and pressure must be checked. Cooling system pressure will have an effect on the cooling system temperature. For an example, refer to Illustration 74. This will show the effect of pressure on the boiling point (steam) of water. This will also show the effect of height above sea level.

### WARNING

**Personal injury can result from hot coolant, steam and alkali.**

**At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.**

**Remove filler cap slowly to relieve pressure only when engine is stopped and radiator cap is cool enough to touch with your bare hand.**

**Cooling System Conditioner contains alkali. Avoid contact with skin and eyes.**

The coolant level must be to the correct level in order to check the coolant system. The engine must be cold and the engine must not be running.

After the engine is cool, loosen the pressure cap in order to relieve the pressure out of the cooling system. Then remove the pressure cap.

The level of the coolant should not be more than 13 mm (0.5 inch) from the bottom of the filler pipe. If the cooling system is equipped with a sight glass, the coolant should be to the proper level in the sight glass.

## Test Tools For Cooling System

Table 14

| Required Tools           |                           |          |
|--------------------------|---------------------------|----------|
| Part Number              | Part Name                 | Quantity |
| 4C-6500                  | Digital Thermometer       | 1        |
| 8T-2700                  | Blowby/Air Flow Indicator | 1        |
| 9S-8140                  | Pressurizing Pump         | 1        |
| 9U-7400                  | Multitach Tool Group      | 1        |
| 1U-7297<br>or<br>1U-7298 | Coolant/Battery Tester    | 1        |

### WARNING

**Making contact with a running engine can cause burns from hot parts and can cause injury from rotating parts.**

**When working on an engine that is running, avoid contact with hot parts and rotating parts.**



Illustration 75

g00286267

4C-6500 Digital Thermometer

The 4C-6500 Digital Thermometer is used in the diagnosis of overheating conditions and in the diagnosis of overcooling conditions. This group can be used to check temperatures in several different parts of the cooling system. Refer to Operating Manual, NEHS0554, "4C-6500 Digital Thermometer Group" for the testing procedure.

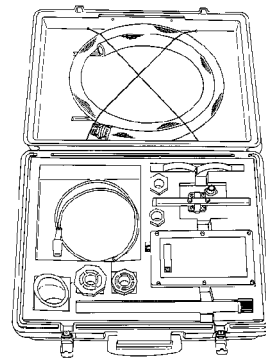


Illustration 76

g00286269

8T-2700 Blowby/Air Flow Indicator

The 8T-2700 Blowby/Air Flow Indicator is used to check the air flow through the radiator core. Refer to Special Instruction, SEHS8712, "Using the 8T-2700 Blowby/Air Flow Indicator" for the test procedure for checking blowby.

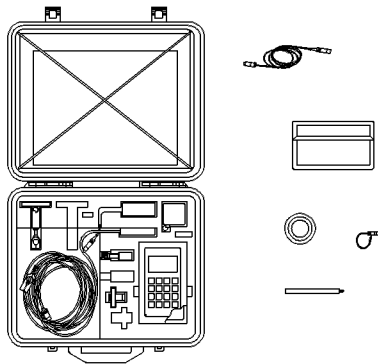


Illustration 77  
9U-7400 Multitach

The **9U-7400** Multitach Tool Group is used to check the fan speed. Refer to Operating Manual, NEHS0605, “**9U-7400** Multitach Tool Group ” for the testing procedure.

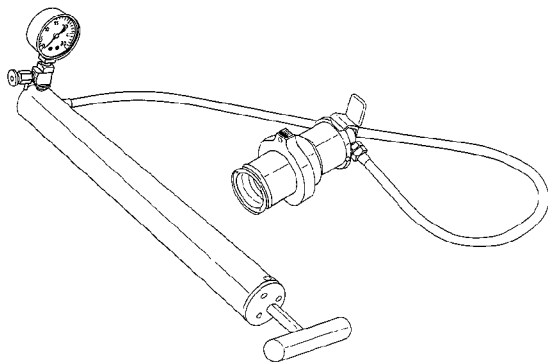


Illustration 78  
9S-8140 Pressurizing Pump

The **9S-8140** Pressurizing Pump is used to test the filler caps. This pressurizing pump is also used to pressure test the cooling system for leaks.

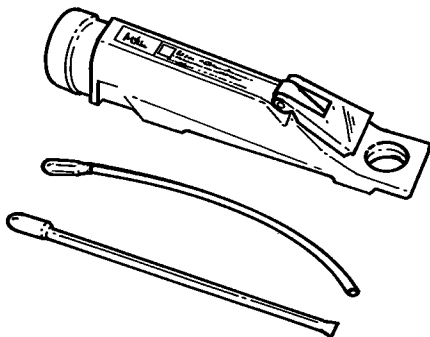


Illustration 79  
1U-7297 Coolant/Battery Tester or 1U-7298 Coolant/Battery Tester

Check the coolant frequently in cold weather for the proper glycol concentration. Use either the **1U-7297** Coolant/Battery Tester or the **1U-7298** Coolant/Battery Tester in order to ensure adequate freeze protection. The testers are identical except for the temperature scale. The testers give immediate, accurate readings. The testers can be used for coolants that contain ethylene or propylene glycol.

### Making the Correct Antifreeze Mixtures

Adding pure antifreeze as a makeup solution for the cooling system top-off is an unacceptable practice. Adding pure antifreeze increases the concentration of antifreeze in the cooling system. This increases the concentration of the dissolved solids and the undissolved chemical inhibitors in the cooling system. Add the coolant/water mixture to the same freeze protection as your cooling system. The following chart assists in determining the concentration of antifreeze to use. Refer to Operation and Maintenance Manual, “General Coolant Information”.

Table 15

| Antifreeze Concentrations     |                              |
|-------------------------------|------------------------------|
| Temperature                   | Concentration                |
| Protection to -15 °C (5 °F)   | 30% antifreeze and 70% water |
| Protection to -23 °C (-10 °F) | 40% antifreeze and 60% water |
| Protection to -37 °C (-34 °F) | 50% antifreeze and 50% water |
| Protection to -51 °C (-60 °F) | 60% antifreeze and 40% water |

### Checking the Filler Cap

Table 16

| Required Tools |                   |          |
|----------------|-------------------|----------|
| Part Number    | Part Name         | Quantity |
| 9S-8140        | Pressurizing Pump | 1        |

One cause for a pressure loss in the cooling system can be a damaged seal on the radiator filler cap.

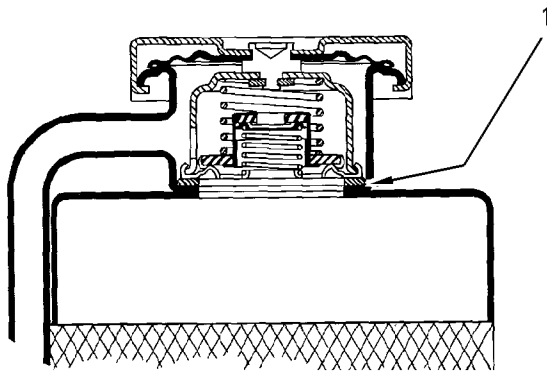


Illustration 80  
Typical schematic of filler cap  
(1) Sealing surface of both filler cap and radiator

### **WARNING**

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove filler cap slowly to relieve pressure only when engine is stopped and radiator cap is cool enough to touch with your bare hand.

Cooling System Conditioner contains alkali. Avoid contact with skin and eyes.

To check for the amount of pressure that opens the filler cap, use the following procedure:

1. After the engine cools, carefully loosen the filler cap. Slowly release the pressure from the cooling system. Then, remove the filler cap.

Carefully inspect the filler cap. Look for any damage to the seals and to the sealing surface. Inspect the following components for any foreign substances:

- Filler cap
- Seal
- Surface for seal

Remove any deposits that are found on these items.

2. Install the filler cap on the 9S-8140 Pressurizing Pump.
3. Look at the gauge for the exact pressure that opens the filler cap.

4. Compare the gauge's reading with the opening pressure that is listed on the filler cap.
5. If the filler cap is damaged, replace the filler cap.

## Testing The Radiator And Cooling System For Leaks

Table 17

| Required Tools |                   |          |
|----------------|-------------------|----------|
| Part Number    | Part Name         | Quantity |
| 9S-8140        | Pressurizing Pump | 1        |

Use the following procedure in order to check the cooling system for leaks:

### **WARNING**

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove filler cap slowly to relieve pressure only when engine is stopped and radiator cap is cool enough to touch with your bare hand.

Cooling System Conditioner contains alkali. Avoid contact with skin and eyes.

1. After the engine is cool, loosen the filler cap slowly and allow pressure out of the cooling system. Then remove the filler cap from the radiator.
2. Ensure that the radiator is full of coolant.
3. Install the 9S-8140 Pressurizing Pump onto the radiator.
4. Take the pressure reading on the gauge to 20 kPa (3 psi) more than the pressure on the filler cap. The pressure on a typical filler cap is 48.3 kPa (7 psi) to 103.42 kPa (15 psi).
5. Check the radiator for leakage on the outside.
6. Check all connection points for leakage, and check the hoses for leakage.

The cooling system does not have leakage only if the following conditions exist:

- You do NOT observe any outside leakage.

- The reading remains steady after five minutes.

**Note:** Check the engine oil for evidence of coolant leakage.

The inside of the engine cooling system has leakage only if the following conditions exist:

- The reading on the gauge goes down.
- You do NOT observe any outside leakage.
- Evidence of coolant on the engine oil gauge.

Make any repairs, as required.

A cooling system pressure test should be performed if the following conditions are met: Coolant is observed on the oil gauge. and Unusually high levels of high sodium are found during a SOS analysis.

The following steps are an outline of the cooling system pressure test:

- Drain the engine oil.
- Remove the oil pan.
- Connect the **9S-8140** Pressurizing Pump.
- Pressurize the system to 20 kPa (3 psi) more than the pressure on the filler cap.
- Inspect the inside of the engine block for coolant.
- Inspect the weep hole.

## Test For The Water Temperature Gauge

Table 18

| Required Tools           |                                          |          |
|--------------------------|------------------------------------------|----------|
| Part Number              | Part Name                                | Quantity |
| 4C-6500<br>or<br>2F-7112 | Digital Thermometer<br>or<br>Thermometer | 1        |

### WARNING

**Personal injury can result from escaping fluid under pressure.**

**If a pressure indication is shown on the indicator, push the release valve in order to relieve pressure before removing any hose from the radiator.**

### WARNING

**Making contact with a running engine can cause burns from hot parts and can cause injury from rotating parts.**

**When working on an engine that is running, avoid contact with hot parts and rotating parts.**

Check the accuracy of the coolant temperature indicator or coolant temperature sensor if you find either of the following conditions:

- The engine runs at a temperature that is too hot, but a normal temperature is indicated. A loss of coolant is found.
- The engine runs at a normal temperature, but a hot temperature is indicated. No loss of coolant is found.

Coolant temperature can also be read on the display screens of the Electronic Service Tool.

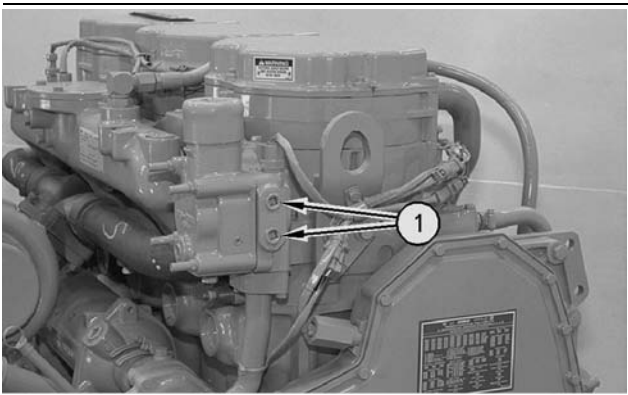


Illustration 81

g00438983

Typical example

(1) Ports

Remove the plug from one of ports (1). Install one of the following thermometers in the open port:

- The **4C-6500** Digital Thermometer
- The **2F-7112** Thermometer

A temperature indicator of known accuracy can also be used to make this check.

Start the engine. Run the engine until the temperature reaches the desired range according to the test thermometer. If necessary, place a cover over part of the radiator in order to cause a restriction of the air flow. The reading on the water temperature indicator should agree with the test thermometer within the tolerance range of the coolant temperature indicator.

i01666401

## Water Temperature Regulator - Test

**SMCS Code:** 1355-081; 1355-081-ON

### **WARNING**

**Personal injury can result from escaping fluid under pressure.**

**If a pressure indication is shown on the indicator, push the release valve in order to relieve pressure before removing any hose from the radiator.**

1. Remove the water temperature regulator from the engine.
2. Heat water in a pan until the temperature of the water is equal to the fully open temperature of the water temperature regulator. Refer to Specifications, "Water Temperature Regulator" for the fully open temperature of the water temperature regulator. Stir the water in the pan. This will distribute the temperature throughout the pan.
3. Hang the water temperature regulator in the pan of water. The water temperature regulator must be below the surface of the water. The water temperature regulator must be away from the sides and the bottom of the pan.
4. Keep the water at the correct temperature for ten minutes.
5. After ten minutes, remove the water temperature regulator. Immediately measure the opening of the water temperature regulator. Refer to Specifications, "Water Temperature Regulator" for the minimum opening distance of the water temperature regulator at the fully open temperature.

If the distance is less than the amount listed in the manual, replace the water temperature regulator.

i01467399

## Water Pump - Test

**SMCS Code:** 1361-081

Table 19

| Tools Needed |                    |          |
|--------------|--------------------|----------|
| Part Number  | Part Name          | Quantity |
| 6V-7775      | Air Pressure Gauge | 1        |

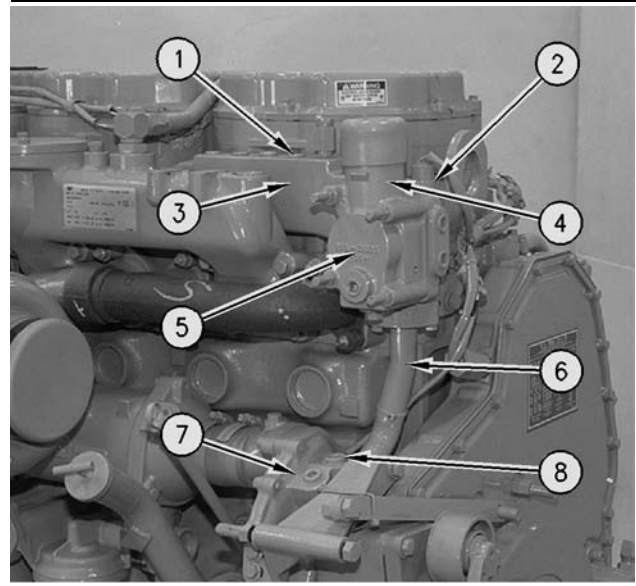


Illustration 82

g00439003

Typical example

- (1) Port
- (2) Coolant temperature sensor
- (3) Water manifold assembly
- (4) Water outlet
- (5) Coolant temperature regulator
- (6) Bypass line
- (7) Water pump
- (8) Port

### **WARNING**

**Making contact with a running engine can cause burns from hot parts and can cause injury from rotating parts.**

**When working on an engine that is running, avoid contact with hot parts and rotating parts.**

Perform the following procedure in order to determine if the water pump is operating correctly:

1. Remove the plug from port (1).
2. Install the 6V-7775 Air Pressure Gauge in port (1).
3. Start the engine. Run the engine until the coolant is at operating temperature.
4. Note the water pump pressure. The water pump pressure should be 100 to 125 kPa (15 to 18 psi).

# Basic Engine

i01124208

## Piston Ring Groove - Inspect

**SMCS Code:** 1214-040

The pistons of the engine have a keystone design ring groove. The piston rings are a keystone ring. The **1U-6431** Piston Ring Groove Gauge is available to check the top ring groove in the piston. Use the **8T-3149** Plug Gauge that is part of this Gauge Group to check the top ring groove on the piston. Refer to the instruction card for correct use of the **1U-6431** Piston Ring Groove Gauge.

i01460477

## Connecting Rod Bearings - Inspect

**SMCS Code:** 1219-040

The connecting rod bearings fit tightly in the bore in the rod. If there is excess movement between the connecting rod and the crankshaft, check the bore size. This can be an indication of wear because of a loose fit. Refer to the Guideline For Reusable Parts, SEBF8009, "Main and Connecting Rod Bearings" or refer to the Guideline For Reusable Parts, SEBF0531, "Main and Connecting Rod Bearings".

Connecting rod bearings are available with 0.508 mm (0.0200 inch) and 0.762 mm (0.0300 inch) smaller inside diameter than the original size bearings. These bearings are for crankshafts that have been ground.

i01125517

## Main Bearings - Inspect

**SMCS Code:** 1203-040

Main bearings are available with 0.508 mm (0.0200 inch) and a 0.762 mm (0.0300 inch) smaller inside diameter than the original size bearings. These bearings are for crankshafts that have been ground.

Main bearings are also available with a larger outside diameter than the original size bearings. These bearings are used for the cylinder blocks with the main bearing bore that is made larger than the bore's original size. The size that is available has a 0.508 mm (0.0200 inch) outside diameter that is larger than the original size bearings.

Refer to Special Instruction, SMHS7606, "Use of **1P-4000** Line Boring Tool Group" for the instructions that are needed to use the **1P-4000** Line Boring Tool Group. The **1P-4000** Line Boring Tool Group is used in order to check the alignment of the main bearing bores. The **1P-3537** Dial Bore Gauge Group can be used to check the size of the bore. Refer to Special Instruction, GMG00981, "**1P-3537** Dial Bore Gauge Group" for the instructions that are needed to use the **1P-3537** Dial Bore Gauge Group.

i01854009

## Cylinder Block - Inspect

**SMCS Code:** 1201-040

Table 20

| Required Tools |                       |          |
|----------------|-----------------------|----------|
| Part Number    | Part Name             | Quantity |
| 1P-3537        | Dial Bore Gauge Group | 1        |

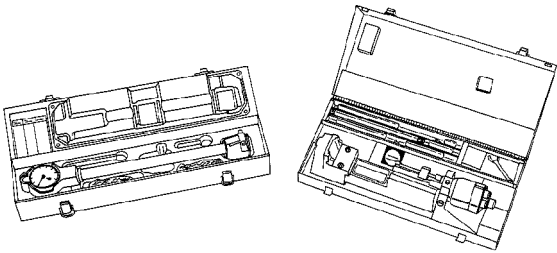


Illustration 83

g00285686

**1P-3537** Dial Bore Gauge Group

If the main bearing caps are installed without bearings, the bore in the block for the main bearings can be checked. Tighten the nuts on the bearing caps to the torque that is given in Specifications, "Cylinder Block". Alignment error in the bores must not be more than 0.08 mm (0.003 inch).

The **1P-3537** Dial Bore Gauge Group can be used to check the size of the bore. Refer to Special Instruction, GMG00981, "**1P-3537** Dial Bore Gauge Group" for the instructions that are needed to use the **1P-3537** Dial Bore Gauge Group.

i01900664

## Cylinder Liner Projection - Inspect

**SMCS Code:** 1216-040

Table 21

| Required Tools |                             |          |
|----------------|-----------------------------|----------|
| Part Number    | Part Name                   | Quantity |
| 8T-0455        | Liner Projection Tool Group | 1        |

1. Clean the cylinder liner flange and the cylinder block surface. Remove any nicks on the top of the cylinder block.

Table 22

| Required Components |             |             |                           |                            |
|---------------------|-------------|-------------|---------------------------|----------------------------|
| Item                | Part Number | Description | Quantity for One Cylinder | Quantity for Six Cylinders |
| 1                   | 8T-4193     | Bolt        | 6                         | 36                         |
| 2                   | 2S-5658     | Hard Washer | 6                         | 36                         |
| 3                   | 8F-1484     | Washer      | 6                         | 36                         |
| 4                   | 7K-1977     | Washer      | 6                         | 36                         |

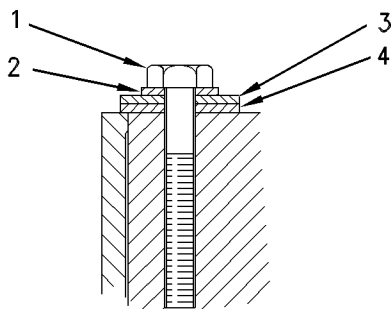


Illustration 84

g00441535

Location of the components

- (1) Bolt
- (2) Washer
- (3) Washer
- (4) Washer

2. The components should be assembled in the order that is shown in Illustration 84. **7K-1977** Washer (4) is made of a cotton fabric that is impregnated with resin. The washer will not damage the sealing surface of the cylinder block.

**Note:** Inspect the washer before measuring the liner projection. Replace the washer if the washer is worn or damaged.

3. Tighten bolts (1) evenly according to the following procedure:

First tighten each bolt to the following torque. .... 27 N·m (20 lb ft)

Then tighten each bolt to the following torque. .... 54 N·m (40 lb ft)

Then tighten each bolt to the following torque. .... 68 N·m (50 lb ft)

Again tighten each bolt to the following torque. .... 68 N·m (50 lb ft)

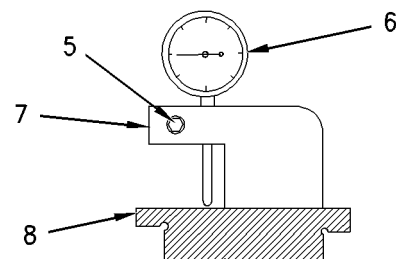


Illustration 85

g00737184

8T-0455 Liner Projection Tool Group

- (5) Bolt
- (6) Dial indicator
- (7) Gauge body
- (8) Gauge block

4. Loosen bolt (5) until dial indicator (6) can be moved. Place gauge body (7) and dial indicator (6) on the long side of gauge.
5. Slide dial indicator (6) into the correct position. When the point of the dial indicator contacts gauge block (8), the dial indicator is in the correct position. Slide the dial indicator until the needle of the gauge makes a quarter of a revolution clockwise. The needle should be in a vertical position. Tighten bolt (5) and zero the dial indicator.

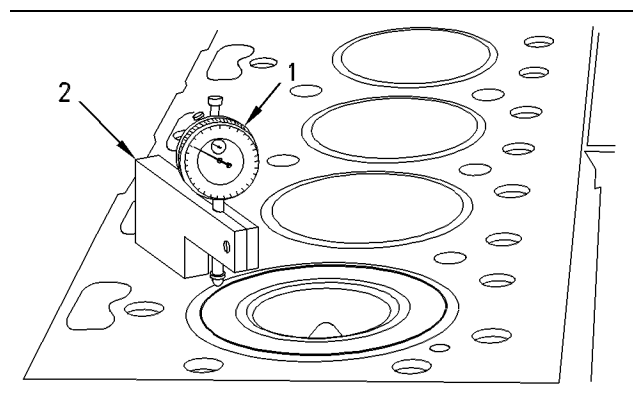


Illustration 86 g00326384

Measure the liner projection.

- (1) Dial indicator  
(2) Gauge body

6. Place gauge body (7) on the plate for the cylinder block. The indicator point should be on the liner flange. Read the dial indicator in order to find the amount of liner projection. Check the projection at four locations (every 90 degrees) around each cylinder liner.

Table 23

| Specifications                                    |                                              |
|---------------------------------------------------|----------------------------------------------|
| Liner Projection                                  | 0.040 to 0.200 mm<br>(0.0016 to 0.0079 inch) |
| Maximum Variation in Each Liner                   | 0.050 mm (0.0020 inch)                       |
| Maximum Average Variation Between Adjacent Liners | 0.050 mm (0.0020 inch)                       |
| Maximum Variation Between Liners                  | 0.100 mm (0.0040 inch)                       |

7. If a liner does not meet the recommended cylinder liner projection specification, check the following parts:
- The depth of the cylinder block bore should be  $100.00 \pm 0.03$  mm ( $3.937 \pm 0.001$  inch).
  - The liner flange should be  $100.12 \pm 0.03$  mm ( $3.942 \pm 0.001$  inch).

If the dimensions for the liner flange do not match the specifications, replace the liner. Then repeat the liner projection measurements. If the dimensions for the depth of the cylinder block bore do not match the specifications, replace the cylinder block. Then repeat the liner projection measurements.

## Flywheel - Inspect

SMCS Code: 1156-040

Table 24

| Required Tools |                |     |
|----------------|----------------|-----|
| Part Number    | Description    | Qty |
| 8T-5096        | Dial Indicator | 1   |

## Face Runout (Axial Eccentricity) of the Flywheel

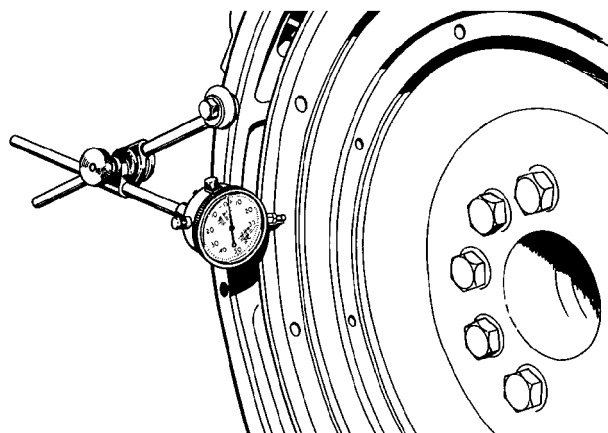


Illustration 87 g00286049

Checking face runout of the flywheel

1. Refer to Illustration 87 and install the dial indicator. Always put a force on the crankshaft in the same direction before the dial indicator is read. This will remove any crankshaft end clearance.
2. Set the dial indicator to read 0.0 mm (0.00 inch).
3. Turn the flywheel at intervals of 90 degrees and read the dial indicator. Refer to Testing and Adjusting, "Finding Top Center Position for No. 1 Piston".
4. Take the measurements at all four points. The difference between the lower measurements and the higher measurements that are performed at all four points must not be more than 0.15 mm (0.006 inch), which is the maximum permissible face runout (axial eccentricity) of the flywheel.

## Bore Runout (Radial Eccentricity) of the Flywheel

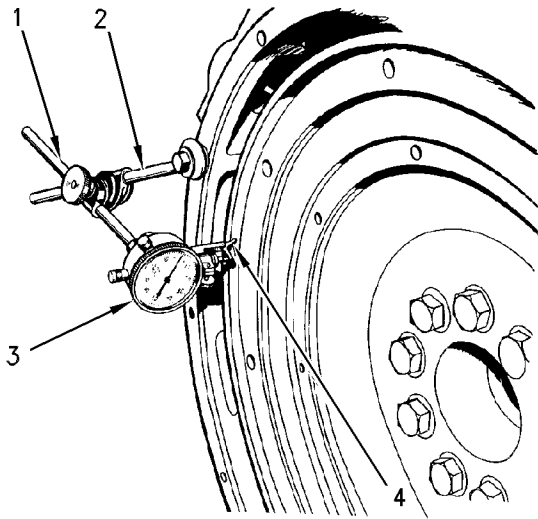


Illustration 88

g00286054

Checking bore runout of the flywheel

- (1) 7H-1945 Holding Rod
- (2) 7H-1645 Holding Rod
- (3) 7H-1942 Dial Indicator
- (4) 7H-1940 Universal Attachment

1. Install **7H-1942** Dial Indicator (3). Make an adjustment of **7H-1940** Universal Attachment (4) so the dial indicator makes contact on the flywheel.
2. Set the dial indicator to read 0.0 mm (0.00 inch).
3. Turn the flywheel at intervals of 90 degrees and read the dial indicator.
4. Take the measurements at all four points. The difference between the lower measurements and the higher measurements that are performed at all four points must not be more than 0.15 mm (0.006 inch), which is the maximum permissible face runout (radial eccentricity) of the flywheel.

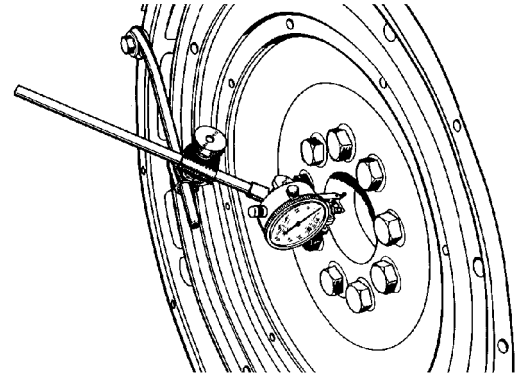


Illustration 89

g00286058

Flywheel clutch pilot bearing bore

5. To find the runout (eccentricity) of the pilot bearing bore, use the preceding procedure.
6. The runout (eccentricity) of the bore for the pilot bearing in the flywheel must not exceed 0.13 mm (0.005 inch).

i01461438

## Flywheel Housing - Inspect

**SMCS Code:** 1157-040

Table 25

| Tools Needed |                |          |
|--------------|----------------|----------|
| Part Number  | Part Name      | Quantity |
| 8T-5096      | Dial Indicator | 1        |

## Face Runout (Axial Eccentricity) of the Flywheel Housing

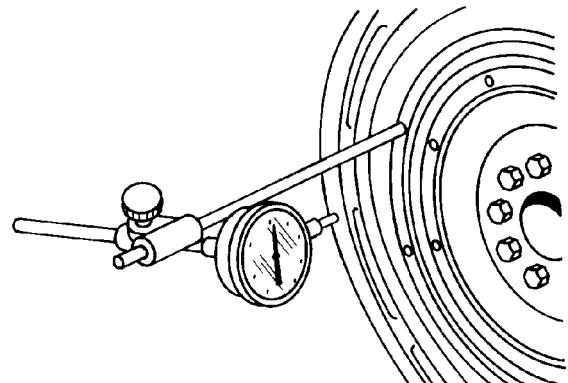


Illustration 90

g00285931

8T-5096 Dial Indicator

If you use any other method except the method that is given here, always remember that the bearing clearance must be removed in order to receive the correct measurements.

1. Fasten a dial indicator to the flywheel so the anvil of the dial indicator will contact the face of the flywheel housing.
2. Use a rubber mallet and tap the crankshaft toward the rear before the dial indicator is read at each point.

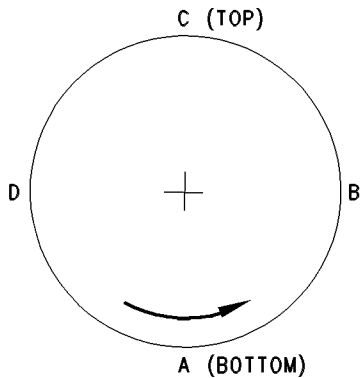


Illustration 91  
Checking face runout of the flywheel housing

3. Turn the flywheel while the dial indicator is set at 0.0 mm (0.00 inch) at location (A). Read the dial indicator at locations (B), (C) and (D).
4. The difference between the lower measurements and the higher measurements that are performed at all four points must not be more than 0.38 mm (0.015 inch), which is the maximum permissible face runout (axial eccentricity) of the flywheel housing.

### Bore Runout (Radial Eccentricity) of the Flywheel Housing

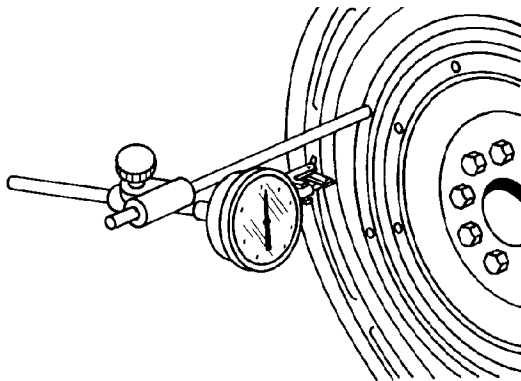


Illustration 92  
8T-5096 Dial Indicator

1. Fasten a dial indicator to the flywheel so the anvil of the dial indicator will contact the bore of the flywheel housing.

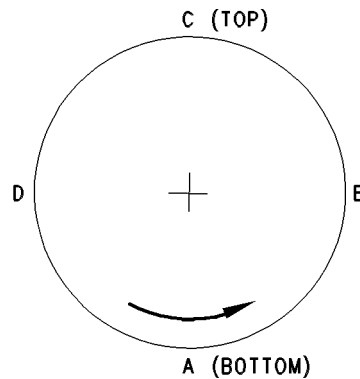


Illustration 93  
Checking bore runout of the flywheel housing

| CHART FOR DIAL INDICATOR MEASUREMENTS |                            |   |   |   |   |
|---------------------------------------|----------------------------|---|---|---|---|
|                                       | Position of dial indicator |   |   |   |   |
|                                       | Line No.                   | A | B | C | D |
| Correction for bearing clearance      | 1                          | 0 |   |   |   |
| Dial Indicator Reading                | 2                          | 0 |   |   |   |
| Total of Line 1 & 2                   | 3                          | 0 |   |   |   |
| the total horizontal eccentricity.    |                            |   |   |   |   |

Illustration 94

2. While the dial indicator is in the position at location (C) adjust the dial indicator to 0.0 mm (0.00 inch). Push the crankshaft upward against the top of the bearing. Refer to Illustration 94. Write the measurement for bearing clearance on line 1 in column (C).

**Note:** Write the measurements for the dial indicator with the correct notations. This notation is necessary for making the calculations in the chart correctly.

3. Divide the measurement from Step 2 by two. Write this number on line 1 in columns (B) and (D).
4. Turn the flywheel in order to put the dial indicator at position (A). Adjust the dial indicator to 0.0 mm (0.00 inch).
5. Turn the flywheel counterclockwise in order to put the dial indicator at position (B). Write the measurements in the chart.

i01892251

6. Turn the flywheel counterclockwise in order to put the dial indicator at position (C). Write the measurement in the chart.
7. Turn the flywheel counterclockwise in order to put the dial indicator at position (D). Write the measurement in the chart.
8. Add the lines together in each column.
9. Subtract the smaller number from the larger number in column B and column D. Place this number on line III. The result is the horizontal eccentricity (out of round). Line III in column C is the vertical eccentricity.

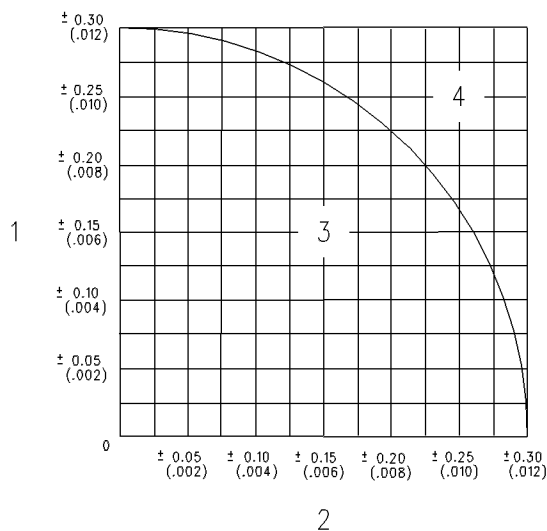


Illustration 95

g00286046

Graph for total eccentricity

- (1) Total vertical eccentricity
- (2) Total horizontal eccentricity
- (3) Acceptable value
- (4) Unacceptable value

10. Find the intersection of the eccentricity lines (vertical and horizontal) in Illustration 95.
11. If the point of the intersection is in the "Acceptable" range, the bore is in alignment. If the point of intersection is in the "Not acceptable" range, the flywheel housing must be changed.

## Vibration Damper - Check

SMCS Code: 1205-535

### Rubber Vibration Damper (If Equipped)

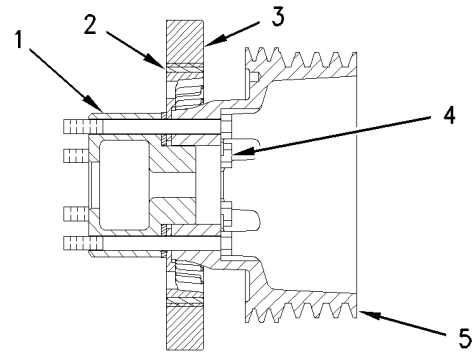


Illustration 96

g00681808

Vibration damper and pulley

Typical example

- (1) Adapter
- (2) Rubber
- (3) Damper assembly
- (4) Bolt
- (5) Crankshaft pulley

The vibration damper is installed on the front of crankshaft. The space in the damper assembly (3) is filled with rubber (2). The vibration damper limits the torsional vibration.

Replace the damper if any of the following conditions exist:

- The damper is dented or cracked.
- The paint on the damper is discolored from heat.
- There is a large amount of gear train wear that is not caused by lack of oil.
- Analysis of the oil has revealed that the front main bearing is badly worn.
- The engine has had a failure because of a broken crankshaft.

### Viscous Vibration Damper (If Equipped)

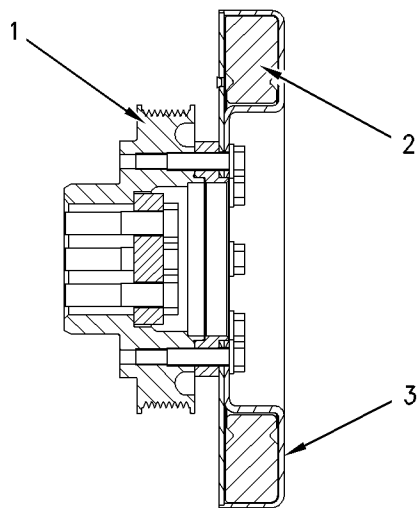


Illustration 97

g00750399

### Viscous vibration damper

Typical example

- (1) Crankshaft pulley
- (2) Weight
- (3) Case

Damage to the vibration damper or failure of the vibration damper will increase vibrations. This will result in damage to the crankshaft.

Replace the damper if any of the following conditions exist:

- The damper is dented, cracked, or fluid is leaking from the damper.
- The paint on the damper is discolored from excessive heat.
- The damper is bent.
- The bolt holes are worn or there is a loose fit for the bolts.
- The engine has had a crankshaft failure due to torsional forces.

## NOTICE

Inspect the viscous vibration damper for signs of leaking and for signs of damage to the case. Either of these conditions can cause the weight to contact the case. This contact can affect damper operation.

# Compression Brake

i01896342

## Slave Piston Lash - Adjust

SMCS Code: 1119-025

### WARNING

To prevent possible injury, do not use the starter to turn the flywheel.

Hot engine components can cause burns. Allow additional time for the engine to cool before measuring valve clearance.

### WARNING

This engine uses high voltage to control the fuel injectors.

Disconnect electronic fuel injector enable circuit connector to prevent personal injury.

Do not come in contact with the fuel injector terminals while the engine is running.

**Note:** The engine valve lash must be adjusted before adjusting the slave piston lash.

For information that relates to the adjustment of the engine valve lash, refer to Testing and Adjusting, "Engine Valve Lash - Inspect/Adjust".

Table 26

| Slave Piston Lash Setting<br>Jake Brake Compression Brakes |                                        |
|------------------------------------------------------------|----------------------------------------|
| Serial Number Prefix                                       | Slave Piston Lash Setting              |
| 3CS1-UP<br>2PN1-UP<br>8YS1-UP                              | 0.64 ± 0.08 mm<br>(0.025 ± 0.003 inch) |
| 2KS1-UP<br>9NS1-UP                                         | 1.02 ± 0.08 mm<br>(0.040 ± 0.003 inch) |
| 1YN1-UP                                                    | 1.14 ± 0.08 mm<br>(0.045 ± 0.003 inch) |
| MBJ1-UP<br>MBL1-UP                                         | 1.40 ± 0.08 mm<br>(0.055 ± 0.003 inch) |

**Note:** Slave piston lash is measured between the slave piston and the actuating pin that is on the exhaust valve bridge. Make all adjustments to the slave piston while the engine is stopped and the exhaust valves are closed.

Use the following procedure to adjust the slave piston lash on cylinders 1, 3, and 5:

- Put the No. 1 piston at the top center position on the compression stroke. Refer to Testing and Adjusting, "Finding Top Center Position for No. 1 Piston".

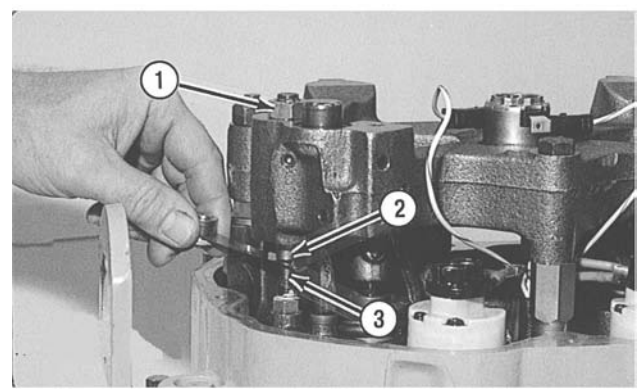


Illustration 98

g00481680

Compression brake

Typical example

- Adjusting screw and locknut
- Slave piston
- Actuating pin

- Loosen adjustment locknut (1).
- Place the appropriate feeler gauge between slave piston (2) and screw assembly (3). Then, turn the adjustment screw in a clockwise direction. Slide the feeler gauge between the slave piston and the screw assembly. Continue turning the adjustment screw until a slight drag is felt on the feeler gauge. Remove the feeler gauge.
- Tighten adjustment locknut (1) to a torque of 35 N·m (25 lb ft). Recheck the slave piston lash after tightening the locknut.
- Continue to check the slave piston lash for each of the cylinders that have been designated for adjustment. After each adjustment, tighten the nut for valve adjustment screw to a torque of 35 N·m (25.0 lb ft) and check the adjustment again.
- Remove the timing bolt and turn the flywheel by 360 degrees in the direction of engine rotation. This will put the No. 6 piston at the top center position on the compression stroke. Install the timing bolt in the flywheel.

To make an adjustment to the slave piston lash on cylinders 2, 4, and 6, follow steps 2 through 5.

- Remove the timing bolt from the flywheel after all adjustments have been performed. Reinstall the timing cover on the flywheel housing.

For information that is related to testing the compression brake circuit, refer to Troubleshooting, "Retarder (Compression Brake) Solenoid Circuit - Test".

Refer to Testing and Adjusting, "Electronic Unit Injector - Adjust" for information that relates to the adjustment of the electronic unit injector.

i01126056

# Control Valve (Compression Brake) - Test

SMCS Code: 1119-081-V4

## Oil Pressure Test

Ensure that the engine oil pressure is within the correct operating range before performing any tests on the compression brake. Low engine oil pressure can cause weak braking or no braking at all. High engine oil pressure may cause the compression brake not to operate at all. Refer to Testing and Adjusting, "Engine Oil Pressure - Test".

To properly troubleshoot the compression brake, the technician must know the oil pressure reading at the compression brake housings. The oil pressure reading on the oil pressure gauge in the instrument panel does not provide the same information as the oil pressure reading at the compression brake housings.

Low oil pressure in the compression brake housings may cause weak braking or no braking at all. If the oil pressure in the compression brake housings is too high, the compression brake may not operate at all. High oil pressure may also prevent the compression brake from shutting off.

## Oil Pressure Test at Control Valve

### WARNING

Work carefully around an engine that is running. Engine parts that are hot, or parts that are moving, can cause personal injury.

Table 27

| Tools Needed |                       |          |
|--------------|-----------------------|----------|
| Part Number  | Part Name             | Quantity |
| 149-6111     | Oil Pressure Test Kit | 1        |

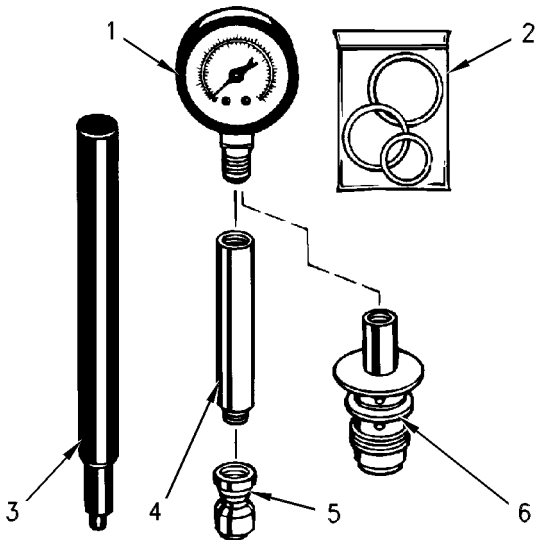


Illustration 99

g00514378

### 149-6111 Oil Pressure Test Kit

- (1) Pressure gauge
- (2) Seals
- (3) Control valve removal tool
- (4) Control valve test body
- (5) Control valve adapter
- (6) Solenoid valve adapter

1. Assemble pressure gauge (1), control valve test body (4), and control valve adapter (5). Hand tighten the adapter assembly.
2. While the engine is not running, remove the control valve. Refer to Disassembly and Assembly, "Compression Brake - Disassemble".

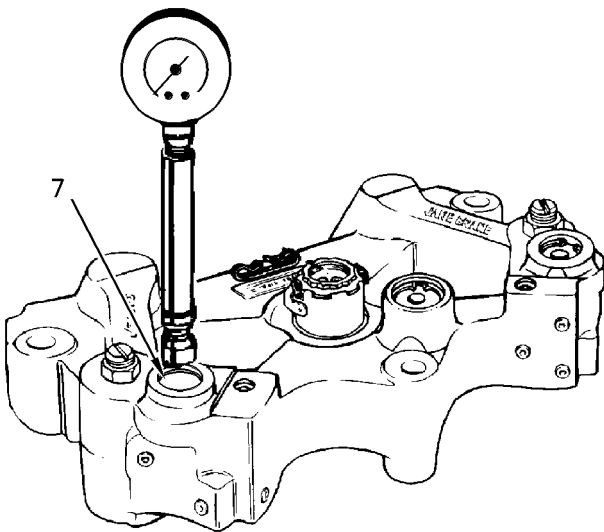


Illustration 100

g00514468

i01126281

3. Insert the adapter assembly into bore (7) for the control valve. Hold the adapter assembly approximately 3.1750 mm (0.125 inch) above the bottom of the bore in order to allow the oil to enter the adapter assembly.
4. Start the engine. Ensure that the switch for the compression brake is in the OFF position.

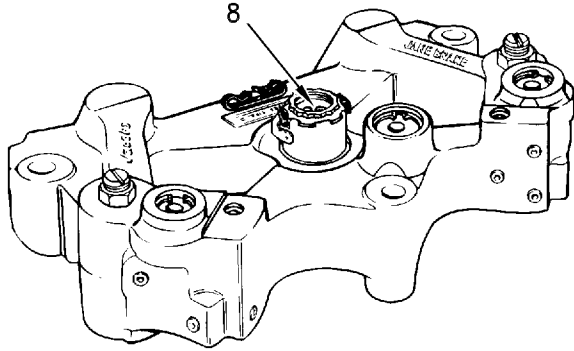


Illustration 101

g00514472

5. Hold the adapter assembly in place and depress solenoid armature (8). Record the oil pressure reading.

**Note:** Hold the solenoid on only long enough to obtain the oil pressure reading. The oil pressure may be too low at idle in order to operate the compression brake. This condition is normal.

6. Increase the engine speed by 400 rpm.
7. Repeat Steps 5 and 6 up to the rated engine rpm.
8. Stop the engine. Install the control valve. Refer to Disassembly and Assembly, "Compression Brake - Assemble".

The oil pressure readings at the control valve above 1000 rpm should be  $275 \pm 105$  kPa ( $40 \pm 15$  psi).

- If the oil pressure readings at the control valve are within the specified range, there is not a problem with the oil pressure in the compression brake.
- If the oil pressure readings at the control valve are below the specified range, proceed to Testing and Adjusting, "Solenoid Valve (Compression Brake) - Test".
- If the oil pressure readings at the control valve are above the specified range, the engine oil pressure is too high. Refer to Testing and Adjusting, "Engine Oil Pressure - Test".

## Solenoid Valve (Compression Brake) - Test

**SMCS Code:** 1119-081-JV

### Oil Pressure Test

Ensure that the engine oil pressure is within the correct operating range before performing any tests on the compression brake. Low engine oil pressure can cause weak braking or no braking at all. High engine oil pressure may cause the compression brake not to operate at all. Refer to Testing and Adjusting, "Engine Oil Pressure - Test".

To properly troubleshoot the compression brake, the technician must know the oil pressure reading at the compression brake housings. The oil pressure reading on the oil pressure gauge in the instrument panel does not provide the same information as the oil pressure reading at the compression brake housings.

Low oil pressure in the compression brake housings may cause weak braking or no braking at all. If the oil pressure in the compression brake housings is too high, the compression brake may not operate at all. High oil pressure may also prevent the compression brake from shutting off.

### Oil Pressure Test at Solenoid Valve

#### **WARNING**

**Work carefully around an engine that is running. Engine parts that are hot, or parts that are moving, can cause personal injury.**

Table 28

| Tools Needed |                       |          |
|--------------|-----------------------|----------|
| Part Number  | Part Name             | Quantity |
| 149-6111     | Oil Pressure Test Kit | 1        |

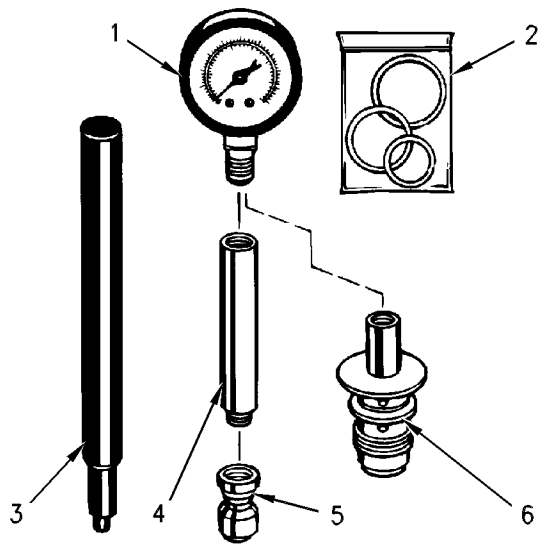


Illustration 102

g00514378

**149-6111 Oil Pressure Test Kit**

- (1) Pressure gauge
- (2) Seals
- (3) Control valve removal tool
- (4) Control valve test body
- (5) Control valve adapter
- (6) Solenoid valve adapter

1. Attach pressure gauge (1) to solenoid valve adapter (6).
2. Remove the solenoid valve and the three seals from the compression brake housing. Refer to Disassembly and Assembly, "Compression Brake - Disassemble".
3. Install three seals (2), which are included in the **149-6111** Oil Pressure Test Kit. Place the smallest seal in the bottom of the bore for the solenoid. Install the other two seals on solenoid valve adapter (6).

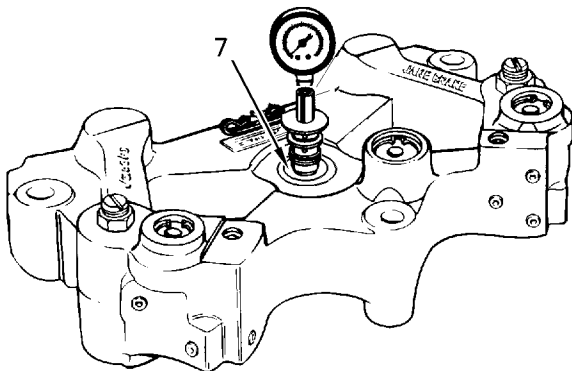


Illustration 103

g00514470

4. Lubricate the seals with clean engine oil and install the adapter assembly in bore (7) for the solenoid valve. Hand tighten the adapter assembly and then tighten the adapter assembly for an additional 1/4 turn.
5. Start the engine. Ensure that the switch for the compression brake is in the OFF position.
6. Record the oil pressure reading.
7. Increase the engine speed by 400 rpm.
8. Repeat Steps 6 and 7 up to the rated engine rpm.
9. Stop the engine. Install the solenoid valve. Refer to Disassembly and Assembly, "Compression Brake - Assemble".

Compare the oil pressure readings from the solenoid valve to the oil pressure readings from the Testing and Adjusting, "Control Valve (Compression Brake) - Test".

- If the oil pressure at the solenoid valve is higher than the oil pressure at the control valve, continue with Steps 10 through 15.
- If the oil pressure at the solenoid valve is the same pressure as the oil pressure at the control valve, the oil pressure to the compression brake is too low. Refer to Testing and Adjusting, "Engine Oil Pressure - Test".

10. Start the engine.
11. Disconnect the wires from each solenoid valve except for the solenoid valve that is being tested in order to check each solenoid valve individually.

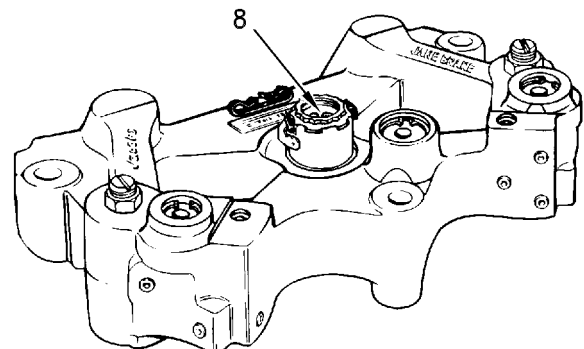


Illustration 104

g00514472

- 12.** Turn the switch for the compression brake to the ON position. Observe the movement of solenoid armature (8). The solenoid armature should snap down fully. Push down on the solenoid armature in order to verify that the solenoid is fully actuated.

If the solenoid valve fully actuates, proceed to Step 14. Otherwise, continue to Step 13.

- 13.** Check the voltage at the terminals on the solenoid valve. There should be at least ten volts at the terminals.

Insufficient voltage at the terminals indicates that there are either poor connections in the circuit or that battery voltage is low. Verify proper battery voltage. Check the condition of the connections and clean the connections.

Replace the solenoid valve if both of the following conditions exist:

- The solenoid valve is receiving the correct voltage.
- The solenoid valve does not actuate fully.

- 14.** Look for excessive oil leakage from the bottom of the compression brake housing opposite the solenoid valve. While the solenoid valve is electrically actuated, leakage should be no more than two drops per second. Replace the solenoid valve if there is excessive leakage.

**Note:** When the solenoid valve is deactivated, oil will dump from the solenoid valve. This is a normal condition that should not be confused with oil leakage while the solenoid valve is electrically actuated.

- 15.** Repeat Steps 11 through 14 for each solenoid valve.

# Electrical System

i01126605

## Battery - Test

**SMCS Code:** 1401-081

Most of the tests of the electrical system can be done on the engine. The wiring insulation must be in good condition. The wire and cable connections must be clean, and both components must be tight.

### **WARNING**

**Never disconnect any charging unit circuit or battery circuit cable from the battery when the charging unit is operated. A spark can cause an explosion from the flammable vapor mixture of hydrogen and oxygen that is released from the electrolyte through the battery outlets. Injury to personnel can be the result.**

---

The battery circuit is an electrical load on the charging unit. The load is variable because of the condition of the charge in the battery.

---

### **NOTICE**

The charging unit will be damaged if the connections between the battery and the charging unit are broken while the battery is being charged. Damage occurs because the load from the battery is lost and because there is an increase in charging voltage. High voltage will damage the charging unit, the regulator, and other electrical components.

---

See Special Instruction, SEHS7633, "Battery Test Procedure" for the correct procedures to use to test the battery. This publication also contains the specifications to use when you test the battery.

i01660532

## Belt Tension Chart

**SMCS Code:** 1357

**Note:** Do not use the belt tension charts for belts with tensioners that are spring loaded.

Table 29

| Belt Tension Chart                                                |                       |                                       |                                       |                          |                         |
|-------------------------------------------------------------------|-----------------------|---------------------------------------|---------------------------------------|--------------------------|-------------------------|
| Size of Belt                                                      | Width of Belt         | Gauge Reading                         |                                       | Kent-Moore Gauge Numbers |                         |
|                                                                   |                       | Belt Tension "Initial" <sup>(1)</sup> | Belt Tension "Used" <sup>(2)(3)</sup> | Number of the Old Gauge  | Number of the New Gauge |
| 3/8                                                               | 10.72 mm (0.422 inch) | 534 ± 111 N (120 ± 25 lb)             | 400 ± 22 N (90 ± 5 lb)                | BT-33-97                 | BT 3397                 |
| 1/2                                                               | 13.89 mm (0.547 Inch) | 578 ± 111 N (130 ± 25 lb)             | 445 ± 44 N (100 ± 10 lb)              | BT-33-97                 | BT 3397                 |
| 5V                                                                | 15.88 mm (0.626 Inch) | 712 ± 111 N (160 ± 25 lb)             | 445 ± 44 N (100 ± 10 lb)              | BT-33-72C                | BT 3372C                |
| 11/16                                                             | 17.48 mm (0.688 Inch) | 712 ± 111 N (160 ± 25 lb)             | 445 ± 44 N (100 ± 10 lb)              | BT-33-72C                | BT 3372C                |
| 3/4                                                               | 19.05 mm (0.750 Inch) | 712 ± 111 N (160 ± 25 lb)             | 445 ± 44 N (100 ± 10 lb)              | BT-33-72C                | BT 3372C                |
| 15/16                                                             | 23.83 mm (0.938 Inch) | 712 ± 111 N (160 ± 25 lb)             | 445 ± 44 N (100 ± 10 lb)              | BT-33-77                 | BT 3372C                |
| 8PK                                                               | 27.82 mm (1.095 inch) | 1068 ± 111 N (240 ± 25 lb)            | 890 ± 44 N (200 ± 10 lb)              | BT-33-109                | BT 33109                |
| 6PK                                                               | 20.94 mm (0.824 Inch) | 801 ± 111 N (180 ± 25 lb)             | 667 ± 44 N (150 ± 10 lb)              | BT-33-109                | BT 33109                |
| Measure the tension of the belt that is farthest from the engine. |                       |                                       |                                       |                          |                         |

(1) Belt tension "Initial" is for a new belt.

(2) Belt tension "Used" is for a belt that has operated for 30 minutes or more of operation at the rated speed.

(3) If a belt falls below the "Used" belt tension, the belt should be tightened again to the high side of the "Used" belt tension.

Table 30

| DAYCO Supplied Belts Only |                          |                                    |
|---------------------------|--------------------------|------------------------------------|
| Size of Belt              | Belt Tension "Initial"   | Belt Tension "Used" <sup>(4)</sup> |
| 0.380 (V-Belt)            | 623 ± 22 N (140 ± 5 lb)  | 245 - 534 N (100 ± 5 lb)           |
| 0.440 (V-Belt)            | 667 ± 22 N (150 ± 5 lb)  | 245 - 534 N (100 ± 5 lb)           |
| 0.500 (V-Belt)            | 712 ± 22 N (160 ± 5 lb)  | 245 - 534 N (100 ± 5 lb)           |
| 0.600 (V-Belt)            | 779 ± 22 N (175 ± 5 lb)  | 245 - 534 N (100 ± 5 lb)           |
| 0.660 (V-Belt)            | 890 ± 22 N (200 ± 5 lb)  | 245 - 534 N (100 ± 5 lb)           |
| 0.790 (V-Belt)            | 890 ± 22 N (200 ± 5 lb)  | 245 - 534 N (100 ± 5 lb)           |
| 4 - RIB PVK               | 623 ± 22 N (140 ± 5 lb)  | 267 - 445 N (90 ± 5 lb)            |
| 5 - RIB PVK               | 779 ± 22 N (175 ± 5 lb)  | 334 - 556 N (100 ± 5 lb)           |
| 6 - RIB PVK               | 934 ± 22 N (210 ± 5 lb)  | 400 - 667 N (130 ± 5 lb)           |
| 8 - RIB PVK               | 1157 ± 22 N (260 ± 5 lb) | 534 - 890 N (180 ± 5 lb)           |
| 10 - RIB PVK              | 1557 ± 22 N (350 ± 5 lb) | 667 - 1112 N (230 ± 5 lb)          |
| 12 - RIB PVK              | 1869 ± 22 N (420 ± 5 lb) | 800 - 1335 N (270 ± 5 lb)          |
| 15 - RIB PVK              | 2336 ± 22 N (525 ± 5 lb) | 1000 - 1669 N (350 ± 5 lb)         |

(4) If a belt falls below the "Used" belt tension, the belt should be tightened again to the "Initial" belt tension.

i01852470

# Charging System - Test

**SMCS Code:** 1406-081

The condition of charge in the battery at each regular inspection will show if the charging system is operating correctly. An adjustment is necessary when the battery is constantly in a low condition of charge or a large amount of water is needed. A large amount of water would be more than one ounce of water per a cell per a week or per every 100 service hours.

When it is possible, make a test of the charging unit and voltage regulator on the engine, and use wiring and components that are a permanent part of the system. Off-engine testing or bench testing will give a test of the charging unit and voltage regulator operation. This testing will give an indication of needed repair. After repairs are made, perform a test in order to prove that the units have been repaired to the original condition of operation.

## Test Tools For The Charging System

Table 31

| Tools Needed              |                                                |          |
|---------------------------|------------------------------------------------|----------|
| Part Number               | Part Name                                      | Quantity |
| 8T-0900                   | Ammeter                                        | 1        |
| 6V-7070<br>or<br>146-4080 | Digital Multimeter<br>or<br>Digital Multimeter | 1        |

### 8T-0900 Ammeter

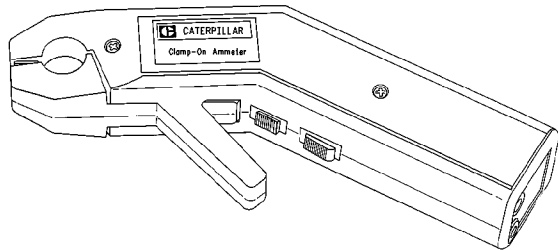


Illustration 105  
8T-0900 Ammeter

g00296441

The **8T-0900** Ammeter is completely portable. This ammeter is a self-contained instrument that measures electrical currents without breaking the circuit and without disturbing the conductor's insulation.

The ammeter contains a digital display that is used to monitor current directly within a range between 1 ampere and 1200 amperes. If an optional **6V-6014** Cable is connected between this ammeter and a digital multimeter, current readings can be viewed directly from the display of the multimeter. This can be accomplished under only one condition:

- the readings are less than 1 ammeter.

A lever opens the ammeter's jaws over a conductor. The conductor's diameter can not be larger than 19 mm (0.75 inch).

The spring loaded jaws close around the conductor for measuring the current. A trigger switch controls the ammeter. The trigger switch can be locked into the ON position or into the OFF position.

After the trigger has been working and the trigger is turned to the OFF position, the reading appears in the digital display for five seconds. This accurately measures currents in areas with a limited access. For example, these areas include areas that are beyond the operator's sight. For DC operation, an ammeter contains a zero control, and batteries inside the handle supply the power.

**Note:** Refer to Special Instruction, SEHS8420 for more information about using the **8T-0900** Ammeter.

### 6V-7070 Digital Multimeter or 146-4080 Digital Multimeter

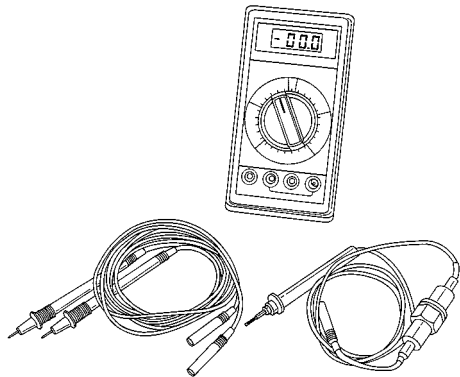


Illustration 106  
6V-7070 Digital Multimeter or 146-4080 Digital Multimeter

g00283566

i01530390

The **6V-7070** Digital Multimeter or the **146-4080** Digital Multimeter is a hand-held service tool with a digital display, that is completely portable. This multimeter is built with extra protection against damage in field applications. The multimeter is equipped with 7 functions and 29 ranges. The **6V-7070** Digital Multimeter or the **146-4080** Digital Multimeter has an instant ohms indicator. This indicator permits checking continuity for a fast inspection of the circuits. The multimeter can also be used for troubleshooting capacitors that have small values.

**Note:** Refer to Special Instruction, SEHS7734 for complete information for the use of the **6V-7070** Digital Multimeter. Refer to Operation Manual, NEHS0678 for complete information for the use of the **146-4080** Digital Multimeter.

i01833081

## Electric Starting System - Test

**SMCS Code:** 1450-081

Most of the tests of the electrical system can be done on the engine. The wiring insulation must be in good condition. The wire and cable connections must be clean, and both components must be tight. The battery must be fully charged. If the on-engine test shows a defect in a component, remove the component for more testing.

The starting system consists of the following four components:

- Keyswitch
- Start relay
- Starting motor solenoid
- Starting motor

Trouble with the starting system could be caused by the battery or by charging system problems. If the starting system is suspect, refer to Service Manual, SENR3581, "37-MT, 41-MT & 42-MT Series Starting Motors". This publication contains troubleshooting for the starting system, test procedures, and specifications.

## Engine Oil Pressure Sensor - Test

**SMCS Code:** 1924-081

Refer to the Troubleshooting, RENR2238, "Engine Oil Pressure Sensor Open Or Short Circuit Test" for information on checking the engine oil pressure sensor.

i01357620

## Pinion Clearance - Adjust

**SMCS Code:** 1454-025

When the solenoid is installed, make an adjustment of the pinion clearance. The adjustment should be made with the starting motor removed.

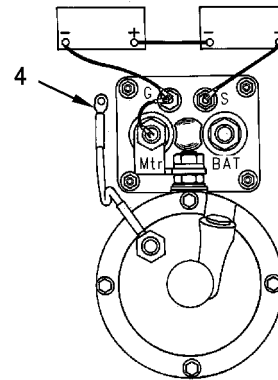


Illustration 107

g00716453

Connections for pinion clearance check

(4) Starting motor negative wire

1. Disconnect the starting motor negative wire (4) from the "G" ground terminal of the solenoid.
2. Connect the batteries to the solenoid per Illustration 107. The Illustration shows a 24 volt system with two 12 volt batteries that are connected in series to a starting motor. Connect four 8 volt batteries in series for a 32 volt system. Connect eight 8 volt batteries in series for a 64 volt system. Connect the positive side of the battery cable to the "S" terminal of the starting motor solenoid. Connect the negative side of the battery cable to the "G" terminal of the starting motor .
3. Temporarily, touch a wire from the "G" terminal to the "Mtr" terminal. The pinion will shift to the crank position and the pinion will stay there until the battery is disconnected.

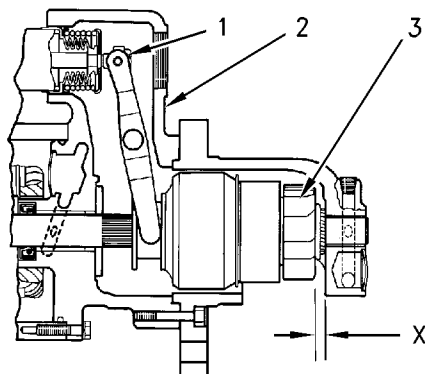


Illustration 108

g00716455

Pinion clearance check

- (1) Adjustment nut
- (2) Pinion drive housing
- (3) Pinion
- (X) pinion clearance

4. Push the pinion toward the commutator in order to remove free movement.
5. Measure the clearance (X) from the pinion to the pinion drive housing. Pinion clearance (X) must be  $9.10 \pm 0.8$  mm ( $0.36 \pm 0.03$  inch).
6. If the clearance is not correct, remove the plug on the shift lever housing (2). Turn the adjustment nut (1) until the clearance is correct. Turning the nut clockwise will decrease the clearance (X).

**Note:** The plunger may turn when the adjustment nut (1) is being turned. If the plunger turns, disconnect the battery from the solenoid. Remove the solenoid from the starting motor. Hold the plunger from turning and adjust the nut (1). This procedure may need to be performed several times until the correct clearance is obtained.

7. Disconnect the batteries and install the plug into the shift lever housing.

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# Disassembly and Assembly

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## **C-10 and C-12 Truck Engines**

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MBJ1-Up (Engine)  
MBL1-Up (Engine)



## Important Safety Information

Most accidents that involve product operation, maintenance and repair are caused by failure to observe basic safety rules or precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards. This person should also have the necessary training, skills and tools to perform these functions properly.

**Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.**

**Do not operate or perform any lubrication, maintenance or repair on this product, until you have read and understood the operation, lubrication, maintenance and repair information.**

Safety precautions and warnings are provided in this manual and on the product. If these hazard warnings are not heeded, bodily injury or death could occur to you or to other persons.

The hazards are identified by the "Safety Alert Symbol" and followed by a "Signal Word" such as "DANGER", "WARNING" or "CAUTION". The Safety Alert "WARNING" label is shown below.



The meaning of this safety alert symbol is as follows:

**Attention! Become Alert! Your Safety is Involved.**

The message that appears under the warning explains the hazard and can be either written or pictorially presented.

Operations that may cause product damage are identified by "NOTICE" labels on the product and in this publication.

**Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. If a tool, procedure, work method or operating technique that is not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that the product will not be damaged or be made unsafe by the operation, lubrication, maintenance or repair procedures that you choose.**

The information, specifications, and illustrations in this publication are on the basis of information that was available at the time that the publication was written. The specifications, torques, pressures, measurements, adjustments, illustrations, and other items can change at any time. These changes can affect the service that is given to the product. Obtain the complete and most current information before you start any job. Caterpillar dealers have the most current information available.



**When replacement parts are required for this product Caterpillar recommends using Caterpillar replacement parts or parts with equivalent specifications including, but not limited to, physical dimensions, type, strength and material.**

**Failure to heed this warning can lead to premature failures, product damage, personal injury or death.**

---

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## Disassembly and Assembly Section

### Fuel Priming Pump - Remove and Install

i01468998

**SMCS Code:** 1258-010

#### Removal Procedure

##### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

##### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

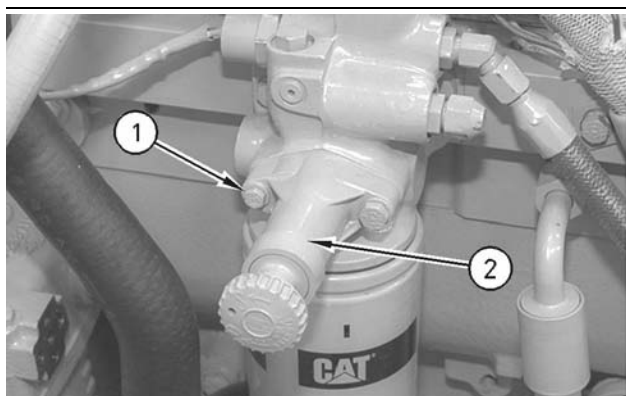


Illustration 1

g00511626

Typical example

1. Remove two bolts (1) and the washers.
2. Remove fuel priming pump (2) and the gasket from the fuel filter base.

### Installation Procedure

##### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

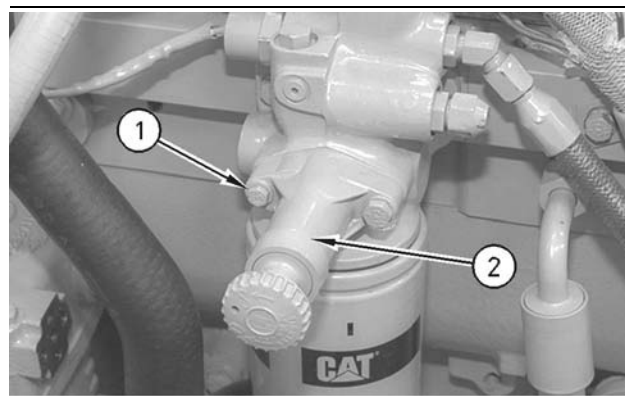


Illustration 2

g00511626

Typical example

1. Inspect the condition of the gasket. Replace the gasket, if necessary.
2. Ensure that both valve assemblies are located in the correct position in the fuel priming pump.
3. Install the gasket and fuel priming pump (2) on the fuel filter base.
4. Install the washers and bolts (1).
5. Check the operation of the fuel priming pump.

i01015037

### Fuel Filter Base - Remove and Install

**SMCS Code:** 1262-010

#### Removal Procedure

Table 1

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 2P-8250     | Strap Wrench Assembly | 1   |

##### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

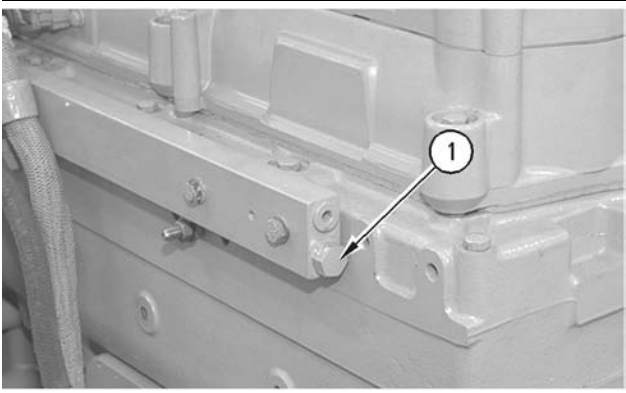


Illustration 3 g00519561

1. Remove plug (1) and drain fuel into a suitable container for storage or disposal.

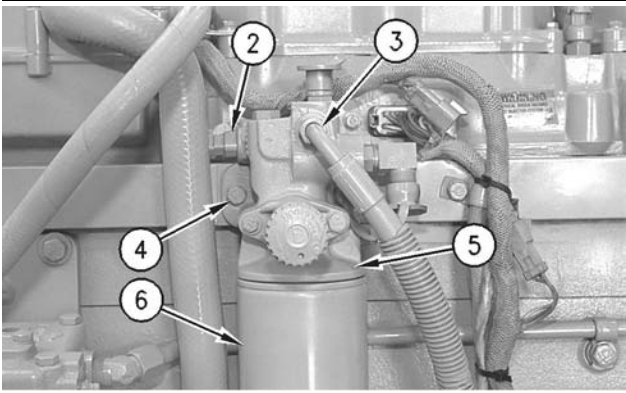


Illustration 4 g00520760  
Typical example

2. Use Tool (A) to remove fuel filter (6).
3. Disconnect the sensor connector from fuel temperature sensor (2).
4. Remove fuel temperature sensor (2).

5. Disconnect hose assembly (3). Plug all openings immediately.
6. Remove two bolts (4) and the washers.
7. Remove fuel filter base (5) and the two O-ring seals.
8. Remove the fuel filter mounting stud from the fuel filter base, if necessary.

**Installation Procedure**

Table 2

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 2P-8250     | Strap Wrench Assembly | 1   |

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Apply 9S-3263 Thread Lock Compound to the threads of the fuel filter mounting stud. Apply the compound to the tapered end of the stud no more than 7.6 mm (.30 inch) from the end.
2. Install the stud and tighten to a torque of  $70 \pm 15$  N·m ( $50 \pm 11$  lb ft).
3. Inspect the condition of the two O-ring seals. Replace the seals, if necessary.

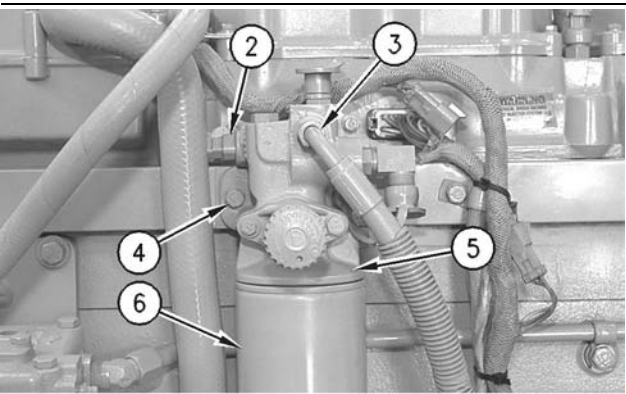


Illustration 5 g00520760  
Typical example

4. Install the two O-rings on fuel filter base (5).
5. Install fuel filter base (5) on the fuel manifold with two bolts (4) and the washers.

6. Connect hose assembly (3).
7. Install fuel temperature sensor (2). Connect the sensor connector to fuel temperature sensor (2).
8. Use Tool (A) in order to install new fuel filter (6). Follow the instructions on the fuel filter.

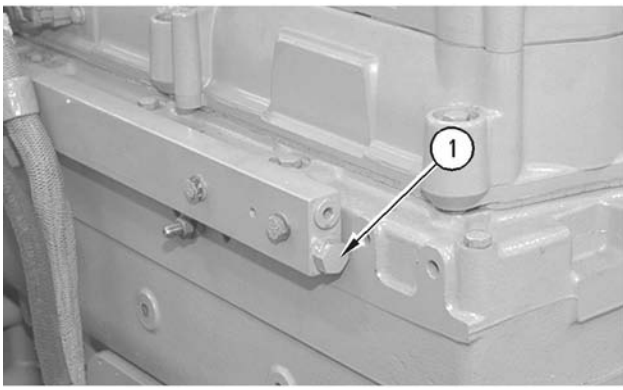


Illustration 6

g00519561

9. Install plug (1).

i01781003

## Fuel Recirculation Valve - Remove and Install

**SMCS Code:** 135V-010

### Removal Procedure

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

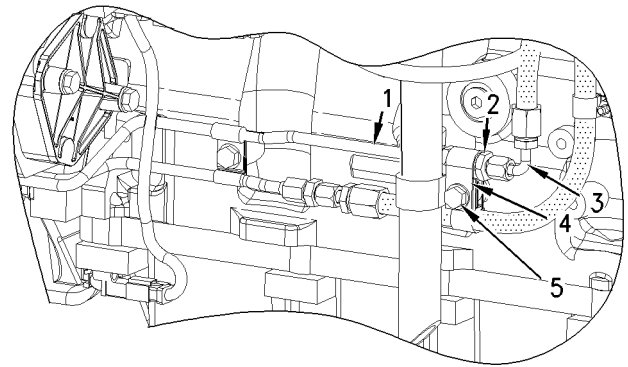


Illustration 7

g00912217

1. Remove bolt (5). Remove clamp (4).
2. Disconnect line (1) from valve (2).
3. Disconnect hose assembly (3) from valve (2).
4. Remove valve (2).

### Installation Procedure

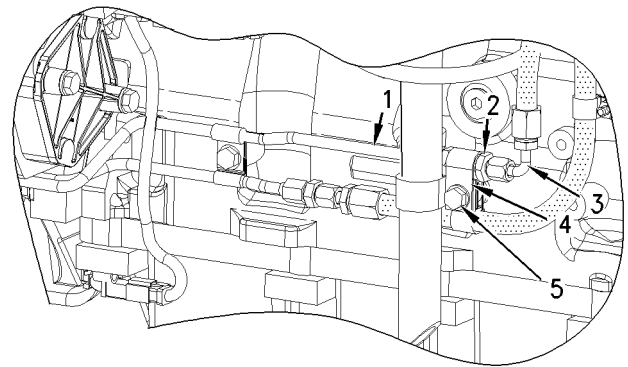


Illustration 8

g00912217

1. Connect valve (2) to line (1).
2. Connect hose assembly (3) to valve (2).
3. Install clamp (4). Secure clamp (4) with bolt (5).

i00999915

## Fuel Transfer Pump - Remove

**SMCS Code:** 1256-011

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

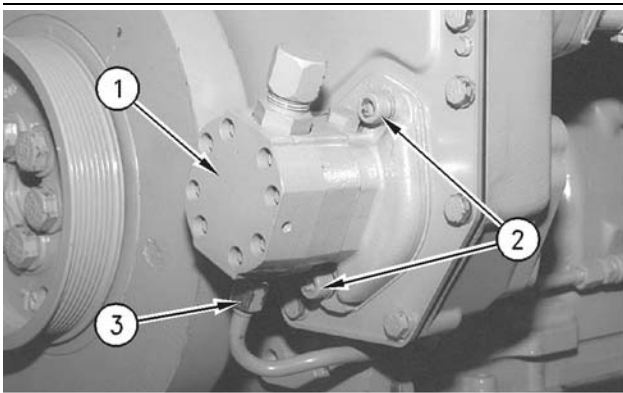


Illustration 9

g00511822

1. Disconnect tube assembly (3). Cover all openings immediately.
2. Remove two bolts (2) and the seal washers.
3. Remove fuel transfer pump (1) and the gasket.

i01770561

## Fuel Transfer Pump - Install

**SMCS Code:** 1256-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

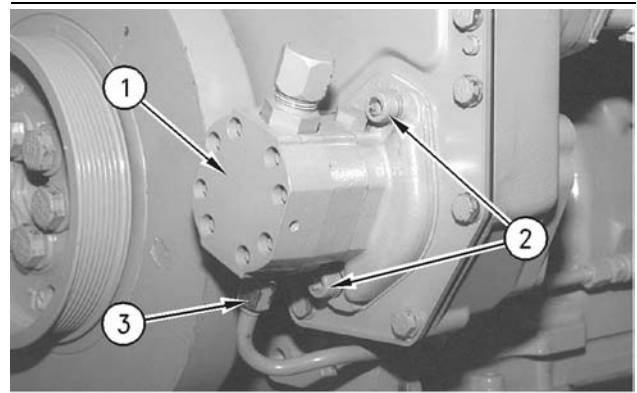


Illustration 10

g00511822

1. Inspect the condition of the gasket. Replace the gasket, if necessary.
  2. Position fuel transfer pump (1) and the gasket on the front housing cover.
- Note:** Do not reuse the seal washers. Use new seal washers during installation.
3. Install bolts (2) and the new seal washers. Tighten the bolts to a torque of  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft).
  4. Connect tube assembly (3).

i01633853

## Fuel Manifold - Remove

**SMCS Code:** 1702-011

### Removal Procedure

#### Start By:

- a. Remove the fuel filter base. Refer to Disassembly and Assembly, "Fuel Filter Base - Remove and Install".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

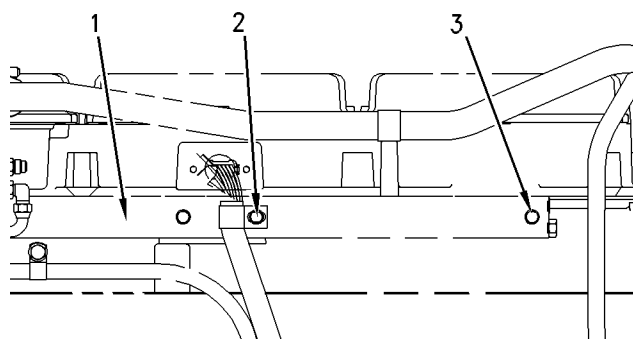


Illustration 11

g00846247

Typical example

1. Remove bolt (2) and move the wiring harness out of the way.
2. Remove bolts (3) and the washers.
3. Remove fuel manifold (1).

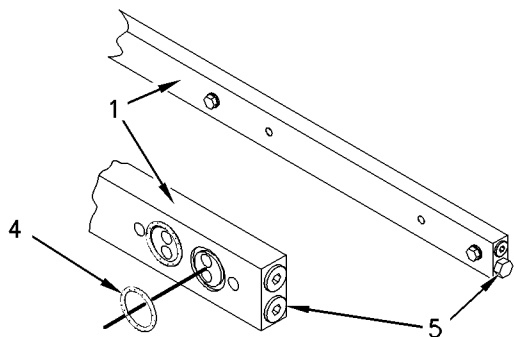


Illustration 12

g00846262

4. Remove six O-ring seals (4).

5. Remove plugs (5) and the O-ring seals, if necessary.

i01633987

## Fuel Manifold - Install

**SMCS Code:** 1702-012

### Installation Procedure

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the O-ring seals. Replace the seals, if necessary.

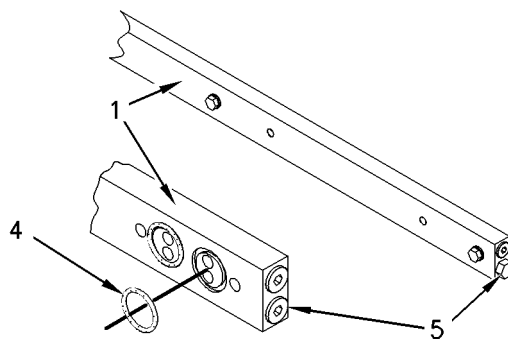


Illustration 13

g00846262

2. Install plugs (5) and the O-ring seals.
3. Install six O-ring seals (4) on fuel manifold (1).

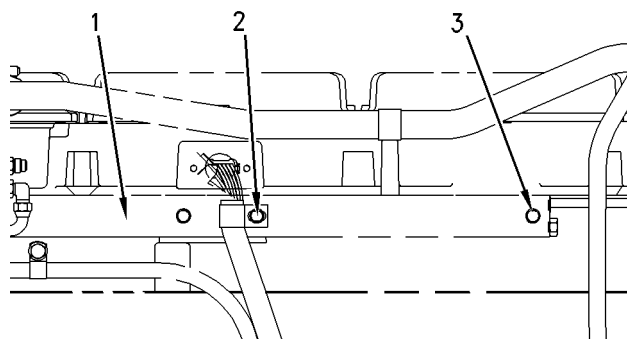


Illustration 14

g00846247

4. Install fuel manifold (1) on the cylinder head with bolts (3) and the washers.

5. Install the wiring harness with bolt (2). Tighten all of the bolts to a torque of  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft).

**End By:**

- a. Install the fuel filter base. Refer to Disassembly and Assembly, "Fuel Filter Base - Remove and Install".

i01770576

# Electronic Unit Injector - Remove

**SMCS Code:** 1290-011

## Removal Procedure

Table 3

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 5F-4764     | Pry Bar          | 1   |

**Start By:**

- a. Remove the rocker shaft and the pushrods. Refer to Disassembly and Assembly, "Rocker Shaft and Pushrod - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

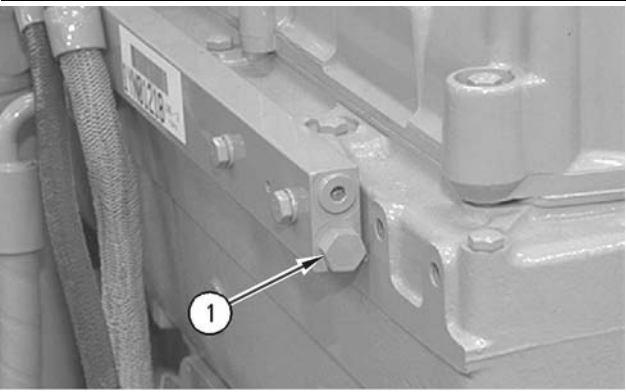


Illustration 15

g00511865

1. Remove plug (1) from the fuel manifold.
2. Drain the fuel from the cylinder head assembly into a suitable container for storage or disposal.

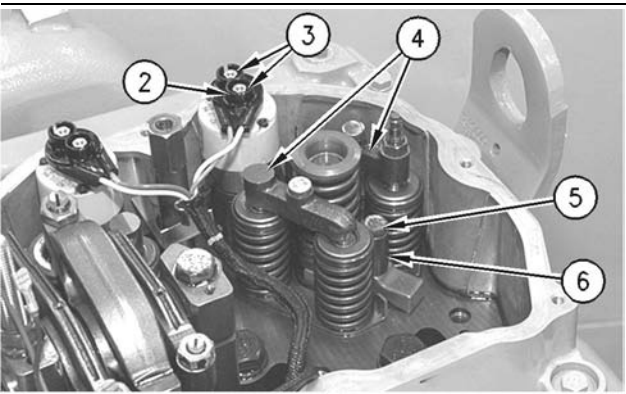


Illustration 16

g00511868

3. Mark bridge assemblies (4) for reference during installation.
4. Loosen nuts (3) and remove cap assembly (2).
5. Remove bolt (5) and spacer assembly (6) that holds the unit injector in place.

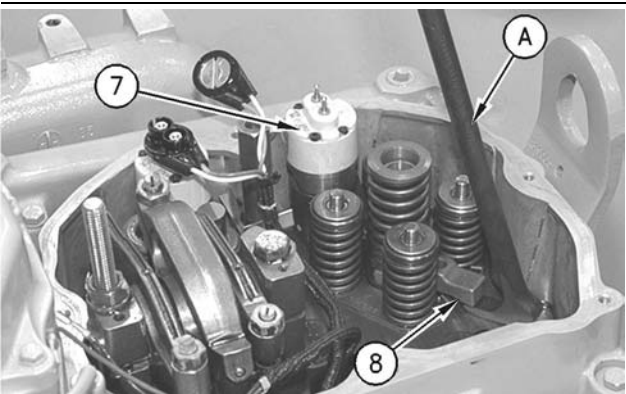


Illustration 17

g00764868

6. Position Tooling (A) under hold-down (8). Carefully pry unit injector (7) out of the cylinder head assembly. Remove unit injector (7).

i01770633

Electronic Unit Injector - Install

SMCS Code: 1290-012

Installation Procedure

Table 4

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 9U-6862     | Tapered Brush    | 1   |
|                | 9U-7237     | Brush Extension  | 1   |
| B              | 9U-6863     | Small Bore Brush | 1   |
|                | 9U-7237     | Brush Extension  | 1   |

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

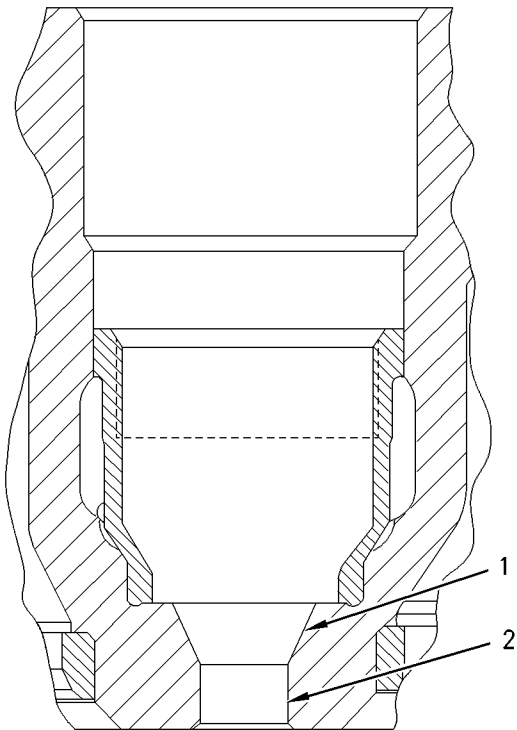


Illustration 18

g00511917

- (1) Injector seating  
(2) Tip bore

1. Use Tooling (A) to clean injector seating (1).  
2. Use Tooling (B) to clean tip bore (2).

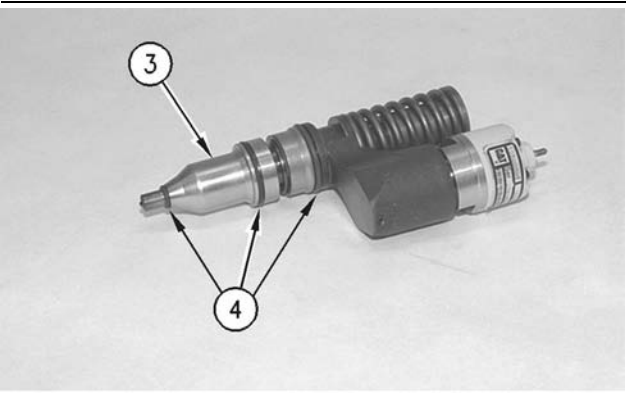


Illustration 19

g00511927

**Note:** Do not reuse seals (4). Use new seals for installation.

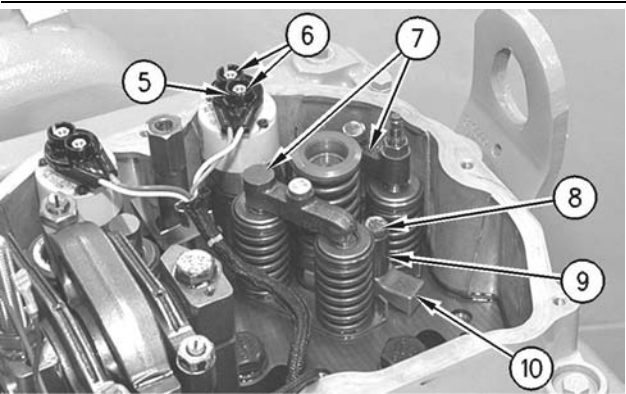


Illustration 20

g00511929

3. Lubricate the bore lightly with a 50/50 mix of clean engine oil and 8T-2998 Lubricant.

**Note:** The injector code will be used later if the injectors have been changed. Write down the injector code which is located on the solenoid.

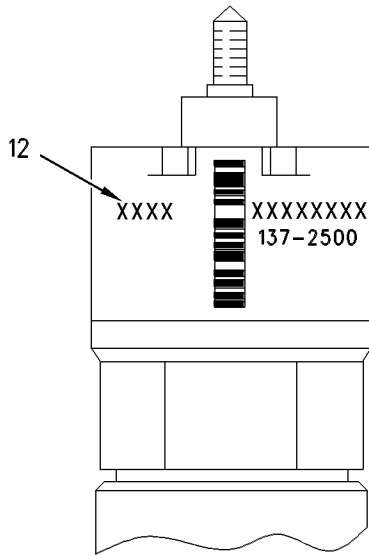


Illustration 21  
(12) Electronic injector code

g00529729

**NOTICE**

When a fuel injector group is serviced, the new fuel injector group's electronic injector code must be programmed into the engine's personality module software by using the calibration menu on the Electronic Service Tool. If the new fuel injector group's electronic code is not entered, the previous fuel injector group's characteristics are assumed.

If it is not possible to immediately reprogram the electronic injector code of the injector into the personality module software, the engine will not be severely harmed. The new electronic injector code should be reprogrammed as quickly as possible in order to optimize engine performance.

4. Position the unit injector (3) in the cylinder head assembly. Place the hold down clamp (10) at the base of the unit injector. Push down on the return spring of the unit injector in order to seat the unit injector.
5. Install spacer assembly (9) and bolt (8) that holds the unit injector in place.
6. Pull the unit injector completely into the bore. Tighten the bolt to a torque of 30 N·m (22 lb ft).
7. Install cap assembly (5). Tighten nuts (6) to a torque of  $2.5 \pm 0.25$  N·m ( $22 \pm 2$  lb in).
8. Install bridge assemblies (7) onto the valve stems.

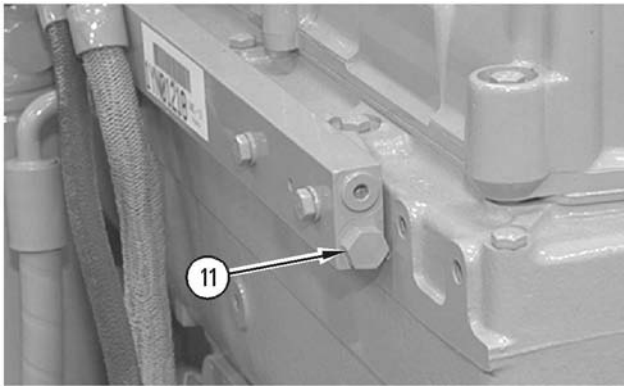


Illustration 22

g00511936

9. Install plug (11) in the fuel manifold.

**End By:**

- a. Install the rocker shaft and the pushrods. Refer to Disassembly and Assembly, "Rocker Shaft and Pushrod - Install".

i01770685

## Electronic Unit Injector Sleeve - Remove

**SMCS Code:** 1713-011

### Removal Procedure

Table 5

| Required Tools   |             |                  |     |
|------------------|-------------|------------------|-----|
| Tool             | Part Number | Part Description | Qty |
| A <sup>(1)</sup> | 221 - 9778  | Puller Stud      | 1   |
|                  | 128 - 7889  | Bridge Puller    | 1   |
|                  | 9U - 6877   | Thrust Bearing   | 1   |
|                  | 5P - 8247   | Hard Washer      | 1   |
|                  | 4K - 0367   | Nut              | 1   |

<sup>(1)</sup> The tools that are listed in the chart are part of the 9U - 6891 Injector Tool Group.

**Start By:**

- a. Remove the electronic unit injectors. Refer to Disassembly and Assembly, "Electronic Unit Injector - Remove".

i01770693

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

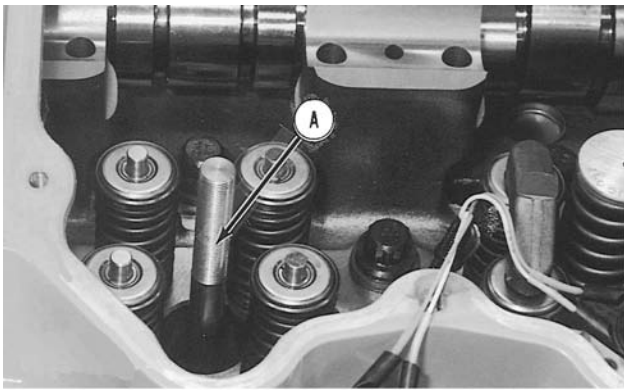


Illustration 23

g00512128

Typical example

1. Install the puller stud from Tooling (A) into the unit injector sleeve.

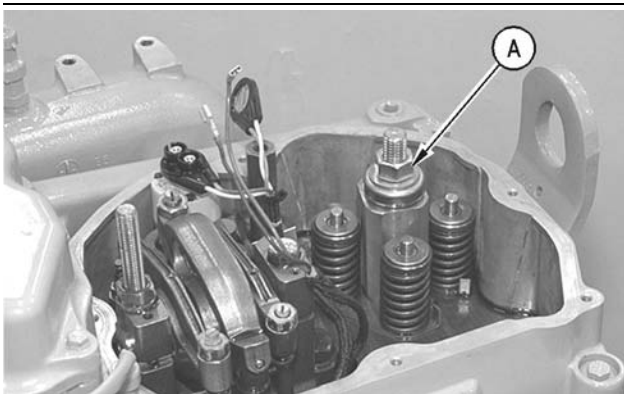


Illustration 24

g00512002

2. Install the following parts from Tooling (A) over the stud: bridge puller, thrust bearing, hard washer, and nut. See Step 1.
3. Tighten the nut until the unit injector sleeve is pulled free of the cylinder head assembly.
4. Remove Tooling (A) from the unit injector sleeve.
5. Repeat Steps 1 through 4 in order to remove the remaining unit injector sleeves.

## Electronic Unit Injector Sleeve - Install

**SMCS Code:** 1713-012

### Installation Procedure

Table 6

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 4C-5552     | Large Bore Brush | 1   |
| B              | 9U-7237     | Brush Extension  | 1   |
|                | 9U-7244     | End Brush        | 1   |
| C              | 9U-7258     | Driver Cap       | 1   |

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

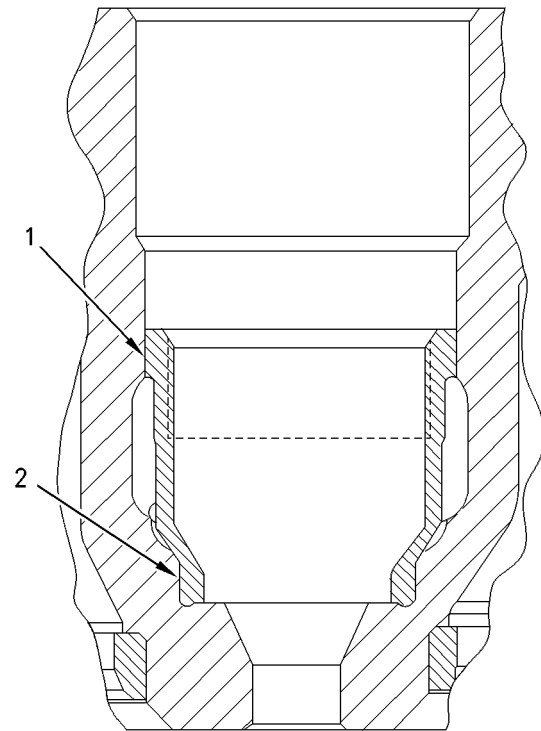


Illustration 25

g00512115

1. Use Tooling (A) to clean the unit injector bore.
2. Use Tooling (B) to clean the unit injector seat.

**Note:** Check the fuel passage in the cylinder head assembly. Remove any debris left from the use of Tooling (A) and Tooling (B).

3. Apply **4C-9507** Retaining Compound to upper land (1) and lower land (2) on the unit injector sleeve.

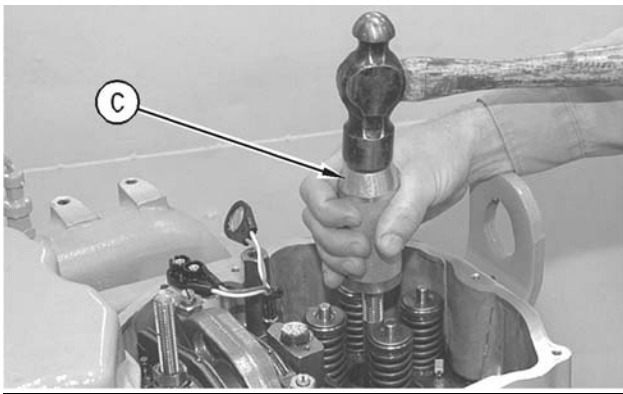


Illustration 26

g00512116

4. Position the unit injector sleeve in the cylinder head assembly. Use Tooling (C) and a hammer to seat the sleeve.
5. Repeat Step 1 through Step 4 in order to install the remaining unit injector sleeves.

#### End By:

- a. Install the electronic unit injectors. Refer to Disassembly and Assembly, "Electronic Unit Injector - Install".

i01781769

## Turbocharger - Remove

**SMCS Code:** 1052-011

### Removal Procedure

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

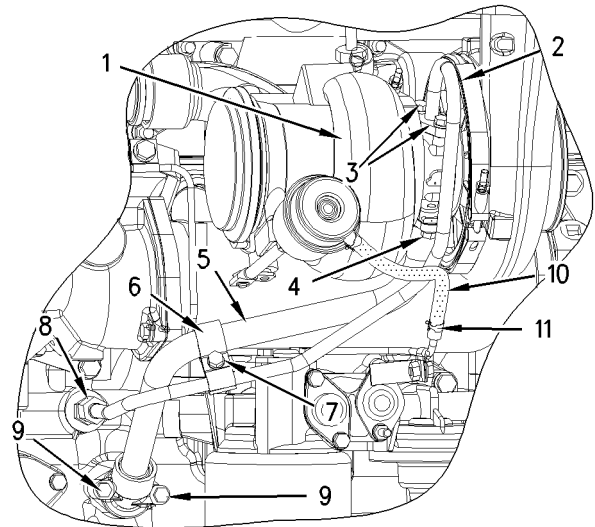


Illustration 27

g00911767

1. Remove bolts (3).
2. Remove bolt (7) with the nut and the washer. Remove saddle clamp (6).
3. Remove bolts (9) and the washers.
4. Disconnect fitting (8).
5. Remove oil supply tube (2) and the gasket.
6. Remove bolts (4) and the washers.
7. Remove oil drain tube (5). Remove the O-ring seal and the gasket from oil drain tube (5).
8. Move hose clamp (11). Disconnect hose (10).

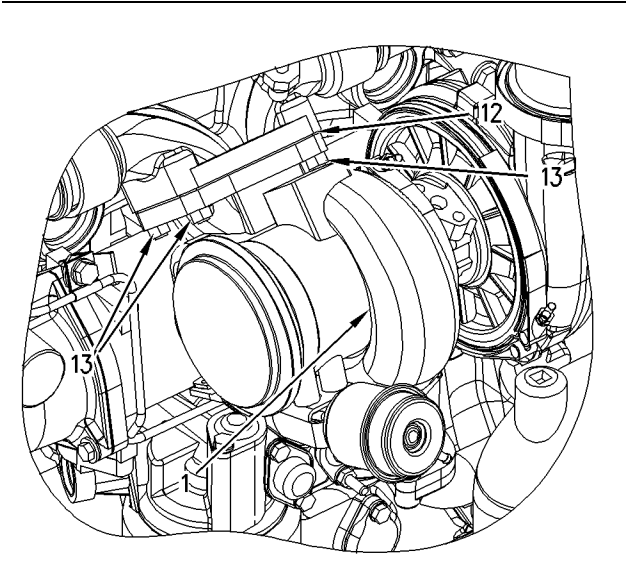


Illustration 28

g00911698

9. Remove four nuts (13).
10. Remove turbocharger (1) and gasket (12) from the exhaust manifold.

Turbocharger - Install

SMCS Code: 1052-012

Installation Procedure

Table 7

| Required Tools |             |                     |     |
|----------------|-------------|---------------------|-----|
| Tool           | Part Number | Part Description    | Qty |
| (A)            | 5P -3931    | Anti-Seize Compound | 1   |

1. Inspect the condition of the gasket between the turbocharger and the exhaust manifold and the O-ring seals. Replace the O-ring seals and the gaskets, if necessary.

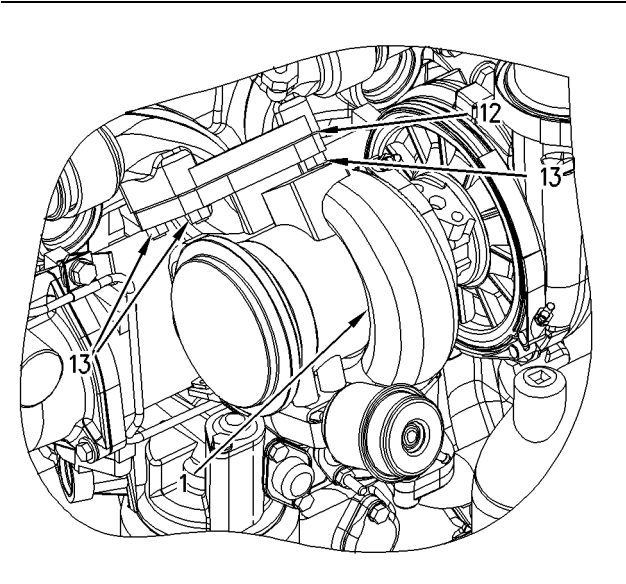


Illustration 29

g00911698

- Note: Refer to the Specifications Manual for the tightening procedure of the clamps on specific turbochargers.
2. Place clean engine oil in the oil supply port of the turbocharger prior to installation. The oil will provide lubrication during starting and protection during storage.
3. Apply Tooling (A) to the turbocharger mounting studs on the exhaust manifold.
4. Install turbocharger (1) and gasket (12) on the exhaust manifold. Install nuts (13). Tighten the nuts to a torque of 70 N·m (52 lb ft).

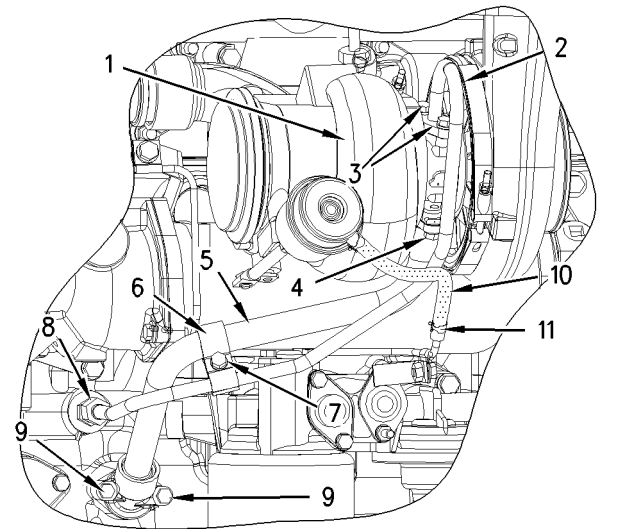


Illustration 30

g00911767

5. Place the gasket and the O-ring seal on oil drain tube (5).
6. Install oil drain tube (5) with two bolts (4) and the washers.
7. Install bolts (9) and the washers.
8. Install oil supply tube (2) and the gasket on turbocharger (1). Install bolts (3) .
9. Connect fitting (8).
10. Install saddle clamp (6) and bolt (7).
11. Install hose assembly (10) onto the tube. Secure the hose assembly with hose clamp (11).

i01781006

## Modulating Valve - Remove and Install

**SMCS Code:** 3139-010

### Removal Procedure

**Note:** Put identification marks on all hose assemblies, on all wires, and on all tube assemblies for installation purposes. Plug all hose assemblies and all tube assemblies. This helps to prevent fluid loss, and this helps to keep contaminants from entering the system.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

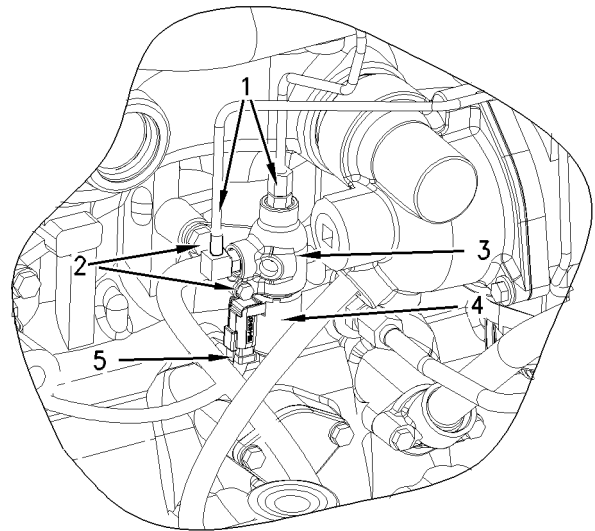


Illustration 31

g00912917

1. Disconnect lines (1) from valve (3).
2. Disconnect wiring harness assembly (5) from solenoid (4).
3. Remove bolts (2). Remove valve (3).
4. Remove solenoid (4) from valve (3).

### Installation Procedure

**Note:** Cleanliness is an important factor. Before assembly, all parts should be thoroughly cleaned in cleaning fluid. Allow the parts to air dry. Wiping cloths or rags should not be used to dry parts. Lint may be deposited on the parts which may cause later trouble. Inspect all parts. If any parts are worn or damaged, use new parts for replacement. All disassembly and all assembly procedures must be performed on a clean work surface and in a clean hydraulic area. Keep cleaned parts covered and protected at all times.

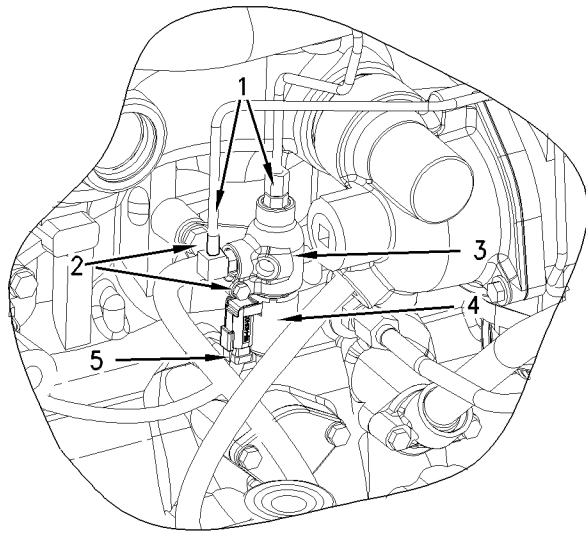


Illustration 32

g00912917

1. Install solenoid (4) onto valve (3). Tighten solenoid (4) to a torque of  $50 \pm 5$  N·m ( $37 \pm 4$  lb ft).
2. Install valve (3) in position. Secure valve (3) with bolts (2).
3. Connect wiring harness assembly (5) to solenoid (4).
4. Connect lines (1) onto valve (3).

i01465377

## Exhaust Manifold - Remove and Install

**SMCS Code:** 1059-010

### Removal Procedure

#### Start By:

- a. Remove the turbocharger. Refer to Disassembly and Assembly, "Turbocharger - Remove".

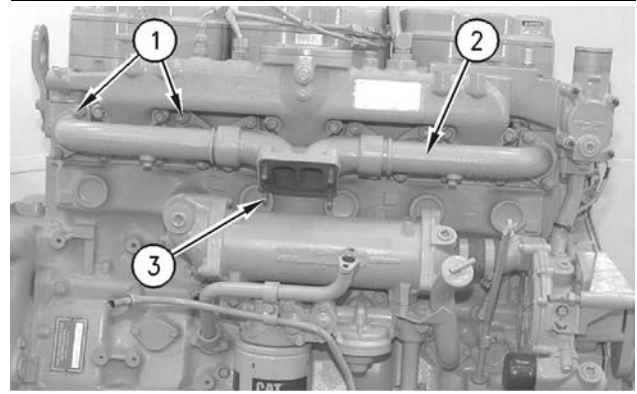


Illustration 33

g00512515

Typical example

1. Remove locknuts (1) with the washers and spacers.
2. Remove exhaust manifold (2) and the gaskets.
3. Remove the studs that hold the exhaust manifold to the cylinder head assembly.
4. Remove studs (3) from the exhaust manifold, if necessary.

### Installation Procedure

1. Inspect the condition of the gasket. Replace the gasket, if necessary.
2. Apply **5P-3931** Anti-Seize Compound to the threads of the studs that hold the exhaust manifold to the cylinder head assembly.
3. Install the studs for the cylinder head assembly. Tighten the studs to a torque of  $35 \pm 5$  N·m ( $26 \pm 4$  lb ft).

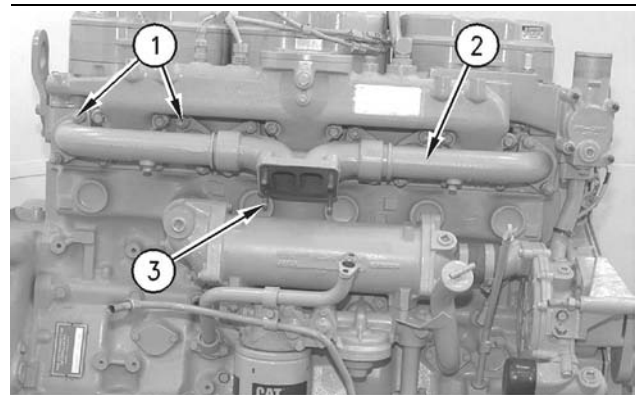


Illustration 34

g00512515

Typical example

4. Position the gaskets with the tabs in the upward position. Install exhaust manifold (2).

5. Apply 5P-3931 Anti-Seize Compound to locknuts (1). Install locknuts (1) with the washers and spacers. Tighten the locknuts to a torque of  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $41 \pm 7 \text{ lb ft}$ ).
6. Apply 5P-3931 Anti-Seize Compound to the threads of studs (3).
7. Install studs (3) in exhaust manifold (2). Tighten the studs to a torque of  $35 \pm 5 \text{ N}\cdot\text{m}$  ( $26 \pm 4 \text{ lb ft}$ ).

**End By:**

- a. Install the turbocharger. Refer to Disassembly and Assembly, "Turbocharger - Install".

i01337457

## Inlet Manifold - Remove

**SMCS Code:** 1058-011

### Removal Procedure

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

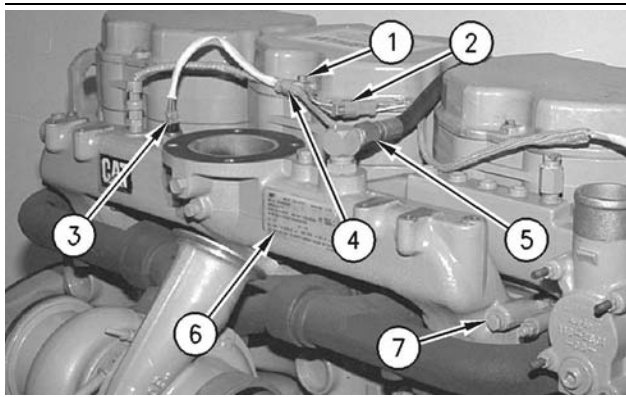


Illustration 35

g00707340

Typical example

1. Disconnect sensor connector (3).
2. Remove bolt (1) and clip (4).
3. Disconnect sensor connector (2).
4. Remove hose assembly (5).
5. Remove bolts (7) and the washers. Remove air inlet manifold (6) and the three gaskets.

i01337547

## Inlet Manifold - Install

**SMCS Code:** 1058-012

### Installation Procedure

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

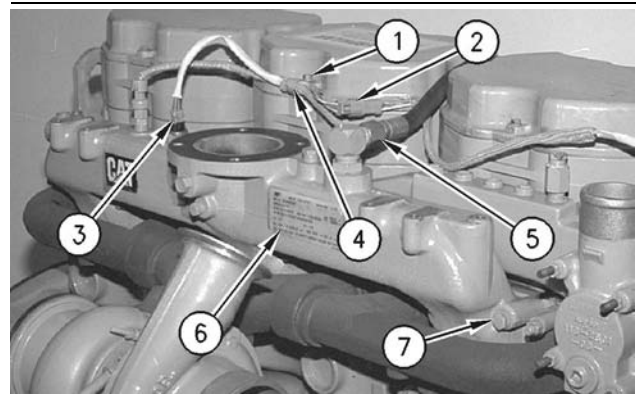


Illustration 36

g00707340

Typical example

1. Inspect the condition of the gaskets. Replace the gaskets, if necessary.
2. Position air inlet manifold (6) and the three gaskets on the cylinder head.
3. Install bolts (7) and the washers. Tighten bolts (7) to a torque of  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $40 \pm 7 \text{ lb ft}$ ).
4. Install hose assembly (5).
5. Connect sensor connector (2).
6. Install clip (4) and bolt (1). Tighten bolt (1) to a torque of  $12 \pm 3 \text{ N}\cdot\text{m}$  ( $9 \pm 2 \text{ lb ft}$ ).
7. Connect sensor connector (3).

i01770734

## Inlet and Exhaust Valve Springs - Remove and Install

SMCS Code: 1108-010

### Removal Procedure

Table 8

| Required Tools |             |                         |     |
|----------------|-------------|-------------------------|-----|
| Tool           | Part Number | Part Description        | Qty |
| A              | 4C-6726     | Valve Spring Compressor | 1   |

#### Start By:

- Remove the valve mechanism cover. Refer to Disassembly and Assembly, "Valve Mechanism Cover - Remove and Install".
- Remove the electronic unit injector. Refer to Disassembly and Assembly, "Electronic Unit Injector - Remove".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**Note:** The following procedure is for the removal of the inlet valve springs and the exhaust valve springs without removing the cylinder head. This procedure can be performed on only one cylinder at a time. This will prevent the inlet valves and the exhaust valves from falling into the cylinder.

- Rotate the crankshaft in order to bring the piston to the top center position in the cylinder.

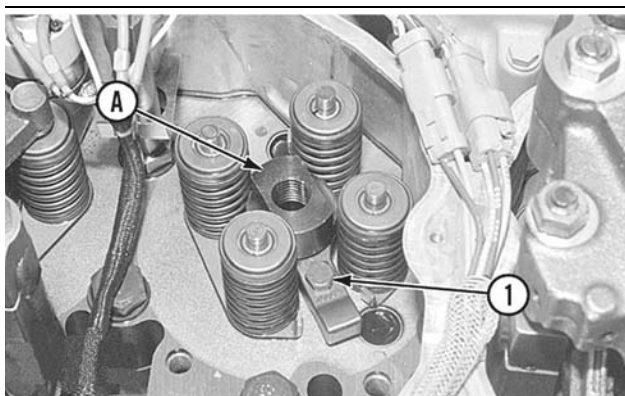


Illustration 37

g00526801

- Place the bottom portion of Tooling (A) on the cylinder head. Install the hold down bolt (1) .

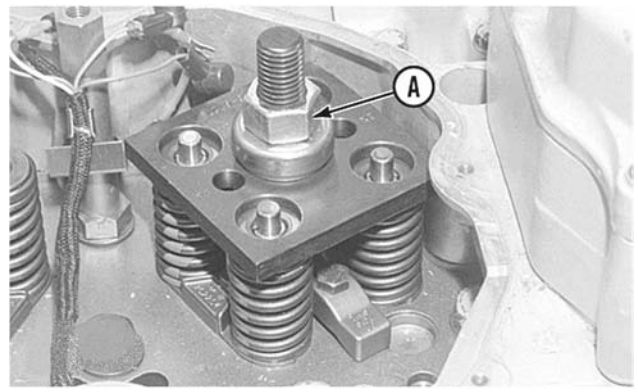


Illustration 38

g00526802

- Install the remainder of Tooling (A).

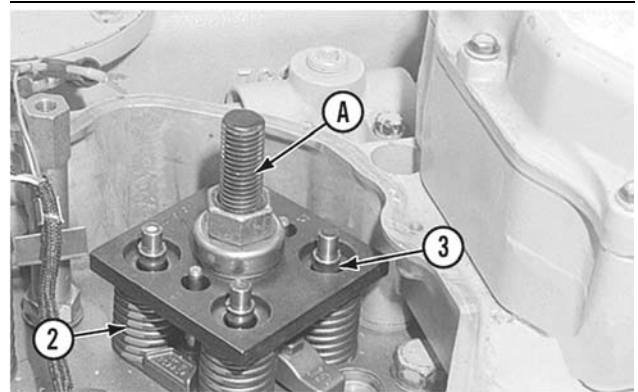


Illustration 39

g00526803

#### WARNING

**Personal injury can result from being struck by parts propelled by a released spring force.**

**Make sure to wear all necessary protective equipment.**

**Follow the recommended procedure and use all recommended tooling to release the spring force.**

- Tighten the nut on Tooling (A) in order to compress the inlet valve springs and the exhaust valve springs (2).
- Remove retainer locks (3).
- Remove Tooling (A).

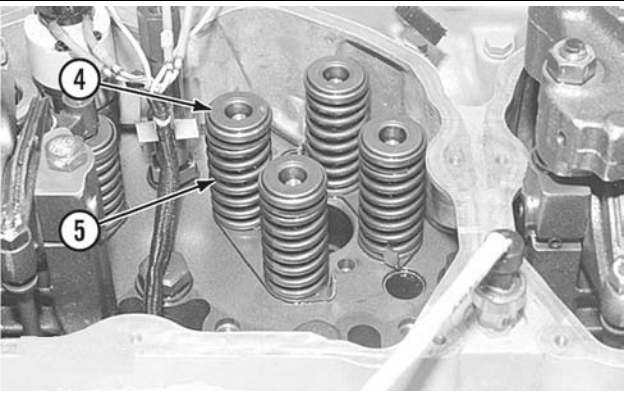


Illustration 40 g00526804

- 7. Remove valve rotators (4).
- 8. Remove the inlet valve springs and the exhaust valve springs (5) from each valve.

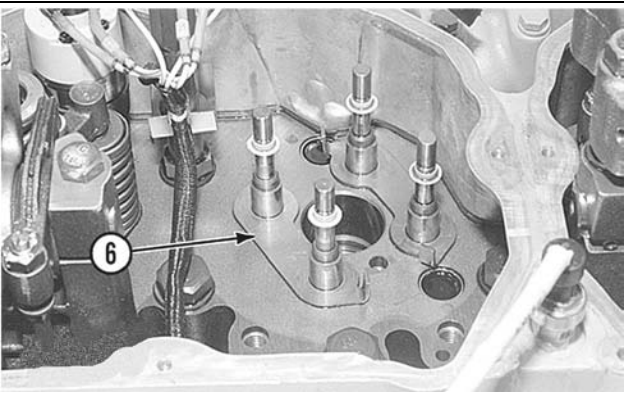


Illustration 41 g00526805

- 9. Remove bases (6) from the cylinder head.
- 10. Install the inlet valve springs and the exhaust valve springs before you rotate the crankshaft.

Installation Procedure

Table 9

| Required Tools |             |                         |     |
|----------------|-------------|-------------------------|-----|
| Tool           | Part Number | Part Description        | Qty |
| A              | 4C-6726     | Valve Spring Compressor | 1   |

NOTICE  
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

Note: Ensure that the piston is at the top center position in order to install the inlet valve springs and the exhaust valve springs.

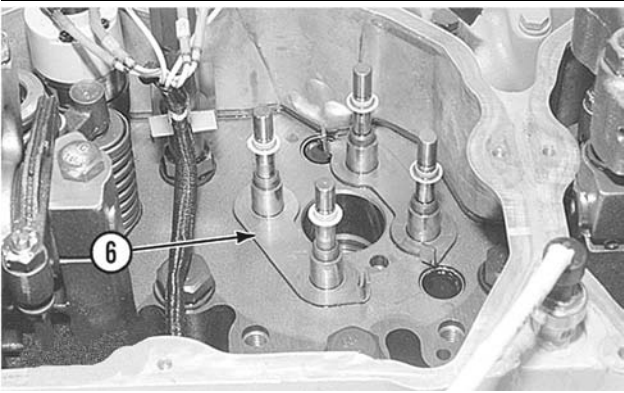


Illustration 42 g00526805

- 1. Install bases (6) for the inlet valve springs and the exhaust valve springs.

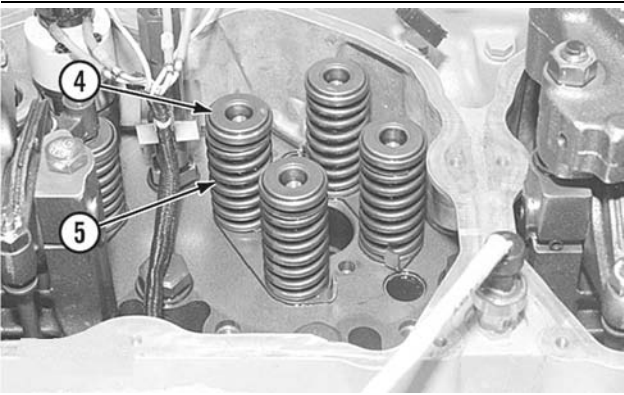


Illustration 43 g00526804

- 2. Install the inlet valve springs and the exhaust valve springs (5).
- 3. Install valve rotators (4).

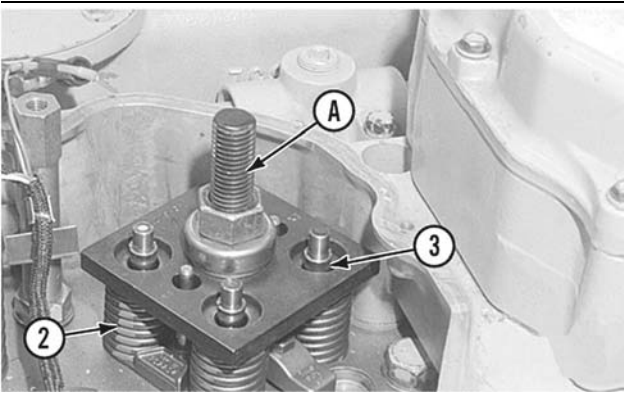


Illustration 44 g00526803

**⚠ WARNING**

Personal injury can result from being struck by parts propelled by a released spring force.

Make sure to wear all necessary protective equipment.

Follow the recommended procedure and use all recommended tooling to release the spring force.

4. Install Tooling (A).
5. Tighten the nut on Tooling (A) in order to compress the inlet valve springs and the exhaust valve springs (2).
6. Install retainer locks (3).
7. Remove Tooling (A).

**End By:**

- a. Install the electronic unit injector. Refer to Disassembly and Assembly, "Electronic Unit Injector - Install".
- b. Install the valve mechanism cover. Refer to Disassembly and Assembly, "Valve Mechanism Cover - Remove and Install".

i01779757

## Inlet and Exhaust Valves - Remove and Install

**SMCS Code:** 1105-010

### Removal Procedure

Table 10

| Required Tools |             |                         |     |
|----------------|-------------|-------------------------|-----|
| Tool           | Part Number | Part Description        | Qty |
| A              | 5S-1330     | Valve Spring Compressor | 1   |

**Start By:**

- a. Remove the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

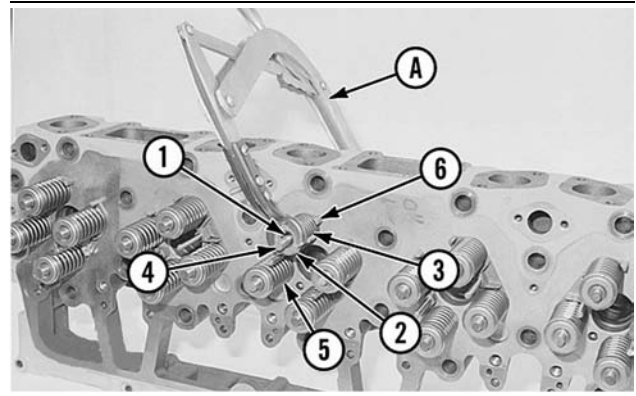


Illustration 45

g00513273

**⚠ WARNING**

Personal injury can result from being struck by parts propelled by a released spring force.

Make sure to wear all necessary protective equipment.

Follow the recommended procedure and use all recommended tooling to release the spring force.

1. Place the cylinder head assembly in an upright position. See Illustration 45.
2. Use Tooling (A) to compress valve springs (3) and valve spring (5). Remove retainer locks (1).
3. Remove Tooling (A). Remove valve rotator (2), valve springs (3) and valve spring (5).
4. Repeat Steps 2 and 3 in order to remove valve springs (3), valve spring (5), and bridge supports (6).
5. Remove valves (4) from the cylinder head.
6. Repeat Steps 2 through 5 in order to remove the remaining inlet valves and exhaust valves.

### Installation Procedure

Table 11

| Required Tools |             |                         |     |
|----------------|-------------|-------------------------|-----|
| Tool           | Part Number | Part Description        | Qty |
| A              | 5S-1330     | Valve Spring Compressor | 1   |

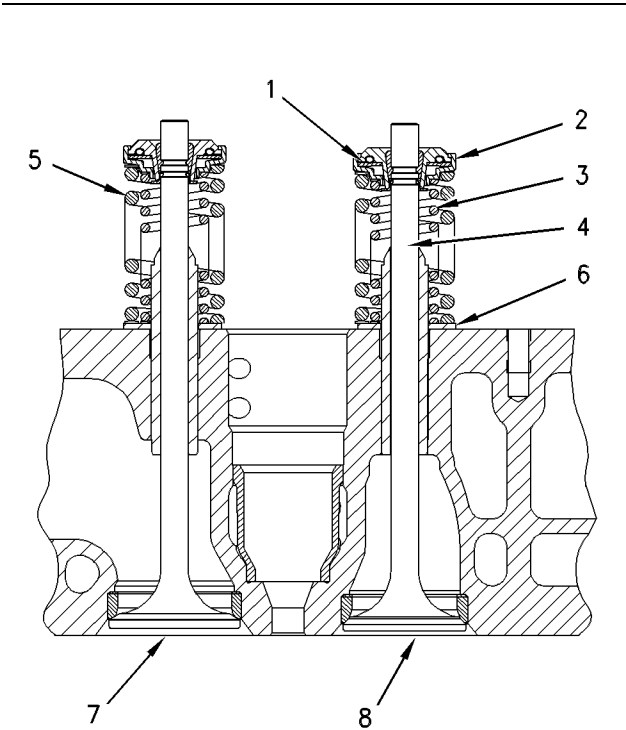


Illustration 46

g00513277

1. Lubricate the inlet valve (7) and the exhaust valve (8) with clean engine oil. Install the inlet valves and exhaust valves in the cylinder head assembly.

**⚠ WARNING**

The valve keepers can be thrown from the valve when the valve spring compressor is released. Ensure that the valve keepers are properly installed on the valve stem. To help prevent personal injury, keep away from the front of the valve keepers and valve springs during the installation of the valves.

Note: A small amount of grease can be used to hold the retainer locks in position during installation.

2. Place the following items on valve stem (4): bridge support (6), valve springs (3), valve spring (5), and valve rotators (2).
3. Use Tooling (A) to compress valve springs (3) and (5). Install retainer locks (1).
4. Repeat Steps 1 through 3 in order to install the remaining inlet valves and exhaust valves.

**End By:**

- a. Install the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Install".

i01771830

## Inlet and Exhaust Valve Guides - Remove and Install

SMCS Code: 1104-010

### Removal Procedure

Table 12

| Required Tools |             |                                 |     |
|----------------|-------------|---------------------------------|-----|
| Tool           | Part Number | Part Description                | Qty |
| A              | 9U-6460     | Valve Guide and Seat Tool Group | 1   |

**Start By:**

- a. Remove the inlet valves and exhaust valves. Refer to Disassembly and Assembly, "Inlet and Exhaust Valves - Remove and Install".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

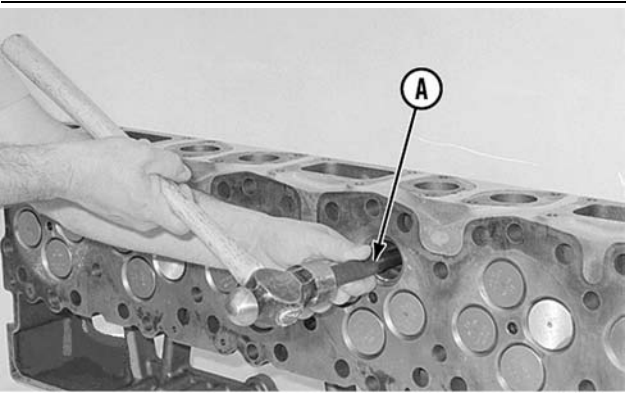


Illustration 47

g00513516

Typical example

1. Use Tooling (A) to remove the valve guides from the cylinder head assembly.

## Installation Procedure

Table 13

| Required Tools |             |                                 |     |
|----------------|-------------|---------------------------------|-----|
| Tool           | Part Number | Part Description                | Qty |
| A              | 9U-6460     | Valve Guide and Seat Tool Group | 1   |

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

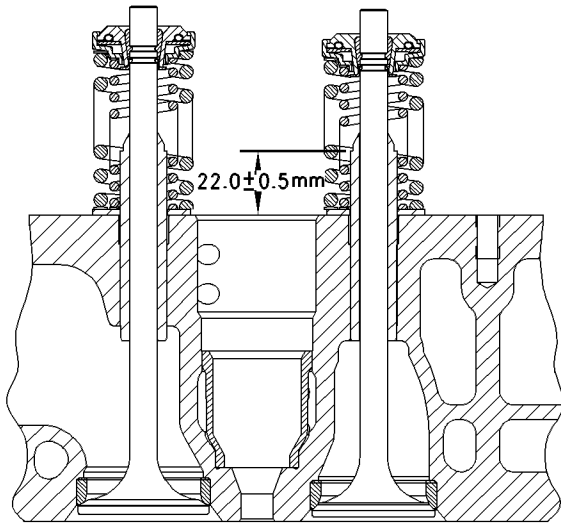


Illustration 48

g00513519

1. Position the valve guide and tap on the top of the valve guide. This is done in order to start the valve guide into the cylinder head assembly.
2. Use Tooling (A) to install the valve guides. Install the valve guides until the protrusion is  $22.0 \pm 0.5$  mm ( $0.87 \pm 0.02$  inch) above the cylinder head assembly.

### End By:

- a. Install the inlet valves and exhaust valves. Refer to Disassembly and Assembly, "Inlet and Exhaust Valves- Remove and Install".

## Inlet and Exhaust Valve Seat Inserts - Remove and Install

i01771898

SMCS Code: 1103-010

## Removal Procedure

Table 14

| Required Tools |             |                                 |     |
|----------------|-------------|---------------------------------|-----|
| Tool           | Part Number | Part Description                | Qty |
| A              | 9U-6460     | Valve Guide and Seat Tool Group | 1   |

### Start By:

- a. Remove the inlet and exhaust valves. Refer to Disassembly and Assembly, "Inlet and Exhaust Valves - Remove and Install".

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

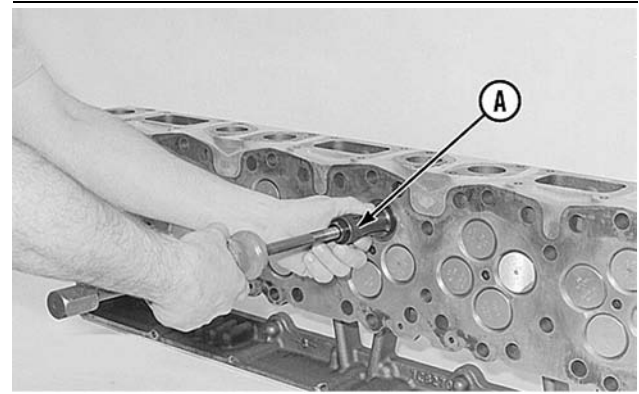


Illustration 49

g00513641

Typical example

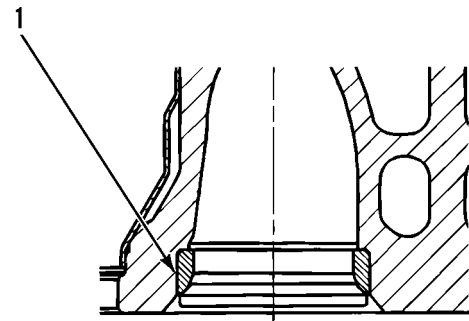


Illustration 50

g00513644

1. Use Tooling (A) to remove valve seat inserts (1).

Installation Procedure

Table 15

| Required Tools |             |                                 |     |
|----------------|-------------|---------------------------------|-----|
| Tool           | Part Number | Part Description                | Qty |
| A              | 9U-6460     | Valve Guide and Seat Tool Group | 1   |

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

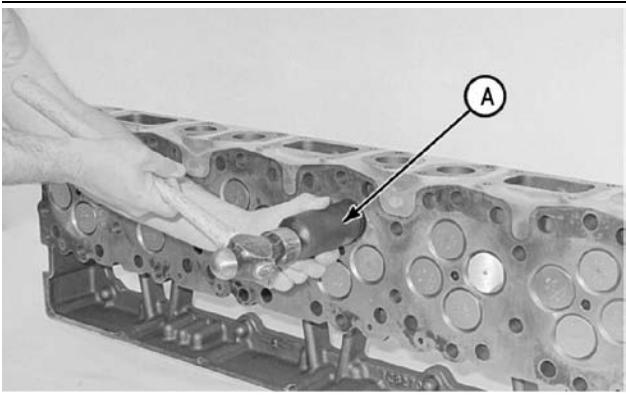


Illustration 51

g00513645

Typical example

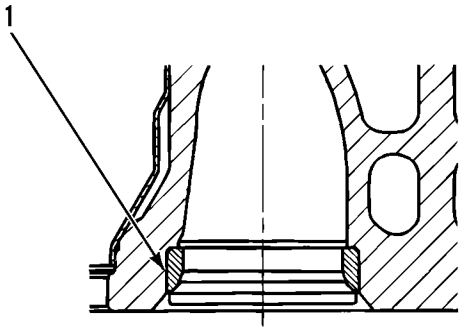


Illustration 52

g00513644

1. Lower the temperature of new valve seat insert (1).
2. Use Tooling (A) to install the new valve seat insert in the cylinder head assembly.

End By:

- a. Install the inlet and exhaust valves. Refer to Disassembly and Assembly, "Inlet and Exhaust Valves - Remove and Install".

i01234011

Engine Centrifugal Oil Filter -  
Remove and Install

SMCS Code: 1328-010

Removal Procedure

Table 16

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 1P-1857     | Retaining Ring Pliers | 1   |

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

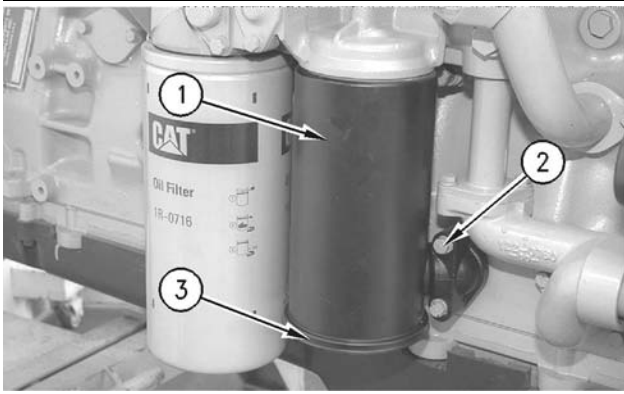


Illustration 53

g00512749

1. Remove filter cover assembly (3) and filter rotor assembly (1) from the body of the filter assembly.
2. Remove two bolts (2) and the washers. Remove the body of the filter assembly.

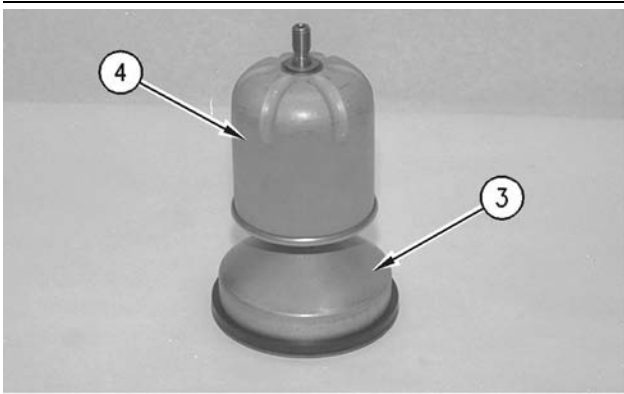


Illustration 54

g00512750

3. Remove filter rotor assembly (4) from filter cover assembly (3).

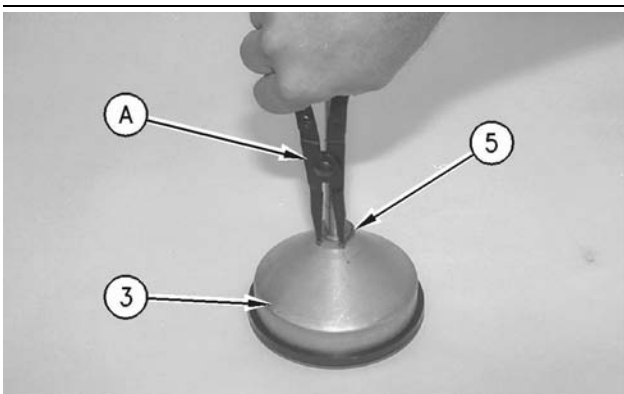


Illustration 55

g00512752

4. Use Tool (A) to remove retaining ring (5) from filter cover assembly (3).

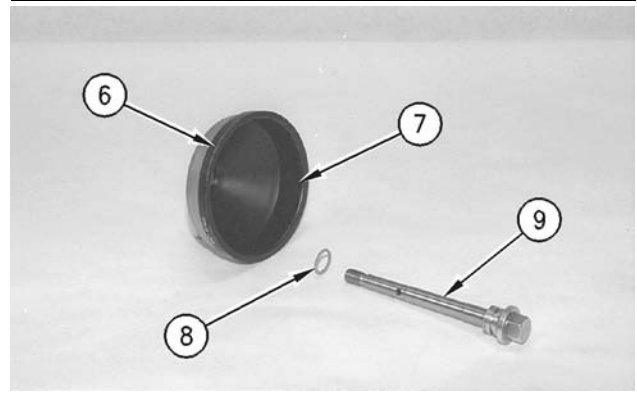


Illustration 56

g00512754

5. Remove the following items from filter cover (7): filter cover bolt (9), O-ring seal (8), and seal (6).

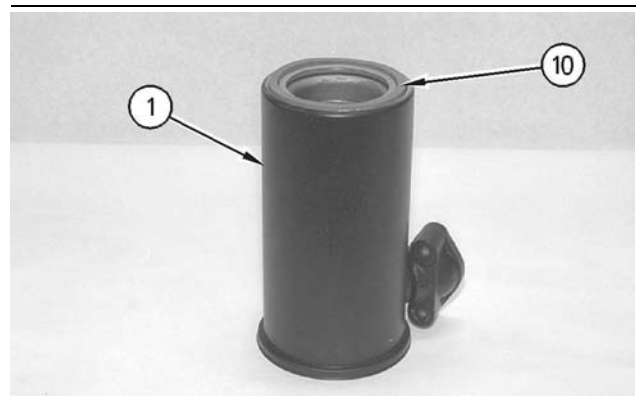


Illustration 57

g00512845

6. Remove seal (10) from the body of the filter assembly (1).

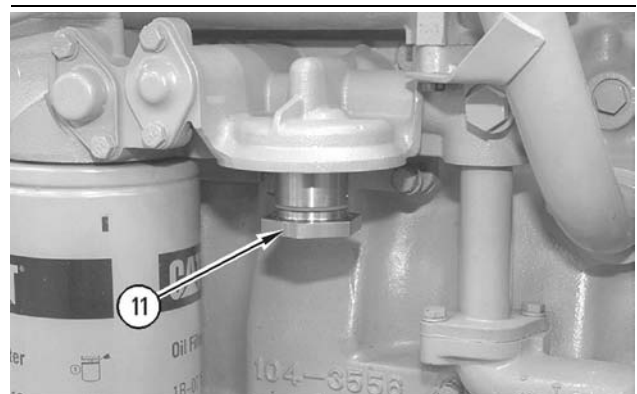


Illustration 58

g00512756

7. Remove filter adapter assembly (11) from the oil filter base.

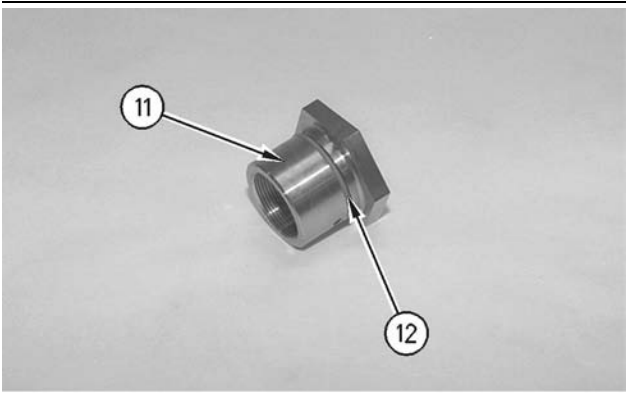


Illustration 59

g00512755

8. Remove O-ring seal (12) from filter adapter assembly (11).

## Installation Procedure

Table 17

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 1P-1857     | Retaining Ring Pliers | 1   |

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the O-ring seals. Replace the seals, if necessary.

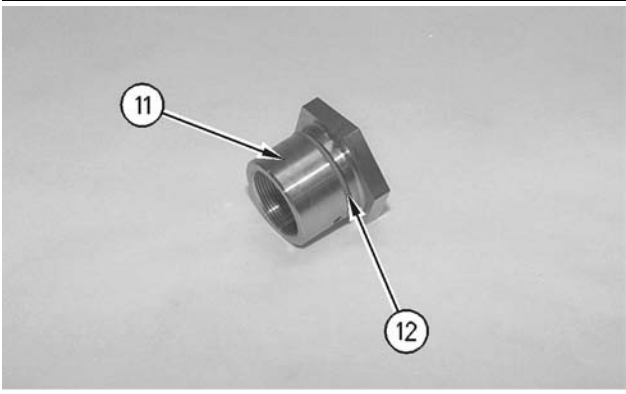


Illustration 60

g00512755

2. Install O-ring seal (12) onto filter adapter assembly (11).

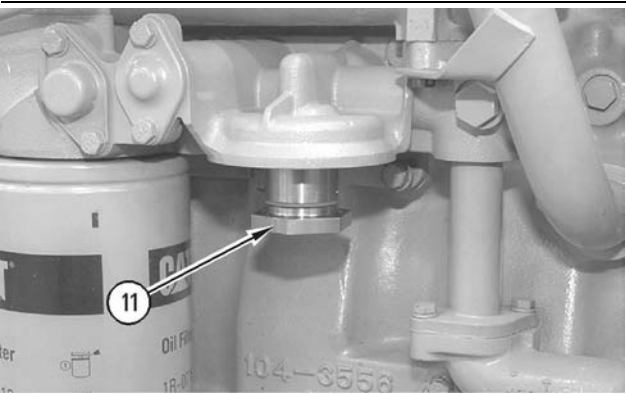


Illustration 61

g00512756

3. Place the filter adapter assembly (11) on the oil filter base. Tighten the adapter to a torque of  $47 \pm 7$  N·m ( $35 \pm 5$  lb ft).

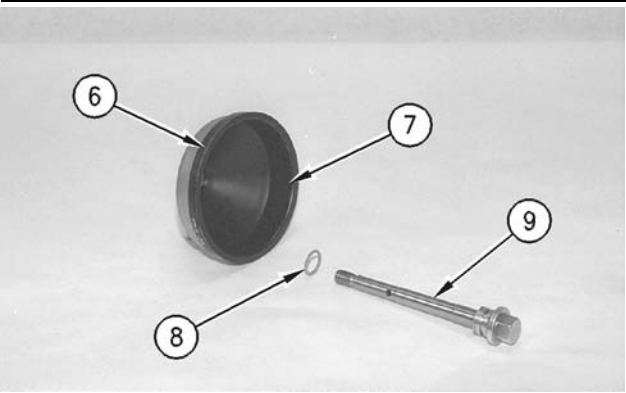


Illustration 62

g00512754

4. Install O-ring seal (8) on filter cover bolt (9).

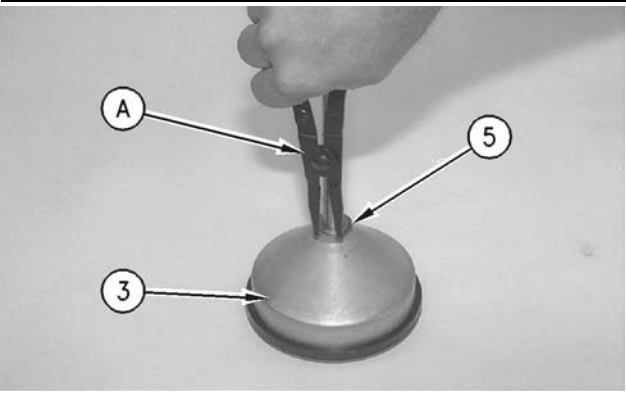


Illustration 63

g00512752

5. Install filter cover bolt (9) in filter cover (6) with retaining ring (5).
6. Install seal (7) on filter cover (6).

i01770753

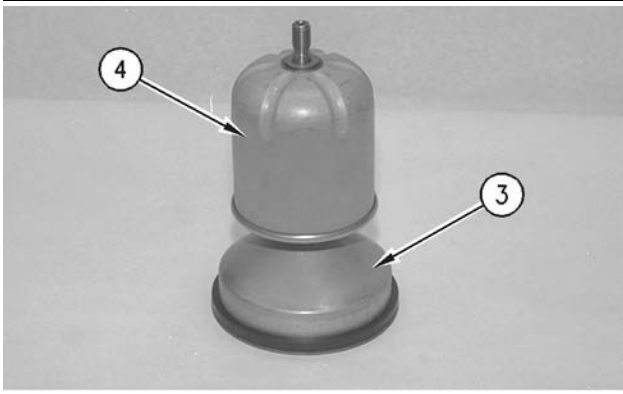


Illustration 64

g00512750

7. Place filter rotor assembly (4) on filter cover assembly (3).

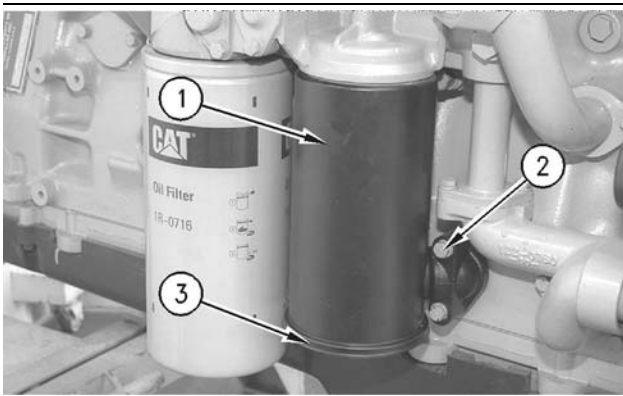


Illustration 65

g00512749

8. Install filter cover assembly (4) in the body of filter assembly (1). Tighten the filter body bolt to a torque of  $20 \pm 3$  N·m ( $15 \pm 2$  lb ft).
9. Install the O-ring seal on the body of filter assembly (1).
10. Install the body of filter assembly (1) with two bolts (2) and the washers. Tighten the bolts to a torque of  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft).

## Engine Oil Filter Base - Remove

**SMCS Code:** 1306-011

### Removal Procedure

Table 18

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 185 - 3630  | Strap Wrench Assembly | 1   |

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

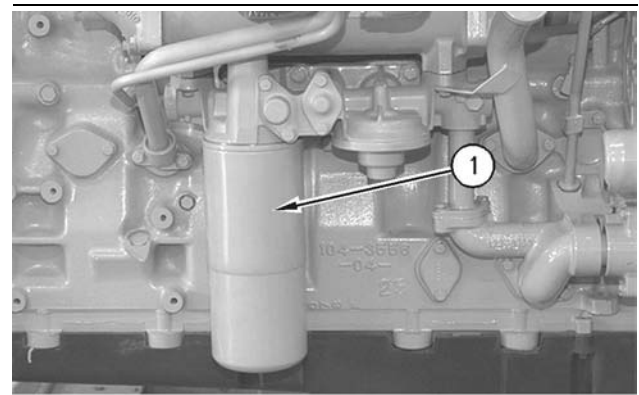


Illustration 66

g00512858

1. Remove oil filter assembly (1) with Tooling (A).

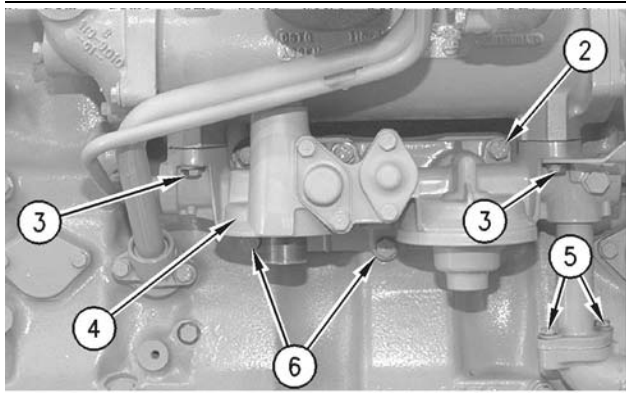


Illustration 67

g00512861

2. Remove bolts (5).

3. Remove bolts (3).

**Note:** Bolts (6) that hold the oil filter base assembly to the cylinder block are marked. This is done for installation purposes.

4. Remove bolts (2).

5. Remove oil filter base assembly (4) and the O-ring seals.

i01465384

## Engine Oil Filter Base - Disassemble

**SMCS Code:** 1306-015

### Disassembly Procedure

#### Start By:

- a. Remove the engine oil filter base. Refer to Disassembly and Assembly, "Engine Oil Filter Base - Remove".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

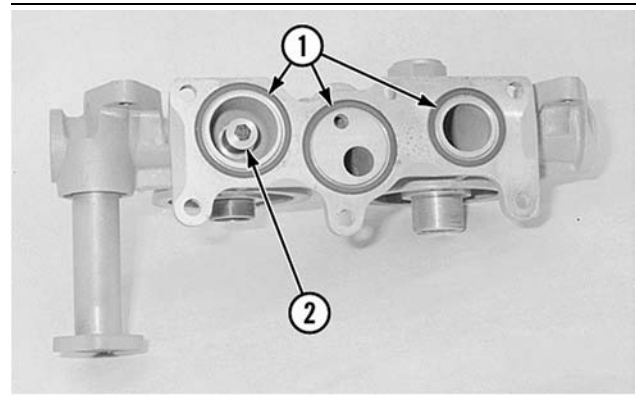


Illustration 68

g00513002

1. Remove three O-ring seals (1) from the oil filter base.

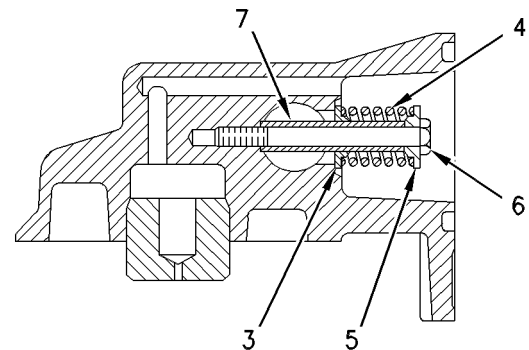


Illustration 69

g00513037

Typical example

2. Remove relief valve assembly (2):

- a. Loosen bolt (6) slowly in order to release the compression on the spring (4).
- b. Remove the following items from the oil filter base: bolt (6), seat of the relief valve spring (5), spring (4), and retainer (3).

- c. Remove spacer (7) from bolt (6).

i01582272

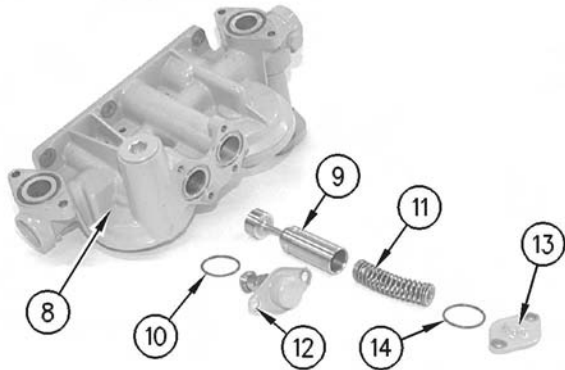


Illustration 70

g00513077

Typical example

3. Remove the two bolts and the washers that hold valve (12).
4. Remove valve (12) and O-ring seal (10) from oil filter base (8).
5. Remove the two bolts and the washers that hold cover (13).
6. Remove the following items from oil filter base (8): cover (13), O-ring seal (14), spring (11), and oil pump bypass valve (9).

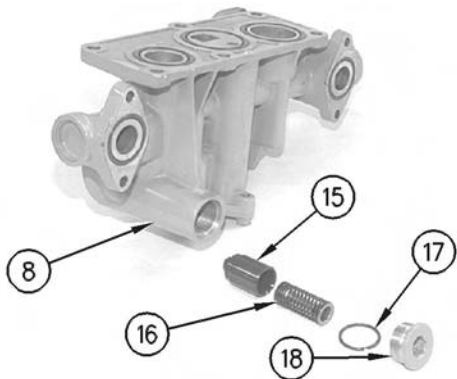


Illustration 71

g00513080

Typical example

7. Remove the following items from oil filter base (8): plug (18), O-ring seal (17), spring (16), and relief valve plunger (15).
8. Remove the stud that holds the oil filter to the oil filter base, if necessary.

## Engine Oil Filter Base - Assemble

SMCS Code: 1306-016

### Assembly Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the O-ring seals. Replace the seals, if necessary.
2. Install the stud that holds the oil filter to the oil filter base. Tighten the stud to a torque of  $68 \pm 7$  N·m ( $50 \pm 5$  lb ft).

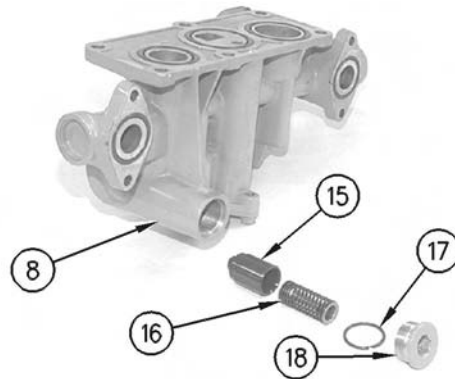


Illustration 72

g00513080

Typical example

3. Install relief valve plunger (15) and spring (16) into oil filter base (8).
4. Place O-ring seal (17) on plug (18) and install the plug in oil filter base (8).

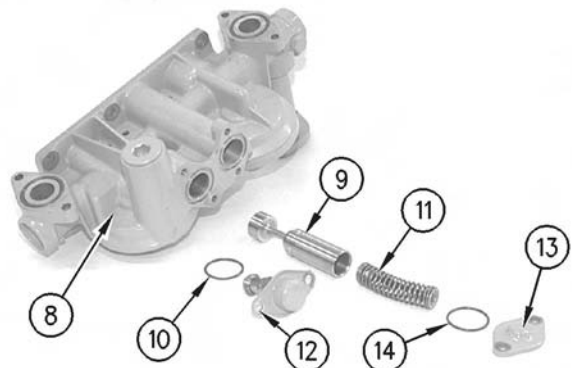


Illustration 73

g00764944

5. Install the following items in oil filter base (8): oil pump bypass valve (9), spring (11), and O-ring seal (14).
6. Install cover (13) and the two bolts with washers that hold the cover in place.
7. Install O-ring seal (10) and valve (12) in oil filter base (8).
8. Install the two bolts with washers that hold valve (12) in place.

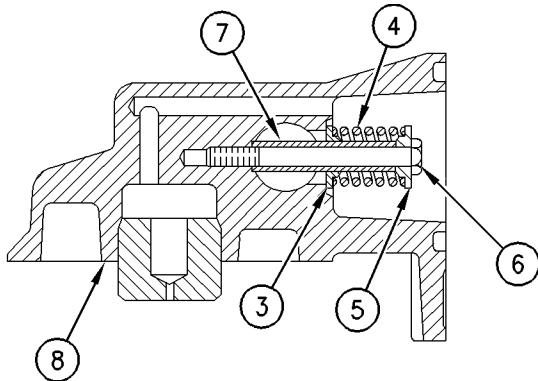


Illustration 74 g00764949

9. Install the following items in oil filter base (8): retainer (3), retainer spring (4), seat of the relief valve spring (5), and spacer (7).
10. Install bolt (6).

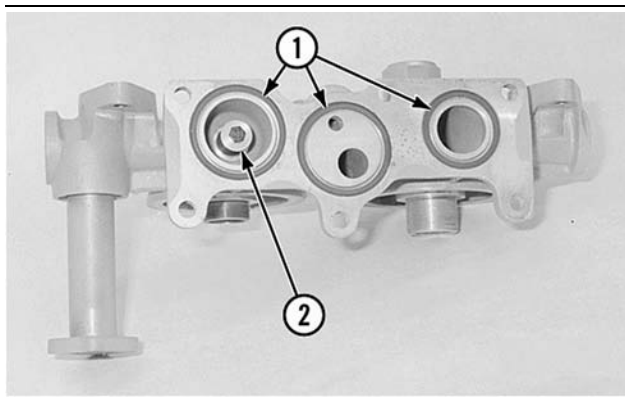


Illustration 75 g00513002

11. Install relief valve assembly (2).
12. Install three O-ring seals (1) in the oil filter base assembly.

#### End By:

- a. Install the engine oil filter base. Refer to Disassembly and Assembly, "Engine Oil Filter Base - Install".

i01770797

## Engine Oil Filter Base - Install

**SMCS Code:** 1306-012

### Installation Procedure

Table 19

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 185-3630    | Strap Wrench Assembly | 1   |

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the O-ring seals. Replace the seals, if necessary.

**Note:** Place all O-ring seals and install all bolts finger tight before tightening to the standard torque.

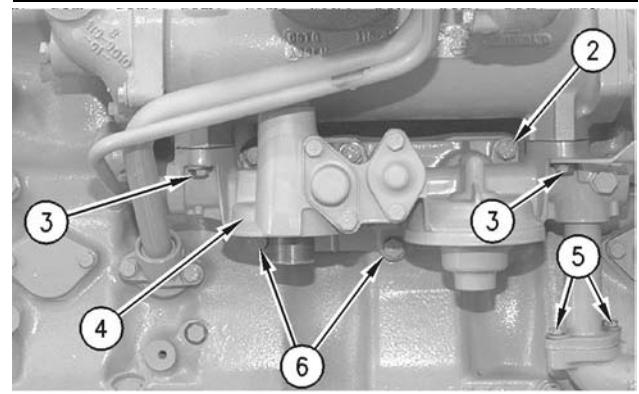


Illustration 76 g00512861

2. Install the O-ring seals in oil filter base assembly (4). Position the oil filter base assembly on the cylinder block. Apply **9S-3263** Thread Lock Compound to bolts (6) that are marked.
3. Install bolts (2).
4. Install bolts (3).
5. Install bolts (5).

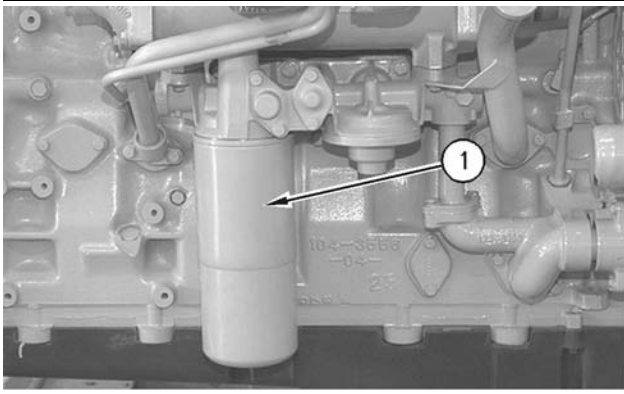


Illustration 77

g00512858

6. Install oil filter assembly (1) with Tooling (A). Follow the instructions with the oil filter assembly for the correct installation.

i01463799

## Engine Oil Cooler - Remove

SMCS Code: 1378-011

### Removal Procedure

#### WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the cooling system into a suitable container for storage or disposal.

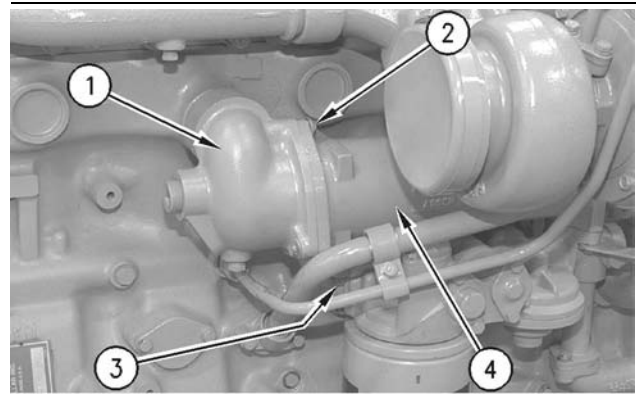


Illustration 78

g00513708

Typical example

2. Remove three bolts (2) and the washers.
3. Remove elbow (1) of the engine oil cooler and the O-ring seals.
4. Remove two bolts (3) from the core of engine oil cooler (4).

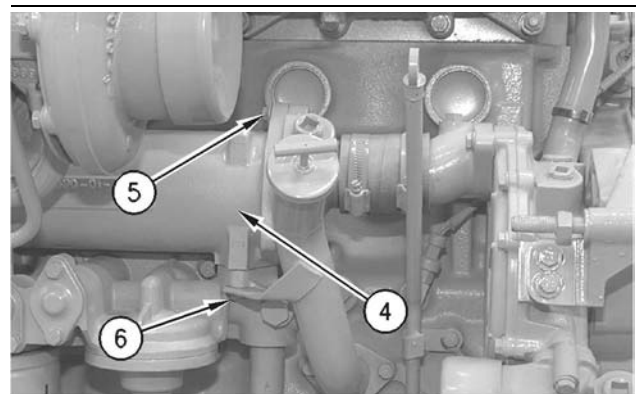


Illustration 79

g00513711

Typical example

5. Remove three bolts (5) and the washers.
6. Remove two bolts (6) and the washers.
7. Remove engine oil cooler (4) and the two O-ring seals from the oil filter base.

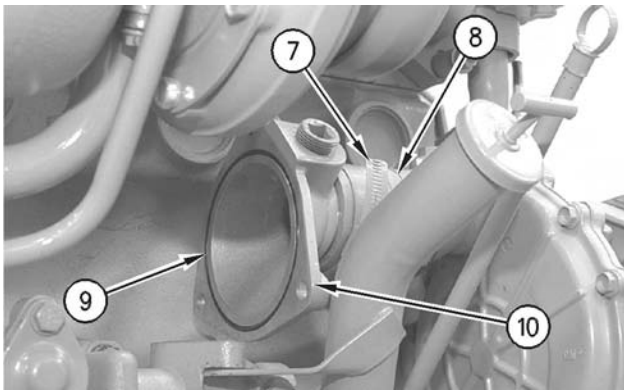


Illustration 80

g00513712

Typical example

8. Loosen hose clamp (7) on hose (8). Remove engine oil cooler bonnet (10) and O-ring seal (9).

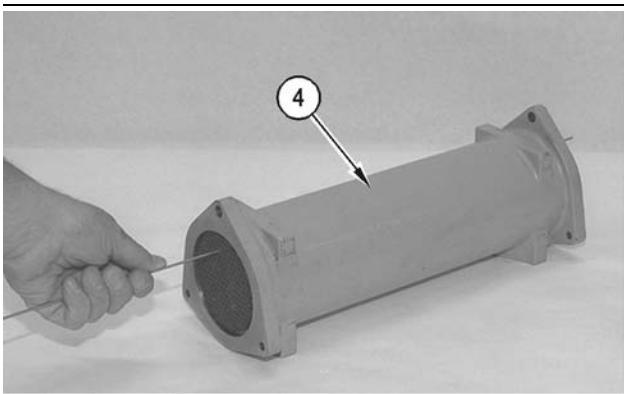


Illustration 81

g00513713

Typical example

9. Clean the cooler tube bundles in engine oil. Clean cooler core (4) with a rod that has a suitable diameter. Thoroughly clean the tube bundles. Check the oil cooler for cracks or leaks.

## Engine Oil Cooler - Install

i01002243

**SMCS Code:** 1378-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the O-ring seals. Replace the seals, if necessary.

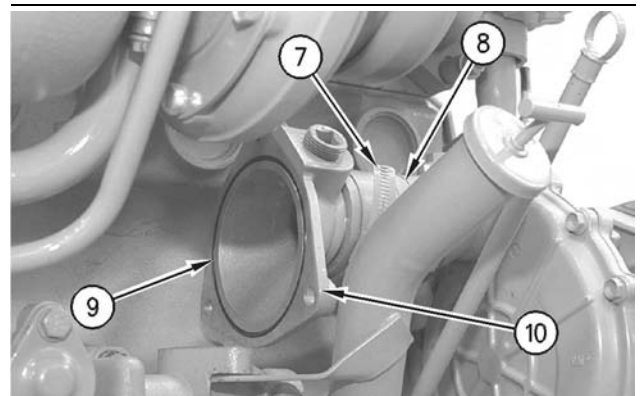


Illustration 82

g00513712

Typical example

2. Install O-ring seal (9). Position engine oil cooler bonnet (10) in hose (8).

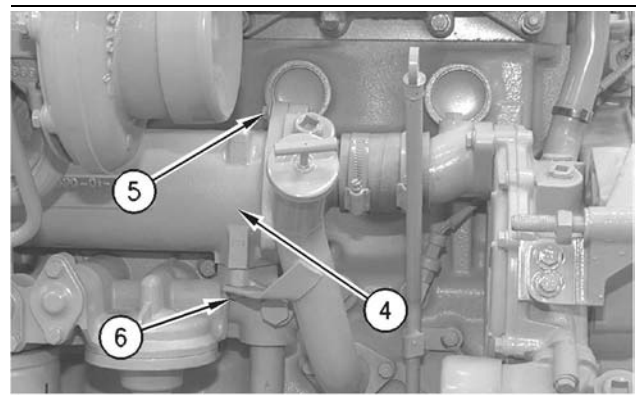


Illustration 83

g00513711

Typical example

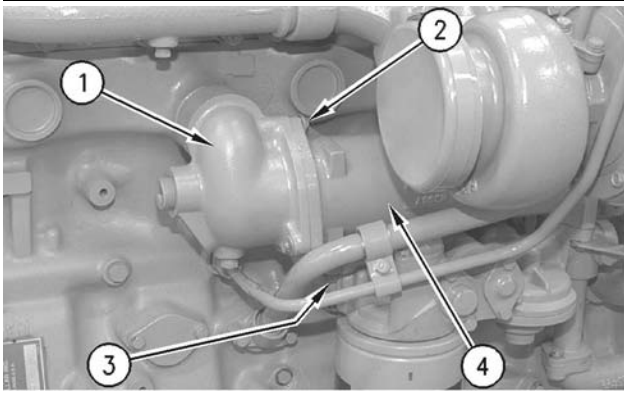


Illustration 84

g00513708

Typical example

3. Install the two O-ring seals on the oil filter base. Position the oil cooler (4) on the oil filter base.
4. Install two bolts (6) with washers and two bolts (3) with washers.
5. Install three bolts (5) with washers. Tighten hose clamp (7).
6. Install the O-ring seals on elbow (1) of the engine oil cooler.
7. Lubricate the bore lightly with glycerin. Position the elbow of the engine oil cooler on engine oil cooler core (4).
8. Install three bolts (2) and the washers.
9. Fill the cooling system with coolant to the correct level. Refer to Operation and Maintenance Manual for the correct filling procedure.

i01000454

## Engine Oil Pump - Remove

**SMCS Code:** 1304-011

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

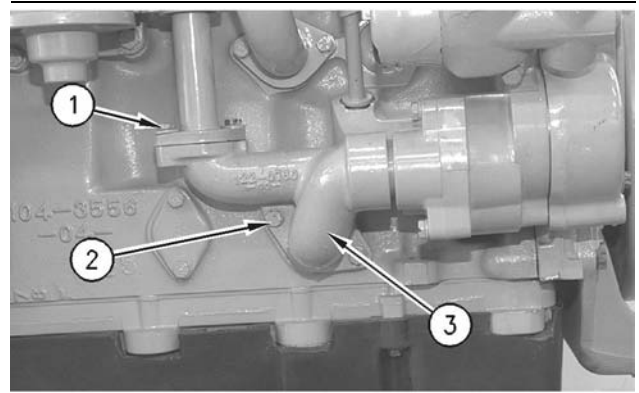


Illustration 85

g00512289

1. Remove bolts (1) and the washers that hold the tube to the elbow of the oil pump.
2. Remove bolts (2) and the washers that hold the elbow of oil pump (3) to the cylinder block.

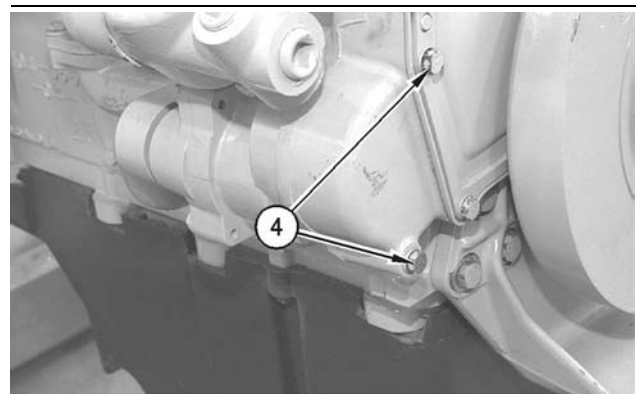


Illustration 86

g00512297

3. Remove bolts (4) and the washers that hold the engine oil pump to the front housing. Remove the engine oil pump and the elbow as a unit.
4. Remove the two O-ring seals. One O-ring seal is on the connection of the tube to the elbow. The second O-ring seal is on the engine oil pump.

i01770805

# Engine Oil Pump - Disassemble

SMCS Code: 1304-015

## Disassembly Procedure

Table 20

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 5F - 7344   | Block-Puller     | 1   |
|                | 7S - 7786   | Puller Leg       | 1   |
|                | 5F - 7345   | Screw            | 1   |

### Start By:

- a. Remove the engine oil pump. Refer to Disassembly and Assembly, "Engine Oil Pump - Remove ".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

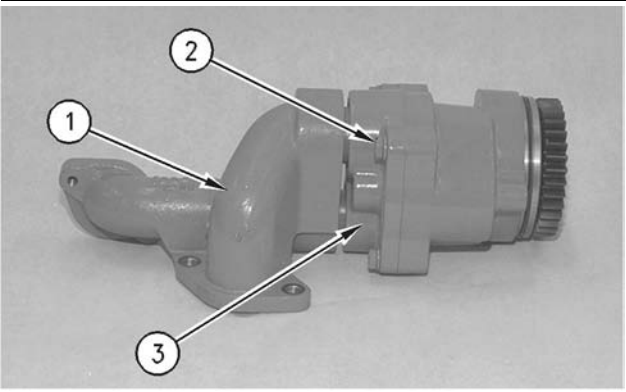


Illustration 87

g00512335

1. Remove elbow (1) from the engine oil pump.
2. Remove the two O-ring seals from the elbow of the engine oil pump.
3. Remove bolts (2).
4. Remove cover (3) from the pump housing.

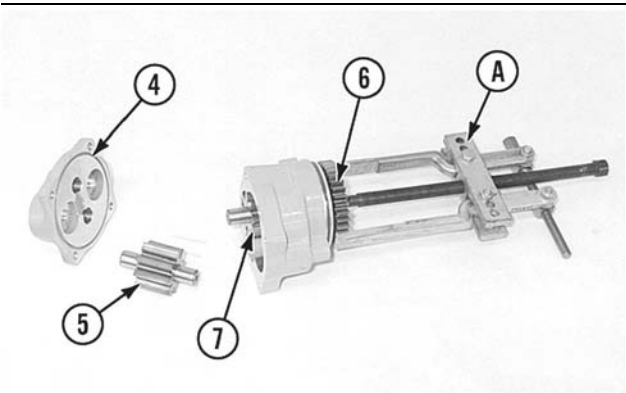


Illustration 88

g00512369

5. Remove O-ring seal (4) and idler gear (5) from the cover.
6. Use Tooling (A) to remove gear (6) from drive shaft (7).

#### NOTICE

Before removing the drive shaft from the pump housing, be sure that no burrs exist on the drive shaft. If the drive shaft has burrs on it, the bores in the pump housing may be scratched.

7. Remove drive shaft (7) from the pump housing.

i01770816

## Engine Oil Pump - Assemble

**SMCS Code:** 1304-016

### Assembly Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the O-rings. Replace O-rings, if necessary.

**Note:** Lubricate all internal parts of the engine oil pump with clean engine oil.

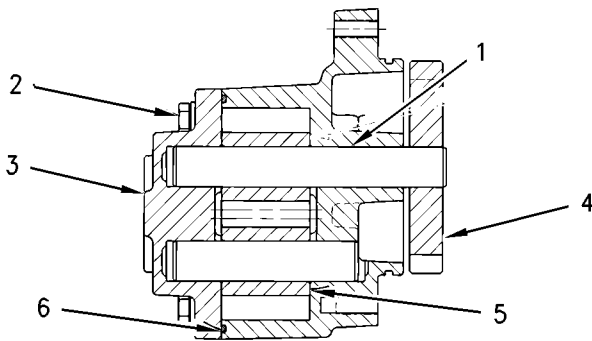


Illustration 89

g00766947

2. Install drive shaft (1) in the pump housing.

**Note:** Heat gear (4) to a maximum temperature of 316 °C (600 °F).

3. Install gear (4) on the end of drive shaft (1). Position the gear so the distance between the outside face of the gear and the step face on the pump body is  $29.40 \pm 0.50$  mm ( $1.158 \pm 0.020$  inch).
4. Install idler gear (5) in the pump housing.
5. Install O-ring seal (6) in cover (3).
6. Install cover (3) and bolts (2).

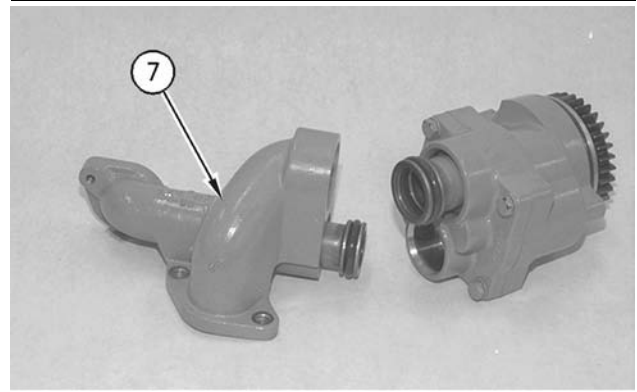


Illustration 90

g00512488

7. Lubricate the two O-ring seals. Install the two O-ring seals on elbow (7) of the oil pump.
8. Install the elbow on the engine oil pump.

#### NOTICE

The pump must rotate freely by hand after assembly.

#### End By:

- a. Install the engine oil pump. Refer to Disassembly and Assembly, "Engine Oil Pump - Install".

i01000464

## Engine Oil Pump - Install

**SMCS Code:** 1304-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

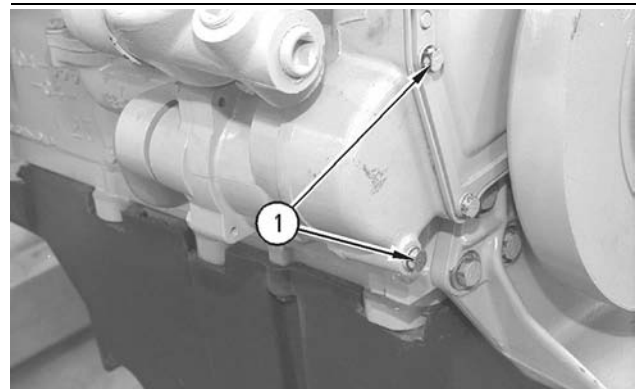


Illustration 91

g00512304

1. Inspect the condition of the O-ring seals. There is one O-ring seal on the connection of the tube to the elbow. The second O-ring seal is on the engine oil pump. Replace the seals, if necessary.
2. Install the O-ring seals. Place the engine oil pump in the correct position in the front housing.

**Note:** Lubricate the bore in the front housing with clean engine oil.

3. Install bolts (1) and the washers that hold the engine oil pump to the front housing.

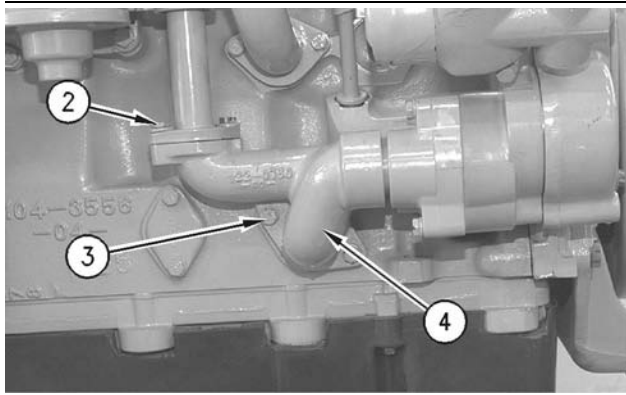


Illustration 92

g00512305

4. Install two bolts (3) that hold the elbow of oil pump (4) to the cylinder block.
5. Install two bolts (2) and the washers that hold the tube to the elbow of the oil pump.

i01463822

## Water Pump - Remove

**SMCS Code:** 1361-011

### Removal Procedure

#### Start By:

- a. Remove the alternator. Refer to Disassembly and Assembly, "Alternator - Remove and Install".
- b. Remove the belt tensioner. Refer to Disassembly and Assembly, "Belt Tensioner - Remove and Install".
- c. Remove the engine oil cooler. Refer to Disassembly and Assembly, "Engine Oil Cooler - Remove".

### WARNING

**Personal injury can result from hot coolant, steam and alkali.**

**At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.**

**Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.**

**Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.**

**Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.**

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the system into a suitable container for storage or disposal.

i01770979

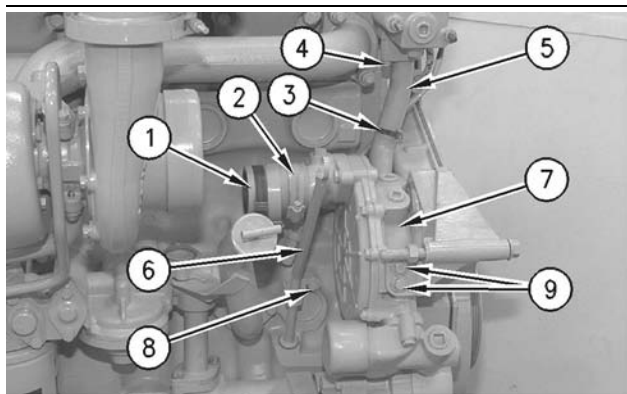


Illustration 93

g00513750

2. Remove two bolts (4) and the washers from tube assembly (5).
3. Cut plastic tie (3) from tube assembly (5).
4. Loosen hose clamp (2) and remove hose (1).
5. Remove bolt (8) and the washer.
6. Remove dipstick tube (6) with the O-ring seal.
7. Remove two bolts (9) and the washers from the alternator bracket.

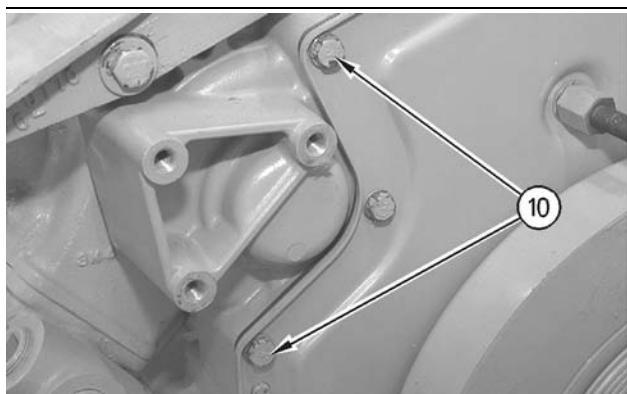


Illustration 94

g00513753

8. Remove two bolts (10) and the washers.
9. Remove water pump (7) and the tube assembly. Remove the flange gasket and the O-ring seal.
10. Remove the tube assembly and O-ring seals from the water pump, if necessary.

## Water Pump - Disassemble

SMCS Code: 1361-015

### Disassembly Procedure

Table 21

| Required Tools |             |                             |     |
|----------------|-------------|-----------------------------|-----|
| Tool           | Part Number | Description                 | Qty |
| A              | 5F - 7344   | Puller Block <sup>(1)</sup> | 1   |
|                | 5F - 7345   | Screw <sup>(1)</sup>        | 1   |
|                | 7S - 7789   | Bumper <sup>(1)</sup>       | 2   |
| B              | 1P - 1861   | Retaining Ring Pliers       | 1   |
| C              | 8B - 7554   | Puller Group                | 1   |

<sup>(1)</sup> Part of 8B - 7554 Bearing Cup Puller

#### Start By:

- a. Remove the water pump. Refer to Disassembly and Assembly, "Water Pump - Remove". The information is in this manual.

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

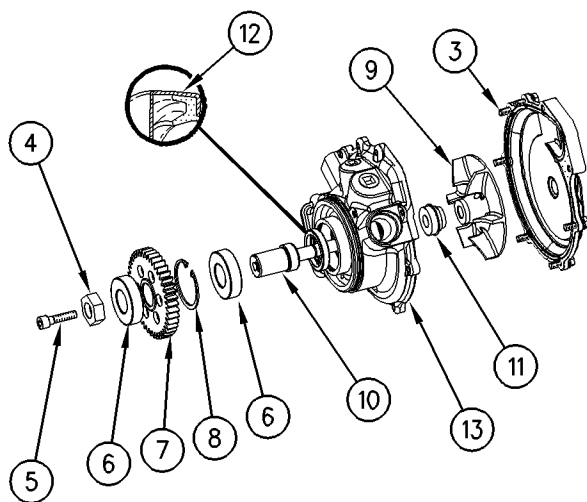


Illustration 95

g00824994

- (3) Cover bolts
- (4) Hex washer
- (5) Bolt
- (6) Bearing
- (7) Gear
- (8) Retaining ring
- (9) Impeller
- (10) Shaft
- (11) Water seal
- (12) Oil seal
- (13) Water pump housing

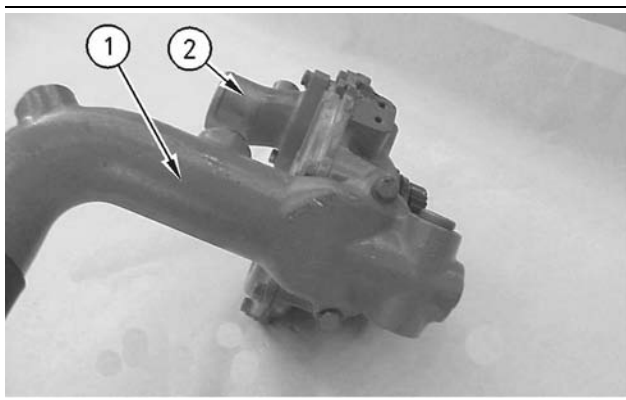


Illustration 96

g00718976

1. Remove water pump elbow (1). Remove elbow (2).

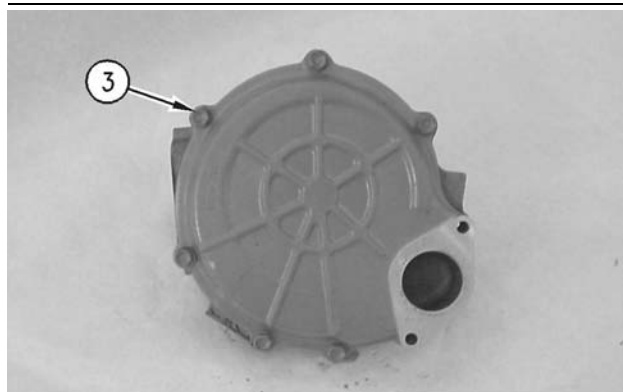


Illustration 97

g00718982

2. Remove cover bolts (3). Carefully separate the housing from the cover.

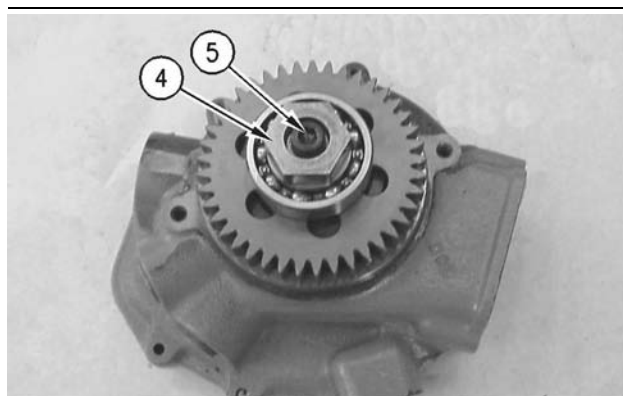


Illustration 98

g00718996

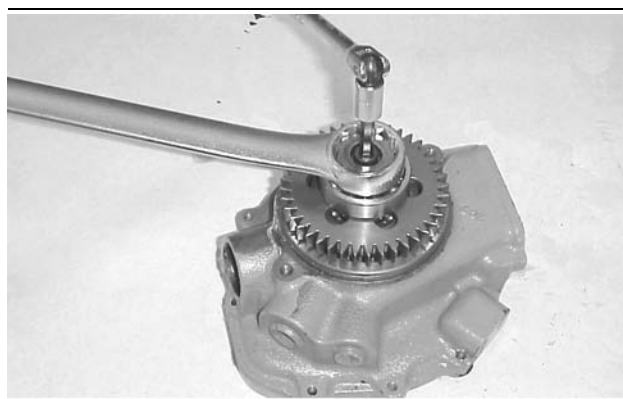


Illustration 99

g00718989

3. Hold hex washer (4). Remove hex socket head bolt (5).

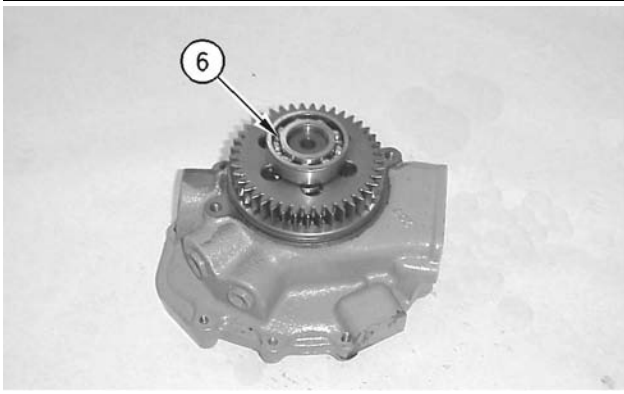


Illustration 100

g00719029

4. Remove bearing (6).

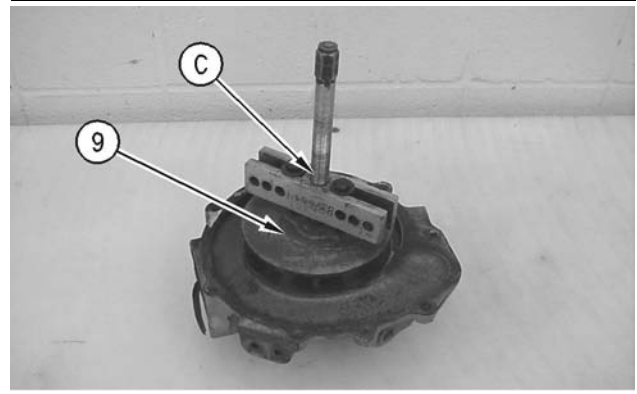


Illustration 103

g00916133

7. Use Tooling (C) to remove impeller (9).

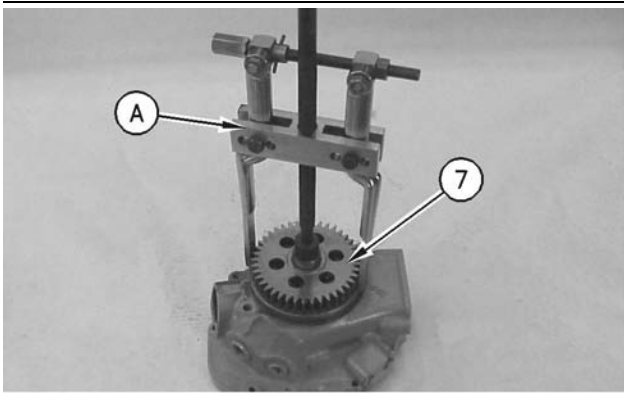


Illustration 101

g00719007

5. Use Tooling (A) to remove gear (7).

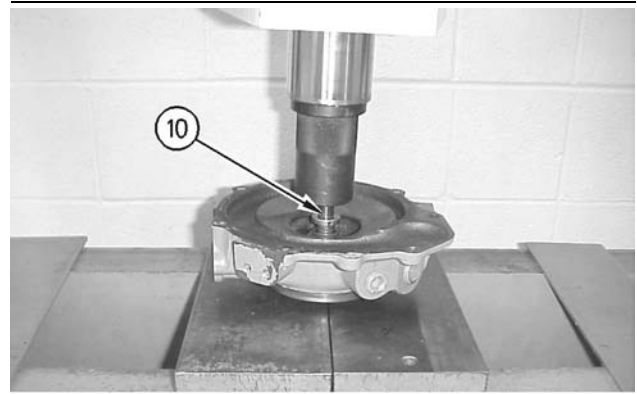


Illustration 104

g00719348

8. Use a hydraulic press and remove shaft assembly (10).

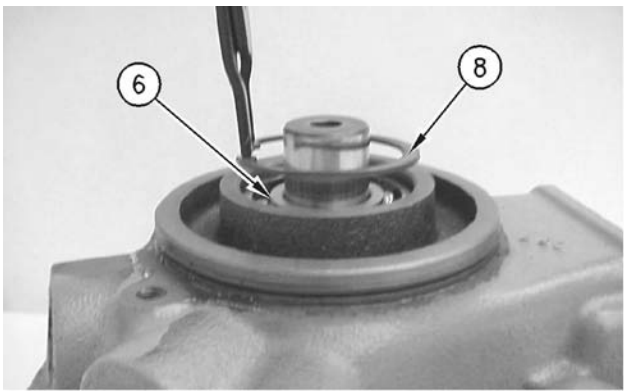


Illustration 102

g00830165

6. Use Tooling (B) to remove retaining ring (8).  
Remove inner bearing (6).

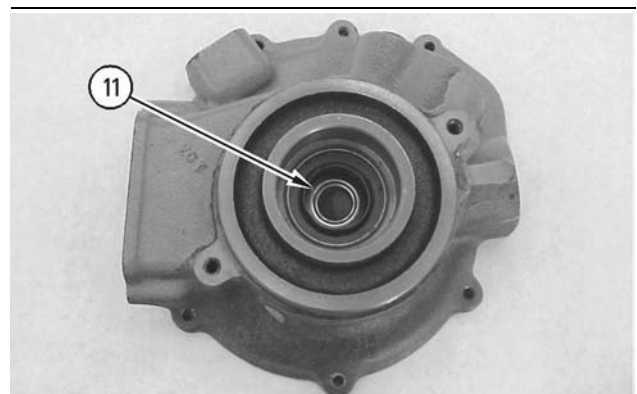


Illustration 105

g00719377

9. Remove water seal (11).

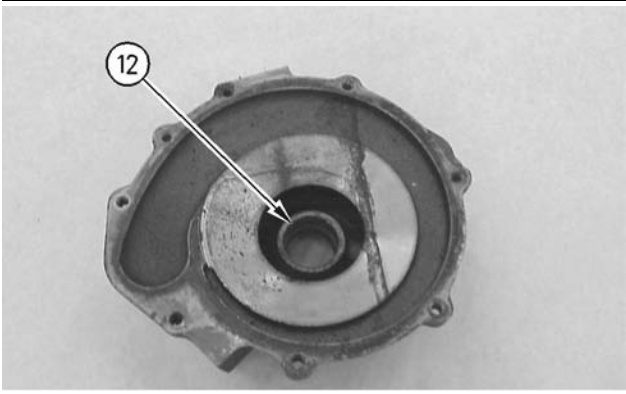


Illustration 106 g00719396

10. Remove oil seal (12).

i01771127

# Water Pump - Assemble

SMCS Code: 1361-016

## Assembly Procedure

Table 22

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Description           | Qty |
| A              | 1P-0766     | Seal Guide            | 1   |
| B              | 183-4028    | Seal Guide            | 1   |
| C              | 6V-2012     | Depth Micrometer      | 1   |
| D              | 169-5464    | Quick Cure Primer     | 1   |
| E              | 1P-1861     | Retaining Ring Pliers | 1   |

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

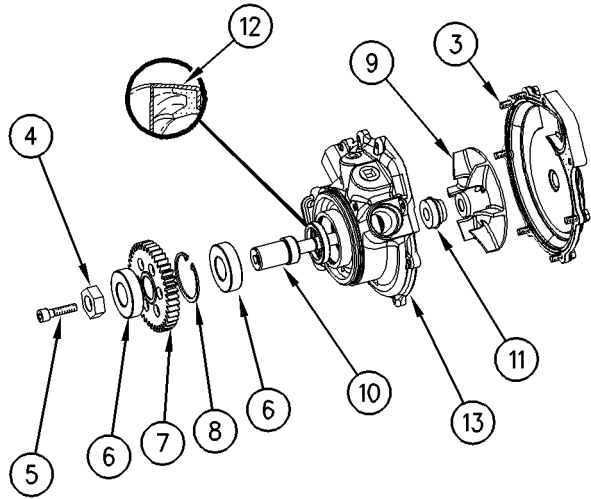


Illustration 107 g00824994

- (3) Cover bolts
- (4) Hex washer
- (5) Bolt
- (6) Bearing
- (7) Gear
- (8) Retaining ring
- (9) Impeller
- (10) Shaft
- (11) Water seal
- (12) Oil seal
- (13) Water pump housing

1. Use Tool (D) to clean the shaft assembly and the counterbore in the housing.

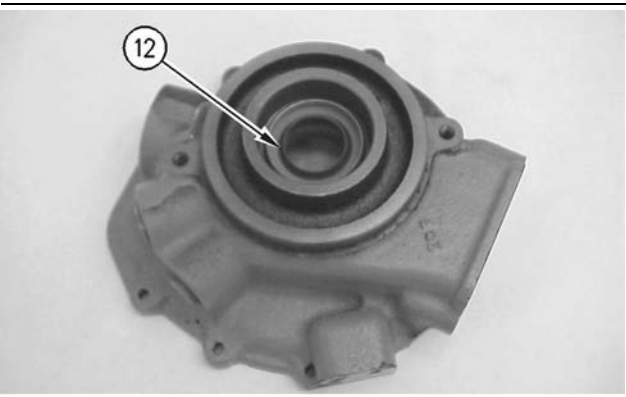


Illustration 108 g00824941

2. Use Tooling (A) to install oil seal (12) into the water pump housing. Install the oil seal with the part number toward the housing. Ensure that the oil seal is flush with the housing. Do not lubricate the seal.

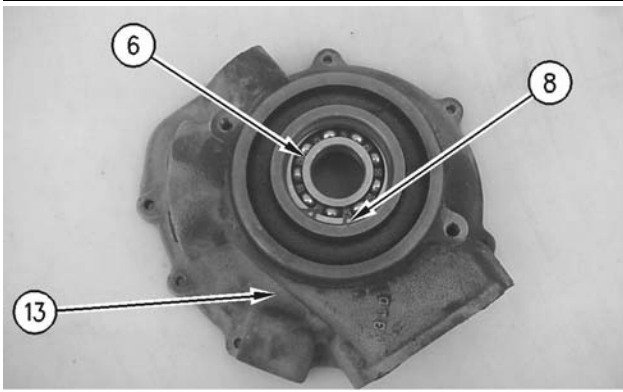


Illustration 109

g00912795

3. Install bearing (6) into water pump housing (13).
4. Use Tooling (E) to install retaining ring (8).

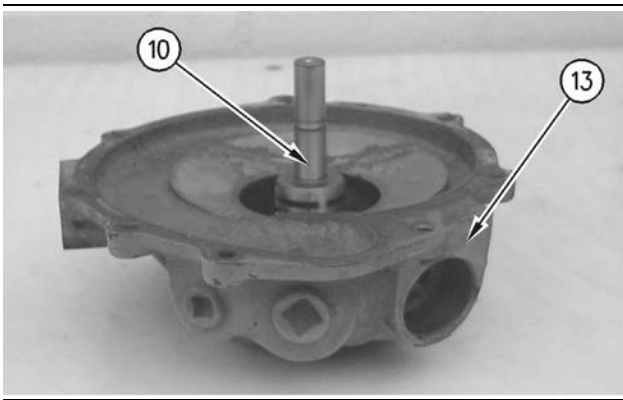


Illustration 110

g00912794

5. Install shaft assembly (10) into water pump housing (13).

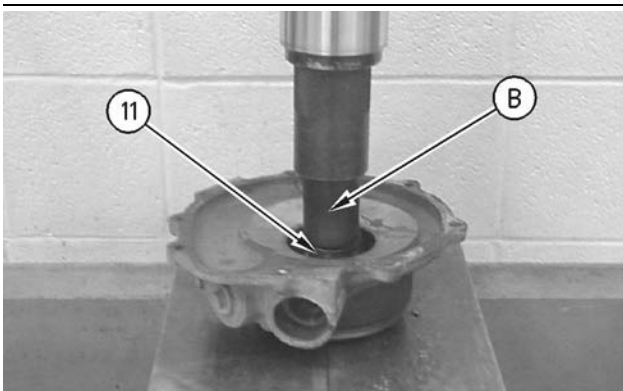


Illustration 111

g00912821

6. Use a hydraulic press with Tooling (B) to install water seal (11) onto shaft assembly (10). Install the water seal with the part number upward. Ensure that the water seal is flush against the pump housing. Do not lubricate the seal.

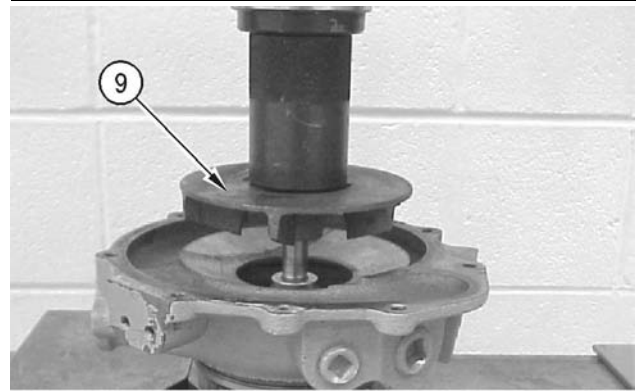


Illustration 112

g00824983

7. Use a hydraulic press to install impeller (9) onto the shaft assembly. The impeller should be flush with the top of the shaft assembly.

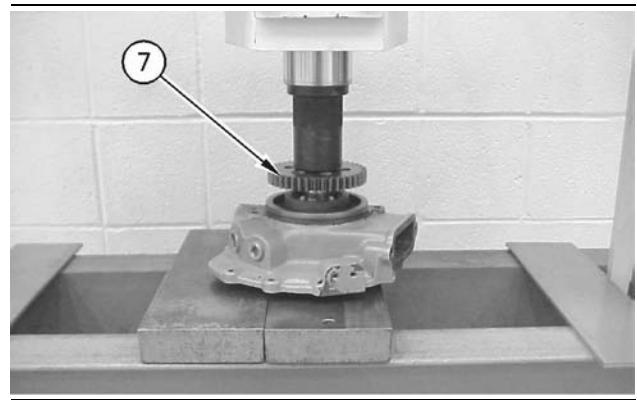


Illustration 113

g00719552

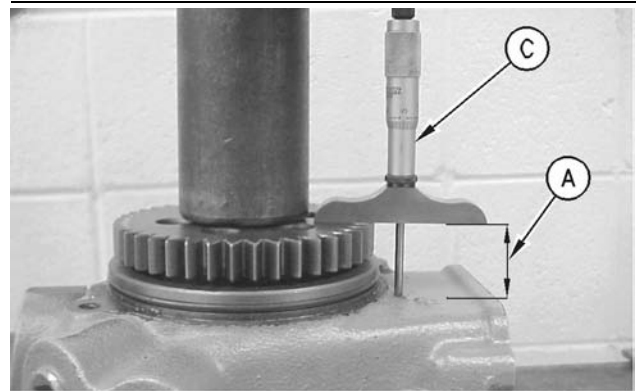


Illustration 114

g00913308

Dimension "A" is 29.3 mm (1.15 inch).

8. Use a hydraulic press to install gear (7). Install the gear to a height of 29.3 mm (1.15 inch). Use Tooling (C) to measure dimension "A".

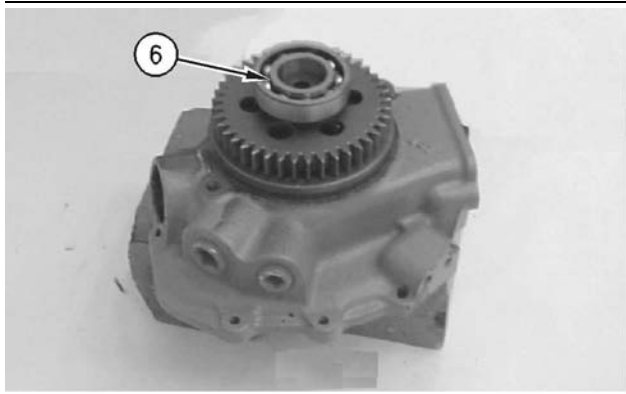


Illustration 115

g00824987

- 9.** Install bearing (6). Ensure that the bearing is flush with the gear.

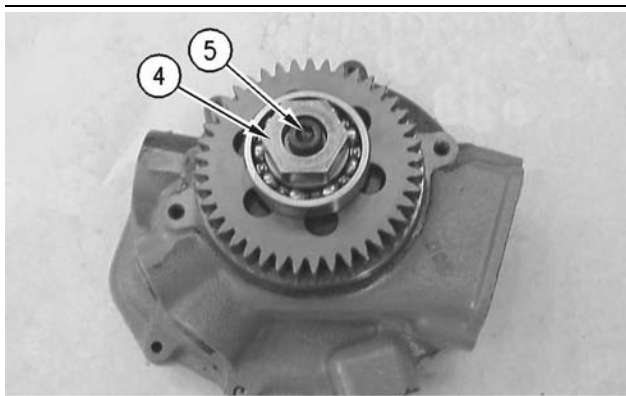


Illustration 116

g00718996

- 10.** Install hex washer (4) and hex socket head bolt (5).

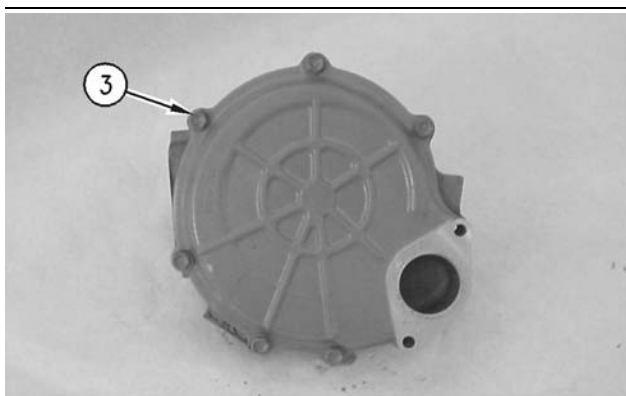


Illustration 117

g00718982

- 11.** Install a new gasket and the cover. Tighten bolts (3) to a torque of  $18 \pm 3$  N·m ( $13 \pm 2$  lb ft).

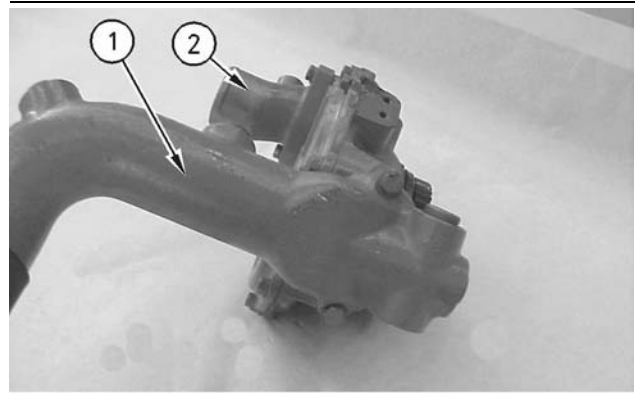


Illustration 118

g00718976

- 12.** Install a new gasket and water pump elbow (1). Install a new gasket and elbow (2).

### End By:

- a.** Install the water pump. Refer to Disassembly and Assembly, "Water Pump - Install". The information is in this manual.

i01463891

## Water Pump - Install

**SMCS Code:** 1361-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

- 1.** Inspect the condition of the O-ring seals and the flange gasket. Replace the seals and the gasket, if necessary.

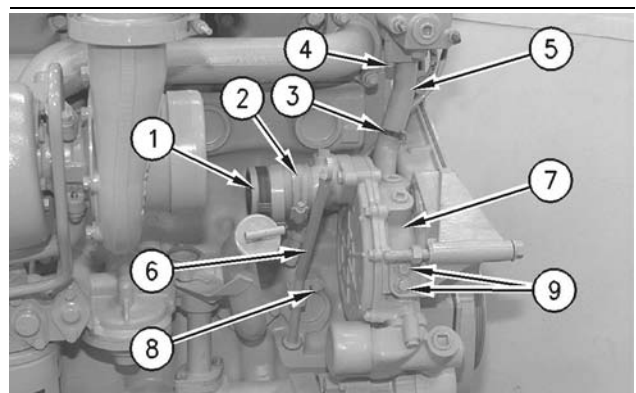


Illustration 119

g00513750

i01771159

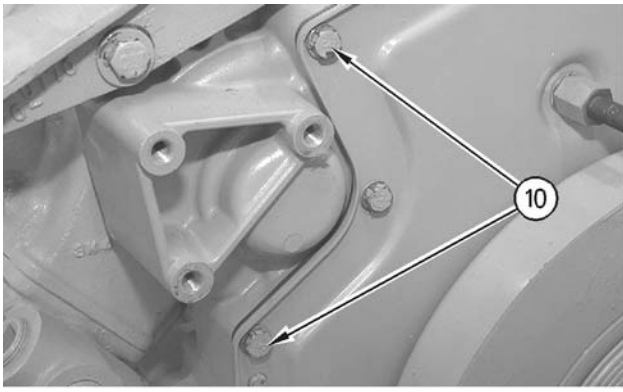


Illustration 120

g00513753

2. Install the O-ring seal on water pump (7). Lubricate the bore lightly with clean engine oil.
3. Lubricate the O-ring seals with coolant and install the O-ring seals on tube assembly (5). Install tube assembly (5) on water pump (7).
4. Position water pump (7) and tube assembly (5) in the front housing. Install two bolts (10) and the washers. Tighten the bolts to a torque of  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft).
5. Position the flange gasket between tube assembly (5) and the water temperature regulator housing. Install two bolts (4) and the washers. Tighten the bolts to a torque of  $28 \pm 7$  N·m ( $21 \pm 5$  lb ft).
6. Install hose (1) and tighten hose clamp (2).
7. Put new plastic tie (3) on tube assembly (5) in order to hold the wiring.
8. Install two bolts (9) and washers into the alternator bracket.
9. Install dipstick tube (6) with bolt (8) and the washer.
10. Fill the cooling system with coolant to the correct level. Refer to the Operation and Maintenance Manual for the correct filling procedure.

**End By:**

- a. Install the engine oil cooler. Refer to Disassembly and Assembly, "Engine Oil Cooler Install".
- b. Install the belt tensioner. Refer to Disassembly and Assembly, "Belt Tensioner - Remove and Install".
- c. Install the alternator. Refer to Disassembly and Assembly, "Alternator - Remove and Install".

## Water Temperature Regulator - Remove and Install

SMCS Code: 1355-010

### Removal Procedure

#### **WARNING**

**Personal injury can result from hot coolant, steam and alkali.**

**At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.**

**Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.**

**Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.**

**Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.**

---

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

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#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the system into a suitable container for storage or disposal.

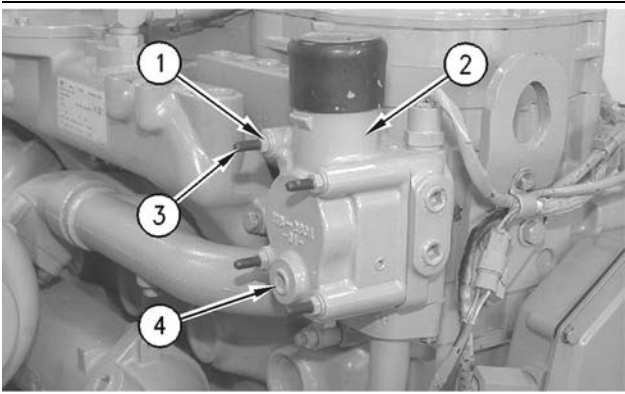


Illustration 121  
Typical example

- 2. Remove nuts (1) from regulator housing (2).
- 3. Remove regulator housing (2). Remove studs (3), if necessary.
- 4. Remove pipe plug (4), if necessary.

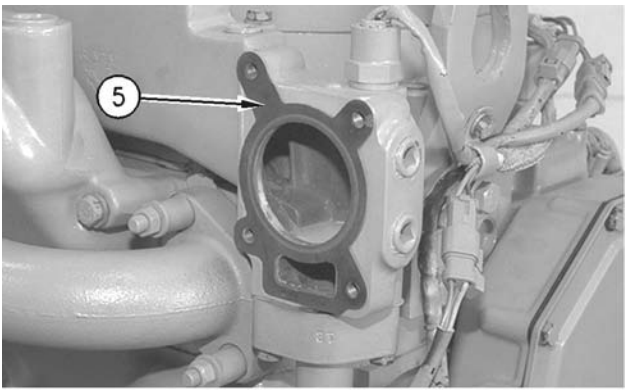


Illustration 122  
Typical example

- 5. Remove gasket (5).

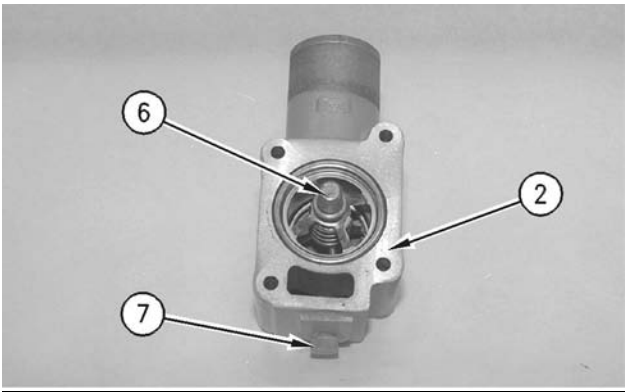


Illustration 123

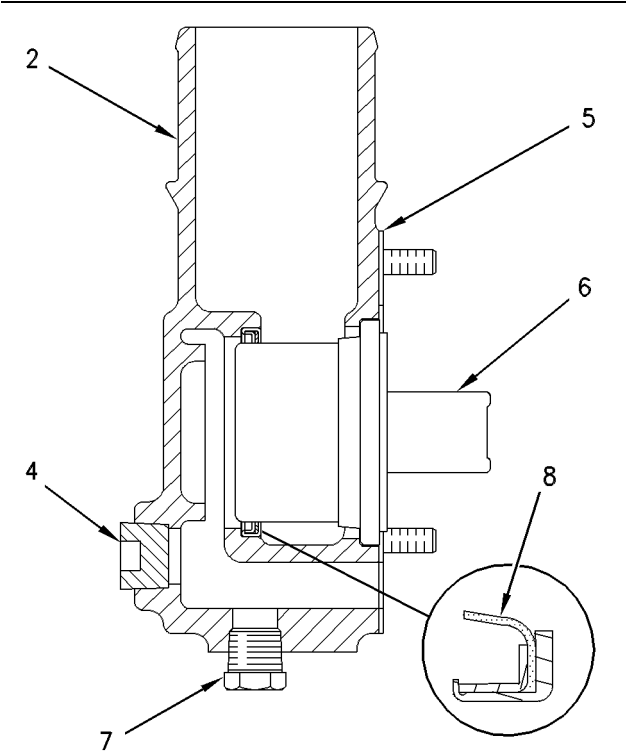


Illustration 124

- 6. Remove water temperature regulator (6). Remove pipe plug (7), if necessary.
- 7. Remove lip seal (8) from regulator housing (2).

Installation Procedure

Table 23

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 1P-0510     | Driver Group     | 1   |

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

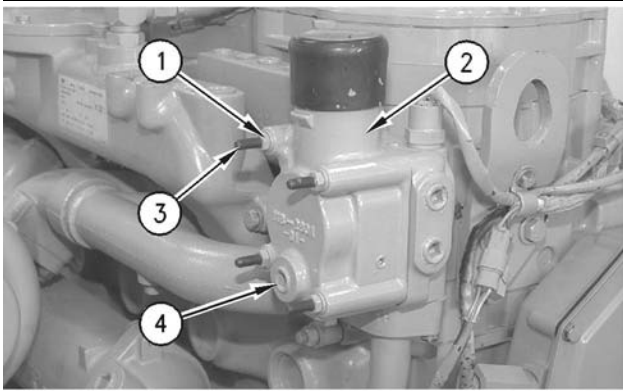


Illustration 125

g00513801

Typical example

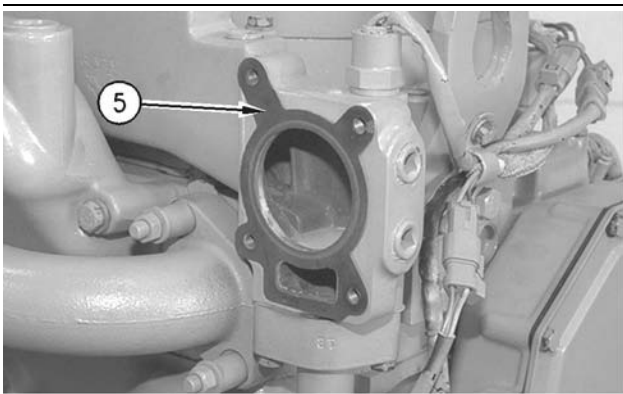


Illustration 126

g00513802

Typical example

1. Install studs (3) in the water outlet manifold.  
Position gasket (5) on studs (3).

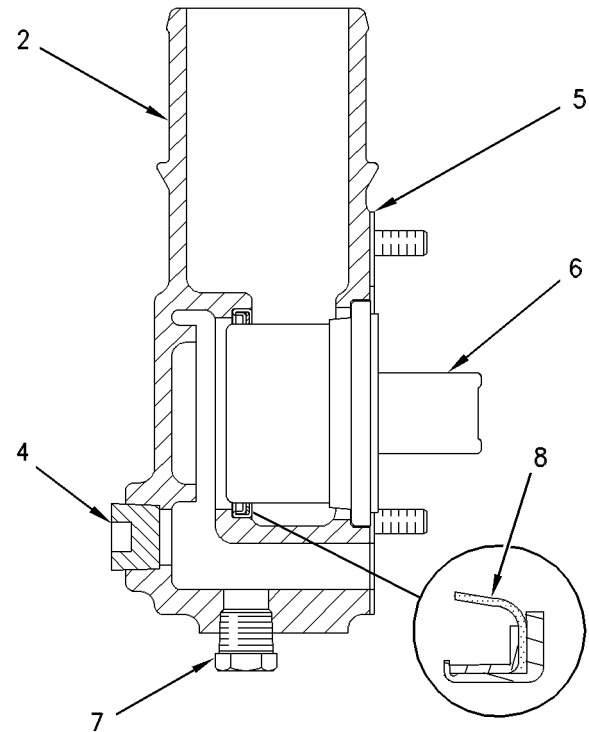


Illustration 127

g00513806

2. Position the seal with the lip toward the inside of the housing. Lubricate the lip of the seal with glycerin. Install lip seal (8) in regulator housing (2) with Tooling (A). Install the seal until contact with the counterbore in the housing.

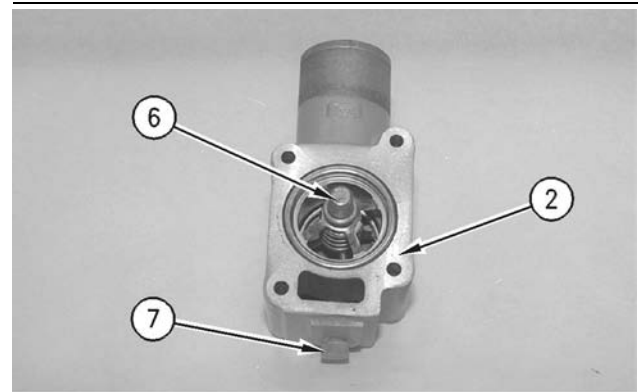


Illustration 128

g00513803

3. Install pipe plug (4) and (7) with 5P-3413 Pipe Sealant.
4. Position water temperature regulator (6) in regulator housing (2).
5. Install regulator housing (2) with nuts (1).
6. Fill the cooling system with coolant to the correct level. Refer to the Operation and Maintenance Manual for the correct filling procedure.

i01005442

## Water Outlet Manifold - Remove

SMCS Code: 1362-011

### Removal Procedure

#### WARNING

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.

Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

1. Drain the level of the coolant below the water outlet manifold. Drain the coolant into a suitable container for storage or disposal.

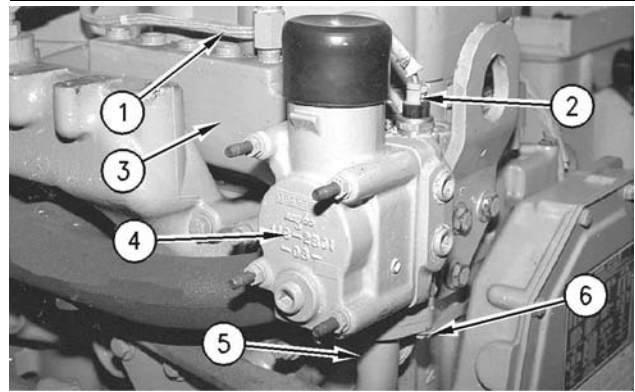


Illustration 129

g00515792

2. Remove two bolts (6) and the washers that hold tube (5) to water outlet manifold (3).
3. Disconnect sensor connector (2) and lay wiring harness (1) aside.

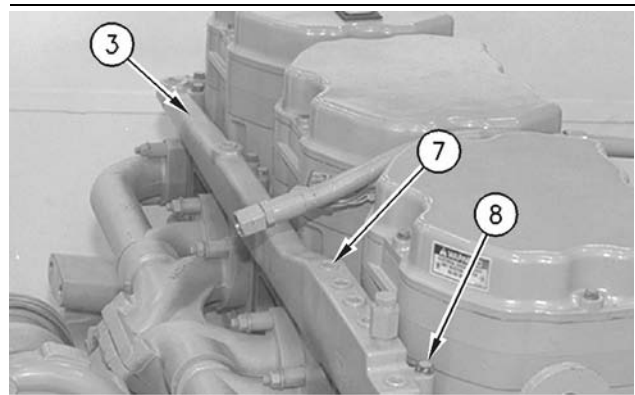


Illustration 130

g00515793

The air inlet manifold has been removed for photo purposes.

4. Remove bolts (8) and the washers.
5. Remove water outlet manifold (3) and water temperature regulator housing (4) as a unit from the cylinder head. Remove the gaskets.
6. Remove plugs (7), if necessary.

i01005443

## Water Outlet Manifold - Install

**SMCS Code:** 1362-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

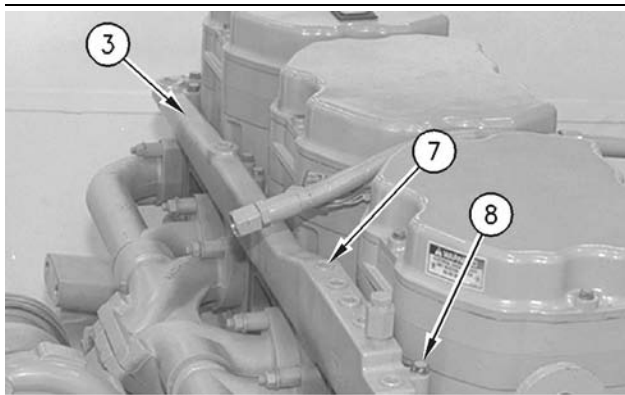


Illustration 131

g00515793

The air inlet manifold has been removed for photo purposes.

1. Inspect the condition of the gaskets. Replace the gaskets, if necessary.
2. Install the gaskets and position water outlet manifold (3) on the cylinder head.
3. Install bolts (8) and the washers.
4. Apply 5P-3413 Pipe Sealant to the threads of plugs (7). Install plugs (7).

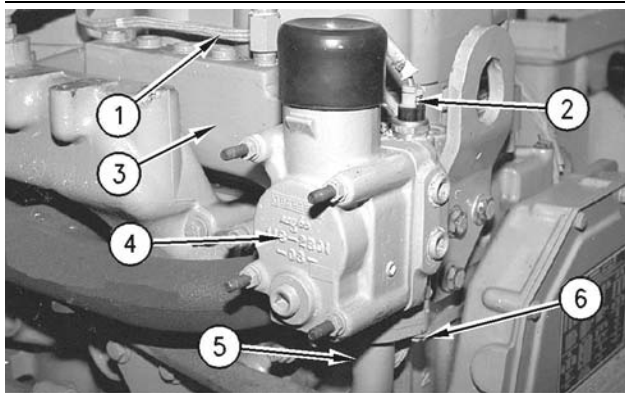


Illustration 132

g00515792

5. Install two bolts (6) and the washers.
6. Connect sensor connector (2).

7. Fill the cooling system with coolant to the correct level. Refer to the Operation and Maintenance Manual for the correct filling procedure.

i01582419

## Flywheel - Remove

**SMCS Code:** 1156-011

### Removal Procedure

Table 24

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 138-7573    | Link Bracket     | 2   |
| B              | FT2711      | Special Bolt     | 2   |
| C              | FT2712      | Guide Stud       | 2   |

1. Support the engine with blocks or a suitable lifting device.

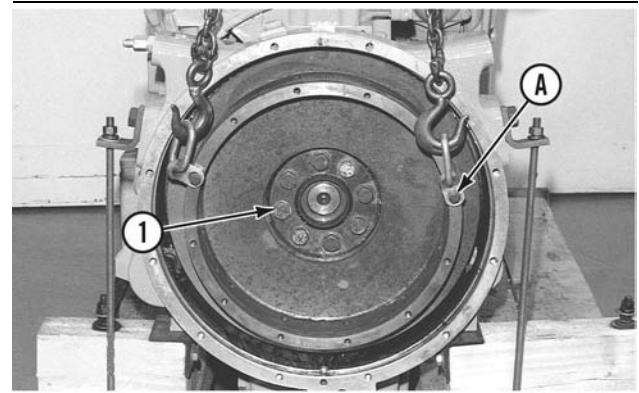


Illustration 133

g00513879

2. Install Tool (A) on the flywheel. Fasten a suitable lifting device to the flywheel.
3. Remove one bolt (1) and the washer. Remove another bolt (1) and the washer that is 180 degrees from the first bolt. Replace the two bolts with Tool (C).
4. Remove the remaining six bolts (1) and the washers.
5. Remove the flywheel. Slide the flywheel off the guide studs. The weight of the flywheel is 57 kg (125 lb).

**Note:** If your engine is equipped with a Rear Power Take-Off RPTO you need to reinstall two bolts (B). This procedure will hold the gear tight against the end of the crankshaft.

6. Inspect the flywheel ring gear. Replace the ring gear, if necessary. Place the flywheel on a wood block. Use a hammer and a punch in order to remove the ring gear.

i01771163

## Flywheel - Install

**SMCS Code:** 1156-012

### Installation Procedure

Table 25

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 138-7573    | Link Bracket     | 2   |
| B              | FT2711      | Special Bolt     | 2   |
| C              | FT2712      | Guide Stud       | 2   |

### WARNING

**Always wear protective gloves when handling parts that have been heated.**

1. Heat the ring gear to a maximum temperature of 315 °C (599 °F). Position the ring gear with the part number toward the crankshaft. Install the ring gear on the flywheel. Allow the ring gear to cool. Use a soft hammer in order to seat the ring gear against the shoulder of the flywheel.

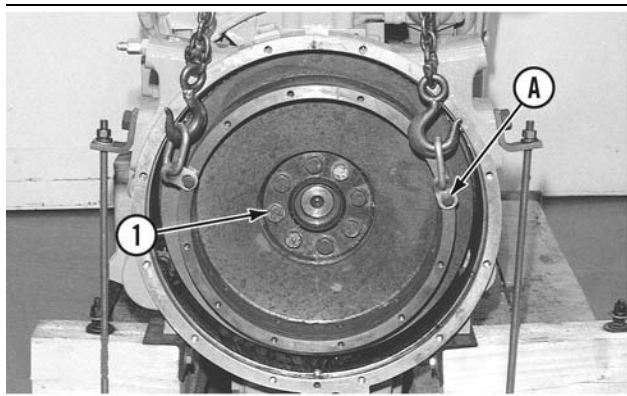


Illustration 134

g00513879

2. Install Tooling (A) on the flywheel. Fasten a suitable lifting device to the flywheel.

**Note:** If your engine is equipped with a Rear Power Take-Off RPTO, remove Tooling (B) from the end of the crankshaft and the crankshaft gear.

3. Place the flywheel in the original position on the crankshaft. Align the arrows on the flywheel and the crankshaft. Slide the flywheel over the guide studs.
4. Apply 7M-7456 Compound to the threads of eight bolts (1).
5. Install six of the eight bolts (1) and the washers. Remove Tooling (C) from the end of the crankshaft and replace with the two remaining bolts. Tighten the bolts evenly to a torque of 300 ± 40 N·m (221 ± 30 lb ft).
6. Check the flywheel runout. Refer to Testing and Adjusting, "Flywheel - Inspect" for the correct procedure.

i01465394

## Crankshaft Rear Seal - Remove

**SMCS Code:** 1161-011

### Removal Procedure

**Start By:**

- a. Remove the flywheel. Refer to Disassembly and Assembly, "Flywheel - Remove".

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

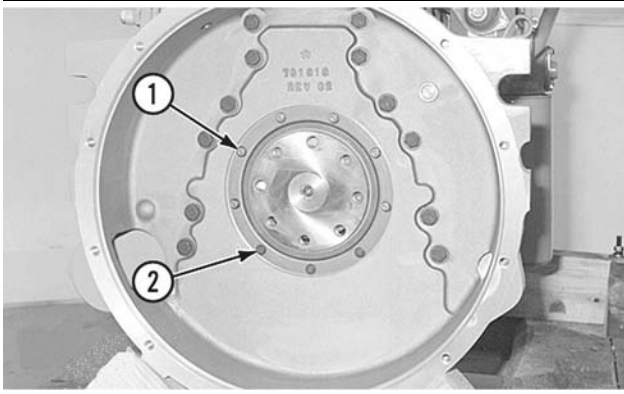


Illustration 135

g00514081

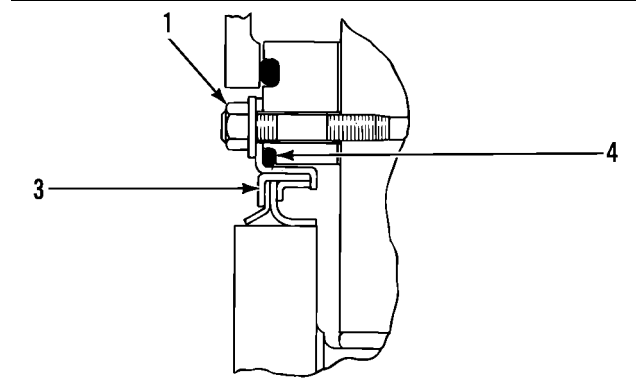


Illustration 137

g00514082

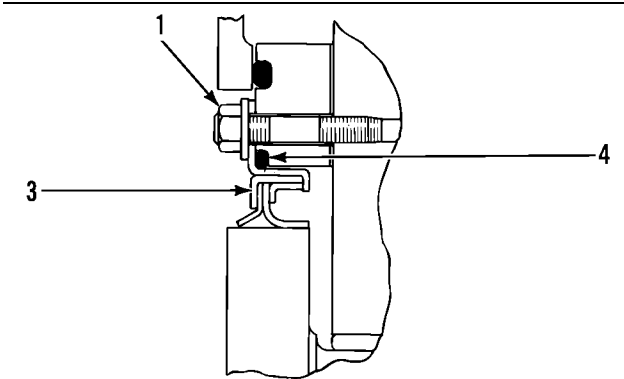


Illustration 136

g00514082

1. Remove six nuts (1) and three bolts (2).
2. Use a screwdriver to remove crankshaft rear seal (3).
3. Remove O-ring seal (4). The O-ring seal is located behind the crankshaft rear seal (3).

i01602316

## Crankshaft Rear Seal - Install

**SMCS Code:** 1161-012

### Installation Procedure

**Note:** Leave the shipping sleeve in place in order to install the crankshaft rear seal. The crankshaft rear seal must be installed dry.

1. Inspect the condition of the O-ring seal. Replace the seal, if necessary.

2. Clean the seal surface of the crankshaft. Lubricate O-ring seal (4) with clean engine oil. Position O-ring seal (4) on the back of crankshaft rear seal (3).
3. Position the crankshaft rear seal (3) on the crankshaft and the studs. Push crankshaft rear seal (3) in place. This will dislodge the shipping sleeve.
4. Install the six nuts (1) and the washers.
5. Apply 7M-7456 Compound to the threads of three bolts (2). Install three bolts (2) at the bottom of crankshaft rear seal (3). Tighten the bolts to a torque of  $12 \pm 3$  N·m ( $106 \pm 27$  lb in).

### End By:

- a. Install the flywheel. Refer to Disassembly and Assembly, "Flywheel - Install".

i01533804

# Crankshaft Rear Seal Carrier - Remove and Install

SMCS Code: 1161-010-C3

## Removal Procedure

Table 26

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 5F-4764     | Pry Bar          | 1   |

### Start By:

- a. Remove the flywheel housing. Refer to Disassembly and Assembly, "Flywheel Housing - Remove and Install".

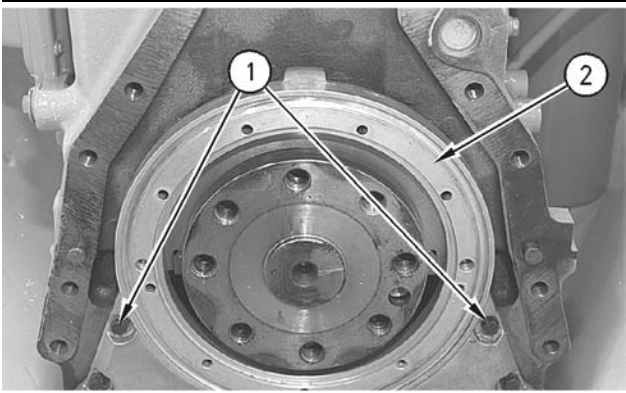


Illustration 139

g00521945

- 1. Remove four bolts (1) and the washers that hold crankshaft rear seal carrier (2) to the cylinder block.

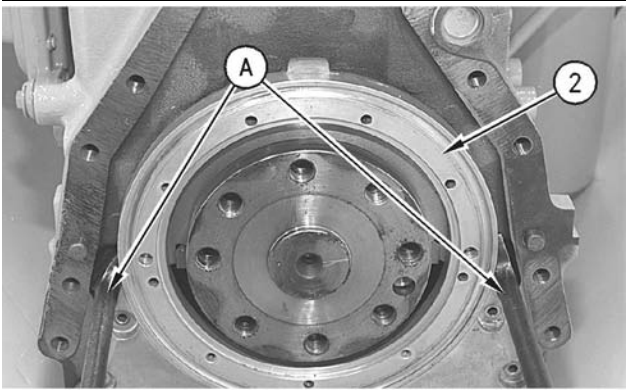


Illustration 140

g00521947

- 2. Use Tool (A) to carefully pry crankshaft rear seal carrier (2) from the crankshaft.

- 3. Remove the gasket between the cylinder block and crankshaft rear seal carrier (2).

## Installation Procedure

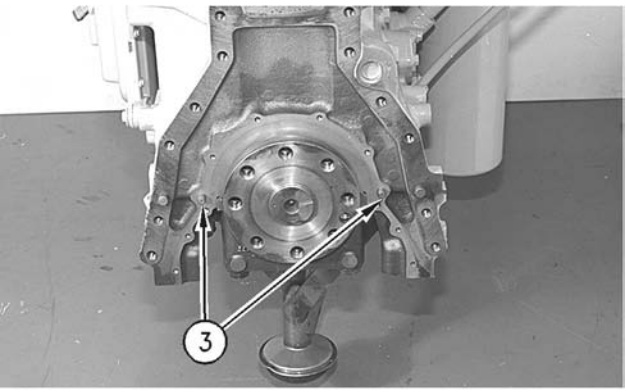


Illustration 141

g00521969

- 1. Place a new gasket over dowels (3).

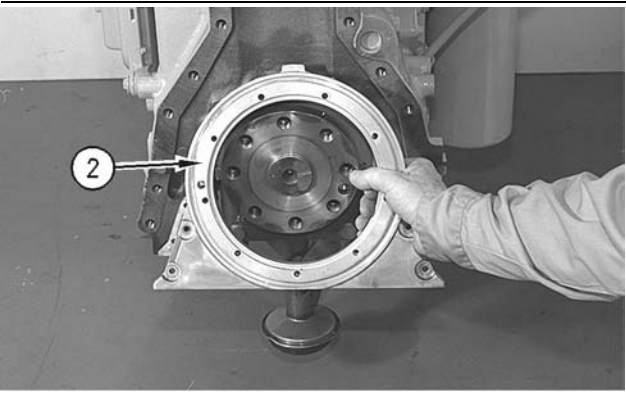


Illustration 142

g00521970

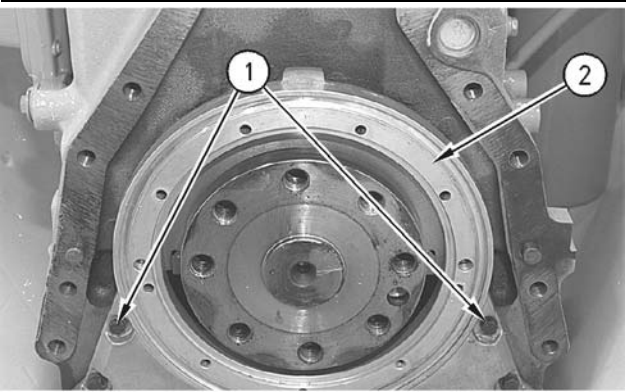


Illustration 143

g00521945

- 2. Place crankshaft rear seal carrier (2) on the crankshaft.
- 3. Install four bolts (1) and the washers.

**End By:**

- a. Install the flywheel housing. Refer to Disassembly and Assembly, "Flywheel Housing - Remove and Install".

i01582698

## Flywheel Housing - Remove and Install

**SMCS Code:** 1157-010

### Removal Procedure

**Start By:**

- a. Remove the flywheel. Refer to Disassembly and Assembly, "Flywheel - Remove".
- b. Remove the crankshaft rear seal. Refer to Disassembly and Assembly, "Crankshaft Rear Seal - Remove".

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Support the engine with blocks or a suitable lifting device.

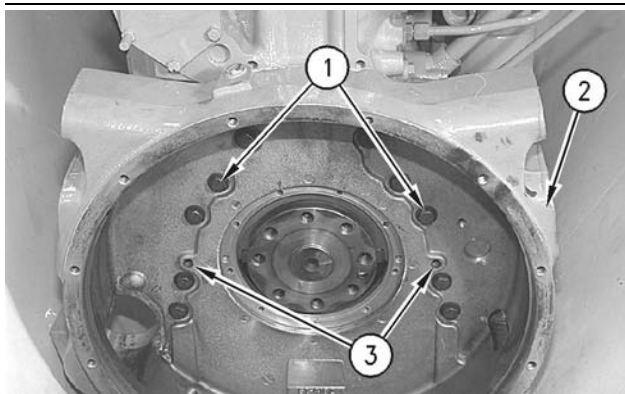


Illustration 144

g00514036

2. Remove twelve bolts (1) and the washers.

3. Attach a suitable lifting device to flywheel housing (2).
4. Lift flywheel housing (2) away from dowel pins (3). The weight of flywheel housing (2) is 23 kg (50 lb).

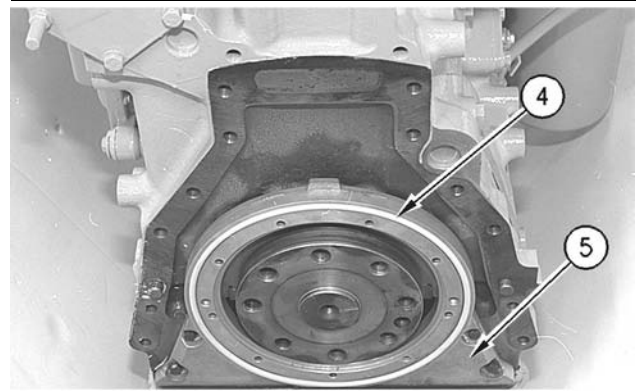


Illustration 145

g00514037

5. Remove O-ring seal (4) from rear seal carrier (5).

### Installation Procedure

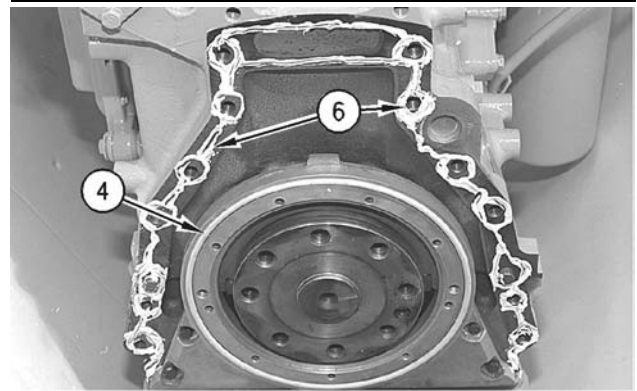


Illustration 146

g00514040

1. Apply 1U-6396 O-Ring Assembly Compound to O-ring seal (4). Install O-ring (4).
2. Apply 1U-8846 Sealant to the face of the cylinder block.

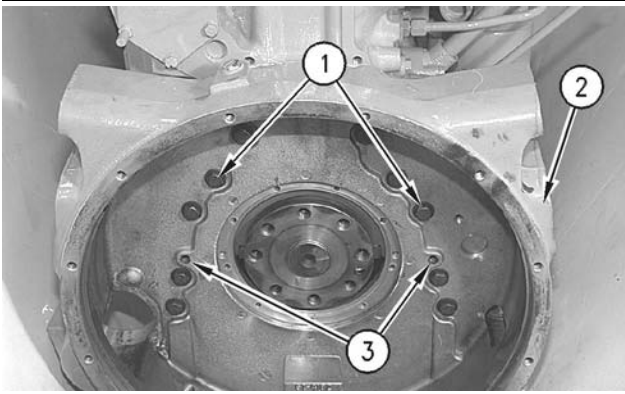


Illustration 147 g00514036

3. Position flywheel housing (2) on the rear of the cylinder block. Install twelve bolts (1) and the washers.

**End By:**

- a. Install the crankshaft rear seal. Refer to Disassembly and Assembly, "Crankshaft Rear Seal - Install" .
- b. Install the flywheel. Refer to Disassembly and Assembly, " Flywheel - Install".

i01771286

# Rear Power Take-Off (RPTO) - Remove

SMCS Code: 1165-011-RE

## Removal Procedure

Table 27

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 8T-0514     | Guide Bolt       | 2   |
| B              | 1P-0520     | Driver Group     | 1   |

**Start By:**

- a. Remove the flywheel. Refer to Disassembly and Assembly, "Flywheel - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

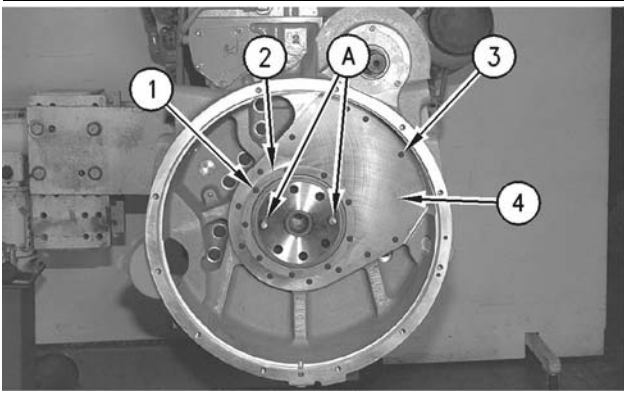


Illustration 148 g00520761

- 1. Install Tooling (A) in the crankshaft.
- 2. Remove bolts (1) and crankshaft seal (2).
- 3. Remove bolts (3), cover (4), and the gasket. Inspect the condition of the gasket. Replace the gasket, if necessary.

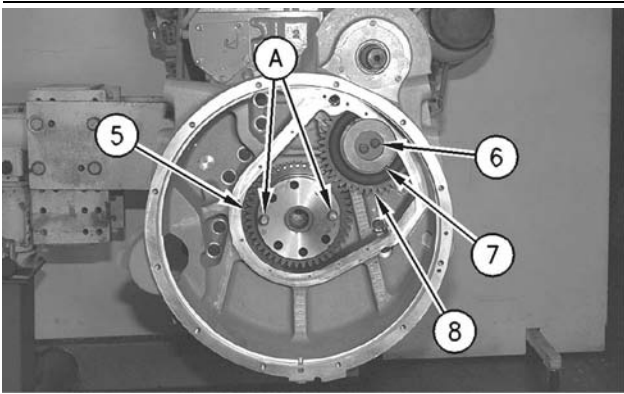


Illustration 149 g00520762

- 4. Remove crankshaft gear (5) from the crankshaft. Remove the O-ring seal from crankshaft gear (5).
- 5. Remove Tooling (A).
- 6. Remove bolts (6) that fasten shaft (7) to the flywheel housing.

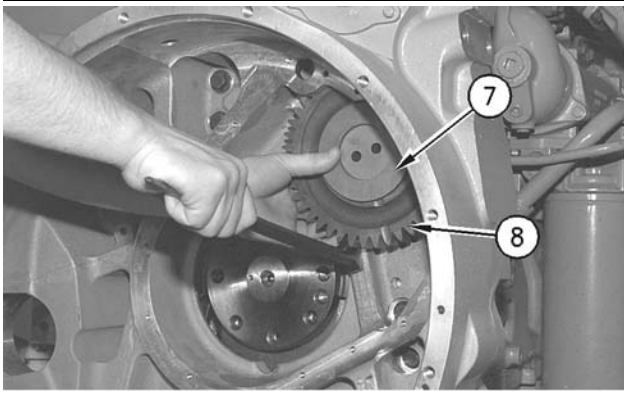


Illustration 150

g00520763

- 7.** Use a pry bar in order to remove shaft (7) and gear assembly (8).

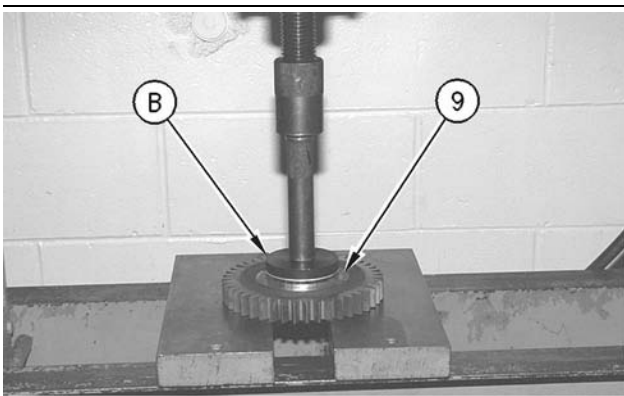


Illustration 151

g00520764

- 8.** Inspect the condition of the bearing in gear assembly (8). If necessary, remove bearing (9) from the gear with Tooling (B).

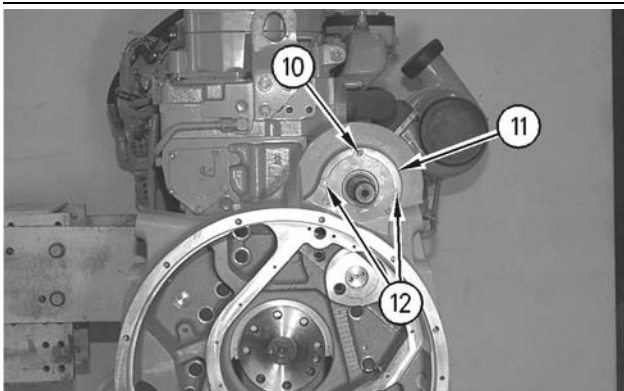


Illustration 152

g00520766

- 9.** Remove bolts (10) that fasten carrier assembly (11) to the flywheel housing. Install two bolts (10) opposite each other at location (12). Use these bolts as forcing screws in order to remove carrier assembly (11).

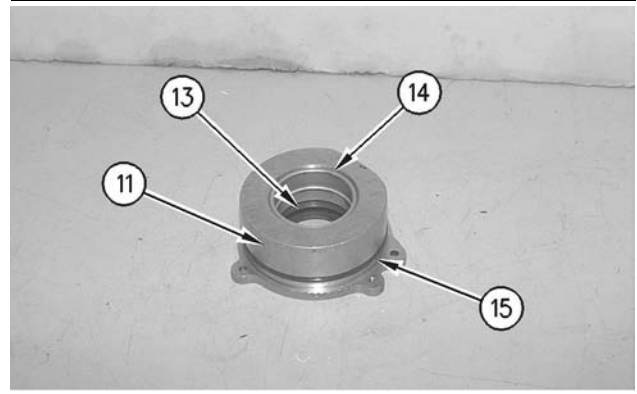


Illustration 153

g00765120

- 10.** Remove O-ring seal (15) and lip type seal (13) from carrier assembly (11).

- 11.** Inspect the condition of bearing (14). If necessary, use Tooling (B) to remove bearing (14) from carrier assembly (11).

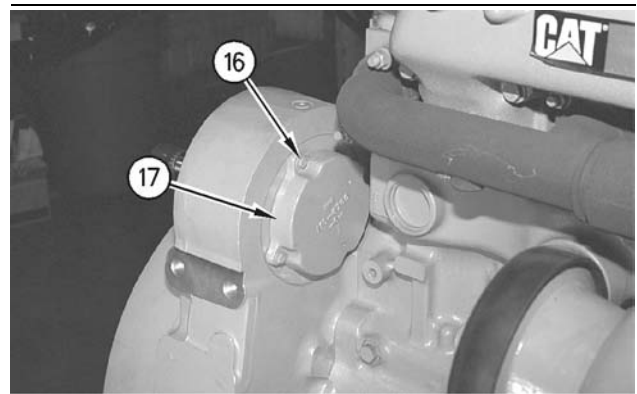


Illustration 154

g00520769

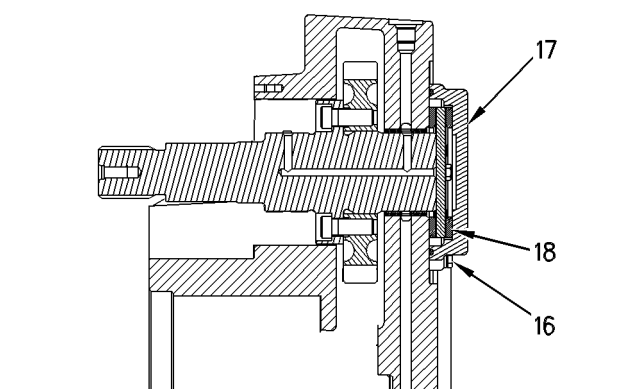


Illustration 155

g00520770

- 12.** Remove bolts (16), cover assembly (17), and thrust washer (18). Replace the O-ring seal.

- 13.** Inspect the condition of thrust washer (18). Replace thrust washer (18), if necessary.

**Note:** The minimum allowable thickness for thrust washer (18) is 4.665 mm (0.1837 inch).

14. Inspect the condition of the dowels for thrust washer (18) in cover assembly (17). Remove the dowels if replacement is necessary.

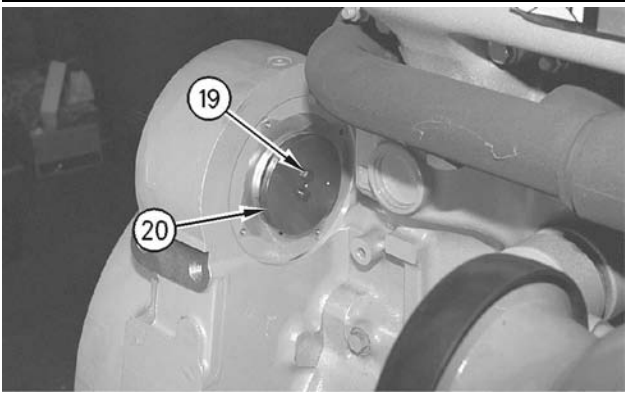


Illustration 156

g00520771

15. Remove bolts (19) and thrust plate (20) from the output shaft.

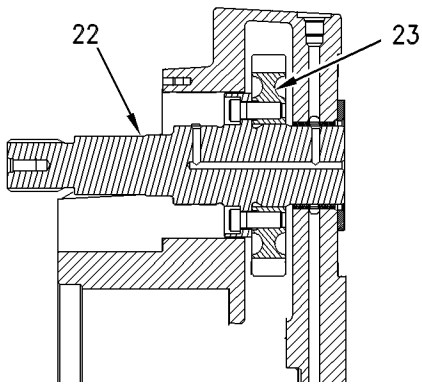


Illustration 157

g00907516

16. Remove hex socket head bolts (21) that fasten gear (23) to output shaft (22).

17. Remove output shaft (22) from gear (23). Remove gear (23) from the flywheel housing.

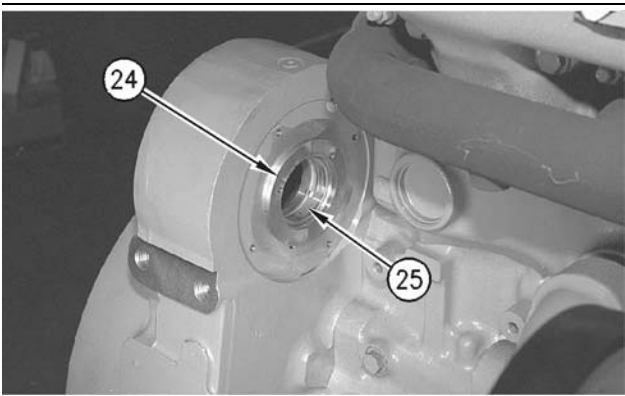


Illustration 158

g00520774

18. Remove thrust washer (24) from the flywheel housing.

19. Inspect the condition of thrust washer (24). Replace thrust washer (24), if necessary.

**Note:** The minimum allowable thickness for thrust washer (24) is 4.665 mm (0.1837 inch).

20. Inspect the condition of the dowels for thrust washer (24) in the flywheel housing. Remove the dowels if replacement is necessary.

21. Inspect the condition of bearing (25). If necessary, remove bearing (25) from the flywheel housing with Tooling (B).

i01771332

## Rear Power Take-Off (RPTO) - Install

**SMCS Code:** 1165-012-RE

### Installation Procedure

Table 28

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 8T-0514     | Guide Bolt       | 2   |
| B              | 1P-0520     | Driver Group     | 1   |

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

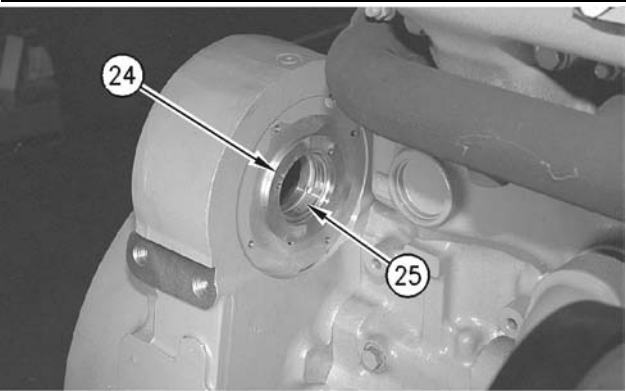


Illustration 159

g00520774

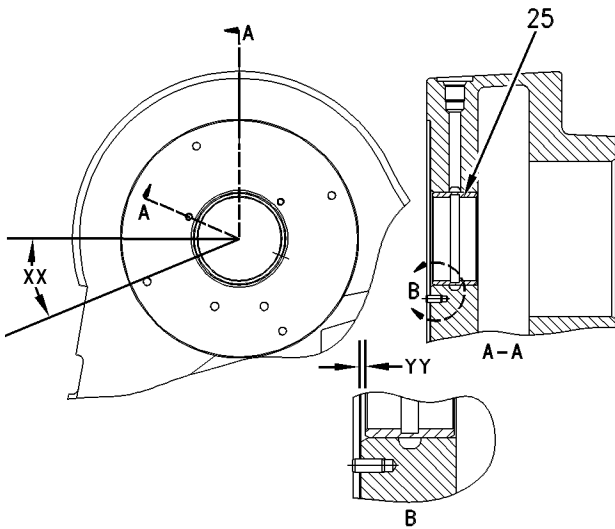


Illustration 160

g00520775

1. Install bearing (25) in the flywheel housing with Tooling (B).

**Note:** The split in the bearing should be located at angle (XX), which is  $23 \pm 10$  degrees from the horizontal centerline. The bearing should be installed to depth (YY), which is  $1.5 \pm 0.5$  mm ( $0.06 \pm 0.02$  inch).

2. Install the dowels for thrust washer (24) in the flywheel housing.

**Note:** The dowels should extend beyond the surface of the flywheel housing by a distance of  $3.0 \pm 0.2$  mm ( $0.12 \pm 0.01$  inch).

3. Install thrust washer (24) on the dowels in the flywheel housing.

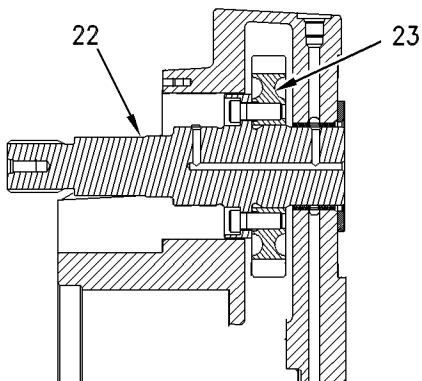


Illustration 161

g00520773

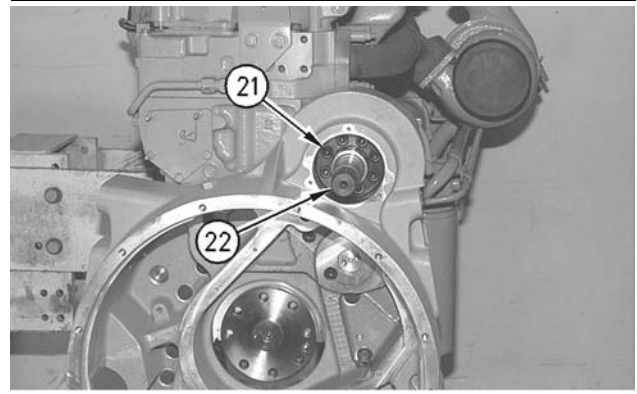


Illustration 162

g00520772

4. Hold gear (23) in the flywheel housing and install output shaft (22) in gear (23).
5. Fasten gear (23) to output shaft (22) with hex socket head bolts (21).

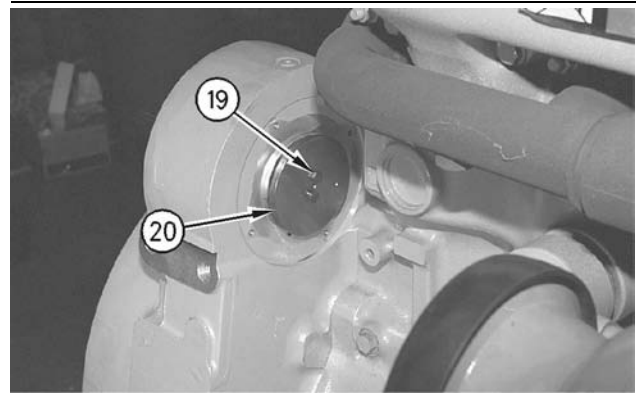


Illustration 163

g00520771

6. Fasten thrust plate (20) to the output shaft with bolts (19).

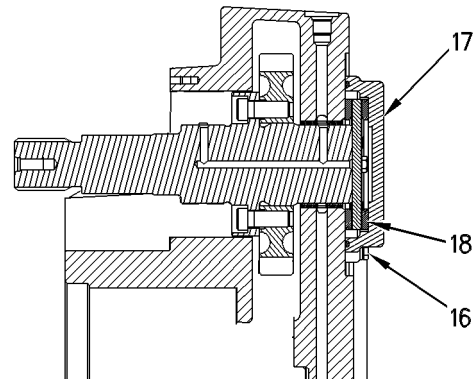


Illustration 164

g00520770

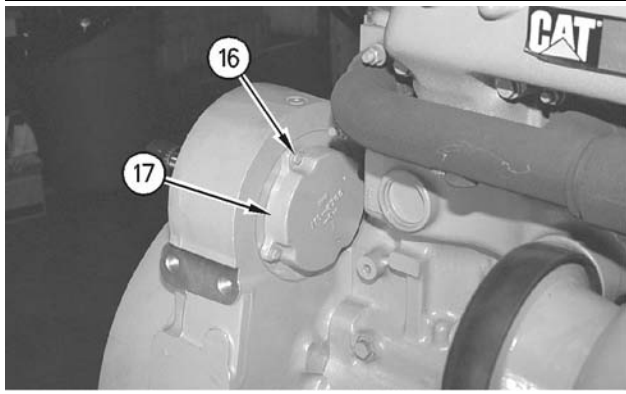


Illustration 165

g00520769

7. Install the dowels for thrust washer (18) in cover assembly (17).

**Note:** The dowels should extend beyond the surface of cover assembly (17) by a distance of  $3.0 \pm 0.2$  mm ( $0.12 \pm 0.01$  inch).

8. Install thrust washer (18) on the dowels in cover assembly (17). Fasten cover assembly (17) to the flywheel housing with bolts (16).

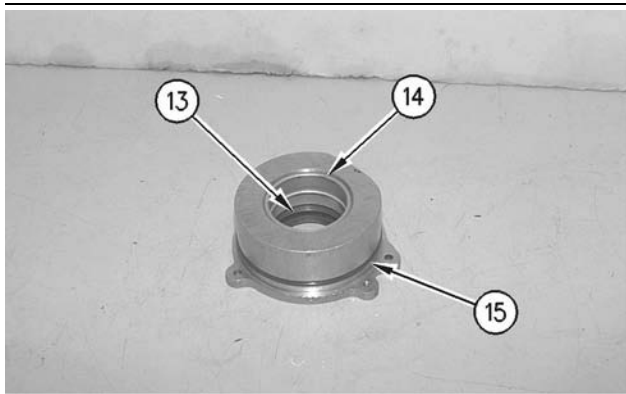


Illustration 166

g00520767

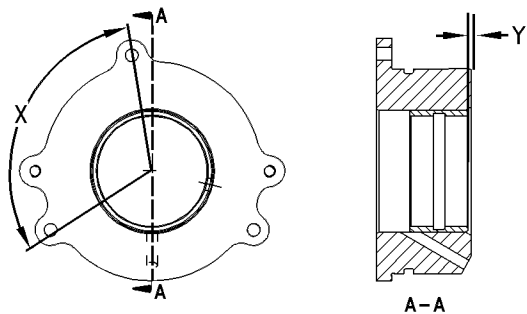


Illustration 167

g00520768

9. Install bearing (14) in the carrier assembly with Tooling (B).

**Note:** The split in the bearing should be located at angle (X), which is  $112 \pm 10$  degrees from the centerline in the upper bolt hole in the carrier assembly. The bearing should be installed to depth (Y), which is  $1.6 \pm 0.5$  mm ( $0.06 \pm 0.02$  inch).

10. Install O-ring seal (15) and lip type seal (13) in the carrier assembly. Lubricate lip type seal (13) lightly with the lubricant that is being sealed.

**Note:** Lip type seal (13) should be installed to a depth of  $2.5 \pm 0.5$  mm ( $0.1 \pm 0.02$  inch).

11. Install the carrier assembly in the flywheel housing. Install bolts (10).

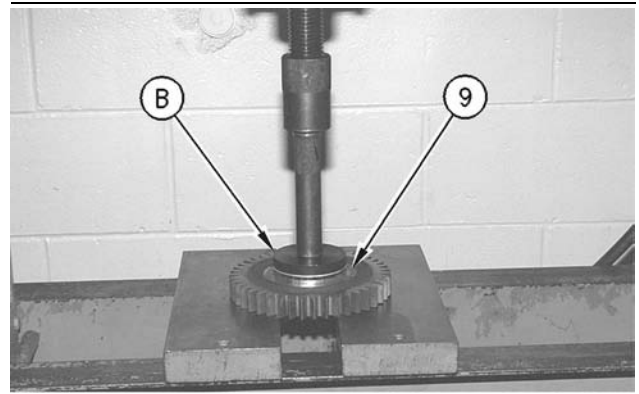


Illustration 168

g00520764

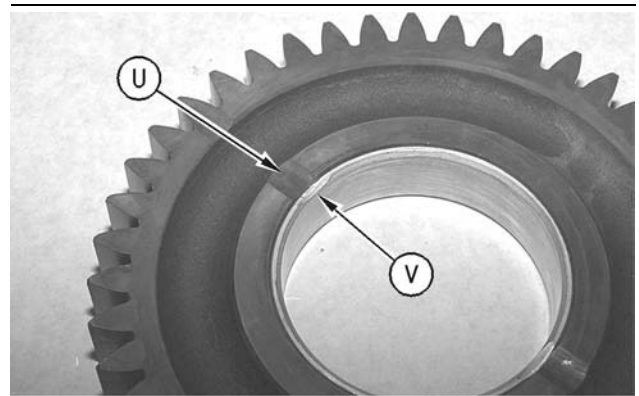


Illustration 169

g00520765

12. Install bearing (9) in gear (8) with Tooling (B).

**Note:** Bearing relief groove (U) must line up with the relief groove in gear (8) within two degrees. Bearing (9) must not extend beyond either face of gear (8).

i01464063

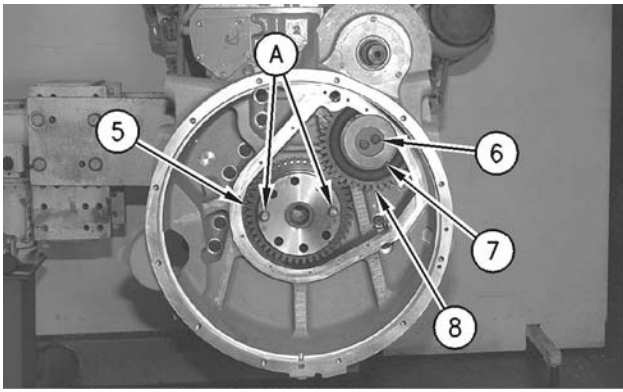


Illustration 170

g00520762

13. Install shaft (7) and gear assembly (8) in the flywheel housing. Fasten shaft (7) to the flywheel housing with bolts (6).
14. Install Tooling (A) in the crankshaft. Install a new O-ring seal on crankshaft gear (5).
15. Install crankshaft gear (5).

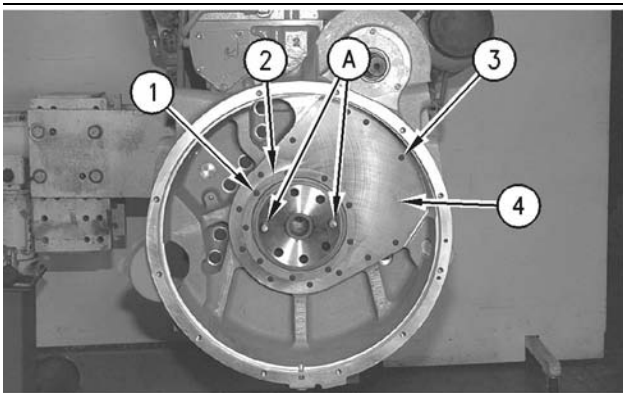


Illustration 171

g00520761

**Note:** Some flywheel housings have locating dowels for cover (4). If the flywheel housing does not have these locating dowels, care must be taken in order to ensure that the cover is properly aligned. Ensure that the gap between the dust lip and the crankshaft gear is uniform in order to avoid damage to the crankshaft seal.

16. Hold cover (4) and the gasket in position on the flywheel housing and install bolts (3).
17. Install crankshaft seal (2) and bolts (1).
18. Remove Tooling (A) from the crankshaft.

#### End By:

- a. Install the flywheel. Refer to Disassembly and Assembly, "Flywheel - Install".

## Vibration Damper and Pulley - Remove and Install

**SMCS Code:** 1205-010

### Removal Procedure

1. Relieve the tension on the belt tensioner and remove the drive belts.

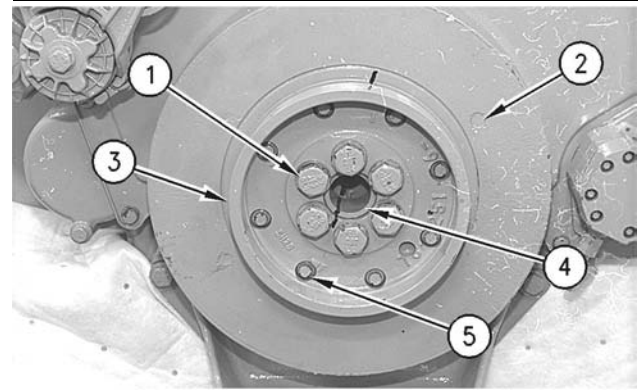


Illustration 172

g00514208

2. Remove six bolts (1) and the washers.
3. Remove the following items from the end of the crankshaft: vibration damper (2), pulley (3), and adapter (4).
4. Remove eight bolts (5).

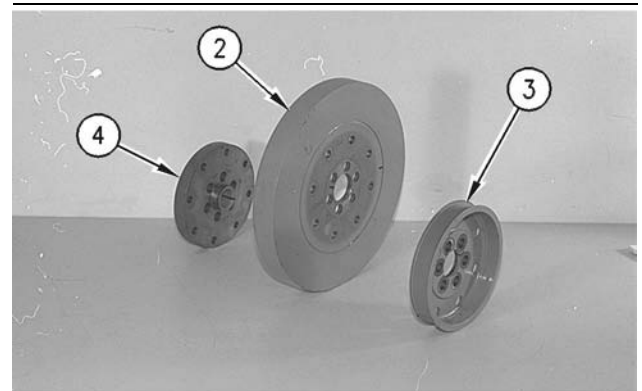


Illustration 173

g00514209

5. Separate pulley (3) and adapter (4) from vibration damper (2).

## Installation Procedure

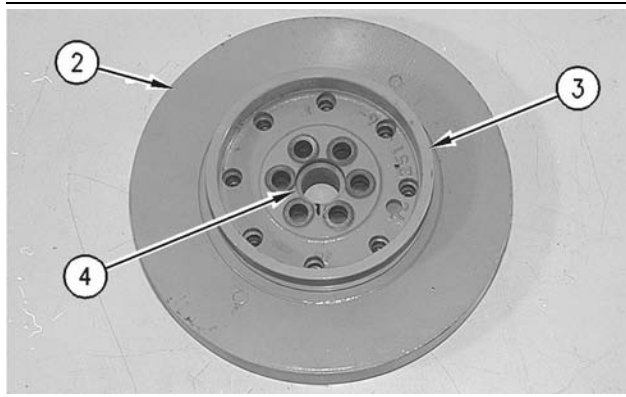


Illustration 174

g00514210

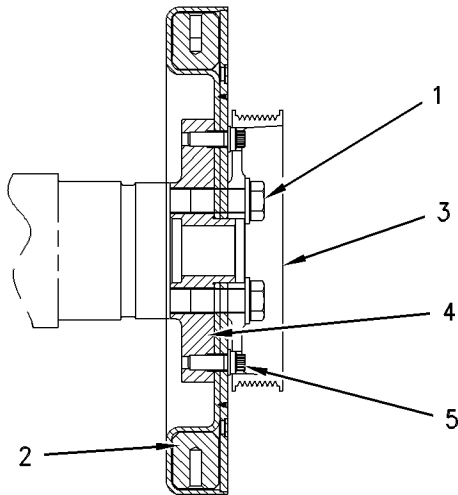


Illustration 175

g00514211

1. Place adapter (4) and pulley (3) on vibration damper (2). Install eight bolts (5). Tighten the bolts to a torque of  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft).
2. Install bolts (1) and the washers. Tighten all bolts evenly to a torque of  $240 \pm 40$  N·m ( $175 \pm 30$  lb ft).
3. Install the drive belt and apply the correct tension.

## Crankshaft Front Seal - Remove

i01464074

**SMCS Code:** 1160-011

### Removal Procedure

#### Start By:

- a. Remove the vibration damper and the pulley. Refer to Disassembly and Assembly, "Vibration Damper and Pulley - Remove and Install".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

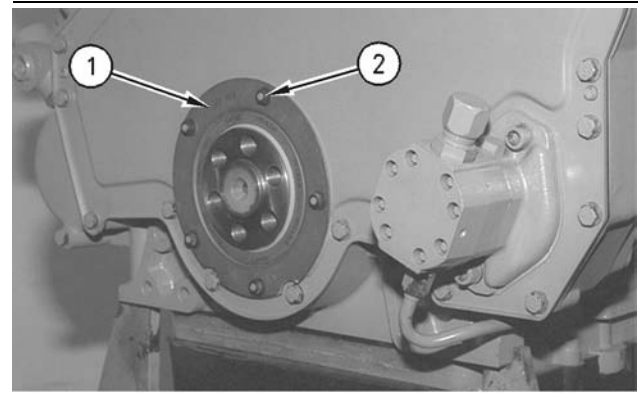


Illustration 176

g00514250

1. Remove nuts (2) and the washers.
2. Remove crankshaft front seal (1).

i01464078

## Crankshaft Front Seal - Install

**SMCS Code:** 1160-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**Note:** Leave the shipping sleeve in place in order to install the crankshaft front seal. The crankshaft front seal must be installed dry.

1. Clean the seal surface on the crankshaft and the front cover.

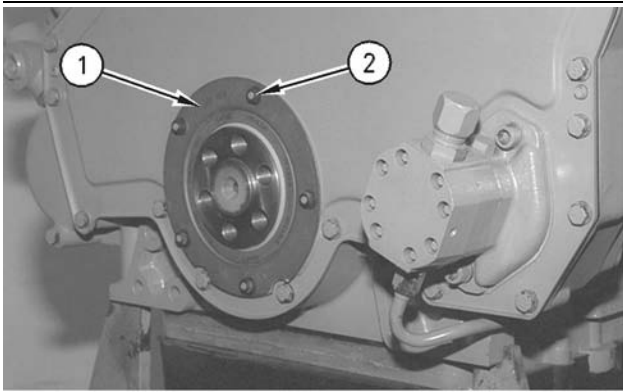


Illustration 177

g00514250

2. Place crankshaft front seal (1) on the crankshaft. Push the crankshaft front seal in place. This will dislodge the shipping sleeve.
3. Install nuts (2) and the washers.

**End By:**

- a. Install the vibration damper and the pulley. Refer to Disassembly and Assembly, "Vibration Damper - Remove and Install".

i01464100

## Front Cover - Remove

**SMCS Code:** 1166-011

### Removal Procedure

**Start By:**

- a. Remove the alternator. Refer to Disassembly and Assembly, "Alternator Remove and Install".
- b. Remove the belt tensioner. Refer to Disassembly and Assembly, "Belt Tensioner - Remove and Install".
- c. Remove the crankshaft front seal. Refer to Disassembly and Assembly, "Crankshaft Front Seal - Remove".
- d. Remove the fuel transfer pump. Refer to Disassembly and Assembly, "Fuel Transfer Pump - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

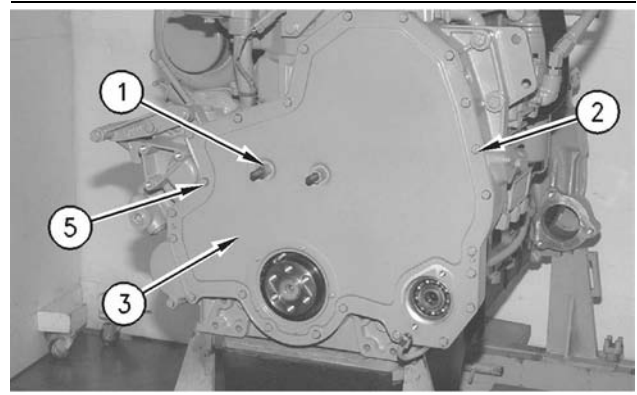


Illustration 178

g00514870

Typical example

1. Remove studs (1) with the nuts and the seal washers.
2. Remove bolts (2) and the washers.
3. Remove bolt (5).
4. Remove cover (3).

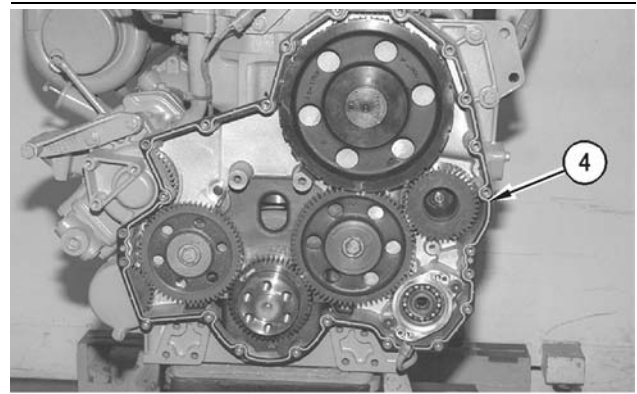


Illustration 179

g00514872

Typical example

5. Remove housing cover gasket (4).

i01464118

# Front Cover - Install

**SMCS Code:** 1166-012

## Installation Procedure

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**Note:** The rear face of the cover and the front face of the housing must be clean.

1. Inspect the condition of the housing cover gasket. Replace the gasket, if necessary.

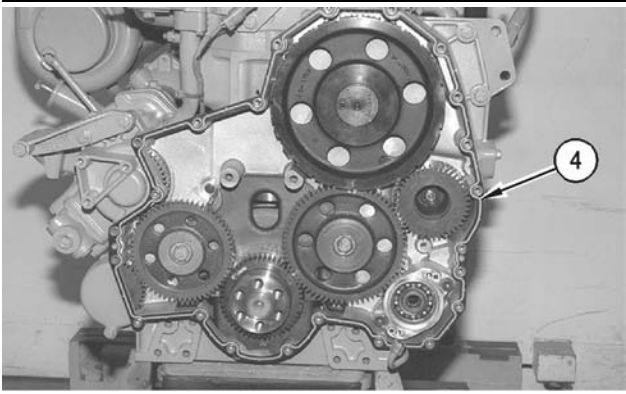


Illustration 180

g00514872

Typical example

2. Install housing cover gasket (4).

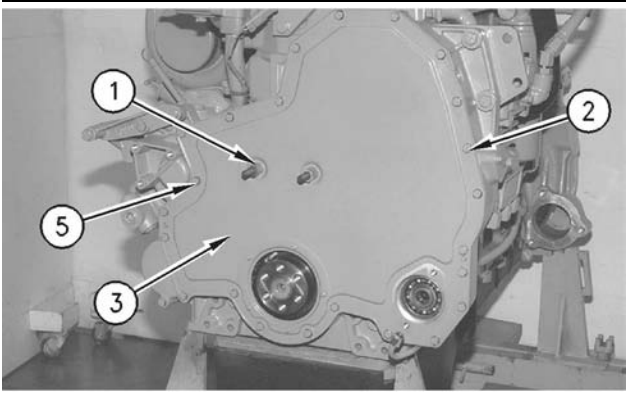


Illustration 181

g00514870

Typical example

**Note:** Bolt (5) is a through bolt. Apply **9S-3263** Thread Lock Compound to threads prior to installation.

3. Install the following items on the front housing: front cover (3), bolts (2), bolt (5), and the washers.

**Note:** Do not reuse the seal washers. Use new seal washers for installation.

4. Install studs (1) with new seal washers and the nuts.

### End By:

- a. Install the fuel transfer pump. Refer to Disassembly and Assembly, "Fuel Transfer Pump - Install".
- b. Install the crankshaft front seal. Refer to Disassembly and Assembly, "Crankshaft Front Seal - Install".
- c. Install the belt tensioner. Refer to Disassembly and Assembly, "Belt Tensioner - Remove and Install".
- d. Install the alternator. Refer to Disassembly and Assembly, "Alternator - Remove and Install".

i01533820

# Gear Group (Front) - Remove

**SMCS Code:** 1206-011

## Removal Procedure

Table 29

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 6V-8214     | Bolt             | 4   |

### Start By:

- a. Remove the water pump. Refer to Disassembly and Assembly, "Water Pump - Remove".
- b. Remove the engine oil pump. Refer to Disassembly and Assembly, "Engine Oil Pump - Remove".
- c. Remove the air compressor. Refer to Disassembly and Assembly, "Air Compressor - Remove and Install".
- d. Remove the alternator. Refer to Disassembly and Assembly, "Alternator - Remove and Install".

- e. Remove the camshaft position sensor. Refer to Disassembly and Assembly, "Camshaft Position Sensor - Remove and Install".
- f. Remove the crankshaft position sensor. Refer to Disassembly and Assembly, "Crankshaft Position Sensor - Remove and Install".
- g. Remove the camshaft. Refer to Disassembly and Assembly, "Camshaft - Remove".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

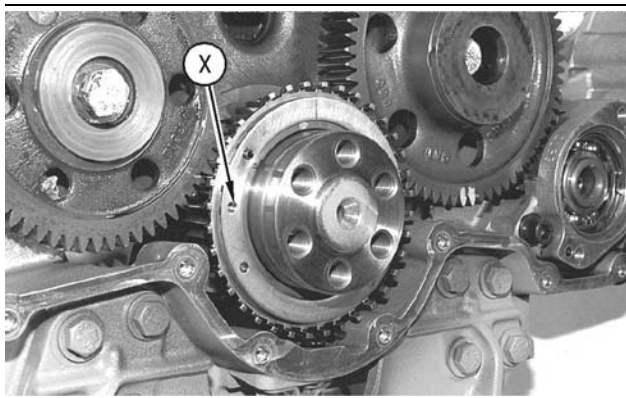


Illustration 182

g00522893

**Note:** The sensor plate must be removed from the crankshaft gear for Engines (S/N: 3CS1-11999) and (S/N: 2KS1-39999) in order to remove the front gear group. The sensor plate can have either two threaded holes to aid in the removal of the sensor plate or the sensor plate can have four threaded holes.

1. Install Tool (A) in bolt holes (X). Turn the bolts evenly in order to separate the sensor plate from the crankshaft gear.

**Note:** Align the timing marks on the idler gear with the marks on the crankshaft gear and the camshaft gear before removal.

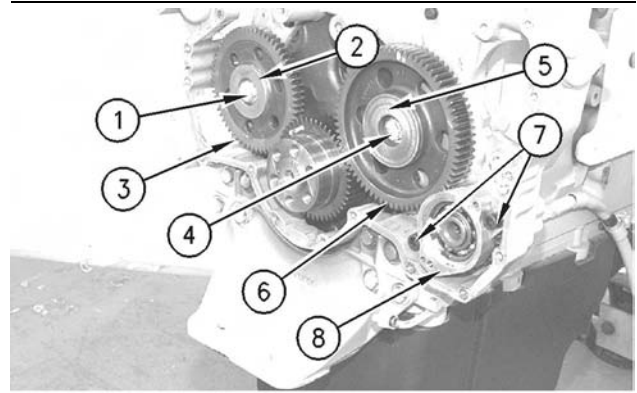


Illustration 183

g00522192

2. Remove the following items from the front housing: bolt (1), shaft (2), gear (3), and the bearing.
3. Next, remove the following items from the front housing: bolt (4), shaft (5), gear (6), and the bearing.

**Note:** During installation, it is possible to install the accessory drive incorrectly. Before removal, mark accessory drive (8) for correct orientation.

4. Remove bolts (7) and accessory drive (8).

i01533832

## Gear Group (Front) - Install

**SMCS Code:** 1206-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

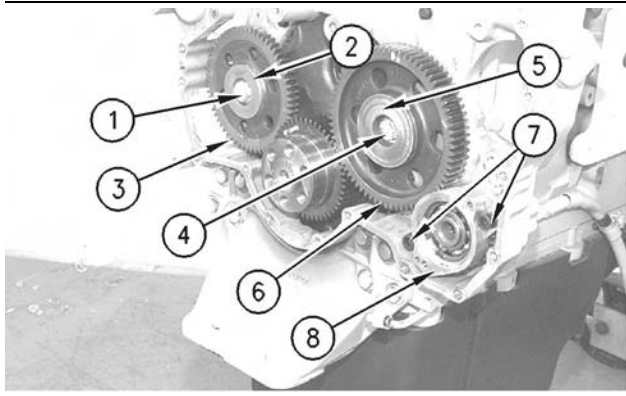


Illustration 184

g00522192

1. Hold accessory drive (8) in position and install two bolts (7). Tighten the bolts to a torque of  $100 \pm 20$  N·m ( $75 \pm 15$  lb ft).
2. Install the following items in the front housing: gear (6), the bearing and shaft (5), and bolt (4). Tighten bolt (4) to a torque of  $240 \pm 40$  N·m ( $175 \pm 30$  lb ft).
3. Install the following items in the front housing: gear (3), the bearing and shaft (2), and bolt (1). Tighten bolt (1) to a torque of  $240 \pm 40$  N·m ( $175 \pm 30$  lb ft).

**Note:** Ensure that the timing marks on the gears are in alignment.

### **WARNING**

**Always wear protective gloves when handling parts that have been heated.**

4. Heat the sensor plate to a temperature of  $210^{\circ}\text{C}$  ( $410^{\circ}\text{F}$ ) for no more than 20 minutes.

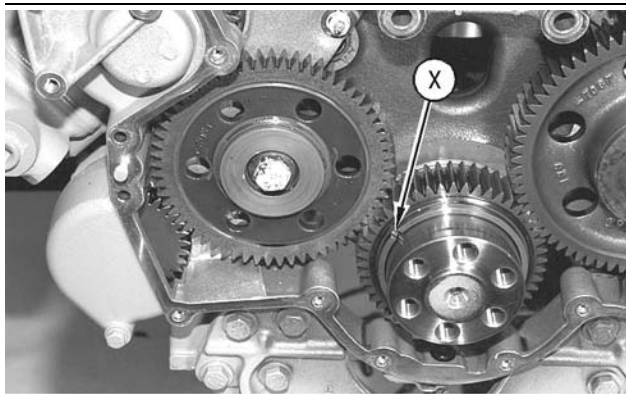


Illustration 185

g00523036

5. Position the hot sensor plate so that the tab on the sensor plate aligns with groove (X) on the crankshaft gear.
6. Push the sensor plate onto the crankshaft gear until the back of the sensor plate contacts the front of the crankshaft gear. Hold the sensor plate on the crankshaft gear in this position until the sensor plate cools and the sensor plate is properly seated.
7. Use a feeler gauge in order to measure the clearance between the back of the sensor plate and the front of the crankshaft gear. The measurement should be taken in three places and the clearance should not exceed 0.25 mm (0.010 inch).

**Note:** Steps 4 through 7 are for Engines (S/N: 3CS1-11999) and (S/N: 2KS1-39999) that have the sensor plate.

### **End By:**

- a. Install the camshaft. Refer to Disassembly and Assembly, "Camshaft - Install".
- b. Install the camshaft position sensor. Refer to Disassembly and Assembly, "Camshaft Position Sensor - Remove and Install".
- c. Install the crankshaft position sensor. Refer to Disassembly and Assembly, "Crankshaft Position Sensor - Remove and Install".
- d. Install the alternator. Refer to Disassembly and Assembly, "Alternator - Remove and Install".
- e. Install the air compressor. Refer to Disassembly and Assembly, "Air Compressor - Remove and Install".
- f. Install the engine oil pump. Refer to Disassembly and Assembly, "Engine Oil Pump - Install".
- g. Install the water pump. Refer to Disassembly and Assembly, "Water Pump - Install".

i01464328

## Housing (Front) - Remove

**SMCS Code:** 1151-011

### Removal Procedure

#### Start By:

- a. Remove the front gear group. Refer to Disassembly and Assembly, "Gear Group (Front) - Remove".

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

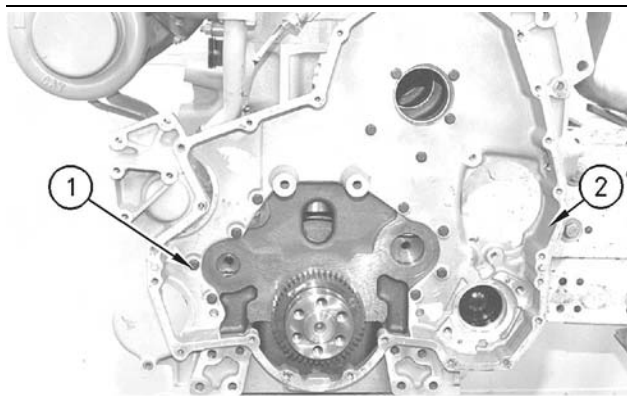


Illustration 186

g00514280

1. Remove thirteen bolts (1) and the washers.
2. Remove front housing (2) and the gasket.

i01783247

## Housing (Front) - Install

**SMCS Code:** 1151-012

### Installation Procedure

**Note:** The face of the housing and the face of the block must be clean prior to installation.

#### NOTICE

Do not use fingers or towels in order to remove the excess 8T-9022 Silicone Gasket. The use of fingers or towels will not leave a flat sealing surface. A leak may develop.

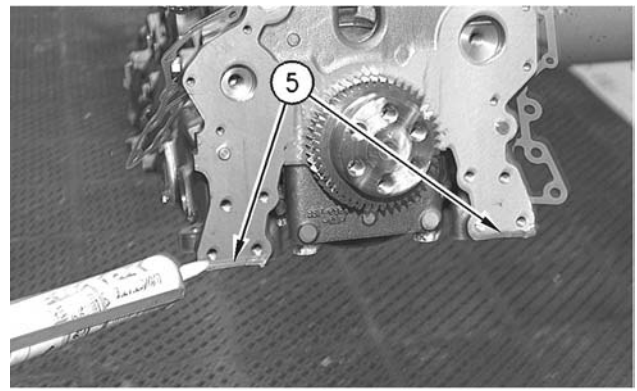


Illustration 187

g00514286

Typical example

1. Apply a bead of 8T-9022 Silicone Gasket to the lower edges (5) of the cylinder block.

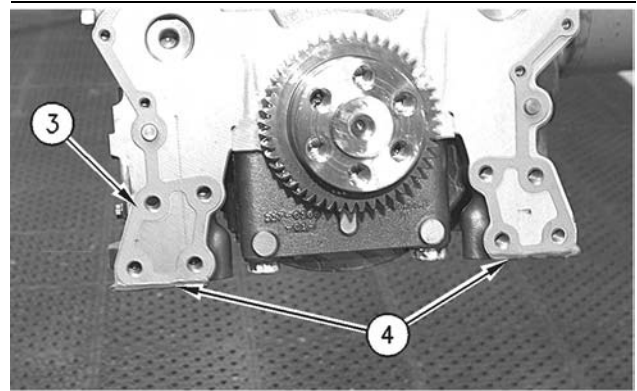


Illustration 188

g00514307

Typical Example

2. Install gasket (3). Apply a bead of 8T-9022 Silicone Gasket to lower edges (4) of gasket (3).

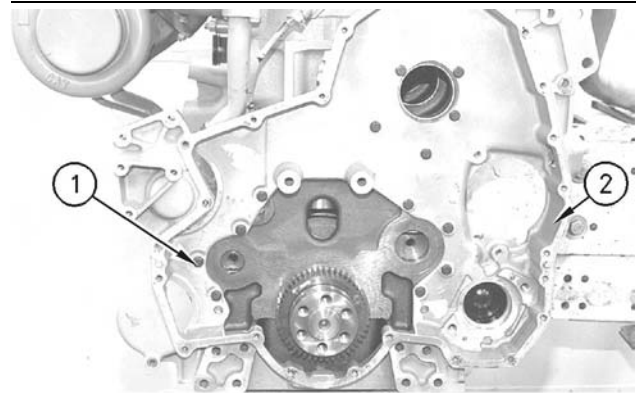


Illustration 189

g00514280

**Note:** Verify that the correct bolts are being used before proceeding to Step 3.

### 3.

- a. The 40 mm (1.57 inch) Bolt requires a torque value of  $35 \pm 8$  N·m ( $26 \pm 6$  lb ft). Thirteen bolts (1) are required to install the front housing (2).
- b. The 30 mm (1.18 inch) Bolt (204-0712 Bolt) requires a torque value of  $40 \pm 8$  N·m ( $30 \pm 6$  lb ft). Thirteen bolts (1) are required to install the front housing (2).

4. Remove the excess 8T-9022 Silicone Gasket from the following locations: cylinder block, front housing, and oil pan. Use a suitable sized flat tool such as a putty knife in order to remove the excess sealant.

#### End By:

- a. Install the front gear group. Refer to Disassembly and Assembly, "Gear Group Front - Install" .

i01667069

## Housing (Front) - Install

**SMCS Code:** 1151-012

### Installation Procedure

The front of the engine block has been modified. This modification requires the use of different bolts that attach the front housing to the engine block. The correct bolts need to be used to attach the front housing. Use Table 30 to identify your type of engine.

Table 30

|            | Engine Model | Serial Number (Type 1) | Serial Number (Type 2) |
|------------|--------------|------------------------|------------------------|
| Industrial | 3176C        | 2AW1-1328              | 2AW1329-Up             |
|            | 3196         | 1DW1-627               | 1DW628-Up              |
|            | C-10         | BCX1-268               | BCX269-Up              |
|            | C-12         | BDL1-236               | BDL237-Up              |
| Machine    | 3176C        | 3PD1-450               | 3PD451-Up              |
|            | 3176C        | 7ZR1-6108              | 7ZR6109-Up             |
|            | 3196         | 5ED1-428               | 5ED429-Up              |
|            | 3196         | 6AR276-2872            | 6AR2873-Up             |
|            | C12          | -                      | JAC1-Up                |
| Marine     | 3176C        | 6BW333-500             | 6BW501-Up              |
|            | 3196         | 2XR1-4150              | 2XR4151-42478          |
|            | C12          | -                      | 9HP1-Up                |
| Truck      | C-10         | 2PN1-Up                | -                      |
|            | C-10         | 8YS1-Up                | -                      |
|            | C-10         | 3CS1-21950             | 3CS21951-Up            |
|            | C-12         | 1YN1-Up                | -                      |
|            | C-12         | 9NS1-Up                | -                      |
|            | C-12         | 2KS1-69602             | 2KS69603-Up            |
|            | C-12         | CPD1-304               | CPD305-Up              |
|            | C-12         | CPN1-323               | CPN324-Up              |
|            | C-12         | 9SM1-862               | 9SM863-Up              |

Table 30 is your guide for selecting your correct type of engine. If your engine is a Type 1 proceed to "Type 1". If your engine is a Type 2 proceed to "Type 2". If you have a remanufactured engine then proceed to "Remanufactured Engines". The following illustrations will provide added information on the installation of the front housing.

**Note:** Check on the availability of a gasket kit for your application. The kit includes both the short bolts and the long bolts.

## Type 1

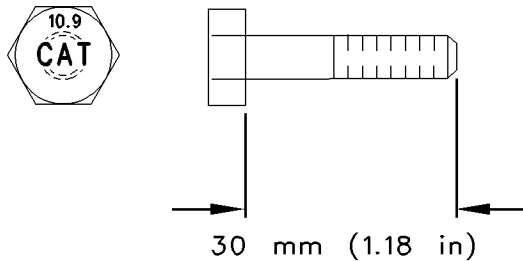


Illustration 190  
6V-2317 Bolt

g00856488

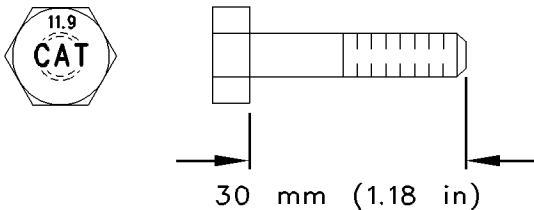


Illustration 191  
204-0712 Bolt

g00856491

The two bolts (Type 1) are 30 mm (1.18 inch) long. Marks on the head of the bolt represent the grade of the bolt. These markings can be seen in illustration 190 and 191. **The 6V-2317 Bolt is replaced by 204-0712 Bolt. The 204-0712 Bolt has a torque value of  $40 \pm 8 \text{ N}\cdot\text{m}$  ( $30 \pm 6 \text{ lb ft}$ ).** 13 bolts are required for the installation of the front housing.

## Type 2

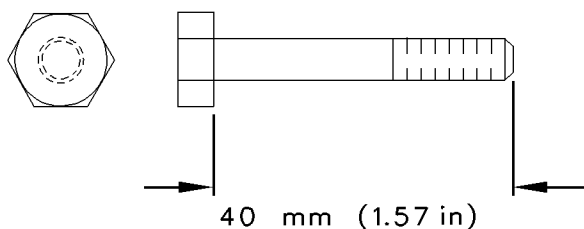


Illustration 192

g00856344

40 mm (1.57 inch) bolt length

The 40 mm (1.57 inch) Bolt is 10 mm (0.39 inch) longer than the 204-0712 Bolt. The longer bolt is needed due to a deeper counterbore and the tapped hole. **Do Not use the 204-0712 Bolt or the 6V-2317 Bolt. This could result in damage to the threads. The torque on the 40 mm (1.57 inch) Bolt is  $35 \pm 8 \text{ N}\cdot\text{m}$  ( $26 \pm 6 \text{ lb ft}$ ).** 13 bolts are required for the installation of the front housing.

## Remanufactured Engines

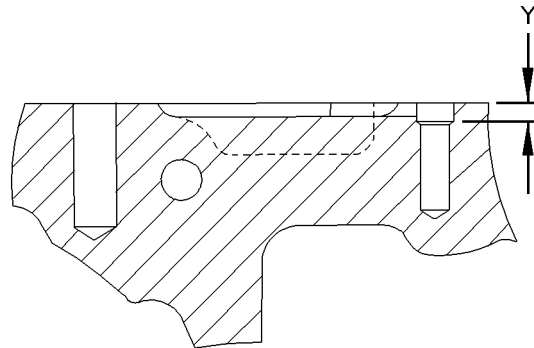


Illustration 193

g00838008

Depth of counterbore (Y) requires engines to use 204-0712 Bolt.

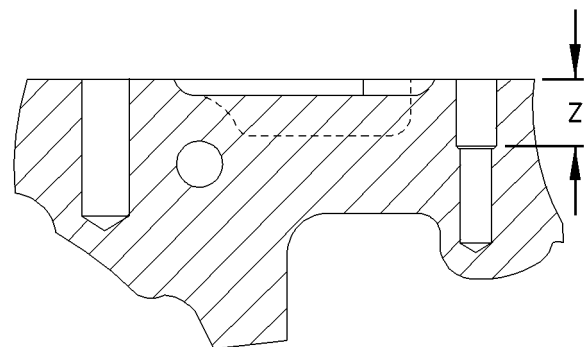


Illustration 194

g00838007

Depth of counterbore (Z) requires engines to use the 40 mm (1.57 inch) Bolt.

Remanufactured engines require special attention in order to use the correct bolt. The counterbore in the engine block must be measured in order to determine the bolts that will be used. If your engine block has a counterbore (Y) of 5 mm (0.20 inch), refer to "Type 1". If your engine block has a counterbore (Z) of 15 mm (0.60 inch), refer to "Type 2".

Table 31

| Dimension | Counterbore Depth    | Bolt Part Number                    |
|-----------|----------------------|-------------------------------------|
| Y         | 5 mm (0.20 inch)     | 204-0712<br>Type 1                  |
| Z         | 15 mm<br>(0.60 inch) | 40 mm<br>(1.57 inch) Bolt<br>Type 2 |

**Note:** The face of the housing and the face of the block must be clean prior to installation.

**NOTICE**

Do not use fingers or towels in order to remove the excess 8T-9022 Silicone Gasket. The use of fingers or towels will not leave a flat sealing surface. A leak may develop.

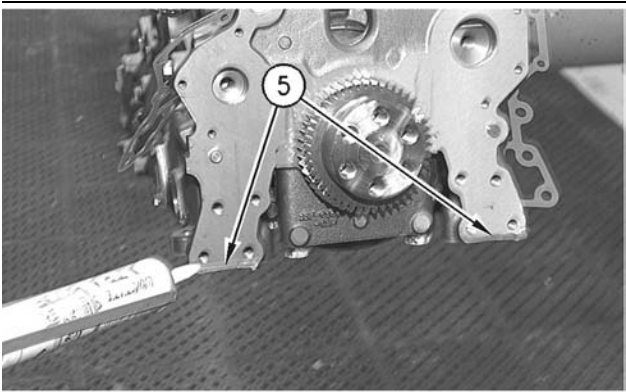


Illustration 195  
Typical example

1. Apply a bead of 8T-9022 Silicone Gasket to the lower edges (5) of the cylinder block.

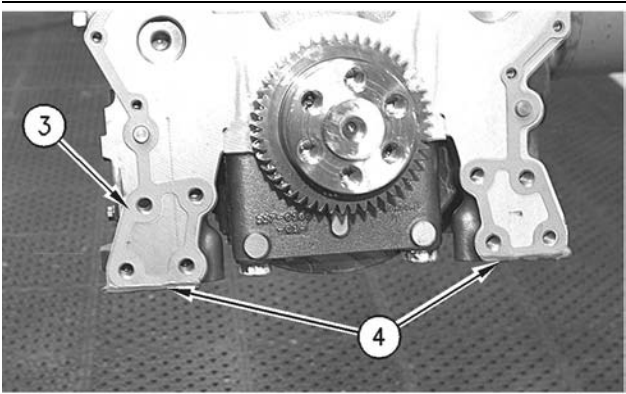


Illustration 196  
Typical example

2. Install gasket (3). Apply a bead of 8T-9022 Silicone Gasket to lower edges (4) of gasket (3).

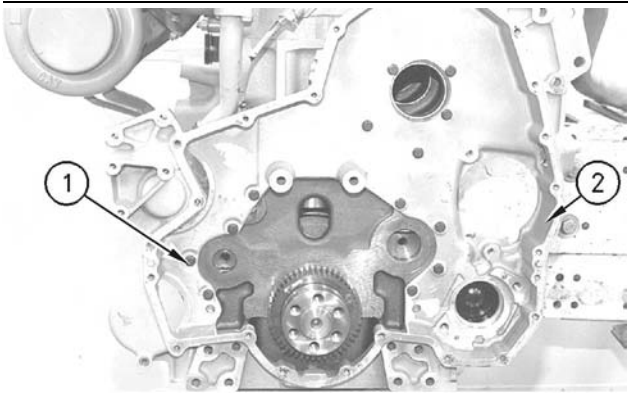


Illustration 197  
g00514280

**Note:** Verify that the correct bolts are being used before proceeding to Step 3.

- 3.
- a. The 40 mm (1.57 inch) Bolt requires a torque value of  $35 \pm 8$  N·m ( $26 \pm 6$  lb ft). Thirteen bolts (1) are required to install the front housing (2).
  - b. The 30 mm (1.18 inch) Bolt (204-0712 Bolt) requires a torque value of  $40 \pm 8$  N·m ( $30 \pm 6$  lb ft). Thirteen bolts (1) are required to install the front housing (2).
4. Remove the excess 8T-9022 Silicone Gasket from the following locations: cylinder block, front housing, and oil pan. Use a suitable sized flat tool such as a putty knife in order to remove the excess sealant.

**End By:**

- a. Install the front gear group. Refer to Disassembly and Assembly, "Gear Group Front - Install" .

i01464331

**Accessory Drive - Remove**

**SMCS Code:** 1207-011

**Removal Procedure**

**Start By:**

- a. Remove the belt tensioner. Refer to Disassembly and Assembly, "Belt Tensioner - Remove and Install".
- b. Remove the front cover. Refer to Disassembly and Assembly, "Front Cover - Remove".

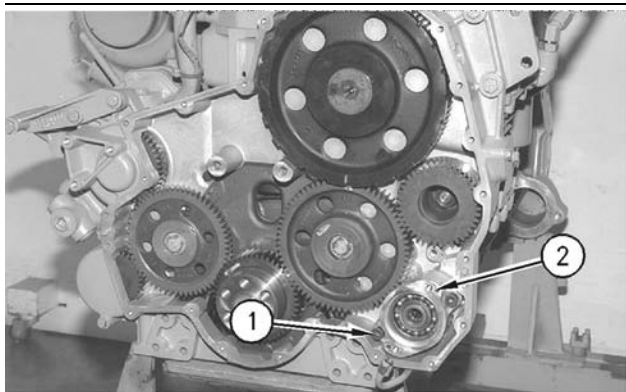


Illustration 198  
Typical example  
g00515001

1. Remove two bolts (1) and the washers.
2. Remove accessory drive (2).

i01771916

## Accessory Drive - Disassemble

SMCS Code: 1207-015

### Disassembly Procedure

Table 32

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 3H-0468     | Puller Plate          | 4   |
|                | 9S-9155     | Spacer                | 1   |
|                | 8B-7550     | Push-Puller Leg       | 2   |
|                | 5F-7344     | Block Puller          | 1   |
|                | 8H-0663     | Bearing Puller        | 1   |
|                | 5F-7345     | Screw                 | 1   |
| B              | 1P-1858     | Retaining Ring Pliers | 1   |

#### Start By:

- a. Remove the accessory drive. Refer to Disassembly and Assembly, "Accessory Drive - Remove".

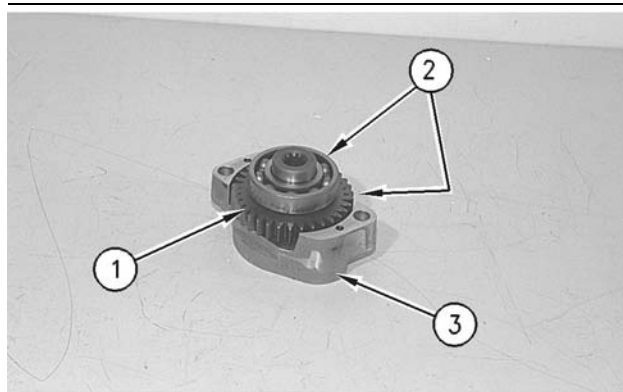


Illustration 199  
g00521981

1. Remove gear and shaft (1) with two bearings (2) from accessory drive housing (3).

**Note:** One end of the gear shaft is longer than the opposite end. Prior to assembly, ensure that the gear shaft is in the correct position.

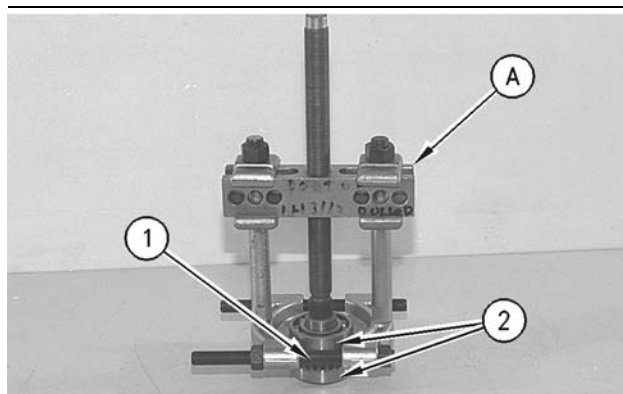


Illustration 200  
g00521983

2. Use Tooling (A) to remove two bearings (2) from gear and shaft (1).

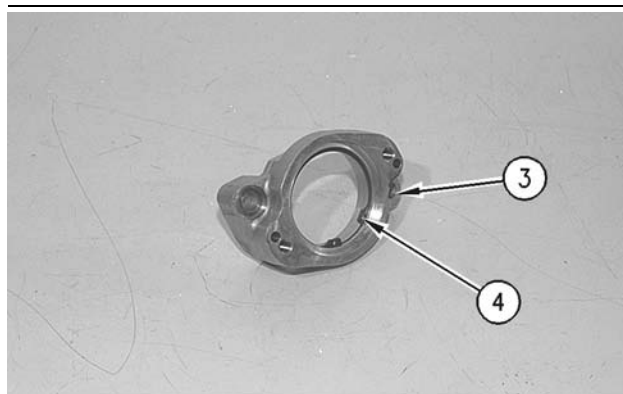


Illustration 201  
g00521986

3. Use Tooling (B) to remove snap ring (4) from accessory drive housing (3).

i01771920

# Accessory Drive - Assemble

SMCS Code: 1207-016

## Assembly Procedure

Table 33

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 1P-1858     | Retaining Ring Pliers | 1   |

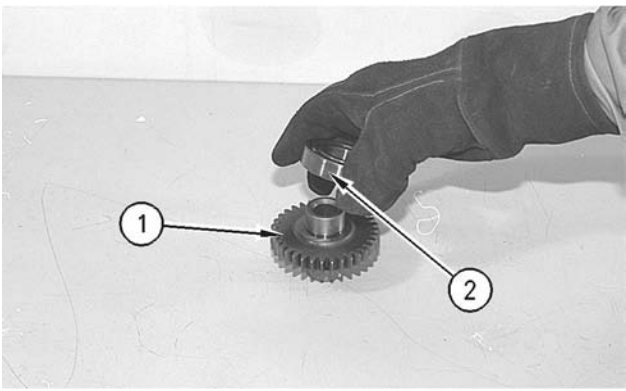


Illustration 202

g00522020

1. Lower the temperature of gear and shaft (1).

### WARNING

Always wear protective gloves when handling parts that have been heated.

2. Heat two bearings (2) in an oven to a temperature of 100 °C (212 °F).
3. Place bearings (2) in the original location on gear and shaft (1).

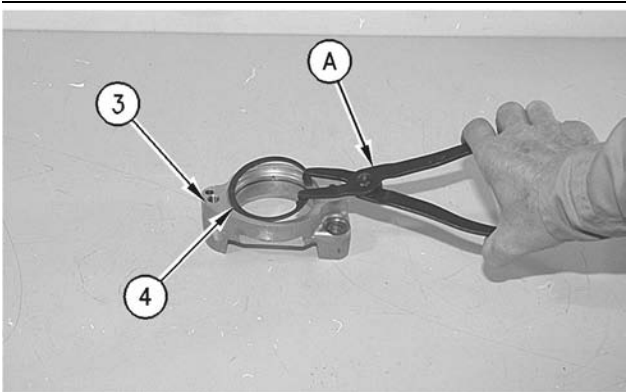


Illustration 203

g00522021

4. Use Tooling (A) to install snap ring (4) into accessory drive housing (3).

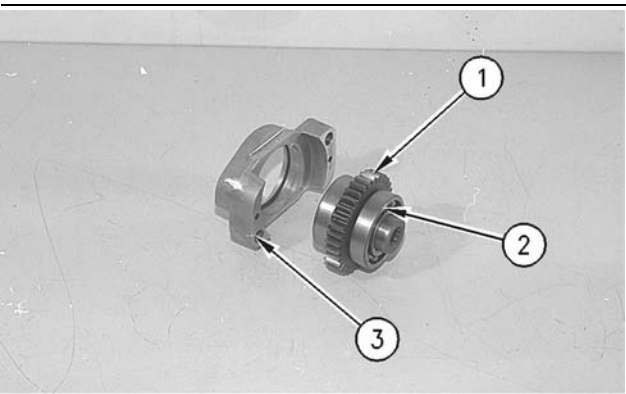


Illustration 204

g00522024

5. Install gear and shaft (1) with bearings (2) into accessory drive housing (3).

### End By:

- a. Install the accessory drive. Refer to Disassembly and Assembly, "Accessory Drive - Install".

i01464334

# Accessory Drive - Install

SMCS Code: 1207-012

## Installation Procedure

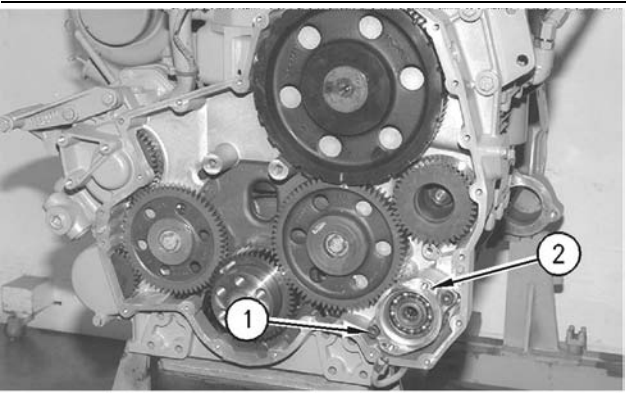


Illustration 205

g00515001

Typical example

1. Install accessory drive (2).
2. Install two bolts (1) and the washers. Tighten the bolts to a torque of 100 ± 20 N·m (74 ± 15 lb ft).

## End By:

- a. Install the front cover. Refer to Disassembly and Assembly, "Front Cover - Install".
- b. Install the belt tensioner. Refer to Disassembly and Assembly, "Belt Tensioner - Remove and Install".

i01004798

## Valve Mechanism Cover - Remove and Install

**SMCS Code:** 1107-010

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

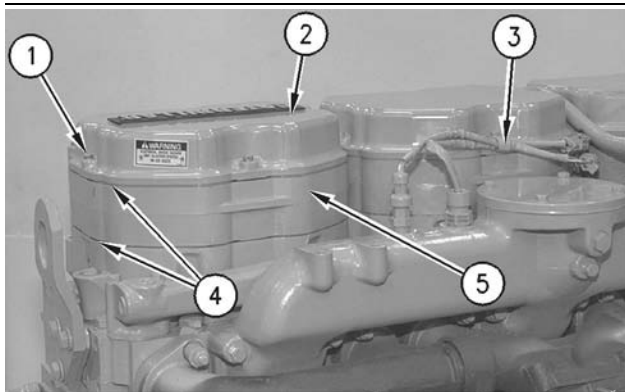


Illustration 206

g00515241

Typical example

1. Remove the bolt and the washer that holds clip (3) to the valve cover base.

2. Move clip (3) and the wiring harness out of the way.

3. Remove bolts (1) and the washers.

**Note:** Spacer (5) is used on engines that are equipped with a compression brake.

4. Remove the following items from the cylinder head: valve cover (2), spacer (5), and seals (4).

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Cut seals (4) to the correct length for the valve cover and spacer. Apply **3S-6252** Sealant to each joint surface after installation.
2. Place the following items on the valve cover base: seals (4), spacer (5), and valve cover (2).
3. Place clip (3) on the valve cover.

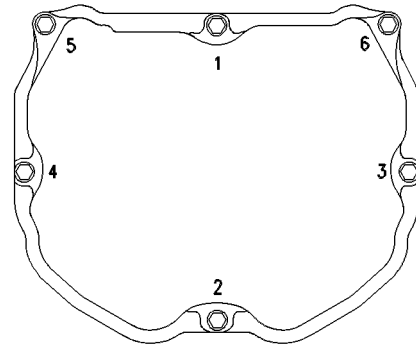


Illustration 207

g00280493

4. Install bolts (1) and the washers. Use the following procedure in order to tighten the bolts:
  - a. Tighten bolts in a numerical sequence to a torque of  $12 \pm 3$  N·m ( $9 \pm 2$  lb ft).
  - b. Tighten bolts again in a numerical sequence to a torque of  $12 \pm 3$  N·m ( $9 \pm 2$  lb ft).

i01465407

## Valve Mechanism Cover Base - Remove and Install

**SMCS Code:** 1120-010

### Removal Procedure

#### Start By:

- a. Remove the valve mechanism covers. Refer to Disassembly and Assembly, "Valve Mechanism Cover - Remove and Install".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

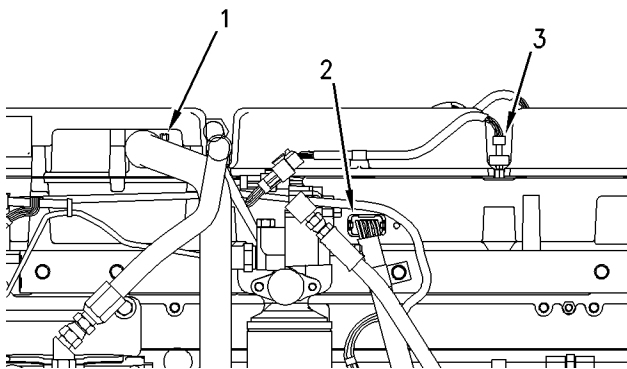


Illustration 208

g00668535

1. Disconnect wiring harness (2) and the clips. Loosen clamp (1) and disconnect the hose from the side of the valve cover base. Disconnect sensor (3) from the valve cover base.

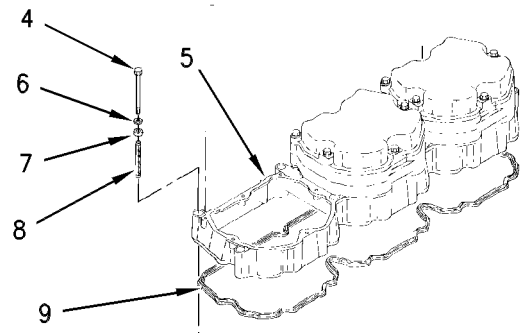


Illustration 209

g00515303

2. Remove bolts (4) and washers (6).
3. Remove isolators (7) and isolator sleeves (8).
4. Remove valve cover base (5) and base gasket (9).

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

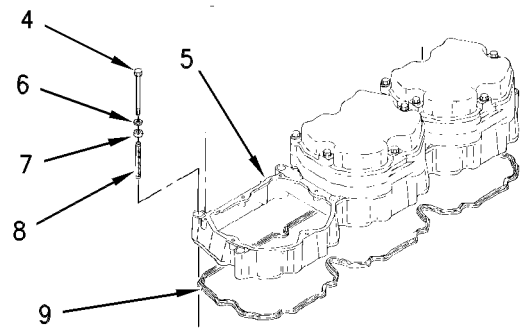


Illustration 210

g00515303

1. Install base gasket (9) and valve cover base (5) on the cylinder head.
2. Install isolator sleeves (8) and isolators (7).
3. Install washers (6) and bolts (4).

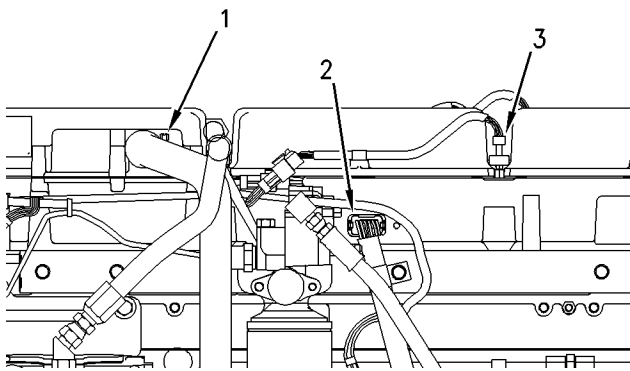


Illustration 211 g00668535

- 4. Install sensor (3).
- 5. Connect wiring harness (2) and the clips.
- 6. Connect the hose to the side of the valve cover base and tighten clamp (1).

**End By:**

- a. Install the valve mechanism covers. Refer to Disassembly and Assembly, “Valve Mechanism Cover - Remove and Install”.

i01464340

**Compression Brake - Remove**

**SMCS Code:** 1119-011

**Removal Procedure**

**Start By:**

- a. Remove the valve mechanism covers. Refer to Disassembly and Assembly, “Valve Mechanism Cover - Remove and Install”.

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Tools and Shop Products Guide” for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

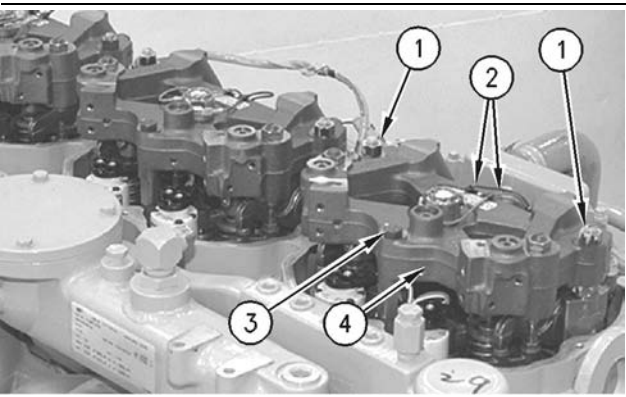


Illustration 212 g00521562

- 1. Disconnect wires (2) from the solenoid valve.
- 2. Remove two nuts (1) and bolt (3).
- 3. Remove compression brake housing (4).

i01465421

**Compression Brake - Disassemble**

**SMCS Code:** 1119-015

**Disassembly Procedure**

Table 34

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 1P- 1858    | Retaining Ring Pliers | 1   |
| B              | 1P- 1854    | Retaining Ring Pliers | 1   |
| C              | 134- 2837   | Piston Compressor     | 1   |

**Start By:**

- a. Remove the compression brake. Refer to Disassembly and Assembly, "Compression Brake - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

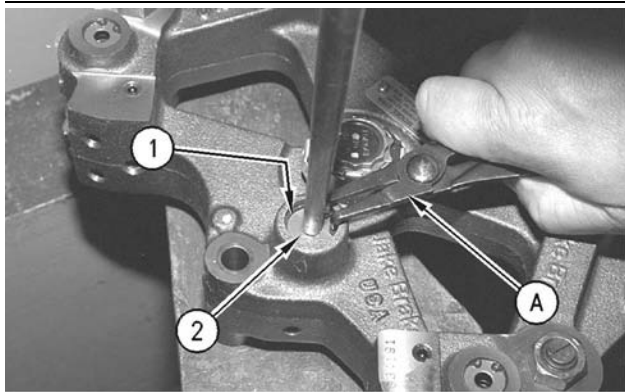


Illustration 213

g00523471

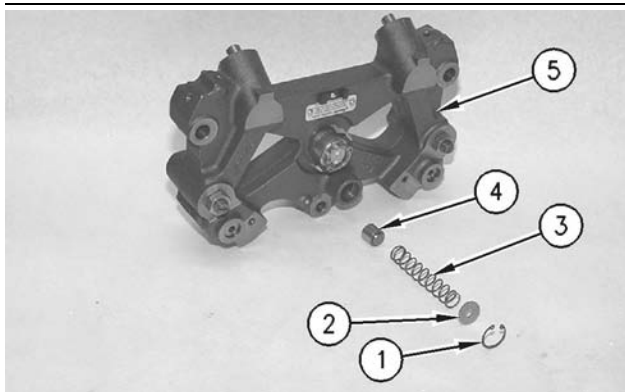


Illustration 214

g00523472

**⚠ WARNING**

**Remove accumulator covers carefully. The accumulator covers are under load from the accumulator springs. Remove with care to avoid personal injury.**

1. Apply downward pressure to accumulator cover (2) with a punch of suitable size. Remove retaining ring (1) with Tool (A).
2. Slowly release the pressure from accumulator cover (2) until the spring pressure ceases. Remove accumulator cover (2) and spring (3).
3. Use a magnet or needle nose pliers to remove accumulator piston (4) from the bore in compression brake housing (5).

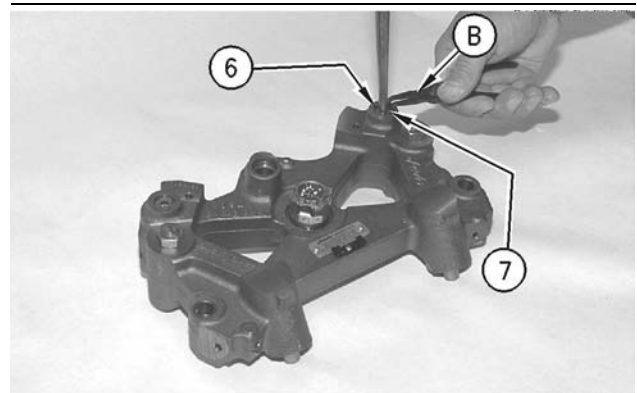


Illustration 215

g00523476

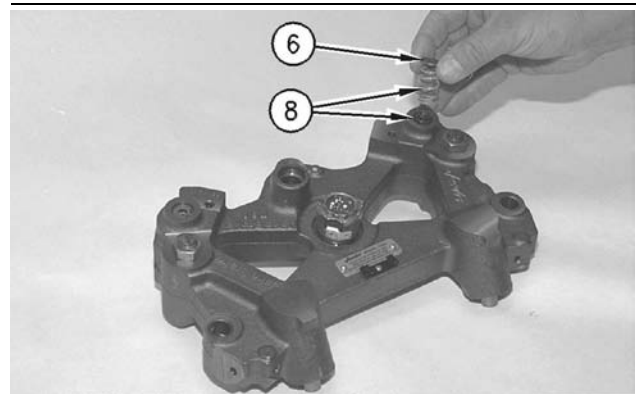


Illustration 216

g00524780

**⚠ WARNING**

**Remove control valve covers carefully. Control valve spring covers are under load from the control valve springs. Remove with care to avoid personal injury.**

4. Apply downward pressure to control valve cover (6) with a punch of suitable size. Remove retaining ring (7) with Tool (B).
5. Slowly release the pressure from control valve cover (6) until the spring pressure ceases. Remove control valve cover (6) and two control valve springs (8).

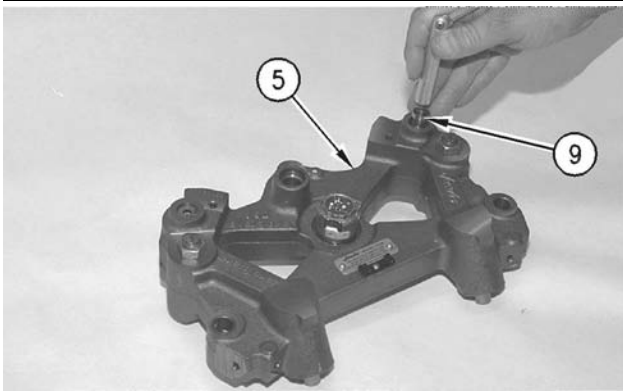


Illustration 217

g00524781

6. Use a magnet or needle nose pliers in order to reach into the bore and remove control valve (9) from compression brake housing (5).

**Note:** Repeat Steps 1 through 6 for the remaining control valve.

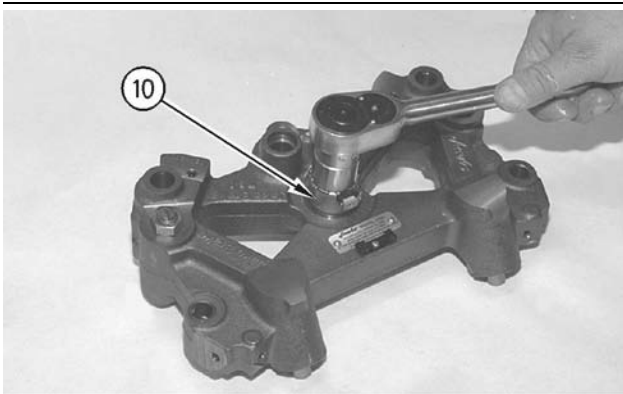


Illustration 218

g00523481

**NOTICE**

Do not disassemble or tamper with the solenoid valve. Engine damage could result.

7. Unscrew solenoid valve (10) from the compression brake housing.

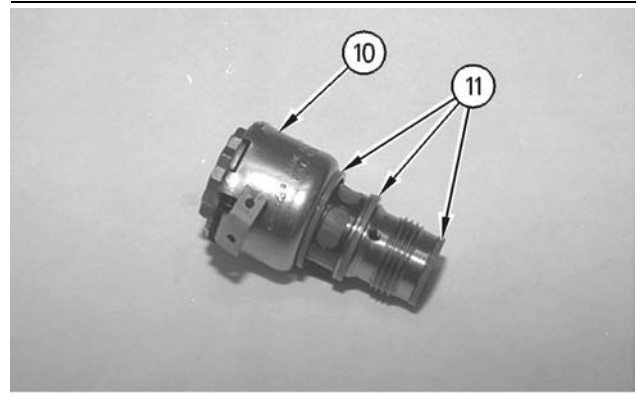


Illustration 219

g00523482

Typical example

8. Remove three O-ring seals (11) from solenoid valve (10). Discard the O-ring seals.

**Note:** The lower O-ring may stay in the bore in the compression brake housing. If this is the case, use a seal pick to remove the O-ring from the compression brake housing.

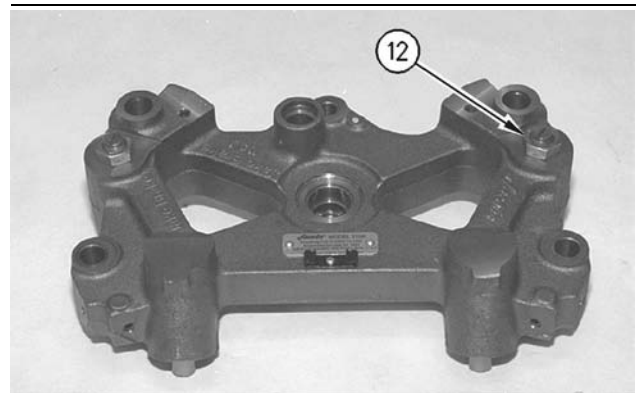


Illustration 220

g00523483

**NOTICE**

Do not disassemble or tamper with slave piston adjusting screw. Engine damage could result.

9. Remove slave piston adjustment screw locknut (12).

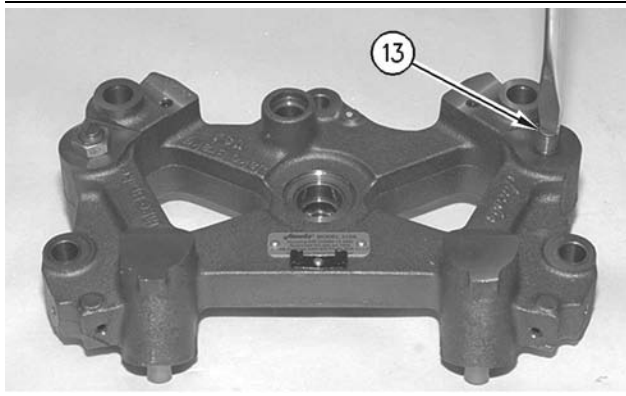


Illustration 221

g00523484

- 10.** Loosen slave piston adjustment screw (13) until the slave piston is fully retracted.

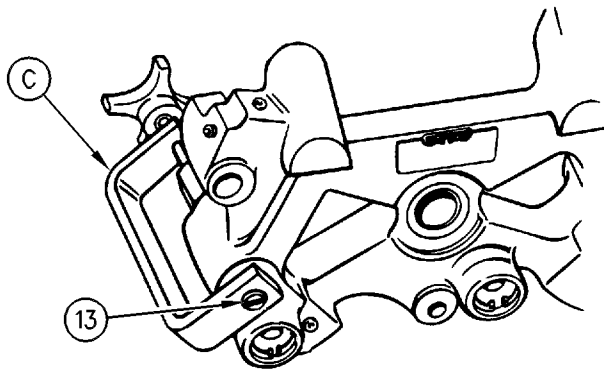


Illustration 222

g00524785

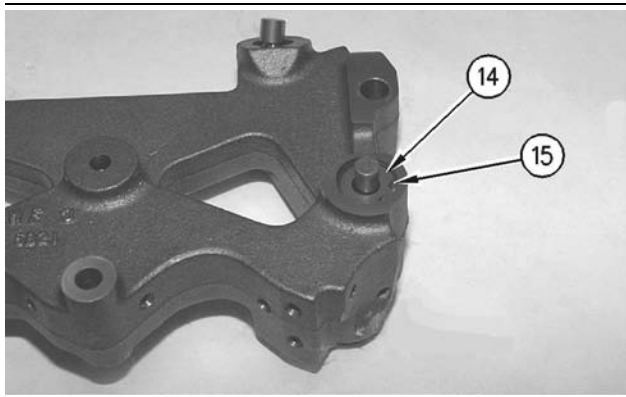


Illustration 223

g00523486

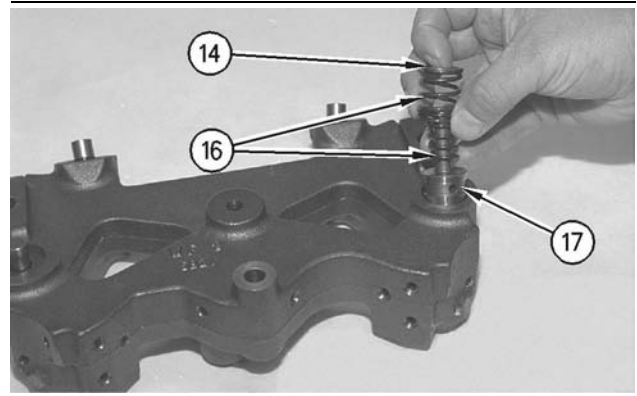


Illustration 224

g00524782

### **WARNING**

**Remove slave pistons carefully. The slave pistons are under spring pressure. Remove with care to avoid personal injury.**

- 11.** Place Tool (C) over slave piston adjustment screw (13). Turn the handle on Tool (C) until spring retainer (14) is depressed approximately 1.0 mm (.04 inch) in order to relieve the pressure from retaining ring (15).
- 12.** Remove retaining ring (15) with Tool (A). Slowly back out the handle on Tool (C) until the spring pressure ceases. Remove Tool (C).
- 13.** Remove spring retainer (14), springs (16), and slave piston (17).

**Note:** Repeat Steps 9 through 13 for the remaining slave piston.

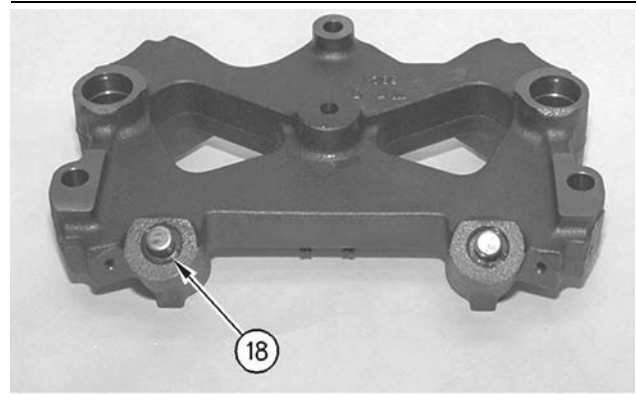


Illustration 225

g00524783

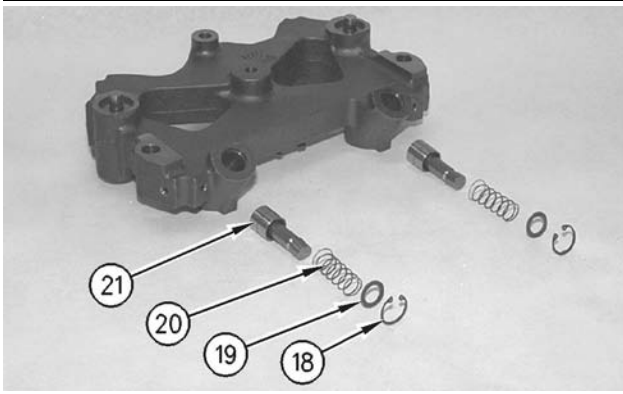


Illustration 226

g00524784

### **WARNING**

**Remove master pistons carefully. The master piston is under spring pressure. Remove with care to avoid personal injury.**

**14.** Apply downward pressure to retaining washer (19) with a punch of suitable size. Remove retaining ring (18) with Tool (B).

**15.** Slowly release the pressure from retaining washer (19) until the spring pressure ceases. Remove retaining washer (19), two springs (20), and master piston (21).

**Note:** Repeat Steps 14 and 15 for the remaining master piston.

i01533855

## Compression Brake - Assemble

**SMCS Code:** 1119-016

### Assembly Procedure

Table 35

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 1P-1858     | Retaining Ring Pliers | 1   |
| B              | 1P-1854     | Retaining Ring Pliers | 1   |
| C              | 134-2837    | Piston Compressor     | 1   |

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

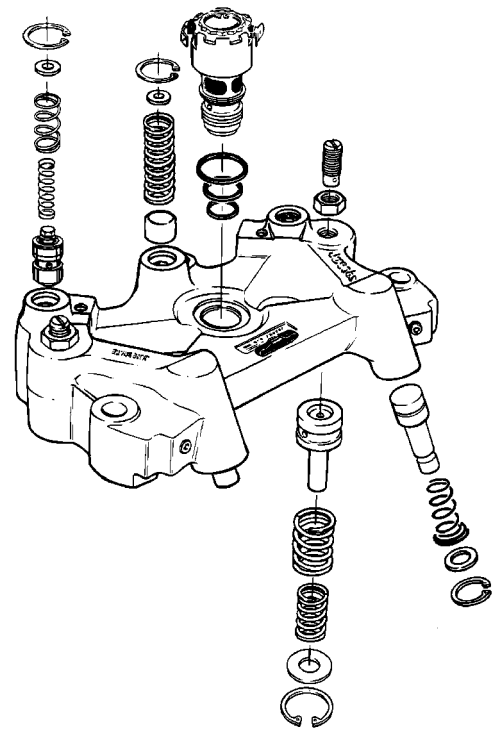


Illustration 227

g00523650

**1.** Inspect the condition of the bores in the compression brake housing. The bores must be free of wear and score marks.

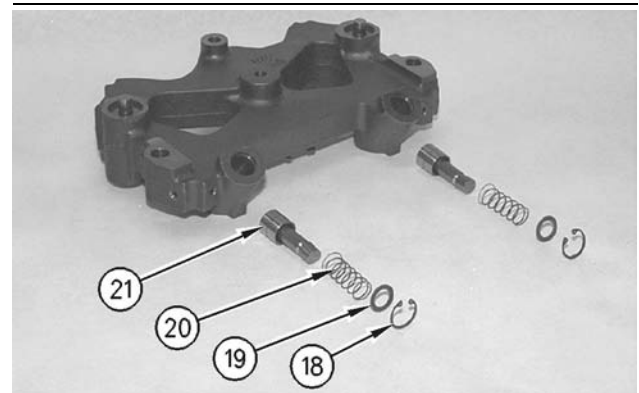


Illustration 228

g00524784

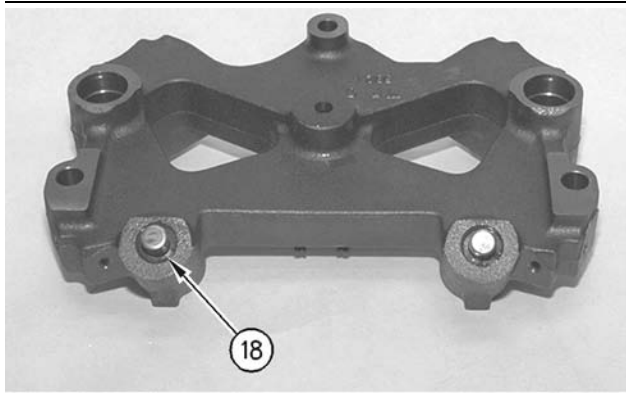


Illustration 229

g00524783

2. Inspect the condition of master piston (20). The master piston must be free of wear and score marks. Replace master piston (21), if necessary.
3. Install master piston (21) in the bore. Install the small end of spring (20) in the bore. Install retaining washer (19), and retaining ring (18) over master piston (21) and compress until retaining washer (19) is below the retaining ring groove. Install retaining ring (18) with Tool (B). Ensure that retaining ring (18) is fully seated in the groove.

**Note:** Repeat Steps 2 and 3 for the remaining master piston.

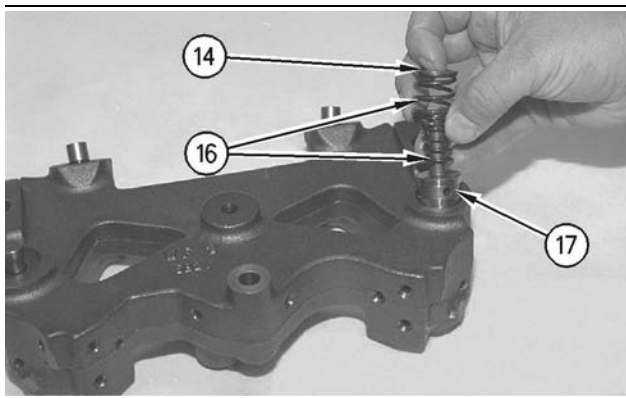


Illustration 230

g00524782

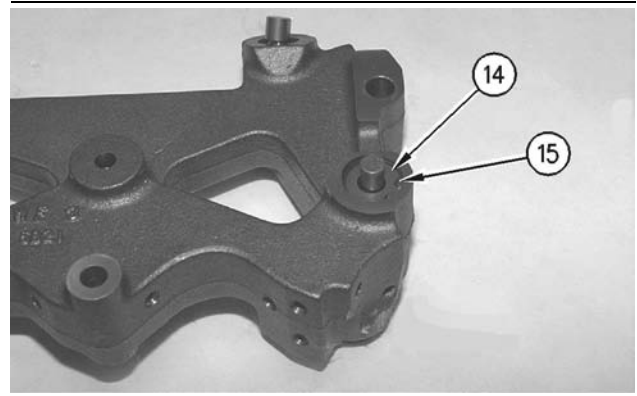


Illustration 231

g00523486

4. Inspect the condition of slave piston (17). The slave piston must be free of wear and score marks. Replace slave piston (17), if necessary.
5. Install slave piston (17), springs (16), spring retainer (14), and retaining ring (15) in the bore.

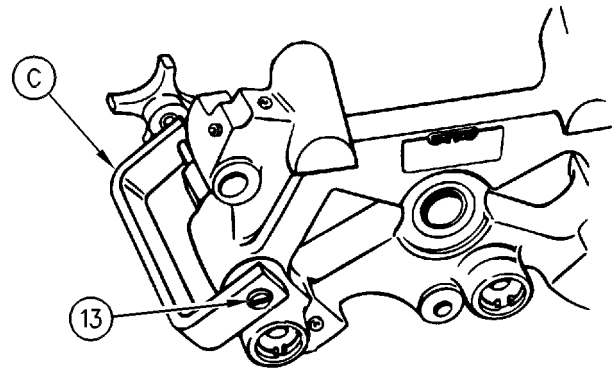


Illustration 232

g00524785

6. Place Tool (C) over slave piston adjustment screw (13) and spring retainer (14). Turn the handle on Tool (C) until spring retainer (14) is approximately 1.0 mm (.04 inch) below the retaining ring groove. Install retaining ring (15) with Tool (A).
7. Remove Tool (C) slowly in order to ensure that retaining ring (15) is properly seated in the groove.

**Note:** Repeat Steps 4 through 7 for the remaining slave piston.

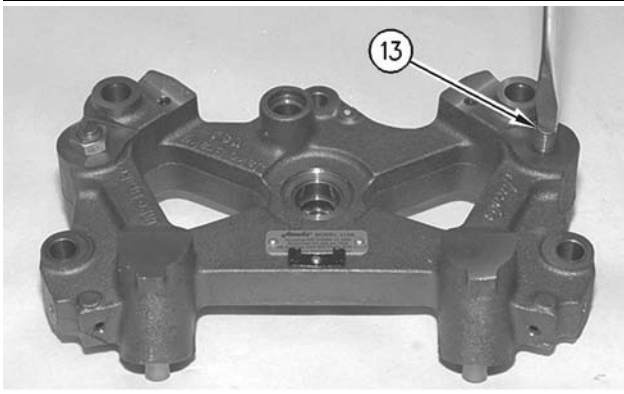


Illustration 233

g00523484

8. Screw in slave piston adjustment screw (13) until contact is made with the slave piston.

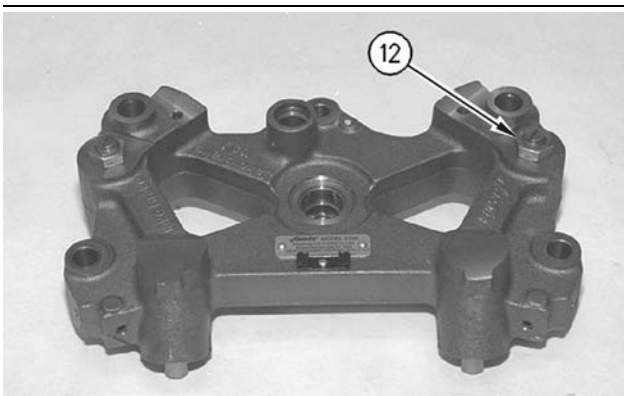


Illustration 234

g00523483

9. Install slave piston adjustment screw locknut (12).

**Note:** Repeat Steps 8 and 9 for the remaining slave piston adjustment screw.

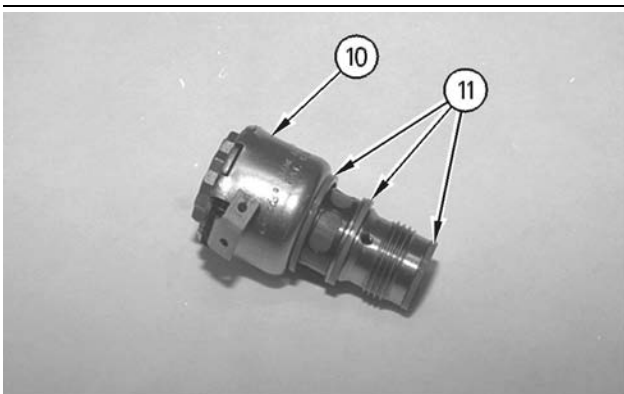


Illustration 235

g00523482

Typical example

10. Coat three new O-ring seals (11) with clean engine oil. Install the upper O-ring seal and the center O-ring seal on solenoid valve (10). Install the small O-ring seal in the bottom of the bore in the compression brake housing.

**Note:** Ensure that O-ring seals (11) are seated properly on solenoid valve (10). Do not twist the seals or unseat the seals during installation of the solenoid valve.

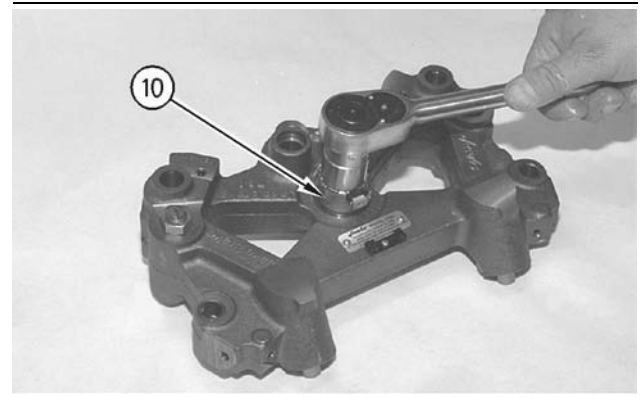


Illustration 236

g00523481

11. Carefully install solenoid valve (10) in the compression brake housing.

There are two different designs of solenoid valves. The difference can be determined by the number of points on the head of the solenoid valve. Use the correct torque value for your solenoid valve.

Tighten the solenoid valve with a twelve point head to the following torque. .... 12.5 N·m  
(9 lb ft)

Tighten the solenoid valve with a six point head to the following torque. .... 20 N·m  
(15 lb ft)

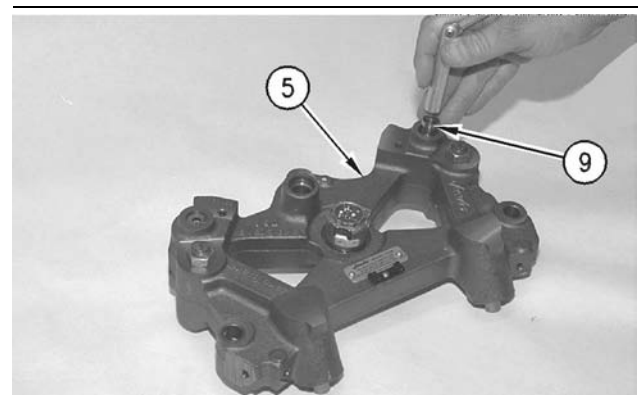


Illustration 237

g00524781

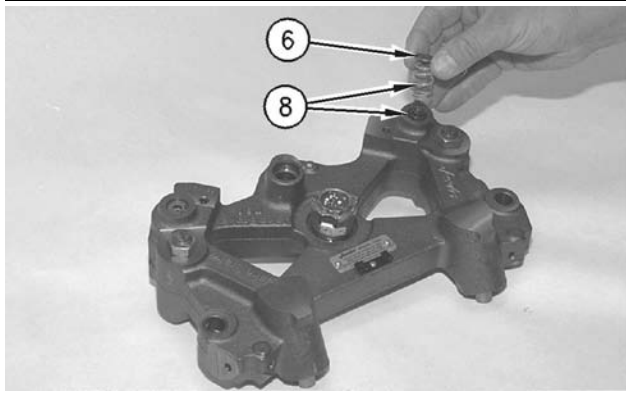


Illustration 238

g00524780

- 12.** Inspect the condition of control valve (9). Insert a wire into the hole in the base of control valve (9) until the wire contacts the check ball. The check ball should rise with light pressure on the wire. If the check ball is stuck, replace control valve (9). Apply clean engine oil to control valve (9).
- 13.** Install control valve (9), two control valve springs (8), and control valve cover (6) in compression brake housing (5).
- 14.** Apply downward pressure to control valve cover (6) with a punch of suitable size. Install retaining ring (7) with Tool (B). Slowly release the pressure from control valve cover (6) in order to ensure that retaining ring (7) is fully seated in the groove.

**Note:** Repeat Steps 12 through 14 for the remaining control valve.

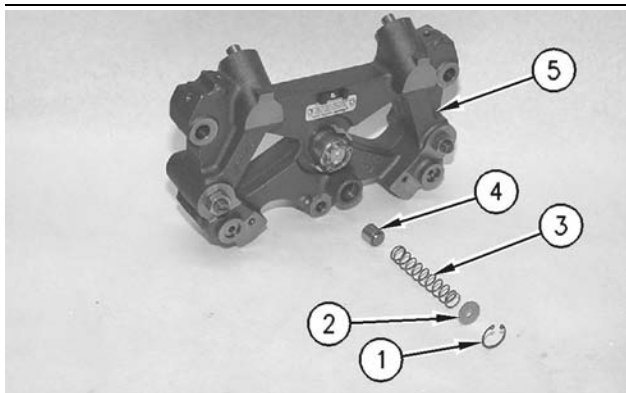


Illustration 239

g00523472

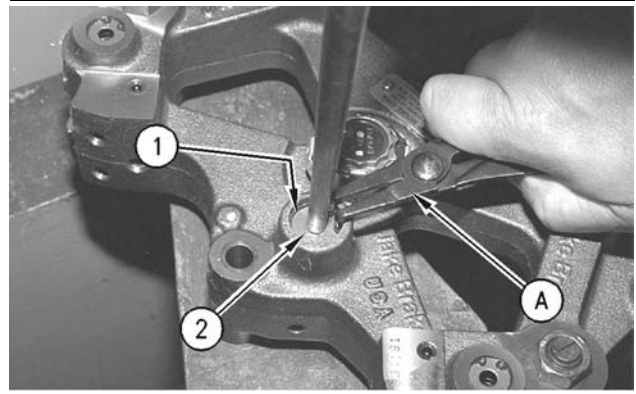


Illustration 240

g00523471

- 15.** Inspect the condition of accumulator piston (4). The accumulator piston should be free of wear and score marks. Apply clean engine oil to accumulator piston (4).
- 16.** Install accumulator piston (4), spring (3), and accumulator cover (2) in the bore in compression brake housing (5).
- 17.** Apply downward pressure to accumulator cover (2) with a punch of suitable size. Install retaining ring (1) with Tool (A). Slowly release the pressure from accumulator cover (2) in order to ensure that retaining ring (1) is fully seated in the groove.

#### End By:

- a.** Install the compression brake. Refer to Disassembly and Assembly, "Compression Brake - Install".

i01018713

## Compression Brake - Install

**SMCS Code:** 1119-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

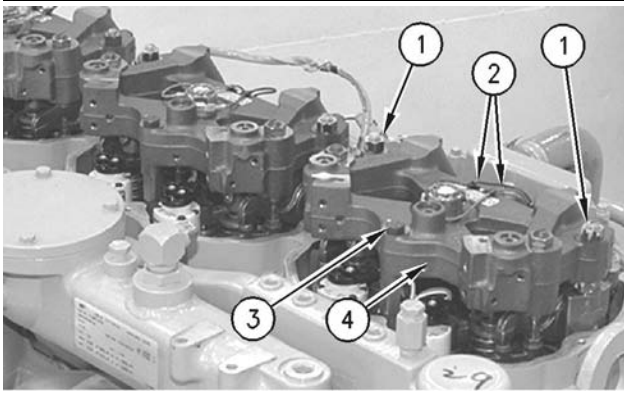


Illustration 241

g00521562

1. Position compression brake housing (4) on the two studs.
2. Install two nuts (1) and tighten to a torque of  $80 \pm 15 \text{ N}\cdot\text{m}$  ( $60 \pm 11 \text{ lb ft}$ ).
3. Install bolt (3) and tighten to a torque of  $55 \pm 10 \text{ N}\cdot\text{m}$  ( $41 \pm 7 \text{ lb ft}$ ).
4. Connect wires (2) to the solenoid valve on the compression brake.

**Note:** The valve lash must be adjusted before adjusting the slave piston lash.

**End By:**

- a. Adjust the valve lash. Refer to the Testing and Adjusting, "Air Inlet and Exhaust System" topic for the correct procedure.
- b. Adjust the slave piston lash. Refer to the Testing and Adjusting, "Compression Brake" topic for the correct procedure.
- c. Install the valve mechanism covers.

i01464346

## Rocker Shaft and Pushrod - Remove

**SMCS Code:** 1102-011; 1208-011

### Removal Procedure

**Start By:**

- a. Remove the valve mechanism cover. Refer to Disassembly and Assembly, "Valve Mechanism Cover - Remove and Install".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

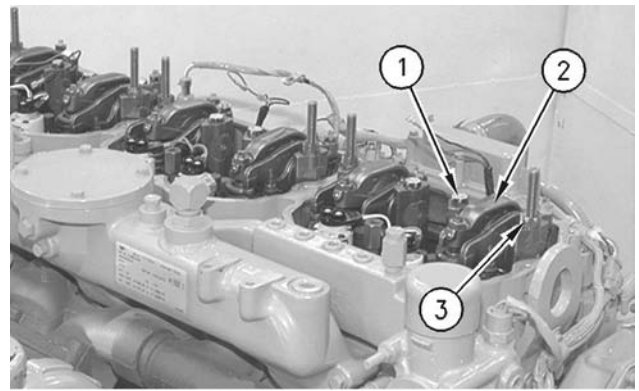


Illustration 242

g00524203

Typical example

1. Put identification marks on each rocker arm in order to identify the proper location in the engine for installation.
2. Remove bolt (1) and stud (3).
3. Remove rocker arm assembly (2) and the push rods. Keep each push rod with the respective rocker arm.

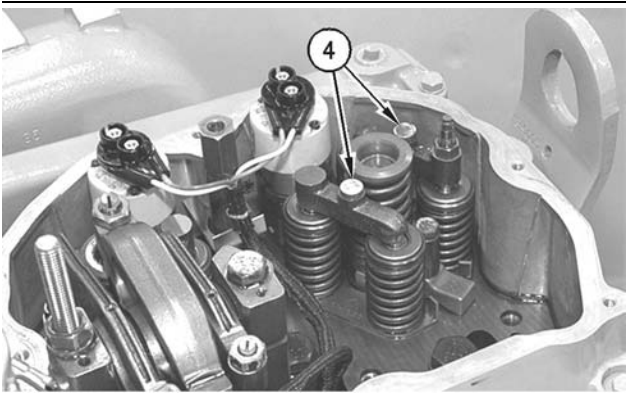


Illustration 243 g00524205

4. Remove valve bridge assemblies (4).

i01464353

## Rocker Shaft and Pushrod - Install

**SMCS Code:** 1102-012; 1208-012

### Installation Procedure

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the following items: rocker arm assemblies, valve bridge assemblies, and push rods. Replace the parts, if necessary.

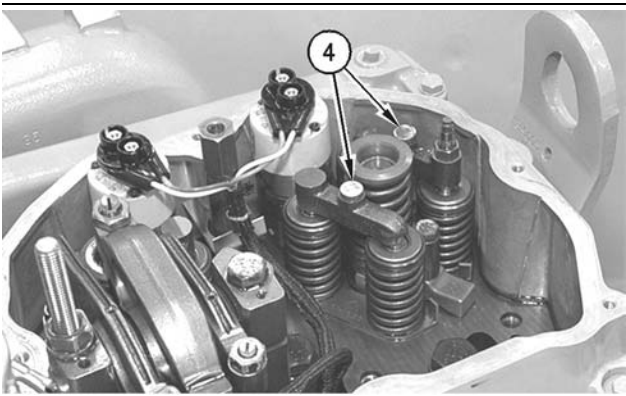


Illustration 244 g00524205

2. Install valve bridge assemblies (4). Apply clean engine oil to the top surface of the valve bridge assemblies.

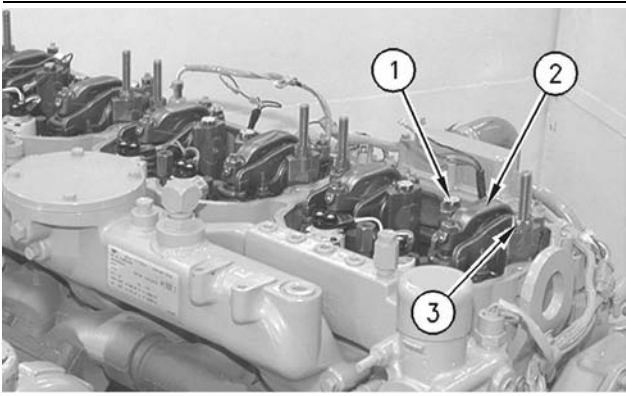


Illustration 245 g00524203

3. Install the push rods in the original location in the engine.
4. Position rocker arm assembly (2) in the original location on the engine. Install stud (3) and bolt (1). Tighten stud (3) and bolt (1) to a torque of  $100 \pm 20 \text{ N}\cdot\text{m}$  ( $75 \pm 15 \text{ lb ft}$ ).

**End By:**

- a. Install the valve mechanism cover. Refer to Disassembly and Assembly, "Valve Mechanism Cover - Remove and Install".
- b. Adjust the engine valve lash. Refer to Systems Operation Testing and Adjusting, "Engine Valve Lash - Inspect/Adjust".

i01591136

## Cylinder Head - Remove

**SMCS Code:** 1100-011

### Removal Procedure

Table 36

| Required Tools |             |                       |     |
|----------------|-------------|-----------------------|-----|
| Tool           | Part Number | Part Description      | Qty |
| A              | 2P-8250     | Strap Wrench Assembly | 1   |

**Start By:**

- a. Remove the exhaust manifold. Disassembly and Assembly, "Exhaust Manifold - Remove and Install".
- b. Remove the inlet manifold. Refer to Disassembly and Assembly, "Inlet Manifold - Remove".

- c. Remove the valve cover base. Refer to Disassembly and Assembly, "Valve Mechanism Cover Base - Remove and Install".
- d. Remove the electronic unit injectors. Disassembly and Assembly, "Electronic Unit Injector - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the cooling system into a suitable container for storage or disposal.

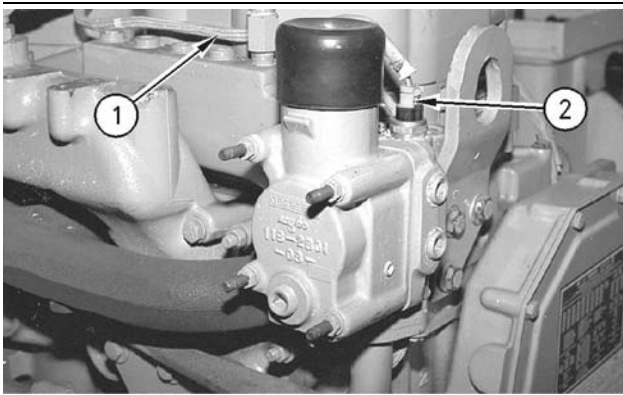


Illustration 246

g00524436

2. Disconnect sensor connector (2). Lay wiring harness (1) out of the way.

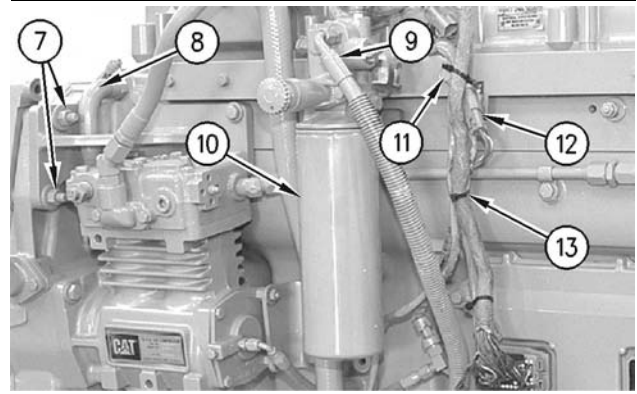


Illustration 247

g00524537

Typical example

3. Remove two nuts (7), the bolts, and the washers.
4. Remove hose (8).
5. Use tool (A) to remove fuel filter (10). Disconnect hose (9). Plug all openings immediately.
6. Remove bolt (11), the washer, and the clip.
7. Disconnect sensor connector (12) and cut tie straps (13).

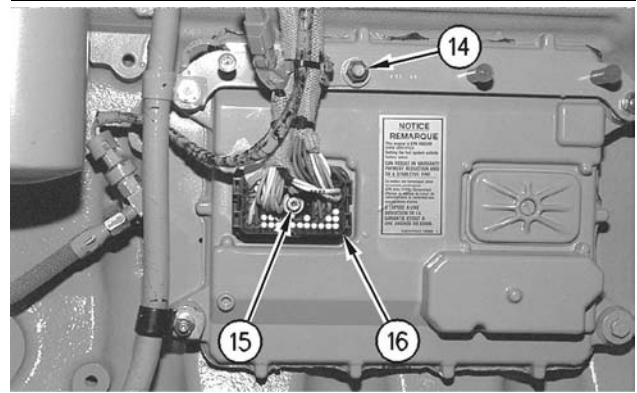


Illustration 248

g00524775

8. Remove nut (14), the washer, and the clip.
9. Loosen bolt (15) until wiring harness (16) can be disconnected. Disconnect wiring harness (16). Lay the wiring harness away from the cylinder head.

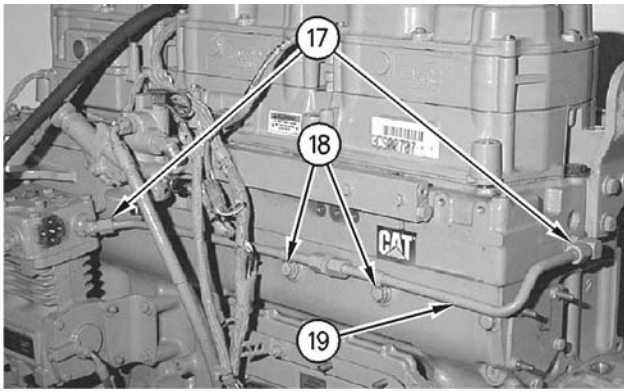


Illustration 249

g00707634

Typical example

- 10.** Disconnect fittings (17). Remove bolts (18) and the washers. Remove line (19).

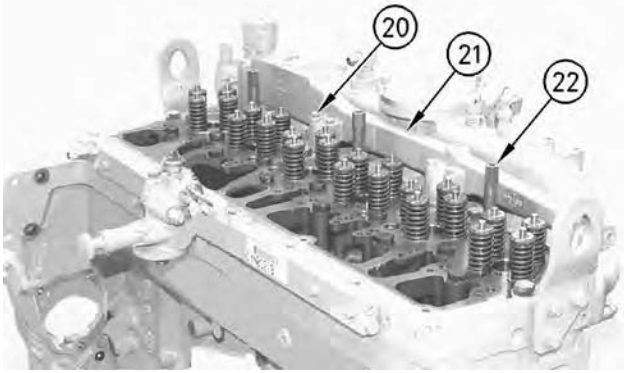


Illustration 250

g00525468

Typical example

- 11.** Remove bolts (20) and the washers. Remove water outlet manifold (21).
- 12.** Remove spacers (22) from the head bolt studs.

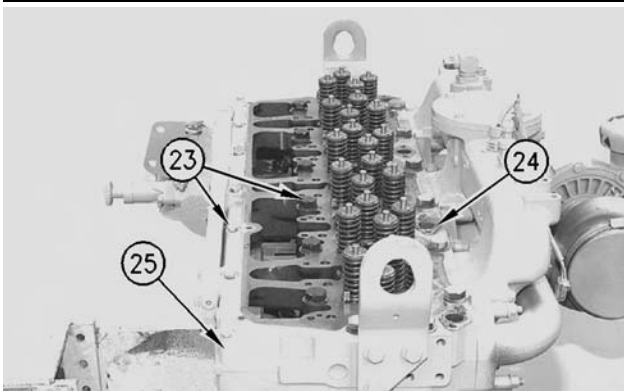


Illustration 251

g00525534

Typical example

**Note:** Engines that are equipped with a compression brake will have three head bolt studs. The three head bolt studs are installed in place of head bolts. Remove the head bolt studs with the head bolts.

- 13.** Remove head bolts (23) and the head bolt studs, if equipped.

- 14.** Remove four gaskets (24) from cylinder head (25).

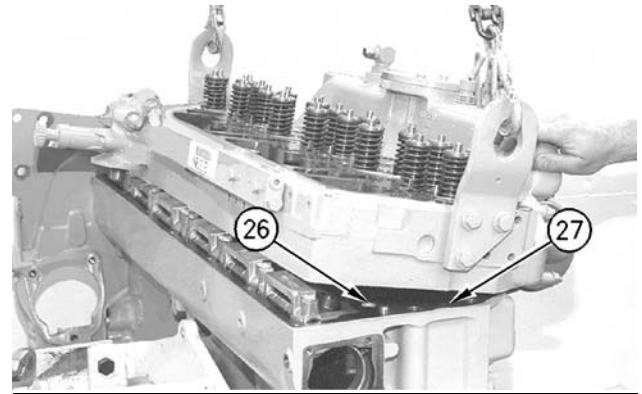


Illustration 252

g00525775

Typical example

- 15.** Attach a suitable lifting device to cylinder head (25). Ensure that all wires, lines, and hoses are free from the cylinder head.
- 16.** Carefully remove cylinder head (25) from the cylinder block. The weight of cylinder head (25) is approximately 135 kg (300 lb).
- 17.** Remove cylinder head gasket (26) and O-ring seals (27) from the cylinder block.

i01772015

## Cylinder Head - Install

SMCS Code: 1100-012

### Installation Procedure

Table 37

| Required Tools |                        |                                              |     |
|----------------|------------------------|----------------------------------------------|-----|
| Tool           | Part Number            | Part Description                             | Qty |
| A              | 185-3630               | Strap Wrench Assembly                        | 1   |
| B              | 1U-5686 <sup>(1)</sup> | 21/32 inch Drill Bit                         | 1   |
| C              | 4C-9732 <sup>(1)</sup> | M16x2.0-D7 Plug Tap                          | 1   |
| D              | 9U-5604                | M16x2.0 32mm Heli-Coil Insert <sup>(2)</sup> | 1   |
| E              | 8T-2799 <sup>(1)</sup> | Heli-Coil Installation Tool                  | 1   |
| F              | 8T-3096                | Plug Tap                                     | 1   |

<sup>(1)</sup> Part of 4C-8367 Thread Repair Kit

<sup>(2)</sup> This is a special length.

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**Reference:** Refer to Video Tape, LEVN5101, "C-10 and C-12 Head Gasket Repair Procedure".

1. Thoroughly clean the mating surfaces of the cylinder head and the cylinder block.
2. Clean all 26 head bolt holes on the cylinder block with Tooling (F).
3. After cleaning the threads, remove any debris from the head bolt holes on the cylinder block with a vacuum hose.
4. Visually inspect the head bolt holes on the cylinder block for thread damage.
5. If a damaged head bolt thread is observed, repair the damaged head bolt thread with a Heli-Coil thread repair kit.
6. Inspect the cylinder head and the cylinder block for flatness. Refer to Guideline for Reusable Parts and Salvage Operations, SEBF8162, "Procedure to Measure and Salvage Cylinder Head Assemblies and Related Components".

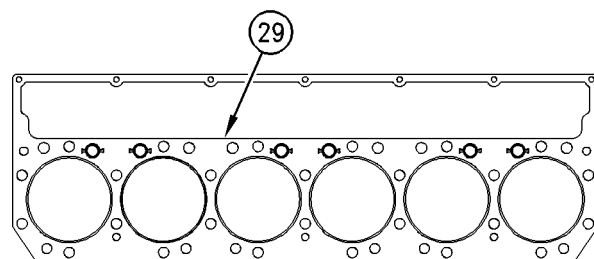


Illustration 253  
Cylinder Head Gasket  
(29) Straight Edge

g00810396

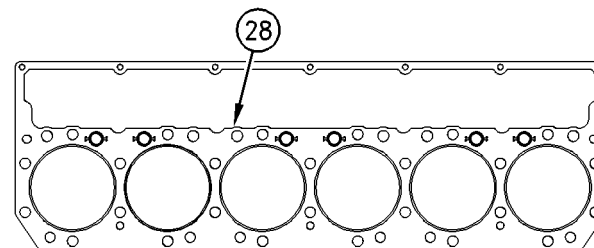


Illustration 254  
Cylinder Head Gasket  
(28) Scalloped Edge

g00810399

7. Determine the correct cylinder head gasket to use. Refer to Illustration 253 and Illustration 254.

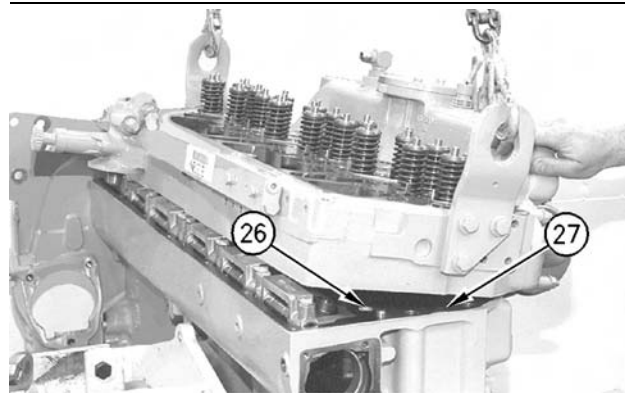


Illustration 255

g00525775

8. Install the correct cylinder head gasket (26) and new O-ring seals (27) on the cylinder block. Inspect the condition of the remaining gaskets. Replace the gaskets, if necessary.

**Note:** To avoid damage to the cylinder head gasket, use guide bolts of an appropriate size to install the cylinder head.

9. Use a suitable lifting device to install the cylinder head on the cylinder block.

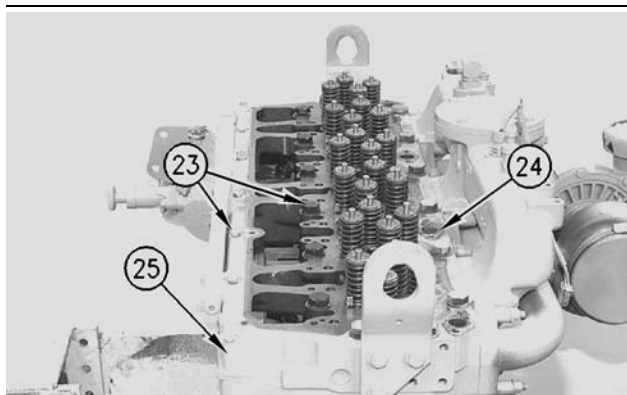


Illustration 256

g00525534

10. Install four gaskets (24) on cylinder head (25).
11. Install new cylinder head bolts (23). If the engine is equipped with a compression brake, then also install new head bolt studs.
12. Lubricate the following components with 6V-4876 Lubricant: the bolt threads, the underside of the bolt heads, and the washers. Follow Step 13 for the correct tightening procedure.

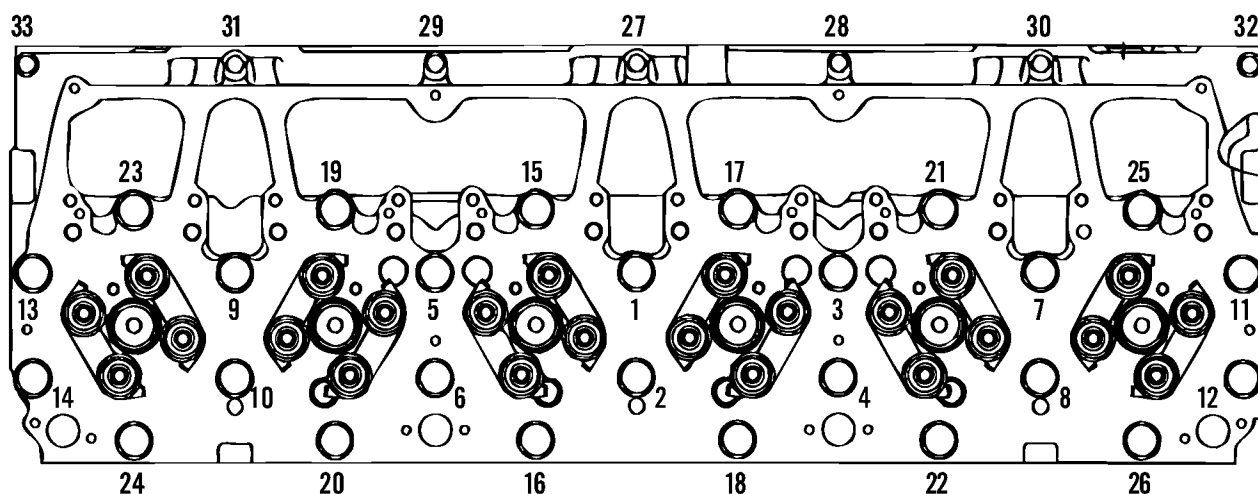


Illustration 257

g00517501

**Note:** Engines that are equipped with a compression brake will have three head bolt studs. Install the head bolt studs in positions (2), (8), and (10). Tighten the head bolt studs in sequence with the cylinder head bolts.

**13.** Tighten the cylinder head bolts according to the following procedure:

- a. Tighten bolt (1) through bolt (26) in a numerical sequence to a torque of  $160 \pm 15$  N·m ( $120 \pm 11$  lb ft).
- b. Tighten bolt (1) through bolt (26) again in a numerical sequence to a torque of  $160 \pm 15$  N·m ( $120 \pm 11$  lb ft).
- c. Place an index mark on each bolt head. Turn bolt (1) through bolt (26) in a numerical sequence for an additional 90 degrees ( $1/4$  turn).
- d. **Loosen bolt (1) through bolt (26) until the washers are loose under the bolt heads.**
- e. Tighten bolt (1) through bolt (26) in a numerical sequence to a torque of  $160 \pm 15$  N·m ( $120 \pm 11$  lb ft).
- f. Tighten bolt (1) through bolt (26) again in a numerical sequence to a torque of  $160 \pm 15$  N·m ( $120 \pm 11$  lb ft).
- g. Place an index mark on each bolt head. Turn bolt (1) through bolt (26) in a numerical sequence for an additional 90 degrees ( $1/4$  turn).
- h. Tighten bolt (27) through bolt (33) in a numerical sequence to a torque of  $28 \pm 7$  N·m ( $20 \pm 5$  lb ft).

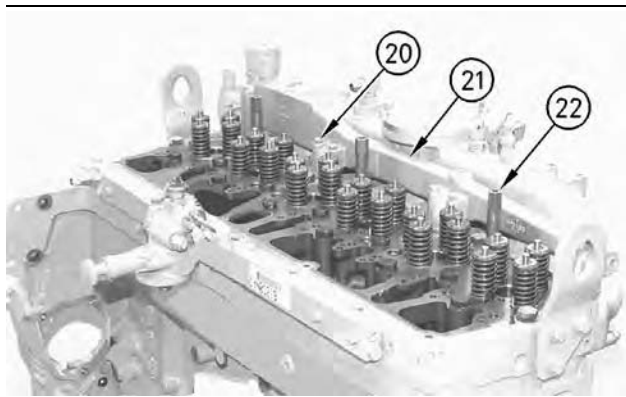


Illustration 258

g00525468

**14.** Install head bolt spacers (22) on the three head bolt studs. Tighten head bolt spacers (22) to a torque of  $100 \pm 20$  N·m ( $75 \pm 15$  lb ft).

**15.** Install water outlet manifold (21) and bolts (20).

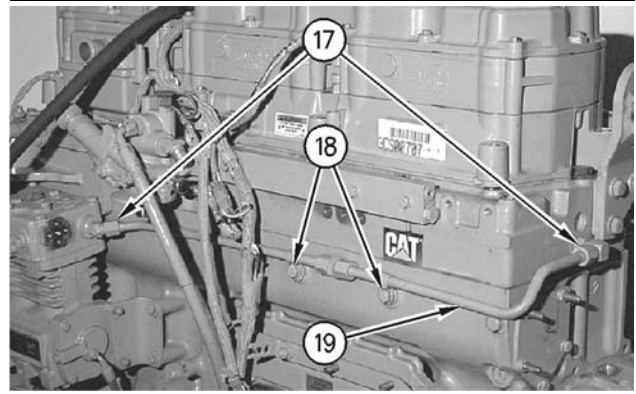


Illustration 259

g00707634

**16.** Hold line (19) in position and connect fittings (17). Install bolts (18).

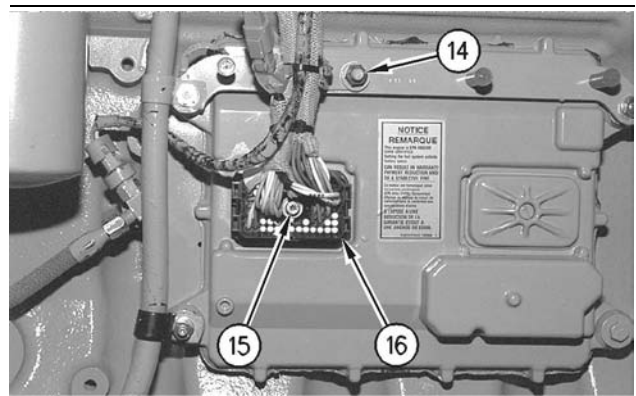


Illustration 260

g00524775

**17.** Connect harness assembly (16). Tighten bolt (15) to a torque of  $6 \pm 1$  N·m ( $53 \pm 9$  lb in).

**18.** Install the clip, the washer, and nut (14).

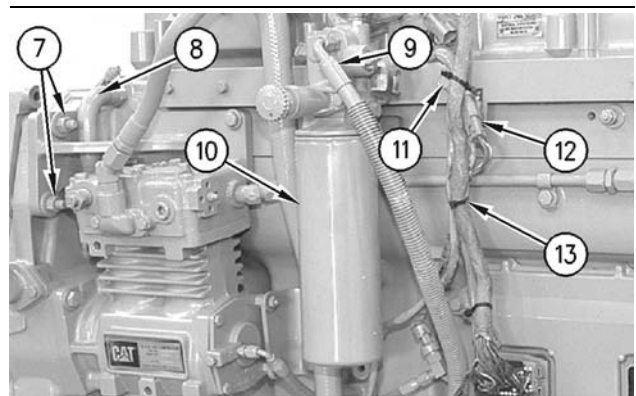


Illustration 261

g00524537

**19.** Connect harness assembly (12). Install the clip, the washer, and bolt (11). Secure the wiring harness with tie straps (13).

**20.** Install fuel filter (10) with Tooling (A). Follow the instructions on the fuel filter.

21. Connect hose assembly (9).
22. Install hose (8).
23. Install the bolts, the washers, and two nuts (7).

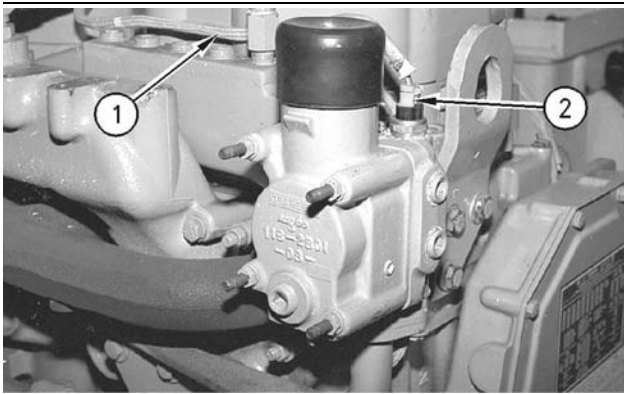


Illustration 262

g00524436

24. Route harness assembly (1) accordingly and connect harness assembly(2).
25. Fill the cooling system with coolant to the correct level. Refer to the Operation and Maintenance Manual for the correct filling procedure.

#### End By:

- a. Install the electronic unit injectors. Refer to Disassembly and Assembly, "Electronic Unit Injector - Install".
- b. Install the valve cover base. Refer to Disassembly and Assembly, "Valve Mechanism Cover Base - Remove and Install".
- c. Install the inlet manifold. Refer to Disassembly and Assembly, "Inlet Manifold - Install".
- d. Install the exhaust manifold. Refer to Disassembly and Assembly, "Exhaust Manifold - Remove and Install".

i01464393

## Lifter Group - Remove and Install

SMCS Code: 1209-010

### Removal Procedure

#### Start By:

- a. Remove the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Remove".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

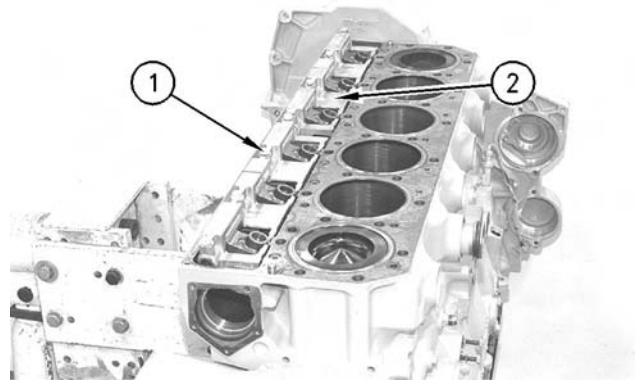


Illustration 263

g00515645

1. Remove bolts (1) and lifter assembly (2). The lifter assembly can be removed in two sections.

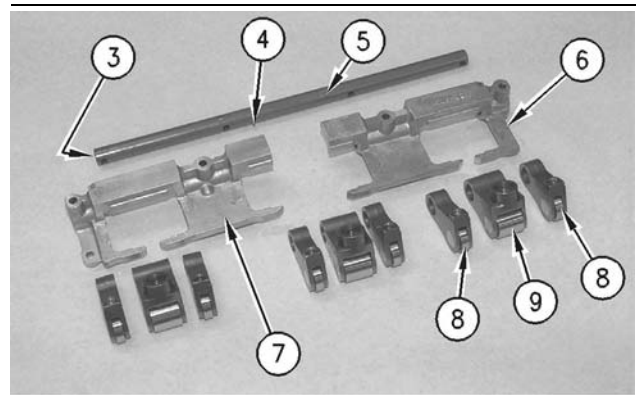


Illustration 264

g00515668

2. Remove the following items from the right end of lifter shaft (5): bracket assembly (6), valve lifters (8), and unit injector lifter (9).
3. Remove the following items from the middle of lifter shaft (5): valve lifters (8) and unit injector lifter (9).
4. Remove the following items from the left end of lifter shaft (5): bracket assembly (7), valve lifters (8), and unit injector lifter (9).
5. Remove dowel (4) and cap plug (3) from lifter shaft (5).
6. Remove the spring pin from bracket (7), if necessary.

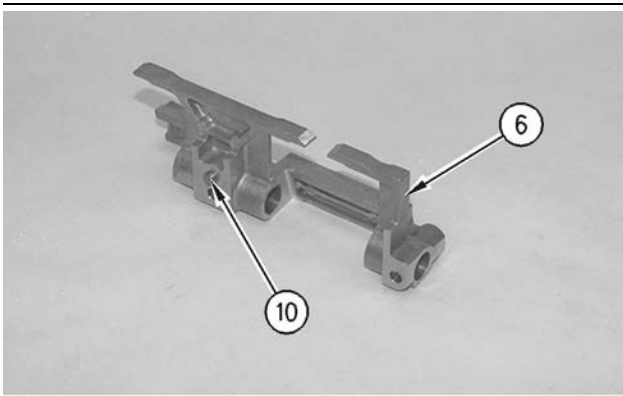


Illustration 265  
Typical example g00515670

7. Remove dowel (10) from bracket (6), if necessary.

## Installation Procedure

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

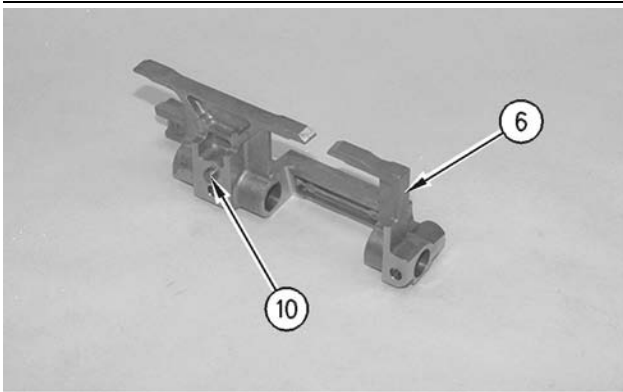


Illustration 266  
Typical example g00515670

1. Install the spring pin and dowel (10) in bracket (7).

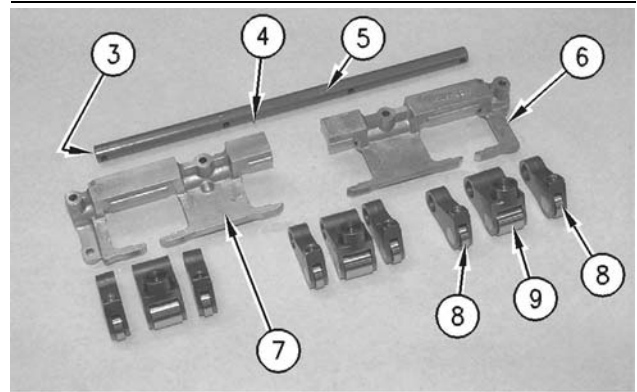


Illustration 267 g00515668

2. Install dowel (4) and cap plug (3) in lifter shaft (5). Lubricate the lifter shaft with clean engine oil prior to assembly.
3. Install the following items on the left end of lifter shaft (5): bracket (7), valve lifters (8), and unit injector lifter (9).
4. Install the following items on the middle of lifter shaft (5): valve lifters (8) and unit injector lifter (9).
5. Install the following items on the right end of lifter shaft (5): bracket (6), valve lifters (8), and unit injector lifter (9).

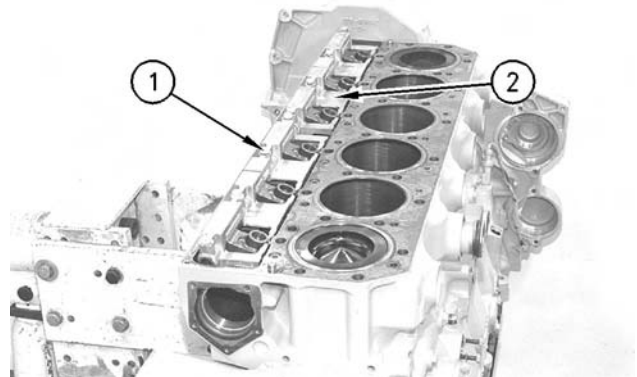


Illustration 268 g00515645

**Note:** Immerse the entire lifter assembly in clean engine oil prior to installation on the engine.

6. Place the lifter assembly (2) on the engine and install bolts (1). Tighten the bolts to a torque of  $30 \pm 3$  N·m ( $22 \pm 2$  lb ft).

### End By:

- a. Install the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Remove".

i01772030

# Camshaft - Remove

SMCS Code: 1210-011

## Removal Procedure

Table 38

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 131 - 1525  | Camshaft Pilot   | 1   |

### Start By:

- a. Remove the rocker shaft and the pushrods. Refer to Disassembly and Assembly, "Rocker Shaft and Pushrod - Remove".
- b. Remove the front cover. Refer to Disassembly and Assembly, "Front Cover - Remove".
- c. Remove the camshaft position sensor. Refer to Disassembly and Assembly, "Camshaft Position Sensor - Remove and Install".

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

- 1. Turn the crankshaft to top center compression stroke for the No. 1 piston. Install the timing bolt in the flywheel. This is for timing during installation.
- 2. Ensure that the timing marks on the following gears are aligned: the camshaft drive gear, the idler gear, and the crankshaft gear.

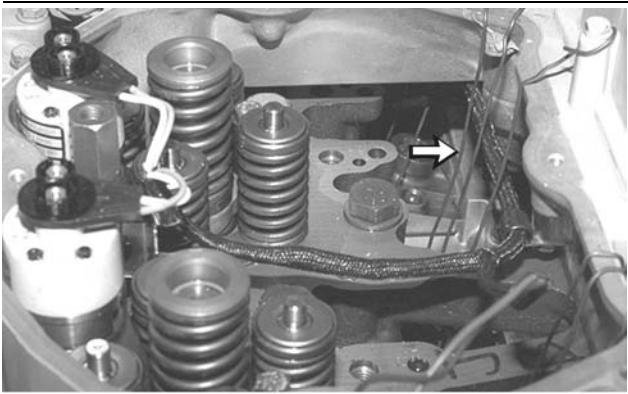


Illustration 269

g00529917

**Note:** It is not necessary to remove the cylinder head for removal of the camshaft.

- 3. Wire the valve lifters away from the camshaft. See Illustration 269.

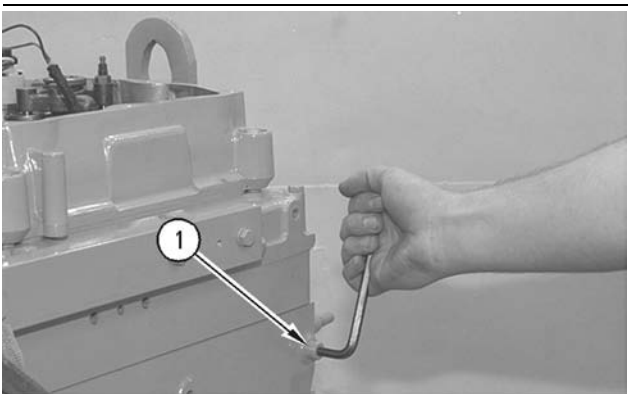


Illustration 270

g00516004

- 4. Remove thrust pin (1).

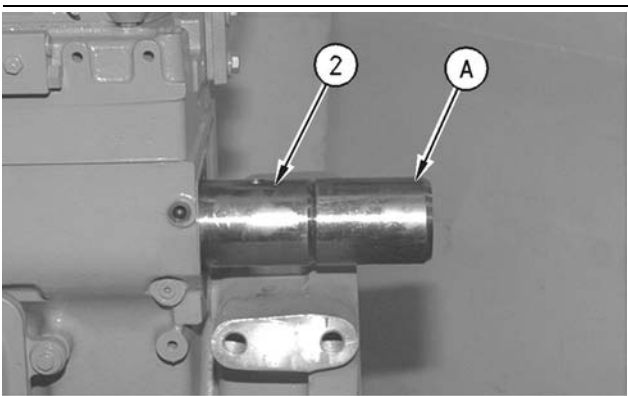


Illustration 271

g00765230

- 5. Install Tooling (A) into camshaft (2).

**Note:** Lubricate Tooling (A) for easier removal of the camshaft. Lubrication also prevents damaging the camshaft lobes and the camshaft bearings.

6. Carefully remove camshaft (2) from the engine.

i01772108

Camshaft - Install

SMCS Code: 1210-012

Installation Procedure

Table 39

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 131 - 1525  | Camshaft Pilot   | 1   |

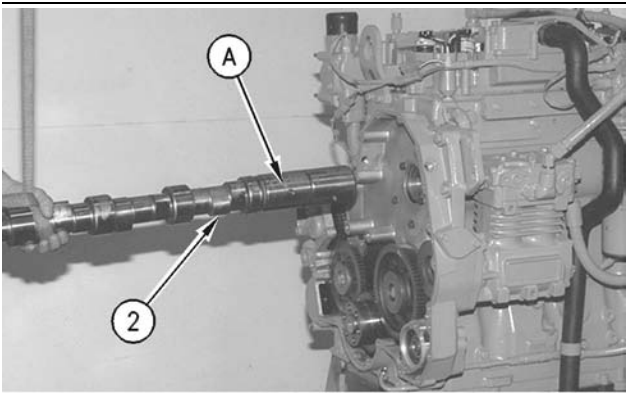


Illustration 272  
Typical example

1. Install Tooling (A).

**Note:** Ensure that camshaft (2) and Tooling (A) are clean. Apply clean engine oil to the lobes and journals of camshaft (2) and Tooling (A).

**Note:** During installation of camshaft (2), rotate the camshaft in both directions in order to prevent binding in the camshaft bearing bores.

2. Carefully install camshaft (2) in the engine.

NOTICE

When installing the camshaft, make sure the number one cylinder is at top center of the compression stroke with the timing bolt installed in the flywheel. The camshaft timing is very important. The timing mark on the camshaft drive gear must line up with the timing mark on the idler gear. Refer to the Specifications Manual for more information.

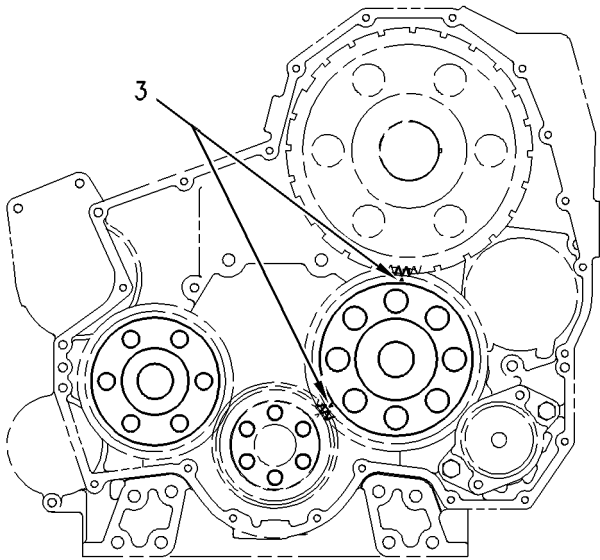


Illustration 273

3. Ensure that the timing marks (3) on the idler gear are aligned with the timing marks on the crankshaft gear and the camshaft drive gear.

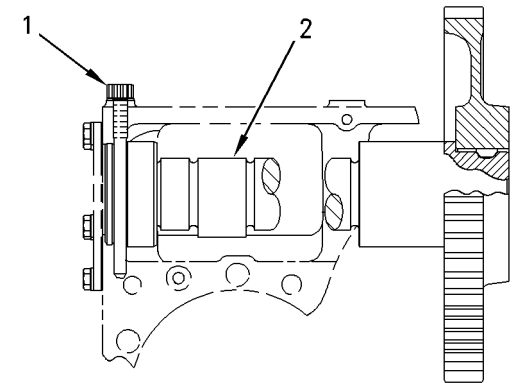


Illustration 274  
(1) Thrust Pin  
(2) Camshaft

4. Install thrust pin (1). Tighten thrust pin (1) to a torque of  $28 \pm 7$  N·m ( $20 \pm 5$  lb ft). Remove the wires that were used to hold the valve lifters.

End By:

- a. Install the camshaft position sensor. Refer to Disassembly and Assembly, "Camshaft Position Sensor - Remove and Install".
- b. Install the front cover. Refer to Disassembly and Assembly, "Front Cover - Install".
- c. Install the rocker shaft and the pushrods. Refer to Disassembly and Assembly, "Rocker Shaft and Pushrod - Install".

i01465422

# Camshaft Gear - Remove and Install

SMCS Code: 1210-010-GE

## Removal Procedure

**Start By:**

- a. Remove the camshaft. Refer to Disassembly and Assembly, "Camshaft - Remove".
- 1. Wrap the camshaft with a protective covering in order to prevent damage to the camshaft.

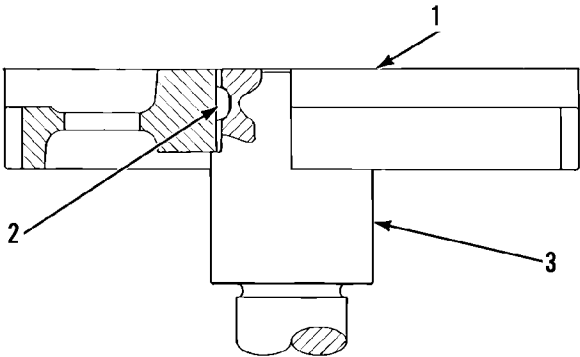


Illustration 275

g00522082

**NOTICE**

Do not allow the camshaft to fall to the floor when pressing it from the drive gear. Also make sure that the camshaft lobes do not catch on the press plates.

- 2. Place camshaft (3) and drive gear (1) in a suitable press. Press camshaft (3) from drive gear (1).
- 3. Remove woodruff key (2) from camshaft (3).

## Installation Procedure

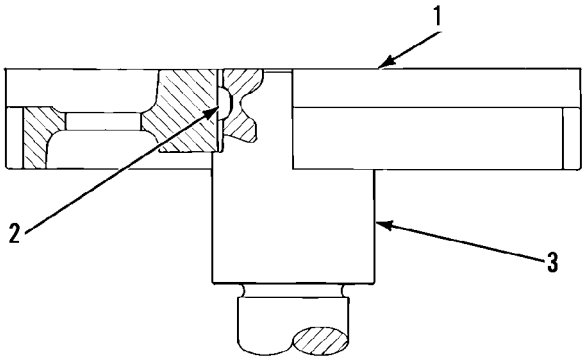


Illustration 276

g00522082

- 1. Install woodruff key (2) in camshaft (3).
- 2. Heat drive gear (1) to a temperature of 300 °C (572 °F) for 30 minutes.
- 3. Install drive gear (1) on the end of camshaft (3).
- 4. Ensure that woodruff key (2) is aligned. Also ensure that drive gear (1) makes contact with the shoulder on the end of camshaft (3).

**End By:**

- a. Install the camshaft. Refer to Disassembly and Assembly, "Camshaft - Install".

i01533953

# Camshaft Bearings - Remove

SMCS Code: 1211-011

## Removal Procedure

Table 40

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 129-6671    | Reaction Sleeve  | 1   |
|                | 129-6672    | Puller Plate     | 1   |
|                | 129-6673    | Bearing Pilot    | 1   |
|                | 129-6674    | Backup Plate     | 1   |
|                | 6F-7032     | Bolt             | 1   |
|                | 1P-5542     | Taper Lock Stud  | 1   |

Start By:

- a. Remove the lifter group. Refer to Disassembly and Assembly, "Lifter Group - Remove and Install".
- b. Remove the camshaft. Refer to Disassembly and Assembly, "Camshaft - Remove".

**Note:** All of the tools that are listed above are used with the 8S-2241 Camshaft Bearing Tool Group.

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

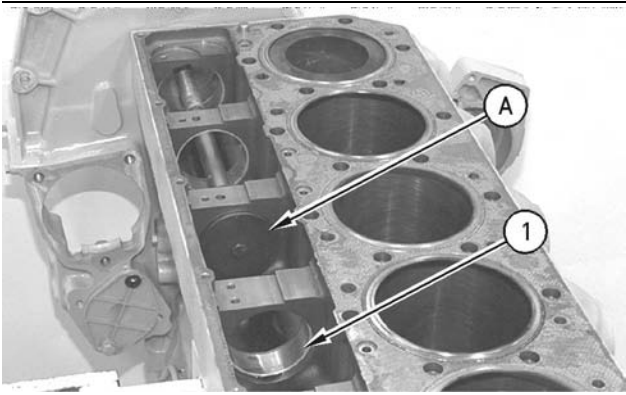


Illustration 277

g00516210

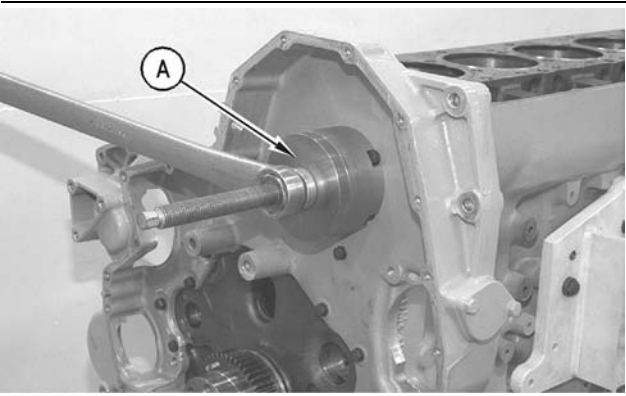


Illustration 278

g00516211

- 1. Install Tool (A). Apply Anti-Seize Compound on the threads.
- 2. Remove camshaft bearings (1).

i01533955

## Camshaft Bearings - Install

**SMCS Code:** 1211-012

### Installation Procedure

Table 41

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 129-6671    | Reaction Sleeve  | 1   |
|                | 129-6672    | Puller Plate     | 1   |
|                | 129-6673    | Bearing Pilot    | 1   |
|                | 129-6674    | Backup Plate     | 1   |
|                | 6F-7032     | Bolt             | 1   |
|                | 1P-5542     | Taper Lock Stud  | 1   |

**Note:** All of the tools that are listed above are used with the 8S-2241 Camshaft Bearing Tool Group.

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

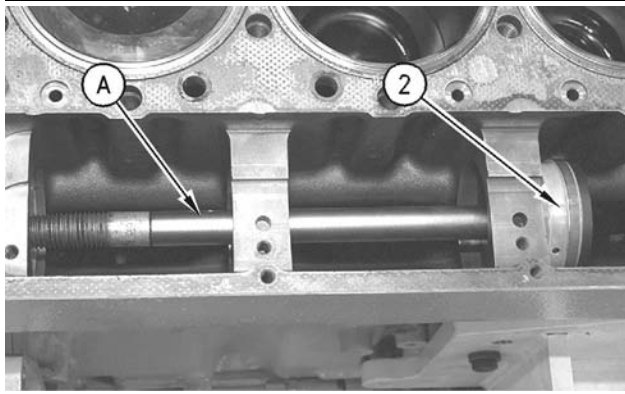


Illustration 279

g00516213

1. Position new bearing (2) in Tool (A) and install the bearing from the rear to the front of the engine.

#### NOTICE

Location of the camshaft bearing lubrication holes is critical. Position the clinch butt joint up for all seven camshaft bearings. The No. 1 camshaft bearing oil hole is located 115 degrees counter clockwise from top center to align with the oil supply passage in the block. Camshaft bearings No. 2 through No. 7 are installed with the oil hole 45 degrees clockwise from top center. All are viewed from the front of the engine. Camshaft bearings 1,2,3,4,5, and 6 are the same width. Camshaft bearing No. 7 has a smaller width. Refer to the Specifications Manual for more information.

#### End By:

- a. Install the camshaft. Refer to Disassembly and Assembly, "Camshaft - Install".
- b. Install the lifter group. Refer to Disassembly and Assembly, "Lifter Group - Remove and Install".

i01183159

## Engine Oil Pan - Remove and Install

SMCS Code: 1302-010

### Removal Procedure

#### WARNING

Hot oil and components can cause personal injury.

**Do not allow hot oil or components to contact skin.**

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Drain the engine oil into a suitable container for storage or disposal.

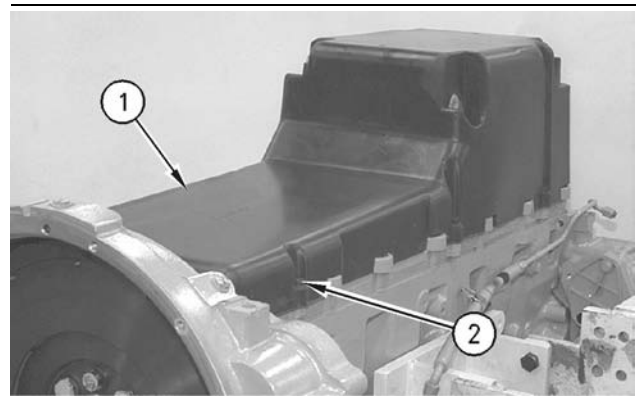


Illustration 280

g00516290

Typical example

2. Remove six bolts (2) and the washers. Remove sound suppression cover (1), if equipped.

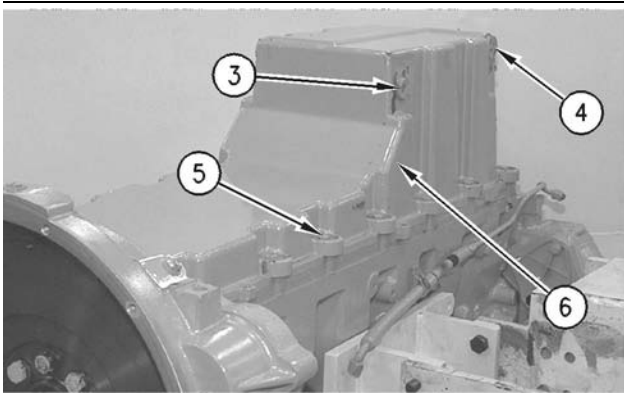


Illustration 281

g00516292

Typical example

3. Remove the following items from the cylinder block: fourteen bolts (5), the washers, the isolators, and the isolator sleeves.
4. Remove engine oil pan (6) and the gasket.
5. Remove pipe plug (3) and drain plug (4) with the O-ring seal, if necessary.

## Installation Procedure

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the gasket. Replace the gasket, if necessary.
2. Clean the mating surface of the engine oil pan and the cylinder block.

**Note:** Apply **8T-9022** Silicone Gasket to the right and the left edges of the cylinder block prior to installation.

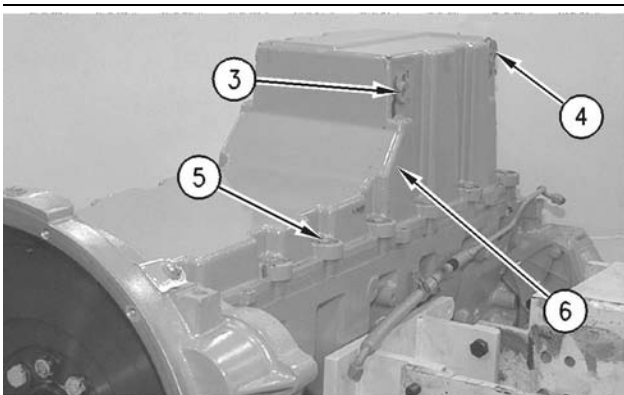


Illustration 282

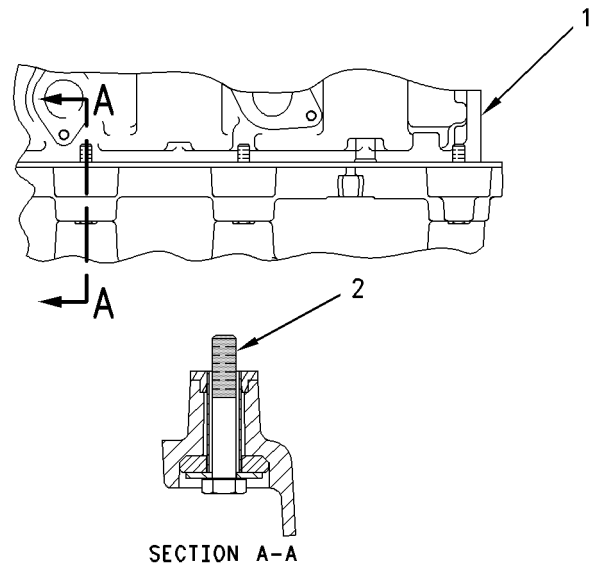
g00516292

Typical example

3. Place the gasket and engine oil pan (6) on the cylinder block.
4. Install the following items: isolator sleeves, isolators, bolts (5), and the washers.

**Note:** Perform the following procedure in order to eliminate the possibility of an oil leak from the bolt for the engine oil pan:

**Note:** The path of the oil leak is through the threads of the bolt for the engine oil pan. The third bolt from the front of the engine block on the right side of the engine is the suspect bolt.



SECTION A-A

Illustration 283

g00633606

Typical example

- (1) Front face of the engine block
- (2) Bolt threads

- a. Clean the bolt hole and dry the bolt hole.
- b. Clean the oil pan bolt and dry the oil pan bolt.
- c. Apply **4C-4030** Thread Lock Compound or **9S-3263** Thread Lock Compound to the threads of the bolt.
- d. Install the oil pan bolt and tighten the bolt to a torque of  $55 \pm 10$  N·m ( $40 \pm 7$  lb ft).
5. Wipe off excess **8T-9022** Silicone Gasket from all gasket joints. Use a flat, straight tool in order to remove the sealant.
6. Apply **9S-3263** Thread Lock Compound to pipe plug (3). Install pipe plug (3). Tighten the plug to a torque of  $80 \pm 11$  N·m ( $60 \pm 8$  lb ft).
7. Install drain plug (4) and the O-ring seal. Tighten the plug to a torque of  $70 \pm 15$  N·m ( $50 \pm 11$  lb ft).

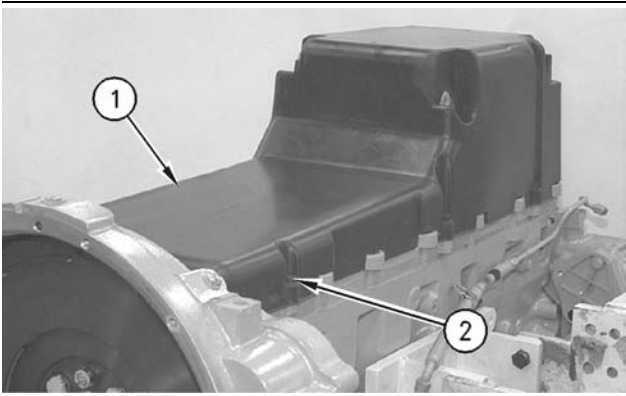


Illustration 284  
Typical example

8. Position the sound suppression cover (1) on the engine oil pan and install six bolts (2) with the washers.
9. Fill the engine with oil to the correct level. See the Operation and Maintenance Manual for the correct filling procedure.

## Cylinder Liner - Remove

SMCS Code: 1216-011

### Removal Procedure

Table 42

| Required Tools |                         |                      |     |
|----------------|-------------------------|----------------------|-----|
| Tool           | Part Number             | Part Description     | Qty |
| A              | 1U-9593 <sup>(1)</sup>  | Cylinder Pack Puller | 1   |
|                | 129-6675 <sup>(2)</sup> | Cylinder Pack Puller | 1   |
|                | 1U-9897                 | Bridge Group         | 1   |
|                | 1U-6319                 | Socket               | 1   |

<sup>(1)</sup> The 1U-9593 Cylinder Pack Puller is for use on C-10 Engines.  
<sup>(2)</sup> The 129-6675 Cylinder Pack Puller is for use on C-12 Engines.

#### Start By:

- a. Remove the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Remove".
- b. Remove the piston cooling jets. Refer to Disassembly and Assembly, "Piston Cooling Jets - Remove and Install".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

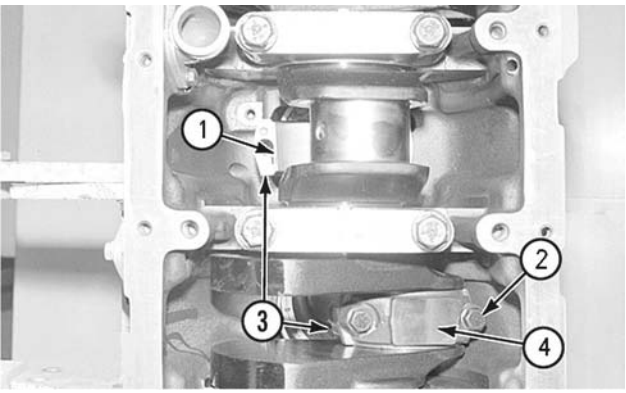


Illustration 285

1. Remove connecting rod cap bolts (2) from each connecting rod. Remove connecting rod caps (4).
2. Tap connecting rod (3) away from the crankshaft with a soft hammer. Remove the top half of connecting rod bearing (1).

i01772116

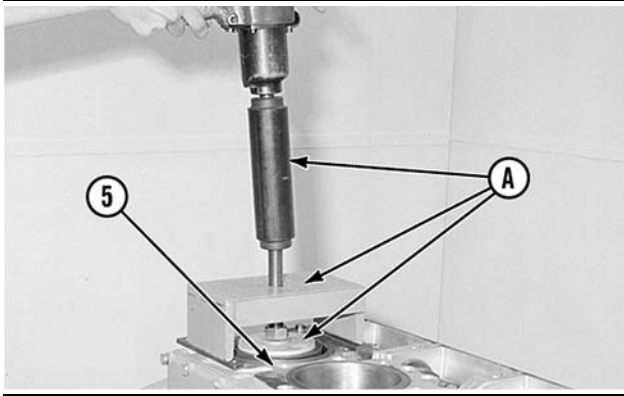


Illustration 286

g00516411

#### NOTICE

When removing the cylinder liner, watch the rod, and prevent it from catching on the bore in the cylinder block as it comes out.

3. Install Tooling (A) and remove cylinder liner (5), piston assembly and connecting rod.
4. Remove Tooling (A) from the cylinder liner.

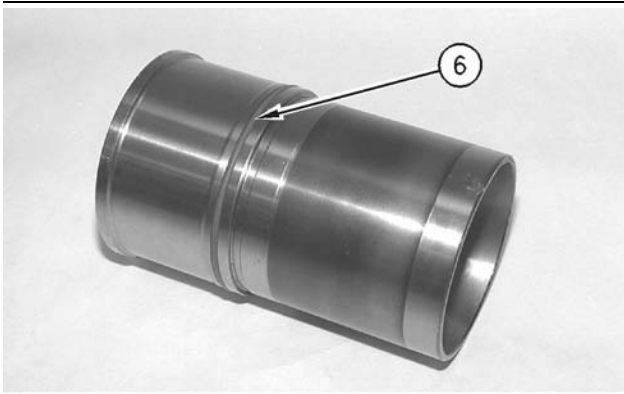


Illustration 287

g00516413

5. Remove the connecting rod and the piston from the cylinder liner. Remove seal (6) from the cylinder liner.

## Cylinder Liner - Install

SMCS Code: 1216-012

### Installation Procedure

Table 43

| Required Tools |             |                                   |   |
|----------------|-------------|-----------------------------------|---|
| Tool           | Part Number | Part Description                  | 1 |
| A              | 2P - 8260   | Cylinder Liner Installation Group | 1 |

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

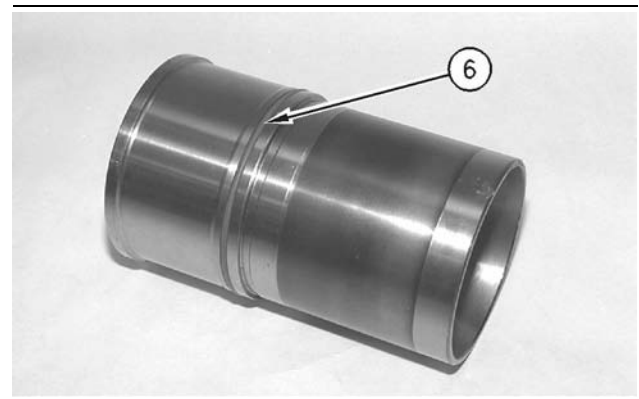


Illustration 288

g00516413

1. Apply 5P - 3975 Rubber Lubricant to the new seal. Install new seal (6) in the machined groove on the cylinder liner.

**Note:** The word "FRONT" may be stamped on the crown of the pistons in some engines. Ensure that the word "FRONT" is toward the front of the engine when the piston is installed. The etched number on the connecting rod must be on the right side of the engine in the corresponding cylinder.

2. Lubricate the lower portion of the cylinder liner with clean engine oil. Ensure that the corresponding crankshaft throw is at bottom center. Position the cylinder liner and piston assembly. Guide the connecting rod in place. Use Tooling (A) to press the cylinder liner in place.

**Note:** Refer to the Testing and Adjusting, "Cylinder Liner Projection" topic in order to check the cylinder liner projection.

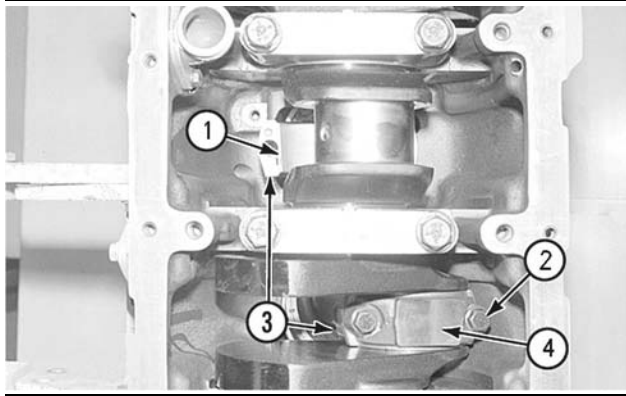


Illustration 289

g00474523

3. Install the upper half of connecting rod bearing (1) in connecting rod (3). Ensure that the bearing tab engages with the slot in the connecting rod. Lubricate the bearing surface with clean engine oil. Tap the piston with a soft hammer until the connecting rod bearing makes contact with the crankshaft.
4. Install connecting rod cap (4). Lubricate connecting rod cap bolts (2) with **185-3989** Anti-Seize Compound. Install connecting rod cap bolts (2).
5. Tighten the bolts to a torque of  $130 \pm 7$  N·m ( $95 \pm 5$  lb ft).
6. Place an index mark on each bolt head. Tighten each bolt for an additional  $60 \pm 5$  degrees ( $1/6$  turn).

**End By:**

- a. Install the piston cooling jets. Refer to Disassembly and Assembly, "Piston Cooling Jets - Remove and Install".
- b. Install the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Install".

i01464716

## Piston Cooling Jets - Remove and Install

**SMCS Code:** 1331-010

### Removal Procedure

**Start By:**

- a. Remove the engine oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

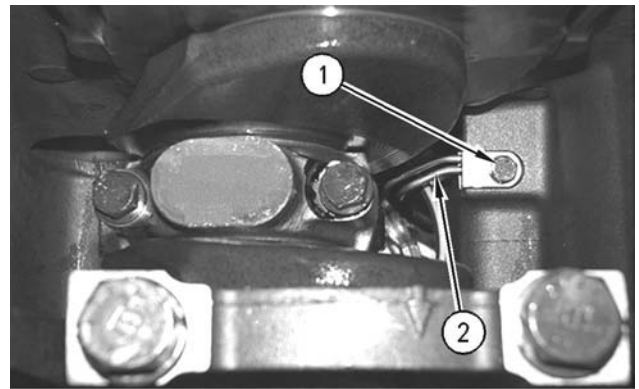


Illustration 290

g00516600

1. Remove bolt (1) and piston cooling jet (2) from each cylinder.

### Installation Procedure

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

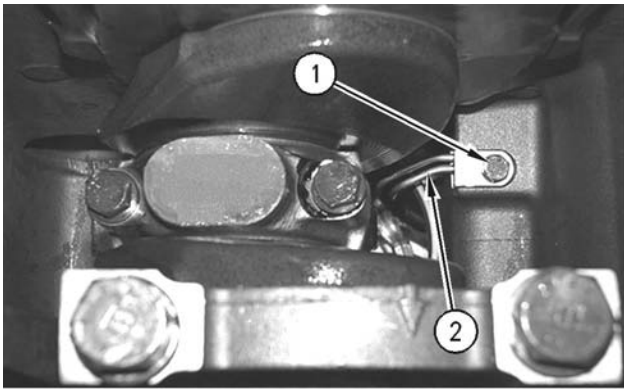


Illustration 291

g00516600

1. Position piston cooling jet (2) and install bolt (1) in each cylinder. Tighten the bolt to a torque of  $35 \pm 3 \text{ N}\cdot\text{m}$  ( $26 \pm 2 \text{ lb ft}$ ).

**End By:**

- a. Install the engine oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".

i01464749

## Pistons and Connecting Rods - Remove

**SMCS Code:** 1225-011

### Removal Procedure

**Start By:**

- a. Remove the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Remove".
- b. Remove the engine oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

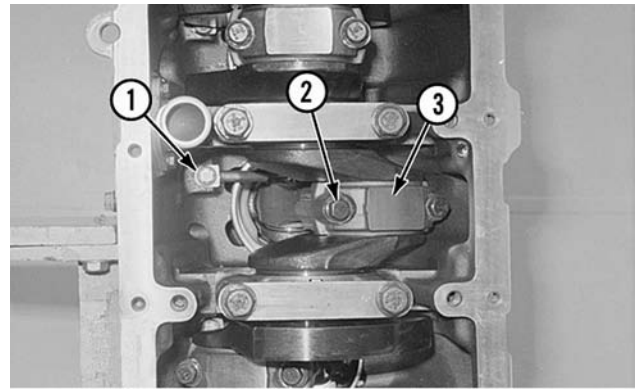


Illustration 292

g00516602

1. Remove the bolts that hold piston cooling jets (1) in position. Remove piston cooling jets (1).
2. Inspect the connecting rod and connecting rod cap (3) for the proper identification mark. The connecting rod and connecting rod cap (3) should have an etched number on the side that is toward the right side of the engine. The number should match the cylinder number. The bearing retainer notch should also be on the right side. Mark the connecting rod and connecting rod cap (3), if necessary.
3. Remove connecting rod cap bolts (2) and connecting rod cap (3). Remove the lower half of the bearing from connecting rod cap (3).

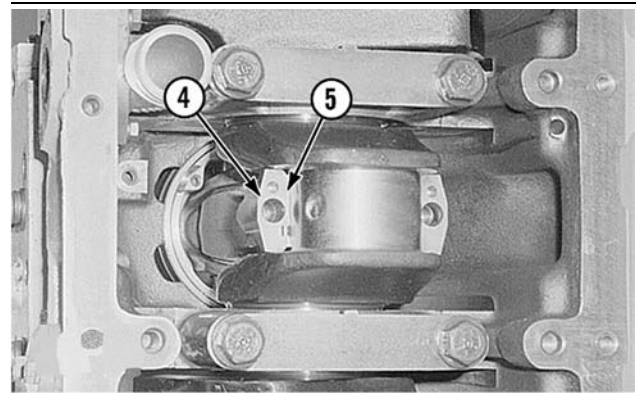


Illustration 293

g00516604

Typical example

4. Push the piston and connecting rod (4) away from the crankshaft. Remove the upper half of connecting rod bearing (5).
5. Remove the piston and connecting rod (4) from the cylinder block.
6. Repeat Steps 1 through 5 for the remaining pistons and connecting rods.

i01772125

# Pistons and Connecting Rods - Disassemble

SMCS Code: 1225-015

## Disassembly Procedure

Table 44

| Required Tools |             |                      |     |
|----------------|-------------|----------------------|-----|
| Tool           | Part Number | Part Description     | Qty |
| A              | 4C-3601     | Piston Ring Expander | 1   |
|                | 5P-8639     | Hydraulic Press      | 1   |
| B              | 8F-0024     | Hose Assembly        | 1   |
|                | 1P-2375     | Connecting Coupler   | 1   |
|                | 1P-2376     | Connecting Coupler   | 1   |
|                | 5P-9725     | Hydraulic Cylinder   | 1   |
|                | 5P-8651     | Spacer               | 1   |
|                | 5P-8649     | Adapter              | 1   |
|                | 5P-8650     | Adapter              | 1   |

### Start By:

- a. Remove the pistons and the connecting rods.  
Refer to Disassembly and Assembly, "Piston and Connecting Rods - Remove".

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

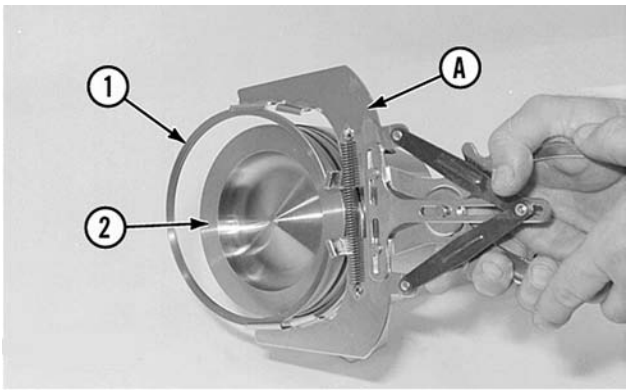


Illustration 294

g00516756

- 1. Use Tooling (A) in order to remove piston rings (1) from piston crown (2).

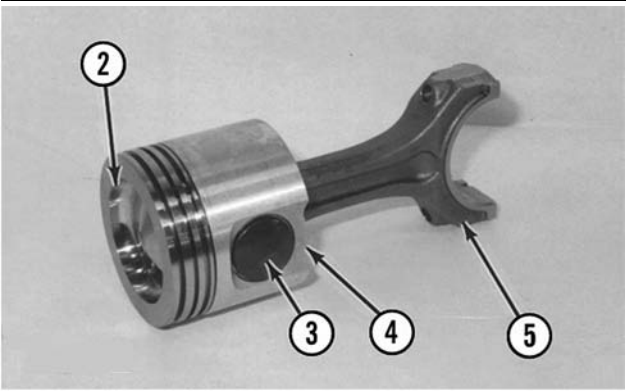


Illustration 295

g00516777

- 2. Remove plug (3) and the piston pin.

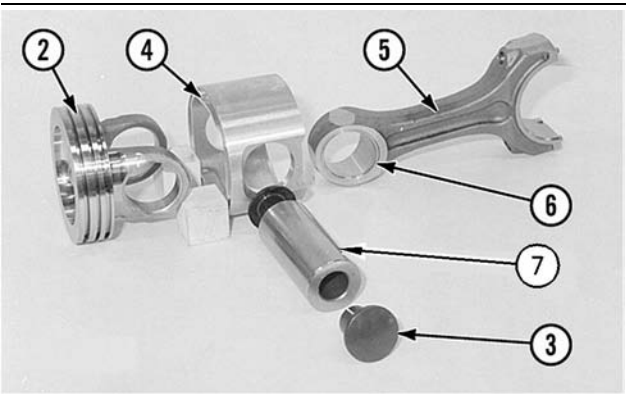


Illustration 296

g00765257

- 3. Separate piston crown (2) from piston skirt (4) and connecting rod (5).

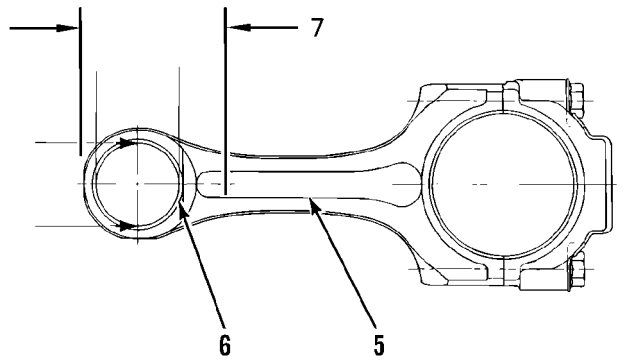


Illustration 297

g00516784

### NOTICE

The connecting rod must be heated for the installation of the piston pin bearing. Do not use a torch.

4. Heat connecting rod (5) in the indicated area (7). The maximum distance of area (7) is 85 mm (3.35 inch). The temperature should be between 175 °C (347 °F) and 260 °C (500 °F).
5. Use Tooling (B) in order to remove old bearing (6) and install new bearing (6) at the same time.
6. The bearing joint must be within 10 degrees of one of the locations that are indicated by the arrows.

i01772139

## Pistons and Connecting Rods - Assemble

**SMCS Code:** 1225-016

### Assembly Procedure

Table 45

| Required Tools   |             |                      |     |
|------------------|-------------|----------------------|-----|
| Tool             | Part Number | Part Description     | Qty |
| A                | 4C-3601     | Piston Ring Expander | 1   |
| B <sup>(1)</sup> | 208-7630    | Ring Groove Gauge    | 1   |

<sup>(1)</sup> Use the tool on the C-12 engine only.

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Verify correct assembly of the pistons and the connecting rods. Ensure that the etched numbers on the rod and the rod cap are in the correct positions. The etched number on the rod and the rod cap correspond to the cylinder in which it should be installed.

#### NOTICE

Do not use a hard surfaced tool to install plug. Do not strike or impact plug during installation. If the plug is dropped, replace the plug with another new plug.

**Note:** The word "FRONT" may be stamped on the crown of the pistons in some engines. Ensure that the word "FRONT" is toward the front of the engine when the piston is installed. The etched number on the connecting rod must be on the right side of the engine in the corresponding cylinder. Ensure that the piston crown and the etched number are correctly positioned.

**Note:** Inspect condition of all piston ring grooves. Use Tooling (B) to verify diameter of top ring groove. Refer to Tool Operating Manual, NEHS0840 for additional information.

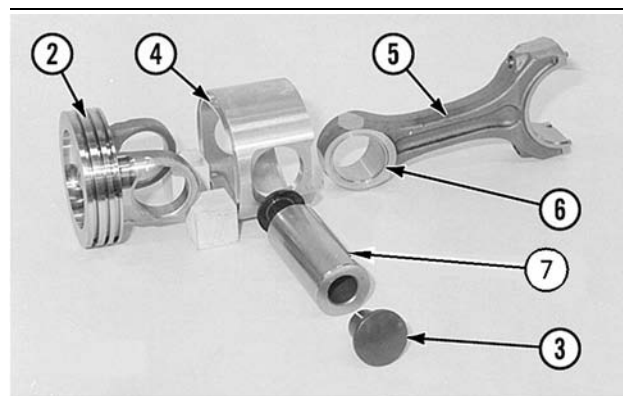


Illustration 298

g00765257

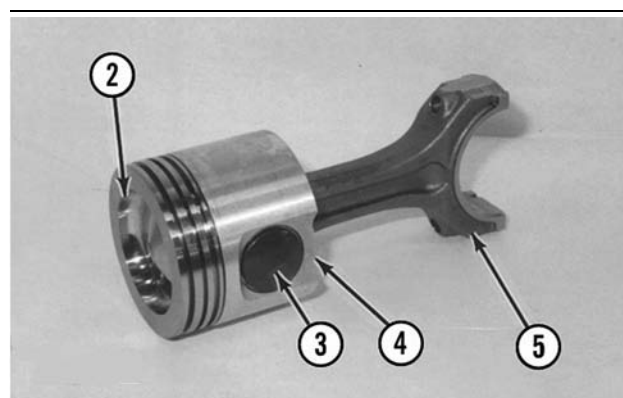


Illustration 299

g00516777

1. Place piston crown (2) in piston skirt (4). Place the piston assembly on connecting rod (5). Put clean engine oil on piston pin (7) and install the pin. Install new plug (3).
2. Check the clearance between the ends of the piston rings. See the Specifications Manual, "Pistons and Rings" topic.

**Note:** The oil ring must be installed over the spring with the end gap 180 degrees from the oil ring spring joint.

3. Install the oil control piston ring. The ends of the spring should be rotated 180 degrees from the ring end gap. The white colored portion of the spring must be visible at the ring end gap.

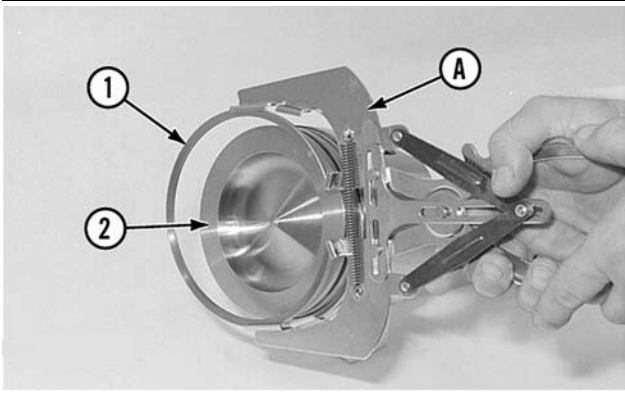


Illustration 300 g00516756

4. Install the intermediate piston ring with the side marked “UP-2” toward the top of the piston. Use Tooling (A) in order to install the ring.
5. Install the top piston ring with the side marked “UP-1” toward the top of the piston. Use Tooling (A) in order to install the ring.
6. After installation, all three of the piston rings should be placed 120 degrees away from each other.

**End By:**

- a. Install the pistons and the connecting rods. Refer to Disassembly and Assembly, “Piston and Connecting Rods - Install”.

i01734968

# Pistons and Connecting Rods - Install

**SMCS Code:** 1225-012

## Installation Procedure

Table 46

| Required Tools |                                                            |                        |     |
|----------------|------------------------------------------------------------|------------------------|-----|
| Tool           | Part Number                                                | Part Description       | Qty |
| A              | 161 -4163 <sup>(1)</sup><br>or<br>161 -4164 <sup>(2)</sup> | Piston Ring Compressor | 1   |

<sup>(1)</sup> Use with C-10 Engines

<sup>(2)</sup> Use with C-12 Engines

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Apply clean engine oil on the following items: the piston, the piston rings, and the cylinder bore.

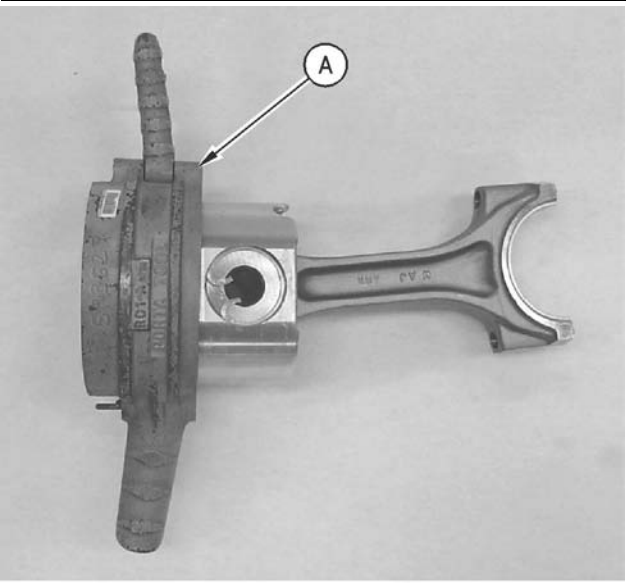


Illustration 301 g00889906

2. Install Tooling (A) in order to compress the piston rings.
3. With the number one crankshaft throw at the bottom center, install the piston and connecting rod in the engine. Ensure that the proper piston and connecting rod are in the corresponding cylinder with the bearing retainer notch toward the right side of the engine.

**Note:** The word “FRONT” may be stamped on the crown of the pistons in some engines. Ensure that the word “FRONT” is toward the front of the engine when the piston is installed. The etched number on the connecting rod must be on the right side of the engine in the corresponding cylinder.

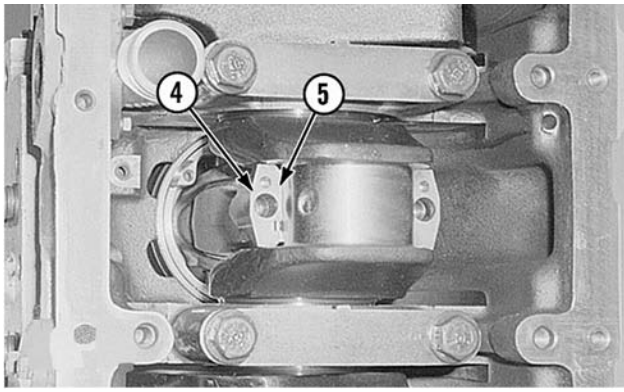


Illustration 302

g00516604

4. Line up the piston and connecting rod (4) with the crankshaft. Use a soft hammer and tap the piston into the cylinder bore until Tooling (A) comes off of the piston.
5. Before the connecting rod comes in contact with the crankshaft, install the upper half of connecting rod bearing (5). Ensure that the bearing tab engages with the slot in the connecting rod.
6. Apply clean engine oil on the surface of the upper half of connecting rod bearing (5). Use a soft hammer in order to tap the piston into the cylinder bore while you guide the connecting rod onto the crankshaft.

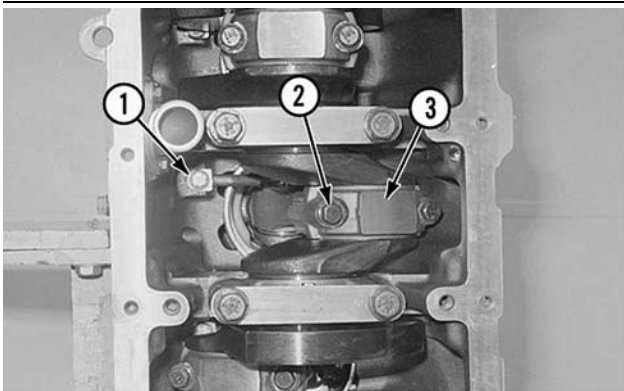


Illustration 303

g00516602

7. Place the lower half of connecting rod bearing (5) in corresponding connecting rod cap (3). Ensure that the bearing tab engages with the groove in connecting rod cap (3).
8. Apply clean engine oil on the surface of the lower half of connecting rod bearing (5). Install connecting rod bearing cap (3) on the connecting rod. Ensure that the number on connecting rod bearing cap (3) matches the number on the connecting rod. Ensure that the numbers are on the same side.

9. Install connecting rod cap bolts (2). Tighten the bolts to a torque of  $130 \pm 7 \text{ N}\cdot\text{m}$  ( $95 \pm 5 \text{ lb ft}$ ).
10. Place an index mark on each bolt head. Tighten each bolt for an additional  $60 \pm 5$  degrees ( $1/6$  turn).
11. Repeat Steps 1 through 10 for the remaining pistons and connecting rods.
12. Install piston cooling jets (1) and the bolts. Tighten the bolts to a torque of  $35 \pm 3 \text{ N}\cdot\text{m}$  ( $26 \pm 2 \text{ lb ft}$ ).

#### End By:

- a. Install the engine oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".
- b. Install the cylinder head. Refer to Disassembly and Assembly, "Cylinder Head - Install".

i01464812

## Connecting Rod Bearings - Remove

**SMCS Code:** 1219-011

### Removal Procedure

#### Start By:

- a. Remove the engine oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

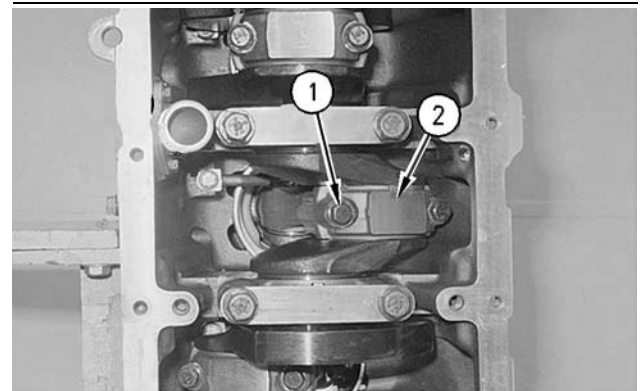


Illustration 304

g00522734

1. Inspect the connecting rod and connecting rod cap (2) for the proper identification mark. The connecting rod and connecting rod cap (2) should have an etched number on the side that is toward the right side of the engine. The number should match the cylinder number. The bearing retainer notch should also be on the right side. Mark the connecting rod and connecting rod cap (2), if necessary.
2. Remove connecting rod cap bolts (1) and connecting rod cap (2).
3. Remove the lower half of the connecting rod bearing from connecting rod cap (2).

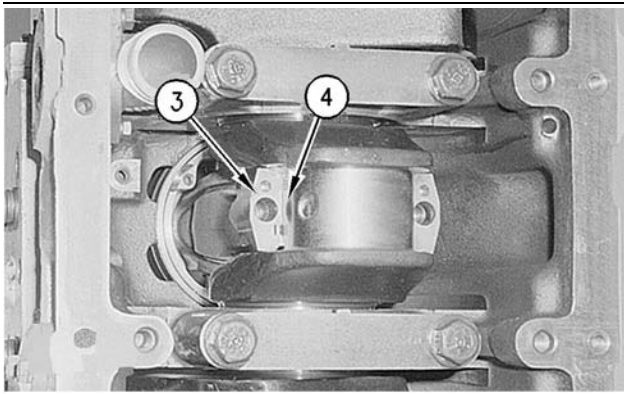


Illustration 305

g00522735

4. Push connecting rod (3) away from the crankshaft. Remove the upper half of connecting rod bearing (4).
5. Repeat Steps 1 through 4 in order to remove the remaining connecting rod bearings.

i01735295

## Connecting Rod Bearings - Install

**SMCS Code:** 1219-012

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

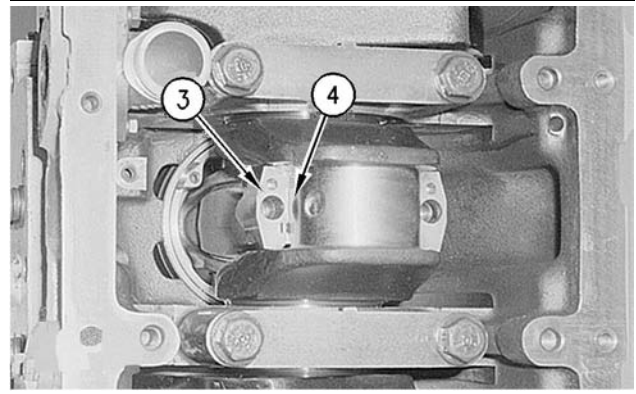


Illustration 306

g00522735

1. Install the upper half of connecting rod bearing (4) into connecting rod (3). Ensure that the bearing tab engages with the slot in the connecting rod.
2. Apply clean engine oil on the surface of the upper half of connecting rod bearing (4).
3. Position connecting rod (3) and connecting rod bearing (4) on the crankshaft.

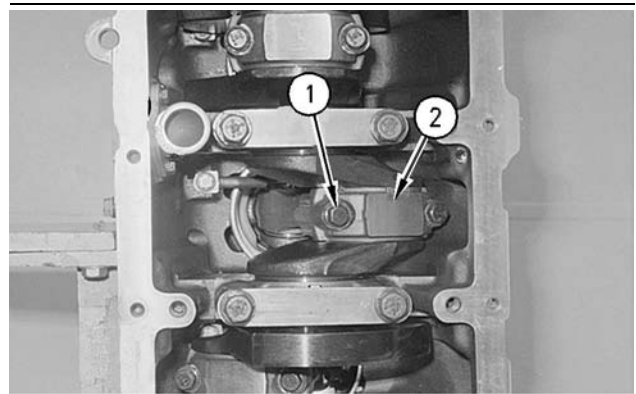


Illustration 307

g00522734

4. Install the lower half of the connecting rod bearing in corresponding connecting rod cap (2). Ensure that the bearing tab engages with the groove in connecting rod cap (2).
5. Apply clean engine oil on the surface of the lower half of the connecting rod bearing.
6. Install connecting rod cap (2) on the corresponding connecting rod. Ensure that the numbers are on the same side.
7. Install connecting rod cap bolts (1). Tighten the bolts to a torque of  $130 \pm 7$  N·m ( $95 \pm 5$  lb ft).

8. Place an index mark on each bolt head. Tighten each bolt for an additional  $60 \pm 5$  degrees (1/6 turn).
9. Repeat Steps 1 through 8 in order to install the remaining connecting rod bearings.

**End By:**

- a. Install the engine oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".

i01772186

## Crankshaft Main Bearings - Remove

**SMCS Code:** 1203-011

### Removal Procedure

Table 47

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 2P-5518     | Bearing Tool     | 1   |

**Start By:**

- a. Remove the oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

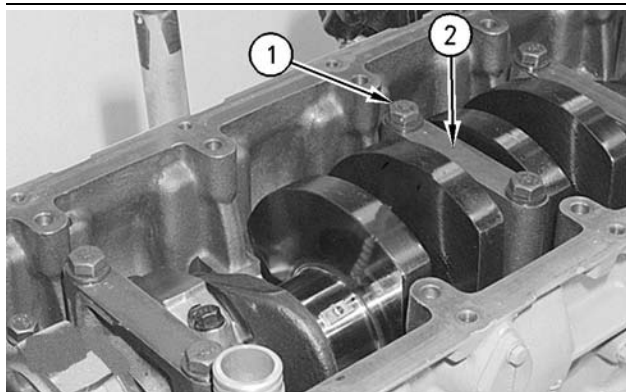


Illustration 308

g00824745

1. Remove bolts (1) that hold crankshaft main bearing cap (2). Remove crankshaft main bearing cap (2).

2. Remove the lower half of the crankshaft main bearing from crankshaft main bearing cap (2).

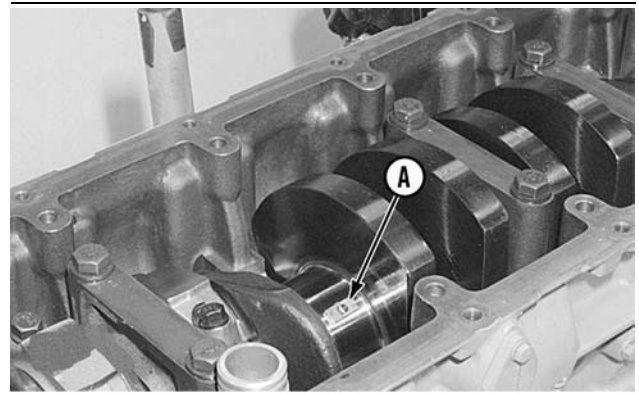


Illustration 309

g00516920

Typical example

**NOTICE**

If the crankshaft is turned in the wrong direction, the tab of the crankshaft main bearing will be pushed between the crankshaft and the cylinder block. This can cause damage to either or both the crankshaft and the cylinder block.

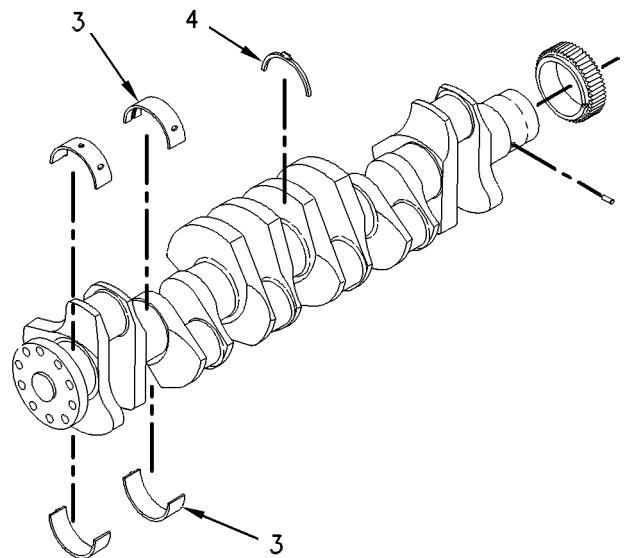


Illustration 310

g00516919

3. Use the following steps in order to remove the upper half of crankshaft main bearings (3):

- a. Turn the crankshaft until Tooling (A) can be installed in the crankshaft journal. Install Tooling (A).
- b. Turn the crankshaft in order to remove the upper half of the crankshaft main bearing. In order to get the tab out first, the crankshaft must be turned in the correct direction.

- c. Check the condition of the crankshaft main bearings. Refer to the Guideline For Reusable Parts, SEBF8009, "Main and Connecting Rod Bearings" or refer to the Guideline For Reusable Parts, SEBV0544, "Engine Bearings and Crankshafts".

4. Remove thrust plate (4).

i01772200

# Crankshaft Main Bearings - Install

SMCS Code: 1203-012

## Installation Procedure

Table 48

| Required Tools |             |                      |     |
|----------------|-------------|----------------------|-----|
| Tool           | Part Number | Part Description     | Qty |
| A              | 2P-5518     | Bearing Tool         | 1   |
| B              | 8T-5096     | Dial Indicator Group | 1   |

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**Note:** Place clean engine oil on the crankshaft main bearings prior to assembly. Ensure that the tabs on the back side of the crankshaft main bearings fit in the grooves of the crankshaft main bearing caps and the cylinder block.

### NOTICE

Ensure that the crankshaft main bearings are installed in the correct locations. Two different part numbers are used in quantities of three and four. The part number having the quantity of four is to be installed on crankshaft main bearings 2, 3, 5, and 6. The part number having the quantity of three is to be installed on crankshaft bearing 1, 4, and 7. The thrust plate is to be installed on the center crankshaft main bearing only.

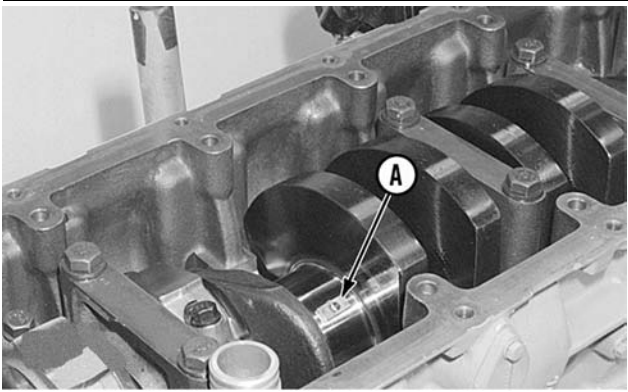


Illustration 311

g00516920

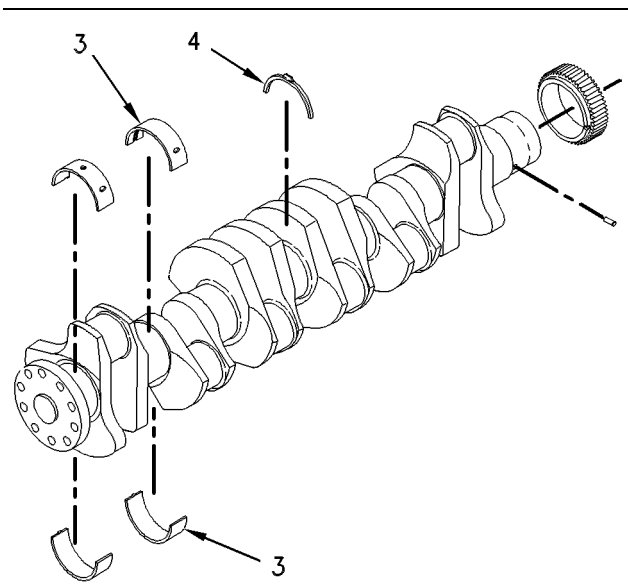


Illustration 312

g00516919

1. Use Tooling (A) and install the new upper halves of crankshaft main bearings (3) in the cylinder block. This half of the bearing has an oil hole. Do not put oil on the back of the crankshaft main bearing.

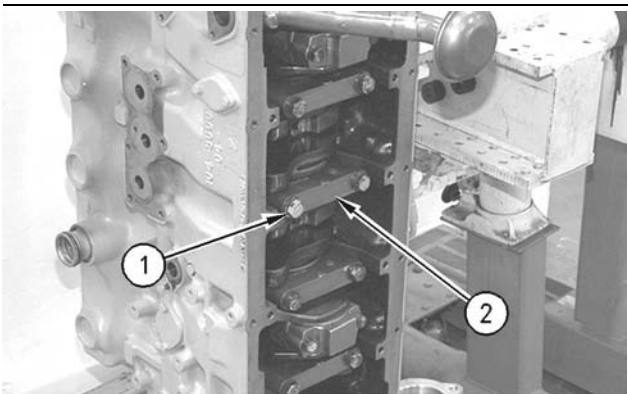


Illustration 313

g00516918

i01465195

2. Install the new lower halves of the crankshaft main bearings in crankshaft main bearing caps (2). Do not put oil on the back of the crankshaft main bearing.

**NOTICE**

Crankshaft main bearing caps should be installed with the part number toward the right side of the engine. Crankshaft main bearing caps are to be identified by stamped numbers 1 through 7 located on the bottom surface. The thrust plate is used on the number 6 crankshaft main bearing only.

3. Place crankshaft main bearing caps (2) in position on the cylinder block. Place clean engine oil or Molylube on the bolt threads and the washer face. Install bolts (1).
4. Tighten the bolts to a torque of  $95 \pm 5$  N·m ( $70 \pm 4$  lb ft). Place an index mark on each bolt head. Tighten each bolt for an additional  $90 \pm 5$  degrees ( $1/4$  turn).

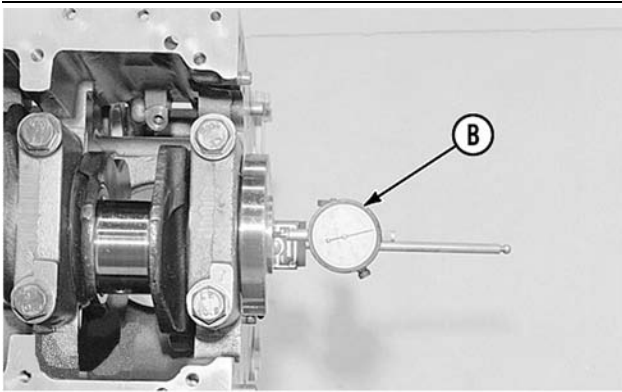


Illustration 314 g00517003

5. Check the end play of the crankshaft with Tool (B). The end play must be 0.10 mm (0.004 inch) to 0.50 mm (0.020 inch).

**Note:** The crankshaft end play is controlled by crankshaft thrust bearings (4). Thrust bearings (4) are located at the center of the crankshaft.

**End By:**

- a. Install the oil pan. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install"

## Crankshaft - Remove

**SMCS Code:** 1202-011

### Removal Procedure

**Start By:**

- a. Remove the engine oil pump. Refer to Disassembly and Assembly, "Engine Oil Pan - Remove and Install".
- b. Remove the front housing. Refer to Disassembly and Assembly, "Housing (Front) - Remove".
- c. Remove the flywheel housing. Refer to Disassembly and Assembly, "Flywheel Housing - Remove and Install".
- d. Remove the pistons and connecting rods. Refer to Disassembly and Assembly, "Pistons and Connecting Rods - Remove".

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

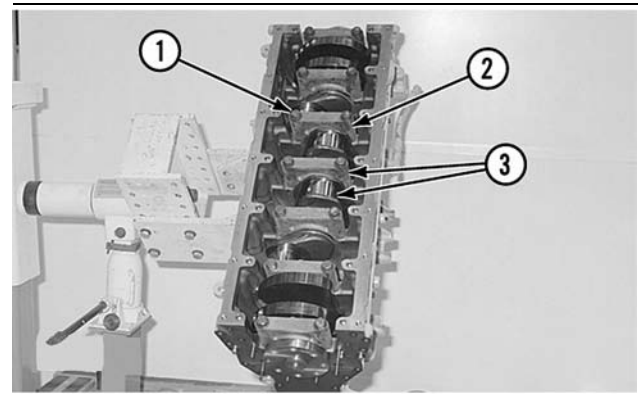


Illustration 315

g00517115

1. Remove bolts (1).
2. Remove crankshaft main bearing caps (2).
3. Remove crankshaft thrust bearings (3).

**Note:** Crankshaft main bearing caps (2) are marked with the numbers 1 through 7. Crankshaft main bearing caps (2) that are not marked should be marked with the corresponding cylinder number on the side that is toward the right side of the engine.

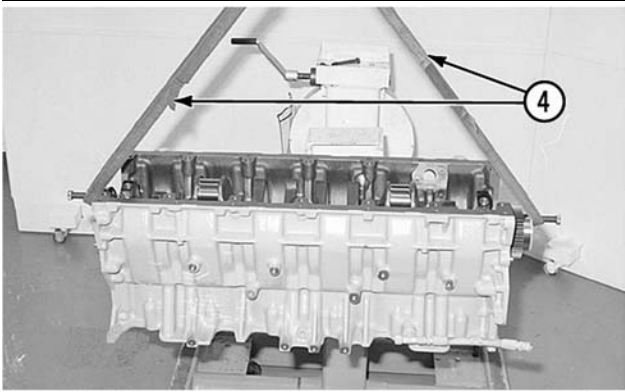


Illustration 316

g00517116

4. Install a suitable bolt into each end of the crankshaft. Fasten suitable lifting device (4) to the bolts in the crankshaft. Remove the crankshaft. The weight of the crankshaft is 129 kg (285 lb).
5. Remove the upper halves of the crankshaft main bearings.

i01465227

## Crankshaft - Install

**SMCS Code:** 1202-012

### Installation Procedure

Table 49

| Required Tools |             |                      |     |
|----------------|-------------|----------------------|-----|
| Tool           | Part Number | Part Description     | Qty |
| A              | 8T-5096     | Dial Indicator Group | 1   |

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Ensure that the crankshaft main bearing tabs engage with the grooves in the block and the crankshaft main bearing cap.

1. Place the upper halves of the crankshaft main bearings in the cylinder block. Place the lower halves of the crankshaft main bearings in crankshaft main bearing caps.

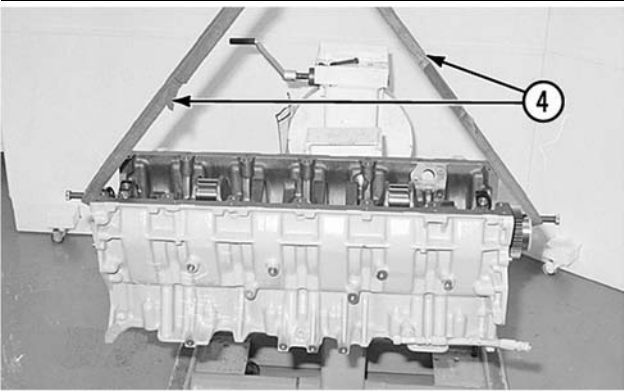


Illustration 317

g00517116

2. Fasten suitable lifting device (4) to the crankshaft and place the crankshaft in the cylinder block.

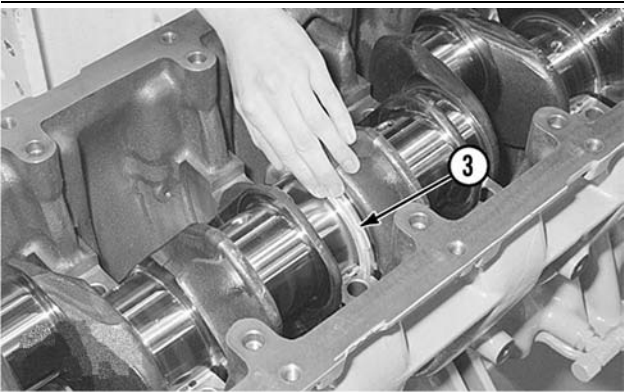


Illustration 318

g00517179

3. Install crankshaft thrust bearings (3).

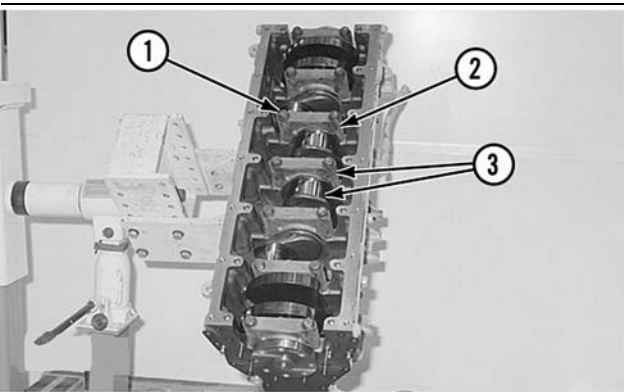


Illustration 319

g00517115

i01772213

4. Install crankshaft main bearing caps (2) with the part numbers toward the right hand side of the cylinder block. Ensure that crankshaft main bearing caps (2) have numbers 1 through 7. The numbers should start at the front of the engine. Place clean engine oil or Molylube on the bolt threads and the washers. Install bolts (1). Tighten the bolts to a torque of  $95 \pm 5$  N·m ( $70 \pm 4$  lb ft).
5. Place an index mark on each bolt head. Turn the bolts for an additional  $90 \pm 5$  degrees ( $1/4$  turn).

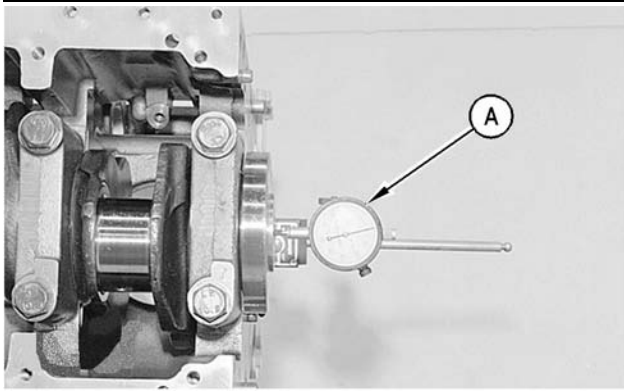


Illustration 320

g00517182

6. Check the end play of the crankshaft with Tool (A). The end play must be 0.10 mm (.004 inch) to 0.50 mm (.020 inch).

#### End By:

- a. Install the pistons and connecting rods. Refer to Disassembly and Assembly, "Piston and Connecting Rods - Install".
- b. Install the flywheel housing. Refer to Disassembly and Assembly, "Flywheel Housing - Remove and Install".
- c. Install the front housing. Refer to Disassembly and Assembly, "Housing (Front) - Install".
- d. Install the engine oil pump. Refer to Disassembly and Assembly, "Engine Oil Pump - Install".

## Crankshaft Gear - Remove and Install

**SMCS Code:** 1204-010-GE

### Removal Procedure

Table 50

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 1U-6400     | Three Jaw Puller | 1   |

#### Start By:

- a. Remove the crankshaft. Refer to Disassembly and Assembly, "Crankshaft - Remove".

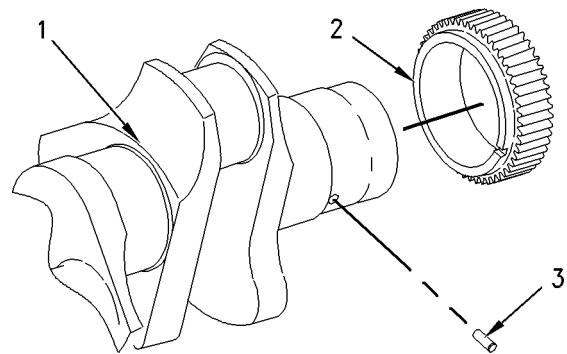


Illustration 321

g00912338

1. Use Tooling (A) in order to remove crankshaft gear (2) from crankshaft (1).
2. Remove drive pin (3).

### Installation Procedure

Table 51

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 6V-2098     | Press Sleeve     | 1   |
| B              | 8H-8581     | Gauge            | 1   |

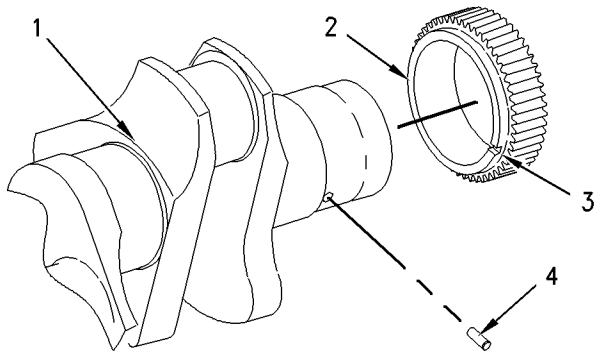


Illustration 322

g00909842

1. Install drive pin (4).

**WARNING**

**Always wear protective gloves when handling parts that have been heated.**

2. Heat crankshaft gear to a temperature of 210 °C (410 °F).
3. Install crankshaft gear (2) on the end of crankshaft (1). Ensure that slot (3) in crankshaft gear (2) is aligned with drive pin (4).

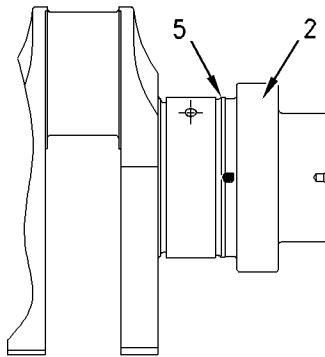


Illustration 323

g00912222

4. Use Tooling (A) to seat crankshaft gear (2) against shoulder (5) of crankshaft (1).
5. Use Tooling (B) to ensure that crankshaft gear (2) is seated against shoulder (5) on crankshaft (1).

**Note:** Check the crankshaft gear in three places with Tooling (B).

**End By:**

- a. Install the crankshaft. Refer to Disassembly and Assembly, "Crankshaft - Install".

# Bearing Clearance - Check

**SMCS Code:** 1203-535; 1219-535

## Measurement Procedure

Table 52

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 198-9142    | Plastic Gauge    | 1   |
|                | 198-9143    | Plastic Gauge    | 1   |
|                | 198-9144    | Plastic Gauge    | 1   |
|                | 198-9145    | Plastic Gauge    | 1   |

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**Note:** Caterpillar does not recommend the checking of the actual bearing clearances particularly on small engines. This is because of the possibility of obtaining inaccurate results and the possibility of damaging the bearing or the journal surfaces. Each Caterpillar engine bearing is quality checked for specific wall thickness.

**Note:** The measurements should be within specifications and the correct bearings should be used. If the crankshaft journals and the bores for the block and the rods were measured during disassembly, no further checks are necessary. However, if the technician still wants to measure the bearing clearances, Tooling (A) is an acceptable method. Tooling (A) is less accurate on journals with small diameters if clearances are less than 0.10 mm (0.004 inch).

**NOTICE**

Lead wire, shim stock or a dial bore gauge can damage the bearing surfaces.

The technician must be very careful to use Tooling (A) correctly. The following points must be remembered:

- Ensure that the backs of the bearings and the bores are clean and dry.
- Ensure that the bearing locking tabs are properly seated in the tab grooves.

i01016545

- The crankshaft must be free of oil at the contact points of Tooling (A).

1. Put a piece of Tooling (A) on the crown of the bearing that is in the cap.

**Note:** Do not allow Tooling (A) to extend over the edge of the bearing.

2. Use the correct torque-turn specifications in order to install the bearing cap. Do not use an impact wrench. Be careful not to dislodge the bearing when the cap is installed.

**Note:** Do not turn the crankshaft when Tooling (A) is installed.

3. Carefully remove the cap, but do not remove Tooling (A). Measure the width of Tooling (A) while Tooling (A) is in the bearing cap or on the crankshaft journal. Refer to Illustration 324.

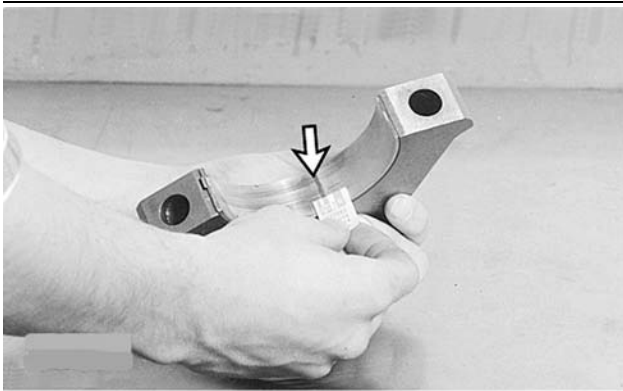


Illustration 324

g00473227

Typical Example

4. Remove all of Tooling (A) before you install the bearing cap.

**Note:** When Tooling (A) is used, the readings can sometimes be unclear. For example, all parts of Tooling (A) are not the same width. Measure the major width in order to ensure that the parts are within the specification range. Refer to Specifications, "Crankshaft" for the correct clearances.

## Camshaft Position Sensor - Remove and Install

**SMCS Code:** 1912-010

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

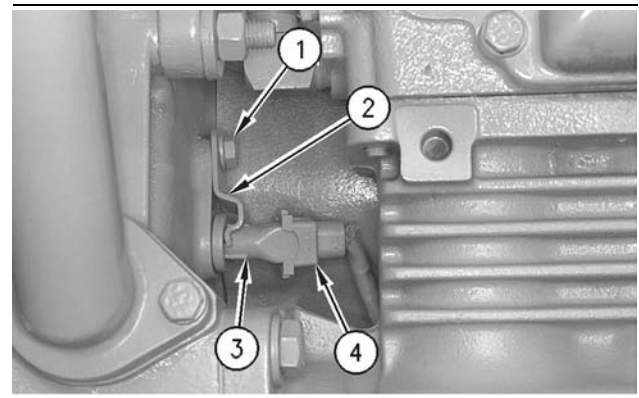


Illustration 325

g00521613

1. Disconnect sensor connector (4) from camshaft position sensor (3).
2. Remove bolt (1) and the washer.
3. Remove camshaft position sensor (3) and bracket (2) as a unit from the front housing.
4. Inspect the condition of the O-ring seal on camshaft position sensor (3). Replace the O-ring seal, if necessary.

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

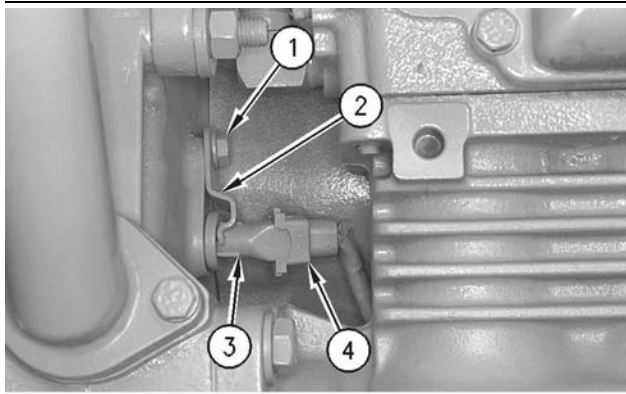


Illustration 326

g00521613

1. Install camshaft position sensor (3) and bracket (2) as a unit into the front housing.
2. Install bolt (1) and the washer.
3. Connect sensor connector (4) to camshaft position sensor (3).
4. Calibrate the engine speed-timing. Refer to Electronic Troubleshooting for more information.

i01018581

## Crankshaft Position Sensor - Remove and Install

**SMCS Code:** 1912-010

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

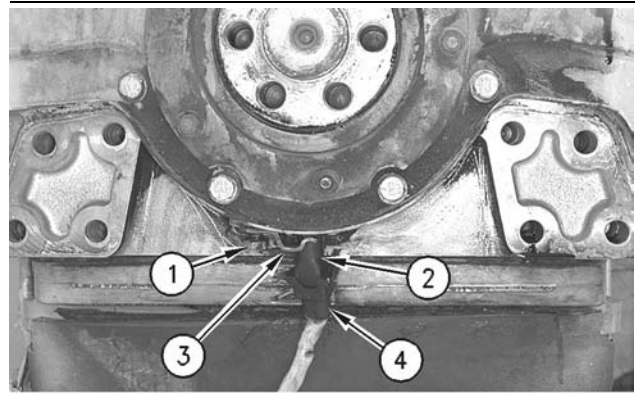


Illustration 327

g00522655

The vibration damper and the front engine support have been removed for photo purposes.

1. Disconnect sensor connector (4) from crankshaft position sensor (2).
2. Remove bolt (1) and the washer.
3. Remove crankshaft position sensor (2) and bracket (3) as a unit from the front housing.
4. Inspect the condition of the O-ring seal on crankshaft position sensor (2). Replace the O-ring seal, if necessary.

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

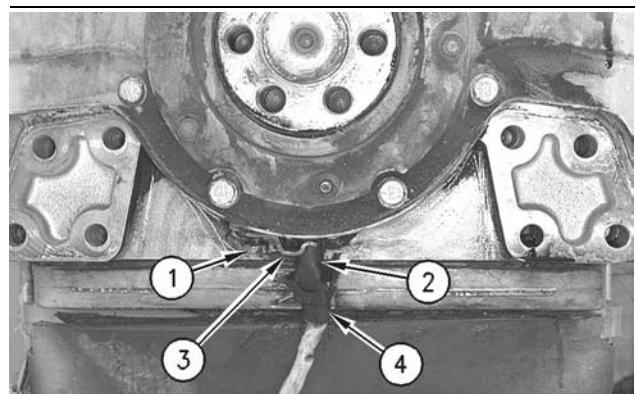


Illustration 328

g00522655

The vibration damper and the front engine support have been removed for photo purposes.

1. Install crankshaft position sensor (2) and bracket (3) as a unit into the front housing.

2. Install bolt (1) and the washer.
3. Connect sensor connector (4) to crankshaft position sensor (2).
4. Calibrate the engine speed-timing. Refer to Electronic Troubleshooting for more information.

i01014165

## Coolant Temperature Sensor - Remove and Install

**SMCS Code:** 1906-010

### Removal Procedure

#### **WARNING**

Personal injury can result from hot coolant, steam and alkali.

At operating temperature, engine coolant is hot and under pressure. The radiator and all lines to heaters or the engine contain hot coolant or steam. Any contact can cause severe burns.

Remove cooling system pressure cap slowly to relieve pressure only when engine is stopped and cooling system pressure cap is cool enough to touch with your bare hand.

**Do not attempt to tighten hose connections when the coolant is hot, the hose can come off causing burns.**

**Cooling System Coolant Additive contains alkali. Avoid contact with skin and eyes.**

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

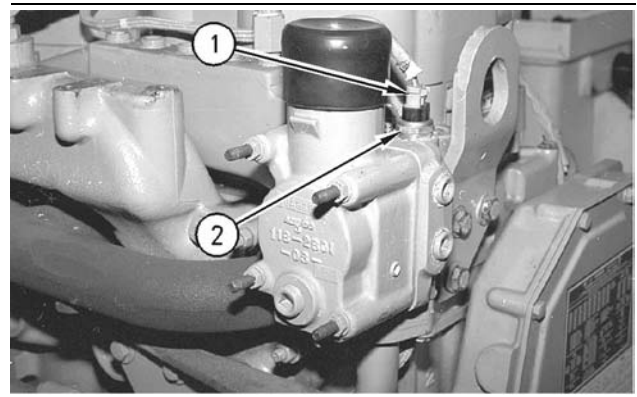


Illustration 329

g00520237

1. Drain the level of the coolant below the coolant temperature sensor. Drain the coolant into a suitable container for storage or disposal.
2. Disconnect sensor connector (1) from coolant temperature sensor (2).
3. Remove coolant temperature sensor (2).

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

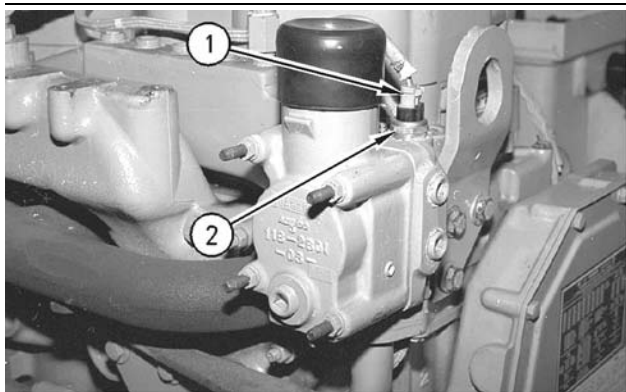


Illustration 330

g00520237

1. Install coolant temperature sensor (2). Tighten coolant temperature sensor (2) to a torque of  $20 \pm 5$  N·m ( $15 \pm 4$  lb ft).
2. Connect sensor connector (1) to coolant temperature sensor (2).
3. Fill the cooling system with coolant to the correct level. Refer to the Operation and Maintenance Manual for the correct procedure.

i01000174

## Engine Oil Pressure Sensor - Remove and Install

**SMCS Code:** 1924-010

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the machine. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide", for tools and supplies suitable to collect and contain fluids in Caterpillar machines.

Dispose of all fluids according to local regulations and mandates.

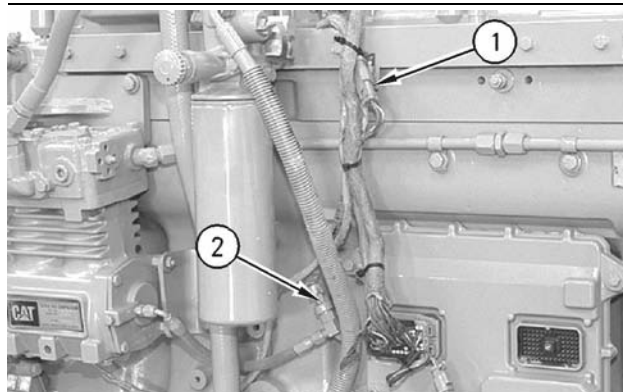


Illustration 331

g00512032

1. Disconnect sensor connector (1). Cut the wire ties that secure the sensor wiring to the engine.
2. Remove engine oil pressure sensor (2).

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

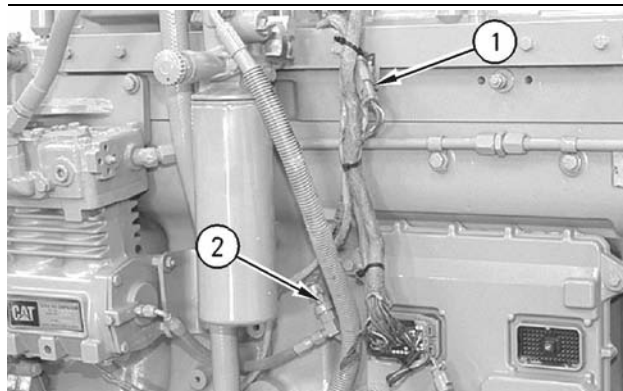


Illustration 332

g00512032

1. Install engine oil pressure sensor (2). Tighten the engine oil pressure sensor to a torque of  $10 \pm 2$  N·m ( $88 \pm 18$  lb in).
2. Connect sensor connector (1). Secure the sensor wiring to the engine with wire ties.

i01428365

## Fuel Pressure Sensor - Remove and Install

**SMCS Code:** 7414-010

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

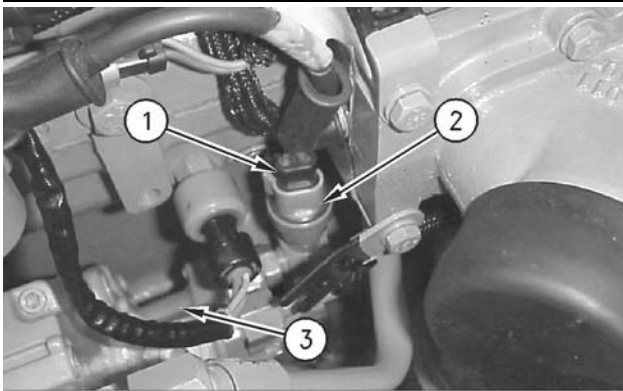


Illustration 333

g00748139

1. Disconnect wiring harness connector (1) from fuel pressure sensor (2).
2. Remove fuel pressure sensor (2) from fuel transfer pump (3).

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

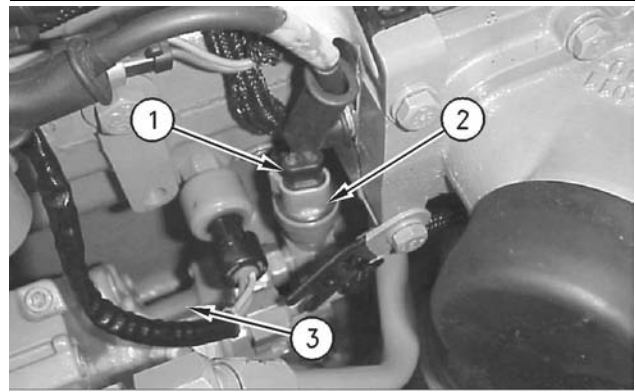


Illustration 334

g00748139

1. Install fuel pressure sensor (2). Tighten fuel pressure sensor (2) to a torque of  $10 \pm 2$  N·m ( $7.3 \pm 1.4$  lb ft).
2. Connect wiring harness connector (1) to fuel pressure sensor (2).

i01426862

## Boost Pressure Sensor - Remove and Install

**SMCS Code:** 1917-010

### Removal Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

#### NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

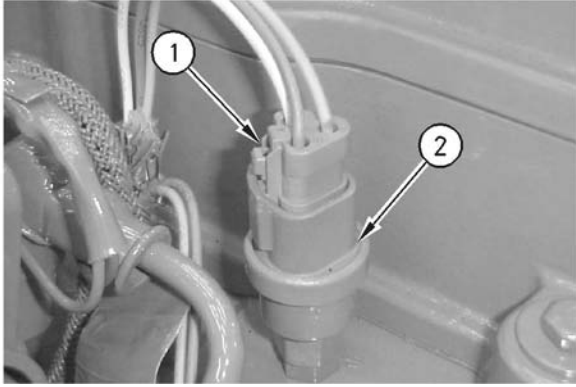


Illustration 335

g00636492

1. Disconnect wiring harness (1) from boost pressure sensor (2).
2. Remove boost pressure sensor (2) and the O-ring seal.

### Installation Procedure

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

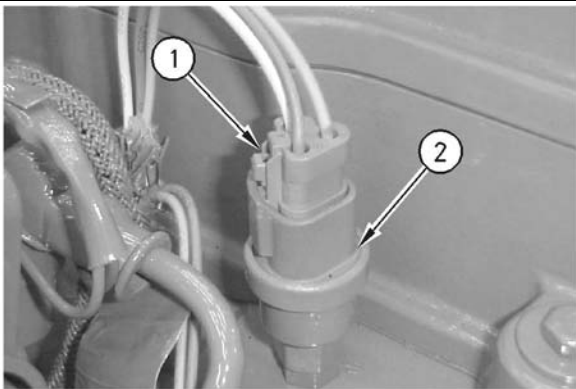


Illustration 336

g00636492

**Note:** Check the condition of the O-ring seal. If the O-ring seal is worn or damaged, use a new part for replacement.

1. Install the O-ring seal on boost pressure sensor (2).
2. Install boost pressure sensor (2). Tighten boost pressure sensor (2) to a torque of  $10 \pm 2$  N·m ( $88 \pm 18$  lb in).
3. Connect wiring harness (1) to boost pressure sensor (2).

## Belt Tensioner - Remove and Install

i01582984

SMCS Code: 1358-010

### Removal Procedure



Illustration 337

g00519255

1. Use a breaker bar to release the tension on belt tightener (1). Remove the drive belt.

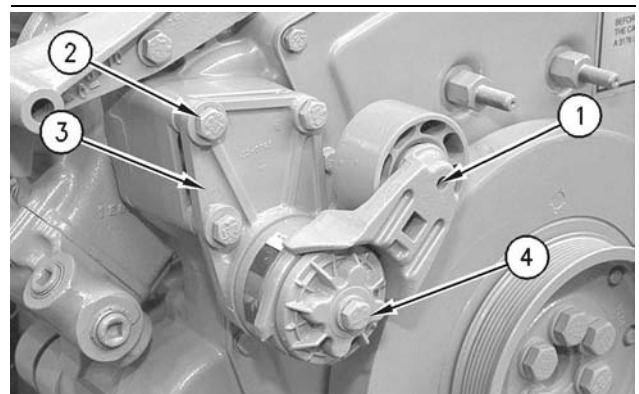


Illustration 338

g00519256

2. Remove bolt (4) and the washer.
3. Remove belt tightener (1).
4. Remove three bolts (2) and the washers.
5. Remove support (3).

### Installation Procedure

1. Inspect the condition of the belt tightener and the drive belt. Replace the tightener and the belt, if necessary.

i01533573

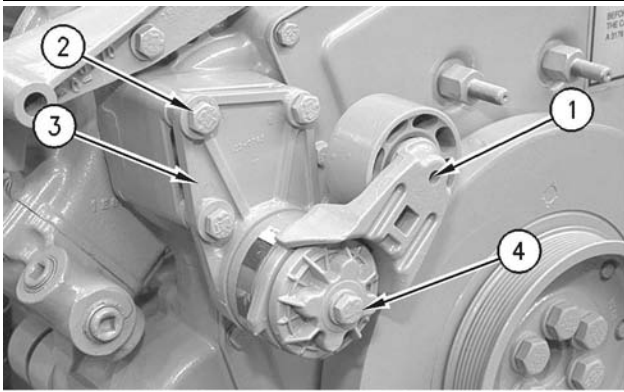


Illustration 339

g00519256

2. Install support (3) on the front housing with three bolts (2) and the washers. Tighten the bolts to a torque of  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft).
3. Install belt tightener (1) with bolt (4) and the washer. Tighten the bolt to a torque of  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft).
4. Use a breaker bar to rotate the belt tensioner. Install the drive belt while the belt tensioner is rotated. Allow the tensioner to rotate in order to apply tension to the belt. Remove the breaker bar from the belt tensioner.

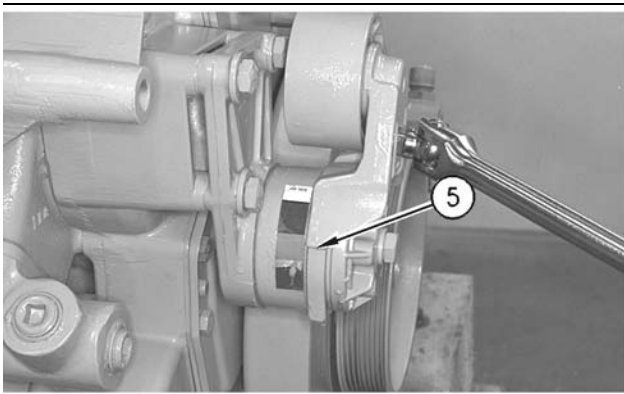


Illustration 340

g00519259

5. If the alternator is equipped with an adjustable support strap, adjust the position of the alternator until the indicator edge (5) is aligned with the green area of the indicator decal.

## Engine Control Module - Remove and Install

SMCS Code: 1901-010

### Removal Procedure

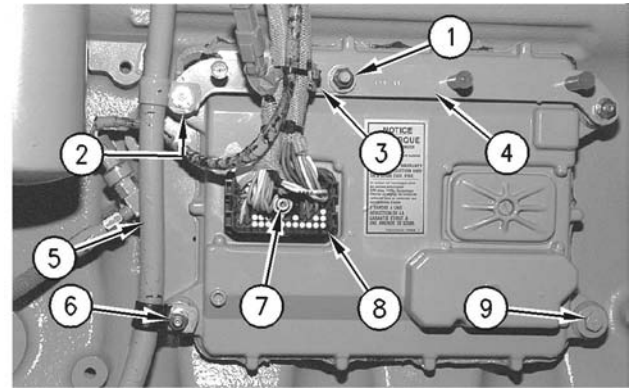


Illustration 341

g00523113

1. Remove nut (1), the washer, and clip (3).
2. Loosen bolt (7) until wiring harness (8) can be disconnected. Disconnect wiring harness (8).
3. Remove three nuts (6) and the washers. Move hose (5) out of the way. Remove plate (4).
4. Remove bolt (9) and the washer. Remove three studs (2) and the washers. Remove the electronic control module from the engine.

### Installation procedure

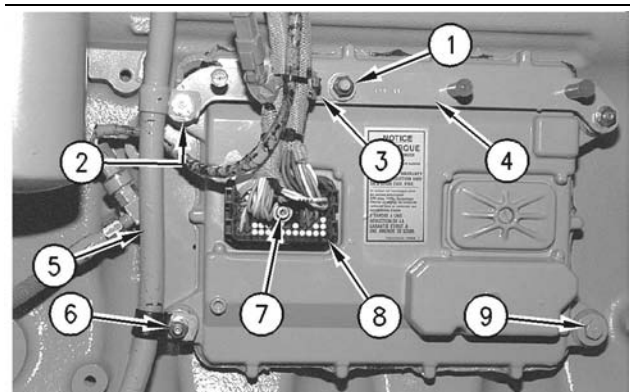


Illustration 342

g00523113

1. Position the electronic control module on the engine and install three studs (2) and the washers.
2. Install bolt (9) and the washer.

3. Install hose (5) and plate (4). Install three nuts (6).
4. Connect wiring harness (8). Tighten bolt (7) to a torque of  $6 \pm 1$  N·m ( $53 \pm 9$  lb in).
5. Install clip (3), the washer, and nut (1).

i01591987

## Alternator - Remove and Install

SMCS Code: 1405-010

### Removal Procedure

#### WARNING

**Personal injury can result from failure to disconnect the battery.**

**First, disconnect the negative battery cable. Then, disconnect the positive battery cable.**

**A positive power lead can cause sparks if the battery is not disconnected. Sparks can possibly result in battery explosion or fire.**

1. Remove the drive belt from the drive pulley on the alternator.
2. Place an index mark on all of the electrical wires that are connected to the alternator. Disconnect the electrical wires from the alternator.

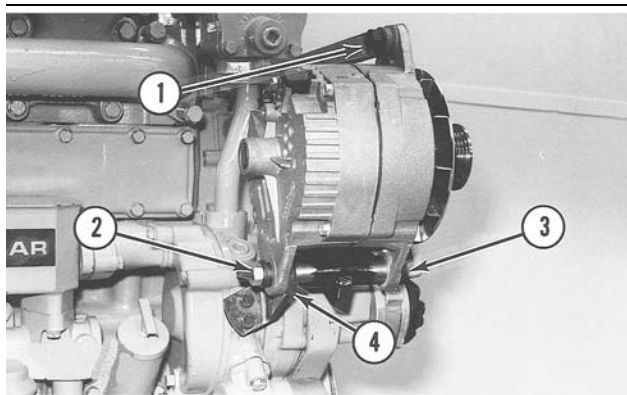


Illustration 343

g00519357

Typical example

3. Remove bolt (1) and the washer.
4. Remove nut (2) with the washer and bolt (3) from mounting bracket (4).
5. Remove the alternator.

### Installation Procedure

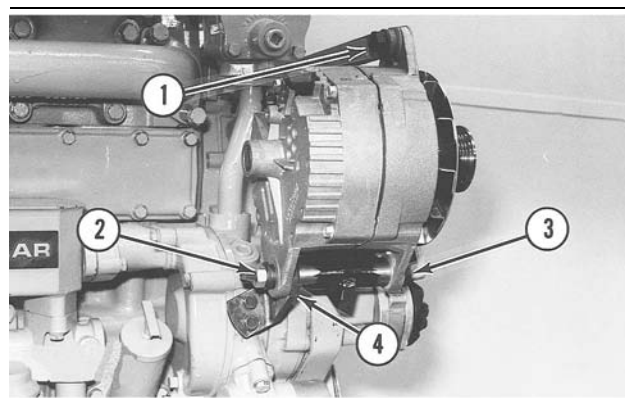


Illustration 344

g00519357

Typical example

1. Place the alternator on mounting bracket (4). Install bolt (3) with the washer and nut (2). Hand tighten nut (2).
2. Install bolt (1) and the washer. Do not tighten.
3. Install the drive belt on the drive pulley of the alternator.
4. Adjust the tension on the drive belt. Refer to the Operation and Maintenance Manual for the correct adjusting procedure.
5. Tighten bolt (1).
6. Tighten nut (2).
7. Connect all wires to the alternator. Use the index marks to correctly connect the wires.

i01748192

## Electric Starting Motor - Remove and Install

SMCS Code: 1453-010

### Removal Procedure

#### WARNING

**Personal injury can result from failure to disconnect the battery.**

**First, disconnect the negative battery cable. Then, disconnect the positive battery cable.**

**A positive power lead can cause sparks if the battery is not disconnected. Sparks can possibly result in battery explosion or fire.**

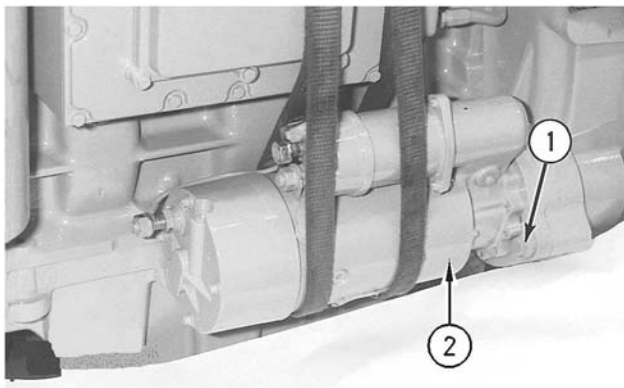


Illustration 345

g00894904

Typical example

1. Place an index mark on all of the electrical wires that are connected to the electric starting motor. Disconnect the electrical wires from the electric starting motor (2).
2. Fasten a suitable lifting device to the electric starting motor.
3. Remove three bolts (1).
4. Remove the electric starting motor and the gasket from the flywheel housing. The weight of the electric starting motor is approximately 29 kg (65 lb).

## Installation Procedure

1. Inspect the condition of the gasket. Replace the gasket, if necessary.

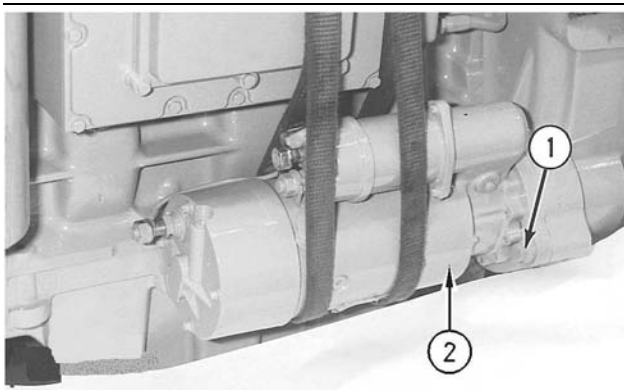


Illustration 346

g00894904

Typical example

2. Install the gasket on the electric starting motor (2).
3. Fasten a suitable lifting device to the electric starting motor.
4. Place the electric starting motor in the flywheel housing. Install three bolts (1).

5. Connect all of the electrical wires to the electric starting motor by using the index marks.

i01024767

## Air Conditioner Support Bracket - Remove and Install

SMCS Code: 1802-010

### Removal Procedure

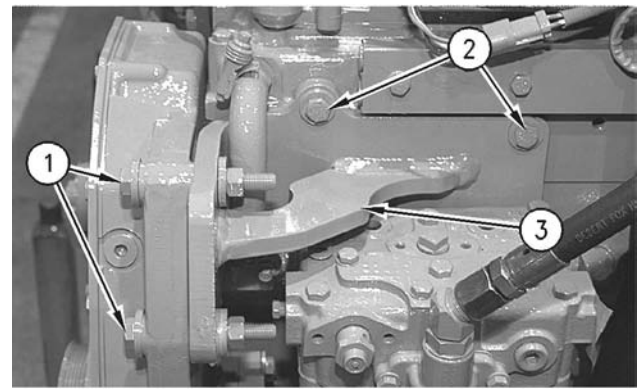


Illustration 347

g00526793

Typical example

1. Remove two bolts (1) with the nuts and the washers.
2. Remove two bolts (2) and the washers.
3. Remove air conditioner compressor support bracket (3).

### Installation Procedure

#### NOTICE

The air conditioner support bracket provides additional support for the mounting pad on the engine front housing. However, due to the increased stiffness of this bracket, extra care is needed when installing the air conditioner compressor support bracket on the engine. Following the correct installation procedure will avoid applying assembly stresses to the engine front housing mounting pad which may lead to a crack.

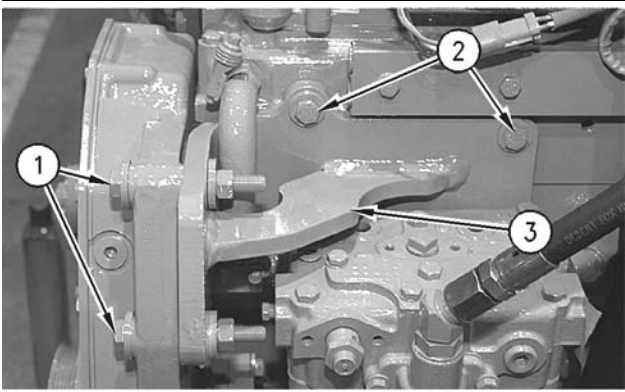


Illustration 348  
Typical example

1. Install air conditioner support bracket (3) with the following procedure.
  - a. Install two bolts (1) with the nuts and the washers. Finger tighten the bolts.
  - b. Install two bolts (2) and the washers. Finger tighten the bolts.

**NOTICE**

Ensure that there is no gap between the air conditioner compressor support bracket and the engine front housing. Ensure that there is no gap between the air conditioner support bracket and the cylinder head. Reposition the air conditioner compressor support bracket in order to eliminate the gap.

- c. Lightly tighten bolts (1) and bolts (2).
- d. Ensure that air conditioner support bracket (3) remains tight against the engine front housing and the cylinder head.
- e. Tighten the bolts to a torque of  $47 \pm 9$  N·m ( $35 \pm 7$  lb ft).

# Air Compressor - Remove and Install

SMCS Code: 1803-010

## Removal Procedure

Table 53

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 138-7573    | Link Bracket     | 1   |

**WARNING**

**Do not disconnect the air lines until the air pressure in the system is at zero. If hose is disconnected under pressure it can cause personal injury.**

**NOTICE**

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

**NOTICE**

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, "Caterpillar Tools and Shop Products Guide" for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

1. Remove the air pressure from the air tank and drain the coolant from the cooling system.

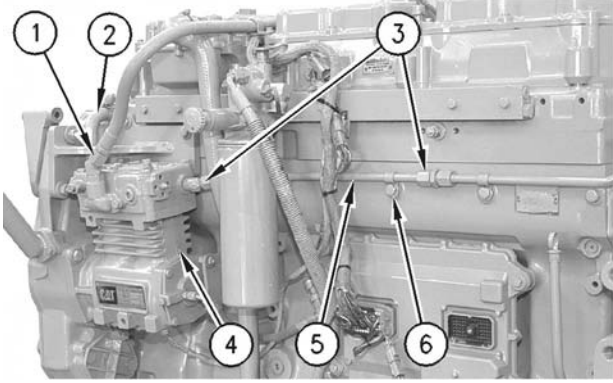


Illustration 349

g00511527

2. Remove bolt (6). Disconnect fittings (3) and remove tube assembly (5).
3. Disconnect hose assembly (1) from the air compressor.
4. Disconnect hose assembly (2).

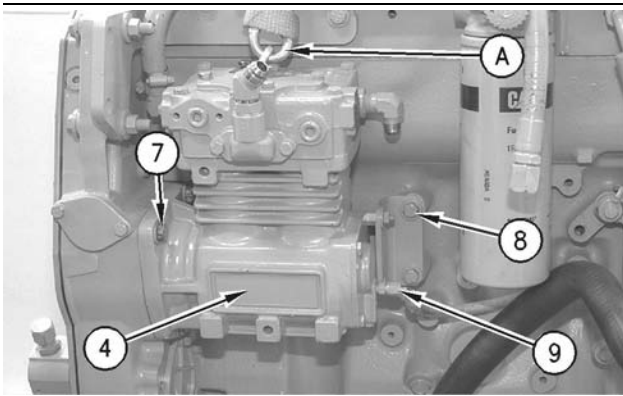


Illustration 350

g00511528

5. Install Tooling (A) and a proper lifting device to the air compressor. The weight of the air compressor is approximately 27 kg (60 lb).
6. Disconnect hose assembly (9) from the air compressor.
7. Remove bolts (7).
8. Use a suitable lifting device. Apply tension to the lifting device and remove bolts (8). Remove air compressor (4) and the gasket from the engine.

## Installation Procedure

Table 54

| Required Tools |             |                  |     |
|----------------|-------------|------------------|-----|
| Tool           | Part Number | Part Description | Qty |
| A              | 138-7573    | Link Bracket     | 1   |

### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

1. Inspect the condition of the gasket. Replace the gasket, if necessary.

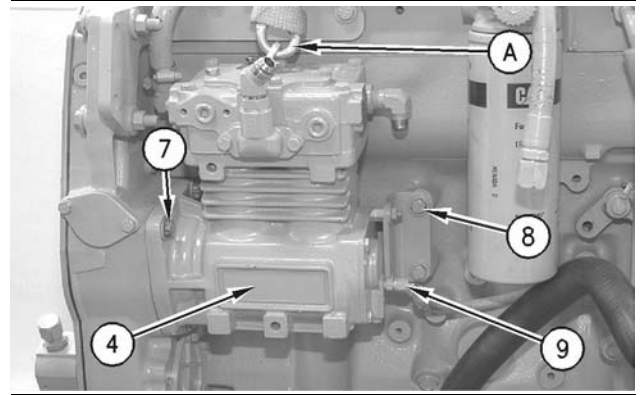


Illustration 351

g00511528

2. Install Tooling (A). Use a suitable lifting device. The weight of the air compressor is approximately 27 kg (60 lb).
3. Position air compressor (4) in the front housing. Ensure that the drive gear engages correctly with the gear in the front housing.
4. Apply 9S-3263 Thread Lock Compound to the threads of bolts (7). Install bolts (7). Tighten bolts (7) to a torque of  $100 \pm 20$  N·m ( $75 \pm 15$  lb ft).
5. Install bolts (8). Tighten bolts (8) to a torque of  $55 \pm 10$  N·m ( $41 \pm 7$  lb ft). Remove Tooling (A).
6. Connect hose assembly (9) to the air compressor.

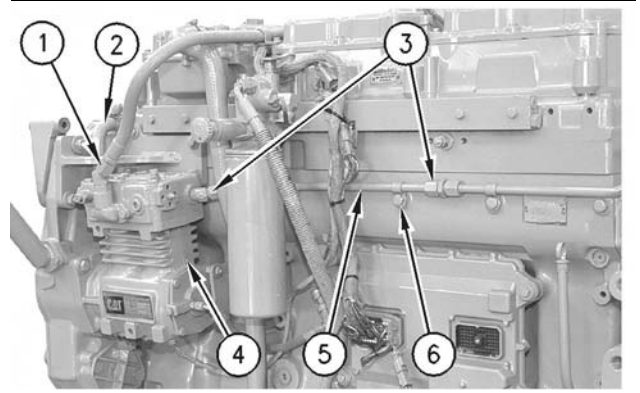


Illustration 352

g00511527

7. Install tube assembly (5) and connect fittings (3).
8. Install bolt (6).

9. Connect hose assembly (2).
10. Connect hose assembly (1).
11. Fill the cooling system with coolant to the correct level. Refer to the Operation and Maintenance Manual for the correct filling procedure.

i01772534

## Air Compressor Drive Gear - Remove

SMCS Code: 1803-011-GE

### Removal Procedure

Table 55

| Required Tools |             |                                             |     |
|----------------|-------------|---------------------------------------------|-----|
| Tool           | Part Number | Part Description                            | Qty |
| A              | 132-5451    | Holding Fixture (Air Compressor Drive Gear) | 1   |

#### Start By:

- a. Remove the air compressor. Refer to Disassembly and Assembly, "Air Compressor - Remove and Install".

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

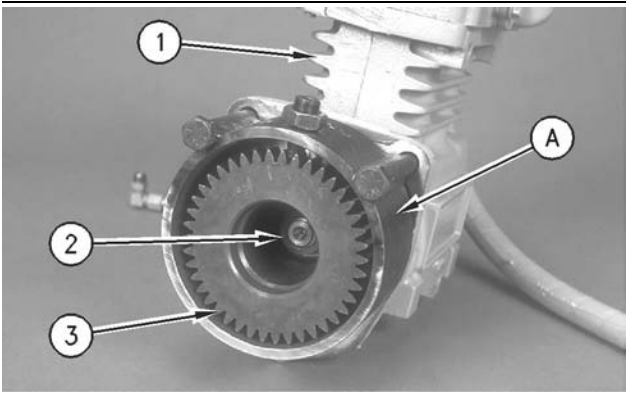


Illustration 353

Typical example

1. Mount air compressor (1) in a bench vise.
2. Install Tooling (A) on air compressor (1), as shown.

3. Remove nut (2).
4. Remove Tooling (A) and air compressor drive gear (3).

i01768768

## Air Compressor Drive Gear - Install

SMCS Code: 1803-012-GE

### Installation Procedure

Table 56

| Required Tools |             |                                             |     |
|----------------|-------------|---------------------------------------------|-----|
| Tool           | Part Number | Part Description                            | Qty |
| A              | 132-5451    | Holding Fixture (Air Compressor Drive Gear) | 1   |

#### NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

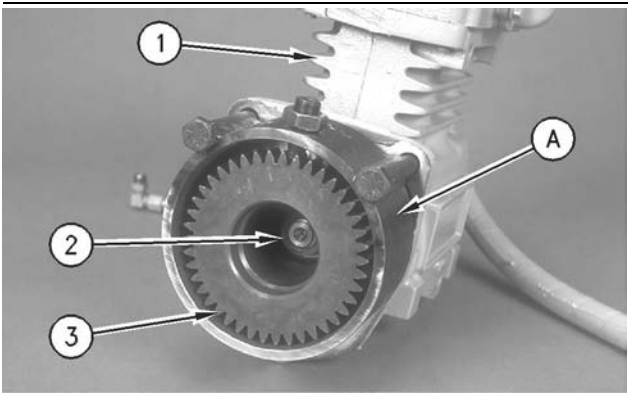


Illustration 354

Typical example

g00524786

1. Mount air compressor (1) in a bench vise.
2. Install air compressor drive gear (3) on the air compressor crankshaft.
3. Install nut (2).
4. Install Tooling (A) on air compressor (1), as shown.
5. Refer to Specifications, "Air Compressor Drive Gear - Install" for the correct torque for the air compressor drive gear.

6. Remove Tooling (A).

**End By:**

- a. Install the air compressor. Refer to Disassembly and Assembly, "Air Compressor - Remove and Install".

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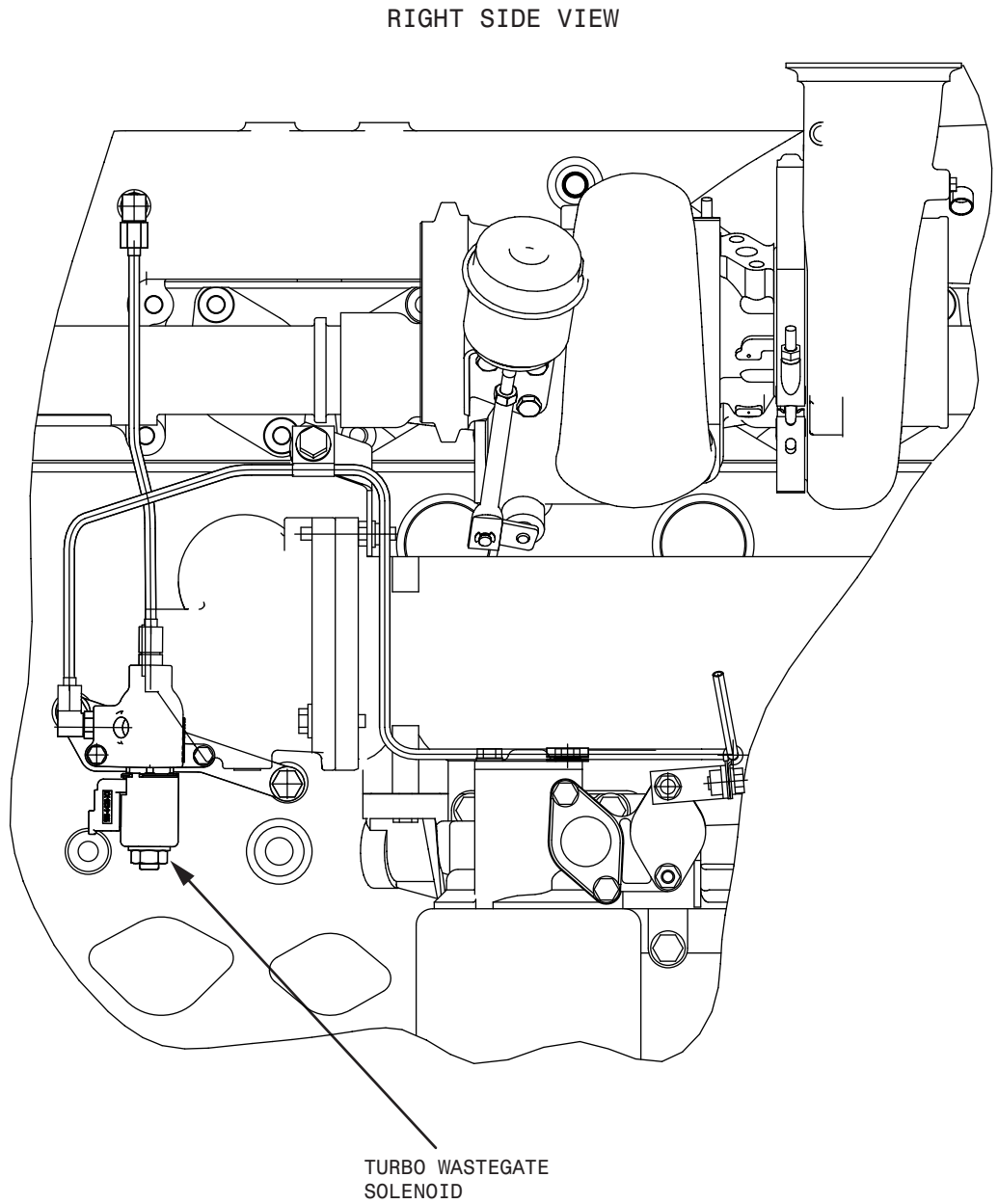
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# Schematic

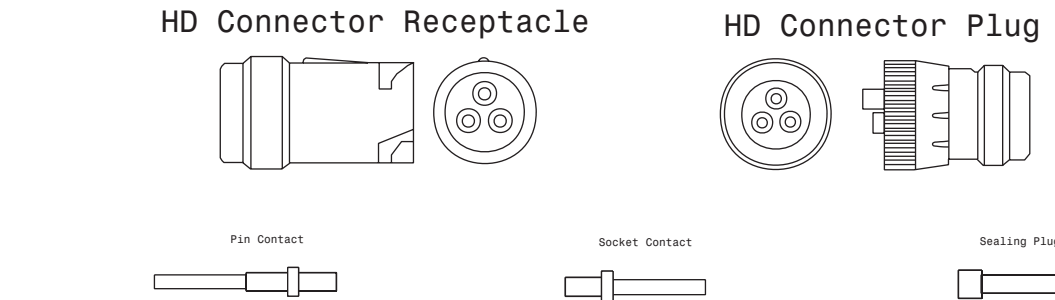
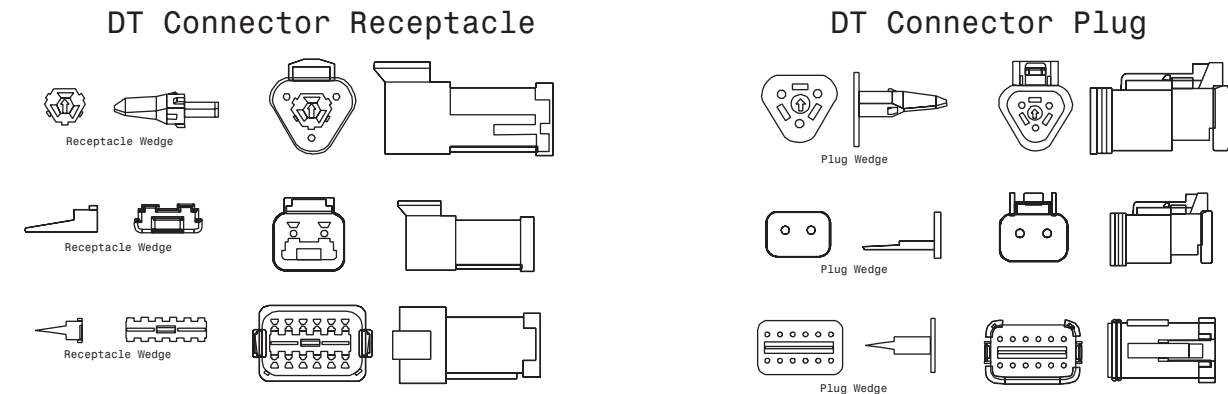
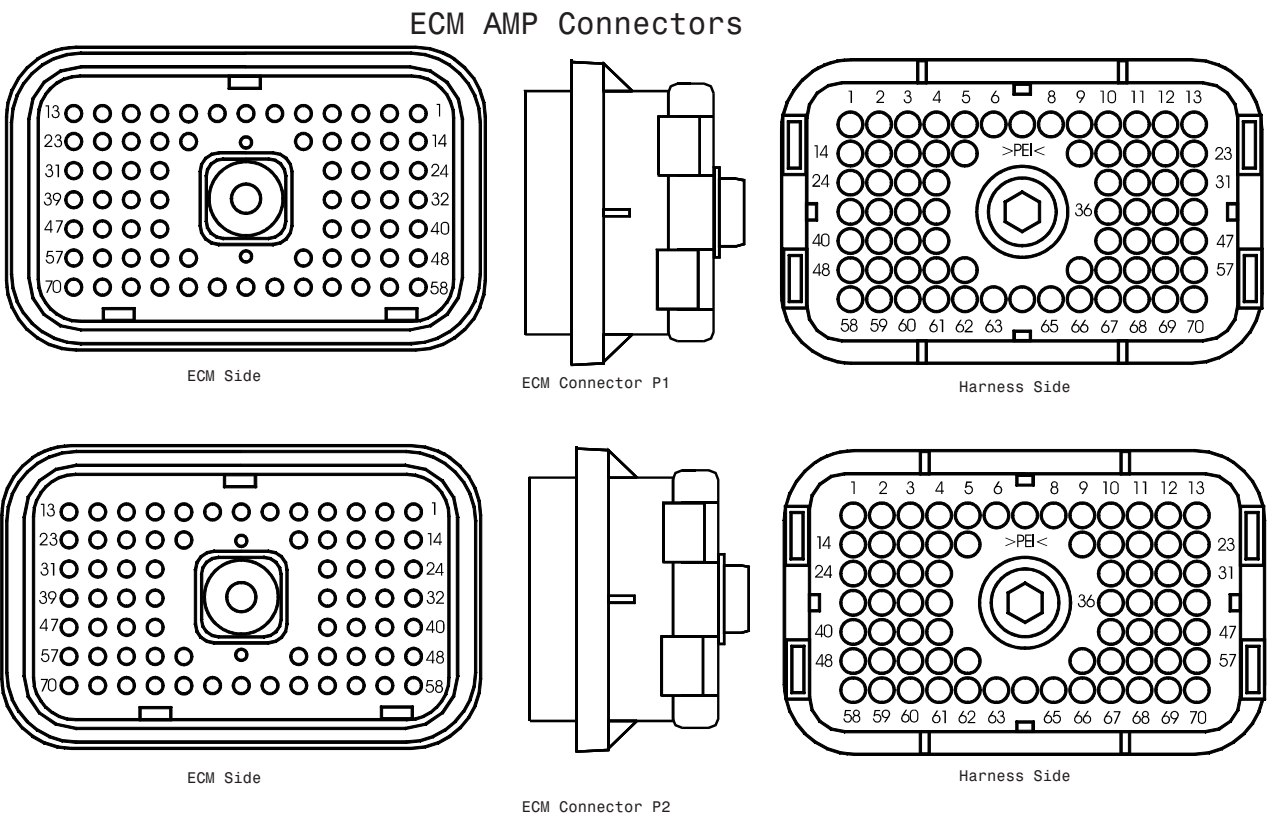
## C-10 and C-12 Truck Engine Electrical System

MBJ1-UP  
MBL1-UP

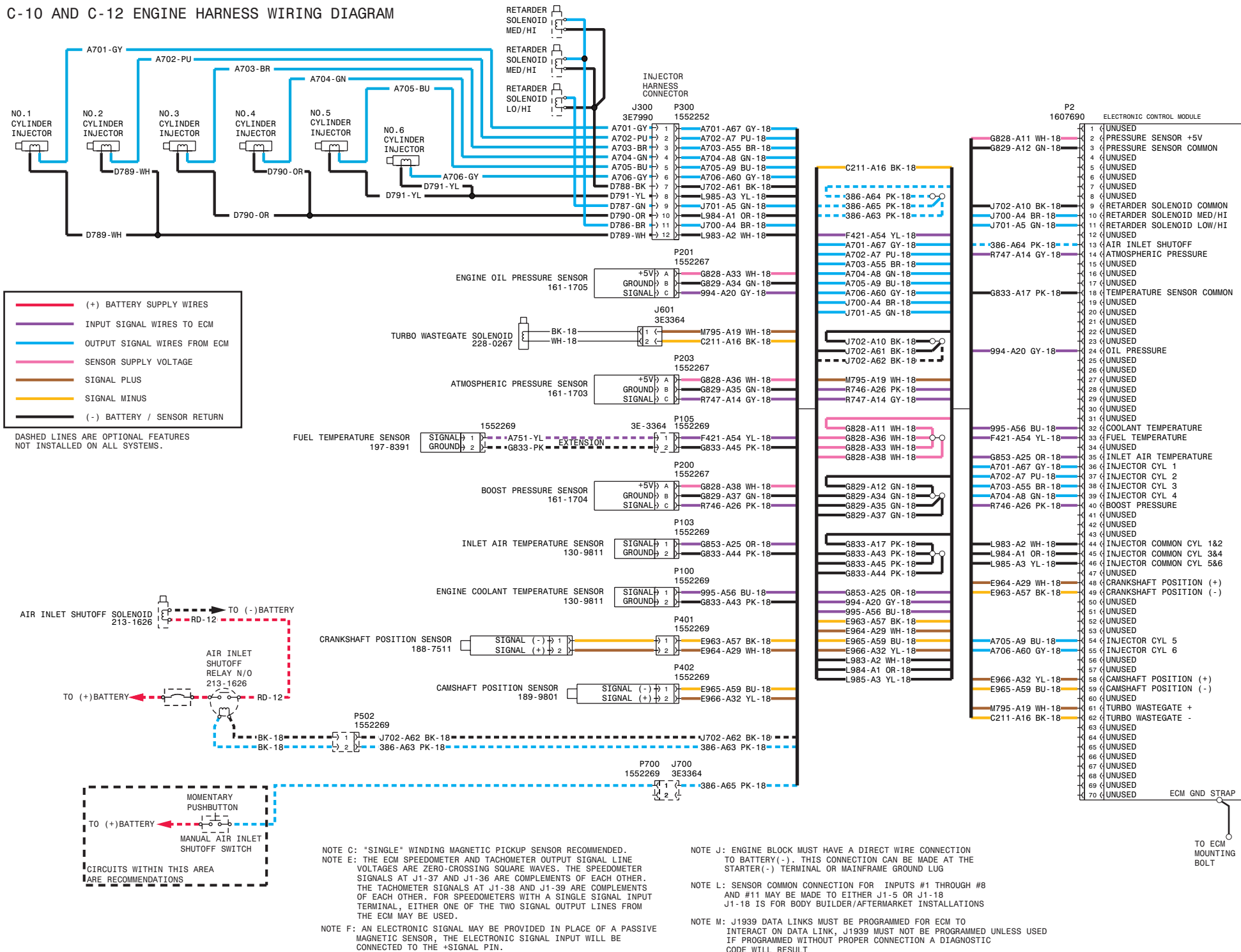
| Diagnostic Code Listing |                                              |             |         |                                                 |             |
|-------------------------|----------------------------------------------|-------------|---------|-------------------------------------------------|-------------|
| PID-FMI                 | Description                                  | Flash Codes | PID-FMI | Description                                     | Flash Codes |
| 0-00                    | No Detected Faults                           | 55          | 105-03  | Inlet Manifold Temperature Sensor Open Circuit  | 38          |
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| 71-01                   | Idle Shutdown Occurrence                     | 47          | 173-00  | High Exhaust Gas Temperature Derate             | 16          |
| 71-14                   | PTO Shutdown Timer Occurrence                | 47          | 173-11  | Very High Exhaust Gas Temperature Derate        | 16          |
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| 100-11                  | Very Low Oil Pressure                        | 46          | 232-04  | 5 Volt Supply Below Normal                      | 21          |
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| 102-02                  | Erratic Boost Pressure                       | 25          | 247-11  | Brake Switch #2 Fault                           | 00          |
| 102-03                  | Boost Pressure Sensor Open Circuit           | 25          | 249-11  | J1922 Data Link Fault                           | 58          |
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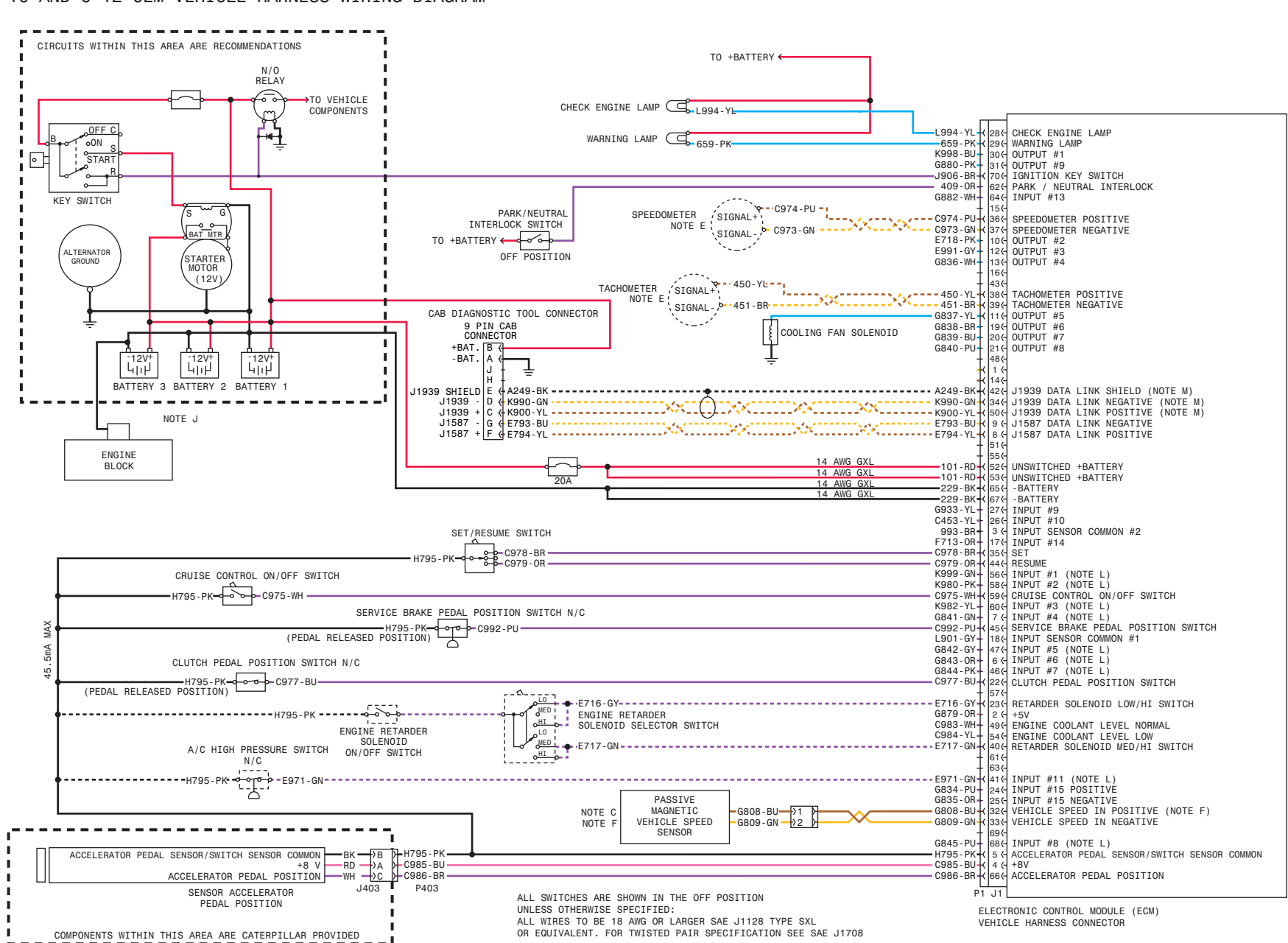
TURBO WASTEGATE SOLENOID



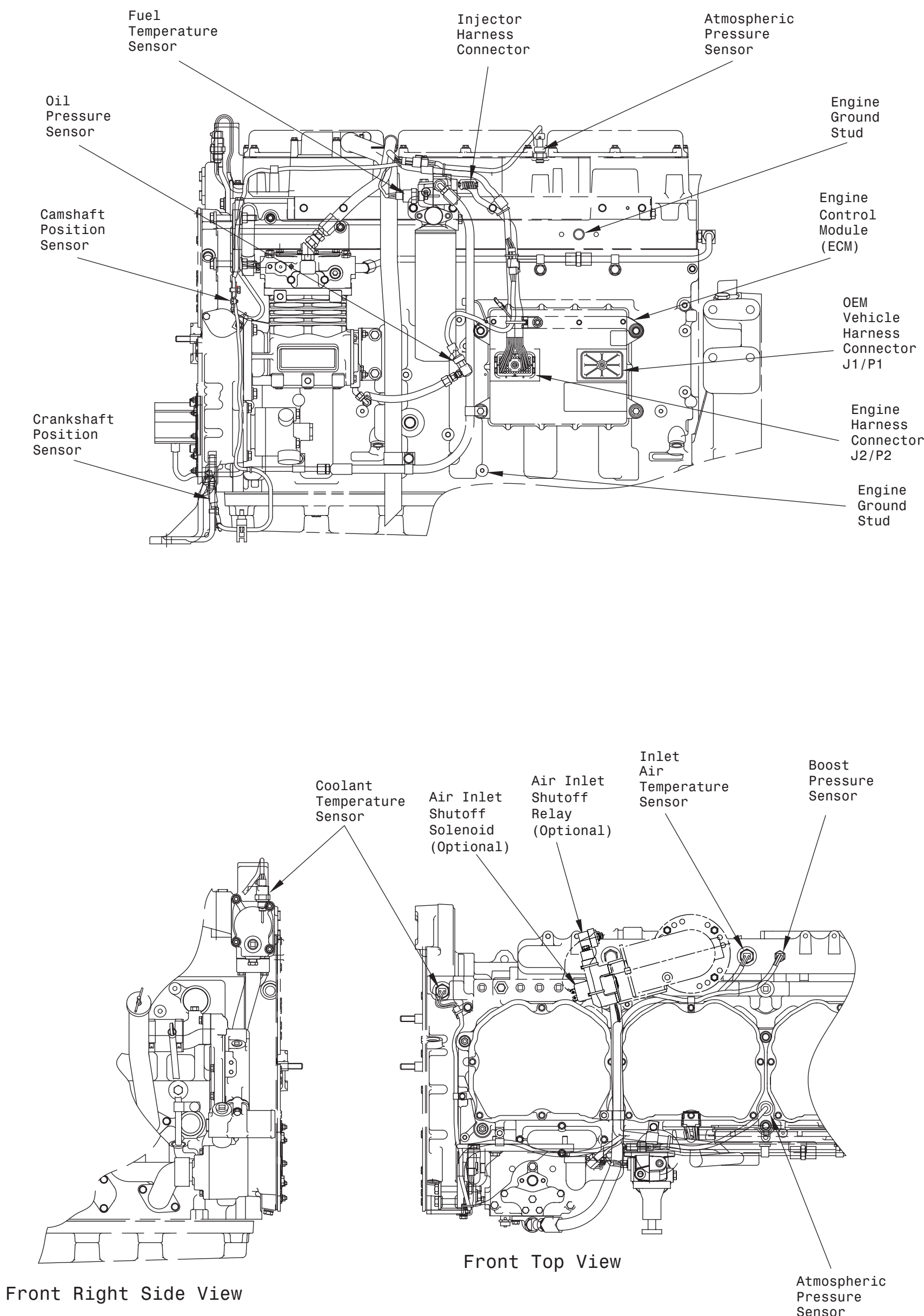
C-10 AND C-12 ENGINE HARNESS WIRING DIAGRAM



C-10 AND C-12 OEM VEHICLE HARNESS WIRING DIAGRAM



Left Side View



Front Top View



Front Right Side View



SENR9667-00